SNN–Transformer Hybrids: Full Master Extraction Table

Complete 60-field record for all 118 included primary studies (P001–P118). One card per paper. "NR" = not reported.

P001

Bishop: Sparsified Bundling Spiking Transformers on Heterogeneous Cores with Error-constrained Pruning

Full Citation	Xu, B., Yin, Y., Iyer, V., & Li, P. (2025). Bishop: Sparsified Bundling Spiking Transformers on Heterogeneous Cores with Error-constrained Pruning. In Proceedings of the 52nd Annual International Symposium on Computer Architecture (ISCA ’25), June 21–25, 2025, Tokyo, Japan. ACM, New York, NY, USA, 14 pages. https://doi.org/10.1145/3695053.3731063
University	University of California
Country	United States
Publication Year	2025
Publication Venue	Proceedings of the 52nd Annual International Symposium on Computer Architecture, ISCA ’25
Publisher	ACM
DOI / URL	https://doi.org/10.1145/3695053.3731063
Publication Type	Conference paper
Database Source	ACM Digital Library
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Efficient inference acceleration for spiking Transformer workloads on image, event-based gesture, and speech-command recognition benchmarks
Data Modality	Static images, event-camera/DVS data, and speech-command data
Signal Dimensionality	2D image/token sequences; 2D spatiotemporal event data; sequence-based speech-command tokens
Signal Characteristics	CIFAR images are 32 × 32 and split into 64 tokens; ImageNet-100 images are 224 × 224 and split into 196 tokens; DVS-Gesture input is 128 × 128 and split into 64 tokens; models use 4–20 timesteps depending on dataset
Dataset Name	CIFAR10, CIFAR100, ImageNet-100, DVS-Gesture-128, Google Speech Command V2
Dataset Type	Public benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Spiking Transformer workload accelerated through a dedicated HW/SW co-design framework; model uses spiking tokenizer, multi-head spiking self-attention, spiking MLP blocks, and LIF-based spike processing
SNN Component	LIF neurons, spiking tokenizer, spiking MLP, spiking Q/K/V projections, spike generator, binary spike activations
Neuron Model	LIF
Transformer Component	Multi-head Spiking Self-Attention block, residual encoder blocks, Q/K/V/O projection layers, spiking MLP block
Attention Type	Multi-head spiking self-attention; binary Q/K/V attention; softmax-free spiking attention
SNN–Transformer Interface Mechanism	Spiking tokenizer converts input into T × N × D spiking token representation; LIF layers generate spike-form Q/K/V; Token-Time Bundle represents spiking tokens across time for hardware processing
Model Depth / Scale	Model 1: CIFAR10, 4 blocks, T=10, N=64, D=384; Model 2: CIFAR100, 4 blocks, T=8, N=64, D=384; Model 3: ImageNet-100, 8 blocks, T=4, N=196, D=128; Model 4: DVS-Gesture, 2 blocks, T=20, N=64, D=128; Model 5: Google SC, 4 blocks, T=8, N=256, D=384
Parameter Count	NR
Design / Contribution Type	New accelerator architecture and HW/SW co-design framework; Token-Time Bundle representation; heterogeneous dense/sparse/attention cores; Bundle-Sparsity Aware training; Error-Constrained TTB Pruning
Input Encoding Method	Spiking tokenizer / token-based spiking representation; exact spike encoding rule for each dataset is NR
Encoding Parameters	Timesteps: 4–20 depending on dataset; token counts: 64, 196, or 256; feature dimensions: 128 or 384; TTB bundle parameters BSt and BSn explored
Encoding Learned or Fixed	NR
Event / Spike Representation	Binary spike tensor; spiking activations arranged as Token-Time Bundles; binary spiking Q/K/V; active and inactive TTBs
Temporal Resolution	4 timesteps for ImageNet-100, 8 for CIFAR100 and Google Speech Command, 10 for CIFAR10, 20 for DVS-Gesture
Output Decoding Method	Global average pooling across tokens and time followed by a linear prediction layer; hardware spike generator produces binary spike outputs
Encoding / Temporal Information Analysis	Yes. The paper analyzes spatiotemporal sparsity, Token-Time Bundle sparsity, active/inactive bundle distribution, bundle size sensitivity, and pruning effects on attention computation
Training Paradigm	Supervised learning for benchmark classification models; HW/SW co-design training pipeline
Training Method	Bundle-Sparsity Aware training and ECP-aware training; loss combines cross-entropy with bundle-level sparsity loss
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	Cross-entropy loss plus bundle-level sparsity regularization: Ltot = LCE + λLbsp
Training Hyperparameters	Batch size 256 and 300 epochs for CIFAR and ImageNet-100; batch size 64 and 100 epochs for DVS-Gesture; λ = 1 for CIFAR10, 0.5 for CIFAR100, 0.3 for ImageNet, 1.0 for DVS-Gesture; ECP pruning threshold 10 for DVS-Gesture and 6 for other models
Primary Performance Metric	Latency speedup and energy efficiency improvement
Primary Metric Value	Average 5.91× speedup and 6.11× energy-efficiency improvement over PTB SNN accelerator; 299× average speedup over NVIDIA Jetson Nano edge GPU
Secondary Metrics	Accuracy, chip area, power, latency, energy dissipation, EDP, pruning ratio, memory access reduction, attention-layer speedup
Baseline Models Compared	NVIDIA Jetson Nano edge GPU; Parallel Time Batching SNN accelerator; PTB; comparison also includes ANN and SNN accuracy references in background
Performance Gain Over Baseline	Bishop+BSA+ECP achieves up to 7.73× speedup over PTB on ImageNet-100 and 4.06× on DVS-Gesture; attention core reduces latency by 10.7–23.3× and gives 1.39–1.96× energy saving over PTB

Statistical Validation	Benchmark-based evaluation and cycle-accurate hardware simulation; no repeated runs, standard deviation, confidence interval, or significance test reported
Ablation Study	Full. Includes BSA effect, ECP threshold analysis, stratification strategy, TTB bundle volume sensitivity, hardware area/power breakdown, and latency/energy comparisons
Training Framework / Code Availability	PyTorch-based training flow; CACTI 7.0 for memory energy estimation; STONNE simulator for sparse core simulation; code availability NR
Target Hardware	Dedicated heterogeneous ASIC-style accelerator; 28 nm CMOS synthesis; dense core, sparse core, attention core, spike generator, GLB/DRAM memory hierarchy
Hardware Deployment Status	RTL synthesis and cycle-accurate simulation; no fabricated chip or real neuromorphic hardware deployment reported
Energy Evidence	Estimated energy using CACTI 7.0, DRAM power model, RTL synthesis, and cycle-accurate accelerator simulation; not measured from fabricated silicon
Latency / Memory / Complexity Evidence	Reports 2.96 mm^2 die area, 627 mW peak power, 500 MHz clock; 144 KB weight GLB; two 12 KB spike TTB GLBs; reports latency, energy, EDP, memory access, and attention computation reduction
Comparison with ANN Baseline	Yes. Uses NVIDIA Jetson Nano edge GPU as an ANN hardware baseline and includes ANN/SNN accuracy comparison table, but the main efficiency comparison is against PTB SNN accelerator
Main Contribution / Key Limitation	Contribution: first dedicated hardware accelerator and HW/SW co-design framework for spiking Transformers using Token-Time Bundles, heterogeneous cores, BSA training, and ECP pruning. Limitation: evaluation is based on simulation and RTL synthesis, not fabricated hardware; dataset focus is mainly classification benchmarks
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4

P002

An Efficient Hybrid Cascade Tracker with Spiking Neural Networks for Event Domain Tracking

Full Citation	Zhou, Y., Yin, H., Tan, C., Wu, Q., Sun, C., & Hong, R. (2026). An Efficient Hybrid Cascade Tracker with Spiking Neural Networks for Event Domain Tracking. ACM Transactions on Multimedia Computing, Communications, and Applications, 22(3), Article 75, 20 pages. https://doi.org/10.1145/3795518
University	Anhui University
Country	China
Publication Year	2026
Publication Venue	ACM Transactions on Multimedia Computing, Communications, and Applications
Publisher	ACM
DOI / URL	https://doi.org/10.1145/3795518
Publication Type	Journal article
Database Source	ACM Digital Library
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Single object tracking / bounding-box prediction in event streams
Data Modality	Event-camera data; simulated event data from RGB frame datasets
Signal Dimensionality	2D spatiotemporal event representation; event voxel/grid-style frame representation
Signal Characteristics	Event stream represented as quadruples (x, y, t, p) with spatial location, timestamp, and polarity; events between adjacent frames are aggregated into an n-bin voxel grid with positive and negative polarity frames
Dataset Name	VisEvent, FE108, COESOT, Event-OTB, Event-VOT
Dataset Type	Public event-based tracking benchmarks; simulated event datasets converted from RGB tracking benchmarks
Dataset Size	VisEvent: 820 videos / 371,127 frames; FE108: 108 videos / 208,672 frames; COESOT: 1,354 videos / 478,721 frames; OTB100/Event-OTB: 100 videos / 59,040 frames; VOT2019/Event-VOT: 60 videos / 19,935 frames
Dataset Balance / Split	Uses public benchmark training/testing protocols; exact class/object balance NR
Hybrid Topology	Parallel/cascade hybrid CNN–SNN–Transformer topology; EHCT uses repeated HCST blocks containing C-SNN and T-SNN branches, connected by a T2C Adapter and fused by CTFM
SNN Component	C-SNN branch, T-SNN branch, LIF layers, spike-based self-attention, spike-based MLP, spike-driven temporal encoding
Neuron Model	LIF
Transformer Component	Transformer-SNN encoder block; spike-based self-attention; spike-based MLP; T-SNN branch
Attention Type	Spike-based self-attention; scaled dot-product attention in CTFM fusion module
SNN–Transformer Interface Mechanism	Q and K are projected then converted into spike sequences using LIF neurons; spike-form Q and K allow dot-product computation through logical AND and summation; T2C Adapter concatenates T-SNN and C-SNN outputs, applies 1 × 1 convolution, then LIF for spike-form compatibility
Model Depth / Scale	HCST block number N tested from 1 to 5; final model uses N = 3 HCST blocks; T-SNN can be stacked by N layers
Parameter Count	NR
Design / Contribution Type	New event-domain hybrid cascade tracker; HCST block; C-SNN and T-SNN dual-branch design; T2C Adapter; CTFM fusion module
Input Encoding Method	Event-to-frame / voxel-grid representation; events between adjacent frames are aggregated into n bins and separated into positive and negative polarity frames
Encoding Parameters	Event tuple (x, y, t, p); n-bin voxel grid; positive and negative event frames; timestep T = 5 for SNN energy analysis; LIF threshold h = 1.0; leakage factor uses $\tau = 2.0$
Encoding Learned or Fixed	Fixed event representation; feature fusion and projection layers are learned
Event / Spike Representation	Event frames, positive/negative polarity frames, spike sequences, spike-form Q and K, LIF spike activations
Temporal Resolution	Event cameras provide microsecond-level temporal resolution; SNN timestep T = 5 in energy analysis
Output Decoding Method	Classifier and regressor heads produce the tracking bounding box
Encoding / Temporal Information Analysis	Partial. The paper analyzes event sparsity, firing rate of LIF layers, spike-driven temporal encoding, and feature visualizations, but does not deeply quantify temporal information loss
Training Paradigm	Supervised learning within Siamese tracking framework
Training Method	Surrogate-gradient direct training using SpikeJelly; SiamFC++/STNet-style tracking loss framework

Surrogate Gradient Function	ATan surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	Stochastic Gradient Descent
Loss Function	Follows SiamFC++ and STNet tracking loss; exact full loss formula NR
Training Hyperparameters	Batch size 32; trained on NVIDIA RTX 3090 GPUs; backbone frozen in first epoch; backbone learning rate 0.1 from epoch 2; other components use learning rate 0.001 to 0.01 during epochs 1–5 then decay 10× every 5 epochs; SGD momentum 0.9; weight decay 5e-5; LIF threshold h = 1.0; τ = 2.0
Primary Performance Metric	Success Rate, Precision Rate, Overlap Precision
Primary Metric Value	VisEvent: SR 39.6, OP0.50 46.5, OP0.75 25.6, PR 55.5; FE108: SR 54.5, OP0.50 69.0, OP0.75 22.9, PR 85.1; COESOT: SR 45.1, OP0.50 53.9, OP0.75 27.2, PR 52.2
Secondary Metrics	Normalized Precision Rate, FPS, firing rate, theoretical energy consumption
Baseline Models Compared	ATOM, DiMP, PrDiMP, TransT, ToMP, STARK, MixFormer, MDNet, CEUTrack, HDETrack, FENet, STNet, DANet, AFNet, HiT, FERMT
Performance Gain Over Baseline	On VisEvent, EHCT achieves best PR 55.5 and OP0.50 46.5; on FE108, EHCT achieves best SR 54.5 and PR 85.1; on Event-OTB and Event-VOT, EHCT outperforms CEUTrack, STNet, AFNet, and HDETrack in SR and PR
Statistical Validation	Public benchmark evaluation and ablation experiments; no standard deviation, confidence interval, repeated-run analysis, or significance test reported
Ablation Study	Full. Includes C-SNN vs T-SNN branches, with/without SNN, T2C Adapter variants, CTFM variants, number of HCST blocks, lightweight backbone, and single-block variants
Training Framework / Code Availability	PyTorch; SpikingJelly; source code stated to be available at GitHub: masac11/EHCT
Target Hardware	GPU/CPU for implementation; future neuromorphic hardware and embedded platforms mentioned
Hardware Deployment Status	Evaluated on PC/GPU; no real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimate using SOP, firing rate, timestep, FLOPs, MAC energy 4.6 pJ, and AC energy 0.9 pJ; average firing rate about 2.1567%; claims approximately 45× energy efficiency over ANN with same structure
Latency / Memory / Complexity Evidence	EHCT runs at 95 FPS on VisEvent; inference requires 2 GB GPU memory and 5 GB CPU RAM; lightweight variants report up to 111 FPS but lower accuracy
Comparison with ANN Baseline	Yes. Compares with RGB/ANN-based trackers such as ATOM, DiMP, PrDiMP, TransT, ToMP, STARK, MixFormer, MDNet, HiT, and FERMT
Main Contribution / Key Limitation	Contribution: proposes EHCT, a hybrid CNN–SNN–Transformer tracker for event-domain object tracking, using HCST blocks, T2C Adapter, and CTFM fusion to combine local spike features and global Transformer context. Limitation: struggles under extreme fast motion and long occlusion; energy evidence is theoretical and no real neuromorphic deployment is reported
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P003

Temporally Dynamic Spiking Transformer Network for Speech Enhancement

Full Citation	Alohali, M. A., Saleem, N., Rhouma, D., Medani, M., Elmannai, H., & Bourouis, S. (2024). Temporally Dynamic Spiking Transformer Network for Speech Enhancement. IEEE Access, 12, 146513–146526. https://doi.org/10.1109/ACCESS.2024.3444596
University	Princess Nourah Bint Abdulrahman University
Country	Saudi Arabia
Publication Year	2024
Publication Venue	IEEE Access
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/ACCESS.2024.3444596
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Speech, Audio, and Auditory Processing
Specific Task	Single-channel speech enhancement / noisy speech reconstruction
Data Modality	Audio signal represented as complex spectrogram
Signal Dimensionality	1D temporal speech waveform converted to 2D time-frequency spectrogram
Signal Characteristics	Input is real and imaginary parts of noisy spectrogram, $Y \in R(2 \times T \times F)$; sampling rate 16 kHz; STFT window length 512 samples; frame shift 16 ms
Dataset Name	WSJ0-SI84; VCTK+DEMAND
Dataset Type	Public benchmark speech enhancement datasets
Dataset Size	WSJ0-SI84: 3000 training utterances, 1000 test utterances, approximately 150 hours of training mixtures; VCTK+DEMAND: 11,572 training mixtures and 824 test mixtures
Dataset Balance / Split	WSJ0-SI84: training utterances from 83 speakers, testing from unseen speakers; SNR −5 dB to 5 dB for training and −3, 0, 3, 6 dB for testing. VCTK+DEMAND: 28 training speakers, 2 test speakers; training SNRs 0, 5, 10, 15 dB; testing SNRs 2.5, 7.5, 12.5, 17.5 dB
Hybrid Topology	CNN encoder-decoder with a dual-branch Spiking Transformer Network bottleneck
SNN Component	Spike neuron layers, spike-based Q/K/V, spiking self-attention, spiking Transformer encoder
Neuron Model	NR
Transformer Component	Spiking Transformer Network bottleneck, spiking self-attention, MLP block, temporal attention branch, frequency attention branch
Attention Type	Spiking self-attention; temporal attention; frequency attention
SNN–Transformer Interface Mechanism	Q, K, and V are first generated using learnable matrices and batch normalization, then transformed into spike sequences through spike neuron layers; spike-based Q/K/V are used in the self-attention computation
Model Depth / Scale	Encoder has four convolutional layers with multi-scale feature blocks; decoder mirrors encoder with four sub-pixel deconvolutional layers; bottleneck uses dual-branch STN

Parameter Count	3.14 million parameters
Design / Contribution Type	New application architecture combining convolutional encoder-decoder, multi-scale feature extraction, and dual-branch Spiking Transformer bottleneck for speech enhancement
Input Encoding Method	Complex spectrogram input; real and imaginary parts of noisy speech spectrogram are processed by CNN encoder, then STN converts projected features into spike-based Q/K/V
Encoding Parameters	Sampling rate 16 kHz; window length 512 samples; frame shift 16 ms; spike Q/K/V generated through spike neuron layers; α and β adaptive branch weights initialized as 1
Encoding Learned or Fixed	Partially learned: spectrogram extraction is fixed, while Q/K/V projections and branch weights are learned
Event / Spike Representation	Spike sequences for Q, K, and V; spike-based attention representations with binary 0/1 spike values
Temporal Resolution	Frame shift 16 ms; temporal frames T in spectrogram; exact SNN timestep NR
Output Decoding Method	Decoder outputs real and imaginary components of the estimated clean spectrum, followed by inverse STFT to reconstruct waveform
Encoding / Temporal Information Analysis	Partial. The paper discusses temporal/spectral dependency modeling and visualizes spectrogram quality, but does not deeply analyze spike encoding information loss or spike temporal preservation
Training Paradigm	Supervised learning
Training Method	Direct end-to-end training of CNN encoder-decoder with Spiking Transformer bottleneck
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Weighted spectro-temporal loss: $L = \upsilon L_{time} + (1-\upsilon) L_{frequency}$, with $\upsilon = 0.4$
Training Hyperparameters	Batch size 16; 100 epochs; Adam with $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, ε = 10■■; learning rate starts at 0.001 and decays to 0.0005, 0.0002, and 0.0001; dropout 0.20
Primary Performance Metric	PESQ, ESTOI, SI-SDR
Primary Metric Value	WSJ0-SI84 seen condition average: ΔSI-SDR = 11.52 dB, ΔESTOI = 39.67%, ΔPESQ = 1.22. VCTK+DEMAND: STOI = 95.0, PESQ = 3.13, SNRSeg = 11.12
Secondary Metrics	STOI, SNRSeg, FW-SNRSeg, Csig, Cbak, Covl, MACs, RTF, memory footprint
Baseline Models Compared	CRN, DPRNN, GCRN, DCCRN, AECNN, CTSNet, GaGNet, CPBNet, PHASEN, RDLNet, DEMUCS, TSTNN, MSSA-TCN, FAFNet, PFRNet, UTFAT, DBS4D, DBCRN, SADN-UNet, NSNet, DPT-FSNet, DPTGAN, DPT-ECA, FullSubNet+, FRCRN, DF-Net
Performance Gain Over Baseline	On WSJ0-SI84, TD-STNet outperforms CRN by 3.83 dB ΔSI-SDR, 7.28% ΔESTOI, and 0.35 ΔPESQ. On VCTK+DEMAND, it improves over several baselines while keeping 3.14M parameters and 0.18 RTF
Statistical Validation	Benchmark evaluation on seen/unseen speakers and noises; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes TD-STNet with traditional Transformer, without STN bottleneck, STN with only FAB, STN with only TAB, and full dual-branch STN
Training Framework / Code Availability	Framework NR; code availability NR
Target Hardware	CPU/GPU evaluation; real-time speech enhancement setting
Hardware Deployment Status	Evaluated computationally on CPU/GPU; no neuromorphic hardware deployment reported
Energy Evidence	No direct energy measurement; efficiency discussed through parameters, MACs, real-time factor, inference time, and memory footprint
Latency / Memory / Complexity Evidence	3.14M parameters; 9.64 G/s MACs; 0.18 RTF; approximately 10.12 MB memory; CPU processing time 0.20 s; PMFP 0.28 G; BMFP 0.29 G; training batch time 0.25 s
Comparison with ANN Baseline	Yes. Compares against many ANN/CNN/RNN/Transformer-based speech enhancement baselines
Main Contribution / Key Limitation	Contribution: proposes TD-STNet, a convolutional encoder-decoder with dual-branch Spiking Transformer bottleneck for temporal and spectral speech enhancement. Limitation: attention only uses current input at each time step, underusing historical temporal information; no real neuromorphic hardware or measured energy evaluation
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P004

A 28nm Spiking Vision Transformer Accelerator with Dual-Path Sparse Compute Core and EMA-free Self-Attention Engine for Embodied Intelligence

Full Citation	Fang, C., Shen, Z., Li, T., Zhao, S., Tian, F., Yang, J., & Sawan, M. (2025). A 28nm Spiking Vision Transformer Accelerator with Dual-Path Sparse Compute Core and EMA-free Self-Attention Engine for Embodied Intelligence. IEEE Transactions on Circuits and Systems for Artificial Intelligence. https://doi.org/10.1109/TCASAI.2025.3596210
University	Westlake University
Country	China
Publication Year	2025
Publication Venue	IEEE Transactions on Circuits and Systems for Artificial Intelligence
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TCASAI.2025.3596210
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Real-time visual cortex acceleration for embodied intelligence tasks, including manipulation and navigation
Data Modality	RGB image frames encoded into spike format
Signal Dimensionality	2D image sequence / spatiotemporal visual input
Signal Characteristics	Uses three consecutive visual frames per inference; input resolutions include 256 × 256 and 128 × 128; RGB input is encoded into spike format with 8 timesteps

Dataset Name	CortexBench tasks; Adroit; MetaWorld; DeepMind Control; ObjectNav; ImageNav; HM3DSem; Gibson; Mujoco-based datasets
Dataset Type	Public embodied AI / robotic manipulation / navigation benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Spiking Vision Transformer visual cortex accelerator; spiking tokenizer/encoder converts RGB input to spike format, followed by spiking Transformer blocks and an off-chip non-spiking policy MLP
SNN Component	Spike encoder layer, spike-format feature maps, spiking Transformer blocks, spike-based Q/K/V, 1-bit spike activations
Neuron Model	Spiking neurons used for input encoding; exact neuron model NR
Transformer Component	Spiking Vision Transformer; spiking Transformer blocks; Spiking Self-Attention; ViT-B and ViT-L visual cortex models
Attention Type	Spiking self-attention; linear attention; softmax-free SSA
SNN–Transformer Interface Mechanism	RGB frames are encoded into spike format with 8 timesteps; patch embeddings are processed by spiking Transformer blocks; Q, K, and V are represented as 1-bit spike matrices in SSA
Model Depth / Scale	ViT-B and ViT-L are mapped; typical target SSA setting uses N = 256 and head dimension d = 64; ideal hardware efficiency supports N ≤ 4096 and D ≤ 4096
Parameter Count	ViT-B: 86M parameters; ViT-L: 307M parameters
Design / Contribution Type	Fabricated 28nm Spiking Vision Transformer accelerator; dual-path sparse compute core; group-wise frame-differential dataflow; EMA-free SSA engine; first-order SSA approximation; KV-first dataflow; transpose engine; unified 1b/8b adder-tree array
Input Encoding Method	RGB-to-spike encoding using an encoder layer with spiking neurons; group-wise frame-differential dataflow uses 1 raw frame and 2 differential frames
Encoding Parameters	Spike timestep = 8; input group size = 3 frames; block-wise CSR compression for differential frames; 1-bit spike representation; INT8 precision for weights/operations
Encoding Learned or Fixed	Partially learned: spike encoder layer is part of the model, while frame-differential dataflow and block-wise CSR compression are hardware/dataflow mechanisms
Event / Spike Representation	1-bit spike feature maps; raw bitmaps for raw frames; block-wise CSR format for differential sparse frames; spike-based Q/K/V matrices
Temporal Resolution	8 SNN timesteps; 3-frame temporal window for embodied intelligence inference
Output Decoding Method	CLS tokens from three frame embeddings are extracted and concatenated, then passed to a 3-layer non-spiking policy MLP to generate the agent action
Encoding / Temporal Information Analysis	Yes. The paper analyzes input sparsity, frame-to-frame similarity, frame-differential sparsity, cosine similarity between approximated and exact SSA, and energy efficiency under different sparsity levels
Training Paradigm	Pretrained visual representation / embodied AI inference setting; exact SNN training paradigm NR
Training Method	NR
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	NR
Optimizer	NR
Loss Function	NR
Training Hyperparameters	NR
Primary Performance Metric	Energy efficiency; energy per inference/frame; task latency; accuracy loss compared with INT8 ViT
Primary Metric Value	Energy efficiency: 57.7 TOPS/W for manipulation, 51.7 TOPS/W for ObjectNav, 41.2 TOPS/W for ImageNav; energy per inference: 1.79 mJ for ViT-B and 3.78 mJ for ViT-L
Secondary Metrics	FPS, EMA reduction, accuracy loss, sparsity, power, voltage/frequency scaling, die area, SRAM size, latency
Baseline Models Compared	Vanilla INT8 ViT models; SOTA ViT accelerators including VLSI’23, ISSCC’22, ISSCC’23, JSSC’24, and VLSI’24 processors
Performance Gain Over Baseline	58% input EMA reduction; 83.6% lower SSA memory space; 46.5% lower SSA computation latency; 4× faster 1-bit matrix multiplication with 19% area overhead; 4.1× lower energy than previous work with same model; 34.8% higher task-level energy efficiency than SOTA ViT accelerator
Statistical Validation	Hardware measurement across multiple EAI tasks; no standard deviation, confidence interval, or significance test reported
Ablation Study	Partial/full hardware ablation. Reports system-level performance with optimizations enabled/disabled, SSA approximation accuracy loss, cosine similarity analysis, sparsity-energy analysis, and area/power breakdown
Training Framework / Code Availability	Hardware implemented in 28nm CMOS; FPGA-based testing platform; code availability NR
Target Hardware	Fabricated 28nm CMOS accelerator chip; FPGA-based test platform; RISC-V controller; 32 compute cores; 640KB SRAM
Hardware Deployment Status	Fabricated and measured silicon chip; validated on real hardware test platform for EAI tasks
Energy Evidence	Measured chip power and energy using an independent power supply / Keysight B2902A DC power analyzer; reports energy per frame and TOPS/W including EMA
Latency / Memory / Complexity Evidence	Die area 6.5 mm²; 640KB SRAM; 0.55–1.1V; 30–200 MHz; power 9.04–222.65 mW; task latency 29.76 ms for ViT-B and 62.77 ms for ViT-L; FPS 33.60, 15.93, and 81.43 across tasks
Comparison with ANN Baseline	Yes. Compares accuracy loss against INT8 vanilla ViT and compares hardware efficiency against non-spiking ViT accelerators
Main Contribution / Key Limitation	Contribution: proposes and fabricates a 28nm Spiking Vision Transformer accelerator for embodied intelligence with dual-path sparse computation, EMA-free SSA, and unified 1b/8b adder-tree acceleration. Limitation: policy MLP remains off-chip on FPGA; evaluation focuses on embodied vision tasks rather than broader SNN–Transformer domains; training details are not reported
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3

P005
An Energy-Efficient Unstructured Sparsity-Aware Deep SNN Accelerator With 3-D Computation Array

Full Citation	Fang, C., Shen, Z., Wang, Z., Wang, C., Zhao, S., Tian, F., Yang, J., & Sawan, M. (2025). An Energy-Efficient Unstructured Sparsity-Aware Deep SNN Accelerator With 3-D Computation Array. IEEE Journal of Solid-State Circuits, 60(3), 977–989. https://doi.org/10.1109/JSSC.2024.3507095
University	Westlake University
Country	China
Publication Year	2025
Publication Venue	IEEE Journal of Solid-State Circuits

Publisher	IEEE
DOI / URL	https://doi.org/10.1109/JSSC.2024.3507095
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Hardware acceleration of direct-trained deep SNNs, including Spikingformer and Spiking ResNet, for image classification and gesture recognition
Data Modality	Static image data and dynamic vision sensor/event data
Signal Dimensionality	2D image inputs converted to spike sequences; event/spike sequences with temporal timesteps
Signal Characteristics	Static images are encoded into spike format using one convolution layer and spiking neuron layer; DVS raw spike sequences are segmented into T = 8 timesteps and accumulated before re-encoding into spike format
Dataset Name	ImageNet, CIFAR-10, DVS-128 Gesture
Dataset Type	Public static image classification and dynamic vision/event-based gesture recognition benchmarks
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Spiking Transformer and Spiking ResNet workloads mapped to a fabricated sparse DSNN accelerator; Spiking Transformer uses spiking tokenizer, spiking Q/K/V generation, and spiking self-attention
SNN Component	Spiking tokenizer, LIF neurons, spike-only operators, spiking convolution, spiking Q/K/V generation, spiking self-attention, single-bit spikes
Neuron Model	LIF
Transformer Component	Spiking Transformer / Spikingformer blocks; spiking tokenizer; S_QKV_Gen; spiking self-attention
Attention Type	Spiking self-attention; softmax-free spike attention
SNN–Transformer Interface Mechanism	LIF neurons are placed before operators so that SCONV, linear layers, and attention receive 1-bit spike inputs; S_QKV_Gen converts input sequence X ∊ R(T×N×Dx) into spike-form Q, K, and V
Model Depth / Scale	Spikingformer-8-768: 8 Transformer blocks, hidden dimension 768; Spiking ResNet-9-32: 9 residual layers, 32 channels; timestep T = 8
Parameter Count	NR
Design / Contribution Type	Fabricated sparse DSNN accelerator with 3-D computation array, unstructured non-zero data fetcher, two-stage fan-in prioritized sparse dataflow, and multimode unified computation scheduler for SCONV, S_QKV_Gen, and SSA
Input Encoding Method	Static images encoded into spike format by a convolution layer and spiking neuron layer; DVS event data segmented into T = 8 timesteps and accumulated, then encoded again using an additional convolution layer
Encoding Parameters	Timestep T = 8; 8-bit weights; single-bit spike activations; two-stage bitmap format using global bitmap and local bitmap
Encoding Learned or Fixed	Partially learned: the convolutional spike encoder is model-based, while two-stage bitmap compression and sparse scheduling are fixed hardware/dataflow mechanisms
Event / Spike Representation	Single-bit spike sequences; global bitmap for positions where all T spikes are zero/non-zero; local bitmap for timestep-level spike information; compressed spike bitmap format
Temporal Resolution	8 timesteps for evaluated models
Output Decoding Method	Classification output from mapped Spikingformer / Spiking ResNet models; exact classifier decoding details NR
Encoding / Temporal Information Analysis	Yes. The paper analyzes timestep-parallel processing, spike sparsity, per-layer sparsity, two-stage bitmap storage efficiency, and energy efficiency under different spike sparsity levels
Training Paradigm	Direct-trained SNN inference workloads; hardware paper does not train models directly
Training Method	NR
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A for the evaluated direct-trained SNN setting
Optimizer	NR
Loss Function	NR
Training Hyperparameters	NR
Primary Performance Metric	Energy efficiency; energy per spike operation; recognition accuracy
Primary Metric Value	Best energy efficiency: 0.078 pJ/SOP; ImageNet accuracy using Spikingformer: 77.6% / 77.7%; ImageNet energy efficiency: 42.8 TOPS/W
Secondary Metrics	TOPS/W, pJ/SOP, task energy, latency, peak performance, area efficiency, power, voltage/frequency scaling, spike sparsity, storage efficiency
Baseline Models Compared	SOTA SNN processors, sparse ANN processors, Spikingformer, Spiking ResNet, prior ISSCC/JSSC accelerators
Performance Gain Over Baseline	68.3% external memory access reduction from timestep-parallel dataflow; 78% SSA operation reduction using sparse attention skipper; 51.1% SCONV latency reduction from memory router; 1.75× higher ImageNet energy efficiency than SOTA SNN processors; 1.4× higher than sparse ANN processors
Statistical Validation	Hardware measurement and benchmark comparison; no standard deviation, confidence interval, or statistical significance testing reported
Ablation Study	Partial hardware ablation/analysis. Includes energy efficiency vs spike sparsity, storage efficiency comparison, load-balancer window-size analysis, per-layer energy/sparsity analysis, and operator-specific scheduling analysis
Training Framework / Code Availability	Fabricated 40-nm CMOS accelerator; FPGA-based testing platform; software/code availability NR
Target Hardware	Fabricated 40-nm CMOS neuromorphic accelerator with 3 computation clusters, 9 sparse cores, 3-D computation arrays, FPGA test platform
Hardware Deployment Status	Fabricated and measured silicon chip
Energy Evidence	Measured chip power using Keysight B2902A DC power analyzer; reports TOPS/W, pJ/SOP, task energy, and voltage/frequency scaling
Latency / Memory / Complexity Evidence	Die area 1.54 × 1.46 mm² / 2.24 mm²; 192 KB SRAM; 50–200 MHz; 0.56–1.1 V; power 2.84–43.58 mW; peak performance 230.4 GOPS; best dense energy efficiency 20.28 TOPS/W without sparsity
Comparison with ANN Baseline	Yes. Compares against sparse ANN processors and discusses ANN MAC baseline versus SNN AC-only processing

Main Contribution / Key Limitation	Contribution: proposes a fabricated 40-nm sparse DSNN accelerator that supports modern direct-trained Spiking Transformers and Spiking ResNets using timestep-parallel computation, non-zero spike fetching, and operator-specific scheduling. Limitation: focuses mainly on vision/image and DVS benchmarks; training details are not reported; first non-spiking convolution/preprocessing for DVS is implemented on FPGA rather than fully on-chip
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3

P006

Advancing Neuromorphic Architecture Toward Emerging Spiking Neural Network on FPGA

Full Citation	Gao, Y., Wang, T., Yang, Y., Gong, L., Chen, X., Wang, C., Li, X., & Zhou, X. (2025). Advancing Neuromorphic Architecture Toward Emerging Spiking Neural Network on FPGA. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 44(9), 3465–3478. https://doi.org/10.1109/TCAD.2025.3547275
University	University of Science and Technology of China
Country	China
Publication Year	2025
Publication Venue	IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TCAD.2025.3547275
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Accelerating Transformer Spiking Neural Network inference on FPGA for CIFAR-10, CIFAR-100, and ImageNet classification
Data Modality	Static image data converted into spike sequences
Signal Dimensionality	2D image sequence represented as X ∊ R(T×C×H×W), then reshaped into spike patches I ∊ R(T×N×D)
Signal Characteristics	Input spikes are represented by 1-bit values; synapse weights are quantized to 8-bit; TSNN uses spike patches, channel dimension, timesteps, and LIF neuron dynamics
Dataset Name	CIFAR-10, CIFAR-100, ImageNet
Dataset Type	Public image classification benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Transformer Spiking Neural Network accelerated by heterogeneous FPGA architecture; TSNN consists of spiking patch splitting, L-block spike encoder, spike-driven self-attention module, spike MLP module, shortcut connections, and classification head
SNN Component	LIF spike neuron layers, binary spikes, spiking patch splitting, spike encoder, spike-driven self-attention, spike MLP, membrane shortcut
Neuron Model	LIF
Transformer Component	Transformer Spiking Neural Network; spike-driven self-attention; spike MLP; spike encoder blocks
Attention Type	Spike-driven self-attention using mask and accumulation operations
SNN–Transformer Interface Mechanism	Spike neuron layers convert inputs into binary spike vectors; spike linear transformation generates Qs, Ks, and Vs; SDSA applies mask operations between Ks and Vs, accumulates along spike patches, then masks with Qs
Model Depth / Scale	CIFAR-10: 10.23M parameters, 9.88G mult-add, spike patches 64, channel 512, 2 blocks, 4 timesteps; CIFAR-100: 10.28M parameters, 9.88G mult-add, spike patches 64, channel 512, 2 blocks, 4 timesteps; ImageNet: 29.69M parameters, 66.09G mult-add, spike patches 196, channel 512, 8 blocks, 4 timesteps
Parameter Count	CIFAR-10: 10.23M; CIFAR-100: 10.28M; ImageNet: 29.69M
Design / Contribution Type	FPGA-based neuromorphic architecture named SpikeTA; heterogeneous hardware engines; DSP-efficient addition tree; depth-aware buffer management; streaming dataflow mapping; split-engine dataflow mapping; latency-aware parallelism search
Input Encoding Method	Spiking patch splitting converts image sequences into spike patch sequences; LIF spike neuron layers convert continuous inputs into spike sequences
Encoding Parameters	1-bit input spikes; 8-bit synapse weights; membrane threshold = 1.0; coefficient τ = 2.0; timestep = 4 for evaluated TSNN models
Encoding Learned or Fixed	Partially learned: TSNN is trained/fine-tuned using backpropagation; hardware mapping/dataflow and quantization are fixed deployment mechanisms
Event / Spike Representation	Binary spike vectors; spike patches; Qs/Ks/Vs spike matrices; 1-bit spike inputs with 8-bit synaptic weights
Temporal Resolution	4 timesteps
Output Decoding Method	Classification head performs final image classification
Encoding / Temporal Information Analysis	Partial. The paper analyzes spike representation, LIF charging/firing/resetting, spike-driven MAAC computation, and hardware buffering/dataflow for timesteps, but does not deeply analyze temporal information preservation
Training Paradigm	Supervised direct training / fine-tuning of TSNN
Training Method	Backpropagation training with surrogate function; quantization optimization within 10 epochs using the training recipe from Spike-driven Transformer
Surrogate Gradient Function	Sigmoid surrogate function
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	PyTorch 2.3.1, CuPy 13.2, SpikingJelly 0.0.0.0.10, CUDA 12.4; quantization optimization within 10 epochs; training time about 11 min for CIFAR-10, 23 min for CIFAR-100, and 2 days for ImageNet on NVIDIA A100; threshold 1.0; τ = 2.0; 1-bit spikes; 8-bit weights
Primary Performance Metric	Architecture performance, speedup, energy efficiency, classification accuracy
Primary Metric Value	Peak performance: 28.99 TOPs; end-to-end execution time: 1.11 s for CIFAR-10, 1.11 s for CIFAR-100, and 68.08 s for ImageNet; accuracy: 95.60% on CIFAR-10, 78.40% on CIFAR-100, and 74.56% on ImageNet
Secondary Metrics	GOPs/W, power consumption, hardware utilization, FPGA resource utilization, DSP/LUT/BRAM/URAM usage, execution latency, performance speedup over CPU/GPU/accelerators

Baseline Models Compared	AMD EPYC 7542 CPU; NVIDIA A100 GPU; FireFly; DeepFire2; E3NE; HeatViT; SSR Transformer accelerator; other SNN and non-spiking Transformer accelerators
Performance Gain Over Baseline	140.73×–1023.53× speedup over AMD EPYC 7542 CPU; 2.97×–7.29× speedup over NVIDIA A100 GPU; 223.21×–2150.18× energy-efficiency gain over CPU; 6.41×–28.04× energy-efficiency gain over GPU; 2.79× over SOTA SNN accelerator; 2.66× over SOTA non-spiking Transformer accelerator
Statistical Validation	Benchmark and hardware comparison across three datasets; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Partial hardware ablation. Includes comparison of computing strategies SpikeTA-op1/op2/op3/final, hardware utilization analysis, resource utilization breakdown, and pipelined latency balancing analysis
Training Framework / Code Availability	PyTorch, CuPy, SpikingJelly, CUDA 12.4 for training; Xilinx Vitis HLS/Vivado for FPGA implementation; code availability NR
Target Hardware	Xilinx Alveo U280 FPGA with xcu280-fsvh2892-2L-e chip; 9024 DSPs, 1304k LUTs, 2607k FFs, 2016 BRAMs, 960 URAMs, and 32 HBM channels
Hardware Deployment Status	Implemented and evaluated on FPGA prototype using Vitis HLS and Vivado
Energy Evidence	Power measured using Xilinx Board Utility xbutil; reported power consumption is 70.93 W, 71.45 W, and 71.76 W on CIFAR-10, CIFAR-100, and ImageNet respectively; energy efficiency reported up to 403.99 GOPs/W
Latency / Memory / Complexity Evidence	Clock frequency 450 MHz; uses three parallel hardware kernels across SLRs; end-to-end times of 1.11 s, 1.11 s, and 68.08 s; total hardware utilization about 91.87%–91.94%; resource use includes 61.87% URAM, 77.98% BRAM, 79.43% DSP, 8.87% LUTRAM, and 63.86% LUTCOM
Comparison with ANN Baseline	Yes. Compares with non-spiking Transformer accelerators and CPU/GPU ANN-style execution baselines
Main Contribution / Key Limitation	Contribution: proposes SpikeTA, the first FPGA-based neuromorphic hardware architecture explicitly designed for TSNN acceleration, using heterogeneous hardware engines, DSP-efficient addition, depth-aware buffering, and optimized dataflow mapping. Limitation: evaluated only on image classification TSNN models; energy efficiency is FPGA-board based rather than ASIC silicon; adaptability to broader TSNN models is left for future work
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4

P007

Spiking Transformer: Introducing Accurate Addition-Only Spiking Self-Attention for Transformer

Full Citation	Guo, Y., Liu, X., Chen, Y., Peng, W., Zhang, Y., & Ma, Z. (2025). Spiking Transformer: Introducing Accurate Addition-Only Spiking Self-Attention for Transformer. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025), pp. 24398–24408. https://doi.org/10.1109/CVPR52734.2025.02272
University	Peking University
Country	China
Publication Year	2025
Publication Venue	IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/CVPR52734.2025.02272
Publication Type	Conference paper
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	Image classification on CIFAR-10, CIFAR-100, and ImageNet-1K
Data Modality	Static RGB image data represented as repeated image sequences over timesteps
Signal Dimensionality	2D images converted into spiking patch sequences, I ∊ R(T×C×H×W), then represented as N flattened spike patches with D-dimensional channels
Signal Characteristics	CIFAR images are 32 × 32; ImageNet-1K images are resized to 224 × 224; images are repeated across T timesteps; main experiments use 4 timesteps
Dataset Name	CIFAR-10, CIFAR-100, ImageNet-1K
Dataset Type	Public image classification benchmark datasets
Dataset Size	CIFAR: 50,000 training images and 10,000 test images; ImageNet-1K: approximately 1.28 million training images and 50,000 test images
Dataset Balance / Split	Uses standard train/test benchmark splits; exact class-balance details NR
Hybrid Topology	Spiking Transformer with Spiking Patch Splitting, L-block Spiking Transformer encoder, Accurate Addition-Only Spiking Self-Attention, MLP block, membrane-potential residual connections, global average pooling, and classification head
SNN Component	LIF spike neuron layers, binary spiking neuron for Q, ternary spiking neuron for V, binary spike hidden states, membrane potential dynamics, spike-based residual structure
Neuron Model	LIF; binary spiking neuron; ternary spiking neuron
Transformer Component	Spiking Transformer encoder block, A2OS2A attention module, MLP block, patch-splitting module, relative position embedding, classification head
Attention Type	Accurate Addition-Only Spiking Self-Attention; binary Q, ReLU/full-precision K, ternary V; softmax-free and scaling-free attention
SNN–Transformer Interface Mechanism	Linear + BN layers generate Q, K, and V; Q is converted by binary spiking neuron, K by ReLU, and V by ternary spiking neuron; Q·K■·V is computed through addition-only operations instead of conventional multiplication-heavy self-attention
Model Depth / Scale	Reported models include Spiking Transformer-2-256, 2-384, 4-256, 4-384, 2-512, 8-384, 6-512, 8-512, and 10-512; main ImageNet models use 4 timesteps
Parameter Count	CIFAR models: 2.59M, 4.15M, 5.76M, 9.32M, and 10.23M depending on model scale; ImageNet models: 16.81M, 23.37M, 29.68M, and 36.01M
Design / Contribution Type	New spiking self-attention mechanism, A2OS2A, designed to reduce information loss while preserving addition-only computation; theoretical analysis of information loss in vanilla spiking self-attention
Input Encoding Method	Spiking Patch Splitting converts repeated image sequence into spike patch embeddings; spike neuron layer converts membrane potentials into spike tensors
Encoding Parameters	Main experiments use T = 4 timesteps; binary Q values {0,1}; ternary V values {-1,0,1}; ReLU/full-precision K; LIF threshold Vth, reset potential Vreset, and decay factor β are defined but exact values NR

Encoding Learned or Fixed	Partially learned: patch projection, Q/K/V projections, BN, RPE, and model weights are learned; spike firing rules are fixed neuron dynamics
Event / Spike Representation	Binary spike tensors; binary Q; ternary V; spike patch embeddings; membrane potential tensors; binary hidden states
Temporal Resolution	4 timesteps in the main reported experiments
Output Decoding Method	Global Average Pooling over final Spiking Transformer encoder output followed by a fully connected classification head
Encoding / Temporal Information Analysis	Yes. The paper provides entropy-based theoretical analysis of information loss in binary spike representations and compares representational capacity of binary spike features versus real-valued features
Training Paradigm	Supervised direct learning
Training Method	Direct training of Spiking Transformer following Spike-driven Transformer experimental setup; backpropagation with surrogate-gradient-style training implied through LIF neuron discussion
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	Main experiments use 4 timesteps; other training configurations follow Spike-driven Transformer; detailed optimizer, learning rate, batch size, and schedule NR
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	Best reported accuracy: 96.42% on CIFAR-10, 79.90% on CIFAR-100, and 78.66% on ImageNet-1K
Secondary Metrics	Parameter count, timestep count, ablation accuracy gain over baseline
Baseline Models Compared	Hybrid training, Diet-SNN, STBP, STBP-NeuNorm, TSSL-BP, STBP-tdBN, TET, Spikformer, Spikingformer, Spike-driven Transformer, Spiking ResNet, QCFS, COS, SEW-ResNet, Attention-SNN
Performance Gain Over Baseline	CIFAR-10/100: Spiking Transformer-2-512 improves over baseline by about 0.91% on CIFAR-10 and 1.07% on CIFAR-100. ImageNet: Spiking Transformer-8-512 reaches 76.28%, improving over Spikingformer-8-512 by 1.49%; Spiking Transformer-10-512 reaches 78.66%
Statistical Validation	Benchmark comparison and ablation study; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full/clear ablation. Compares baseline Spike-driven Transformer variants against the proposed Spiking Transformer on CIFAR-10 and CIFAR-100 for 2-256 and 2-512 models
Training Framework / Code Availability	Training framework NR; code availability NR
Target Hardware	Neuromorphic hardware motivation discussed; no specific hardware deployment target used in experiments
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	No measured energy result; efficiency is argued theoretically through addition-only computation, removal of softmax/scaling, and SNN event-driven properties
Latency / Memory / Complexity Evidence	Reports parameter counts and timesteps; no measured latency, memory footprint, or computational complexity table reported
Comparison with ANN Baseline	Partial. Compares against ANN-to-SNN methods and SNN baselines; does not directly evaluate a matched ANN Transformer baseline in experiments
Main Contribution / Key Limitation	Contribution: proposes A2OS2A, a more accurate addition-only spiking self-attention mechanism using binary Q, ReLU K, and ternary V to reduce information loss while avoiding softmax and scaling. Limitation: no hardware implementation or measured energy/latency results; several training details are not reported and are referred to prior baseline settings
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ1

P008

A Structured-Sparsity-Based Design Approach for Energy-Efficient Spiking Transformer Processing

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University	Korea University
Country	South Korea
Publication Year	2026
Publication Venue	IEEE Transactions on Very Large Scale Integration (VLSI) Systems
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TVLSI.2026.3675299
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Structured-sparsity-based acceleration of Spiking Transformer SSA blocks for image classification, object detection, and semantic segmentation
Data Modality	Static image data represented through spike activations in Spiking Transformer models
Signal Dimensionality	Spiking token tensors with timestep, token, and channel dimensions; SSA inputs represented as X ∈ {0,1}^(T×N×Cin), with Q/K/V split into attention heads
Signal Characteristics	Binary spike activations; sparse SSA intermediate matrix A and output matrix Z; rowwise sparsity in A and columnwise sparsity in Z; low-spike-rate neurons firing at most once across timesteps are filtered during sparsity detection
Dataset Name	ImageNet-1K, CIFAR-10, CIFAR-100, COCO 2017, ADE20K
Dataset Type	Public image classification, object detection, and semantic segmentation benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Spiking Transformer SSA block with QKV generation, SSA/FSSA mechanism, and projection layer; proposed fully spiking SSA inserts additional LIF neuron after K■V to increase structured sparsity

SNN Component	LIF neurons, spike activations, FSSA operator, low-spike-rate neuron filtering, sparse spike tensors, structured sparsity detection
Neuron Model	LIF neuron with learnable threshold voltage
Transformer Component	Spiking self-attention block; QKV generation layer; attention heads; projection layer; Spikingformer, Spike-driven Transformer v2, and QKFormer models
Attention Type	Spiking self-attention; fully spiking SSA; structured-sparsity-aware SSA
SNN–Transformer Interface Mechanism	Q, K, and V are generated as spike tensors using spiking neuron layers; FSSA adds LIF neuron to transform intermediate K■V attention matrix into sparse spike activation before Q·A computation
Model Depth / Scale	Evaluated ST models include Spikingformer, Spike-driven Transformer v2, QKFormer-224, and QKFormer-384; exact block counts and parameter counts for each model NR in this paper
Parameter Count	Object detection SDT v2: 34.9M; ADE20K semantic segmentation SDT v2: 16.5M; classification model parameter counts NR
Design / Contribution Type	Algorithm–hardware co-design introducing FSSA, low-spike-rate neuron filtering, sparsity-aware two-stage dataflow, and sparsity-aware hardware add-on for runtime computation and memory-transfer skipping
Input Encoding Method	Uses pretrained Spiking Transformer models with binary spike activations; exact image-to-spike input encoding method NR
Encoding Parameters	FSSA LIF threshold initialized to vth = 4; neuron filtering treats neurons firing at most once over all timesteps as sparse; fine-tuning for 10 epochs
Encoding Learned or Fixed	Partially learned: FSSA LIF thresholds are learnable and QKV generation weights are fine-tuned; sparsity detection/filtering logic is fixed
Event / Spike Representation	Binary spike activations; spike tensors Q, K, V; sparse A and Z matrices; compact binary sparsity vectors for active/inactive channels
Temporal Resolution	T timesteps used in evaluated ST models; exact timestep varies by model/task and is not fully reported; COCO uses timestep 1, ADE20K uses timestep 4
Output Decoding Method	Classification head / downstream task heads from the underlying ST models; exact decoding details NR
Encoding / Temporal Information Analysis	Yes. The paper analyzes SSA structured sparsity, rowwise sparsity in A, columnwise sparsity in Z, low-spike-rate neurons across timesteps, and sparsity effects on computation and memory skipping
Training Paradigm	Supervised fine-tuning of pretrained Spiking Transformer models
Training Method	Lightweight fine-tuning restricted mainly to QKV generation layers and learnable FSSA LIF thresholds; 10 epochs of fine-tuning
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	FSSA threshold initialized to 4; fine-tuned for 10 epochs; QKV generation weights updated; up to 80% fewer fine-tuned parameters than full-model fine-tuning; other optimizer and learning-rate details NR
Primary Performance Metric	Effective throughput and energy efficiency of SSA block processing
Primary Metric Value	Proposed design with filtering achieves 234.44 GOP/s effective throughput and 319.12 GOP/J energy efficiency
Secondary Metrics	Accuracy, structured sparsity, row-A sparsity, column-Z sparsity, area overhead, power overhead, mAP, mIoU
Baseline Models Compared	Baseline SSA/ST models; Spikingformer, Spike-driven Transformer v2, QKFormer; baseline SNN accelerator/Prosperity; PTB; Bishop; SpikeX; Phi
Performance Gain Over Baseline	FSSA increases structured sparsity by about 1.8× with less than 1% accuracy loss; neuron filtering adds about 1.5× more sparsity over FSSA; proposed hardware achieves 24.3% throughput gain and 28.2% energy-efficiency gain over baseline SNN accelerator
Statistical Validation	Benchmark evaluation and postsynthesis simulation; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes FSSA direct implementation, FSSA suppression-method comparison, FSSA fine-tuning, neuron-filtering fine-tuning, downstream detection/segmentation evaluation, hardware overhead, and accelerator comparison
Training Framework / Code Availability	Hardware implemented in Verilog RTL; synthesized with Synopsys Design Compiler in 28 nm process; evaluated using open-source cycle-accurate simulator for Prosperity; training framework/code availability NR
Target Hardware	28 nm SNN accelerator-style hardware integrated with sparsity-aware hardware modules; baseline modeled using Prosperity accelerator framework
Hardware Deployment Status	RTL synthesis and cycle-accurate simulation; no fabricated chip reported
Energy Evidence	Energy estimated from postsynthesis simulation and synthesized hardware power; proposed sparsity-aware hardware consumes 2.23 mW with 0.24% power overhead over baseline
Latency / Memory / Complexity Evidence	Proposed sparsity-aware hardware area 0.0057 mm²; 1.08% area overhead; 500 MHz frequency; 128 processing elements; skips redundant computations and weight transfers in Q generation and projection layers
Comparison with ANN Baseline	Partial. Motivation compares ST accuracy/efficiency with ANN transformers, but experiments mainly compare against SNN/ST models and SNN accelerators rather than a matched ANN baseline
Main Contribution / Key Limitation	Contribution: identifies and exploits structured sparsity in SSA through FSSA, low-spike-rate neuron filtering, and sparsity-aware hardware/dataflow to improve ST throughput and energy efficiency. Limitation: hardware results are based on RTL synthesis and simulation rather than fabricated silicon; optimization focuses mainly on SSA blocks rather than full end-to-end accelerator deployment
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4

P009

C-Transformer: An Energy-Efficient Homogeneous DNN-Transformer/SNN-Transformer Processor for Large Language Models

Full Citation	Kim, S., Kim, S., Jo, W., Kim, S., Hong, S., Lee, N., Lee, J., & Yoo, H.-J. (2025). C-Transformer: An Energy-Efficient Homogeneous DNN-Transformer/SNN-Transformer Processor for Large Language Models. IEEE Journal of Solid-State Circuits, 60(10), 3802–3815. https://doi.org/10.1109/JSSC.2025.3554699
University	Korea Advanced Institute of Science and Technology
Country	South Korea
Publication Year	2025
Publication Venue	IEEE Journal of Solid-State Circuits
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/JSSC.2025.3554699

Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Language modeling, translation, and summarization using GPT-2, mT5, T5, and FSMT
Data Modality	Text/token sequence data; spike trains used internally for SNN-Transformer mode
Signal Dimensionality	1D language token sequences; encoding uses 1024 input tokens and decoding generates one output token in the measurement setup
Signal Characteristics	Tokenized language input processed by Transformer models; INT8 activations/weights in HDSC; rate-coded spike trains in spiking-Transformer mode; outlier inputs handled by INT16 1-D SIMD
Dataset Name	WikiText, IWSLT, CNN/DailyMail; FSMT translation dataset NR
Dataset Type	Public NLP benchmark datasets for language modeling, translation, and summarization
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Homogeneous DNN-Transformer/SNN-Transformer processor using complementary DNN-style allocation; workload dynamically switches between DNN-Transformer mode and spiking-Transformer mode based on input magnitude
SNN Component	Spiking-Transformer mode, rate encoding, spike trains, integrate-and-fire operations, output spike speculation, spike-level zero skipping
Neuron Model	Integrate-and-fire style spike generation; exact neuron model beyond threshold-based integrate-and-fire NR
Transformer Component	DNN-Transformer and spiking-Transformer execution for GPT-2, mT5, T5, and FSMT language models; attention and feed-forward blocks processed in HDSC
Attention Type	Transformer self-attention / multi-self-masked attention; spiking-Transformer computations use spike-domain operations; exact SNN attention formulation NR
SNN–Transformer Interface Mechanism	Input loader converts scalar input values into spike trains in ST-mode; HMAU reconfigures between MAC operation for DNN-Transformer mode and accumulation operation for spiking-Transformer mode; workload allocator selects efficient network type based on input magnitude
Model Depth / Scale	GPT-2 Large: 708M parameters, 36 blocks; GPT-2 Small: 12 blocks; mT5 Large: 24 blocks; mT5 Small: 8 blocks; T5 Large: 24 blocks; T5 Small: 6 blocks
Parameter Count	GPT-2 Large: 708M; T5 Large: 402M; other model parameter counts partly NR
Design / Contribution Type	Fabricated ASIC processor with homogeneous DNN/SNN Transformer core, hybrid multiplication/accumulation unit, output spike speculation, spike-level zero skipping, big–little network compression, implicit weight generation, and extended sign compression
Input Encoding Method	Rate encoding converts input values into spike trains in spiking-Transformer mode
Encoding Parameters	INT8 input/weight precision in HDSC; INT16 outlier handling; threshold set to match DNN LSB resolution; time window configured as 2^(w–1); output spike speculation sampling ratios selected from {0, 25, 50, 75}
Encoding Learned or Fixed	Fixed rate-encoding and hardware-level spike speculation; model parameters/compression components are trained or generated separately
Event / Spike Representation	Rate-coded spike train; 1-bit spike activity; sampled output spike probability; speculated output spike patterns for remaining timesteps
Temporal Resolution	Time-window-based spike processing; exact total timestep depends on word length and fixed-point configuration; layer-wise sampling ratios include 25%, 50%, and 75%
Output Decoding Method	Transformer language-model decoding generates output tokens; output spike speculation estimates full-time-step spike patterns from sampled timesteps
Encoding / Temporal Information Analysis	Partial. The paper analyzes rate-coding spike probability, sampled timestep behavior, output spike speculation, and layer-wise sampling ratios, but does not deeply analyze temporal information preservation
Training Paradigm	Hardware inference paper; uses pretrained/compressed language models
Training Method	Big–little network and implicit weight generation are adopted for compression; exact LLM training/fine-tuning procedure NR
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	NR
Optimizer	NR
Loss Function	For implicit weight generation, losses are calculated from differences between parameters, feature maps, and logits; exact formulas NR
Training Hyperparameters	NR
Primary Performance Metric	System energy consumption, computation energy, EMA energy, latency, energy efficiency
Primary Metric Value	GPT-2 language modeling: 35.9 mJ and 656 ms; mT5 translation: 21.3 mJ and 359 ms; T5 summarization: 15.0 mJ and 307 ms; FSMT translation: 5.5 mJ and 93 ms
Secondary Metrics	Perplexity, BLEU, ROUGE-L, TOPS/W, TOPS, external bandwidth, memory-access reduction, compression ratio, throughput
Baseline Models Compared	Baseline without proposed features; previous Transformer ASICs from ISSCC 2022/2023; C-DNN processor; DDR3 EMA baseline
Performance Gain Over Baseline	Achieves 0.21×–0.33× computation energy and 0.31×–0.41× EMA energy compared with baseline; HDSC gives 1.32× chip energy-efficiency gain, spike-level zero skipping 1.21×, and output spike speculation 1.19×; total system energy reduced to 0.43×
Statistical Validation	Chip measurement and benchmark comparison; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full hardware feature analysis. Includes HDSC, spike-level zero skipping, output spike speculation, BLN, IWG, ESC, layer-wise energy/latency, compression performance, and sampling-ratio analysis
Training Framework / Code Availability	Fabricated Samsung 28-nm 1P8M CMOS ASIC; evaluation board with DDR3 DRAM, flash memory, FPGA, and host PC; code availability NR
Target Hardware	Fabricated 28-nm Samsung 1P8M CMOS ASIC with 48 homogeneous DNN/SNN Transformer cores, two weight generators, NoC, top controller, 1-D SIMD unit, and 500 KB SRAM
Hardware Deployment Status	Fabricated and measured silicon chip with demonstration system
Energy Evidence	Measured chip/system energy; includes EMA energy estimation using DDR3 interface; chip consumes 47.5 mW at 50 MHz/0.7 V and 469.2 mW at 200 MHz/1.1 V
Latency / Memory / Complexity Evidence	Die area 20.25 mm²; 0.7–1.1 V; 50–200 MHz; peak performance 3.41 TOPS; energy efficiency 22.9–47.8 TOPS/W; 1.6 GB/s external bandwidth; 500 KB SRAM

Comparison with ANN Baseline	Yes. Compares against DNN-Transformer processing, previous non-spiking Transformer ASICs, and baseline Transformer execution with EMA energy
Main Contribution / Key Limitation	Contribution: proposes and fabricates C-Transformer, a homogeneous DNN-Transformer/SNN-Transformer ASIC for on-device language models using reconfigurable HMAU, output spike speculation, and three-stage weight compression to reduce computation and external-memory energy. Limitation: spiking mechanism is mainly hardware-level rate-coded processing rather than a fully reported SNN training method; some training and dataset details are NR
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3

P010

ESTU: Enabling Spiking Transformers on Ultra-Low-Power FPGAs

Full Citation	Leone, G., Busia, P., Orrù, M., Raffo, L., & Meloni, P. (2025). ESTU: Enabling Spiking Transformers on Ultra-Low-Power FPGAs. IEEE Transactions on Circuits and Systems II: Express Briefs, 72(12), 2027–2031. https://doi.org/10.1109/TCSII.2025.3626209
University	University of Cagliari
Country	Italy
Publication Year	2025
Publication Venue	IEEE Transactions on Circuits and Systems II: Express Briefs
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TCSII.2025.3626209
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Hardware-focused
E&QA; Score	4
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	Spiking Transformer inference on low-power FPGA; sEMG gesture classification on NinaPro DB-5; benchmarking of spiking Transformer models for fast prosthetic hand control and epileptic seizure detection/prediction
Data Modality	Sensor time-series converted into spike trains; sEMG signals; EEG/seizure-related benchmark model; generic spike-token inputs
Signal Dimensionality	Temporal sensor streams represented as spike sequences; model uses token/patch sequence length, attention heads, embedding dimensions, and timesteps
Signal Characteristics	sEMG data are converted into spikes using delta modulation; sEMG model uses 32 input channels, embedding size 64, 4 attention heads, 200 timestep memory for K and V, and projection size 16; 91% neurons inactive on average during test inference
Dataset Name	NinaPro DB-5; benchmark models from prior fast prosthetic hand control and epileptic seizure detection/prediction studies
Dataset Type	Public biomedical sensor dataset and prior benchmark spiking Transformer models
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Microcode-programmable Spiking Transformer accelerator with configurable encoding slot, ESTU accelerator core, configurable decoding slot, and RISC-V softcore; supports embedding, encoder/SSA, and classifier stages
SNN Component	LIF neuron module, spike trains, spike memory, spike-weight and spike-spike operations, sparsity stack memory, event-sparsity exploitation
Neuron Model	LIF
Transformer Component	Spiking Transformer encoder; lightweight spiking self-attention; Q/K/V projections; multi-head attention; classifier stage
Attention Type	Spiking self-attention; softmax-free attention using binary Q, K, and V projections
SNN–Transformer Interface Mechanism	Sensor data are converted into spike trains by a configurable encoding slot; ESTU executes dense, sum, and matrix multiplication operators over spike/integer data; operation results can be passed through LIF to generate binary spike outputs for subsequent Transformer layers
Model Depth / Scale	sEMG model: dense-based embedding plus one encoder stage, 4 attention heads, K/V memory of 200 timesteps, projection size 16, 32 input channels, embedding size 64. Benchmarked models include configurations with 8 heads/32 patch/150 sequence length and 1 head/8 patch/24 sequence length
Parameter Count	NR
Design / Contribution Type	Ultra-low-power FPGA architecture named ESTU; microcode-programmable accelerator; operator fusion and hardware reuse; sparsity-aware execution; RISC-V-controlled system; configurable encoding/decoding slots
Input Encoding Method	Delta modulation for sEMG spike feature extraction; configurable encoding slot for sensor-to-spike conversion
Encoding Parameters	sEMG model uses 32 input channels; K/V memory length 200 timesteps; embedding size 64; projection size 16; 8-bit integer quantized model parameters; 20-bit fixed-point membrane potentials
Encoding Learned or Fixed	Fixed/preprocessing-based encoding; model weights trained separately; ESTU hardware/microcode is configurable
Event / Spike Representation	Spike trains; grouped spikes in blocks of four for stack-memory sparsity tracking; binary spike memory; LIF-generated spikes
Temporal Resolution	sEMG model uses 200 timestep memory for K and V; other benchmark models use sequence lengths of 150 and 24; exact sampling rate NR
Output Decoding Method	Configurable decoding slot; spike-rate evaluation mentioned as an example for classification outcome generation
Encoding / Temporal Information Analysis	Partial. The paper analyzes spike grouping, event sparsity exploitation, inactive-neuron rate, and real-time constraints, but does not deeply analyze temporal information preservation
Training Paradigm	Supervised training for sEMG Spiking Transformer model; hardware inference deployment
Training Method	SpikingJelly used to train the sEMG Spiking Transformer; post-training uniform-scale quantization applied for deployment
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	8-bit integer post-training quantization for parameters; 20-bit fixed-point membrane potentials; other training hyperparameters NR
Primary Performance Metric	Inference power consumption and real-time execution capability
Primary Metric Value	Active inference power: 3.76 mW; standby power: 0.23 mW; sEMG classification accuracy: 87.21% before quantization and 86.97% after quantization

Secondary Metrics	Execution time, energy per inference, throughput per operator, sparsity, resource utilization, accuracy improvement, standby power
Baseline Models Compared	Vanilla SNN model for sEMG classification; RISC-V microcontroller baseline; FPGA-based spiking Transformer accelerators from prior work
Performance Gain Over Baseline	sEMG Spiking Transformer improves accuracy by over 2% compared with vanilla SNN; 91% inactive neurons provide 29% speedup compared with non-sparse execution; energy efficiency is reported as 2.6× higher than one prior work and 2× higher than another FPGA spiking Transformer accelerator
Statistical Validation	Benchmark and hardware measurements reported; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Partial. Includes sparsity grouping trade-off, operator throughput comparison, benchmark model execution, quantization impact, sparsity speedup, and power measurement
Training Framework / Code Availability	SpikingJelly for model training; ESTU is stated as open-source; hardware synthesized with OSS CAD Suite, Yosys, and nextpnr; GitHub repository provided in the paper
Target Hardware	Lattice iCE40UP5K ultra-low-power FPGA with RISC-V SERV softcore
Hardware Deployment Status	Implemented and measured on FPGA evaluation board
Energy Evidence	Measured using shunt resistor on the VCORE track of the iCEBreaker evaluation board; active power 3.76 mW, standby 0.23 mW; at 12 MHz, average power 1.03 mW and 4.28 µJ per inference for sEMG
Latency / Memory / Complexity Evidence	Maximum clock 21 MHz; resource usage: 4,301 logic cells, 30 BRAMs, 2 SPRAMs, 6 DSPs; memory limits: <1 Mb integer memory for attention/synaptic weights, 32 Kb spike memory, 768 LIF neurons; sEMG inference about 0.65 ms with 91% sparsity
Comparison with ANN Baseline	No direct ANN baseline; compares mainly with vanilla SNN, RISC-V software execution, and FPGA spiking Transformer accelerators
Main Contribution / Key Limitation	Contribution: proposes ESTU, an ultra-low-power, microcode-programmable Spiking Transformer FPGA architecture for near-sensor edge inference, validated on sEMG and benchmark models with measured milliwatt-level power. Limitation: limited model size due to small FPGA memory; low peak performance compared with high-end FPGA accelerators; training details are not fully reported
RQ Mapping	Primary: RQ5; Secondary: RQ1, RQ2, RQ3

P011

FireFly-T: High-Throughput Sparsity Exploitation for Spiking Transformer Acceleration with Dual-Engine Overlay Architecture

Full Citation	Li, T., Li, J., Shen, G., Zhao, D., Zhang, Q., & Zeng, Y. (2026). FireFly-T: High-Throughput Sparsity Exploitation for Spiking Transformer Acceleration with Dual-Engine Overlay Architecture. IEEE Transactions on Computers. https://doi.org/10.1109/TC.2026.3672901
University	University of Chinese Academy of Sciences
Country	China
Publication Year	2026
Publication Venue	IEEE Transactions on Computers
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TC.2026.3672901
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	FPGA acceleration of spiking convolutional networks and Spiking Transformer inference, especially sparse activation processing and spiking attention acceleration
Data Modality	Static image data converted into spike-based feature/token representations
Signal Dimensionality	2D image input represented as spike feature maps and spiking token sequences
Signal Characteristics	Uses 1-bit spike activations, sparse spike feature maps, binary Q/K/V attention operands, and multi-bit weights; CIFAR-Net input begins from 3 × 32 × 32 images
Dataset Name	CIFAR-10, CIFAR-100, ImageNet
Dataset Type	Public image classification benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Spiking Transformer mapped onto a dual-engine FPGA overlay; model side follows SPS + stacked Spiking Transformer encoder blocks + classification head; hardware side uses sparse engine for convolution/linear layers and binary engine for spiking attention
SNN Component	LIF neurons, spiking patch splitting, spiking convolution, sparse spike activations, neuronal dynamics module, spike-based MLP, spike-form Q/K/V
Neuron Model	LIF
Transformer Component	Spiking Transformer encoder, Spiking Self-Attention, MLP block, Spikingformer variants, classification head
Attention Type	Binary spiking self-attention; spiking attention using AND-PopCount operations
SNN–Transformer Interface Mechanism	Spike-form Q, K, and V are processed by the binary engine; sparse engine handles spiking convolution/linear operations, while binary engine accelerates QK■ and QK■V attention operations through LUT6-optimized AND-PopCount logic
Model Depth / Scale	Evaluated models include CIFAR-Net, Spikingformer-4-256 for CIFAR-10, Spikingformer-2-512 for CIFAR-100, and Spikingformer-8-512 for ImageNet; hardware parallelism configured as (PTs × PFx, PCi, PCo) = (8 × 16, 64)
Parameter Count	NR
Design / Contribution Type	Dual-engine overlay architecture; high-throughput sparse engine; binary spiking-attention engine; flexible orchestrator; multi-lane sparse decoder; bank-conflict-free workload balancing; LUT6-optimized AND-PopCount; implicit transpose/dataflow manipulation
Input Encoding Method	Spiking Patch Splitting / spiking convolutional tokenizer; exact raw image-to-spike encoding rule NR
Encoding Parameters	1-bit spike activations; binary attention operands; configurable timestep, sequence length, embedding dimension, kernel size, channel parallelism, memory banks, and sparse decoder throughput; exact neuron threshold values NR
Encoding Learned or Fixed	Partially model-based and hardware-configurable; exact encoding learning status NR

Event / Spike Representation	Binary spike feature maps; sparse spike bitmap/index representations; spike-form Q/K/V; fine-grained sparse activations
Temporal Resolution	Symbolic timestep parameter Ts supported by the architecture; exact timestep used in each evaluated network NR
Output Decoding Method	Classification head produces final image classification output
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike sparsity, fine-grained versus coarse-grained sparsity, multidimensional parallelism, sparse decoder throughput, binary attention dataflow, and resource/energy trade-offs
Training Paradigm	Hardware inference acceleration; model training is not the main focus
Training Method	NR
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	NR
Primary Performance Metric	Energy efficiency, throughput, DSP efficiency, and classification accuracy
Primary Metric Value	FireFly-T reports 978.61 GOP/s/W on CIFAR-Net/CIFAR-10; Spikingformer results include 696.64 GOP/s/W on CIFAR-10, 762.87 GOP/s/W on CIFAR-100, and 781.13 GOP/s/W on ImageNet; reported accuracies include 93.60%, 94.45%, 78.38%, and 71.55% across the evaluated configurations
Secondary Metrics	GOP/s, GOP/s/DSP, FPS, LUT usage, register usage, BRAM usage, DSP usage, clock frequency, resource breakdown
Baseline Models Compared	E3NE, SyncNN, DeepFire2, FireFly v2, HeatViT, SSR, SpikeTA, and other SNN / Transformer accelerators
Performance Gain Over Baseline	Achieves 1.39× higher energy efficiency and 4.21× higher DSP efficiency than FireFly v2; achieves 2.40× higher energy efficiency and 7.10× higher DSP efficiency than SpikeTA; also reports 1.47×–13.87× higher energy efficiency than DeepFire2 and 1.03×–1.39× higher than FireFly v2 across CIFAR-10, CIFAR-100, and ImageNet
Statistical Validation	Hardware benchmark comparison and resource/performance evaluation; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full hardware analysis. Includes sparsity support comparison, sparse decoder analysis, load-balancing analysis, parallelism/sparsity comparison, resource breakdown, and comparison with SOTA accelerators
Training Framework / Code Availability	FPGA implementation on Zynq UltraScale+ MPSoC / KV260 platform; code availability NR
Target Hardware	Xilinx Zynq UltraScale+ MPSoC / KV260 FPGA platform; xczu5ev device
Hardware Deployment Status	Implemented and evaluated as an FPGA overlay/IP accelerator on commercial FPGA edge hardware
Energy Evidence	Energy efficiency and power/resource metrics are reported from FPGA implementation/Vivado-style evaluation; no fabricated ASIC measurement
Latency / Memory / Complexity Evidence	FireFly-T uses 46 kLUTs, 117 kREGs, 100 BRAMs, and 304 DSPs at 300 MHz for the G=4 configuration; sparse engine uses 40 kLUTs and 288 DSPs, while binary engine uses 5 kLUTs and 16 DSPs
Comparison with ANN Baseline	Partial. The paper compares against non-spiking Transformer accelerators such as HeatViT and SSR, but it does not provide a matched ANN Transformer model comparison under the same task/model setting
Main Contribution / Key Limitation	Contribution: FireFly-T is a practical FPGA accelerator for Spiking Transformers that combines fine-grained sparsity exploitation, multidimensional parallelism, and a dedicated binary attention engine. Limitation: evaluation is FPGA-based rather than fabricated ASIC; experiments focus mainly on image-classification workloads; training and encoding details are not deeply reported
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3

P012

Motion-Oriented Hybrid Spiking Neural Networks for Event-Based Motion Deblurring

Full Citation	Liu, Z., Wu, J., Shi, G., Yang, W., Dong, W., & Zhao, Q. (2024). Motion-Oriented Hybrid Spiking Neural Networks for Event-Based Motion Deblurring. IEEE Transactions on Circuits and Systems for Video Technology, 34(5), 3742–3754. https://doi.org/10.1109/TCSVT.2023.3317976
University	University, Xi'an
Country	China
Publication Year	2024
Publication Venue	IEEE Transactions on Circuits and Systems for Video Technology
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TCSVT.2023.3317976
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Event-based motion deblurring / sharp image reconstruction from blurry image and event stream
Data Modality	Blurry RGB/image frames and event-camera data
Signal Dimensionality	Multimodal 2D image + 3D spatiotemporal event/spike representation
Signal Characteristics	Event stream represented as asynchronous events (x, y, t, p); event cameras provide microsecond-level temporal resolution; spike features are represented in spatiotemporal form S ∈ R^(n×n×T×C); GoPro images are 1280 × 720; real-event images are cropped to 320 × 256
Dataset Name	GoPro, REBlur, and a custom real-event dataset captured using DAVIS346
Dataset Type	Mixed: public synthetic event-based deblurring dataset, public real-event deblurring dataset, and custom real-event dataset
Dataset Size	GoPro: 3,214 image pairs, with 1,111 test samples; REBlur: 1,389 pairs, with 486 training and 903 testing pairs; custom real-event dataset: 6,309 pairs, with 2,075 test samples
Dataset Balance / Split	GoPro and custom real-event datasets use about one-third of samples for testing; REBlur uses 486 train / 903 test pairs; class balance N/A

Hybrid Topology	Hybrid SNN-CNN two-stream deblurring network; SNN branch extracts motion features from events, CNN branch extracts background/image features, TSST computes spike motion intensity, and HFEE fuses SNN motion features with CNN features for reconstruction
SNN Component	SNN event feature extractor, spiking convolution, spiking pooling, LIF neurons, Temporal-local-Spatio Spiking Transformer, spike temporal attention
Neuron Model	LIF
Transformer Component	Temporal-local-Spatio Spiking Transformer; spiking temporal attention map; attention over local spatial windows and temporal spike sequences
Attention Type	Spiking temporal attention; temporal-local-spatio self-attention; channel attention in the CNN fusion block
SNN–Transformer Interface Mechanism	SNN extracts spike features from events; TSST computes motion intensity maps from local spike release rates and temporal attention; spike features are scaled by the motion-attention map and converted/mapped into floating-point features for CNN fusion through HFEE
Model Depth / Scale	Deblurring network uses 4-layer encoder and 4-layer decoder; CNN encoder feature dimensions are [32, 64, 128, 256]; SNN branch dimensions are [8, 16, 32, 64]; SNN experiments use 10 temporal sequences; TSST window sizes tested from 16 to 128, with best reported setting around 32
Parameter Count	Full model on real-event dataset: 7.11M parameters; TSST SNN branch parameters reported as 40.938 KB in ablation
Design / Contribution Type	New event-based motion deblurring architecture; first use of SNN-CNN hybrid fusion for image deblurring; proposed TSST for motion intensity extraction; proposed HFEE for motion-background feature fusion
Input Encoding Method	Event-to-spike temporal sequence encoding; raw event temporal sequence is compressed into 10 spike timesteps; event spikes are sampled to millisecond scale for GoPro; blurry image is processed by CNN branch
Encoding Parameters	LIF threshold Vth = 1; leak factor τ = 0.5; initial membrane potential u(0)=0; SNN temporal sequence length Tout = 10; TSST spatial window sizes [32, 16, 8, 4] across encoder layers; GoPro events are temporally extended by doubling timestamps
Encoding Learned or Fixed	Partially fixed and learned: event-to-spike temporal binning and LIF dynamics are fixed; SNN/CNN feature extraction and fusion weights are learned end-to-end
Event / Spike Representation	Binary spike sequences; sparse event stream; spatiotemporal spike tensor; spike motion intensity map; mapped floating-point motion feature after TSST
Temporal Resolution	Event camera has microsecond-level event triggering; model training uses 10 SNN temporal sequences; GoPro maximum timestamp around 40,000; custom real-event maximum timestamp around 300,000
Output Decoding Method	CNN decoder reconstructs the final sharp RGB image from fused SNN-CNN features
Encoding / Temporal Information Analysis	Yes. The paper analyzes motion intensity from spike density and temporal attention, TSST window-size effects, spike-to-CNN feature fusion, and the importance of preserving temporal motion information during deblurring
Training Paradigm	Supervised end-to-end image reconstruction
Training Method	Joint SNN-CNN optimization using backpropagation through time for SNN modules and surrogate-gradient training for spike non-differentiability
Surrogate Gradient Function	Triangle surrogate gradient function
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Charbonnier loss plus edge loss: L = Lchar(Z, GT) + λLedge(Z, GT)
Training Hyperparameters	GoPro: random 512 × 512 crops, batch size 16, Adam, learning rate decreases from 3 × 10■■ to 1 × 10■■. Real Events: GoPro pretrained model initialization except SNN neurons, Adam, learning rate decreases from 1 × 10■■ to 1 × 10■■, batch size 16. Loss weight λ = 0.05, ε = 1 × 10■^{3}
Primary Performance Metric	PSNR and SSIM
Primary Metric Value	GoPro: 35.58 dB PSNR; Real Event dataset: 34.67 dB PSNR and 0.968 SSIM; REBlur: 40.15 dB PSNR
Secondary Metrics	Parameter count, ablation PSNR/SSIM, visual deblurring quality, comparison with image-only and event-based baselines
Baseline Models Compared	DMPHN, MPRNet, MIMO-UNet++, Restormer, D2Nets, NAFNet, DS-Deblur, ERDNet, eSL-Net, EFNet, Concatenate baseline, SNN TATT, SNN SATT, LSTM-event baseline
Performance Gain Over Baseline	On GoPro, improves over EFNet by 0.12 dB PSNR and over NAFNet by 1.89 dB. On the custom real-event dataset, improves over EFNet and other SNN-based variants. On REBlur, achieves 40.15 dB PSNR, improving over EFNet by 2.03 dB
Statistical Validation	Benchmark comparison and extensive ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes event pathway removal, LSTM event baseline, TSST vs Concatenate/SNN TATT/SNN SATT, TSST window-size analysis, parameterized vs non-parameterized fusion block, and Charbonnier/edge-loss ablation
Training Framework / Code Availability	PyTorch; trained using two Intel Xeon Gold 6226R CPUs and four NVIDIA RTX 3090 GPUs; code and pretrained models available at GitHub: XDULzx/MotionSNN
Target Hardware	No direct deployment; neuromorphic chip potential discussed for future work
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No direct energy measurement; paper motivates SNN efficiency and notes that full SNN benefits require brain-like/neuromorphic chip deployment
Latency / Memory / Complexity Evidence	Reports parameter counts and model size in ablations; no measured latency, memory footprint, FLOPs, or hardware energy table reported
Comparison with ANN Baseline	Yes. Compares with CNN/Transformer/image-restoration baselines such as Restormer, NAFNet, MPRNet, MIMO-UNet++, and event-based ANN deblurring methods
Main Contribution / Key Limitation	Contribution: proposes TSST and HFEE to fuse event-based SNN motion features with CNN background features for motion deblurring, showing strong PSNR/SSIM on GoPro, REBlur, and real-event data. Limitation: low-power and high-temporal-resolution advantages of SNNs are not fully demonstrated because no neuromorphic chip deployment or energy measurement is provided
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P013

SparseSpikformer: A Co-Design Framework for Token and Weight Pruning in Spiking Transformer

Full Citation	Liu, Y., Xiao, S., Li, B., & Yu, Z. (2024). SparseSpikformer: A Co-Design Framework for Token and Weight Pruning in Spiking Transformer. In ICASSP 2024: IEEE International Conference on Acoustics, Speech and Signal Processing, pp. 6410–6414. https://doi.org/10.1109/ICASSP48485.2024.10446631
University	Sun Yat-sen University

Country	China
Publication Year	2024
Publication Venue	ICASSP 2024: IEEE International Conference on Acoustics, Speech and Signal Processing
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/ICASSP48485.2024.10446631
Publication Type	Conference paper
Database Source	IEEE Xplore
Paper Category	Efficiency-focused
E&QA; Score	4
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Image classification and neuromorphic event-based classification with token pruning and weight pruning
Data Modality	Static image data and neuromorphic/event-based image data
Signal Dimensionality	2D image sequence represented as spiking patch tokens $I \in R^{(T\times C\times H\times W)}$ and $X' \in R^{(T\times N\times D)}$
Signal Characteristics	Input images are converted into spike token sequences; SparseSpikformer selects foreground/relevant image tokens using average spike firing rate and prunes redundant model weights
Dataset Name	CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS-128
Dataset Type	Public static image and neuromorphic/event-based benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Spiking Transformer based on Spikformer; uses Spiking Patch Splitting, Spiking Token Selector, Spiking Self-Attention, MLP, encoder blocks, and linear classification head
SNN Component	Spike neuron layers, spike sequences, Spiking Patch Splitting, Spiking Token Selector, spike firing-rate-based pruning
Neuron Model	NR
Transformer Component	Spikformer / Spiking Transformer encoder block, Spiking Self-Attention, MLP, classification head
Attention Type	Spiking Self-Attention with token pruning before attention calculation
SNN–Transformer Interface Mechanism	SPS converts input image sequence into spike tokens; STS selects top-ρ% important tokens based on spike firing rate; selected spike tokens are passed into SSA and MLP calculations
Model Depth / Scale	Encoder block × L; token selector inserted after the 2nd, 3rd, and 4th encoder layers; exact L and hidden dimension NR
Parameter Count	Exact original parameter count NR; paper reports that SparseSpikformer can retain only about 10% of parameters after weight pruning
Design / Contribution Type	Co-design framework for token pruning and weight pruning; Spiking Token Selector; Lottery Ticket Hypothesis-based weight pruning; hardware-friendly SparseSpikformer
Input Encoding Method	Spiking Patch Splitting converts image sequence into spike patches/tokens; neuromorphic datasets use event-derived spike input following Spikformer setting
Encoding Parameters	Token keep ratio ρ tested from 0.5 to 0.9; token selectors preserve [ρ, ρ^2, ρ^3] tokens across selected encoder layers; LTH pruning uses $p = 25$ and $K = 15$; fixed token keep rate $\rho = 0.7$ used in weight-pruning experiment
Encoding Learned or Fixed	Partially learned: token selector uses MLP and Gumbel-Softmax for differentiable token decision; spike firing-rate criterion guides token selection; weight pruning uses fixed LTH/IMP procedure
Event / Spike Representation	Spike sequence; spike tokens; binary keep/drop token decision $D \in \{0,1\}^N$; sparse weight masks $M \in \{0,1\}$
Temporal Resolution	T timesteps used in spike image sequence, but exact timestep value NR
Output Decoding Method	Fully connected linear classification head outputs class prediction
Encoding / Temporal Information Analysis	Partial. The paper visualizes spike firing-rate attention maps and token pruning behavior, but does not deeply analyze temporal information preservation or encoding loss
Training Paradigm	Supervised learning with pruning-based model compression
Training Method	Follows original Spikformer training principles; token selector trained with Gumbel-Softmax; weight pruning uses Lottery Ticket Hypothesis with Iterative Magnitude Pruning and retraining
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	Experiments conducted on one NVIDIA Tesla V100 GPU; token selector inserted after 2nd, 3rd, and 4th encoder blocks; keep ratios 0.5–0.9; IMP uses $p = 25$, $K = 15$; other training details follow Spikformer but are NR
Primary Performance Metric	Top-1 accuracy, throughput, FLOPs, and weight sparsity
Primary Metric Value	With token pruning, SparseSpikformer maintains competitive accuracy: CIFAR-10 up to 95.22%, CIFAR-100 up to 77.81%, CIFAR10-DVS up to 79.3%, and DVS-128 up to 98.26–98.95 depending on pruning setting
Secondary Metrics	Keep ratio, throughput, GFLOPs, sparsity level, ablation accuracy, IMP/EB/RR pruning comparison
Baseline Models Compared	Original Spikformer; Random Selector; Early-Bird pruning; Iterative Magnitude Pruning; Random Re-initialization
Performance Gain Over Baseline	Reduces model parameters by about 90% and GFLOPs by about 20%; token pruning reduces FLOPs from 3.74G to 2.75G on CIFAR settings and from 7.78G to 6.03G on DVS settings at $\rho = 0.5$; throughput improves across datasets
Statistical Validation	Benchmark evaluation and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full for pruning focus. Includes different token keep ratios, weight sparsity levels, Random Selector vs Spiking Token Selector, and IMP vs Early-Bird vs Random Re-initialization
Training Framework / Code Availability	Training framework NR; code availability NR
Target Hardware	Edge devices / resource-constrained deployment are the motivation; no specific hardware platform deployed
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	No direct energy measurement; efficiency is inferred from reduced FLOPs, throughput improvement, and parameter pruning
Latency / Memory / Complexity Evidence	Reports throughput and FLOPs under different keep ratios; reports up to about 90% weight pruning; does not report measured latency, memory footprint, or hardware energy
Comparison with ANN Baseline	No direct ANN baseline; comparison is mainly against Spikformer and pruning variants

Main Contribution / Key Limitation	Contribution: proposes SparseSpikformer, a co-design pruning framework that combines spike firing-rate-based token selection and LTH-based weight pruning to reduce computation and storage while maintaining accuracy. Limitation: short conference paper with limited training details, no real hardware deployment, and no measured energy/latency results
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4

P014

SLTNet: Efficient Event-based Semantic Segmentation with Spike-driven Lightweight Transformer-based Networks

Full Citation	Long, X., Zhu, X., Guo, F., Zhang, W., Gu, Q., Chen, C., & Gu, F. (2025). SLTNet: Efficient Event-based Semantic Segmentation with Spike-driven Lightweight Transformer-based Networks. In 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 4331–4338. https://doi.org/10.1109/IROS60139.2025.11247662
University	Chongqing University
Country	China
Publication Year	2025
Publication Venue	IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2025
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/IROS60139.2025.11247662
Publication Type	Conference paper
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Event-based semantic segmentation using only event-camera data
Data Modality	Event-camera streams
Signal Dimensionality	3D spatiotemporal event representation converted into voxel/event tensor format
Signal Characteristics	Raw event stream represented as (x, y, t, p) with polarity $p \in \{-1,+1\}$; DDD17 uses DAVIS event data at 346 × 260 resolution; DSEC-Semantic uses 440 × 640 event/RGB driving sequences; event interval Δt = 50 ms
Dataset Name	DDD17; DSEC-Semantic
Dataset Type	Public event-based semantic segmentation benchmark datasets
Dataset Size	DDD17: 15,950 training and 3,890 testing pairs with 6 semantic categories. DSEC-Semantic: 8,082 training frames from 8 sequences and 2,809 testing frames from 3 sequences with 11 categories
Dataset Balance / Split	Uses official or commonly adopted train/test splits; class balance NR
Hybrid Topology	Single-branch hierarchical SNN-based encoder-decoder with spike-driven convolution blocks and spike-driven Transformer blocks; first three encoder stages use SCBs, final stage uses two STBs for long-range context
SNN Component	Evolutionary Spiking Neuron, LIF-based spiking neurons, Spike-driven Convolution Blocks, Spike-LD modules, binary spike transmission, membrane shortcut
Neuron Model	Evolutionary Spiking Neuron based on LIF
Transformer Component	Spike-driven Transformer Block; spike-driven multi-head self-attention; two-layer MLP inside STB
Attention Type	Spike-driven multi-head self-attention; binary mask / Hadamard-product attention
SNN–Transformer Interface Mechanism	Event streams are converted into voxel/event tensors and processed through ESN-based spike blocks; STB generates spike-driven Q, K, and V through re-parameterization convolution and spiking neurons; SDMSA computes attention using $\mathrm{SUM_c}(Q \otimes K) \otimes V$
Model Depth / Scale	Four-stage encoder; initial block with three spike convolution layers; stages 1–3 use SCBs with three Spike-LD modules each; stage 4 uses two STBs; decoder has three upsampling stages with Spike-LD modules and two segmentation heads
Parameter Count	DDD17: 0.41M parameters; DSEC-Semantic: 1.67M parameters
Design / Contribution Type	New lightweight event-based semantic segmentation architecture; Spike-driven Convolution Block; Spike-LD module; Spike-driven Transformer Block; SDMSA attention; membrane shortcut; efficient single-branch SNN segmentation design
Input Encoding Method	Event voxel-grid encoding followed by event tensor conversion for SNN processing
Encoding Parameters	Event interval Δt = 50 ms; membrane time constant T = 1; LIF reset value $U_{reset} = 0$; membrane time constant $\tau = 0.25$; spike threshold $U_{th} = 1.0$; EvAF uses $K_{min} = 1$ and $K_{max} = 10$
Encoding Learned or Fixed	Fixed event voxel-grid encoding; spike-driven feature extraction is learned end-to-end
Event / Spike Representation	Event voxel grid; event tensor; binary spike activations; membrane potentials; spike-driven Q/K/V
Temporal Resolution	Event interval Δt = 50 ms; model uses timestep/membrane constant setting T = 1; event cameras provide microsecond-level sensing but model processing uses binned event representation
Output Decoding Method	Lightweight spiking decoder and two segmentation heads generate prediction probability maps P1 and P2
Encoding / Temporal Information Analysis	Partial. The paper analyzes event representation, spike-driven feature extraction, binary spike attention, energy estimation, and segmentation visualizations, but does not deeply analyze temporal information loss
Training Paradigm	Supervised end-to-end semantic segmentation
Training Method	Direct spike-driven training with surrogate-gradient backpropagation using evolutionary surrogate activation function
Surrogate Gradient Function	Evolutionary surrogate gradient function / EvAF
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Cross-entropy semantic loss with Online Hard Example Mining plus early-stage cross-entropy loss: Loss = $\lambda 1 l1 + \lambda 2 l2$
Training Hyperparameters	DDD17: 300 epochs, learning rate 1e-3, weight decay 1e-4, batch size 64, StepLR step size 5, γ = 0.92. DSEC-Semantic: 500 epochs, learning rate 1e-3, weight decay 1e-4, batch size 16, StepLR step size 10, γ = 0.92. Loss weights: $\lambda 1$ = 1.0, $\lambda 2$ = 0.4, OHEM top-k ratio k = 0.7
Primary Performance Metric	Mean Intersection over Union, mIoU
Primary Metric Value	DDD17: 51.93% mIoU, 0.41M parameters, 1.96G FLOPs, 114.29 FPS. DSEC-Semantic: 47.91% mIoU, 1.67M parameters, 6.97G FLOPs, 114.29 FPS
Secondary Metrics	Parameters, FLOPs, FPS, energy consumption, MAC/ACC operation counts, qualitative segmentation results
Baseline Models Compared	Ev-SegNet, Evdistill, ESS, Spiking-DeepLab, Spiking-FCN, EvSegSNN, SCGNet, MFS-EFS, SDSA2, SDSA3, Spike-driven Transformer, Spikformer

Performance Gain Over Baseline	On DDD17, improves over Spiking-DeepLab by 7.30% mIoU, Spiking-FCN by 9.06%, EvSegSNN by 6.39%, and SCGNet by 2.63%. On DSEC-Semantic, improves over Spiking-DeepLab by 3.33%, Spiking-FCN by 9.39%, and MFS-EFS by 1.81%. Uses 57.93× fewer parameters and 22.54× fewer FLOPs than Ev-SegNet on DDD17
Statistical Validation	Benchmark comparison and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes SDMSA comparison with SDSA2, SDSA3, Spike-driven Transformer, and Spikformer; shortcut ablation with MS, VS, and SEW; architecture ablation without FE, STB, and fusion; loss-design ablation with OHEM and early loss
Training Framework / Code Availability	Trained on a single NVIDIA RTX 4090 GPU with 24 GB memory; code available at GitHub: longxianlei/SLTNet-v1.0
Target Hardware	Resource-constrained edge/mobile robotics platforms are the motivation; no specific neuromorphic hardware target deployed
Hardware Deployment Status	No real hardware deployment reported; evaluated on GPU
Energy Evidence	Theoretical energy estimation using MAC and ACC energy assumptions; reports SLTNet energy consumption of 1.42 mJ on DDD17, lower than ANN and SNN baselines
Latency / Memory / Complexity Evidence	Reports 114.29 FPS inference speed; DDD17 model uses 0.41M parameters and 1.96G FLOPs; DSEC-Semantic model uses 1.67M parameters and 6.97G FLOPs; energy comparison table reports 131M MAC and 1830M ACC for SLTNet
Comparison with ANN Baseline	Yes. Compares with ANN-based event segmentation methods Ev-SegNet, Evdistill, and ESS
Main Contribution / Key Limitation	Contribution: proposes SLTNet, a lightweight event-only SNN–Transformer segmentation network combining SCBs, Spike-LD, STBs, SDMSA, and membrane shortcuts to improve mIoU, FPS, and energy efficiency. Limitation: still has an accuracy gap compared with some ANN/event+frame methods; energy results are theoretical, and no neuromorphic hardware deployment is reported
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P015

SpikeHCD: Spiking Transformer With Parallel Neurons and Memory-Enhanced Attention for Hyperspectral Change Detection

Full Citation	Mei, Z., Li, J., Zhang, B., Wang, C., Guo, L., & Qian, J. (2025). SpikeHCD: Spiking Transformer With Parallel Neurons and Memory-Enhanced Attention for Hyperspectral Change Detection. IEEE Transactions on Geoscience and Remote Sensing, 63, Article 5534712. https://doi.org/10.1109/TGRS.2025.3636865
University	Ningbo University
Country	China
Publication Year	2025
Publication Venue	IEEE Transactions on Geoscience and Remote Sensing
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TGRS.2025.3636865
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Remote Sensing, Geospatial, and Environmental Monitoring
Specific Task	Hyperspectral change detection between multitemporal hyperspectral images
Data Modality	Multitemporal hyperspectral images
Signal Dimensionality	3D hyperspectral image cube with spectral–spatial dimensions; bitemporal inputs represented as X1, X2 ∊ R^(C×H×W) and converted into 4-timestep patch sequences
Signal Characteristics	High spectral dimensionality; Farmland has 155 valid spectral bands after noise removal; River has 198 valid spectral bands; Hermiston has 154 valid spectral bands; patch size set to 3 × 3
Dataset Name	Farmland, River, Hermiston
Dataset Type	Public hyperspectral change-detection benchmark datasets
Dataset Size	Farmland: 450 × 140 pixels, 155 valid bands; River: 463 × 241 pixels, 198 valid bands; Hermiston: 307 × 241 pixels, 154 valid bands
Dataset Balance / Split	3% training, 2% validation, remaining samples for testing
Hybrid Topology	Spiking Transformer architecture for hyperspectral CD; includes Spiking Patch Embedding, dual-branch memory-enhanced spiking Transformer block, Spiking Difference Module, and classification layer
SNN Component	Probability-driven parallel spiking neurons, spiking patch embedding, spiking attention, spiking difference module, spike sequences
Neuron Model	Probability-driven Parallel Spiking Neuron based on LIF dynamics; traditional LIF and PSN used as comparison
Transformer Component	Memory-enhanced spiking Transformer block with MSA and MLP modules; dual spectral–temporal and spatial–temporal branches
Attention Type	Memory-enhanced spiking attention; spatial–temporal attention; spectral–temporal attention; compared with SSA and QK attention
SNN–Transformer Interface Mechanism	Multitemporal hyperspectral patches are duplicated into 4-timestep image sequences and embedded by SPE; PPSN generates spike-form features; MSA uses spike-form Q and K plus memory feature M to model historical temporal information
Model Depth / Scale	Input duplicated to 4 timesteps; patch size 3 × 3; dual-branch memory-enhanced spiking Transformer block; embedded sequence dimension 128 for Farmland and Hermiston, 256 for River
Parameter Count	Farmland: 0.56M; River: 2.05M; Hermiston: 0.56M
Design / Contribution Type	New Spiking Transformer for hyperspectral CD; first SNN specifically designed for hyperspectral change detection; proposes PPSN, MSA, and SDM
Input Encoding Method	Bitemporal hyperspectral images are duplicated into four-timestep image sequences, divided into p × p patches, and processed by spiking patch embedding
Encoding Parameters	Timestep T = 4; patch size p = 3; MSA decay parameter λ initialized at 0.2 and learned; PPSN uses probability-driven reset based on membrane-potential extrema
Encoding Learned or Fixed	Partially learned: timestep duplication and patch extraction are fixed, while SPE, MSA, PPSN-based network parameters, and λ are learned
Event / Spike Representation	Binary spike sequences; membrane potential; spike firing probability matrix; spike-form Q and K; memory representation M; temporal difference spike features

Temporal Resolution	4 timesteps for main experiments; ablation tests timestep lengths from 4 to 128
Output Decoding Method	SDM features are passed to a fully connected classification layer for binary changed/unchanged prediction
Encoding / Temporal Information Analysis	Yes. The paper analyzes temporal memory in MSA, parallel spike computation in PPSN, timestep length effects, temporal difference strategies in SDM, and patch-size effects
Training Paradigm	Supervised binary change detection
Training Method	End-to-end training with backpropagation through time across spatial and temporal domains
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Cross-entropy classification loss for changed/unchanged binary labels
Training Hyperparameters	150 epochs; learning rate 0.0004; batch size 128; patch size 3 × 3; 3% train, 2% validation, remaining test; λ initialized to 0.2; timestep T = 4
Primary Performance Metric	Overall Accuracy and Kappa coefficient
Primary Metric Value	Farmland: 98.27% OA, 95.80 KC, 1.16 mJ, 0.56M parameters. River: 97.61% OA, 84.09 KC, 3.67 mJ, 2.05M parameters. Hermiston: 97.21% OA, 91.95 KC, 1.22 mJ, 0.56M parameters
Secondary Metrics	Energy consumption, parameter count, inference speedup ratio, OA under timestep variation, ablation OA/Kappa
Baseline Models Compared	CVA, SVM, CDFormer, CSANet, SSTFormer, GLAFormer, MFCEN, BTCDNet, AIWSEN, MsFNet, LIF-neuron variant, PSN variant, SSA, QK attention
Performance Gain Over Baseline	Achieves best OA/KC on Farmland and River while using lowest energy; on Hermiston achieves second-highest accuracy but much lower parameters and energy than MFCEN; PPSN improves efficiency over LIF by parallelizing timestep computation
Statistical Validation	Benchmark comparison and extensive ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes model size, PPSN vs LIF/PSN, timestep length, dual-branch structure, MSA, patch size, and SDM temporal difference strategies
Training Framework / Code Availability	PyTorch; NVIDIA GTX 3090 GPUs; code available at GitHub: mzhcode/HCD_snn
Target Hardware	Resource-constrained satellite/remote-sensing devices are the motivation; energy estimated using 45-nm MAC/AC assumptions
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Theoretical energy estimation based on SOP, firing rate, timestep, FLOPs, and 45-nm operation energy assumptions: MAC = 4.6 pJ, AC = 0.9 pJ
Latency / Memory / Complexity Evidence	Reports parameter count, energy consumption, and PPSN speedup ratio; no real measured hardware latency or memory footprint reported
Comparison with ANN Baseline	Yes. Compares with multiple ANN and Transformer-based hyperspectral CD methods including CDFormer, CSANet, SSTFormer, GLAFormer, MFCEN, BTCDNet, AIWSEN, and MsFNet
Main Contribution / Key Limitation	Contribution: proposes SpikeHCD, the first SNN/Spiking Transformer designed for hyperspectral change detection, using PPSN for parallel spike computation, MSA for temporal memory, and SDM for change-feature extraction. Limitation: energy results are theoretical, no real neuromorphic or satellite hardware deployment is reported, and performance is validated only on three hyperspectral CD datasets
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P016

IML-Spikeformer: Input-Aware Multilevel Spiking Transformer for Speech Processing

Full Citation	Song, Z., Zhang, S., Chou, Y., Wu, J., & Li, H. (2026). IML-Spikeformer: Input-Aware Multilevel Spiking Transformer for Speech Processing. IEEE Transactions on Neural Networks and Learning Systems, 37(3), 1377–1389. https://doi.org/10.1109/TNNLS.2025.3615971
University	The Hong Kong Polytechnic University
Country	China
Publication Year	2026
Publication Venue	IEEE Transactions on Neural Networks and Learning Systems
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TNNLS.2025.3615971
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Speech, Audio, and Auditory Processing
Specific Task	Automatic speech recognition, speaker identification, and speaker verification
Data Modality	Speech/audio signals represented as acoustic feature sequences
Signal Dimensionality	1D temporal speech sequences; acoustic inputs represented using 80-dimensional log-Mel filterbank features
Signal Characteristics	Variable-length speech sequences; speech features are downsampled by two 3 × 3 convolutional layers with stride 2; model targets multiscale temporal dependencies from phoneme-level to utterance-level speech structure
Dataset Name	AiShell-1, LibriSpeech-960, VoxCeleb1, VoxCeleb2
Dataset Type	Public speech recognition and speaker recognition benchmark datasets
Dataset Size	AiShell-1: 170 hours Mandarin speech; LibriSpeech-960: approximately 960 hours English audiobook speech; VoxCeleb1: over 100,000 utterances from 1,251 speakers; VoxCeleb2: over 1,000,000 utterances from 6,112 speakers; VoxCeleb speaker identification split: 138,361 training utterances and 8,251 test utterances
Dataset Balance / Split	Uses ESPnet recipes for ASR; VoxCeleb1 official split for speaker identification; VoxCeleb1-O for speaker verification; exact class balance NR
Hybrid Topology	Spiking Transformer architecture with IML-Spikeformer blocks; each block contains HD-RepSSA for token mixing and Spiking ChannelMLP for channel mixing

SNN Component	Input-aware multilevel spike firing neurons, multilevel spikes, binary spike trains for inference, spike-driven computation, spiking ChannelMLP, spike-driven attention
Neuron Model	Input-aware multilevel spike firing neuron based on IF/LIF-style spiking dynamics with adaptive thresholding
Transformer Component	IML-Spikeformer block, HD-RepSSA, RepSSA, HDM, ChannelMLP, Transformer encoder for speech tasks
Attention Type	Hierarchical decay reparameterized spiking self-attention; softmax RepSSA and linear RepSSA variants; HDM-modulated spiking attention
SNN–Transformer Interface Mechanism	IMLS converts presynaptic speech features into multilevel spikes during training and equivalent binary spike trains during inference; HD-RepSSA computes attention maps using reparameterized query-key weights and modulates them with hierarchical decay masks
Model Depth / Scale	ASR models tested at 30.35M, 46.11M, and 61.89M parameters; AiShell uses 4 or 6 timesteps; LibriSpeech uses 4 or 6 timesteps; speaker identification uses 0.6M, 1.18M, and 1.76M parameter models
Parameter Count	Main AiShell model: 30.35M; larger ASR models: 46.11M and 61.89M; LibriSpeech model: 99.36M; speaker identification models: 0.6M, 1.18M, and 1.76M
Design / Contribution Type	New Spiking Transformer for large-scale speech processing; proposes IMLS firing, input-aware adaptive thresholding, RepSSA, HDM, and HD-RepSSA
Input Encoding Method	80-dimensional log-Mel filterbank acoustic features; SpecAugment and speed perturbation used for ASR; features are downsampled by convolutional subsampling before Transformer encoding
Encoding Parameters	IMLS uses time windows such as 4 and 6 timesteps; adaptive threshold scaling factor λ is computed from channelwise maximum presynaptic inputs; running average uses momentum α; HDM uses layer-specific decay function $\phi(l) = 1 – 2^{(-5–l)}$
Encoding Learned or Fixed	Partially learned/adaptive: acoustic feature extraction is fixed preprocessing; IMLS threshold scaling is input-aware during training and fixed during inference using running average; model weights are learned
Event / Spike Representation	Multilevel spike during training; equivalent binary spike train during inference; spike firing rates; spike-driven synaptic operations; attention maps modulated by HDM
Temporal Resolution	Main reported IML-Spikeformer settings use 4 or 6 timesteps; IMLS simulates multitimestep firing within a single training timestep
Output Decoding Method	ASR uses Transformer decoder for token/character or word prediction; speaker identification uses classification output; speaker verification uses speaker embedding comparison and EER evaluation
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike firing-rate distributions, attention maps across layers, hierarchical temporal dependencies, and training efficiency/memory reduction from IMLS
Training Paradigm	Supervised learning for ASR, speaker identification, and speaker verification
Training Method	Direct training of Spiking Transformer using IMLS firing; generalized straight-through estimator for multilevel spike gradients; HD-RepSSA training with reparameterization
Surrogate Gradient Function	Generalized straight-through estimator
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	ASR and speaker task losses follow ESPnet / speaker-recognition recipes; exact loss formulas NR
Training Hyperparameters	ASR follows ESPnet recipes; ASR uses 80-D log-Mel features, SpecAugment, speed perturbation, two 3 × 3 stride-2 convolutional subsampling layers, and a fixed six-layer Transformer decoder; IMLS tested with 4 and 6 timesteps; detailed learning rate, batch size, and epoch settings NR
Primary Performance Metric	CER/WER for ASR; accuracy for speaker identification; EER for speaker verification; theoretical inference energy
Primary Metric Value	AiShell-1: 5.9% CER on test set for 30.35M model with 6 timesteps; LibriSpeech-960: 3.4% WER on clean test set for 99.36M model with 6 timesteps; speaker identification: up to 74.34% accuracy; speaker verification: 2.70% EER when trained on VoxCeleb2
Secondary Metrics	Energy consumption, energy saving ratio, parameter count, spike firing rate, training time per epoch, GPU memory cost
Baseline Models Compared	ANN Transformer, VGG-BiLSTM, Binary S4D, LIF SMLP, GSU SMLP, Spiking TCN, Spike-driven Transformer/SDSA-3, Spiking-LEAF
Performance Gain Over Baseline	On AiShell-1, IML-Spikeformer reduces CER from 11.9% in spike-driven Transformer to 5.9%; achieves comparable CER to ANN Transformer with 4.64× theoretical energy reduction. On LibriSpeech-960, achieves 4.32× energy saving compared with ANN Transformer. IMLS reduces per-epoch training time by 2.24× and GPU memory cost by 3.31× compared with iterative multitimestep firing
Statistical Validation	Benchmark comparison and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes firing mechanism ablation, HD-RepSSA component ablation, linear vs softmax attention, spike firing-rate analysis, attention-map analysis, and training-efficiency analysis
Training Framework / Code Availability	ESPnet used for ASR recipes; code and model checkpoints available at GitHub: Pooookeman/IML-Spikeformer
Target Hardware	Neuromorphic hardware motivation; Loihi and Speck compatibility discussed for multilevel-to-binary spike deployment; no direct chip experiment
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using operation counts; 45-nm CMOS assumptions with MAC = 4.6 pJ and AC = 0.9 pJ; includes synaptic and neuronal operation energy estimation
Latency / Memory / Complexity Evidence	Reports theoretical inference energy, energy-saving ratios, training-time reduction, and GPU memory reduction; no measured hardware latency reported
Comparison with ANN Baseline	Yes. Compares directly with ANN Transformer and VGG-BiLSTM baselines across ASR and speaker tasks
Main Contribution / Key Limitation	Contribution: proposes IML-Spikeformer, a scalable Spiking Transformer for large-scale speech processing using IMLS firing and HD-RepSSA to improve performance, training efficiency, and theoretical energy efficiency. Limitation: energy results are theoretical; no real neuromorphic deployment is reported; optimizer and some training details are not fully specified
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P017

DESTformer: Energy-Efficient Monocular Depth Estimation with Spiking Transformer for Edge Devices

Full Citation	Tumpa, S. A., Devulapally, A., Lee, C., Das, C. R., & Narayanan, V. (2025). DESTformer: Energy-Efficient Monocular Depth Estimation with Spiking Transformer for Edge Devices. In 2025 International Conference on Neuromorphic Systems (ICONS), pp. 26–33. https://doi.org/10.1109/ICONS69015.2025.00015
University	Pennsylvania State University

Country	United States
Publication Year	2025
Publication Venue	2025 International Conference on Neuromorphic Systems, ICONS
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/ICONS69015.2025.00015
Publication Type	Conference paper
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Event-based monocular dense depth estimation
Data Modality	Event-camera data; voxel-grid event representation; spike input generated from RGB frames for one experiment
Signal Dimensionality	3D spatiotemporal event tensor; event stream represented as spatial dimensions H × W with temporal bins B
Signal Characteristics	Event stream represented as (x, y, t, p) with polarity p ∊ {+1, −1}; uses 50 ms event windows and 5 temporal bins; inputs resized to 224 × 224 for Transformer encoder
Dataset Name	MVSEC; DENSE
Dataset Type	Public event-based monocular depth estimation benchmark datasets; MVSEC real event-camera dataset; DENSE synthetic event dataset from CARLA
Dataset Size	MVSEC: Outdoor Day2 used for training; Outdoor Night1, Night2, Night3, and Day1 used for testing; 8,523 training samples, 1,826 validation samples, and approximately 5,000 testing samples. DENSE: 5 training sequences / 5,000 samples, 2 validation sequences / 2,000 samples, and 1 test sequence / 1,000 samples
Dataset Balance / Split	MVSEC follows prior split using Outdoor Day2 for training and Outdoor Night1/2/3 and Day1 for testing. DENSE uses Towns 01–05 for training, Towns 06–07 for validation, and Town10 for testing
Hybrid Topology	Hybrid SNN–Transformer encoder with non-spiking lightweight convolutional decoder; spiking Transformer encoder extracts sparse global event features, decoder reconstructs dense depth map
SNN Component	LIF neurons, MultiStepLIFNode, spiking patch embedding, spiking Transformer encoder, spike-based activations, spiking MLP
Neuron Model	LIF
Transformer Component	QKFormer-based hierarchical spiking Transformer encoder; Spiking Patch Embedding with Deformed Shortcut; Q-K attention block; Spiking MLP
Attention Type	Q-K attention in hierarchical Spiking Transformer / QKFormer-style attention
SNN–Transformer Interface Mechanism	Event voxel-grid or spike input is divided into patches through spiking patch embedding; spike activations are processed by QKFormer blocks; final spiking activations are averaged across timesteps before entering the non-spiking depth decoder
Model Depth / Scale	Transformer encoder depth = 10; Stage 1 has 1 spiking Transformer unit, Stage 2 has 2 units, and Stage 3 has 7 units
Parameter Count	17.00M parameters
Design / Contribution Type	New event-based monocular depth estimation framework using Spiking Transformer encoder and lightweight decoder; edge deployment on NVIDIA Jetson Orin Nano
Input Encoding Method	Spatiotemporal voxel-grid encoding for event input; Poisson spike generation for spike-input DENSE experiment
Encoding Parameters	Event window ΔT = 50 ms; temporal bins B = 5; input resized to 224 × 224; LIF membrane time constant τ = 2.0; threshold voltage = 1.0; reset voltage = 0
Encoding Learned or Fixed	Fixed event voxel-grid preprocessing and Poisson spike generation; spiking Transformer weights learned through supervised training
Event / Spike Representation	Event voxel grid; spike input; sparse spiking activations; temporally aggregated spiking feature map
Temporal Resolution	50 ms event accumulation window with 5 temporal bins
Output Decoding Method	Final encoder spiking activations are averaged across timesteps, passed through convolutional and upsampling decoder layers, and output as a single-channel inverse depth map
Encoding / Temporal Information Analysis	Partial. The paper uses voxel-grid temporal bins and timestep aggregation, and reports qualitative depth maps, but does not deeply analyze temporal information preservation or spike encoding loss
Training Paradigm	Supervised depth regression
Training Method	Direct supervised training of hybrid spiking Transformer encoder and lightweight decoder
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Combined depth loss: L1 loss, normal loss, multiscale gradient matching loss, and scale-invariant logarithmic loss
Training Hyperparameters	100 epochs; batch size 8; learning rate 3 × 10■■; AdamW optimizer; transformer encoder depth 10; loss weights α = 1.0, β = 0.5, γ = 0.25, ζ = 10; LIF τ = 2.0, threshold = 1.0, reset = 0
Primary Performance Metric	Depth estimation error and edge deployment efficiency
Primary Metric Value	Edge deployment on Jetson Orin Nano: 65.24 MB model size, 17.00M parameters, 4.5 GB RAM, 0.83 s latency, and 7.40 J average energy. DENSE spike-input Town10: Abs Rel 0.27, Sq Rel 0.39, RMSE log 0.40, SI log 0.11, δ<1.25 = 0.69
Secondary Metrics	MAE at 10 m, 20 m, and 30 m cutoff; Abs Rel; Sq Rel; RMSE log; SI log; δ accuracy thresholds; model size; RAM usage; latency; energy
Baseline Models Compared	Zhu et al.; E2Depth; StereoSpike; EReFormer; Spike-T; Zhang et al.; U-Net; FCN decoder; RRDB decoder
Performance Gain Over Baseline	On Jetson Orin Nano, DESTformer achieves 3.76× speedup and 4.2× lower energy compared with existing methods; compared with Spike-T, latency improves from 3.14 s to 0.83 s and energy reduces from 30.93 J to 7.40 J; compared with EReFormer, latency improves from 2.10 s to 0.83 s and energy reduces from 13.21 J to 7.40 J
Statistical Validation	Benchmark comparison, edge-device deployment, and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full for design choices. Includes decoder architecture, scale-invariant logarithmic loss, input skip connection, and SNN threshold value
Training Framework / Code Availability	PyTorch; SpikingJelly; MultiStepLIFNode; trained on NVIDIA A100 80GB PCIe GPUs; code availability NR
Target Hardware	NVIDIA Jetson Orin Nano edge device; future neuromorphic hardware such as Intel Loihi mentioned
Hardware Deployment Status	Deployed and measured on NVIDIA Jetson Orin Nano; no neuromorphic chip deployment reported
Energy Evidence	Measured edge-device energy on Jetson Orin Nano; DESTformer reports 7.40 J average energy per inference

Latency / Memory / Complexity Evidence	Reports 65.24 MB model size, 17M parameters, 4.5 GB RAM usage, 0.83 s latency, and 7.40 J energy on Jetson Orin Nano
Comparison with ANN Baseline	Yes. Compares with ANN/event-based depth estimation methods such as E2Depth and EReFormer, plus spike-based baselines
Main Contribution / Key Limitation	Contribution: proposes DESTformer, an event-based monocular depth estimation model using a QKFormer-style spiking Transformer encoder and lightweight decoder, with real edge-device deployment showing lower latency and energy. Limitation: decoder remains non-spiking; performance is not consistently best across all MVSEC/DENSE metrics; no neuromorphic hardware deployment or measured neuromorphic energy is reported
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P018

AutoST: Training-Free Neural Architecture Search for Spiking Transformers

Full Citation	Wang, Z., Zhao, Q., Cui, J., Liu, X., & Xu, D. (2024). AutoST: Training-Free Neural Architecture Search for Spiking Transformers. In ICASSP 2024: IEEE International Conference on Acoustics, Speech and Signal Processing, pp. 3455–3459. https://doi.org/10.1109/ICASSP48485.2024.10445971
University	North Carolina State University
Country	United States
Publication Year	2024
Publication Venue	ICASSP 2024: IEEE International Conference on Acoustics, Speech and Signal Processing
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/ICASSP48485.2024.10445971
Publication Type	Conference paper
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Training-free neural architecture search for Spiking Transformers; image classification on static and neuromorphic datasets
Data Modality	Static image data and neuromorphic/event-based image data
Signal Dimensionality	2D image sequences represented as I ∈ R^(T×C×H×W) and converted into spiking patch tokens X ∈ R^(T×N×D)
Signal Characteristics	Uses binary spike trains; image input is processed through spiking patch embedding, Spiking Transformer blocks, and classification head; timestep value of 4 selected as preferred efficiency–accuracy trade-off
Dataset Name	CIFAR-10, CIFAR-100, ImageNet, CIFAR10-DVS
Dataset Type	Public static image classification and neuromorphic vision benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Pure Spiking Transformer architecture searched by AutoST; consists of Spiking Patch Embedding, multiple Spiking Transformer blocks, Spiking Self-Attention, Spiking MLP, Global Average Pooling, and fully connected classifier
SNN Component	LIF neurons, binary spike trains, spiking patch embedding, spiking self-attention, spiking MLP
Neuron Model	LIF
Transformer Component	Spiking Transformer blocks; Spiking Self-Attention; Spiking MLP; patch embedding; classification head
Attention Type	Spiking Self-Attention
SNN–Transformer Interface Mechanism	Spiking Patch Embedding converts input image sequence into spiking patches; patches are processed by stacked SSA and SMLP blocks with residual connections
Model Depth / Scale	Search spaces: AutoST-tiny depth 1–8, embedding 192–384; AutoST-small depth 2–12, embedding 256–512; AutoST-base depth 4–15, embedding 384–768. Preferred timestep setting: 4
Parameter Count	AutoST-tiny: 4.20M on CIFAR, 5.99M on ImageNet; AutoST-small: 11.52M on CIFAR, 14.68M on ImageNet; AutoST-base: 29.64M on CIFAR, 34.44M on ImageNet
Design / Contribution Type	First training-free NAS method for Spiking Transformers; uses FLOPs as a training-free performance predictor; evolutionary search over embedding size, head number, MLP ratio, and depth
Input Encoding Method	Spiking Patch Embedding converts 2D image sequence into spiking patches
Encoding Parameters	Search variables include embedding size, MLP ratio, number of heads, and depth; timestep values analyzed as 1, 2, 4, 8, and 16; selected setting uses 4 timesteps for main experiments
Encoding Learned or Fixed	Partially learned: architecture is searched using FLOPs without training; selected model weights are trained for evaluation; spike neuron dynamics are fixed LIF-based
Event / Spike Representation	Binary spike trains; spiking patch tokens; spike outputs from LIF neurons
Temporal Resolution	Main experiments use 4 timesteps; CIFAR10-DVS comparison uses 16 timesteps
Output Decoding Method	Global Average Pooling followed by fully connected classification head
Encoding / Temporal Information Analysis	Partial. The paper analyzes timestep impact on accuracy and efficiency, but does not deeply analyze spike encoding information or temporal representation quality
Training Paradigm	Training-free neural architecture search followed by supervised training/evaluation of selected architectures
Training Method	Evolutionary search using FLOPs as performance metric; selected architectures are trained for classification evaluation
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	100-epoch training used for sampled architecture correlation analysis; 200 samples searched on CIFAR-100 with a single GPU; detailed final training hyperparameters NR
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	CIFAR-10: AutoST-tiny 95.14%, AutoST-small 96.03%, AutoST-base 96.21%. CIFAR-100: AutoST-tiny 76.29%, AutoST-small 79.44%, AutoST-base 79.69%. ImageNet: AutoST-tiny 63.80%, AutoST-small 71.02%, AutoST-base 74.54%. CIFAR10-DVS: 81.6%

Secondary Metrics	FLOPs, parameter count, timesteps, Kendall correlation, Spearman correlation, search time, energy consumption
Baseline Models Compared	Spikformer, AutoSNN, SNASNet-Bw, TET, DSR, SpikeDHS, Diet-SNN, RMP, SEW-ResNet, tdBN, LIAF-Net, PLIF, Dspike
Performance Gain Over Baseline	Compared with other NAS-searched SNN models, AutoST improves accuracy by 3.06% on CIFAR-10, 4.34% on CIFAR-100, and 9.10% on CIFAR10-DVS. AutoST-base reaches 96.21% on CIFAR-10 and 79.69% on CIFAR-100, outperforming Spikformer-8-512 with similar parameters
Statistical Validation	Correlation analysis using Kendall and Spearman coefficients; benchmark comparisons and topology/timestep analysis; no standard deviation, confidence interval, repeated-run analysis, or significance test reported
Ablation Study	Partial. Includes metric-correlation comparison, timestep analysis, depth analysis, embedding-width analysis, and comparison across tiny/small/base search spaces
Training Framework / Code Availability	Full code, model, and data are released at GitHub: AlexandreWANG915/AutoST
Target Hardware	Edge/resource-constrained SNN deployment is the motivation; no specific hardware platform deployed
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	Partial. Energy consumption is compared visually for CIFAR-10 models and discussed through timesteps/FLOPs, but no measured hardware energy or detailed energy-estimation formula is reported
Latency / Memory / Complexity Evidence	Reports FLOPs-based complexity and parameter ranges; search time for 200 samples on CIFAR-100 with one GPU; no measured latency or memory footprint reported
Comparison with ANN Baseline	Partial. Motivation discusses the gap with ANN counterparts, but experimental comparison is mainly against SNN, Spiking Transformer, and NAS-based SNN baselines
Main Contribution / Key Limitation	Contribution: proposes AutoST, a training-free NAS framework that uses FLOPs to search efficient Spiking Transformer architectures without costly supernet training, achieving strong results on static and neuromorphic datasets. Limitation: it is a short conference paper with limited training details, no hardware deployment, and energy evidence is not directly measured
RQ Mapping	Primary: RQ2; Secondary: RQ4, RQ5

P019

Knowledge Distillation-Based Spiking Neural Network for Online Video Action Understanding

Full Citation	Wang, H., Zhang, S., Han, X., Pang, K., & Zhang, Q. (2026). Knowledge Distillation-Based Spiking Neural Network for Online Video Action Understanding. IEEE Transactions on Industrial Informatics, 22(3), 2553–2564. https://doi.org/10.1109/TII.2025.3641291
University	Yanshan University
Country	China
Publication Year	2026
Publication Venue	IEEE Transactions on Industrial Informatics
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TII.2025.3641291
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Industrial Monitoring, Fault Diagnosis, and Infrastructure Inspection
Specific Task	Online action detection and action anticipation in video streams
Data Modality	RGB video features and optical-flow features
Signal Dimensionality	Temporal video feature sequences; long-term memory ML ∈ R^(D×NL) and short-term memory MS ∈ R^(D×NS)
Signal Characteristics	Streaming video features are divided into long-term and short-term memories; long-term memory captures distant video context, while short-term memory contains recent action information; RGB and optical-flow features are processed separately
Dataset Name	THUMOS14; EPIC-Kitchens 100
Dataset Type	Public online action detection and action anticipation video benchmark datasets
Dataset Size	THUMOS14: 413 untrimmed videos with 20 action classes; 200 validation videos used for training and 213 test videos for evaluation. EPIC-Kitchens 100: over 100 hours of egocentric activity video, about 90K action segments, 20K narrations, 97 verb classes, and 300 noun classes
Dataset Balance / Split	THUMOS14 follows prior OAD split: 200 validation videos for training and 213 test videos for testing. EPIC-Kitchens 100 uses benchmark split; exact class balance NR
Hybrid Topology	Knowledge-distillation-based SNN student network with BaseNet, multiscale spike CNN blocks, spike Transformer blocks, and classification head; ANN teacher network guides the SNN student through feature and logit distillation
SNN Component	LIF neurons, spike CNN block, spike Transformer block, spike self-attention, spike MLP, spike-driven temporal feature processing
Neuron Model	LIF
Transformer Component	Spike Transformer block for action context and anticipation context; dual-stage spike self-attention; spike MLP
Attention Type	Spike self-attention without softmax; progressive spike self-attention for action-context and anticipation-context querying
SNN–Transformer Interface Mechanism	Video features are transformed by BaseNet, then converted into spike representations through LIF layers inside spike CNN and spike Transformer blocks; spike self-attention projects Q/K/V through ConvBN and LIF, then computes QS(KS■VS)
Model Depth / Scale	Long-term memory length NL = 256; short-term memory length NS = 32; spike Transformer heads/hidden dimensions set to 4/512 for THUMOS14 and 4/1024 for EPIC-Kitchens 100; main timestep setting is 4
Parameter Count	THUMOS14 OAD: 35.0M parameters; EPIC-Kitchens 100 OAD: 116.3M parameters
Design / Contribution Type	First SNN baseline/framework for online action detection and anticipation; introduces spike CNN block, spike Transformer block, and optimal-transport-based ANN-to-SNN feature distillation
Input Encoding Method	RGB and optical-flow frame features are extracted using MMAction2 checkpoints, then processed by BaseNet and converted into spike features through LIF-based spike CNN and spike Transformer modules
Encoding Parameters	Timesteps = 4; long-term memory frames = 256; short-term memory frames = 32; scale-level best setting = 5; cascaded spike CNN depth best setting ND = 2
Encoding Learned or Fixed	Partially learned: RGB/flow features are extracted from fixed pretrained feature extractors; SNN student, spike CNN, spike Transformer, and distillation-guided representations are learned

Event / Spike Representation	Binary spike activations; spike values {0,1} after LIF; spike-driven action context and anticipation context; temporal average of spike features before classification
Temporal Resolution	4 SNN timesteps in main experiments; video memory uses 256 long-term and 32 short-term frames
Output Decoding Method	Action and anticipation contexts are averaged over timestep dimension, then passed to a shared fully connected classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes long/short-term memory interaction, spike CNN multiscale temporal compression, spike Transformer context querying, LIF output histograms, and OAD/anticipation behavior
Training Paradigm	Supervised learning with knowledge distillation from pretrained ANN teacher to SNN student
Training Method	Direct SNN student training using feature distillation, logit distillation, and task loss; ANN teacher is pretrained MAT and is not retrained
Surrogate Gradient Function	Alternative/surrogate gradient using differentiable functions such as Sigmoid to replace the Heaviside function during backpropagation
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Total loss: Ltotal = Ldistill + Ltask; distillation loss includes optimal-transport/Wasserstein feature distillation and temperature-softened logit distillation; task loss combines action detection loss and anticipation loss
Training Hyperparameters	Learning rate: 7e-5 for THUMOS14 and EPIC-Kitchens 100; weight decay: 5e-5 for THUMOS14 and 1e-4 for EPIC-Kitchens 100; batch size: 8 for THUMOS14 and 32 for EPIC-Kitchens 100; experiments conducted on three NVIDIA TITAN GPUs
Primary Performance Metric	mAP for online action detection and action anticipation; energy consumption
Primary Metric Value	THUMOS14 OAD: 67.8% mAP, 35.0M parameters, 28.2 mJ energy. THUMOS14 action anticipation: average mAP 52.5. EPIC-Kitchens 100 OAD: verb 46.7, noun 45.0, action 24.1, 39.4 mJ energy
Secondary Metrics	Parameter count, FPS, energy, action anticipation mAP at different τ values, verb/noun/action recall on EK100, ablation mAP
Baseline Models Compared	TRN, OadTR, LSTR, TeSTra, MAT, CMeRT, RULSTM, TTPP, Meta-SpikeFormer-based SNN baseline, SVFormer, spike-driven Transformer, spike-driven Transformer V2, Spikingformer
Performance Gain Over Baseline	On THUMOS14, KDSNN uses only 27.1% of MAT's energy and 37.0% of MAT's parameters. On EPIC-Kitchens 100, it uses only 10.0% of MAT's energy and 27.7% of MAT's parameters. Compared with the SNN baseline, KDSNN improves THUMOS14 OAD mAP from 61.4 to 67.8 and action anticipation average mAP from 5.2 to 52.5
Statistical Validation	Benchmark comparison and extensive ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes distillation loss, feature/logit distillation, OT/Wasserstein loss comparison with MSE/CE/KL/BRD, spike Transformer variants, spike CNN depth, scale-level, teacher network type, teacher selection, ε, p, FPS, and parameter/energy comparison
Training Framework / Code Availability	RGB/flow features extracted using MMAction2; PyTorch-OpCounter/thop used for FLOPs; training framework otherwise NR; code availability NR
Target Hardware	Resource-constrained industrial edge devices and neuromorphic processors are the motivation; Loihi and TrueNorth are discussed conceptually
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Theoretical/operation-based energy estimation using FLOPs for ANN methods and AC/MAC operation assumptions for SNNs; no measured chip or board-level energy
Latency / Memory / Complexity Evidence	Reports FPS and parameters. THUMOS14 OAD: KDSNN achieves 55.2 FPS with timestep 4; when timestep is set to 1, FPS increases to 102.6. Parameter count is 35.0M on THUMOS14 and 116.3M on EK100
Comparison with ANN Baseline	Yes. Compares with ANN-based OAD/anticipation methods including TRN, OadTR, LSTR, TeSTra, MAT, CMeRT, RULSTM, and TTPP
Main Contribution / Key Limitation	Contribution: proposes the first SNN-based framework for online action detection and anticipation, combining spike CNN, spike Transformer, and optimal-transport-based ANN-to-SNN knowledge distillation. Limitation: energy is estimated rather than measured on real neuromorphic hardware; performance still trails the best ANN models on some THUMOS14 settings; input relies on pretrained RGB/flow features
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P020
Masked Spiking Transformer

Full Citation	Wang, Z., Fang, Y., Cao, J., Zhang, Q., Wang, Z., & Xu, R. (2023). Masked Spiking Transformer. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV 2023), pp. 1761–1771. https://doi.org/10.1109/ICCV51070.2023.00169
University	The Hong Kong University of Science and Technology
Country	China
Publication Year	2023
Publication Venue	IEEE/CVF International Conference on Computer Vision, ICCV 2023
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/ICCV51070.2023.00169
Publication Type	Conference paper
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Static image classification and neuromorphic/event-based image classification using Masked Spiking Transformer
Data Modality	Static image data and event-camera / neuromorphic vision data
Signal Dimensionality	Static 2D image inputs; neuromorphic event streams converted into frame-based spatiotemporal tensors
Signal Characteristics	Static images are processed through ANN-to-SNN conversion; neuromorphic events are represented by spatial coordinates, timestamp, and polarity, then transformed into high-rate frame sequences of size (n × C, H, W)
Dataset Name	CIFAR-10, CIFAR-100, ImageNet, CIFAR10-DVS, N-Caltech101, N-Cars, Action Recognition, ASL-DVS
Dataset Type	Public static image classification and neuromorphic/event-based vision benchmark datasets
Dataset Size	NR

Dataset Balance / Split	NR
Hybrid Topology	Fully spiking Transformer converted from Swin Transformer using ANN-to-SNN conversion; includes spiking self-attention, spiking MLP, IF neurons, BN-based normalization, QCFS training activation, and Random Spike Masking during inference/training
SNN Component	IF neurons, binary spikes, spike matrices for Q/K/attention, Random Spike Masking, spike-based self-attention, spiking MLP
Neuron Model	Integrate-and-Fire neuron
Transformer Component	Swin Transformer backbone; spiking self-attention module; Transformer MLP block; shifted-window-style Transformer architecture inherited from Swin-T
Attention Type	Spiking self-attention; spike-based query-key attention; Random Spike Masking applied to query, key, attention matrices, and MLP spike outputs
SNN–Transformer Interface Mechanism	ANN Swin Transformer is trained with QCFS activation and then converted to SNN by replacing QCFS with IF neurons; linear/normalization outputs are converted into spike matrices, and spiking Q/K attention is computed in the spiking domain
Model Depth / Scale	Main backbone is Swin-T with BN; static dataset model uses around 27.6M–28.5M parameters; experiments report 64, 128, 256, and 512 timesteps depending on dataset
Parameter Count	CIFAR-10/CIFAR-100 MST: 27.6M; ImageNet MST: 28.5M; parameter counts for neuromorphic dataset variants NR
Design / Contribution Type	New Masked Spiking Transformer framework; first full self-attention Spiking Transformer using ANN-to-SNN conversion; proposes Random Spike Masking for spike pruning and power reduction
Input Encoding Method	ANN-to-SNN conversion for static images; frame-based event representation for neuromorphic datasets; QCFS activation during ANN training before conversion to IF spiking neurons
Encoding Parameters	Static datasets use timesteps of 64, 128, 256, and 512; neuromorphic datasets use timesteps from 32 to 512 depending on dataset; RSM masking ratios tested from 0% to 100%, with key results at 50% and 75%
Encoding Learned or Fixed	Partially learned: ANN backbone is trained with QCFS before conversion; spike masking is a fixed/random pruning mechanism during training and inference
Event / Spike Representation	Binary spike trains; input spike matrices; spiking Q/K/attention matrices; masked spike matrices; event-frame tensor representation for neuromorphic datasets
Temporal Resolution	Static datasets require multiple SNN timesteps, commonly 64–512; neuromorphic experiments use 32–512 timesteps depending on dataset
Output Decoding Method	Classification head inherited from Swin Transformer; final prediction based on accumulated spiking activity / converted model output
Encoding / Temporal Information Analysis	Partial. The paper analyzes timestep effects, attention maps under different masking ratios, spike redundancy, and power reduction, but does not deeply analyze temporal information preservation
Training Paradigm	ANN-to-SNN conversion with post-conversion spike masking and fine-tuning
Training Method	Train ANN with QCFS activation and BN replacement, convert QCFS activations to IF neurons for SNN inference, then fine-tune masked model using unmasked model as teacher
Surrogate Gradient Function	N/A for main ANN-to-SNN conversion pipeline; surrogate-gradient direct training is discussed only as background
ANN-to-SNN Conversion Method	Yes. QCFS-based ANN-to-SNN conversion using IF neurons
Optimizer	NR
Loss Function	NR; knowledge-distillation-style fine-tuning is used for masked model, but exact loss formula is NR
Training Hyperparameters	Detailed training settings are referred to supplementary material; masking ratios include 0%, 50%, and 75%; timesteps include 64/128/256 for CIFAR and 128/256/512 for ImageNet; optimizer, learning rate, batch size, and epochs NR
Primary Performance Metric	Top-1 classification accuracy and power consumption
Primary Metric Value	CIFAR-10: 97.27% at 256 timesteps; CIFAR-100: 86.91% at 256 timesteps; ImageNet: 78.51% at 512 timesteps; CIFAR10-DVS: 88.12%; N-Caltech101: 91.38%; N-Cars: 97.28%; Action Recognition: 88.21%; ASL-DVS: 99.10%
Secondary Metrics	Time steps, model parameters, power ratio, masking ratio, attention-map visualization, standard deviation over 10 runs for masking experiments
Baseline Models Compared	Diet-SNN, tdBN, TET, DSR, Spikformer, RMP, RNL, QCFS, TA-SNN, PLIF, Dspike, NDA, SALT, CarSNN, STCA, Mb-SNN, Meta-SNN, Swin-T ANN baseline
Performance Gain Over Baseline	MST improves over prior SOTA SNNs by 1.21% on CIFAR-10, 7.3% on CIFAR-100, and 3.7% on ImageNet. On neuromorphic datasets, it improves over previous SOTA by 4.95% on CIFAR10-DVS, 7.68% on N-Caltech101, 5.38% on N-Cars, and 3.06% on ASL-DVS
Statistical Validation	Benchmark comparison and masking sensitivity experiments; standard deviation shown for some masking experiments over 10 runs; no formal significance test or confidence interval reported
Ablation Study	Full. Includes RSM masking ratio on self-attention and MLP, masking on ResNet-18 and VGG-16, number of masked blocks, power/accuracy comparison, and attention-map visualization under 0%, 50%, and 75% masking
Training Framework / Code Availability	SpikingJelly used for energy/masking experiments; code available at GitHub: bic-L/Masked-Spiking-Transformer
Target Hardware	Neuromorphic hardware motivation; power estimated for neuromorphic-chip-style spike activity; no specific deployed chip used
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Theoretical power estimation based on total transmitted spikes; assumes spike activity consumes α joules and one timestep takes 1 ms; 75% masking reduces MST power from 3.9Gα W to 2.9Gα W
Latency / Memory / Complexity Evidence	Reports timesteps, parameters, power proxy, and masking-ratio effects; no measured latency, runtime memory, or hardware deployment metrics reported
Comparison with ANN Baseline	Yes. Compares against Swin-T BN ANN baseline and several ANN-to-SNN conversion baselines, but MST still remains below the ANN accuracy on some datasets
Main Contribution / Key Limitation	Contribution: proposes MST, a high-performing ANN-to-SNN Spiking Transformer using Swin-T backbone and Random Spike Masking to reduce redundant spikes and power while preserving accuracy. Limitation: requires relatively long timesteps compared with direct-trained SNNs; energy evidence is theoretical; no real neuromorphic deployment is reported
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ5

P021

Rethinking Spiking Self-Attention Mechanism: Implementing α-XNOR Similarity Calculation in Spiking Transformers

Full Citation	Xiao, Y., Wang, S., Zhang, D., Wei, W., Shan, Y., Liu, X., Jiang, Y., & Zhang, M. (2025). Rethinking Spiking Self-Attention Mechanism: Implementing α-XNOR Similarity Calculation in Spiking Transformers. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025), pp. 5444–5454. https://doi.org/10.1109/CVPR52734.2025.00512

University	University of Electronic Science and Technology of China
Country	China
Publication Year	2025
Publication Venue	IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/CVPR52734.2025.00512
Publication Type	Conference paper
Database Source	IEEE Xplore
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	Image classification and redesign of spiking self-attention similarity calculation
Data Modality	Static images and event-based/DVS image data
Signal Dimensionality	2D image inputs represented as spike-token sequences; spike Q/K/V matrices represented across timestep, token, and embedding dimensions
Signal Characteristics	ImageNet input size 224 × 224; CIFAR datasets use 32 × 32 images; spike trains are sparse binary vectors where non-spikes are represented as 0 and spikes as 1
Dataset Name	ImageNet, CIFAR-10, CIFAR-100, CIFAR10-DVS
Dataset Type	Public static image classification and neuromorphic/event-based image classification benchmarks
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Spiking Transformer with α-XNOR Spiking Self-Attention inserted into single-stage ViT and multi-stage ViT/Swin-style Spiking Transformer architectures
SNN Component	LIF neurons, spiking neuron layers, binary spike Q/K/V, spike-form attention output, α-XNOR spike similarity calculation
Neuron Model	LIF
Transformer Component	Spiking self-attention module, single-stage ViT, multi-stage ViT/Swin-style Spiking Transformer, Transformer encoder blocks
Attention Type	α-XNOR Spiking Self-Attention; softmax-free spiking self-attention; spike-train similarity-based attention
SNN–Transformer Interface Mechanism	Q, K, and V are generated through linear projection, batch normalization, and spiking neuron layers; α-XNOR replaces dot-product similarity between spike-form Q and K; attention score is linearly transformed using k and b before multiplying with V and passing through LIF
Model Depth / Scale	ImageNet models include α-SSA-ViT-8-384, α-SSA-ViT-8-512, α-SSA-Swin-10-384, and α-SSA-Swin-10-512; all reported main models use 4 timesteps
Parameter Count	ImageNet: 16.8M, 29.7M, 31.8M, and 54.6M depending on model. CIFAR S-ViT: 9.32M; CIFAR M-ViT: 6.50M; CIFAR10-DVS S-ViT: 2.57M; CIFAR10-DVS M-ViT: 1.54M
Design / Contribution Type	New α-XNOR similarity measure for spike trains; α-XNOR Spiking Self-Attention module; custom gradient for α-XNOR operation; entropy-based theoretical analysis of spike similarity
Input Encoding Method	Spike-form Q/K/V generation using spiking neuron layers; exact dataset-level image/event input encoding method is NR
Encoding Parameters	Main experiments use 4 timesteps; α = 0.3 for ImageNet; α = 0.3 and 0.5 tested on CIFAR/CIFAR10-DVS; α sweep uses [0, 0.2, 0.4, 0.6, 0.8, 1]
Encoding Learned or Fixed	Partially learned: Q/K/V projections are learned; α is treated as a fixed hyperparameter; k and b are derived attention-score transformation constants
Event / Spike Representation	Binary spike trains; spike-form Q, K, V; α-XNOR similarity scores; spiking attention output Attn
Temporal Resolution	4 timesteps in reported main experiments
Output Decoding Method	Classification output through the underlying Spiking Transformer classifier head; exact decoding/readout detail NR
Encoding / Temporal Information Analysis	Yes. The paper analyzes dot-product information loss, spike/non-spike matching, entropy of similarity scores, α-value sensitivity, firing rates, attention-score distributions, and attention heatmaps
Training Paradigm	Supervised direct learning
Training Method	Direct training of Spiking Transformer architectures with α-SSA; custom gradients are defined for α-XNOR similarity; other training settings follow existing baseline architectures
Surrogate Gradient Function	Custom gradient for α-XNOR operation is defined; LIF neuron surrogate-gradient function NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	ImageNet: 224 × 224 input, batch size 128, 300 epochs, α = 0.3, 4 timesteps. CIFAR/CIFAR10-DVS: α = 0.3 and 0.5 tested; other optimizer, learning rate, and batch-size details NR
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	Best reported results: ImageNet 80.62% with α-SSA-Swin-10-512; CIFAR-10 96.15% with M-ViT + α-SSA; CIFAR-100 81.08% with M-ViT + α-SSA; CIFAR10-DVS 82.24% with M-ViT + α-SSA
Secondary Metrics	Parameter count, timestep count, firing rates of Q/K/V/attention output, α sensitivity, entropy, attention heatmaps, ablation accuracy
Baseline Models Compared	SEW ResNet, MS ResNet, Att-MS-ResNet, PLIF, tdBN, DSpike, TET, Spikformer, Spikingformer, Spike-driven Transformer, SpikingResformer, Meta-SpikeFormer
Performance Gain Over Baseline	On CIFAR-100, Spikformer improves from 78.21% to 80.22% with α-SSA, a +2.01% gain. On ImageNet, α-SSA-Swin-10-512 reaches 80.62%, exceeding Meta-SpikeFormer-512 at 79.70%. On CIFAR10-DVS, M-ViT + α-SSA improves from 80.0% to 82.24%
Statistical Validation	Benchmark comparison and ablation study; no standard deviation, confidence interval, repeated-run analysis, or significance test reported
Ablation Study	Full. Includes α = 0, α = 1, and α = 0.3 ablation; α-value sweep; attention heatmap comparison; entropy and distribution analysis
Training Framework / Code Availability	NR
Target Hardware	Neuromorphic hardware motivation discussed; no specific hardware target used in experiments
Hardware Deployment Status	No hardware deployment reported

Energy Evidence	No measured energy result; efficiency is argued through spike-based operation and improved attention similarity, but energy consumption is not quantified
Latency / Memory / Complexity Evidence	Reports parameter count and timestep count; no measured latency, memory footprint, or computational complexity results reported
Comparison with ANN Baseline	Partial. The paper discusses the performance gap between Spiking Transformers and ANN Transformers, but experiments mainly compare against SNN and Spiking Transformer baselines
Main Contribution / Key Limitation	Contribution: proposes α-XNOR similarity and α-SSA to fix dot-product weakness in spike-train similarity by giving weighted importance to matching spikes and matching non-spikes. Limitation: no hardware deployment, no measured energy/latency, limited reporting of optimizer/loss/training details, and no explicit limitation section
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ1

P022

Spiking Transformer Hardware Accelerators in 3D Integration

Full Citation	Xu, B., Hwang, J., Vanna-iampikul, P., Lim, S. K., & Li, P. (2024). Spiking Transformer Hardware Accelerators in 3D Integration. In IEEE/ACM International Conference on Computer-Aided Design, ICCAD '24, October 27–31, 2024, New York, NY, USA. ACM, 9 pages. https://doi.org/10.1145/3676536.3676826
University	University of California
Country	United States
Publication Year	2024
Publication Venue	IEEE/ACM International Conference on Computer-Aided Design, ICCAD '24
Publisher	IEEE
DOI / URL	https://doi.org/10.1145/3676536.3676826
Publication Type	Conference paper
Database Source	IEEE Xplore
Paper Category	Hardware-focused
E&QA; Score	4
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Hardware acceleration of Spiking Transformer MLP and spiking self-attention workloads using 3D integration
Data Modality	Event-based vision / DVS spike-stream data
Signal Dimensionality	2D event streams converted into spiking token sequences with timestep, token, head, and feature dimensions
Signal Characteristics	Uses binary spiking activations; spiking Q/K/V; evaluates neuromorphic datasets including CIFAR10-DVS and DVS-Gesture; DVS-Gesture has 11 gesture categories from 29 individuals under 3 illumination conditions
Dataset Name	CIFAR10-DVS; DVS-Gesture
Dataset Type	Public neuromorphic / event-based vision benchmark datasets
Dataset Size	DVS-Gesture: 11 hand gesture categories from 29 individuals under 3 illumination conditions; CIFAR10-DVS sample count NR
Dataset Balance / Split	NR
Hybrid Topology	Spiking Transformer architecture with spiking tokenizer, repeated spiking encoder blocks, spiking MLP, multi-head spiking self-attention, and classification head, mapped to a 3D accelerator
SNN Component	LIF/IF neurons, spiking generators, binary spike activations, spiking MLP layers, spiking Q/K/V, spike-based attention output
Neuron Model	LIF; IF model also discussed as simplified LIF for converted SNNs
Transformer Component	Spiking Transformer encoder block, spiking MLP, multi-head spiking self-attention, spiking tokenizer, classification head
Attention Type	Multi-head spiking self-attention using binary Q, K, and V; AND-and-accumulate style spike attention
SNN–Transformer Interface Mechanism	Spiking tokenizer converts event/image input into spiking tokens; spiking Q/K/V are processed in a reconfigurable attention array; attention computation uses binary query/key/value reuse and LIF-based spike generation
Model Depth / Scale	Exact network depth NR; hardware arrays include MLP accelerators with 16×128 and 64×16 arrays, and attention accelerators with 16×16 and 16×8 arrays
Parameter Count	NR
Design / Contribution Type	First dedicated 3D accelerator architecture and physical design methodology for Spiking Transformers; memory-on-logic and logic-on-logic F2F 3D integration; spatiotemporal systolic array; kernel fusion; reconfigurable spiking attention array
Input Encoding Method	Event-stream/spike-based input processed through a spiking tokenizer; exact dataset-level spike encoding method NR
Encoding Parameters	Evaluated quantization settings include 1/8/16-bit and 1/4/12-bit for spike activation / synaptic weight / synaptic integration respectively
Encoding Learned or Fixed	NR
Event / Spike Representation	1-bit spiking activations; binary spiking Q, K, and V; multi-bit spiking attention maps; multi-bit synaptic integration
Temporal Resolution	Uses T timesteps in the Spiking Transformer formulation; exact timestep count NR
Output Decoding Method	Classification head; exact spike readout/decoding method NR
Encoding / Temporal Information Analysis	Partial. The paper analyzes spatial and temporal reuse of spikes, weights, Q/K/V, attention maps, and synaptic integration, but does not deeply analyze temporal information preservation or encoding quality
Training Paradigm	Hardware inference / physical design evaluation using trained and quantized Spiking Transformer models
Training Method	NR
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	NR
Primary Performance Metric	Power-performance-area improvement, effective frequency, area footprint, power consumption, memory access latency, and memory access power

Primary Metric Value	For spiking MLP workload: 7.0% higher effective frequency, 50% area reduction, 7.8% power reduction, 68.3% memory access latency reduction, and 69.5% memory access power reduction versus 2D CMOS. For spiking self-attention: 6.3% higher effective frequency, 50% area reduction, 1.5% power reduction, 74.2% memory access latency reduction, and 49.3% memory access power reduction
Secondary Metrics	Accuracy, bitwidth, wire length, number of cells, internal power, switching power, leakage power, SRAM access latency, SRAM access power
Baseline Models Compared	2D CMOS implementation of the same accelerator; Spiking VGG and Spiking ResNet for model accuracy comparison
Performance Gain Over Baseline	3D design reduces layout footprint by about 50% and improves memory-access efficiency compared with 2D. Spiking Transformer accuracy: CIFAR10-DVS 81.2% at 1/8/16-bit and 80.5% at 1/4/12-bit; DVS-Gesture 98.26% at 1/8/16-bit and 97.92% at 1/4/12-bit
Statistical Validation	Physical-design and benchmark comparison; no repeated-run statistics, confidence interval, or significance testing reported
Ablation Study	Partial hardware ablation. Includes 2D vs 3D layout comparison, array-size comparison, bitwidth comparison, memory-access analysis, and wire-length distribution analysis
Training Framework / Code Availability	RTL synthesized using Synopsys Design Compiler; physical synthesis with Cadence Innovus; commercial 28nm PDK; SRAM generated with commercial memory compiler; code availability NR
Target Hardware	Face-to-face bonded 3D IC using commercial 28nm PDK; memory-on-logic and logic-on-logic stacked accelerator
Hardware Deployment Status	Post-synthesis / physical-design evaluation; no fabricated chip reported
Energy Evidence	Estimated from RTL synthesis, SRAM compiler, physical synthesis, and PPA analysis; no measured silicon energy reported
Latency / Memory / Complexity Evidence	Reports effective frequency up to 1.85 GHz for MLP and 1.93 GHz for attention design; 3D memory access latency reduced as low as 19–58 ps for MLP design points and 72–100 ps for attention design points; uses 3072×128b SRAM for global buffers and 96×128b / 96×256b SRAM macros for local Q/K/V/S/X buffers
Comparison with ANN Baseline	No direct ANN baseline; comparison is mainly against 2D hardware implementation and SNN baselines
Main Contribution / Key Limitation	Contribution: proposes the first 3D integrated accelerator design for Spiking Transformers, exploiting spatial-temporal reuse and F2F 3D stacking to reduce area, memory latency, and memory power. Limitation: evaluation is based on physical-design estimation rather than fabricated silicon; training details and broader application benchmarks are limited
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3

P023

Scaling Spike-Driven Transformer With Efficient Spike Firing Approximation Training

Full Citation	Yao, M., Qiu, X., Hu, T., Hu, J., Chou, Y., Tian, K., Liao, J., Leng, L., Xu, B., & Li, G. (2025). Scaling Spike-Driven Transformer With Efficient Spike Firing Approximation Training. IEEE Transactions on Pattern Analysis and Machine Intelligence, 47(4), 2973–2990. https://doi.org/10.1109/TPAMI.2025.3530246
University	Chinese Academy of Sciences
Country	China
Publication Year	2025
Publication Venue	IEEE Transactions on Pattern Analysis and Machine Intelligence
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TPAMI.2025.3530246
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Core architecture
E&QA; Score	5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification, object detection, semantic segmentation, neuromorphic human action recognition, and scalable Spiking Transformer training
Data Modality	Static RGB images and event-based/DVS vision data
Signal Dimensionality	2D image inputs; event-based spatiotemporal inputs; spike feature maps represented across timestep, channel, height, and width
Signal Characteristics	Static ImageNet input resolution mainly 224 × 224, with enlarged 384 × 384 inference for some models; HAR-DVS uses DAVIS346 event-camera data with 346 × 260 spatial resolution; spike maps are represented as 4D tensors [T, C, H, W]
Dataset Name	ImageNet-1K, COCO, ADE20K, HAR-DVS, CIFAR10-DVS
Dataset Type	Public image classification, object detection, semantic segmentation, and neuromorphic/event-based benchmark datasets
Dataset Size	ImageNet-1K: 1.3M training images, 50K validation images, 1K classes; COCO: 118K train2017 images and 5K val2017 images; ADE20K: 20K training images, 2K validation images, 150 categories; HAR-DVS: 107,646 samples, 300 classes; CIFAR10-DVS size NR
Dataset Balance / Split	Uses standard benchmark training/validation splits; class-balance details NR
Hybrid Topology	Efficient Spike-Driven Transformer architecture, E-SpikeFormer, containing Conv-based SNN stages and Transformer-based SNN stages; uses SpikeSepConv, Efficient Spike-Driven Self-Attention, channel mixer modules, SFA training, and MIM-SNN pretraining
SNN Component	Spiking neuron layers, binary spikes, spike firing approximation, IF-SR/LIF/IF neuron mechanisms, SpikeSepConv, Spike Sparse Convolution, spike-driven convolution, spike-driven self-attention
Neuron Model	LIF, IF, IF-SR, LIF-HR, LIF-SR, IF-HR, IF-SR discussed; SFA spike-driven inference is implemented using IF-SR with Vth = 1
Transformer Component	E-SpikeFormer, Transformer-based SNN block, Efficient Spike-Driven Self-Attention, channel MLP, masked image modeling Transformer decoder during pretraining
Attention Type	Efficient Spike-Driven Self-Attention; spike-driven self-attention using spike-form Q, K, and V; softmax-free spike attention
SNN–Transformer Interface Mechanism	Q, K, and V are generated through linear layers and spiking neuron layers; E-SDSA computes spike-driven attention using QsKs■Vs; SFA trains integer activations and converts them into spike trains during inference
Model Depth / Scale	E-SpikeFormer scales from small to large models: 5.1M, 10M, 19M, 52M, 83M, and 173M parameters. Figure 4 shows a representative 4-stage design with Conv-based SNN blocks in early stages and Transformer-based SNN blocks in later stages
Parameter Count	ImageNet models: 5.1M, 10M, 19M, 52M, 83M, 173M. COCO detection model: 38.7M. ADE20K segmentation model: 11.0M. HAR-DVS model: 18.7M
Design / Contribution Type	Spike Firing Approximation training; Efficient Spike-Driven Transformer architecture; SpikeSepConv; Efficient Spike-Driven Self-Attention; Spike Sparse Convolution for masked image modeling; scalable SNN training and inference strategy

Input Encoding Method	Static image tasks use one-timestep SFA integer training and spike-driven inference expansion; event-based/dynamic tasks use multi-timestep direct SFA training; MIM pretraining uses masked image input with Spike Sparse Convolution
Encoding Parameters	SFA uses integer activation upper bound D; static ImageNet commonly uses step 1 × 4 or 1 × 8; HAR-DVS uses 8 × 4 and 8 × 8 settings; IF-SR inference uses Vth = 1; ImageNet default input 224 × 224, enlarged input 384 × 384 for selected inference results
Encoding Learned or Fixed	Partially learned/adaptive. SFA firing pattern is training-driven; Q/K/V projections and model weights are learned; spike firing conversion rule is fixed during inference
Event / Spike Representation	Binary spike trains; integer activations during SFA training; spike feature maps; spike-form Q, K, V; spike sparse convolution feature maps
Temporal Resolution	Static tasks: one training timestep expanded to D inference timesteps, commonly 1 × 4 or 1 × 8. Dynamic neuromorphic tasks: T × D setting such as 8 × 4 or 8 × 8
Output Decoding Method	Classification head for image/HAR classification; detection and segmentation heads fine-tuned from ImageNet-pretrained backbone; exact spike readout rule NR
Encoding / Temporal Information Analysis	Yes. The paper analyzes binary spike firing flaws, spatial representation error, temporal dynamics limitations, spike firing approximation error, gradient error, spike distribution, network spike firing rate, feature-rank scaling, and neuromorphic chip implementation
Training Paradigm	Supervised direct training; SFA integer training with spike-driven inference; self-supervised masked image modeling pretraining; transfer learning for detection and segmentation
Training Method	Spike Firing Approximation; one-timestep SFA training for static vision; multi-timestep SFA training for dynamic vision; masked image modeling pretraining with Spike Sparse Convolution; distillation during fine-tuning
Surrogate Gradient Function	Rectangular window surrogate function over the [0, D] range
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	MSE loss for MIM pretraining reconstruction; distillation loss used during fine-tuning; task-specific classification/detection/segmentation losses NR
Training Hyperparameters	ImageNet training comparison uses 200 epochs on 8 NVIDIA A100-40G GPUs; SFA settings include 1 × 4 and 1 × 8 for static tasks; HAR-DVS uses 8 × 4 and 8 × 8; other optimizer, learning rate, and batch-size details are in supplementary material but not fully reported in the main text
Primary Performance Metric	Top-1 accuracy on ImageNet-1K; also mAP for COCO, mIoU for ADE20K, and accuracy for HAR-DVS
Primary Metric Value	ImageNet-1K: 78.5% with 10M model, 79.8% with 19M, 84.0% with 83M, and 86.2% with 173M. COCO: up to 58.8 mAP@50. ADE20K: up to 41.4% mIoU. HAR-DVS: up to 49.2% accuracy
Secondary Metrics	Parameter count, power/energy cost in mJ, training time, spike firing rate, network spike firing rate, effective rank, feature-map visualization, detection mAP, segmentation mIoU
Baseline Models Compared	ANN baselines: DeiT, PVT, Swin, CAFormer, T2T-ViT, RepLKNet, FocalNet, ConvNeXt, CrossFormer. SNN baselines: SEW-ResNet, TET-ResNet, DSR-ResNet, MS-ResNet, Att-MS-ResNet, Spikformer, Spike-driven Transformer, Spikformer V2, Meta-SpikeFormer, ANN-to-SNN methods, Spiking-YOLO, Spike Calibration, Fast-SNN
Performance Gain Over Baseline	The 10M E-SpikeFormer outperforms the best existing SNN by 7.2% on ImageNet; compared with the 10M Meta-SpikeFormer baseline, SFA + E-SpikeFormer improves accuracy from 71.3% to 78.5%, reduces power from 11.9 mJ to 3.7 mJ, and gives about 5.3× training-time acceleration
Statistical Validation	Large-scale benchmark validation and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or significance testing reported
Ablation Study	Full. Includes SFA vs vanilla training, architecture ablation, power/performance analysis, spike distribution visualization, network spike firing rate, MIM scaling analysis, effective-rank analysis, and supplementary MIM ablations
Training Framework / Code Availability	Code available at GitHub: BICLab/Spike-Driven-Transformer-V3
Target Hardware	Neuromorphic chips discussed, including synchronous and asynchronous implementations; Tianjic and Speck are discussed as possible deployment platforms
Hardware Deployment Status	No real neuromorphic hardware deployment reported; hardware implementation is discussed conceptually
Energy Evidence	Reports estimated power/energy in mJ across ImageNet, COCO, ADE20K, and HAR-DVS; no measured silicon or neuromorphic-chip energy result
Latency / Memory / Complexity Evidence	Reports training-time acceleration on 8 NVIDIA A100-40G GPUs; discusses training memory reduction through one-timestep SFA training; inference energy/power reported in mJ; detailed GPU memory acceleration is referred to supplementary material
Comparison with ANN Baseline	Yes. Compares against classic ANN vision backbones such as DeiT, PVT, Swin, ConvNeXt, FocalNet, CrossFormer, and others, showing competitive accuracy with lower reported power
Main Contribution / Key Limitation	Contribution: proposes SFA to solve binary spike firing limitations, designs E-SpikeFormer for scalable spike-driven vision backbones, and introduces SSC-based MIM pretraining for large SNN scaling. Limitation: no fabricated hardware or real neuromorphic deployment; energy is estimated rather than chip-measured; most validation is vision-focused
RQ Mapping	Primary: RQ4; Secondary: RQ2, RQ3, RQ1, RQ5

P024

Efficient Automatic Modulation Classification in Nonterrestrial Networks With SNN-Based Transformer

Full Citation	Zeng, D., Xiao, Y., Liu, W., Du, H., Zhang, E., Zhang, D., Wang, Y., Zhang, M., & Chen, W. (2025). Efficient Automatic Modulation Classification in Nonterrestrial Networks With SNN-Based Transformer. IEEE Internet of Things Journal, 12(3), 2562–2573. https://doi.org/10.1109/JIOT.2024.3467284
University	University of Electronic Science and Technology of China
Country	China
Publication Year	2025
Publication Venue	IEEE Internet of Things Journal
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/JIOT.2024.3467284
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Communication, Cybersecurity, and Network Intelligence

Specific Task	Automatic modulation classification for nonterrestrial network communication signals
Data Modality	I/Q radio signal samples converted into polar-grid temporal images and spike inputs
Signal Dimensionality	1D complex temporal signal transformed into 2D grid-like image sequences across SNN timesteps
Signal Characteristics	Each signal in RadioML2018.01A has 2048 samples: 1024 in-phase and 1024 quadrature samples. Dataset includes 24 modulation types and SNR values from –20 dB to +28 dB in 2-dB increments
Dataset Name	RadioML2018.01A
Dataset Type	Public wireless modulation classification benchmark dataset
Dataset Size	2.5 million data points; 24 modulation types; 2048 samples per data point
Dataset Balance / Split	NR
Hybrid Topology	SNN-based Transformer architecture with signal preprocessing, spiking convolutional tokenization, spiking Transformer encoder, and MLP classification head
SNN Component	LIF neurons, spiking convolutional layers, spike trains, binary spike activations, spiking Q/K/V generation
Neuron Model	LIF
Transformer Component	Spiking Transformer encoder, Q/K/V attention block, residual connection, feed-forward network / MLP head
Attention Type	Spiking self-attention using Hadamard product between spike-form Q and K, column-sum attention compression, and spike-based masking of V
SNN–Transformer Interface Mechanism	Processed signal images are tokenized by spiking convolutional layers; Q, K, and V are generated by linear projection, BN, and LIF neurons; attention scores are computed from spike-form Q and K and used to mask V before residual and MLP processing
Model Depth / Scale	Uses spiking convolutional tokenization followed by spiking Transformer encoder; exact number of encoder blocks NR. Time-step experiments use T = 2, 4, 6, 8, and 16
Parameter Count	Full-precision model: 648K parameters; binarized model: 648K parameters
Design / Contribution Type	SNN-based Transformer for AMC; temporal signal-to-spike representation; polar-grid temporal signal processing; binary weight quantization for deployment-efficient NTN edge inference
Input Encoding Method	Polar feature transformation from I-Q plane to r-θ plane; projection into 36 × 36 grid-like images; temporal segmentation into T groups; direct encoding into spike trains through the first spiking convolutional layer
Encoding Parameters	Grid-like image size 36 × 36; signal length 1024 complex temporal samples; time steps tested from 2 to 16; best reported time-step setting around T = 4; direct encoding; first spiking convolutional layer treated as automatic spike encoding layer
Encoding Learned or Fixed	Partially learned. Polar-grid transformation and temporal grouping are fixed preprocessing; spike encoding through first spiking convolutional layer is learned/model-based
Event / Spike Representation	Binary spike trains; temporal grid-like signal images; spike-form Q, K, V; binarized weights in compressed model
Temporal Resolution	Time steps T = 2, 4, 6, 8, and 16 tested; temporal signal is divided into T segments, each segment aggregated into one image for one SNN timestep
Output Decoding Method	MLP head predicts the modulation class
Encoding / Temporal Information Analysis	Yes. The paper analyzes temporal dynamics through time-step experiments, temporal-dynamics ablation, SNR-dependent accuracy, confusion matrices, firing rate, and energy consumption across timesteps
Training Paradigm	Supervised learning
Training Method	Surrogate-gradient training of SNN-based Transformer; binary quantization training for lightweight model
Surrogate Gradient Function	Triangular surrogate function
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	PyTorch implementation; single NVIDIA GeForce RTX 4090 GPU; 200 GB RAM; grid size 36 × 36; time steps tested from 2 to 16; binarized model uses Hardtanh activation in binarized layers; all convolutional and fully connected layers are binarized except first and last layers
Primary Performance Metric	Average classification accuracy and maximum classification accuracy
Primary Metric Value	Full-precision model: average accuracy 55.03%, maximum accuracy 99.87%. Binarized model: average accuracy 54.26%, maximum accuracy 99.75%
Secondary Metrics	Model size, parameter count, accuracy across SNR, confusion matrix, firing rate, energy consumption, power-consumption reduction ratio
Baseline Models Compared	AMC-CNN, VGG, ResNet, M-CNN, MCNet, SCGNet, WSMF, RanNet, LSM-AMR, OFDMsym-Net, LLDLM, PDARTS-AMC, MCMBNN, CNN-SM, DARTS, CNN-LSTM
Performance Gain Over Baseline	Full-precision model achieves higher maximum accuracy than RanNet, 99.87% vs 94.84%, and higher average accuracy than LSM-AMR, 55.03% vs 53.79%. Binarized model reduces model size from 19.78 MB to 0.96 MB with only 0.77% average-accuracy loss
Statistical Validation	Benchmark evaluation, SNR analysis, confusion-matrix comparison, timestep analysis, ablation study, and simulated energy analysis; no confidence interval, standard deviation, repeated-run statistics, or significance test reported
Ablation Study	Full/clear ablation. Includes temporal-dynamics ablation, full-precision vs binarized comparison, timestep study, SNR analysis, and energy-consumption simulation
Training Framework / Code Availability	PyTorch; code availability NR
Target Hardware	NTN edge devices and low-power neuromorphic/edge deployment are targeted conceptually; no specific hardware chip used
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Simulated energy consumption using operation counts, firing rates, 45-nm CMOS energy assumptions: 4.6 pJ for 32-bit MAC and 0.9 pJ for addition. At 2–4 timesteps, reported energy is about 4.5%–9.5% of ANN baseline
Latency / Memory / Complexity Evidence	Full-precision model size 19.78 MB; binarized model size 0.96 MB; binarized model is about 1/20 of full-precision size; parameter reduction ratio against baseline reported as 95.2%; power-consumption reduction ratio reported as 95.5%
Comparison with ANN Baseline	Yes. Compares against multiple ANN-based AMC models, including CNN, ResNet, VGG, MCNet, RanNet, and others
Main Contribution / Key Limitation	Contribution: proposes an SNN-based Transformer for AMC in nonterrestrial networks, combining temporal signal processing, spiking Transformer encoding, and binary weight quantization to reduce memory, bandwidth, and energy cost. Limitation: no real NTN edge or neuromorphic hardware deployment; energy evidence is simulated; training details such as optimizer and loss are not fully reported
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P025

EECD-Net: An Energy-Efficient Crack Detection Framework Integrating Super-Resolution, Spiking Neural Networks, and Gated Attention

Full Citation	Zhang, S., Tong, S., & Pang, K. (2025). EECD-Net: An Energy-Efficient Crack Detection Framework Integrating Super-Resolution, Spiking Neural Networks, and Gated Attention. IEEE Access, 13, 217349–217363. https://doi.org/10.1109/ACCESS.2025.3647946
University	Yanshan University
Country	China
Publication Year	2025
Publication Venue	IEEE Access
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/ACCESS.2025.3647946
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Industrial Monitoring, Fault Diagnosis, and Infrastructure Inspection
Specific Task	Pixel-level crack detection / crack segmentation under resource-constrained deployment
Data Modality	Static RGB crack images enhanced through super-resolution and encoded into sparse spike representations
Signal Dimensionality	2D image segmentation; RGB image input converted into high-resolution image features and sparse temporal spike sequences
Signal Characteristics	CrackVision12K images are standardized to 256 × 256 pixels; SR module upsamples low-resolution inputs; sparse spike firing rate reported around 0.09 in CIF module; optimal timestep setting T = 6
Dataset Name	CrackVision12K
Dataset Type	Public crack detection / crack segmentation benchmark dataset aggregated from 13 public crack datasets
Dataset Size	12,000 images; split into 9,600 training images, 1,200 validation images, and 1,200 testing images
Dataset Balance / Split	8:1:1 train/validation/test split; also evaluates 50:50 train-test split for generalization analysis
Hybrid Topology	Multi-stage EECD-Net framework combining SRCNN super-resolution, Spike Convolution Unit with CIF neurons, and Gated Attention Transformer for multi-scale feature fusion
SNN Component	Spike Convolution Unit, CIF neurons, sparse spike sequences, integer-coded spikes, spike firing rate map, sparse accumulation operations
Neuron Model	Continuous Integrate-and-Fire neuron
Transformer Component	Gated Attention Transformer; Gated Attention Coding; Multi-scale Intersection Transformer; attention layers for global-local feature fusion
Attention Type	Gated attention; self-attention using softmax(QK■/√d); multi-scale attention-based feature fusion
SNN–Transformer Interface Mechanism	SRCNN-enhanced images are encoded by CIF-based SCU into sparse pulse sequences; CNN extracts local features from spikes; GAT processes these features through gated attention and multi-scale Transformer fusion to produce final crack segmentation output
Model Depth / Scale	Modular architecture with SRCNN, SCU, and GAT; exact number of GAT blocks/layers NR
Parameter Count	NR
Design / Contribution Type	New crack detection framework integrating super-resolution preprocessing, CIF-based SNN encoding, and Gated Attention Transformer for energy-efficient segmentation
Input Encoding Method	SRCNN enhances RGB image resolution; CIF module maps continuous membrane potentials to integer values and unfolds them into sparse binary spike sequences
Encoding Parameters	CIF integer expansion factor D; optimal operating timestep T = 6; example D = 3 shown for spike expansion; EAC = 0.9 pJ based on 45 nm CMOS; image split standardized to 256 × 256
Encoding Learned or Fixed	Partially learned: SRCNN, convolutional feature extraction, and GAT are learned; CIF spike conversion and integer-to-spike expansion are fixed mechanisms during inference
Event / Spike Representation	Sparse binary spike sequences; integer-coded spikes; spike firing maps; sparse pulse sequences S[t, d]
Temporal Resolution	Time-step settings T = 1, 2, 4, 6, 7, 8, and 9 analyzed; T = 6 selected as the optimal accuracy-energy trade-off
Output Decoding Method	Decoder converts final weighted features Ffinal into crack detection output / segmentation mask
Encoding / Temporal Information Analysis	Yes. The paper analyzes timestep–accuracy–energy trade-off, sparse spike firing rate distribution, spike maps, CIF ablation, latency/throughput, and energy consumption across timesteps
Training Paradigm	Supervised learning; DNN-to-SNN conversion-style CIF training/inference pipeline
Training Method	ReLU-based continuous training mapped to CIF spiking inference; CUDA-optimized spike simulation; Adam optimizer training; baseline models retrained from scratch
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	Yes. The CIF framework extends DNN-to-SNN conversion by mapping ReLU activations to membrane potentials, quantizing them into integers, and unfolding them into sparse spike sequences
Optimizer	Adam
Loss Function	Dice loss + Cross-Entropy loss
Training Hyperparameters	Batch size 128; learning rate 1 × 10■■; NVIDIA V100 32GB GPU; PyTorch; custom CUDA kernels for CIF-SNN spike simulation; convergence around 43 epochs
Primary Performance Metric	Accuracy and F1 score for crack segmentation
Primary Metric Value	Full EECD-Net: 98.6% accuracy, 84.6% precision, 79.5% recall, 82.0% F1 score, and 5.6 mJ energy per image
Secondary Metrics	Precision, recall, energy consumption, latency, throughput/FPS, convergence loss, spike firing rate, qualitative segmentation quality
Baseline Models Compared	FCN, UNet, DeepCrack2, SegFormer, AttnUNet, TransUNet, SwinUNet, HrSegNet, Hybrid-Segmentor
Performance Gain Over Baseline	Compared with Hybrid-Segmentor, EECD-Net improves accuracy from 97.1% to 98.6% and reduces energy consumption from 8.4 mJ to 5.6 mJ, about 33% lower energy. EECD-Net also outperforms TransUNet and SwinUNet, which consume over 40 mJ per image
Statistical Validation	Benchmark comparison, ablation study, 50:50 split generalization test, qualitative analysis, and latency/throughput analysis; no standard deviation, confidence interval, repeated-run analysis, or significance test reported

Ablation Study	Full. Includes SR module, CIF module, GAT module, inter-module dependency analysis, timestep-energy trade-off, latency/throughput, spike firing rate visualization, and 50:50 split robustness test
Training Framework / Code Availability	PyTorch; CUDA kernels for spike simulation; code availability NR; CrackVision12K dataset link reported
Target Hardware	Resource-constrained mobile/embedded inspection devices; UAV or vehicle-based inspection systems; low-power edge platforms
Hardware Deployment Status	No real edge/neuromorphic hardware deployment reported; experiments run on GPU with theoretical energy estimation
Energy Evidence	Theoretical inference energy based on AC operation count and 45 nm CMOS energy constant EAC = 0.9 pJ; reports 5.6 mJ per image at T = 6
Latency / Memory / Complexity Evidence	At T = 6, latency is 22.70 ms and throughput is 44 FPS; Hybrid-Segmentor latency is 16.65 ms and 60.1 FPS; EECD-Net remains above 30 FPS real-time requirement
Comparison with ANN Baseline	Yes. Compares against CNN, Transformer, and hybrid ANN segmentation baselines including FCN, UNet, SegFormer, TransUNet, SwinUNet, HrSegNet, and Hybrid-Segmentor
Main Contribution / Key Limitation	Contribution: proposes EECD-Net, a crack detection framework combining SRCNN, CIF-SNN sparse encoding, and GAT for high accuracy and lower energy. Limitation: energy is theoretical rather than measured on hardware; CIF uses multiple timesteps, creating latency trade-off; some fine cracks are missed and complex textures can cause over-detection
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P026

A Neuromorphic Transformer Architecture Enabling Hardware-Friendly Edge Computing

Full Citation	Zhou, P. J., Ma, R. C., Chen, Y. C., Liu, Z. T., Liu, C. Y., Meng, L. W., Qiao, G. C., Liu, Y., Yu, Q., & Hu, S. G. (2025). A Neuromorphic Transformer Architecture Enabling Hardware-Friendly Edge Computing. IEEE Transactions on Circuits and Systems I: Regular Papers, 72(6), 2676–2689. https://doi.org/10.1109/TCSI.2025.3557955
University	University of Electronic Science and Technology of China
Country	China
Publication Year	2025
Publication Venue	IEEE Transactions on Circuits and Systems I: Regular Papers
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TCSI.2025.3557955
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Hardware-friendly neuromorphic Transformer inference for edge tasks such as CIFAR-10 image classification and speech/audio classification
Data Modality	Image data and audio/spiking audio datasets represented as token/spike inputs
Signal Dimensionality	Token-based Transformer input; supports 64 input tokens with maximum dimension 384; spike/event representations distributed over token and channel dimensions
Signal Characteristics	Binary spike activations; high spike sparsity, typically above 80% in evaluated simple edge tasks; event packets encode valid spikes only using AER-style addressing
Dataset Name	CIFAR-10, Speech Commands, SHD, N-Tidigits 18, FSD
Dataset Type	Public image classification and audio/speech/spiking benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Neuromorphic Transformer architecture with spike-based self-attention, spike-based feed-forward network, Q/K/V generator, attention core, FC cores, residual membrane-potential memories, and event-driven sparse computation
SNN Component	LIF neurons, spike-driven LIF model, AER-style spike events, binary spikes, event-based synaptic processing, membrane-potential residuals, spike encoder
Neuron Model	Simplified hardware-friendly LIF neuron with linear leakage; spike-driven LIF neuron model
Transformer Component	Neuromorphic Transformer block, spike-based self-attention, spike-based feed-forward network, Q/K/V generation, FC layers, residual paths
Attention Type	Spike-based self-attention; softmax-free self-attention; AND-accumulate-based attention using binary Q, K, and V
SNN–Transformer Interface Mechanism	Input X is projected into Q, K, and V through spike neuron layers; binary Q/K/V are processed using AND-accumulate operations; valid spikes are encoded as event packets and transmitted/processed through AER-style event-driven hardware
Model Depth / Scale	Hardware integrates 2 Transformer blocks; each block supports 64 input tokens with maximum dimension 384 and 4× FFN expansion, equivalent to FC dimensions 384 × 384 × 1536 × 384
Parameter Count	NR
Design / Contribution Type	Hardware/software co-design for neuromorphic Transformer acceleration; scaling-factor absorption into weights; transposition-free calculation method; membrane-potential residual structure; event-event and event-non-event processing units; spike-driven sparse computation
Input Encoding Method	Original data are embedded into tokens off-chip; valid spikes are encoded using AER-style event packets containing time-synchronization and synaptic-address information
Encoding Parameters	8-bit integer weight quantization; event packet uses most significant bit for time synchronization and remaining bits for synaptic address; supports four-timestep X matrices in Q/K/V generator memory
Encoding Learned or Fixed	Fixed hardware/event encoding during inference; weights are pre-trained and quantized before deployment
Event / Spike Representation	Binary spikes; event packets; AER-style sparse spike representation; Q, K, V, and X matrices represented as binary spike/event data
Temporal Resolution	Supports timestep-based SNN processing; Q/K/V generator stores four-timestep X matrices; exact model timestep per dataset NR
Output Decoding Method	Output buffer stores final Transformer processor result; task-specific classification output is evaluated, but exact classifier readout rule NR

Encoding / Temporal Information Analysis	Partial. The paper analyzes spike sparsity by layer and dataset, event-packet encoding, sparse spike-driven computation, and memory-access/dataflow effects, but does not deeply analyze temporal-information preservation
Training Paradigm	Hardware inference evaluation using trained neuromorphic Transformer models
Training Method	NR
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	NR
Optimizer	NR
Loss Function	NR
Training Hyperparameters	NR
Primary Performance Metric	Hardware energy efficiency, computing efficiency, power, area, and classification accuracy
Primary Metric Value	Accuracy: CIFAR-10 94.03%, Speech Commands 90.29%, SHD 86.13%, N-Tidigits 18 91.27%, FSD 95.67%. Best Q/K/V generator energy efficiency: 0.34 pJ/SOP at 50 MHz and 80% spike sparsity
Secondary Metrics	Die area, SRAM size, logic area, spike sparsity, GSOP/s, GSOP/s/W, power consumption, operation reduction ratio, throughput
Baseline Models Compared	Traditional Transformer architecture; neuromorphic Transformer with spike residual; neuromorphic Transformer with membrane-potential residual; prior processors including JSSC'22, JSSC'23, JSSC'24, TBCAS'19, JSSC'24, and NSR'24 designs
Performance Gain Over Baseline	Eliminates exponential, division, and matrix-transposition operations; reduces multiplications by 99.96% and additions by 80.02%; pipeline attention improves efficiency by 32.02% on CIFAR-10 and 37.75% on Speech Commands
Statistical Validation	Hardware evaluation across multiple datasets and post-layout analysis; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Partial hardware/architecture ablation. Includes comparison of traditional Transformer, neuromorphic Transformer with spike residual, neuromorphic Transformer with membrane-potential residual, and proposed architecture; also analyzes spike sparsity, area breakdown, and energy/computing efficiency
Training Framework / Code Availability	Hardware designed using Design Compiler, Formality, Innovus, PrimeTime, Redhawk, and Virtuoso with 28 nm CMOS process library; code availability NR
Target Hardware	28 nm CMOS neuromorphic Transformer chip architecture for edge computing
Hardware Deployment Status	Prototype/layout verified using 28 nm CMOS process library and EDA flow; no fabricated silicon measurement or real edge-device deployment reported
Energy Evidence	Estimated/post-layout energy and power from 28 nm CMOS design flow; logic power 1.35 mW at 50 MHz; whole-chip power 105.64 mW at 50 MHz and 798.59 mW at 400 MHz; best reported energy efficiency 0.33–0.34 pJ/SOP depending on core/frequency/sparsity
Latency / Memory / Complexity Evidence	Die area 23.28 mm^2 with all memory on-chip; total SRAM 5.13 MB; transformer processor logic area 0.18 mm^2; frequency up to 400 MHz; throughput 3.35 KTokens/s per Transformer processor at 400 MHz and 80% sparsity
Comparison with ANN Baseline	Yes. Compares against traditional ANN Transformer computational requirements and prior ANN Transformer processors; also reports SNN accuracy compared with ANN accuracy in background discussion
Main Contribution / Key Limitation	Contribution: proposes a hardware-friendly neuromorphic Transformer architecture that removes high-cost Transformer operators and exploits spike sparsity for edge computing. Limitation: evaluated mainly on relatively simple edge tasks; limited network depth due to two-block chip design; no fabricated silicon measurement; complex datasets such as SSC showed unsatisfactory accuracy due to limited network size
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3

P027

Spatial–Temporal Spiking Feature Pruning in Spiking Transformer

Full Citation	Zhou, Z., Che, K., Niu, J., Yao, M., Li, G., Yuan, L., Luo, G., & Zhu, Y. (2025). Spatial–Temporal Spiking Feature Pruning in Spiking Transformer. IEEE Transactions on Cognitive and Developmental Systems, 17(3), 644–658. https://doi.org/10.1109/TCDS.2024.3500018
University	Peking University
Country	China
Publication Year	2025
Publication Venue	IEEE Transactions on Cognitive and Developmental Systems
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TCDS.2024.3500018
Publication Type	Journal article
Database Source	IEEE Xplore
Paper Category	Training-efficiency focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Spatial–temporal spiking feature/token pruning to reduce Spiking Transformer training cost while maintaining classification accuracy
Data Modality	Static image data and event-based/DVS vision data
Signal Dimensionality	Spiking input stream represented as $I \in S$^(T×C×H×W); spiking token feature represented as $X \in S$^(T×N×C) or S^(T×N×D)
Signal Characteristics	Binary spike features/tokens; spatial spiking token saliency; temporal spiking feature redundancy; spike accumulation value used to identify high-information tokens and timesteps
Dataset Name	CIFAR10-DVS, DVS128 Gesture, N-Caltech 101, CIFAR-10, CIFAR-100, ImageNet-1K, MS COCO
Dataset Type	Public neuromorphic/event-based classification datasets, public static image classification datasets, and object detection benchmark
Dataset Size	CIFAR-10/CIFAR-100: 50,000 training images and 10,000 test images; ImageNet-1K: over 1.28M training images and 50,000 test images; N-Caltech 101: 8,831 DVS images; DVS128 Gesture: 11 gesture classes from 29 participants; CIFAR10-DVS and MS COCO size NR in paper
Dataset Balance / Split	N-Caltech 101 uses 80% training and 20% validation split; CIFAR uses standard train/test split; ImageNet uses standard train/validation split; other dataset split details NR
Hybrid Topology	Spiking Transformer with Spiking Patch Splitting, multiple Spiking Transformer encoder blocks, spiking self-attention, MLP block, and classification head; proposed STSFP prunes temporal features and spatial tokens before encoder processing

SNN Component	LIF neurons, spike-form tokens, spike accumulation value, spike features, temporal spike sequences, binary spike activations
Neuron Model	LIF neuron with hard reset
Transformer Component	Spikformer, Spike-Driven Transformer, Spikingformer, Spikformer V2, Meta Spike-driven Transformer V2 variant, Spiking Transformer blocks in YOLO-style detection
Attention Type	Spiking self-attention using spike-form Q, K, and V without softmax; Softmatch also uses a spiking attention map for compensation
SNN–Transformer Interface Mechanism	Spiking Patch Splitting converts input into spike-form token sequence; STSFP ranks timesteps and tokens using spike accumulation value; high-SAV features/tokens are retained for Spiking Transformer encoder blocks, while low-SAV ones are pruned or merged through Softmatch
Model Depth / Scale	Evaluated models include Spikformer-2-256, Spikformer-4-384, Spikformer V2-8-384, Spikformer V2-8-512, ST-8-512, Spike-Driven Transformer, Spikingformer, and YOLO-style model with 4 and 2 Spiking Transformer layers in later stages
Parameter Count	Spikformer V2-8-384: about 29M; Spikformer V2-8-512: about 51M; other model parameter counts NR
Design / Contribution Type	Parameter-free spatial–temporal spiking feature pruning method; spike accumulation value-based pruning; spatial spiking token pruning; temporal spiking feature pruning; Softmatch compensation strategy
Input Encoding Method	Uses the original Spiking Transformer input pipeline; event streams or images are converted to spike-form tokens through Spiking Patch Splitting; pruning is applied to spike features after SPS by default
Encoding Parameters	Spatial pruning ratio Ps; temporal pruning ratio Pt; retained features Tp = T(1–Pt) and Np = N(1–Ps); Softmatch threshold θ; scaling factor α in Softmatch
Encoding Learned or Fixed	Fixed/parameter-free pruning rule based on spike accumulation value; underlying Spiking Transformer weights are learned
Event / Spike Representation	Binary spike tensors; spike tokens; spike accumulation values; pruned spike features; retained high-SAV spiking tokens; Softmatch-merged spike features
Temporal Resolution	Dataset/model-dependent timesteps; temporal pruning can retain only selected timesteps. Examples include CIFAR10-DVS achieving 81.10% accuracy with only 4 timesteps when temporal pruning ratio is 0.7
Output Decoding Method	Global average pooling followed by linear classification head for classification tasks; YOLO-style detection head for object detection experiment
Encoding / Temporal Information Analysis	Yes. The paper analyzes spatial spiking token saliency, temporal spiking feature redundancy, spike accumulation value distribution, ratio of information entropy, MS-SSIM, pruning ratios, timestep redundancy, and feature visualizations
Training Paradigm	Supervised direct training of Spiking Transformers with feature pruning embedded into the training process
Training Method	Backpropagation-through-time style direct SNN training with surrogate gradients inherited from baseline Spiking Transformer models; STSFP reduces training cost by pruning redundant spike features/tokens
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW for ImageNet experiments; optimizer for other datasets follows baseline settings and is NR
Loss Function	Classification loss implied; exact loss function NR
Training Hyperparameters	ImageNet: 200 epochs, batch size 512, initial learning rate 0.0005, weight decay 0.05, AdamW optimizer, cosine learning-rate decay, 10-epoch linear warmup, RandAugment, random erasing, stochastic depth, 8 GPUs; fine-tuning after pruning uses 10 epochs
Primary Performance Metric	Classification accuracy and training-time reduction
Primary Metric Value	ImageNet Spikformer V2-8-512: training time reduced from 181.41 h to 127.59 h with fine-tuning, while accuracy remains comparable at 83.07% vs 83.13%; CIFAR100 best STSFP result 78.01% vs 77.38% baseline
Secondary Metrics	Throughput, GPU memory, FLOPs, pruning ratio, RIE, MS-SSIM, mAP50 for object detection, training time
Baseline Models Compared	Spikformer, Spike-Driven Transformer, Spikingformer, Spikformer V2, Meta Spike-driven Transformer V2, previous SNN pruning methods including NDSNN, IMP, Grad R, RCMO-SNN
Performance Gain Over Baseline	On CIFAR100, STSFP improves accuracy from 77.38% to 78.01% while reducing FLOPs and GPU memory. On CIFAR10-DVS with Pt = Ps = 0.6, throughput increases from 4032 to 4855 frames/s, FLOPs reduce from 6.14G to 4.69G, and GPU memory decreases by 27.27%. On ImageNet 8-512, training time drops from 181.41 h to 127.59 h after 10-epoch fine-tuning with almost identical accuracy
Statistical Validation	Extensive benchmark experiments and ablation studies; no standard deviation, confidence interval, repeated-run statistics, or significance test reported
Ablation Study	Full. Includes pruning strategies, Softmatch compensation, Softmatch threshold, pruning module position, training-only vs inference-only pruning, activation epoch, temporal sequence order, and evaluation across multiple Spiking Transformer variants
Training Framework / Code Availability	GPU-based training; computing support from Pengcheng Cloud Brain; specific framework and code availability NR
Target Hardware	GPU training acceleration; neuromorphic-chip deployment motivation discussed, but no specific hardware platform targeted for deployment
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	No direct energy measurement; energy efficiency is motivated through reduced FLOPs, reduced spike features, reduced GPU memory, and SNN AC-operation characteristics
Latency / Memory / Complexity Evidence	Reports throughput, GPU memory, FLOPs, pruning overhead, and training time. Example: CIFAR100 throughput increases from 474 to 785 images/s under Ps = 0.3 and Pt = 0.5; GPU memory decreases from 20 GB to 12 GB; FLOPs decrease from 3.8G to 2.5G
Comparison with ANN Baseline	Partial. Compares pruning behavior with ANN-Transformer token pruning conceptually and uses ANN-Transformer low-SAV pruning analysis, but experiments mainly compare against SNN and Spiking Transformer baselines
Main Contribution / Key Limitation	Contribution: proposes STSFP, a parameter-free pruning method that exploits spatial and temporal spike redundancy to reduce Spiking Transformer training cost with minimal accuracy loss. Limitation: mainly targets training acceleration; object detection results show performance drop; feature pruning for more complex visual tasks is left for future work; no real hardware energy measurement
RQ Mapping	Primary: RQ4; Secondary: RQ2, RQ3, RQ5, RQ1

P028

Spiking Transformer with Spatial-Temporal Spiking Self-Attention

Full Citation	Zhou, Z., Niu, J., Zhang, Y., Yuan, L., & Zhu, Y. (2025). Spiking Transformer with Spatial-Temporal Spiking Self-Attention. In ICASSP 2025: IEEE International Conference on Acoustics, Speech and Signal Processing. IEEE. https://doi.org/10.1109/ICASSP49660.2025.10890026

University	Peking University
Country	China
Publication Year	2025
Publication Venue	IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2025
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/ICASSP49660.2025.10890026
Publication Type	Conference paper
Database Source	IEEE Xplore
Paper Category	Core architecture
E&QA; Score	4
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image/event classification using improved spatial-temporal spiking self-attention
Data Modality	Static images, sequential image inputs, and event-based/DVS visual data
Signal Dimensionality	2D images/event frames represented as spiking token sequences with temporal, token, and channel dimensions
Signal Characteristics	Input features are spike-form tensors; sequential CIFAR images are processed column-by-column to test long-term temporal dependency; neuromorphic datasets contain dynamic spatial-temporal event information
Dataset Name	Sequential CIFAR-10, Sequential CIFAR-100, CIFAR-10, CIFAR-100, CIFAR10-DVS, N-Caltech 101, DVS128 Gesture
Dataset Type	Public static image classification and neuromorphic/event-based classification benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Spiking Transformer with Patch Splitting, Relative Position Embedding, Spatial-Temporal Spiking Self-Attention, MLP, and classification head
SNN Component	LIF neurons, spike-form Q/K/V, Representative Spiking Temporal Tokens, temporal window masking, spike-based temporal feature fusion
Neuron Model	LIF
Transformer Component	Spiking Transformer block, Spiking Self-Attention, Spatial-Temporal Spiking Self-Attention, MLP, Relative Position Embedding, classification head
Attention Type	Spatial-Temporal Spiking Self-Attention; softmax-free spiking self-attention; temporal-window-masked spiking attention
SNN–Transformer Interface Mechanism	Q, K, and V are generated through linear projection, batch normalization, and LIF spiking neuron layers; representative temporal tokens are extracted from Q and inserted between Q and K to fuse temporal and spatial dependencies
Model Depth / Scale	Uses Spikformer as the main baseline and tests generalization on Spikingformer variants; exact layer depth and model scale NR
Parameter Count	NR
Design / Contribution Type	New STSSA module; Representative Spiking Temporal Token extraction; k-order Temporal Window Masking; Multi-dimensional Learnable Scaling Factor
Input Encoding Method	Spiking Patch Splitting converts input into spike-form token sequences; sequential CIFAR images are fed into SNNs column-by-column
Encoding Parameters	Neuromorphic experiments report time steps of 10 and 16; RSTT uses k and n settings, including k = 2 and n = 4 for neuromorphic experiments; ablation uses k = 12 and n = 1; MLSF uses learnable α and β with initial value 0
Encoding Learned or Fixed	Partially learned: Q/K/V projections and MLSF are learned; RSTT extraction and temporal window masking are fixed algorithmic operations
Event / Spike Representation	Binary spike tensors; spike-form Q, K, V; Representative Spiking Temporal Tokens; temporal-window-masked spike tokens
Temporal Resolution	Time steps 10 and 16 are reported for neuromorphic datasets; sequential CIFAR uses temporal column-wise image presentation; exact timestep setting for all static experiments NR
Output Decoding Method	Classification head produces final class prediction; exact readout rule NR
Encoding / Temporal Information Analysis	Yes. The paper analyzes long-term temporal dependency, temporal window masking, sequential CIFAR performance, neuromorphic temporal modeling, and ablation of STSSA/TWM/MLSF
Training Paradigm	Supervised direct training of Spiking Transformer models
Training Method	Direct SNN training following baseline Spikformer/Spikingformer settings; STSSA replaces SSA inside the Spiking Transformer
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	Experimental settings such as optimizer and epoch number are stated to follow baseline settings; MLSF learning rate for α and β is set to 0.01 times the network learning rate; other details NR
Primary Performance Metric	Classification accuracy
Primary Metric Value	STSSA achieves 84.27% on Sequential CIFAR-10 and 63.82% on Sequential CIFAR-100 in ablation results; on neuromorphic datasets, STSSA with 16 time steps achieves 83.80% on CIFAR10-DVS, 81.65% on N-Caltech 101, and 99.30% on DVS128 Gesture
Secondary Metrics	Ablation accuracy, convergence curves, comparison across timesteps, computational complexity
Baseline Models Compared	Spikformer, Spikingformer, Spike-Driven Transformer, STSA, TIM, SSA baseline
Performance Gain Over Baseline	Compared with SSA baseline, STSSA improves Sequential CIFAR-10 by 6.34% and Sequential CIFAR-100 by 5.70%; on CIFAR10-DVS, STSSA improves over Spikformer SSA from 80.90% to 83.80% at 16 time steps; on N-Caltech 101, it improves over TIM by 2.19%
Statistical Validation	Benchmark comparison and ablation study; no standard deviation, confidence interval, repeated-run statistics, or significance test reported
Ablation Study	Clear ablation study. Evaluates STSSA, Temporal Window Masking, and Multi-dimensional Learnable Scaling Factor on Sequential CIFAR-10 and Sequential CIFAR-100
Training Framework / Code Availability	NR
Target Hardware	Neuromorphic hardware motivation discussed; no specific hardware platform targeted
Hardware Deployment Status	No hardware deployment reported

Energy Evidence	No measured energy evidence; efficiency is argued through low-complexity spiking attention and reduced spatial-temporal attention overhead
Latency / Memory / Complexity Evidence	Provides computational complexity comparison: original SSA complexity is OSSA = $2TN^2D$, while STSSA complexity is OSTSSA = T(10 + 4kn)ND; for N = 64 and k = n = 4, STSSA avoids additional computational overhead while adding temporal modeling
Comparison with ANN Baseline	No direct ANN baseline; comparisons are mainly with SNN and Spiking Transformer baselines
Main Contribution / Key Limitation	Contribution: proposes STSSA to overcome the temporal-modeling limitation of standard SSA by inserting representative temporal tokens into attention computation. Limitation: short conference paper with limited training details, no hardware deployment, no energy measurement, and evaluation mainly restricted to classification tasks
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ1

P029
LightSpikformer: A compression framework for spiking transformer via low-rank decomposition of tensor networks

Full Citation	Chen, S., Chen, Z., Zhao, J., Li, L., Zhang, T., & Feng, X. (2025). LightSpikformer: A compression framework for spiking transformer via low-rank decomposition of tensor networks. Neurocomputing, 653, 131169. https://doi.org/10.1016/j.neucom.2025.131169
University	Shandong University of Science and Technology
Country	China
Publication Year	2025
Publication Venue	Neurocomputing
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neucom.2025.131169
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Efficiency-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Compressing Spiking Transformer models for classification while preserving accuracy and reducing parameters, memory, and energy cost
Data Modality	Static images and neuromorphic/event-camera data
Signal Dimensionality	2D image sequences represented as spatiotemporal spike tensors, $I \in R(T\times C\times H\times W)$; event-based 2D spatiotemporal inputs
Signal Characteristics	Static images are converted into spike-form feature vectors; neuromorphic datasets contain positive and negative polarity events; DVS128 Gesture uses 128 × 128 event data; spike feature maps are sparse and low-rank
Dataset Name	CIFAR-10, CIFAR-100, ImageNet, DVS128 Gesture, CIFAR10-DVS
Dataset Type	Public static image and neuromorphic vision benchmark datasets
Dataset Size	NR; paper states DVS128 Gesture contains thousands of gesture samples but does not give exact sample counts
Dataset Balance / Split	NR
Hybrid Topology	Compressed Spiking Transformer topology; LightSpikformer replaces original SSA and MLP linear layers with LRLayer-based low-rank tensor network modules, forming LR-SSA and LR-MLP
SNN Component	LIF neurons, spike activation layers, spiking patch splitting, sparse spike feature maps, spiking self-attention, spiking MLP
Neuron Model	LIF
Transformer Component	Spiking Transformer / Spikingformer encoder block, Spiking Self-Attention, MLP block, SPS module, classification head
Attention Type	Spiking Self-Attention using spike-form Q, K, and V
SNN–Transformer Interface Mechanism	Input image sequence is projected into spike-form patch features through SPS; LRLayer generates compressed spike-form Q, K, and V inside SSA; global average pooling feeds the final classification head
Model Depth / Scale	Evaluated mainly on Spikingformer-style 4-block models; static datasets use feature dimensions such as 384/256; neuromorphic settings use patch embedding dimension 256, patch size 16 × 16, and 8 LR-SSA heads; neuromorphic experiments report T = 10 and T = 16
Parameter Count	Varies by configuration. Reported examples: CIFAR-10 baseline 9.33M and compressed full CDT model 2.28M; DVS128 Gesture baseline 3.18M and compressed full CDT model 1.66M; neuromorphic configurations report around 1.1M–1.7M parameters
Design / Contribution Type	Low-rank tensor-network compression framework using Matrix Product State and Matrix Product Operator; LRLayer; LR-SSA; LR-MLP; channel-independent dynamic learning
Input Encoding Method	Spiking patch splitting with 2D convolution, batch normalization, LIF/spike neuron layers, max pooling, and spiking positional embedding; first convolutional gateway is treated as rate-coding based in the energy analysis
Encoding Parameters	Neuromorphic experiments: T = 10 and T = 16; patch size 16 × 16; embedding dimension 256; 8 attention heads; LIF threshold and exact static-image timestep settings NR
Encoding Learned or Fixed	Partially learned: SPS, LRLayer, Q/K/V projections, and classifier are learned; LIF spike-generation dynamics and tensor-rank structure are fixed/configured
Event / Spike Representation	Sparse binary spike feature maps; spike sequences; spike-form Q, K, and V; low-rank tensor-factorized spike feature transformation
Temporal Resolution	T = 10 and T = 16 for neuromorphic experiments; exact temporal resolution for all static-dataset settings NR
Output Decoding Method	Global average pooling followed by fully connected classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike sparsity, low-rank spike feature maps, spatiotemporal redundancy, noisy spike propagation, compression accuracy loss, memory reduction, and energy reduction
Training Paradigm	Supervised learning
Training Method	Direct training with BPTT-style SNN learning; channel-independent dynamic learning; low-rank tensor-factor training for LRLayer, LR-SSA, and LR-MLP
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW reported for neuromorphic datasets; optimizer for all settings not fully detailed
Loss Function	NR; label smoothing coefficient 0.1 is reported for static-dataset training

Training Hyperparameters	Static datasets: batch size 64, mixed-precision training, 400 epochs, label smoothing 0.1. Neuromorphic datasets: batch size 16, AdamW optimizer, 200 epochs, patch embedding dimension 256, patch size 16 × 16, 8 attention heads
Primary Performance Metric	Classification accuracy and parameter compression ratio
Primary Metric Value	Reported headline results: 84.49% parameter compression on ImageNet; memory consumption reduced to 42.29% of the original model; 94.72% accuracy on CIFAR-10; 97.8% accuracy on DVS128 Gesture
Secondary Metrics	Parameter count, compression rate, accuracy loss, memory consumption, FLOPs, SOPs, theoretical energy consumption, training time
Baseline Models Compared	Spikformer, Spikingformer, SpikformerV2, PLIF, STBP-tdBN, ADMM, ESL-SNN, QP-SNN, TT-SNN, TR-SNN, ANN ViT baseline
Performance Gain Over Baseline	LightSpikformer_MPS achieves 75.54% compression on CIFAR-10 with –0.89% accuracy loss; LightSpikformer_MPO achieves 74.03% compression on CIFAR-10 with –0.25% accuracy loss; DVS128 Gesture shows 50.37% compression with no loss for MPS and 46.88% compression with +0.70% accuracy gain for MPO
Statistical Validation	Benchmark comparison and ablation experiments; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes LR-SSA, LR-MLP, channel-independent dynamic training, MPS vs MPO comparison, compression-rate analysis, parameter analysis, memory analysis, and energy analysis
Training Framework / Code Availability	Training framework NR; code availability NR
Target Hardware	Resource-constrained / low-energy neuromorphic deployment is the target motivation; no specific chip or FPGA deployment
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Theoretical energy estimation using SOPs/FLOPs and 45 nm process assumptions: MAC = 4.6 pJ and AC = 0.9 pJ
Latency / Memory / Complexity Evidence	Reports memory reduction to 42.29% of original model; MPS reduces transformer-block FLOPs from 3.84 × 10■ to 1.26 × 10■ and SOPs from 6.14 × 10■ to 2.03 × 10■ in the analyzed setting
Comparison with ANN Baseline	Partial. Includes ANN ViT in theoretical energy comparison and discusses differences between dense ANN feature maps and sparse SNN spike feature maps
Main Contribution / Key Limitation	Contribution: proposes LightSpikformer, a low-rank tensor-network compression framework using MPS/MPO to reduce Spiking Transformer parameters, memory, and computation while preserving competitive accuracy. Limitation: low-rank compression can cause information loss and accuracy reduction compared with the original baseline; no real hardware validation is provided
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4

P030

SQKformer: Spiking sparse QKformer with adaptive batch normalization for membrane potential

Full Citation	Chen, Y., Xie, Z., Zhong, J., Chen, P., & Xiao, J. (2026). SQKformer: Spiking sparse QKformer with adaptive batch normalization for membrane potential. Neurocomputing, 671, 132666. https://doi.org/10.1016/j.neucom.2026.132666
University	Guangdong University of Technology
Country	China
Publication Year	2026
Publication Venue	Neurocomputing
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neucom.2026.132666
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification on static and neuromorphic datasets using a more efficient Spiking Transformer architecture
Data Modality	Static RGB images and event-camera / neuromorphic data
Signal Dimensionality	2D image data represented as spatiotemporal spike tensors; input described as I ∈ R(T×C×H×W)
Signal Characteristics	ImageNet input size 224 × 224; CIFAR image classification; CIFAR10-DVS and DVS128 Gesture use event-based spatiotemporal data; model uses binary spike sequences during inference
Dataset Name	ImageNet-1K, CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public static image and neuromorphic vision benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Hierarchical Spiking Sparse QKformer architecture based on QKFormer; three-stage structure with downsampling, Spiking Sparse Q-K Attention, SGSA feedforward block, and classification head
SNN Component	LIF neurons, spike neuron layers, membrane potential adaptive batch normalization, binary spike sequences, spike-driven Q-K attention, SGSA block
Neuron Model	LIF
Transformer Component	QKFormer-style hierarchical Transformer block; Sparse QKformer block; Q-K attention; SGSA feedforward replacement; classification head
Attention Type	Spiking Sparse Q-K Attention; Q-K Token Attention; information-enhanced channel attention; spiking gated spatial attention
SNN–Transformer Interface Mechanism	Linear layers generate Q and K, followed by BN/MABN and spike neuron layers; Q-K attention produces token masks, while IECA adds channel-attention information; SGSA replaces the standard Spiking MLP
Model Depth / Scale	ImageNet uses three stages with block distribution {1, 2, 7}; CIFAR uses 4 blocks with distribution {1, 1, 2}; neuromorphic experiments use {0, 1, 1}; main ImageNet variants include SQK-10–384 and SQK-10–512
Parameter Count	SQK-10–384: 14.69M; SQK-10–512: 25.75M; SQK-4–384: 6.23M; SQK-2–256: 1.90M; SQK-10–256: 6.71M
Design / Contribution Type	New Spiking Transformer architecture with three key modules: MABN for membrane-potential-aware normalization, IECA for channel attention, and SGSA as a low-parameter feedforward replacement
Input Encoding Method	Static images are processed through downsampling / patch projection and spike neuron layers; SFA converts integer-valued activations during training into binary spike sequences during inference

Encoding Parameters	ImageNet input size 224 × 224; first downsampling patch size 4 × 4, later stages use 2 × 2; virtual timestep D = 4 for SFA; CIFAR uses 4 timesteps; DVS datasets use 16 timesteps; MABN uses learnable threshold θ and ε = 1e–5
Encoding Learned or Fixed	Partially learned: spike firing dynamics are fixed, but MABN threshold θ, gating coefficient α, and network parameters are learned
Event / Spike Representation	Binary spike sequences; firing-rate approximation during training; spike-form Q and K; token attention spike mask; membrane potential representation
Temporal Resolution	4 timesteps for ImageNet/CIFAR experiments; 16 timesteps for CIFAR10-DVS and DVS128 Gesture
Output Decoding Method	Final fully connected classification head after the last Sparse QKformer block
Encoding / Temporal Information Analysis	Yes. The paper analyzes membrane potential effects, silent values, firing-rate reduction, loss landscape smoothness, gradient distribution, and spike firing-rate behavior under MABN
Training Paradigm	Supervised learning
Training Method	Direct training using Spiking Firing Approximation; integer-valued firing-rate training with spike-driven inference
Surrogate Gradient Function	Rectangular window surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Cross-entropy loss with MABN-related sparse regularization term: Ltotal = LCE + $\lambda\Sigma(1–M)$
Training Hyperparameters	ImageNet: batch size 256, 200 epochs, 4 NVIDIA RTX 3090 GPUs, AdamW, base learning rate 6 × 10■■, virtual timestep 4, random augmentation, blending, and random depth. CIFAR: 400 epochs, batch size 64. DVS128 Gesture: 200 epochs. CIFAR10-DVS: 106 epochs
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	CIFAR-10: 97.30%; CIFAR-100: 82.51%; DVS128 Gesture: 99.0%; CIFAR10-DVS: 84.5%; ImageNet-1K: 80.54% for SQK-10–512
Secondary Metrics	Parameter count, energy consumption, average firing rate, loss landscape, gradient distribution, ablation accuracy, timestep count
Baseline Models Compared	PVT, DeiT, QCFS, Fast-SNN, MST, STA, Spikformer, S-Transformer, SpikingResformer, QKFormer, S-Transformer v2, SNN-ViT, M-ViT+α-SSA, STAtten+, CML, STMixer, STSA
Performance Gain Over Baseline	On CIFAR-10, SQKformer improves over QKFormer by 1.09% while reducing parameters by 0.5M. On CIFAR-100, it improves over QKFormer by 1.36% with fewer parameters. On CIFAR10-DVS, it improves over QKFormer by 0.5%. On ImageNet, SQK-10–512 reaches 80.54% with 25.75M parameters and 6.65 mJ
Statistical Validation	Benchmark comparison and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes MABN, IECA, SGSA, kernel-size comparison in IECA, firing-rate comparison, loss landscape, gradient distribution, BN-method comparison, and comparison with QKFormer under the same setup
Training Framework / Code Availability	Training framework NR; code availability NR; data availability states data will be made available on request
Target Hardware	Neuromorphic hardware motivation; theoretical energy evaluation references SyNAPSE platform and 45 nm operation-energy assumptions
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Theoretical energy estimation using SOPs/FLOPs; assumes 45 nm MAC = 4.6 pJ and AC = 0.9 pJ; ImageNet SQK-10–512 reports 6.65 mJ and SQK-10–384 reports 4.22 mJ
Latency / Memory / Complexity Evidence	Reports parameter reduction and energy estimates; SQK-10–512 has 25.75M parameters versus QKFormer HST-10–512 with 29.08M; SQK-10–384 has 14.69M versus HST-10–384 with 16.47M; latency/memory footprint NR
Comparison with ANN Baseline	Yes. Compares against ANN models such as PVT and DeiT, and ANN-to-SNN methods such as QCFS, Fast-SNN, MST, and STA
Main Contribution / Key Limitation	Contribution: proposes SQKformer with MABN, IECA, and SGSA to improve Spiking Transformer accuracy, reduce firing-rate instability, and lower parameter count. Limitation: the model is not entirely spike-driven because QK attention still requires a small amount of multiplication; larger ImageNet embedding dimensions and tasks such as segmentation/detection remain future work
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ5

P031

TE-Spikformer: Temporal-enhanced spiking neural network with transformer

Full Citation	Gao, S., Fan, X., Deng, X., Hong, Z., Zhou, H., & Zhu, Z. (2024). TE-Spikformer: Temporal-enhanced spiking neural network with transformer. Neurocomputing, 602, 128268. https://doi.org/10.1016/j.neucom.2024.128268
University	Shanghai University
Country	China
Publication Year	2024
Publication Venue	Neurocomputing
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neucom.2024.128268
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Core architecture
E&QA; Score	4
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Event-stream / neuromorphic dataset classification
Data Modality	Event-camera / neuromorphic vision data
Signal Dimensionality	2D spatiotemporal event data represented over multiple timesteps
Signal Characteristics	Neuromorphic event streams with temporal dimension; experiments use DVS-CIFAR10, DVS128 Gesture, and N-Caltech101; model is designed to improve temporal information extraction and reduce timestep-wise AFR imbalance
Dataset Name	DVS128 Gesture, CIFAR10-DVS, N-Caltech101
Dataset Type	Public neuromorphic/event-based vision benchmark datasets

Dataset Size	DVS128 Gesture: 11 gestures from 29 participants under three lighting conditions; CIFAR10-DVS: 10 categories with 1000 examples per category; N-Caltech101: 100 object classes plus one background class after excluding “Faces”
Dataset Balance / Split	CIFAR10-DVS uses 9000 samples for training and 1000 samples for testing; DVS128 Gesture and N-Caltech101 split details NR
Hybrid Topology	TE-Spikformer follows Spikformer architecture but replaces BN with Batch Group Normalization and replaces SSA with Spike Spatio-Temporal Attention; structure includes SPS, SSTA, MLP, skip connections, and classification head
SNN Component	LIF neurons, spiking patch splitting, spiking position embedding, spike neuron layers, spike-based temporal attention, spike-based spatial attention
Neuron Model	LIF
Transformer Component	Spikformer encoder block; Spike Spatio-Temporal Attention; MLP block; classification head
Attention Type	Spike Spatio-Temporal Attention combining Spike Spatial Attention and Spike Temporal Attention
SNN–Transformer Interface Mechanism	Spiking inputs are processed through SPS and Transformer-style blocks; Q/K/V are used for spatial attention, while QT/KT/VT are derived for temporal attention; SSA and STA outputs are summed to form SSTA
Model Depth / Scale	Main SOTA setup uses four SPS modules, patch embedding dimension 512, patch size 16, one Transformer encoding module, and 16 detection heads; ablation also tests 1-512 and 2-256 architectures
Parameter Count	NR
Design / Contribution Type	New temporal-enhanced Spikformer variant; proposes Batch Group Normalization to address temporal covariate shift and Spike Spatio-Temporal Attention to improve temporal feature extraction
Input Encoding Method	Event-based neuromorphic data are processed through Spiking Patch Splitting; exact event-to-frame/bin encoding procedure NR
Encoding Parameters	Patch size 16; four SPS modules; timestep settings T = 10, 16, and 20 depending on dataset/ablation; SSTA scaling factor 0.25
Encoding Learned or Fixed	Partially learned: SPS, attention projections, BGN parameters, and model weights are learned; spike-generation dynamics are fixed LIF-based
Event / Spike Representation	Binary spike activations; timestep-wise spike matrices; AFR computed from final spiking neuron layer before classification head
Temporal Resolution	T = 16 and T = 20 for DVS128 Gesture and CIFAR10-DVS ablations; T = 10 and T = 16 for N-Caltech101 ablations
Output Decoding Method	Classification head after TE-Spikformer encoder; exact readout formula NR
Encoding / Temporal Information Analysis	Yes. The paper analyzes average firing-rate imbalance across timesteps, temporal covariate shift, AFR variance, and how BGN/SSTA stabilize temporal activation patterns
Training Paradigm	Supervised learning
Training Method	Surrogate-gradient direct training
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Mean Squared Error is used in the theoretical BGN analysis; exact experimental classification loss NR
Training Hyperparameters	NVIDIA 3090 GPU; SpikingJelly framework; AdamW optimizer; initial learning rate 0.1 with cosine decay; batch size 16 for DVS128 Gesture and CIFAR10-DVS; epochs: 210 for DVS128 Gesture and CIFAR10-DVS, 160 for N-Caltech101; SSTA scaling factor 0.25
Primary Performance Metric	Classification accuracy
Primary Metric Value	DVS128 Gesture: 99.30%; CIFAR10-DVS: 89.60%; N-Caltech101: 87.42%
Secondary Metrics	Average Firing Rate across timesteps; AFR variance; ablation accuracy under different timesteps and model scales
Baseline Models Compared	RG-CNNs, MLS, LIAF-Net, MVF-Net, Switched-BN + SALT, STBP-tdBN, BNTT, Event Transformer, TA-SNN, PMLIF, PLIF, STS-Transformer, TCSA, Spikingformer + CML, STCA-SNN, TCJA-TET-SNN, Spikformer
Performance Gain Over Baseline	TE-Spikformer improves over Spikformer on CIFAR10-DVS from 80.90% to 89.60%; improves on DVS128 Gesture from 98.30% to 99.30%; ablation shows Spikformer-BGN and TE-Spikformer progressively improve accuracy over baseline
Statistical Validation	Benchmark comparison and ablation experiments; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes Spikformer baseline, Spikformer-BGN, TE-Spikformer, different model depths/embedding dimensions, T = 10/16/20 timestep settings, and AFR imbalance visualization
Training Framework / Code Availability	SpikingJelly; NVIDIA 3090 GPU; code availability NR; data availability states data will be made available on request
Target Hardware	Neuromorphic / low-power SNN motivation only; no specific hardware deployment target
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	Energy efficiency is discussed conceptually through SNN spike/addition operations, but no direct energy measurement or theoretical energy table is reported
Latency / Memory / Complexity Evidence	NR
Comparison with ANN Baseline	Yes. Compares accuracy with ANN event-based methods such as RG-CNNs, MLS, LIAF-Net, and MVF-Net
Main Contribution / Key Limitation	Contribution: proposes TE-Spikformer with BGN and SSTA to preserve temporal characteristics, reduce AFR imbalance across timesteps, and improve neuromorphic classification accuracy. Limitation: evaluation is limited to neuromorphic classification datasets; no static image experiments, hardware validation, parameter count, latency, or energy measurement reported
RQ Mapping	Primary: RQ3; Secondary: RQ1, RQ2, RQ4

P032

Spiking Trans-YOLO: A range-adaptive energy-efficient bridge between YOLO and Transformer

Full Citation	Huo, Y., Ge, H., Tang, G., Gao, S., & Xu, J. (2025). Spiking Trans-YOLO: A range-adaptive energy-efficient bridge between YOLO and Transformer. Neurocomputing, 645, 130407. https://doi.org/10.1016/j.neucom.2025.130407
University	Dalian University of Technology
Country	China
Publication Year	2025
Publication Venue	Neurocomputing
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neucom.2025.130407
Publication Type	Journal article

Database Source	ScienceDirect
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Object detection on static images and event-camera data
Data Modality	Static RGB images and event-camera / neuromorphic event streams
Signal Dimensionality	2D image data and 2D spatiotemporal event data represented as $X \in R(T\times C\times H\times W)$
Signal Characteristics	Static images are repeated along the temporal dimension using direct input encoding; event input is represented as event tuples $e = (x, y, t, p)$, where polarity $p \in \{-1, 1\}$; event streams are aggregated into fixed time windows and converted into image-like 2D representations
Dataset Name	COCO 2017; Gen1
Dataset Type	Public static object-detection dataset and public event-camera automotive object-detection dataset
Dataset Size	COCO 2017: 118K training images, 5K validation images, 80 classes. Gen1: 39 hours of driving/event-camera data, ATIS sensor resolution 304 × 240, 2 classes: pedestrians and cars
Dataset Balance / Split	COCO uses training and validation split of 118K / 5K. Gen1 split details NR
Hybrid Topology	YOLO–Transformer bridge architecture. Spiking Trans-YOLO uses a YOLO-style backbone and detection head, with a Top-Attention Hybrid Feature Fusion module that applies Transformer-style spiking attention only to the highest-level feature map S5, then fuses it with lower-level features S3 and S4
SNN Component	Integer Leaky Integrate-and-Fire neuron, spike-driven inference, spiking convolutional blocks, MSFF, spike-based feature maps, spike self-attention, SpikeMLP
Neuron Model	Integer Leaky Integrate-and-Fire / I-LIF
Transformer Component	Transformer-based SNN block inside TAHFF; Range-Adaptive Spiking Attention; SpikeMLP
Attention Type	Range-Adaptive Spiking Attention; spike-based self-attention with adaptive scaling factors
SNN–Transformer Interface Mechanism	YOLO backbone produces multi-scale spiking features S3, S4, and S5. RASA performs intra-scale spiking attention on the highest-level feature S5 only, generating F5. The attended high-level feature is then fused with lower-level convolutional features before detection
Model Depth / Scale	Main models include 26.3M-parameter and 90.3M-parameter Spiking Trans-YOLO variants. T × D settings include 1 × 1, 1 × 4, 4 × 1, 2 × 1, 4 × 2, and other ablation settings. T is timestep and D is the maximum integer spike value / virtual expansion factor
Parameter Count	Main model: 26.3M parameters. Large model: 90.3M parameters. Ablation variants include 23.3M, 26.4M, 26.5M, 26.9M, 27.2M, 28.0M, and 28.3M
Design / Contribution Type	New Spiking YOLO–Transformer object detector; Top-Attention Hybrid Feature Fusion; Range-Adaptive Spiking Attention; range-adaptive scaling for I-LIF integer-valued training; MSFF with dilated multi-scale feature fusion
Input Encoding Method	Static images use direct input encoding by repeating images over timesteps; event streams are aggregated over fixed temporal windows into image-like 2D representations
Encoding Parameters	Event stream total sequence $\Gamma = T \times dt$, where dt is the fixed time window and T is the number of timesteps. I-LIF uses D as maximum integer spike value. RASA scaling factors use D, firing rate of Qs, firing rate of attention map, attention dimension, and spatial token size
Encoding Learned or Fixed	Partially learned: network features and attention parameters are learned; direct input repetition and event-window aggregation are fixed preprocessing mechanisms; RASA scaling uses firing-rate estimates
Event / Spike Representation	Integer spikes during training; binary spike sequences during inference; spike feature maps; spike firing-rate heatmaps; event tuples converted into temporal image-like representations
Temporal Resolution	COCO uses T × D settings such as 1 × 1, 1 × 4, and 4 × 1. Gen1 uses settings such as 2 × 1, 4 × 1, and 4 × 2. Exact dt for event aggregation NR
Output Decoding Method	YOLO-style detection head outputs object classes and bounding boxes
Encoding / Temporal Information Analysis	Yes. The paper analyzes timestep T, integer spike range D, quantization error, spike firing-rate heatmaps, attention placement, energy distribution, and the effect of D on accuracy and power consumption
Training Paradigm	Supervised direct training
Training Method	Direct SNN training using I-LIF integer-valued training and spike-driven inference; surrogate-gradient-based training is discussed as the direct-training foundation
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	SGD
Loss Function	YOLO-style detection losses are shown in training curves: box loss, classification loss, and DFL loss; exact full loss formula NR
Training Hyperparameters	COCO: Mosaic augmentation, SGD optimizer, learning rate 0.01, decay factor $\beta = 0.25$, 300 epochs, batch size 40, four NVIDIA V100 GPUs. Gen1: 50 epochs, other hyperparameters kept consistent with COCO. RASA firing rates are estimated using EMA momentum 0.999
Primary Performance Metric	mAP@50, mAP@50:95, and energy consumption
Primary Metric Value	COCO 2017: best 67.7% mAP@50 and 50.7% mAP@50:95 using 90.3M parameters, T × D = 1 × 4. Gen1: best 68.6% mAP@50 and 42.1% mAP@50:95 using 26.3M parameters, T × D = 4 × 2
Secondary Metrics	Parameter count, Power45, Power7, spike firing rate, attention heatmaps, training loss curves, timestep/integer-spike ablation
Baseline Models Compared	PVT, DETR, YOLOv5, Spiking-YOLO, Bayesian Optim, Spike Calibration, EMS-YOLO, Meta-SpikeFormer, SpikeYOLO, YOLOv3-tiny, SpikeYOLO-ANN, VGG-11+SDD, MobileNet-64+SSD, DenseNet121-24+SSD, Tr-Spiking-YOLO
Performance Gain Over Baseline	On COCO, Spiking Trans-YOLO reaches 67.7% mAP@50, improving over SpikeYOLO 68.8M by +1.5% mAP@50 and +1.8% mAP@50:95. On Gen1, it reaches 68.6% mAP@50, improving over SpikeYOLO by +1.4% mAP@50. Compared with similar ANN architecture on Gen1, it achieves +2.3% mAP@50 and 3.1× higher energy efficiency at 45 nm
Statistical Validation	Benchmark comparison and ablation experiments; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes T and D analysis, MSFF ablation, RASA removal/replacement with SDSA, RASA placement, number of RASA modules, RASA in image classification, attention heatmaps, training curves, and energy distribution
Training Framework / Code Availability	Code available at GitHub: https://github.com/s1110/Spiking-Trans-YOLO; training hardware uses four NVIDIA V100 GPUs
Target Hardware	Neuromorphic hardware motivation; theoretical energy analysis based on 45 nm and 7 nm CMOS assumptions
Hardware Deployment Status	No real neuromorphic hardware or chip deployment reported
Energy Evidence	Theoretical energy estimation using operation counts, firing rate, timestep, D, and assumed operation energy. Uses 45 nm: MAC = 4.6 pJ, AC = 0.9 pJ; 7 nm: MAC = 1.69 pJ, AC = 0.38 pJ

Latency / Memory / Complexity Evidence	Reports parameter count and theoretical energy. Attention complexity remains $O(N^2D)$. No measured latency or memory footprint reported
Comparison with ANN Baseline	Yes. Compares with PVT, DETR, YOLOv5, YOLOv3-tiny, SpikeYOLO-ANN, and Spiking Trans-YOLO-ANN
Main Contribution / Key Limitation	Contribution: proposes Spiking Trans-YOLO as an efficient bridge between YOLO and Transformer for SNN object detection, using TAHFF and RASA to improve accuracy while controlling power. Limitation: RASA is not compatible with negative-spike neurons; attention complexity remains $O(N^2D)$; framework is less generalizable than Meta-SpikeFormer; no real hardware deployment is reported
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P033

Energy-efficient intrusion detection with a protocol-aware transformer–spiking hybrid model

Full Citation	Karthik, M. G., Keerthika, V., Mantena, S. V., Siri, D., Yeluri, L. P., Lella, K. K., & RamaGanesh, B. (2026). Energy-efficient intrusion detection with a protocol-aware transformer–spiking hybrid model. Scientific Reports, 16, 7095. https://doi.org/10.1038/s41598-026-37367-4
University	GITAM University-Bengaluru Campus
Country	India
Publication Year	2026
Publication Venue	Scientific Reports
Publisher	Springer Nature
DOI / URL	https://doi.org/10.1038/s41598-026-37367-4
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Communication, Cybersecurity, and Network Intelligence
Specific Task	Intrusion detection and attack classification, including normal, DoS, Probe, R2L, and U2R traffic classes
Data Modality	Tabular network-traffic features converted into flow tokens and spike trains
Signal Dimensionality	High-dimensional tabular feature vectors; pseudo-flow token sequences; discrete spike trains over simulation timesteps
Signal Characteristics	NSL-KDD connection records with 41 features, including protocol type, service, flag, byte counts, duration, traffic-count features, and host-based features; categorical and continuous features are normalized, embedded, reconstructed into pseudo-flows, and encoded into sparse spike events
Dataset Name	NSL-KDD, KDDTest+21, CICIDS-2017 subset
Dataset Type	Public benchmark intrusion-detection datasets
Dataset Size	NSL-KDD uses 41 features; exact sample counts for training/testing are NR in the paper
Dataset Balance / Split	Evaluated using 70/30, 60/40, and 80/20 train-test splits; class imbalance addressed using CPSMOTE-Tomek; exact class counts NR
Hybrid Topology	Transformer-Augmented Spiking Neural Network. The model combines protocol-aware preprocessing, FlowToken Transformer Encoder, Spike-Temporal Conv, LIF SNN, Cross-Modal Gating, and dual Transformer/SNN heads
SNN Component	Spike encoding, Spike-Temporal Conv, LIF neurons, lateral inhibition, spike-gated currents, spike-count readout, homeostatic spike-rate control
Neuron Model	Leaky Integrate-and-Fire neuron with surrogate-gradient training
Transformer Component	FlowToken Transformer Encoder with 2–4 layers; reported final setting uses 3 encoder layers, embedding dimension 128, 4 attention heads, and feed-forward hidden dimension 256
Attention Type	Transformer self-attention; attention rollout for feature importance; cross-modal attention/gating through XMG
SNN–Transformer Interface Mechanism	Transformer produces contextual vector hTR and attention maps; hTR is converted/gated into the SNN pathway through Cross-Modal Gating, which scales SNN membrane input currents. Transformer and SNN heads are also aligned through knowledge distillation and saliency consistency
Model Depth / Scale	Transformer: 3 encoder layers, 128 embedding dimension, 4 heads, 256 feed-forward hidden dimension. SNN: two hidden LIF layers with 128 and 64 neurons, followed by linear readout
Parameter Count	NR
Design / Contribution Type	New hybrid IDS framework with PAAN, PRE, CPSMOTE-Tomek, PFR, MASE, EDC, FTE, STC-LIF SNN, XMG, SAIF, and SPS
Input Encoding Method	Multi-Scale Adaptive Spike Encoding, Eventified Delta Coding, TTFS encoding, population coding, rank-order coding for categorical embeddings
Encoding Parameters	TTFS uses learnable αj and $\tau 0$; population coding uses Gaussian receptive fields and Poisson rates; EDC uses learnable thresholds θl; spike-rate target around 0.05–0.1 spikes/neuron/ms; STC kernel size $k \in \{3,5\}$; exact final timestep T is NR
Encoding Learned or Fixed	Partially learned: MASE parameters, EDC thresholds, categorical embeddings, Transformer/SNN weights, and gating are learned; protocol-aware normalization and pseudo-flow grouping rules are fixed or rule-based
Event / Spike Representation	Sparse spike trains generated from tabular features; TTFS spikes, population-code spikes, eventified delta spikes, rank-order categorical spikes, accumulated SNN readout spikes
Temporal Resolution	Discrete simulation horizon T; exact timestep count and physical sampling interval are NR
Output Decoding Method	Transformer head uses softmax over contextual vector; SNN head accumulates readout spikes over time and applies linear + softmax classification
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike count per neuron, latency, energy proxy, spike-gating effect, pseudo-flow context, attention heatmaps, and feature importance through attention rollout
Training Paradigm	Supervised learning with hybrid Transformer and SNN heads
Training Method	Surrogate-gradient training for LIF neurons; optional STDP warm-up discussed but not used in final reported results; knowledge distillation between Transformer and SNN outputs
Surrogate Gradient Function	Sigmoid surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Total loss combines Transformer focal loss, SNN focal loss, KL distillation loss, saliency alignment loss, homeostatic loss, spike-energy loss, and gate-entropy regularization

Training Hyperparameters	Adam optimizer; learning rate 1 × 10■^{3}; focal loss for imbalance; Transformer: 3 layers, 128 embedding, 4 heads, 256 FFN hidden units; SNN hidden layers: 128 and 64 neurons; experiments repeated 5 times with different random seeds
Primary Performance Metric	Accuracy, Macro-F1, AUC, energy proxy
Primary Metric Value	Accuracy: 96.7% for 70/30 split, 96.1% for 60/40 split, 97.2% for 80/20 split. Macro-F1: 0.92 for 70/30, 0.91 for 60/40, 0.93 for 80/20
Secondary Metrics	Precision, recall, F1-score per class, Micro precision/recall, ROC, DET, GAR@FAR=1%, spike count, latency, SpikeOps/sec, adversarial robustness
Baseline Models Compared	SVM, Random Forest, CNN, Transformer, CNN-LSTM, existing ML/DL IDS approaches
Performance Gain Over Baseline	TASNN improves over Transformer accuracy from 93.4% to 96.7% on 70/30, 92.6% to 96.1% on 60/40, and 93.9% to 97.2% on 80/20. It also improves rare-class detection, with U2R F1 = 0.80 and R2L F1 = 0.86
Statistical Validation	Repeated experiments with 5 random seeds; multiple train-test splits; robustness tests under noise, imbalance, cross-dataset transfer, and FGSM adversarial perturbation. Full k-fold cross-validation not conducted
Ablation Study	Full. Includes removal/analysis of PAAN, PFR, MASE/EDC, XMG, SAIF; simplified baselines; noise robustness; imbalance stability; cross-dataset generalization; adversarial robustness
Training Framework / Code Availability	PyTorch; scikit-learn; NumPy; Pandas; imbalanced-learn; code availability NR; datasets available from corresponding author on reasonable request
Target Hardware	Edge and neuromorphic IDS motivation; mentions platforms such as Intel Loihi and SpiNNaker as future/target hardware context
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Software-simulated neuromorphic energy proxy based on spike operations per second; no direct hardware energy measurement
Latency / Memory / Complexity Evidence	Average latency: 4.802 ms for 70/30, 5.122 ms for 60/40, 4.513 ms for 80/20. Average energy proxy: about 198k SpikeOps/sec. Average spike count per neuron: 1.785, 1.899, and 1.69 across splits
Comparison with ANN Baseline	Yes. Compares against SVM, Random Forest, CNN, Transformer, and CNN-LSTM-style IDS baselines
Main Contribution / Key Limitation	Contribution: proposes TASNN, a protocol-aware Transformer–SNN hybrid IDS with adaptive spike encoding, cross-modal gating, and spike-aware feature selection for improved rare-attack detection and energy-aware inference. Limitation: evaluation is benchmark-based, neuromorphic efficiency is simulated rather than hardware-measured, and Transformer components still introduce computational overhead
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P034

Low-power and lightweight spiking transformer for EEG-based auditory attention detection

Full Citation	Lan, Y., Wang, Y., Zhang, Y., & Zhu, H. (2025). Low-power and lightweight spiking transformer for EEG-based auditory attention detection. Neural Networks, 183, 106977. https://doi.org/10.1016/j.neunet.2024.106977
University	University of Electronic Science and Technology of China
Country	China
Publication Year	2025
Publication Venue	Neural Networks
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neunet.2024.106977
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	EEG-based auditory attention detection / classification of attended auditory direction
Data Modality	EEG signals converted into 2D alpha-band spectro-spatial/topological images
Signal Dimensionality	Multichannel EEG time series transformed into 2D spatial EEG images and then token sequences
Signal Characteristics	Raw EEG is segmented using sliding windows; FFT is applied; alpha-band cortical activity is mapped from 3D electrode positions to a 2D plane using azimuthal equidistant projection; decision windows include 0.1, 0.2, 0.5, 1, 2, 5, and 10 s
Dataset Name	KUL; DTU
Dataset Type	Public EEG auditory attention detection datasets
Dataset Size	KUL: 16 normal-hearing subjects, 8 trials per subject, 6 min each. DTU: 18 normal-hearing subjects, 60 trials per subject, 50 s each
Dataset Balance / Split	Dataset labels are divided by auditory attention direction; exact train/test split and class balance NR
Hybrid Topology	Encoder-only binary Spiking Transformer. Pipeline: EEG preprocessing → 2D EEG image generation → spiking convolutional tokenization → N binary spiking Transformer encoders → MLP classification head
SNN Component	LIF neurons, binary spike trains, spiking convolutional patchify stem, spiking Q/K/V, spiking self-attention, spike-based MLP
Neuron Model	Leaky Integrate-and-Fire
Transformer Component	Binary spiking Transformer encoder blocks; spiking self-attention; MLP head
Attention Type	Spiking self-attention using Hadamard product between binary Q and K, column-sum attention score, and column-mask filtering of V; softmax-free attention
SNN–Transformer Interface Mechanism	EEG image tokens are processed by spiking Transformer blocks. Linear + BN + LIF layers generate binary Q, K, and V spike sequences; Q ■ K produces spike attention scores, which mask V before projection and residual connection
Model Depth / Scale	Experiments test 2, 3, and 4 Transformer blocks with token numbers 64, 128, and 256. Main comparison setting uses 4 blocks, 128 tokens, and timestep T = 8
Parameter Count	NR; model size is reported instead of parameter count
Design / Contribution Type	First reported spiking Transformer for EEG-based auditory attention detection; combines spiking Transformer design with post-training binary weight quantization for low-power and lightweight edge deployment
Input Encoding Method	FFT-based EEG preprocessing, alpha-band topological projection, sliding-window segmentation, patch tokenization, and automatic spike encoding through the first spiking convolutional layer
Encoding Parameters	Main timestep T = 8; LIF time constant = 0.5; firing threshold = 1; batch size 64; patch size p used for tokenization but exact value NR

Encoding Learned or Fixed	Partially learned: EEG-to-2D projection and FFT preprocessing are fixed; spiking convolutional tokenization and Transformer weights are learned; weights are binarized after training
Event / Spike Representation	Binary spike trains; binary Q, K, V matrices; 1-bit activation spikes; binarized weights in inference
Temporal Resolution	Main SNN timestep T = 8; timestep study evaluates T = 1, 2, 4, 8, 16, and 32; decision windows range from 0.1 s to 10 s
Output Decoding Method	MLP classification head predicts the attended auditory direction/class
Encoding / Temporal Information Analysis	Yes. The paper analyzes timestep effect on accuracy, firing-rate stability, layer-wise firing rates, SOPs, and energy reduction from spike-based computation
Training Paradigm	Supervised direct training
Training Method	Spatio-temporal backpropagation / STBP-style training with surrogate gradient; post-training quantization for binary weights
Surrogate Gradient Function	Isosceles triangle surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	TET loss used in experiments; cross-entropy loss averaged over timesteps is also formulated
Training Hyperparameters	PyTorch; Kaiming normal initialization; AdamW optimizer; initial learning rate 0.0003; batch size 64; first 20 epochs warm-up at learning rate 0.00001; cosine decay over the next 280 epochs; LIF time constant 0.5; threshold 1; main T = 8
Primary Performance Metric	Classification accuracy; model size reduction; theoretical energy reduction
Primary Metric Value	KUL best accuracy: 98.07% at 10 s decision window. DTU best accuracy: 78.7% at 10 s decision window. Model size reduction exceeds 21× across tested settings
Secondary Metrics	Firing rate, FLOPs, SOPs, energy drop, model size, accuracy after binarization, block/token hyperparameter results
Baseline Models Compared	De Cheveigné et al. linear/CCA model; Vandecappelle CNN; Cai CNN; Su attention model; SVM; LDA; Cai et al. SNN attention model
Performance Gain Over Baseline	On KUL, the proposed model reaches 98.07% at 10 s, exceeding the strongest listed ANN attention baseline at 93.9%. On DTU, it reaches 78.7% at 10 s, exceeding the strongest listed ANN attention baseline at 75.8%. The 4-block/128-token model reduces size from 33.26 MB to 1.31 MB with only a small accuracy drop from 98.55% to 98.07%
Statistical Validation	Benchmark comparison and hyperparameter study; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Partial. Includes timestep analysis, firing-rate analysis, block/token scale comparison, FLOPs/SOPs analysis, model-size/accuracy comparison after binarization, and module-level neuron/firing/SOP statistics
Training Framework / Code Availability	PyTorch; code availability NR
Target Hardware	Edge devices and neuromorphic/low-power hardware motivation; future FPGA deployment mentioned
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Theoretical energy calculation using MAC = 4.6 pJ and ADD = 0.9 pJ from Horowitz-style chip assumptions; no measured hardware energy
Latency / Memory / Complexity Evidence	Model size reduced by more than 21×. Example: 4-block/256-token model decreases from 132.47 MB to 4.67 MB, a 28.4× reduction. Reported energy drop ranges around 82.52%–87.04% depending on model scale. Latency NR
Comparison with ANN Baseline	Yes. Compares against linear, CNN, and attention-based ANN AAD baselines, and compares SNN energy with an ANN counterpart where spiking neurons are replaced by ANN neurons
Main Contribution / Key Limitation	Contribution: proposes a lightweight binary Spiking Transformer for EEG-based auditory attention detection, achieving strong AAD accuracy with low firing rates, theoretical energy savings, and large model-size reduction. Limitation: energy savings are theoretical only; no real FPGA/neuromorphic deployment; evaluation is limited to KUL and DTU datasets
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P035

Spiking-PhysFormer: Camera-based remote photoplethysmography with parallel spike-driven transformer

Full Citation	Liu, M., Tang, J., Chen, Y., Li, H., Qi, J., Li, S., Wang, K., Gan, J., Wang, Y., & Chen, H. (2025). Spiking-PhysFormer: Camera-based remote photoplethysmography with parallel spike-driven transformer. Neural Networks, 185, 107128. https://doi.org/10.1016/j.neunet.2025.107128
University	Tsinghua University
Country	China
Publication Year	2025
Publication Venue	Neural Networks
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neunet.2025.107128
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	Remote pulse-wave and heart-rate estimation from facial videos
Data Modality	RGB facial video and physiological PPG/BVP ground-truth signals
Signal Dimensionality	3D spatiotemporal video input represented as RGB frames, spatiotemporal tube tokens, and spike tensors
Signal Characteristics	Input video is preprocessed to 128 × 128 × 160 clips; facial color changes caused by cardiac pulse are used to estimate pulse wave and HR; model processes spatiotemporal facial features and spike firing-rate attention maps
Dataset Name	PURE, UBFC-rPPG, UBFC-Phys, MMPD
Dataset Type	Public remote photoplethysmography / physiological video benchmark datasets
Dataset Size	After preprocessing: PURE 750 video segments, UBFC-rPPG 351, UBFC-Phys 3939, MMPD 7260. Original datasets: PURE 10 subjects; UBFC-rPPG 42 subjects; UBFC-Phys 56 subjects; MMPD 33 subjects and 660 one-minute videos
Dataset Balance / Split	Cross-dataset evaluation: trained on UBFC-rPPG or PURE and tested on other datasets. Exact train/validation split details NR
Hybrid Topology	Hybrid neural network with ANN-based patch embedding block, SNN-based parallel spike-driven Transformer blocks, and ANN-based predictor head

SNN Component	LIF neurons, direct spike encoding, spike tensors, simplified spiking self-attention, spike-driven Transformer block, spike firing-rate attention map
Neuron Model	Leaky Integrate-and-Fire
Transformer Component	Parallel spike-driven Transformer block, simplified spiking self-attention, multi-head S3A, MLP sub-block, residual/parallel block structure
Attention Type	Simplified Spiking Self-Attention; multi-head S3A; hard attention based on spike firing rate; softmax-free spike attention
SNN–Transformer Interface Mechanism	ANN patch embedding extracts tube tokens; direct encoding replicates tube features across spike timesteps; LIF neurons convert them into spike sequences; after SNN Transformer blocks, membrane potentials are averaged over spike timesteps and passed to ANN predictor head
Model Depth / Scale	Main model uses 4 parallel spike-driven Transformer blocks, SNN timestep Ts = 4, channel dimension D = 96, input size 128 × 128 × 160. Ablation also tests 4, 6, 8, 10, and 12 Transformer blocks
Parameter Count	2.99M parameters
Design / Contribution Type	First SNN/HNN model for rPPG; proposes Spiking-PhysFormer, parallel spike-driven Transformer block, and simplified spiking self-attention that removes the value projection to reduce computation
Input Encoding Method	Haar face detection and DiffNormalized preprocessing; ANN patch embedding extracts spatiotemporal tube tokens; direct encoding replicates features across spike timesteps; fixed-parameter LIF neuron converts features into spikes
Encoding Parameters	Input clip: 160 frames, 128 × 128 resolution; SNN timestep Ts = 4; channel dimension D = 96; LIF α = 2, Vreset = 0, Vth = 1; spike fire rate is used for energy and attention analysis
Encoding Learned or Fixed	Partially learned: ANN patch embedding and Transformer/SNN weights are learned; direct encoding and LIF spike dynamics are fixed; PE block is initialized from a pretrained PhysFormer
Event / Spike Representation	Binary spike sequence S $\in$ {0,1}; spike-form Q, K, and V; spike firing-rate maps; spike tensor representations in S3A
Temporal Resolution	Input sequence length 160 video frames; SNN timestep Ts = 4; predictor outputs 1D pulse wave of length T
Output Decoding Method	Mean accumulation over spike temporal dimension followed by ANN predictor head with temporal upsampling to output a 1D pulse waveform
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike fire rates in Q/K/V/input tensors, SFR-based spatiotemporal attention maps, pulse-wave peak alignment, and energy consumption from FLOPs/SOPs
Training Paradigm	Supervised learning for pulse-wave / heart-rate estimation
Training Method	Hybrid ANN–SNN training using surrogate-gradient learning; PE block initialized from pretrained PhysFormer; Spiking-PhysFormer trained end-to-end after pretraining
Surrogate Gradient Function	Arctangent surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	PhysFormer loss: Negative Pearson temporal loss, frequency cross-entropy loss, and label distribution loss
Training Hyperparameters	PyTorch; SpikingJelly; rPPG-Toolbox; Adam optimizer; initial learning rate 3e-3; weight decay 5e-5; batch size 4; PhysFormer pretrained for 10 epochs; Spiking-PhysFormer trained for 10 epochs; GeForce RTX 4090 GPU
Primary Performance Metric	MAE, MAPE, Pearson correlation, and computational energy
Primary Metric Value	Power consumption: 1.34 mJ per 128 × 128 frame; 10.1% lower than PhysFormer. Transformer block energy is 12.2× lower than PhysFormer. Cross-dataset example: training on UBFC-rPPG and testing on PURE gives MAE 3.83 ± 0.74, MAPE 5.70 ± 0.74, ρ 0.83 ± 0.06
Secondary Metrics	FLOPs, SOPs, parameter count, spike fire rate, Bland–Altman plots, pulse-wave output examples, ablation MAE/MAPE/ρ
Baseline Models Compared	TS-CAN, PhysNet, DeepPhys, EfficientPhys-C, PhysFormer, iBVPNet, rFaceNet, DiffPhys, PhysNet-XY, PhysNet-UV
Performance Gain Over Baseline	Compared with PhysFormer, energy is reduced by 10.1% overall and 12.2× in the Transformer block. When trained on UBFC-rPPG and tested on PURE, MAE improves from PhysFormer's 12.92 to 3.83. The proposed model uses 2.99M parameters versus PhysFormer's 7.03M
Statistical Validation	Cross-dataset testing and ablation studies are reported; some results include mean ± variation. Full statistical significance testing NR
Ablation Study	Full. Includes number of Transformer blocks, SNN timesteps, parallel vs non-parallel sub-blocks, attention projection variants, and ANN/SNN component replacement
Training Framework / Code Availability	PyTorch, SpikingJelly, and rPPG-Toolbox; code availability for this specific model NR
Target Hardware	Edge devices and neuromorphic/low-power deployment are the target motivation; theoretical energy assumes 45 nm technology
Hardware Deployment Status	No real hardware deployment reported; experiments run on Linux/GPU platform
Energy Evidence	Theoretical energy estimation using FLOPs/SOPs and 45 nm operation energy assumptions: MAC = 4.6 pJ and AC = 0.9 pJ
Latency / Memory / Complexity Evidence	2.99M parameters; 290M ANN FLOPs from PE and predictor head; 3.7M SNN SOPs; 1.34 mJ frame energy. No measured latency or memory footprint reported
Comparison with ANN Baseline	Yes. Compares against ANN-based rPPG models and PhysFormer; also compares Transformer block energy with ANN PhysFormer
Main Contribution / Key Limitation	Contribution: proposes the first Spiking/HNN Transformer for rPPG, using ANN feature extraction and prediction with SNN-based parallel spike-driven Transformer blocks and S3A to reduce energy while preserving accuracy. Limitation: tested on Linux/GPU rather than practical edge or neuromorphic hardware; robustness still needs broader validation across motion, skin tone, and lighting conditions
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P036

ESTSformer: Efficient spatio-temporal spiking transformer

Full Citation	Lu, C., Du, H., Wei, W., Sun, Q., Wang, Y., Zeng, D., Chen, W., Zhang, M., & Yang, Y. (2025). ESTSformer: Efficient spatio-temporal spiking transformer. Neural Networks, 191, 107786. https://doi.org/10.1016/j.neunet.2025.107786
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Country	China
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Publication Venue	Neural Networks
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neunet.2025.107786

Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image/event classification on neuromorphic datasets and static image datasets
Data Modality	Event-camera / neuromorphic data and static RGB image data
Signal Dimensionality	2D static images and 2D spatiotemporal event streams represented as spike tokens over time
Signal Characteristics	Neuromorphic datasets use positive and negative event polarities; N-MNIST and DVS-CIFAR10 are event/spiking variants of MNIST and CIFAR-10; DVS128 contains hand gestures under different illumination conditions; CIFAR-10/100 use 32 × 32 color images
Dataset Name	N-MNIST, DVS128 Gesture, DVS-CIFAR10, CIFAR-10, CIFAR-100
Dataset Type	Public neuromorphic vision and static image classification benchmark datasets
Dataset Size	N-MNIST: 60,000 training and 10,000 test samples. DVS-CIFAR10: 60,000 images, 10 categories, 9:1 train-test split. DVS128 Gesture: 11 gesture classes from 29 individuals under 3 illumination conditions. CIFAR-10/100: 60,000 images, 50,000 train and 10,000 test
Dataset Balance / Split	Standard train/test splits for N-MNIST, CIFAR-10, and CIFAR-100; DVS-CIFAR10 uses 9:1 train-test split; exact class-balance details NR
Hybrid Topology	Encoder-only Spiking Transformer using Spiking Patch Splitting, ESTSA-based encoder blocks, MLP module, global average pooling, and classification head
SNN Component	LIF neurons, binary spike trains, spiking patch splitting, spike-driven residual connections, spike-based temporal and spatial attention heads
Neuron Model	Leaky Integrate-and-Fire
Transformer Component	Efficient Spatio-Temporal Spiking Transformer encoder; ESTSA module; MLP module; multi-head temporal/spatial attention split
Attention Type	Efficient Spatio-Temporal Self-Attention; attention heads are divided into temporal heads and spatial heads, then concatenated and fused
SNN–Transformer Interface Mechanism	Input data are tokenized by SPS into spike tokens; Q, K, and V are generated through Linear + BN + LIF layers; attention heads are split into spatial and temporal groups; outputs are concatenated, passed through LIF/linear fusion, and used in spike-driven Transformer blocks
Model Depth / Scale	Reported architectures include ESTSformer-2-256, ESTSformer-4-256, and ESTSformer-4-384 for neuromorphic tasks; static datasets include 4-256 and 2-384 models; timestep settings include 4, 10, and 16 depending on dataset
Parameter Count	Varies by configuration. Reported examples: ESTSformer-2-256 has 1.96M parameters in DVS-CIFAR10 efficiency comparison; CIFAR-10 ESTSformer-4-256 has 3.59M; CIFAR-100 ESTSformer-4-256 has 3.89M; ESTSformer-2-384 has 5.47M in static dataset comparison
Design / Contribution Type	New efficient spatio-temporal attention mechanism and Spiking Transformer architecture; redesigns STSA into ESTSA to reduce computation/storage while preserving temporal and spatial feature extraction
Input Encoding Method	Spiking Patch Splitting module processes static and dynamic input data into spike tokens; neuromorphic event streams use polarity channels; static images are processed into spike-based token representations
Encoding Parameters	Threshold = 1.0; membrane decay factor = 0.5; timestep values: 10 and 16 for neuromorphic experiments, 4 for CIFAR-10/100; DVS-CIFAR10 images resized to 128 × 128; N-MNIST resized to 32 × 32
Encoding Learned or Fixed	Partially learned: SPS, Q/K/V projections, ESTSA, MLP, and classification head are learned; LIF spike dynamics and timestep settings are fixed hyperparameters
Event / Spike Representation	Binary spike trains; spike tokens; Q/K/V spike tensors; separate temporal and spatial spike-attention representations
Temporal Resolution	T = 10 and T = 16 for main neuromorphic experiments; T = 4 for static CIFAR experiments
Output Decoding Method	Global average pooling followed by classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes temporal and spatial attention complexity, timestep impact, temporal/spatial head allocation, ESTSA efficiency, FLOPs, parameter count, and energy consumption
Training Paradigm	Supervised direct training
Training Method	Surrogate-gradient learning with backpropagation through time; direct training using AdamW and TET loss
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	TET loss
Training Hyperparameters	N-MNIST: learning rate 0.01, 100 epochs, batch size 32. DVS128: learning rate 0.01, 200 epochs, batch size 32. DVS-CIFAR10: learning rate 0.001, 300 epochs, batch size 64. CIFAR-10/100: learning rate 0.01, 400 epochs, batch size 64. Threshold 1.0 and decay factor 0.5 for all datasets
Primary Performance Metric	Classification accuracy and computational/energy efficiency
Primary Metric Value	N-MNIST: 99.77%. DVS128: 98.26% with 16 timesteps. DVS-CIFAR10: 84.90% with ESTSformer-4-384 and 16 timesteps. CIFAR-10: 95.66%. CIFAR-100: 78.50%
Secondary Metrics	Parameter count, FLOPs, energy consumption, attention energy, timestep sensitivity, temporal/spatial head allocation ratio
Baseline Models Compared	Slayer, NeuNorm, NaturalConvSNN, STSResNet, LIAF-Net, TA-SNN, tdBN, PLIF, SEW-ResNet, Spikformer, Spikingformer, Spikeformer, SD-Transformer, STS-Transformer, Chen et al. hybrid SNN
Performance Gain Over Baseline	On DVS-CIFAR10, ESTSformer-2-256 reaches 83.20% at 10 timesteps versus STS-Transformer 78.96%, improving by 4.24%. At 16 timesteps, it reaches 83.50% versus 79.93%, improving by 3.57%. On CIFAR-10 with 2-384, ESTSformer achieves 95.66% with lower reported energy than Spikformer and SD-Transformer
Statistical Validation	Benchmark comparison and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes ESTSA vs pure spatial vs pure temporal attention, comparison with STS-Transformer, temporal/spatial head allocation ratios, FLOPs/parameter comparison, and energy-consumption analysis
Training Framework / Code Availability	PyTorch and SpikingJelly; code availability NR
Target Hardware	Neuromorphic / low-power hardware motivation; theoretical energy analysis assumes 45 nm hardware operation costs
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Theoretical energy estimation using MAC = 4.6 pJ and AC = 0.9 pJ under 45 nm assumptions

Latency / Memory / Complexity Evidence	ESTSA reduces attention complexity from STSA's $O((NT)^2)$ style cost to $O(NT(N+T))$; on DVS-CIFAR10, ESTSformer-2-256 reduces FLOPs from 2.14G to 1.74G at T = 10 and from 3.83G to 2.79G at T = 16 compared with STS-Transformer; CIFAR-10 energy is reported as 0.191 mJ for ESTSformer versus 0.569 mJ for Spikformer and 0.301 mJ for SD-Transformer
Comparison with ANN Baseline	Partial. The main comparisons are against SNN and Spiking Transformer baselines; includes a hybrid SNN baseline on CIFAR datasets but no matched ANN Transformer baseline experiment
Main Contribution / Key Limitation	Contribution: proposes ESTSA and ESTSformer to efficiently capture temporal and spatial spike dependencies while reducing computation, storage, parameters, and theoretical energy. Limitation: no real neuromorphic hardware validation; energy is theoretical; experiments focus on classification tasks only
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ5, RQ1

P037

SpikeBERT: A language spikformer learned from BERT with knowledge distillation

Full Citation	Lv, C., Li, T., Qiao, W., Wang, X., Wu, M., Liu, W., Dou, S., Zheng, X., & Huang, X. (2026). SpikeBERT: A language spikformer learned from BERT with knowledge distillation. Neural Networks, 197, 108482. https://doi.org/10.1016/j.neunet.2025.108482
University	Fudan University
Country	China
Publication Year	2026
Publication Venue	Neural Networks
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neunet.2025.108482
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Natural Language Processing and Language Models
Specific Task	Text classification and language understanding using a spiking BERT/Spikformer-style model
Data Modality	Text/token sequences converted into spike representations
Signal Dimensionality	1D textual token sequences represented as spike tensors of shape T × L × D, where T is timestep, L is sentence length, and D is hidden dimension
Signal Characteristics	Discrete textual input is embedded into floating-point token embeddings, then converted into binary spike matrices through a convolution-based token encoder and spiking neuron layer
Dataset Name	Wikipedia 20220301.en, BookCorpus, Chinese Wikipedia dump, GLUE benchmark, SST-2, MRPC, RTE, QNLI, MNLI, QQP, CoLA, STS-B, MR, Subj, SST-5, ChnSenti, Waimai
Dataset Type	Public NLP pretraining corpus and public text classification / language understanding benchmark datasets
Dataset Size	ChnSenti: about 7000 Chinese hotel reviews; Waimai: about 12,000 food-delivery reviews; SST-5: 11,855 sentences; other dataset sizes NR in the main paper
Dataset Balance / Split	Uses standard benchmark splits; exact class balance NR
Hybrid Topology	Spikformer-based language model with convolution-based token encoder, stacked Spike Transformer encoder blocks, Not-Xor spiking self-attention, MLP blocks, and task-specific classification head
SNN Component	LIF neurons, binary spike activations, convolution-based spike token encoder, spiking self-attention, spike Transformer blocks
Neuron Model	Leaky Integrate-and-Fire
Transformer Component	Language-adapted Spikformer / SpikeBERT encoder blocks; BERT-like teacher model used for distillation; Not-Xor SSA replaces standard Spikformer SSA
Attention Type	Not-Xor Spiking Self-Attention; token-pair spiking self-attention; binary spike-based attention
SNN–Transformer Interface Mechanism	Text embeddings are converted into spike matrices using Conv2D plus spiking neuron layer; Q, K, and V are generated by linear projection, batch normalization, and spiking neuron layers; token similarity is computed using NOT-XOR instead of conventional dot-product attention
Model Depth / Scale	Main SpikeBERT uses 12 encoder blocks, timestep T = 4, sentence length 256. Ablation also evaluates different depths including 6, 12, and 18 layers
Parameter Count	109M parameters reported for SpikeBERT in energy comparison
Design / Contribution Type	New language Spikformer architecture and two-stage knowledge-distillation training method from BERT to SpikeBERT
Input Encoding Method	Token embedding followed by convolution-based token encoder; Conv2D output is passed through spiking neurons to produce binary spike tensor $X' \in \{0,1\}T\times L\times D$
Encoding Parameters	Timestep T = 4; sentence length = 256; common spiking neuron threshold Uthr = 1.0; SSA neuron threshold = 0.25; membrane decay β = 0.9; scaling factor τ = 0.125; surrogate α = 2
Encoding Learned or Fixed	Partially learned: token embeddings, convolutional token encoder, Q/K/V projections, and Transformer weights are learned; LIF spike dynamics and threshold settings are fixed hyperparameters
Event / Spike Representation	Binary spike activations with values 0/1; spike matrices Xs, Qs, Ks, Vs; time-varying spike trains across T timesteps
Temporal Resolution	4 timesteps in the main experiments; ablation tests 1, 2, 4, 8, and 12 timesteps
Output Decoding Method	Task-specific classification head over SpikeBERT output; exact pooling/readout formulation NR
Encoding / Temporal Information Analysis	Yes. The paper analyzes timestep effect, decay-rate effect, model-depth effect, attention maps, energy consumption, and feature alignment between ANN teacher and SNN student
Training Paradigm	Two-stage knowledge distillation plus supervised task-specific fine-tuning
Training Method	Stage 1: pre-training distillation from BERT using large unlabelled corpora and feature/embedding alignment. Stage 2: task-specific distillation from fine-tuned BERT using augmented labelled data, logits loss, cross-entropy loss, and feature/embedding alignment
Surrogate Gradient Function	Arctangent-like surrogate gradient
ANN-to-SNN Conversion Method	N/A; the model is trained through knowledge distillation rather than direct ANN-to-SNN conversion
Optimizer	AdamW
Loss Function	Stage 1: feature alignment loss + embedding alignment loss. Stage 2: feature alignment loss + embedding alignment loss + KL logits loss + cross-entropy loss

Training Hyperparameters	Stage 1: batch size 128, AdamW, learning rate 5e–4, weight decay 5e–3, σ1 = 1.0, σ2 = 1.0, 4 NVIDIA A100 GPUs. Stage 2: batch size 32, learning rate 5e–5, λ1 = 0.1, λ2 = 0.1, λ3 = 1.0, λ4 = 0.1, p_mask = 0.1, p_pos = 0.1, p_ng = 0.25, 4 NVIDIA RTX 3090 GPUs
Primary Performance Metric	Classification accuracy / GLUE task metrics and theoretical energy consumption
Primary Metric Value	Average score across reported tasks: SpikeBERT 67.27 versus SNN-TextCNN 54.56 and DT SpikeBERT 56.15. Energy consumption averages about 27.82% of fine-tuned BERT energy
Secondary Metrics	F1, Matthews correlation, Spearman correlation, FLOPs/SOPs, energy reduction, ablation accuracy, attention-map visualization
Baseline Models Compared	Fine-tuned BERT, DistilBERT, Q2BERT, SNN-TextCNN, LIF-BERT, directly trained SpikeBERT, RoBERTa teacher variant
Performance Gain Over Baseline	Compared with SNN-TextCNN, SpikeBERT improves average performance by 12.71 percentage points and up to 23.30% on some tasks. Compared with fine-tuned BERT, SpikeBERT reduces theoretical energy by about 70.49%–73.63% across reported text-classification datasets
Statistical Validation	Main results are averaged across 10 random seeds. Ablation studies are reported, but formal significance testing NR
Ablation Study	Full. Includes removing Stage 1, removing Stage 2, removing data augmentation, removing feature alignment, removing embedding alignment, removing logits loss, removing cross-entropy loss, timestep variation, depth variation, decay-rate variation, and teacher-model comparison
Training Framework / Code Availability	PyTorch-based SnnTorch and SpikingJelly; HuggingFace BERT teachers; code available at GitHub: Lvchangze/SpikeBERT
Target Hardware	Neuromorphic hardware motivation; theoretical evaluation assumes 45 nm neuromorphic hardware
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using MAC = 4.6 pJ and AC = 0.9 pJ under 45 nm assumptions; no measured hardware energy
Latency / Memory / Complexity Evidence	Reports FLOPs/SOPs and energy per sample. Example: SST-2 fine-tuned BERT uses 102.24 mJ, while SpikeBERT uses 28.54 mJ, giving 72.09% energy reduction. GPU memory limitation is discussed because SNNs add timestep dimension T
Comparison with ANN Baseline	Yes. Compares against fine-tuned BERT, DistilBERT, Q2BERT, and RoBERTa-teacher variants
Main Contribution / Key Limitation	Contribution: proposes SpikeBERT, a language Spikformer trained by two-stage BERT knowledge distillation, achieving much stronger NLP performance than previous SNN text models with substantially lower theoretical energy than BERT. Limitation: still has a performance gap on harder GLUE tasks, depends on BERT teacher quality, uses additional feature-alignment modules during training, lacks real neuromorphic hardware validation, and is limited by GPU memory due to timestep dimension
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P038

An energy-efficient spike-aware training architecture for spiking transformers

Full Citation	Ma, Y., Lin, Y., Li, M., Quan, P., Zhou, C., Jia, W., Zhu, X., Meng, Q., Zhou, H., & An, F. (2026). An energy-efficient spike-aware training architecture for spiking transformers. Microelectronics Journal, 172, 107136. https://doi.org/10.1016/j.mejo.2026.107136
University	Southern University of Science and Technology
Country	China
Publication Year	2026
Publication Venue	Microelectronics Journal
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.mejo.2026.107136
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Full BPTT-based training acceleration for Spiking Transformers, evaluated using Spikingformer training workload on ImageNet-1K
Data Modality	Static image data represented as spike-token sequences during Spikingformer training
Signal Dimensionality	2D image inputs unfolded into spatiotemporal spike tensors; input frame tensor I ∊ R(T×C×H×W), token sequence X ∊ R(T×N×D)
Signal Characteristics	ImageNet-1K input size 224 × 224; static image input uses C = 3; model uses temporal spike trains, membrane potentials, spike-gradient masks, FP16 activations/gradients, and binary spike tensors
Dataset Name	ImageNet-1K
Dataset Type	Public large-scale image classification benchmark
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Spikingformer training workload mapped to a spike-aware accelerator; model contains spiking tokenizer, multiple Spiking Transformer blocks, PSSA module, spiking MLP, global average pooling, and classification head
SNN Component	LIF neurons, binary spikes, spike-gradient masks, membrane potentials, SOMA operation, GRAD operation, spike-aware MM, spike-gated accumulation
Neuron Model	Leaky Integrate-and-Fire
Transformer Component	Spikingformer / Spiking Transformer blocks; pre-activation spiking self-attention; Conv1DBN Q/K/V layers; spiking MLP; classification head
Attention Type	Pre-activation Spiking Self-Attention; softmax-free spike-based QK■V attention
SNN–Transformer Interface Mechanism	Spiking tokenizer produces spike tokens; PSSA applies spike neuron layers before Q/K/V generation; Q, K, and V are binary spike tensors, while training architecture supports binary spike accumulation in FP/WG and FP16 dense MAC in BP
Model Depth / Scale	Analysis setting: 8 attention heads, patch size 14, 8 blocks, feature dimension 512, timestep T = 4, batch size 16, maximum bitwidth 16; MM architecture uses 64 × 64 rPE array
Parameter Count	Spikingformer model comparison reports 66.34M parameters
Design / Contribution Type	Dedicated spike-aware training architecture; spiking systolic array with reuse processing elements; unified SOMA/GRAD/RES resource-reuse module; FP/BP batch-normalization modules; cross-layer simulation framework; dataflow optimization for full FP/BP/WG training
Input Encoding Method	Static image input is encoded by the first layer of the spiking tokenizer into spike-based token sequences

Encoding Parameters	Timestep T = 4; input size 224 × 224; patch size 14; batch size 16; 1-bit spike operands in FP/WG; FP16 weights, gradients, activations, and outputs
Encoding Learned or Fixed	Partially learned: Spikingformer weights are trained; spike dynamics and accelerator dataflow/memory hierarchy are fixed architectural choices
Event / Spike Representation	Binary spike tensors, membrane potential U, spike S, spike gradient ∇S, potential gradient ∇U, spike-gradient mask, potential-gradient mask
Temporal Resolution	4 timesteps in the evaluated Spikingformer workload
Output Decoding Method	Global average pooling followed by fully connected classification head in Spikingformer
Encoding / Temporal Information Analysis	Yes. The paper explicitly models temporal spike trains, membrane potentials, spike-gradient masks, BPTT state storage, checkpointing/recomputation, spike sparsity, and gradient sparsity in architecture simulation
Training Paradigm	Supervised BPTT-based training of Spiking Transformers
Training Method	Backpropagation Through Time with FP, BP, and WG phases; spike-aware matrix multiplication, SOMA, GRAD, BN, and residual operations are mapped to the proposed architecture
Surrogate Gradient Function	Spike-gradient mask / surrogate derivative is modeled, but exact surrogate function NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	T = 4, batch size = 16, 8 heads, 8 blocks, dmodel = 512, patch size = 14, maximum bitwidth = 16; optimizer and learning-rate details NR
Primary Performance Metric	Training energy efficiency, throughput, power, latency, and dataflow energy/latency
Primary Metric Value	Optimized OS_C dataflow achieves estimated 3.4 TFLOPS at 1.44 W under 28 nm energy model and 500 MHz assumption, corresponding to 2.36 TFLOPS/W with 83% array utilization
Secondary Metrics	SRAM area/energy trade-off, DRAM/SRAM/register access energy, FP/BP/WG energy breakdown, latency breakdown, FPGA synthesis latency, DSP/BRAM/LUT/FF usage
Baseline Models Compared	SIGMA, SVLSI'20 AI processor, H2Learn, SATA, COSA, TRETA, ArXiv'25 SNN training architecture, Xpikeformer, VESTA, TPU, GPU V100; also compares ViT-B/16, Spikformer, and Spikingformer model energy on ImageNet-1K
Performance Gain Over Baseline	Reported 2.36 TFLOPS/W is 4.9× higher than SIGMA and 1.7× higher than SVLSI'20 under comparable discussion; also better than cited TPU 0.15 TFLOPS/W and GPU V100 0.053 TFLOPS/W baselines
Statistical Validation	Architecture-level simulation, dataflow exploration, CACTI-based energy modeling, and FPGA HLS synthesis of key modules; no statistical significance testing or repeated training accuracy runs reported
Ablation Study	Partial hardware/dataflow ablation. Includes nine dataflow schemes, OS_C selection, energy/latency breakdown across FP/BP/WG, SRAM energy-area trade-off, HLS validation for 16 × 16 and 32 × 32 MM arrays, and RRM sizes 16/32/64
Training Framework / Code Availability	Cross-layer simulation framework using spike-property simulator, ZigZag matrix-computing framework, latency model, CACTI, Allo HLS, Vitis HLS 2022.1, Xilinx VCU128 FPGA synthesis; code availability NR
Target Hardware	28 nm training accelerator architecture; 64 × 64 spike-aware systolic array; SRAM hierarchy with Spike SRAM, FP16 SRAM, Parameter SRAM; FPGA validation on Xilinx VCU128
Hardware Deployment Status	Architecture-level simulation plus FPGA HLS synthesis of representative MM and RRM modules; no full RTL implementation or fabricated ASIC reported
Energy Evidence	Estimated using 28 nm process energy model, CACTI memory characterization, per-operation energy modeling, and cross-layer simulation; not measured from silicon
Latency / Memory / Complexity Evidence	24 MB on-chip SRAM selected as Pareto point; 16 MB FP16 SRAM, 7 MB Parameter SRAM, 1 MB Spike SRAM; OS_C gives lowest energy and about 10%–28% latency reduction over other dataflows; HLS 32 × 32 MM prototype latency 1956 cycles; RRM 64-size latency 86 cycles
Comparison with ANN Baseline	Yes. Compares against ANN/Transformer accelerators and standard ViT-B/16 model energy; also compares with GPU/TPU training-efficiency baselines
Main Contribution / Key Limitation	Contribution: proposes a dedicated spike-aware training architecture and simulation framework for full Spiking Transformer training, supporting FP/BP/WG with spike-gated computation, FP16 gradients, BPTT state handling, and optimized OS_C dataflow. Limitation: results are mainly simulation/model-based; only key modules are FPGA-synthesized; no complete RTL, ASIC fabrication, or real end-to-end hardware training measurement yet
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4

P039

Spike-driven Incepformer: A hierarchical spiking transformer with inception-inspired feature learning

Full Citation	Meng, W., Huang, Z., Liu, Q., Cai, M., Chen, K., & Ma, L. (2025). Spike-driven Incepformer: A hierarchical spiking transformer with inception-inspired feature learning. Neurocomputing, 648, 130727. https://doi.org/10.1016/j.neucom.2025.130727
University	Wuhan University of Technology
Country	China
Publication Year	2025
Publication Venue	Neurocomputing
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neucom.2025.130727
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Static image classification and event-based vision classification
Data Modality	Static RGB images and neuromorphic/event-camera data
Signal Dimensionality	2D static images and 2D spatiotemporal event streams represented as spike sequences over timesteps
Signal Characteristics	Static datasets include 224 × 224 ImageNet images and 32 × 32 CIFAR images; neuromorphic datasets include CIFAR10-DVS and DVS128 Gesture event streams with 128 × 128 spatial resolution, downsampled to 64 × 64 in experiments
Dataset Name	ImageNet-1K, CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS128 Gesture

Dataset Type	Public static image and neuromorphic/event-based vision benchmark datasets
Dataset Size	ImageNet-1K: 1.28M training images and 50,000 validation images across 1000 classes; CIFAR-10/100: 50,000 training and 10,000 test images; CIFAR10-DVS: 10,000 samples, 1000 per class; DVS128 Gesture: 1342 samples from 11 gestures
Dataset Balance / Split	CIFAR10-DVS uses first 100 samples per category for testing and remaining 900 for training; ImageNet and CIFAR use standard splits; exact class-balance details NR
Hybrid Topology	Hierarchical fully spike-driven Transformer architecture with convolutional stem, progressive downsampling, SDFA attention blocks, Spiking Inception Modules replacing MLP/FFN, Diverse Branch Convolution before average pooling, and final classifier
SNN Component	LIF neurons, binary spike activations, membrane shortcut, spike-driven convolutional layers, SDFA, SIM, DBC, spike-based attention maps
Neuron Model	Leaky Integrate-and-Fire
Transformer Component	Hierarchical spike-driven Transformer encoder with SDFA attention and SIM feed-forward replacement
Attention Type	Spike-driven Feature Attention; feature-based binary spike attention; multi-head feature attention
SNN–Transformer Interface Mechanism	Input spikes are processed by feature convolution to generate feature encoding matrix E and feature decoding matrix D; SDFA multiplies original spike input S with feature matrices E and D using binary spike matrix operations, avoiding conventional Q/K/V self-attention
Model Depth / Scale	Six architectures: S, M, L, CIFAR, DVS, and DVSG. ImageNet variants use 3-stage hierarchy with channel settings S: 128/384/512, M: 256/512/768, L: 384/768/1024. CIFAR uses 64/256/384. DVS uses 32/96/192. Main timestep T = 4 for static datasets and T = 10/16 for neuromorphic datasets
Parameter Count	ImageNet: S = 16.07M, M = 30.96M, L = 57.33M. CIFAR model = 8.40M. CIFAR10-DVS model = 1.78M. DVS128 Gesture model = 1.02M
Design / Contribution Type	New spike-driven hierarchical Transformer architecture; SDFA attention; Spiking Inception Module; Diverse Branch Convolution; progressive downsampling; feature-convolution-based multi-scale attention
Input Encoding Method	Static images are encoded into spike form through the first convolutional/stem layer; neuromorphic event streams are downsampled and processed as spike/event inputs; feature convolution extracts multi-scale spike features for attention
Encoding Parameters	ImageNet input 224 × 224; CIFAR input 32 × 32; CIFAR10-DVS and DVS128 Gesture downsampled to 64 × 64; ImageNet/CIFAR timestep T = 4; CIFAR10-DVS T = 10 or 16; DVS128 Gesture T = 16; feature convolution uses p and 2p branches
Encoding Learned or Fixed	Partially learned: stem convolution, feature convolution, SDFA, SIM, DBC, and classifier are learned; LIF spike-generation dynamics and timestep settings are fixed hyperparameters
Event / Spike Representation	Binary spike tensors; original spike input S; feature encoding matrix E; feature decoding matrix D; spike attention map; binary spike activations across timesteps
Temporal Resolution	T = 4 for static image experiments; T = 10 and T = 16 for CIFAR10-DVS; T = 16 for DVS128 Gesture; DVS128 Gesture has microsecond-level temporal resolution
Output Decoding Method	Diverse Branch Convolution followed by average pooling and linear classifier
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike-driven binary computation, information loss from final neuron layers, timestep effects, feature-attention maps using Grad-CAM, SOPs, firing-based energy, and SDFA/SDSA stability
Training Paradigm	Supervised direct training
Training Method	Direct SNN training with surrogate-gradient-based learning; TET used for neuromorphic datasets
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	SoftTargetCrossEntropy for training; CrossEntropy for testing; TET loss adjustment used for CIFAR10-DVS and DVS128 Gesture
Training Hyperparameters	ImageNet: batch size 64, lr 0.001, weight decay 0.01, 300 epochs, CosineAnnealingLR. CIFAR-10/100: batch size 128, lr 0.002, weight decay 0.06, 400 epochs. CIFAR10-DVS: batch size 16, lr 0.01, StepLR, 240 epochs, TET. DVS128 Gesture: batch size 32, lr 0.001, 200 epochs, TET. Minimum lr 1e–5 and 3 warm-up epochs for cosine scheduler settings
Primary Performance Metric	Top-1 classification accuracy and theoretical energy consumption
Primary Metric Value	ImageNet-1K: 80.41% with Spike-driven Incepformer-L. CIFAR-10: 96.26%. CIFAR-100: 80.37%. CIFAR10-DVS: 84.1% at T = 16. DVS128 Gesture: 98.6%
Secondary Metrics	Parameter count, SOPs, theoretical power/energy, timestep sensitivity, attention-map visualization, ablation accuracy
Baseline Models Compared	ResNet, ViT, Swin Transformer, CrossFormer++, SEW-ResNet, MS-ResNet, Spikformer, Spike-driven Transformer, SpikingResformer, Meta-SpikeFormer, SWformer, STBP-tdBN, DSR, Dspike, TIM
Performance Gain Over Baseline	On ImageNet-1K, Incepformer-L reaches 80.41%, exceeding Meta-SpikeFormer-8-512’s 80.0% while using about half the reported energy. On CIFAR-100, it improves over SWformer by 1.07%. On CIFAR10-DVS, it reaches 84.1% with 1.78M parameters, outperforming SWformer’s 83.9%
Statistical Validation	Benchmark comparison and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or formal statistical significance test reported
Ablation Study	Full. Includes feature convolution sensitivity, SDFA versus SDSA, with/without DBC, timestep analysis, SDFA attention-map visualization, and SIM versus alternative FFN/convolution structures
Training Framework / Code Availability	PyTorch and SpikingJelly; code stated to be available at GitHub: 2ephyrus/SDIncepformer
Target Hardware	Neuromorphic / low-energy spike-driven hardware motivation; theoretical energy assumes 45 nm hardware operation costs
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using SOPs/FLOPs with 45 nm assumptions: MAC = 4.6 pJ and AC = 0.9 pJ; no measured hardware energy
Latency / Memory / Complexity Evidence	Reports parameters, SOPs, and theoretical power/energy. ImageNet examples: Incepformer-S 9.41G SOPs and 8.47 mJ; Incepformer-M 19.20G SOPs and 17.28 mJ; Incepformer-L 39.19G SOPs and 35.27 mJ. Latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares against ANN baselines including ResNet, ViT, Swin Transformer, CrossFormer++, and ANN Transformer-4-384 on CIFAR
Main Contribution / Key Limitation	Contribution: proposes Spike-driven Incepformer, combining SDFA, SIM, DBC, and progressive hierarchical downsampling to improve spike-driven Transformer accuracy while balancing parameters and theoretical energy. Limitation: energy is theoretical rather than hardware-measured; no real neuromorphic deployment; further work is needed on high-performance low-power spiking attention mechanisms
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P040

Neural-Transformer: A brain-inspired lightweight mechanical fault diagnosis method under noise

Full Citation	Wang, C., Tian, B., Yang, J., Jie, H., Chang, Y., & Zhao, Z. (2024). Neural-Transformer: A brain-inspired lightweight mechanical fault diagnosis method under noise. Reliability Engineering & System Safety, 251, 110409. https://doi.org/10.1016/j.ress.2024.110409
University	Harbin Institute of Technology
Country	China
Publication Year	2024
Publication Venue	Reliability Engineering & System Safety
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.ress.2024.110409
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Industrial Monitoring, Fault Diagnosis, and Infrastructure Inspection
Specific Task	Fault classification under clean and noisy conditions
Data Modality	Vibration signals transformed into 2D time-frequency representations using Frequency-Slice Wavelet Transform
Signal Dimensionality	1D vibration time series converted into 2D time-frequency images, then represented as spike-form patch sequences
Signal Characteristics	UConn gearbox signals sampled at 20 kHz with 3600 sampling points per sample; Lab bearing signals sampled at 20 kHz with 1024 points per sample; REA waterjet propulsion samples contain 1024 points and multiple operating speeds
Dataset Name	UConn gearbox dataset; Lab bearing dataset; REA marine waterjet propulsion machinery dataset
Dataset Type	One public gearbox dataset, one laboratory bearing dataset, and one real engineering application dataset
Dataset Size	UConn: 9 gear conditions, 104 samples per fault type. Lab: 1400 samples. REA: 4 impeller states, 300 samples per state, 3 rotary speeds.
Dataset Balance / Split	UConn has 104 samples per class. Lab dataset is divided into training/validation/testing using 7:2:1 split. REA exact split NR
Hybrid Topology	Neural-Transformer with FSWT preprocessing, three SMST layers, spike-form relative position embedding, stacked Neural-Transformer encoder blocks, MHSSSA attention, MLP layer, global average pooling, and classification head
SNN Component	LIF spike neurons, spike neuron layers, spike-form patches, spike-form relative position embedding, spike-based Q/K/V, pulse-form attention
Neuron Model	Leaky Integrate-and-Fire
Transformer Component	Transformer encoder-style blocks with Multi-Head Spatiotemporal Spiking Self-Attention and MLP
Attention Type	Multi-Head Spatiotemporal Spiking Self-Attention; non-softmax spike-form attention
SNN–Transformer Interface Mechanism	FSWT converts vibration signals into 2D time-frequency images; SMST projects images into spike-form patch features; linear + BN + spike neuron layers generate spike-form q, k, and v; MHSSSA performs multi-head spatiotemporal spike attention before MLP and classification
Model Depth / Scale	Neural-Transformer variants use 4, 6, or 8 encoder blocks. Architecture uses three SMST layers with output channels 32, 64, and 128; final feature size 128 × 32 × 32 before encoder blocks
Parameter Count	Encoder Block 4: 3.08M; Encoder Block 6: 3.94M; Encoder Block 8: 4.23M
Design / Contribution Type	New lightweight brain-inspired fault diagnosis model; FSWT preprocessing; Separable Multiscale Spiking Tokenizer; Multi-Head Spatiotemporal Spiking Self-Attention; spatiotemporal interaction mechanism for noise robustness
Input Encoding Method	Frequency-Slice Wavelet Transform converts vibration signals to 2D time-frequency images; SMST converts time-frequency images into spike-form features
Encoding Parameters	Input size 3 × 256 × 256; SMST uses 3 × 3, 5 × 5, and 7 × 7 separable convolution kernels; max pooling 2 × 2 with stride 2; main timestep T = 4
Encoding Learned or Fixed	Partially learned: FSWT preprocessing is fixed; SMST convolutional tokenizer, spike-form features, MHSSSA, MLP, and classifier are learned
Event / Spike Representation	Spike-form patch sequence x ∊ R(T×N×D); spike-form q, k, v; binary pulse features generated by LIF neurons
Temporal Resolution	Main SNN timestep T = 4; timestep sensitivity tested with T = 1, 2, 4, 6, and 8
Output Decoding Method	Global Average Pooling followed by classification head for fault-category prediction
Encoding / Temporal Information Analysis	Yes. The paper analyzes FSWT time-frequency representation, spatiotemporal spike interaction, timestep influence, noise robustness, and attention behavior under noisy vibration signals
Training Paradigm	Supervised learning
Training Method	Direct training of Neural-Transformer with spike neuron layers; training uses labelled fault classes and 95% confidence interval evaluation over ten runs
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam selected in best hyperparameter setting; SGD also tested in sensitivity analysis
Loss Function	NR
Training Hyperparameters	Best reported hyperparameter combination: learning rate 1e–3, kernel sizes 3/5/7, Adam optimizer, timestep T = 4. Hardware/software: Python 3.7, Intel i7-11800H CPU, NVIDIA RTX 3050 GPU
Primary Performance Metric	Diagnostic accuracy, F1 score, theoretical energy consumption
Primary Metric Value	UConn best: 99.72% accuracy and 99.25% F1 with 8 blocks. Lab best: 99.16% accuracy and 99.03% F1 with 8 blocks. REA best: 93.14% accuracy and 92.68% F1 with 8 blocks
Secondary Metrics	Parameter count, FLOPs, SOPs, inference time, confidence interval, robustness accuracy under SNR noise, timestep-energy trade-off
Baseline Models Compared	ViT, VGG, ResNet, DSRSN, SDRN-ResNet34, SDRN-ResNet50, SpikingFormer
Performance Gain Over Baseline	On UConn, Neural-Transformer-8 reaches 99.72%, higher than ViT 99.17% and SpikingFormer 99.12%. On Lab, it reaches 99.16%, higher than SpikingFormer 98.40%. On REA, it reaches 93.14%, higher than DSRSN 92.47%, SDRN-ResNet50 92.53%, and SpikingFormer 91.97%
Statistical Validation	Results are reported using 95% confidence intervals over ten runs; noise robustness is evaluated under 6 dB, 2 dB, 0 dB, –2 dB, and –6 dB SNR

Ablation Study	Full. Includes FSWT vs CWT vs STFT, SMC2d ablation, MHSSSA ablation, spatiotemporal interaction ablation under noise, timestep sensitivity, and hyperparameter sensitivity
Training Framework / Code Availability	Python 3.7 environment; code availability NR
Target Hardware	Energy-efficient neuromorphic/edge diagnosis motivation; no specific hardware chip deployed
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation based on SOPs; each SOP is assumed to consume 77 fJ. No measured chip or FPGA energy reported
Latency / Memory / Complexity Evidence	Reports inference time, FLOPs, SOPs, parameters, and energy. Example REA: Neural-Transformer-4 uses 3.08M parameters, 0.63G FLOPs, 3.45G SOPs, 0.27 mJ, and 1.6 s inference time; Neural-Transformer-8 uses 4.23M parameters, 0.67 mJ, and 2.36 s inference time
Comparison with ANN Baseline	Yes. Compares with ANN baselines including ViT, VGG, and ResNet
Main Contribution / Key Limitation	Contribution: proposes Neural-Transformer, a lightweight spiking Transformer fault diagnosis model using FSWT, SMST, and MHSSSA to improve noise-robust mechanical diagnosis with low theoretical energy. Limitation: energy is theoretical only; no real neuromorphic hardware validation; future work is needed for cross-domain diagnosis in practical engineering
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P041

Spiking neural P models with dynamic potential reset mechanism for remote sensing image classification

Full Citation	Yan, J., Xiao, J., & Peng, H. (2026). Spiking neural P models with dynamic potential reset mechanism for remote sensing image classification. Neurocomputing, 679, 133235. https://doi.org/10.1016/j.neucom.2026.133235
University	Xihua University
Country	China
Publication Year	2026
Publication Venue	Neurocomputing
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neucom.2026.133235
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Remote Sensing, Geospatial, and Environmental Monitoring
Specific Task	Remote sensing scene classification on UCM, RSSCN7, and AID datasets
Data Modality	Remote sensing RGB image data converted into spike sequences
Signal Dimensionality	2D remote sensing images represented as time-encoded spiking patch sequences
Signal Characteristics	UCM images are 256 × 256 with 0.3 m spatial resolution; RSSCN7 images are 400 × 400; AID images are approximately 600 × 600 with spatial resolution from 0.5 m to 8 m
Dataset Name	UCM, RSSCN7, AID
Dataset Type	Public remote sensing scene classification benchmark datasets
Dataset Size	UCM: 2100 images, 21 classes, 100 images per class. RSSCN7: 2800 images, 7 classes, 4 scales, 100 images per scale. AID: about 10,000 images, 30 scene classes, 220–420 images per class
Dataset Balance / Split	Each dataset is split into 80% training and 20% testing; UCM and RSSCN7 have balanced class/sample descriptions; AID has 220–420 images per class
Hybrid Topology	NSNP-Former architecture with Spiking Patch Splitting, Spiking Position Embedding, stacked NSNP-Former encoder blocks, NSNP-SSA, ConvTC-NSNP block, and classification head
SNN Component	TC-NSNP neurons, spike sequences, dynamic membrane potential reset, spiking patch splitting, spiking position embedding, ConvTC-NSNP feature block
Neuron Model	TC-NSNP neuron with dynamic membrane potential reset mechanism inspired by nonlinear spiking neural P systems
Transformer Component	Transformer-like NSNP-Former encoder; NSNP-SSA attention block; Q/K/V spiking attention structure
Attention Type	NSNP-like Spiking Self-Attention; nonlinear spiking self-attention without softmax
SNN–Transformer Interface Mechanism	Remote sensing images are converted into spiking patch sequences by SPS; SPE adds spike-form positional information; NSNP-SSA generates spike-form Q, K, and V using Conv + BN + TC-NSNP neurons, then applies nonlinear spiking interaction and TC-NSNP restoration
Model Depth / Scale	NSNP-Former encoder uses N = 8 stacked blocks; SPS uses four Conv2D + BN + TC-NSNP + MaxPool2D blocks; experiments evaluate time steps T = 2, 4, and 8
Parameter Count	NR
Design / Contribution Type	New TC-NSNP neuron model with dynamic potential reset; new NSNP-SSA module; new NSNP-Former model for low-latency remote sensing image classification
Input Encoding Method	Time encoding using TC-NSNP neurons; SPS progressively downsamples image features and converts them into spike patch sequences
Encoding Parameters	Time steps tested: 2, 4, and 8; SPS uses four convolution-pooling stages; SPE uses Conv2D kernel size 3; NSNP-Former encoder block count N = 8; exact threshold values NR
Encoding Learned or Fixed	Partially learned: convolutional layers, NSNP-SSA, ConvTC-NSNP, and classifier are learned; TC-NSNP dynamic reset rule is fixed by the proposed neuron formulation
Event / Spike Representation	Time-encoded spike sequences; sparse spike-form Q, K, and V; membrane potential states with dynamic reset; standard spike outputs after TC-NSNP transformation
Temporal Resolution	Low-latency inference with T = 2, 4, and 8 time steps; main low-latency claims emphasize T = 2
Output Decoding Method	Classification head after stacked NSNP-Former encoder blocks
Encoding / Temporal Information Analysis	Yes. The paper analyzes TC-NSNP temporal dynamics, dynamic reset behavior, low-time-step performance, convergence curves, and neuron/reset ablations
Training Paradigm	Supervised learning

Training Method	Direct training of NSNP-Former using PyTorch and SpikingJelly; exact surrogate-gradient training details NR
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	NR
Training Hyperparameters	Data augmentation includes random scaling/cropping, color dithering, vertical flipping, and random rotation; 80/20 train-test split; Adam optimizer; implemented on A40 server; time steps T = 2, 4, 8
Primary Performance Metric	Classification accuracy
Primary Metric Value	UCM: 99.76% at T = 4 and T = 8; RSSCN7: 98.04% at T = 4 and 96.96% at T = 2; AID: 97.35% at T = 2
Secondary Metrics	Time step, training loss convergence, ablation accuracy, low-latency performance
Baseline Models Compared	AleNet+MSCP, VGG_VD16+MSCP, Scenario I/II, DCA, VGG_VD16+IFK, DNN-SNN, MD-R-SNN-ResNet-34, SNN-VGG-15, S-ResNet18, S-ResNet34, VGG_VD16, CaffeNet, GoogleNet, DBN, FV+HCV, VGG-S+SVM, VGG-15, Fusion+concatenation
Performance Gain Over Baseline	On UCM, NSNP-Former reaches 99.76%, outperforming all listed baselines. On RSSCN7, it reaches 96.96% at T = 2 and 98.04% at T = 4, but remains below S-ResNet34's 99.00% at T = 4. On AID, it reaches 97.35% at T = 2, but remains below S-ResNet34's 98.81% at T = 4
Statistical Validation	Benchmark comparisons and ablation experiments are reported; no standard deviation, confidence interval, repeated-run analysis, or formal significance test reported
Ablation Study	Clear ablation study. Includes NSNP-SSA versus standard SSA, TC-NSNP versus IF/LIF neuron variants, hard reset versus soft reset versus dynamic reset, and convergence analysis
Training Framework / Code Availability	PyTorch and SpikingJelly; hardware platform: A40 server; code availability NR; data availability states the data used are confidential
Target Hardware	Resource-constrained and low-power deployment motivation; no specific neuromorphic chip, FPGA, or edge hardware evaluated
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	NR; the paper discusses low-power and computational efficiency qualitatively but does not report measured or theoretical energy consumption values
Latency / Memory / Complexity Evidence	Reports low-latency time-step performance and states NSNP-SSA reduces attention complexity to $O(Nd^2)$; no measured latency, memory footprint, FLOPs, SOPs, or energy table reported
Comparison with ANN Baseline	Yes. Compares with conventional ANN/CNN remote-sensing classifiers such as VGG, GoogleNet, CaffeNet, DCA, and related feature-fusion CNN methods
Main Contribution / Key Limitation	Contribution: proposes TC-NSNP neurons with dynamic reset and builds NSNP-Former with NSNP-SSA for low-time-step remote sensing classification. Limitation: no hardware or energy measurement, no parameter/latency table, and performance does not surpass the strongest S-ResNet baseline on RSSCN7 and AID
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4

P042

A novel energy-efficient spike transformer network for depth estimation from event cameras via cross-modality knowledge distillation

Full Citation	Zhang, X., Han, L., Davies, S., Sobeih, T., Han, L., & Dancey, D. (2025). A novel energy-efficient spike transformer network for depth estimation from event cameras via cross-modality knowledge distillation. Neurocomputing, 658, 131745. https://doi.org/10.1016/j.neucom.2025.131745
University	Manchester Metropolitan University
Country	United Kingdom
Publication Year	2025
Publication Venue	Neurocomputing
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neucom.2025.131745
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Monocular dense depth estimation from event-camera spike streams
Data Modality	Event-camera spike streams, paired RGB images for teacher distillation, and dense depth/disparity maps
Signal Dimensionality	2D spatiotemporal event streams represented as spike-frame sequences; dense pixel-wise depth map output
Signal Characteristics	DENSE spike streams contain 128 binary spike frames of size 346 × 260. DSEC samples contain 16 spike frames of size 480 × 640. Event cameras encode temporal brightness changes as asynchronous binary spikes
Dataset Name	DENSE spike dataset; DSEC dataset
Dataset Type	Synthetic event-camera depth dataset and real-world stereo/event-camera driving dataset
Dataset Size	DENSE spike: 8 sequences, 5 for training and 3 for evaluation; each sequence has 999 samples. DSEC: 41 sequences; 29 sequences for training and 12 for evaluation; each sequence has 200–900 samples
Dataset Balance / Split	DENSE: 5 training sequences and 3 evaluation sequences. DSEC: 70% training sequences and 30% evaluation sequences. Exact scene/category balance NR
Hybrid Topology	Spike-Driven Transformer backbone with Spiking Patch Embedding, 4 Spiking Transformer blocks, Spiking Self-Attention, Spiking MLP, hybrid Fusion Depth Estimation Head, and DINOv2-based cross-modality knowledge distillation
SNN Component	Multi-step LIF neurons, spike-form patches, binary spike Q/K/V, spike-driven residual connections, Spiking MLP, spike-based feature extraction
Neuron Model	Multi-step Leaky Integrate-and-Fire
Transformer Component	Spike-Driven Transformer backbone; Spiking Transformer blocks; Spiking Self-Attention; Transformer-stage feature fusion
Attention Type	Spiking Self-Attention; softmax-free spike-based attention using binary Q, K, and V
SNN–Transformer Interface Mechanism	Event sequence is converted into spike-form patches by ConvBN, MaxPool, and MLIF. Spiking Transformer blocks generate binary Q/K/V through ConvBN and MLIF, apply SSA, and feed multi-stage features into the fusion depth head

Model Depth / Scale	Backbone uses 4 Spiking Transformer blocks with embedding dimension 384; event patch sequence has N = H/8 × W/8 tokens; fusion head combines features F1–F4 from multiple Transformer stages
Parameter Count	Proposed model: 20.55M parameters
Design / Contribution Type	First Transformer-based SNN for event-camera depth estimation; proposes spike-driven Transformer backbone, hybrid fusion depth estimation head, and single-stage cross-modality knowledge distillation from DINOv2
Input Encoding Method	Event streams are processed through Spiking Patch Embedding using ConvBN, MaxPool, and MLIF to produce spike-form patches
Encoding Parameters	DENSE spike input: 128 spike frames, 346 × 260. DSEC input: 16 spike frames, 480 × 640. Patch embedding downsamples to H/8 × W/8. ConvBN uses 3 × 3 convolution with stride 1. Backbone embedding dimension 384
Encoding Learned or Fixed	Partially learned: ConvBN patch embedding, Transformer blocks, SSA, MLP, and fusion head are learned; MLIF neuron dynamics and spike-frame representation are fixed/model-defined
Event / Spike Representation	Binary spike frames; spike-form patches $X \in R(T \times N \times D)$; binary Q, K, V spike tensors with values 0/1; firing-rate-based sparse activations
Temporal Resolution	DENSE uses 128 spike frames generated at 128 × 30 FPS; DSEC uses 16 spike frames per sample
Output Decoding Method	Hybrid fusion depth estimation head progressively upsamples and fuses multi-stage Transformer features, then applies sigmoid to generate dense depth map
Encoding / Temporal Information Analysis	Yes. The paper analyzes event-stream temporal properties, mean firing rate, spike-based computation, multi-stage feature fusion, low-light depth examples, and energy consumption from spike sparsity
Training Paradigm	Supervised depth-estimation training with cross-modality knowledge distillation
Training Method	Direct SNN training with single-stage knowledge distillation from frozen DINOv2 teacher; combines student event features with RGB teacher features and ground-truth depth supervision
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Feature perceptual loss between teacher and student features plus scale-invariant L2 depth loss
Training Hyperparameters	NVIDIA RTX A6000 GPU; 200 epochs; batch size 4; AdamW optimizer; learning rate 1e–5; weight decay 0.05; cosine annealing schedule
Primary Performance Metric	Depth-estimation error metrics and theoretical energy consumption
Primary Metric Value	DENSE: Abs Rel 0.80, Sq Rel 8.32, RMS log 0.17, SI log 0.46, $\delta<1.25$ = 0.53, energy 12.43 mJ. DSEC: Abs Rel 1.000, Sq Rel 0.999, RMS log 0.105, SI log 0.212, $\delta<1.25$ = 0.387
Secondary Metrics	Parameter count, mean firing rate, $\delta<1.25^2$, $\delta<1.25^3$, visual depth quality, energy reduction, model-size reduction
Baseline Models Compared	U-Net, E2Depth, Spike-T, Linear FCN Head variant, proposed model without knowledge distillation
Performance Gain Over Baseline	Compared with Spike-T on DENSE, Abs Rel is reduced from 1.57 to 0.80 and Sq Rel from 39.77 to 8.32. Energy is reduced from 41.77 mJ to 12.43 mJ, and parameters are reduced from 35.68M to 20.55M
Statistical Validation	Benchmark evaluation on synthetic and real-world datasets plus ablation studies; no standard deviation, confidence interval, repeated-run analysis, or formal statistical significance test reported
Ablation Study	Clear ablation study. Includes Linear FCN head versus fusion head, full model without knowledge distillation, and full proposed model with fusion head and knowledge distillation
Training Framework / Code Availability	Source code and pretrained weights stated to be made publicly available at GitLab: han-research/spike-transformer-for-depth
Target Hardware	Neuromorphic and resource-constrained deployment motivation; future deployment on platforms such as SpiNNaker, BrainScaleS, or TrueNorth discussed
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using 45 nm assumptions: MAC = 4.6 pJ and AC = 0.9 pJ. Proposed model reports 12.43 mJ per inference
Latency / Memory / Complexity Evidence	Proposed model has 20.55M parameters and mean firing rate 0.35. Energy is 12.43 mJ versus U-Net 72.93 mJ, E2Depth 59.25 mJ, and Spike-T 41.77 mJ. Measured runtime latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares with ANN/CNN baselines U-Net and E2Depth, and uses DINOv2 ANN foundation model as teacher
Main Contribution / Key Limitation	Contribution: proposes an energy-efficient spike-driven Transformer for event-camera depth estimation, combining SSA, spike-driven residual learning, fusion depth head, and DINOv2 cross-modality knowledge distillation. Limitation: the fusion head is not fully spike-based, energy evidence is theoretical rather than hardware-measured, and performance depends heavily on DINOv2 knowledge distillation
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P043/110

SGSSA: Spatio-temporal granular-threshold spiking self-attention for spiking neural networks

Full Citation	Zhang, H., Geng, L., Yang, G., & Zheng, Y. (2026). SGSSA: Spatio-temporal granular-threshold spiking self-attention for spiking neural networks. Neurocomputing, 678, 132988. https://doi.org/10.1016/j.neucom.2026.132988
University	Tianjin University of Technology and Education
Country	China
Publication Year	2026
Publication Venue	Neurocomputing
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neucom.2026.132988
Publication Type	Journal article
Database Source	ScienceDirect
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Event-stream classification on gesture, digit, and object-recognition datasets
Data Modality	Event-camera streams converted into frame-based spike/event representations

Signal Dimensionality	2D event frames with temporal sequence dimension; event tensors with polarity channels over multiple time steps
Signal Characteristics	Event streams are captured at 128 × 128 spatial resolution with two channels representing positive and negative polarity events; event streams are partitioned into fixed temporal windows and accumulated into frame sequences
Dataset Name	DVS-Gesture, MNIST-DVS, CIFAR10-DVS
Dataset Type	Public event-camera / neuromorphic vision benchmark datasets
Dataset Size	DVS-Gesture: 11 gesture classes, 1176 training samples, 288 test samples. MNIST-DVS: 30,000 data points across 3 scales, with scale 4 used. CIFAR10-DVS: 10,000 event sequences across 10 object categories
Dataset Balance / Split	DVS-Gesture uses default train/test split. MNIST-DVS uses 9:1 train-test split. CIFAR10-DVS uses 8:2 train-test split with balanced class distribution
Hybrid Topology	LIF-SGSSA module integrated into Conv-SNN and Meta-SpikeFormer backbones; SGSSA acts as a plug-and-play spatiotemporal attention/remapping unit before attention heads or within convolutional SNN layers
SNN Component	LIF neurons, multi-threshold spiking neuron layer, spike tensors, membrane potential modulation, decay-factor modulation, voting layer, rate-based decoding
Neuron Model	Leaky Integrate-and-Fire with SGSSA-based temporal modulation
Transformer Component	Meta-SpikeFormer backbone; Transformer-based SNN blocks; Spike-Driven Self-Attention branch enhanced by LIF-SGSSA
Attention Type	Granular-Threshold Spiking Self-Attention; Spatio-Temporal Granular-Threshold Spiking Self-Attention; Hadamard-based spike attention; cyclic summation attention
SNN–Transformer Interface Mechanism	In Meta-SpikeFormer, LIF-SGSSA is inserted before attention heads in Transformer-based SNN blocks. It aggregates spike features from recent historical time steps, produces a modulation tensor, and uses it to enhance spike features before Q/K/V generation in the original attention block
Model Depth / Scale	DVS-Gesture Conv-SGSSA uses three convolutional blocks and a voting layer. CIFAR10-DVS uses MetaSpikeFormer-style hierarchical backbone with MS_ConvBlock and MS_Block stages. Default timestep T = 20 across datasets
Parameter Count	DVS-Gesture Conv-SGSSA: 2.49M. CIFAR10-DVS MetaSpikeFormer-SGSSA: 0.89M. MNIST-DVS parameter count NR
Design / Contribution Type	New GSSA mechanism; multi-threshold spiking neuron layer; group-wise cyclic fusion; LIF-SGSSA temporal modulation module; joint membrane-potential and decay-factor injection
Input Encoding Method	Frame-based event preprocessing: event streams are partitioned into fixed time windows Δt, and events are accumulated by pixel location and polarity to form evenly spaced input frames
Encoding Parameters	Default T = 20. Δt: 125 ms for DVS-Gesture, 25 ms for MNIST-DVS, 50 ms for CIFAR10-DVS. Sequence length L: 2500 ms, 500 ms, and 1000 ms respectively. Thresholds include Vth values 0.75, 0.8, and 0.7, with granular thresholds Vth1 = 0.35, Vth2 = 0.45, and Vth3 = 0.65 / 0.55 depending on dataset
Encoding Learned or Fixed	Partially learned: convolutional/backbone weights and attention parameters are learned; frame-window preprocessing, timestep settings, and selected threshold values are fixed hyperparameters
Event / Spike Representation	Binary spike tensors, spiking feature maps, positive/negative polarity event frames, multi-threshold spike tensors Q1…Qm, historical spike-feature maps over a temporal window
Temporal Resolution	20 simulation time steps; event windows use Δt = 125 ms, 25 ms, or 50 ms depending on dataset
Output Decoding Method	Rate-based voting layer; each class is represented by multiple neurons, and classification is based on the highest spike firing rate over time
Encoding / Temporal Information Analysis	Yes. The paper analyzes quantization error from spike binarization, multi-threshold error suppression, temporal evidence aggregation, membrane-potential and decay-factor modulation, Poisson noise robustness, and spatiotemporal feature visualization
Training Paradigm	Supervised direct training
Training Method	Spatio-temporal backpropagation using surrogate-gradient approximation; same STBP training pipeline is used for Conv-SGSSA and MetaSpikeFormer-SGSSA
Surrogate Gradient Function	Arctan-based surrogate function with slope parameter α = 2
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Mean squared error loss over rate-based voting outputs; MetaSpikeFormer variant uses the classification loss of its output voting layer
Training Hyperparameters	DVS-Gesture: 100 epochs, batch size 4, learning rate 0.0005, T = 20. MNIST-DVS: 150 epochs, batch size 32, learning rate 0.001, T = 20. CIFAR10-DVS: 200 epochs, batch size 64, learning rate 0.0003, T = 20. Exponential decay strategy used
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	DVS-Gesture: 99.3%. MNIST-DVS: 99.3%. CIFAR10-DVS: 85.4%
Secondary Metrics	Average accuracy, parameter count, Ops, inference latency, noise robustness, confusion matrix, Top-1 accuracy under ablation settings
Baseline Models Compared	Evs-B, SSFE, SCTFA, LIAF-Net, Spikformer, Spike-driven Transformer, TCJA, STSC, QKFormer, MetaSpikeFormer baseline, STFA, CTFA, SCTFA
Performance Gain Over Baseline	On CIFAR10-DVS, MetaSpikeFormer-SGSSA improves over MetaSpikeFormer baseline from 81.8% to 85.4%, a +3.6 percentage-point gain. On DVS-Gesture, Conv-SGSSA improves over Conv-BL from 96.5% to 99.3%. On MNIST-DVS, Conv-SGSSA improves over Conv-BL from 98.4% to 99.3%
Statistical Validation	Benchmark comparison, component ablation, within-component ablation, confusion matrices, and noisy-event robustness tests are reported; no standard deviation, confidence interval, or formal statistical significance test reported
Ablation Study	Full. Includes BL, STFA, CTFA, SCTFA, and SGSSA comparisons; number of thresholds m = 1, 3, 5; fusion pathway V-only, D-only, and V+D; Conv-SNN and MetaSpikeFormer backbones; noise robustness under Poisson noise
Training Framework / Code Availability	PyTorch; NVIDIA 3080 GPU for DVS-Gesture and MNIST-DVS; dual NVIDIA 3090 GPUs for CIFAR10-DVS; code availability NR
Target Hardware	Neuromorphic / real-time event-based computing motivation; no specific hardware platform deployed
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No measured hardware energy and no mJ-level energy table reported. Efficiency is evaluated indirectly through AC-dominant operations, Ops, parameters, latency, spike sparsity, and reduced MAC use
Latency / Memory / Complexity Evidence	DVS-Gesture Conv-SGSSA: 2.49M parameters, 0.287G Ops, 5.70 ms latency. CIFAR10-DVS MetaSpikeFormer-SGSSA: 0.89M parameters, 0.135G Ops, 0.81 ms latency. Compared with baselines, SGSSA adds moderate overhead but improves accuracy
Comparison with ANN Baseline	Mostly no. The main experiments compare SNN and spiking-attention baselines; ANN comparison is discussed conceptually but not evaluated as a matched experimental baseline

Main Contribution / Key Limitation	Contribution: proposes SGSSA/LIF-SGSSA, a multi-threshold spiking self-attention and temporal modulation mechanism that reduces quantization error, improves spatiotemporal modeling, and boosts event-stream classification accuracy. Limitation: added attention complexity increases latency/operations; no measured hardware energy; evaluation is limited to event-stream classification rather than detection, segmentation, or tracking
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ5, RQ1

P044

A Hybrid Spiking-Attention Transformer Model for Robust and Efficient Speech Emotion Recognition on Multi-Dataset Benchmarks

Full Citation	Ali, S. A., & Abbas, J. M. (2025). A Hybrid Spiking-Attention Transformer Model for Robust and Efficient Speech Emotion Recognition on Multi-Dataset Benchmarks. In Proceedings of the 13th International Conference on Applied Innovations in IT (ICAIIT), August 2025, pp. 135–141.
University	University of Diyala
Country	Iraq
Publication Year	2025
Publication Venue	Proceedings of the 13th International Conference on Applied Innovations in IT, ICAIIT 2025
Publisher	Anhalt University of Applied Sciences
DOI / URL	https://www.scopus.com/pages/publications/105032701822?origin=resultslist
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Speech, Audio, and Auditory Processing
Specific Task	Speech Emotion Recognition classification across multiple emotion categories
Data Modality	Speech/audio signals converted into MFCC feature representations
Signal Dimensionality	1D temporal audio transformed into 2D time–feature MFCC representation
Signal Characteristics	Audio normalized to 16 kHz; Butterworth bandpass filtering from 300 Hz to 3400 Hz; MFCC-based features; the paper reports 40 MFCCs and maximum 128 frames, while the training section also reports input features of size 62 × 200
Dataset Name	TESS, SAVEE, RAVDESS, CREMA-D
Dataset Type	Public speech emotion recognition benchmark datasets consolidated from Kaggle
Dataset Size	Total approximately 12,162 audio samples: TESS 2,800 recordings; SAVEE 480 recordings; RAVDESS 1,440 recordings; CREMA-D 7,442 recordings
Dataset Balance / Split	Initial split: 80% training and 20% testing; training set further split into 80% training and 20% validation. Overall effective split is approximately 64% training, 16% validation, and 20% testing. Class balance NR
Hybrid Topology	Sequential hybrid SER pipeline combining MFCC preprocessing, SNN component, temporal attention mechanism, Transformer encoder, dense layers, and softmax classification
SNN Component	Spiking Neural Network module for sparse/event-driven pattern modeling; exact layer structure NR
Neuron Model	NR
Transformer Component	Transformer encoder with self-attention for long-range temporal dependency modeling
Attention Type	Temporal attention and Transformer self-attention
SNN–Transformer Interface Mechanism	MFCC features are passed through a hybrid architecture where SNN-based feature processing is combined with temporal attention and Transformer encoder modeling; exact spike-to-token or token-to-spike interface NR
Model Depth / Scale	Input feature size reported as 62 × 200; number of Transformer layers, attention heads, SNN layers, hidden dimensions, and dense-layer sizes NR
Parameter Count	NR
Design / Contribution Type	Application adaptation of a hybrid SNN–Temporal Attention–Transformer model for multi-dataset speech emotion recognition
Input Encoding Method	MFCC feature extraction from audio; explicit spike encoding method NR
Encoding Parameters	16 kHz sampling rate; 300–3400 Hz Butterworth bandpass filter; 40 MFCCs; maximum 128 frames reported; model input size also reported as 62 × 200
Encoding Learned or Fixed	MFCC extraction and preprocessing appear fixed; attention and Transformer representations are learned; SNN spike encoding mechanism NR
Event / Spike Representation	Spike/event-driven representation is claimed through SNN component, but exact spike tensor, firing rule, timestep, or membrane representation NR
Temporal Resolution	Audio frame length/time step NR; maximum 128 MFCC frames reported; model input also reported as 200 time positions
Output Decoding Method	Softmax layer outputs emotion probabilities
Encoding / Temporal Information Analysis	Partial. The paper discusses temporal attention and emotional frame weighting, but does not deeply analyze spike encoding quality, temporal information preservation, or information loss
Training Paradigm	Supervised learning
Training Method	End-to-end supervised training with Adam optimizer; optional early stopping and learning-rate scheduling
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Cross-Entropy Loss
Training Hyperparameters	100 epochs; learning rate 0.001; optional early stopping and learning-rate scheduling; other hyperparameters NR
Primary Performance Metric	Classification accuracy
Primary Metric Value	Accuracy = 98.0%
Secondary Metrics	Precision = 97.8%; recall = 98.1%; F1-score = 97.9%; macro F1 = 0.979 for the full model
Baseline Models Compared	SNN only; SNN + Attention; Transformer only; Transformer + Attention; literature baselines including CNN + LSTM, CNN with attention, CNN-Transformer, transfer-learning CNN, and other SER models
Performance Gain Over Baseline	Full hybrid model improves over SNN-only model from 91.0% to 98.0%; improves over Transformer-only model from 94.0% to 98.0%; improves over Transformer + Attention from 96.2% to 98.0%

Statistical Validation	No standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Yes. Ablation compares SNN only, SNN + Attention, Transformer only, Transformer + Attention, and full SNN + Attention + Transformer model
Training Framework / Code Availability	Framework NR; code availability NR
Target Hardware	Real-time and resource-limited environments are discussed as motivation, but no specific hardware platform is evaluated
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	NR; energy efficiency is claimed qualitatively based on SNN properties, but no measured or calculated energy result is provided
Latency / Memory / Complexity Evidence	NR; paper mentions compact design and real-time suitability but does not report latency, memory use, FLOPs, parameter count, or complexity
Comparison with ANN Baseline	Yes, partially. The paper compares against ANN/deep-learning SER approaches in literature and ablation variants, but does not provide a controlled matched ANN baseline
Main Contribution / Key Limitation	Contribution: proposes a hybrid SNN–Temporal Attention–Transformer SER model evaluated on TESS, SAVEE, RAVDESS, and CREMA-D, reaching 98% accuracy. Limitation: datasets are clean/controlled; robustness to noisy spontaneous speech is not validated; model requires further optimization for memory, latency, and edge deployment; SNN implementation details are limited
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P045

SITRA: Exploiting Temporal Silence in Spiking Transformers for Fast & Energy-efficient Inference

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University	Yale University
Country	United States
Publication Year	2025
Publication Venue	IEEE/ACM International Symposium on Low Power Electronics and Design, ISLPED 2025
Publisher	IEEE
DOI / URL	https://opendata.uni-halle.de//bitstream/1981185920/124026/1/2-8-ICAIIT_2025_13%284%29.pdf
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Hardware-focused
E&QA; Score	4
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Energy and latency evaluation of SNN Transformer inference using temporal silence, silent input sparsity, and silent output detection
Data Modality	Static image data and text/token sequence data
Signal Dimensionality	2D image patch-token sequences and 1D language token sequences represented as T × N × d spiking Transformer tensors
Signal Characteristics	Binary spike inputs over multiple timesteps; SNN Transformer uses T = 4 timesteps; spike inputs/outputs are 1-bit, weights are 8-bit, and membrane potentials are 8-bit
Dataset Name	CIFAR-10, ImageNet, SST-2
Dataset Type	Public image classification and sentiment/language benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	SNN Transformer inference engine with timestep-parallel SIMD processing, IF activation, PIM-based QK■ computation, silent input skipping, and preemptive silent output detection
SNN Component	IF neuron, binary spike trains, spiking VMM/matmul operations, spike-based Q/K/V, silent input and silent output sparsity
Neuron Model	IF
Transformer Component	Encoder-only Spiking Transformer models including Spikformer, Spike-driven Transformer, and SpikingBERT; Transformer encoder operations include Q/K/V, QK■, projection, MLP-I, and MLP-II
Attention Type	Spiking multi-head self-attention; softmax-free attention where AttentionSNN = QK■V
SNN–Transformer Interface Mechanism	Transformer VMMs are followed by IF activation to generate T × N × d spike tensors; Q, K, and V are spike-based; QK■ is computed using PIM while other spiking linear/matmul operations are processed using SIMD AC operations
Model Depth / Scale	Evaluated models include Spikformer-4-384-64, Spikformer-8-384-197, SDT-2-512-64, and SpikingBERT-6-768-512; T = 4 timesteps; 256 SIMD PEs; 256 × 256 PIM array
Parameter Count	NR
Design / Contribution Type	Hardware evaluation engine named SITRA; timestep-parallel SNN Transformer inference; silent input exploitation; PIM-based preemptive silent output detection; PIM reuse for QK■ operation
Input Encoding Method	Direct coding is described for converting real-world inputs into spike trains across T timesteps
Encoding Parameters	T = 4 timesteps; IF threshold Vth = 0.5; spike input precision 1-bit; output spike precision 1-bit; weights 8-bit; membrane potential 8-bit; τ threshold used for silent output detection
Encoding Learned or Fixed	Fixed hardware-level spike processing and silence detection; τ is calibrated using a batch of training samples; underlying SNN Transformer models are pretrained
Event / Spike Representation	1-bit binary spike tensors; T × N × d spike matrices; silent inputs are inputs that remain zero across all timesteps; silent outputs are outputs that remain zero after IF activation
Temporal Resolution	4 timesteps
Output Decoding Method	SNN inference runs across T timesteps and averages classifier outputs for final prediction
Encoding / Temporal Information Analysis	Yes. The paper analyzes temporal silence, silent input sparsity, silent output sparsity, effective sparsity, timestep-parallelism, and the effect of silence on latency and energy
Training Paradigm	Hardware inference evaluation using pretrained SNN Transformer models
Training Method	NR for model training; τ threshold for silent output detection is calibrated using training samples
Surrogate Gradient Function	NR

ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	T = 4; τ selected per linear/matmul operation using approximately 512 training examples; τ chosen near the mean cumulative PIM column output to limit accuracy degradation to about 1–2%
Primary Performance Metric	Energy-efficiency improvement and inference latency reduction
Primary Metric Value	Up to 6.36× energy reduction / energy-efficiency improvement and 3.41× latency reduction for SpikingBERT-6-768-512 on SST-2 compared with ANN baseline
Secondary Metrics	Test accuracy, accuracy loss, silent input sparsity, silent output sparsity, effective sparsity, SRAM/DRAM access energy, SIMD/PIM energy, hardware area, SRAM buffer sensitivity
Baseline Models Compared	ANN baseline; SNN Baseline-I without silent input/output sparsity; SNN Baseline-II with silent input sparsity only; BNTT, TDBN, Neuro-CIM, SpikeSim, SpinalFlow, LoAS; quantized and token-pruned ANN baselines
Performance Gain Over Baseline	With silent input and output sparsity, SITRA achieves 4.65× energy reduction and 1.62× latency reduction on Spikformer-CIFAR10; 5.82× and 2.37× on Spikformer-ImageNet; 5.0× and 1.53× on SDT-CIFAR10; 6.36× and 3.41× on SpikingBERT-SST-2
Statistical Validation	Hardware simulation and benchmark comparison; no repeated-run standard deviation, confidence interval, or statistical significance test reported
Ablation Study	Full hardware ablation. Compares ANN baseline, SNN without sparsity, SNN with silent input only, and SNN with silent input plus silent output detection; also varies token number, SRAM buffer size, quantization, and token pruning
Training Framework / Code Availability	Hardware metrics derived using Synopsys Design Compiler with 65 nm CMOS; PIM metrics from 8T-SRAM crossbar implementation; code availability NR
Target Hardware	65 nm CMOS hardware evaluation engine with SIMD PE array, SRAM buffers, PIM array, and softmax baseline unit
Hardware Deployment Status	Simulated/synthesized hardware evaluation; no fabricated chip or FPGA prototype reported
Energy Evidence	Energy estimated from 65 nm CMOS synthesis and 8T-SRAM PIM modeling; reports SRAM/DRAM/SIMD/PIM/softmax energy breakdown; no measured silicon energy
Latency / Memory / Complexity Evidence	Total SRAM buffer 624 KB; DRAM bandwidth 64 Gb/s; SRAM bandwidth 128 Gb/s; 256 SIMD PEs; PIM area 0.12 mm²; softmax area 0.25 mm²; SNN hardware area 27.32 mm², comparable to ANN baseline area 27.44 mm²
Comparison with ANN Baseline	Yes. Direct comparison with ANN Transformer baseline using energy, latency, and accuracy metrics
Main Contribution / Key Limitation	Contribution: introduces SITRA, a hardware evaluation engine that exploits timestep parallelism, silent input sparsity, and preemptive silent output detection for fast and energy-efficient SNN Transformer inference. Limitation: results are based on simulation/synthesis rather than measured hardware; SNN Transformer accuracy remains lower than ANN baselines; τ-based thresholding introduces around 1–2% accuracy loss
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3

P046

FE-SpikeFormer: A Camera-based Facial Expression Recognition Method for Hospital Health Monitoring

Full Citation	Dong, Z., Zhu, L., Zhou, S., Ji, X., Lai, C. S., Chen, M., & Ji, J. (2025). FE-SpikeFormer: A Camera-based Facial Expression Recognition Method for Hospital Health Monitoring. IEEE Journal of Biomedical and Health Informatics. https://doi.org/10.1109/JBHI.2025.3589267
University	Hangzhou Dianzi University
Country	China
Publication Year	2025
Publication Venue	IEEE Journal of Biomedical and Health Informatics
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/JBHI.2025.3589267
Publication Type	Journal article
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	Camera-based facial expression recognition across seven emotion classes
Data Modality	Facial images and hospital camera video frames
Signal Dimensionality	2D facial image sequence converted into spike-form image patches; internal representation uses T × C × H × W spiking feature maps
Signal Characteristics	Facial images are processed as spike-form patches; model uses T = 4 timesteps, dmodel = 384, and seven facial expression categories: angry, disgust, fear, happy, sad, surprise, and neutral
Dataset Name	FER2013, AffectNet, and a hospital deployment dataset collected at Lishui Central Hospital
Dataset Type	Public facial expression benchmark datasets plus custom real-world hospital dataset
Dataset Size	FER2013: 36,157 images; AffectNet: 420,299 images; hospital dataset: 40 hospitalized patients and 204 facial expression video segments
Dataset Balance / Split	FER2013 uses official split: 28,709 training, 3,859 validation, and 3,589 testing images. AffectNet uses 416,299 training and 4,000 validation images. Hospital dataset has balanced age/gender distribution but emotion distribution is imbalanced
Hybrid Topology	Initial-Conv → M Spiking Extraction Blocks → N Spiking Integration Blocks → classification head; combines ResConv-based SNN, Pairwise-Attention SNN, Spiking-Attention SNN, and spiking MLP modules
SNN Component	LIF neurons, ResConv-based SNN, Multi-Stage SNN Conv, spiking MLP, Pairwise-Attention SNN, Spiking-Attention SNN
Neuron Model	LIF
Transformer Component	Spiking attention block with Q/K/V projections; pairwise-attention and spiking-attention modules; Transformer-inspired global attention integration
Attention Type	Dual attention: pairwise attention for local/intra-channel feature modeling and spiking attention for global cross-channel modeling
SNN–Transformer Interface Mechanism	Initial-Conv converts image input into spike-form patches; LIF converts feature maps into spike representations; Q, K, and V are generated using learnable matrices and transformed through LIF-based spiking layers before attention computation
Model Depth / Scale	Final selected model uses M = 2 Spiking Extraction Blocks and N = 2 Spiking Integration Blocks; dmodel = 384; T = 4 timesteps

Parameter Count	6.93M
Design / Contribution Type	New application architecture; dual attention-based SNN architecture; spiking feature integration strategy; hospital deployment validation
Input Encoding Method	Initial convolution transforms facial image sequence into spike-form patches; LIF-based spike activation is used inside feature extraction and attention modules
Encoding Parameters	T = 4 timesteps; dmodel = 384; M = 2; N = 2; LIF threshold and membrane time constant are formulated but exact numerical values are NR
Encoding Learned or Fixed	Partially learned: Initial-Conv and attention projections are learned; LIF firing dynamics are fixed by neuron model equations
Event / Spike Representation	Binary spike signals; spike-form patches; T × C × H × W spiking feature maps; spike-form Q/K/V representations
Temporal Resolution	4 SNN timesteps
Output Decoding Method	Classification head produces emotion scores; final class is selected from the highest score
Encoding / Temporal Information Analysis	Yes. The paper analyzes attention maps, spike-form workflow, timestep ablation, t-SNE feature separation, confusion matrices, noise robustness, and hospital deployment latency
Training Paradigm	Supervised learning
Training Method	End-to-end supervised training of FE-SpikeFormer using PyTorch and AdamW
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	Batch size 48 per GPU; base learning rate $1 \times 10^{■3}$; actual learning rate = BatchSize / 256 × $10^{■3}$; M = 2; N = 2; dmodel = 384; T = 4; 300 epochs; trained on dual NVIDIA RTX 4090 GPUs
Primary Performance Metric	Average classification accuracy
Primary Metric Value	FER2013 accuracy: 73.58%; AffectNet accuracy: 59.90%; hospital environment average accuracy: 73.57%
Secondary Metrics	Energy consumption, parameter count, confusion matrix, anti-noise accuracy, response time, latency, processing time
Baseline Models Compared	CLCM, RTEC, IdentiFace, Ad-Corre, VGGNet, SENet, PF-ViT, OAENet, ESR-FER, DENet, ConvSNN, Spikformer, Meta-SpikeFormer
Performance Gain Over Baseline	On FER2013, FE-SpikeFormer achieves the highest reported accuracy of 73.58%, outperforming Spikformer by 2.88 percentage points and Meta-SpikeFormer by 1.76 points. On AffectNet, it reaches 59.90%, outperforming Spikformer and Meta-SpikeFormer while using fewer parameters and lower energy than ESR-FER and DENet
Statistical Validation	Benchmark evaluation, confusion matrices, hospital validation, and clinician annotation are provided; no standard deviation, confidence interval, repeated-run analysis, or significance test reported
Ablation Study	Full. Includes component ablation, block-number ablation, timestep ablation, t-SNE visualization, and anti-noise analysis
Training Framework / Code Availability	PyTorch; code availability NR
Target Hardware	Hospital edge deployment using Logitech C525 remote camera and HS140 local edge computing system
Hardware Deployment Status	Deployed and validated in a real hospital environment; no neuromorphic-chip deployment reported
Energy Evidence	Reported computational energy consumption is 7.40 mJ; exact energy measurement or calculation protocol is not fully detailed
Latency / Memory / Complexity Evidence	6.93M parameters; 7.40 mJ energy; hospital response time ≈ 0.25 s; latency ≈ 0.10 s; processing time ≈ 0.03 s
Comparison with ANN Baseline	Yes. Compared against ANN-based FER models including CLCM, RTEC, IdentiFace, Ad-Corre, VGGNet, SENet, PF-ViT, OAENet, ESR-FER, and DENet
Main Contribution / Key Limitation	Contribution: proposes FE-SpikeFormer, a dual-attention SNN-based facial expression recognition model for hospital health monitoring, achieving a good balance between accuracy, parameter count, energy, and deployment feasibility. Limitation: lower accuracy for negative emotions due to imbalanced clinical data; no neuromorphic hardware measurement; future work aims to build balanced clinical datasets and integrate physiological signals
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P047

SGSAFormer: Spike Gated Self-Attention Transformer and Temporal Attention

Full Citation	Gao, S., Qin, Y., Zhu, R., Zhao, Z., Zhou, H., & Zhu, Z. (2025). SGSAFormer: Spike Gated Self-Attention Transformer and Temporal Attention. Electronics, 14(1), Article 43. https://doi.org/10.3390/electronics14010043
University	Shanghai University
Country	China
Publication Year	2025
Publication Venue	Electronics, 14(1), Article 43
Publisher	MDPI
DOI / URL	https://doi.org/10.3390/electronics14010043
Publication Type	Journal article
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Event-stream classification / neuromorphic image and gesture classification
Data Modality	Event-camera / DVS data
Signal Dimensionality	2D spatiotemporal event-frame streams represented over timesteps
Signal Characteristics	DVS data are integrated into temporal frames; frames with more events are treated as more informative; model processes event streams using spike-based temporal representation
Dataset Name	CIFAR10-DVS, DVS128Gesture, N-Caltech101
Dataset Type	Public neuromorphic vision benchmark datasets

Dataset Size	DVS128Gesture: 1464 samples, 1176 training and 288 testing; CIFAR10-DVS: 10,000 event streams, 10 classes, 1000 samples per class; N-Caltech101: 101 object categories, exact sample size NR
Dataset Balance / Split	CIFAR10-DVS uses 9:1 split, first 900 samples per class for training and remaining 100 for testing; DVS128Gesture uses 1176/288 train-test split; N-Caltech101 split NR
Hybrid Topology	Spikformer-based architecture with Spiking Patch Splitting, SGSAFormer encoder blocks, Spike Gated Self-Attention, Spike Gated Linear Unit, Temporal Attention preprocessing, and linear classification head
SNN Component	LIF neurons, spike neuron layers, Spiking Patch Splitting, spike-form Q/K/V/G, SGLU, SGSA, spike-firing-rate-based computation
Neuron Model	LIF
Transformer Component	Spikformer-style encoder block; Spike Gated Self-Attention module; Spike Gated Linear Unit replacing MLP; classification head
Attention Type	Spike Gated Self-Attention; Temporal Attention; spiking self-attention
SNN–Transformer Interface Mechanism	Input event frames are processed by Spiking Patch Splitting; spike neuron layers generate spike-form features; Q, K, V, and gate matrix G are produced through Linear + TBN + spike neuron layers; SGSA gates the SSA attention output using G
Model Depth / Scale	SGSAFormer-1-512 for CIFAR10-DVS; SGSAFormer-2-256 for DVS128Gesture and N-Caltech101; experiments use 16 timesteps; four SPS modules and 16 detection heads are reported
Parameter Count	NR
Design / Contribution Type	New Spiking Transformer architecture introducing Spike Gated Linear Unit, Spike Gated Self-Attention, and Temporal Attention module for filtering less informative event frames
Input Encoding Method	DVS event streams are integrated into temporal frames; Temporal Attention assigns frame weights according to the number of events and filters frames below a threshold
Encoding Parameters	Main SGSAFormer experiments use T = 16 timesteps; TA experiments reduce effective timesteps from 16 to 5; TA threshold selected around 0.7; energy comparison also reports T = 4 for SGSAFormer-8-512
Encoding Learned or Fixed	Partially fixed and partially learned: TA weighting/filtering is formula-based and threshold-based; SGSA/SGLU projections and gating parameters are learned
Event / Spike Representation	Binary spike sequences; spike-form Q, K, V, and G matrices; DVS temporal frame stream; spike firing rate used for energy estimation
Temporal Resolution	16 timesteps in the main neuromorphic dataset experiments; TA can reduce effective computation to 5 timesteps; energy table also reports 4-timestep models
Output Decoding Method	Linear classification head produces final class prediction
Encoding / Temporal Information Analysis	Yes. The paper analyzes Temporal Attention weighting, event-count-based frame importance, spike firing rate before/after TA, timestep effects, attention maps, and energy–accuracy trade-off
Training Paradigm	Supervised direct training
Training Method	BPTT-based direct training using surrogate-gradient optimization
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	PyTorch 1.13.1; SpikingJelly 0.0.0.0.12; NVIDIA GeForce RTX 3090; AdamW optimizer; initial learning rate 0.01; 200 epochs; 16 timesteps for main experiments
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	CIFAR10-DVS: 85.0 ± 0.5%; DVS128Gesture: 99.0 ± 0.03%; N-Caltech101: 83.9 ± 0.2%
Secondary Metrics	Energy consumption, spike firing rate, timestep sensitivity, ablation accuracy, attention-map visualization
Baseline Models Compared	RG-CNNs, MLS, LIAF-Net, MVF-Net, tdBN, SEW-ResNet, Spikformer, Spikingformer, S-Transformer, STSA/STSformer, SGLFormer, DECOLLE, EventTransformer, STCA-SNN, TCJA-SNN, ANN Transformer
Performance Gain Over Baseline	On CIFAR10-DVS, SGSAFormer-1-512 improves over SGLFormer by 2.4%, STSformer by 5.1%, and S-Transformer by 5.0%. On DVS128Gesture, SGSAFormer-2-256 improves over Spikformer and Spikingformer by 0.7%, STSformer by 0.3%, and SGLFormer by 0.4%. On N-Caltech101, it improves over TCJA-SNN by 1.4%
Statistical Validation	Reports mean ± variation for SGSAFormer results; no detailed confidence interval, significance testing, or repeated-run protocol reported
Ablation Study	Full. Includes TA threshold analysis, TA effect on SCNN and Spikformer, spike-firing-rate comparison, SGLU ablation, SGSA ablation, timestep ablation, architecture-scale comparison, and attention-map visualization
Training Framework / Code Availability	PyTorch and SpikingJelly; code availability NR
Target Hardware	Neuromorphic hardware is discussed as a potential deployment target, including TrueNorth, Loihi, and Tianjic
Hardware Deployment Status	No actual neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using 45 nm hardware assumptions: EMAC = 4.6 pJ and EAC = 0.9 pJ. SGSAFormer-8-512 consumes 12.68 mJ compared with 11.58 mJ for Spikformer and 38.34 mJ for ANN Transformer
Latency / Memory / Complexity Evidence	TA reduces effective computation by filtering low-information frames; SCNN-TA reduces energy from 1.272 mJ to 0.601 mJ; Spikformer-TA reduces energy from 3.055 mJ to 1.253 mJ; exact latency, memory footprint, and parameter count NR
Comparison with ANN Baseline	Yes. Compares with ANN baselines on neuromorphic datasets and reports ANN Transformer energy consumption of 38.34 mJ versus 12.68 mJ for SGSAFormer
Main Contribution / Key Limitation	Contribution: proposes SGSAFormer, introducing gated mechanisms into Spiking Transformers through SGLU and SGSA, plus a TA module that filters less informative event frames to reduce energy. Limitation: gating adds extra computation compared with Spikformer; no real neuromorphic hardware deployment; parameter count and detailed latency/memory results are NR
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P048

MSVIT: Improving Spiking Vision Transformer Using Multi-scale Attention Fusion

Full Citation	Hua, W., Zhou, C., Wu, J., Chua, Y., & Shu, Y. (2025). MSVIT: Improving Spiking Vision Transformer Using Multi-scale Attention Fusion. In Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI-25), pp. 5399–5407.
University	China Nanhu Academy of Electronics and Information Technology
Country	China
Publication Year	2025

Publication Venue	Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25
Publisher	IJCAI
DOI / URL	https://github.com/Nanhu-AI-Lab/MSViT
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification and event-stream classification
Data Modality	Static RGB images and neuromorphic/event-camera data
Signal Dimensionality	2D image inputs and 2D spatiotemporal event-frame inputs represented as T × C × H × W spike tensors
Signal Characteristics	Static RGB datasets use T = 1 and C0 = 3 before spike processing; neuromorphic datasets use T = T0 and C0 = 2; events are aggregated into frames using positive, negative, and summed event channels
Dataset Name	ImageNet-1K, CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public static image and neuromorphic vision benchmark datasets
Dataset Size	ImageNet-1K: standard large-scale benchmark; CIFAR-10/100: standard benchmark datasets; exact sample counts NR in the paper. CIFAR10-DVS and DVS128 Gesture are used as neuromorphic benchmarks; exact sample counts NR
Dataset Balance / Split	Uses standard benchmark splits; exact class balance NR
Hybrid Topology	Hierarchical Spiking Vision Transformer with three stages; uses Spiking Patch Embedding with Multi-scale Feature Fusion, MSFormer blocks, Multi-scale Spiking Attention in early stages, Spiking Self-Attention in final stage, and classification head
SNN Component	Spiking neuron layers, binary spike activations, Spiking Patch Embedding, Spiking MLP, MSSA, SSA, spike-form Q/P/V representations
Neuron Model	Spiking neuron layer used; exact neuron model NR
Transformer Component	Hierarchical Vision Transformer-style encoder; MSFormer block; MSSA token mixer; SMLP channel mixer; SSA in final stage
Attention Type	Multi-scale Spiking Attention; Spiking Self-Attention; hybrid MSSA–SSA attention
SNN–Transformer Interface Mechanism	Input is transformed into spike-form tokens using SPEMSF; Q, P, and V are generated as spike-form representations through learnable projections, BN, and spiking neuron layers; MSSA uses column-wise summation and multi-scale fusion instead of conventional dot-product attention
Model Depth / Scale	ImageNet-1K model uses three stages with layer configuration {l1 = 1, l2 = 2, l3 = 7}; MSVIT-10-384, MSVIT-10-512, and MSVIT-10-768 are reported. CIFAR model uses 4 blocks distributed as {1,1,2}. Neuromorphic lightweight model uses {0,1,1}
Parameter Count	ImageNet: 17.69M, 30.23M, and 69.80M depending on model size; CIFAR: 7.59M; neuromorphic model: 1.67M
Design / Contribution Type	New hierarchical Spiking Vision Transformer architecture; Multi-scale Spiking Attention; Spiking Patch Embedding with Multi-scale Feature Fusion; hybrid attention integration
Input Encoding Method	Static images are treated as image inputs with T = 1 before spike-form embedding; neuromorphic event streams are aggregated into frames with positive, negative, and summed event channels; SPEMSF converts inputs into spike-form tokens
Encoding Parameters	Patch size 4 × 4 in Stage 1; Stage 2 and Stage 3 reduce tokens to H/8 × W/8 and H/16 × W/16; ImageNet uses 4 timesteps; CIFAR uses 4 timesteps; neuromorphic experiments use 10 or 16 timesteps
Encoding Learned or Fixed	Partially learned: SPEMSF uses learnable convolutional projections; event aggregation is a fixed preprocessing step
Event / Spike Representation	Binary spike-form tensors; spike-form Q, P, V; spike-form token embeddings; positive/negative/summed event-frame channels for neuromorphic inputs
Temporal Resolution	Static datasets mainly use 4 SNN timesteps after spike processing; neuromorphic experiments use 10 or 16 timesteps
Output Decoding Method	Classification head outputs top-1/top-5 class predictions
Encoding / Temporal Information Analysis	Yes. The paper analyzes multi-scale spike representation, low-level and high-level feature fusion, attention computation complexity, energy cost, and ablation of MSSA/SSA combinations
Training Paradigm	Supervised direct training from scratch
Training Method	Direct training of hierarchical Spiking Transformer using AdamW-based optimization
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	ImageNet: 200 epochs, 8 NVIDIA A100 GPUs, batch size 512, synchronized AdamW, base learning rate 6 × 10■■ per batch size 512, warmup, half-period cosine decay, RandAugment, random erasing, stochastic depth, gradient accumulation. CIFAR: 400 epochs, batch size 128. DVS128 Gesture: 200 epochs. CIFAR10-DVS: 106 epochs
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	ImageNet-1K best: 85.06% top-1 and 97.58% top-5 with MSVIT-10-768. CIFAR-10: 96.53%. CIFAR-100: 81.98%. DVS128 Gesture: 98.80%. CIFAR10-DVS: 84.30%
Secondary Metrics	Top-5 accuracy, parameter count, theoretical energy consumption, timestep count, ablation accuracy
Baseline Models Compared	DeiT, ViT-B/16, Swin Transformer, Spikformer, Spikformer V2, Spike-driven Transformer, Spike-driven Transformer v2, QKFormer, SDT, CML, ResNet-19, Transformer, MST-T
Performance Gain Over Baseline	On ImageNet-1K, MSVIT-10-768 reaches 85.06%, outperforming QKFormer-10-768 by 0.84%, Spike-driven v2-10-768 by 5.06%, and Spikformer-8-768 by 10.25%. On CIFAR100, MSVIT improves over Spikformer by 3.77% and QKFormer by 0.86%. On CIFAR10-DVS, MSVIT improves over Spikformer by 3.4% at 16 timesteps
Statistical Validation	Benchmark comparison and ablation experiments are provided; no standard deviation, confidence interval, repeated-run protocol, or statistical significance test reported
Ablation Study	Yes. Includes hybrid MSSA–SSA integration, different feature-fusion dimensions, MSSA variants using P/Q features, and parameter–accuracy trade-off analysis
Training Framework / Code Availability	Code available at GitHub: Nanhu-AI-Lab/MSViT; exact software framework NR
Target Hardware	Neuromorphic/edge hardware motivation; no specific hardware platform evaluated
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	Theoretical energy estimation reported. On ImageNet, MSVIT-10-768 consumes 45.88 mJ per image, compared with QKFormer-10-768 at 38.91 mJ, SDT-v2-10-768 at 52.40 mJ, Swin-S at 216.20 mJ, DeiT-B at 80.50 mJ, and ViT-B/16 at 254.84 mJ

Latency / Memory / Complexity Evidence	MSSA reduces attention complexity to linear O(TND) / O(ND), compared with $O(N^2D)$ for vanilla or standard spiking self-attention. Memory and latency values NR
Comparison with ANN Baseline	Yes. Compares against ANN Transformer models including DeiT, ViT-B/16, Swin-T/S, ResNet-19, and Transformer baselines
Main Contribution / Key Limitation	Contribution: proposes MSVIT, a hierarchical Spiking Vision Transformer using Multi-scale Spiking Attention and SPEMSF to improve spike-based multi-scale feature extraction and achieve strong ImageNet-scale performance. Limitation: no real hardware deployment; energy is theoretical; MSSA increases parameters compared with simpler spiking attention designs; limited discussion of failure cases
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P049

Towards High-performance Spiking Transformers from ANN to SNN Conversion

Full Citation	Huang, Z., Shi, X., Hao, Z., Bu, T., Ding, J., Yu, Z., & Huang, T. (2024). Towards High-performance Spiking Transformers from ANN to SNN Conversion. In Proceedings of the 32nd ACM International Conference on Multimedia (MM '24), October 28–November 1, 2024, Melbourne, VIC, Australia, pp. 10688–10697. ACM. https://doi.org/10.1145/3664647.3680620
University	Peking University
Country	China
Publication Year	2024
Publication Venue	ACM International Conference on Multimedia
Publisher	ACM
DOI / URL	https://doi.org/10.1145/3664647.3680620
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	ImageNet-1K image classification using converted Spiking Vision Transformers
Data Modality	Static RGB image data
Signal Dimensionality	2D image inputs split into patch-token sequences for Vision Transformer processing
Signal Characteristics	Images are processed through ViT-style patch embeddings, positional embeddings, Transformer encoder blocks, attention modules, MLP modules, and classification head; SNN inference uses multiple timesteps
Dataset Name	ImageNet-1K
Dataset Type	Public large-scale image classification benchmark
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Pre-trained ANN Vision Transformer converted to SNN using Expectation Compensation Modules and Multi-Threshold Neurons; Transformer encoder retains attention and MLP structure but replaces nonlinear modules with EC-based equivalents
SNN Component	Integrate-and-Fire neurons, Multi-Threshold Neurons, spike communication, reset-by-subtraction dynamics, threshold-based spike emission, parallel parameter normalization
Neuron Model	IF neuron and Multi-Threshold Neuron
Transformer Component	Vision Transformer encoder, patch embedding, CLS token, position embedding, self-attention, LayerNorm, GELU, softmax, matrix product, MLP, classification head
Attention Type	Converted ViT self-attention; Matrix Product-EC and Softmax-EC are used to preserve attention behavior during ANN-to-SNN conversion
SNN–Transformer Interface Mechanism	EC modules replace nonlinear Transformer operations such as LayerNorm, GELU, Softmax, and matrix product; MT neurons are inserted before linear and matrix-product layers; parallel parameter normalization converts ANN weights into SNN weights across multiple threshold channels
Model Depth / Scale	Converted architectures include ViT-S/16, ViT-B/16, ViT-L/16, and EVA. Time steps evaluated include T = 1, 2, 4, 6, 8, 10, and 12
Parameter Count	ViT-S/16: 22M; ViT-B/16: 86M; ViT-L/16: 307M; EVA: 1074M
Design / Contribution Type	New ANN-to-SNN conversion method for Transformers; Expectation Compensation Module; Matrix Product-EC; Multi-Threshold Neuron; Parallel Parameter Normalization
Input Encoding Method	Static images are processed through the original ViT patch-embedding pipeline; spike communication is introduced through MT neurons during converted SNN inference
Encoding Parameters	MT neuron threshold number uses n = 8 for ViT-S/16, ViT-B/16, and ViT-L/16; n = 6 for EVA; threshold percentile p = 99; evaluated with up to 12 timesteps
Encoding Learned or Fixed	Fixed post-training conversion from pretrained ANN models; thresholds are calibrated from ANN activation statistics; no additional SNN training required
Event / Spike Representation	Multi-threshold spike channels; each original connection is expanded into 2n threshold channels; at each time, only one of the 2n channels can emit a spike
Temporal Resolution	Converted SNNs evaluated at T = 1, 2, 4, 6, 8, 10, and 12 timesteps; strong results reported at T = 4 to T = 10 depending on model scale
Output Decoding Method	Classification head outputs ImageNet class probabilities using accumulated SNN inference over timesteps
Encoding / Temporal Information Analysis	Yes. The paper analyzes ANN-SNN conversion error, expectation compensation across previous timesteps, effects of timestep count, threshold number, firing rates across thresholds, and energy ratio across timesteps
Training Paradigm	ANN-to-SNN conversion using pretrained ANN Transformers
Training Method	Training-free SNN conversion; pretrained ViT/EVA models are converted using ECMT without additional SNN training
Surrogate Gradient Function	N/A
ANN-to-SNN Conversion Method	Expectation Compensation and Multi-Threshold conversion method, ECMT
Optimizer	N/A for SNN conversion; optimizer for original pretrained ANN models NR
Loss Function	N/A for conversion; original ANN training loss NR

Training Hyperparameters	No SNN training. Conversion uses threshold percentile p = 99; MT neuron n = 8 for ViT-S/B/L and n = 6 for EVA; evaluated on 224-resolution ViT/EVA models
Primary Performance Metric	Top-1 classification accuracy and energy ratio compared with original ANN
Primary Metric Value	EVA converted SNN achieves 88.60% top-1 accuracy at T = 4 with energy ratio 0.35; ViT-L/16 achieves 83.20% at T = 4 and 84.60% at T = 8; ViT-B/16 achieves 79.40% at T = 8 and 80.12% at T = 10; ViT-S/16 achieves 76.03% at T = 8 and 77.07% at T = 10
Secondary Metrics	Energy ratio, timestep latency, firing rate, threshold sensitivity, parameter count, comparison with direct training and conversion baselines
Baseline Models Compared	Spikingformer, Spike-driven Transformer, Spikeformer, RMP, SNM, TS, QFFS, QCFS, SRP, MST, STA, CNN-to-SNN conversion methods, Transformer-to-SNN conversion methods
Performance Gain Over Baseline	ECMT-EVA reaches 88.60% at T = 4, outperforming previous ImageNet SNN and Transformer-to-SNN conversion baselines. The method achieves near-ANN accuracy with much lower latency than previous Transformer-to-SNN methods such as MST and STA
Statistical Validation	Benchmark evaluation and ablation experiments; no standard deviation, confidence interval, repeated-run protocol, or significance test reported
Ablation Study	Yes. Includes ablation on number and size of MT neuron thresholds, firing-rate distribution across thresholds, timestep sensitivity, and energy estimation
Training Framework / Code Availability	Source code available at GitHub: h-z-h-cell/Transformer-to-SNN-ECMT
Target Hardware	Neuromorphic hardware motivation; Loihi, TrueNorth, and Tianjic are discussed as relevant platforms, but no specific hardware deployment is performed
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using EMAC = 4.6 pJ and EAC = 0.9 pJ. EVA at T = 4 consumes only 35% of the original ANN Transformer energy while retaining 88.60% accuracy
Latency / Memory / Complexity Evidence	Low-latency inference is shown through small timestep counts. One-percent accuracy loss occurs at T = 10 for ViT-S/16, T = 8 for ViT-B/16, T = 6 for ViT-L/16, and T = 4 for EVA. Detailed memory footprint NR
Comparison with ANN Baseline	Yes. Direct comparison with original ANN ViT-S/16, ViT-B/16, ViT-L/16, and EVA models using accuracy and energy ratio
Main Contribution / Key Limitation	Contribution: proposes ECMT, the first successful high-accuracy, low-latency Transformer-to-SNN conversion method for complex ImageNet-scale Vision Transformers, using Expectation Compensation and Multi-Threshold Neurons. Limitation: the converted model still requires a small amount of multiplication and cannot be implemented only with accumulations; no real hardware deployment is reported
RQ Mapping	Primary: RQ4; Secondary: RQ2, RQ3, RQ5

P050

Lightweight Fault Diagnosis for EV Battery Packs via SpikingFormer and Frequency Slice Wavelet Transform

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Country	China
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Publication Type	Journal article
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Industrial Monitoring, Fault Diagnosis, and Infrastructure Inspection
Specific Task	Fault level classification and time-to-fault prediction for EV battery packs
Data Modality	Multivariate battery operational time-series: cell voltage, pack current, temperature, and state-of-charge
Signal Dimensionality	1D multivariate time-series converted into 2D time-frequency representations using FSWT
Signal Characteristics	Data sampled at 1 Hz; 10 min non-overlapping windows with 600 data points per segment; signals include 98 single-cell voltages, pack current, module temperatures, and SoC
Dataset Name	Real-world EV battery dataset collected from 100 EVs
Dataset Type	Private/confidential real-world EV battery operational dataset
Dataset Size	100 EVs; 9800 individual battery cells; 6–12 months of data per vehicle; 74,563 valid samples: 60,884 Normal, 7313 Fault level 1, 6219 Fault level 2, and 147 Fault level 3
Dataset Balance / Split	Train/validation/test split = 7:2:1; dataset is imbalanced, especially Fault level 3 with only 147 total samples
Hybrid Topology	FSWT preprocessing + lightweight SpikingFormer encoder + classification head + regression head
SNN Component	SpikingFormer, LIF neurons, spike trains, separable multiscale convolution, spike-form relative position embedding, multi-head spatiotemporal spiking self-attention
Neuron Model	LIF
Transformer Component	SpikingFormer encoder blocks, multi-head spatiotemporal spiking self-attention, classification head, regression head
Attention Type	Multi-head spatiotemporal spiking self-attention
SNN–Transformer Interface Mechanism	FSWT converts raw battery signals into 2D time-frequency representations; SMC extracts multiscale features and converts them into spike-based representations; MHSSSA then models spatiotemporal dependencies using spike-based attention
Model Depth / Scale	Optimal model uses 6 SpikingFormer encoder blocks; proposed model has 2.98M parameters
Parameter Count	2.98M
Design / Contribution Type	New application framework combining FSWT-based time-frequency feature extraction with lightweight SpikingFormer for real-time EV battery fault diagnosis
Input Encoding Method	Frequency Slice Wavelet Transform converts voltage/current/SoC/temperature signals into 2D time-frequency representations; SpikingFormer converts extracted features into spike trains

Encoding Parameters	Sampling frequency 1 Hz; 10 min window length; 600 points per sample; FSWT frequency slicing and wavelet convolution; SMC kernel size selected as [1, 3, 7]
Encoding Learned or Fixed	FSWT is fixed signal preprocessing; SMC, MHSSSA, classification head, and regression head are learned
Event / Spike Representation	Spike trains generated inside SpikingFormer; spike-based Q/K/V representations in MHSSSA; sparse event-driven computation
Temporal Resolution	Original data resolution = 1 s; input window = 10 min / 600 time points; SNN simulation timestep NR
Output Decoding Method	Classification head predicts Normal/Fault levels 1–3; regression head predicts time-to-fault
Encoding / Temporal Information Analysis	Yes. The paper analyzes FSWT time-frequency representations, energy-domain visualization, attention activation regions, feature separability, and time-to-fault prediction behavior
Training Paradigm	Supervised learning
Training Method	End-to-end supervised training of FSWT–SpikingFormer framework for joint classification and regression
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	NR
Training Hyperparameters	200 epochs; learning rate 1 × 10$^{■3}$; kernel size [1, 3, 7]; Adam optimizer; 6 SpikingFormer encoder blocks selected as optimal
Primary Performance Metric	F1 score and mean accuracy
Primary Metric Value	F1 score = 94.3%; mean accuracy = 93.1%; time-to-fault error = 3.4 min; inference time = 0.25 s
Secondary Metrics	RMSE for time-to-fault, SOPs, FLOPs, parameter count, energy consumption, inference time, confusion matrix
Baseline Models Compared	DeiT, VGG, DenseNet, DSRSN, SDRN; also compares FSWT against STFT, CWT, EMD, and VMD
Performance Gain Over Baseline	Proposed method improves F1 score over SDRN from 90.8% to 94.3%; improves mean accuracy from 89.1% to 93.1%; reduces time-to-fault error from 4.6 min to 3.4 min; reduces inference time from 0.88 s to 0.25 s; reduces energy from 1.80 mJ to 0.26 mJ
Statistical Validation	Uses train/validation/test split, confusion matrix, ablation studies, visualization analysis, and SOTA comparison; no confidence interval, standard deviation, repeated-run protocol, or significance test reported
Ablation Study	Full. Includes FSWT vs STFT/CWT/EMD/VMD, encoder block number, hyperparameter combinations, SMC ablation, MHSSSA ablation, and visualization analysis
Training Framework / Code Availability	Implemented on Intel Core i7-11800H CPU and NVIDIA GeForce RTX 3080Ti GPU; software framework and code availability NR; dataset is confidential
Target Hardware	Real-time EV battery management / edge or cloud-based BMS is discussed; experiments use RTX 3080Ti GPU
Hardware Deployment Status	No embedded BMS or neuromorphic hardware deployment; evaluated offline/on GPU workstation
Energy Evidence	Estimated energy using SOPs; each SOP assumed to consume 77 fJ. Proposed method reports 0.26 mJ energy in SOTA comparison; ablations report 0.09 mJ for the full SMC/MHSSSA configuration
Latency / Memory / Complexity Evidence	Proposed method: 2.98M parameters, 0.64G FLOPs, 3.43G SOPs, 0.26 mJ energy, 0.25 s inference time
Comparison with ANN Baseline	Yes. Compared with ANN/deep models including DeiT, VGG, and DenseNet, and SNN-based DSRSN/SDRN baselines
Main Contribution / Key Limitation	Contribution: proposes a lightweight FSWT–SpikingFormer framework that combines multi-signal time-frequency coupling with event-driven spiking attention for accurate, efficient EV battery fault diagnosis. Limitation: dataset is private; subtle/early-stage faults such as micro short circuits are not fully evaluated; implementation uses RTX 3080Ti GPU, which is not practical for embedded onboard BMS; sensor-fault and internal-fault coupling remains future work
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P051

SpikedAttention: Training-Free and Fully Spike-Driven Transformer-to-SNN Conversion with Winner-Oriented Spike Shift for Softmax Operation

Full Citation	Hwang, S., Lee, S., Park, D., Lee, D., & Kung, J. (2024). SpikedAttention: Training-Free and Fully Spike-Driven Transformer-to-SNN Conversion with Winner-Oriented Spike Shift for Softmax Operation. In Proceedings of the 38th Conference on Neural Information Processing Systems, NeurIPS 2024.
University	Daegu Gyeongbuk Institute of Science and Technology
Country	South Korea
Publication Year	2024
Publication Venue	Conference on Neural Information Processing Systems
Publisher	NeurIPS
DOI / URL	https://www.scopus.com/pages/publications/105000520731?origin=resultslist
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	5
Application Domain	Natural Language Processing and Language Models
Specific Task	Image classification, GLUE text classification, and SQuAD question answering
Data Modality	Static image data and text/token sequence data
Signal Dimensionality	2D image patch/token sequences for Swin Transformer; 1D language token sequences for BERT/MA-BERT
Signal Characteristics	Converts pretrained Transformer activations into spike-based representations; external inputs use binary coding, while intermediate features use one-spike phase coding to reduce energy
Dataset Name	ImageNet, CIFAR-10, CIFAR-100, GLUE benchmark, SQuAD
Dataset Type	Public benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Uses standard benchmark evaluation settings; exact dataset balance/split details NR
Hybrid Topology	Training-free Transformer-to-SNN conversion; converts pretrained Swin Transformer and BERT-style Transformer into fully spike-driven SpikedAttention without changing the original attention architecture
SNN Component	LIF neurons, one-spike phase-coded neurons, binary-coded input neurons, trace-driven dot-product neurons, WOSS-softmax neurons

Neuron Model	LIF
Transformer Component	Swin Transformer; MA-BERT/BERT-style Transformer; vanilla self-attention; softmax attention module
Attention Type	Fully spike-based vanilla self-attention; trace-driven matrix multiplication; Winner-Oriented Spike Shift softmax
SNN–Transformer Interface Mechanism	Transformer Q, K, and V projections are converted into spike representations; QK■ and attention-value multiplication are computed through trace-driven spike dot products; softmax is approximated using WOSS spike timing
Model Depth / Scale	Main vision model: Swin-T with 28.7M parameters; NLP model: MA-BERT/base-style model with 110M parameters; timestep varies by task, including 16, 24, 40, and 48 in reported experiments
Parameter Count	Swin-T SpikedAttention: 28.7M; MA-BERT/BERT-style SpikedAttention: 110M; converted MST pretrained models: 27.5M–28.3M
Design / Contribution Type	New training-free Transformer-to-SNN conversion method; fully spike-driven attention; trace-driven matrix multiplication; winner-oriented spike-shift softmax; preservation of vanilla attention architecture
Input Encoding Method	Binary coding for external input; one-spike phase coding for intermediate features; token embedding output is binary coded for BERT conversion
Encoding Parameters	Phase-coding base B selected from search space approximately (1.0, 1.5]; main Swin-T ImageNet conversion uses T = 40; GLUE tasks use T = 16 or 24; SQuAD uses T = 24; MST comparison uses T = 24 for CIFAR and T = 48 for ImageNet
Encoding Learned or Fixed	Fixed/training-free conversion; encoding base is selected using an error-estimation procedure, not learned by gradient training
Event / Spike Representation	Binary spike tensors; one-spike temporal/phase-coded spikes; signed and unsigned spike representations depending on model setting
Temporal Resolution	Timestep-based spike simulation; reported values include T = 16, 24, 40, and 48 depending on task/model
Output Decoding Method	Final spike-based network output is decoded through the original Transformer task head/classifier; exact decoding rule beyond converted classifier output NR
Encoding / Temporal Information Analysis	Yes. The paper analyzes phase-coding error, base selection, activation–decoded-spike correlation, and the timestep–accuracy–energy trade-off; it shows that longer timesteps reduce accuracy loss but increase energy consumption
Training Paradigm	ANN-to-SNN / Transformer-to-SNN conversion; supervised pretrained ANN Transformer is converted to SNN
Training Method	Training-free conversion; no post-training or conversion-aware training for the converted SpikedAttention models
Surrogate Gradient Function	N/A
ANN-to-SNN Conversion Method	Direct Transformer-to-SNN conversion using one-spike phase coding, trace-driven matrix multiplication, and WOSS spike-based softmax
Optimizer	N/A for conversion; optimizer for original pretrained ANN models NR
Loss Function	N/A for conversion; original ANN training loss NR
Training Hyperparameters	No additional SNN training; conversion hyperparameters include timestep T, phase-coding base B, signed/unsigned neuron setting, and WOSS threshold modification
Primary Performance Metric	Classification accuracy, GLUE task accuracy/Matthews correlation, F1 score for SQuAD, and energy consumption
Primary Metric Value	ImageNet: 80.0% accuracy with 28.7M parameters; GLUE: average accuracy loss only 0.3%; SQuAD: F1 score 87.2%; energy reduction: 42% vs Swin-T, 58% on GLUE, and 59.5% on SQuAD
Secondary Metrics	Parameter count, timestep, energy in mJ, accuracy loss, conversion loss, WOSS hardware overhead
Baseline Models Compared	Swin-T ANN, MST, Spikformer, SDSA, Meta-SpikeFormer, MA-BERT, SpikingBERT
Performance Gain Over Baseline	Compared with Swin-T ANN, SpikedAttention reduces energy by 42% with small accuracy loss; compared with MST, it uses much shorter timesteps and far lower energy CIFAR-10 21.3 mJ vs 1089.9 mJ and CIFAR-100 23.7 mJ vs 1157.1 mJ
Statistical Validation	Benchmark-based validation only; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full/strong analysis. Includes timestep effect, base B selection, accuracy-energy trade-off, ReLU vs non-ReLU conversion, WOSS-softmax behavior, and SQuAD/GLUE extension
Training Framework / Code Availability	Implemented with SpikingJelly and PyTorch; experiments use four NVIDIA GeForce RTX 4090 GPUs; code is reported as available on GitHub
Target Hardware	Neuromorphic/event-driven hardware motivation; hardware-realistic energy model used, but no specific chip deployment
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Estimated energy using hardware-realistic metrics that consider weight accumulation, neuron potential update, and on-chip data movement; no measured chip-level energy
Latency / Memory / Complexity Evidence	Reports energy in mJ, timestep count, parameter count, and WOSS neuron overhead; WOSS support adds 9.88% area and 12.35% power overhead compared with LIF-only hardware
Comparison with ANN Baseline	Yes. Compares directly with Swin-T ANN and MA-BERT ANN; also compares with ANN-to-SNN and directly trained SNN Transformer baselines
Main Contribution / Key Limitation	Contribution: proposes SpikedAttention, a training-free, fully spike-driven Transformer conversion method that preserves vanilla self-attention and implements softmax through WOSS. Limitation: still needs longer timesteps than some directly trained SNNs; GeLU and LayerNorm are not directly supported, so the BERT experiment relies on MA-BERT-style replacements
RQ Mapping	Primary: RQ4; Secondary: RQ2, RQ3, RQ5, RQ1

P052

Spike2Former: Efficient Spiking Transformer for High-performance Image Segmentation

Full Citation	Lei, Z., Yao, M., Hu, J., Luo, X., Lu, Y., Xu, B., & Li, G. (2025). Spike2Former: Efficient Spiking Transformer for High-performance Image Segmentation. In Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI-25), pp. 1364–1372.
University	University of Chinese Academy of Sciences
Country	China
Publication Year	2025
Publication Venue	AAAI Conference on Artificial Intelligence
Publisher	Association for the Advancement of Artificial Intelligence
DOI / URL	10.1609/aaai.v39i2.32126
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4.5

Application Domain	Computer Vision and Image Understanding
Specific Task	Semantic segmentation on ADE20K, Pascal VOC2012, and CityScapes
Data Modality	Static RGB image data
Signal Dimensionality	2D image data processed through multi-scale feature maps and segmentation masks
Signal Characteristics	Input images are resized/cropped to 512 × 512 for ADE20K and Pascal VOC2012, and 512 × 1024 for CityScapes; output is dense semantic segmentation mask
Dataset Name	ADE20K, Pascal VOC2012, CityScapes, ImageNet-1K for backbone pretraining
Dataset Type	Public image segmentation benchmark datasets; ImageNet-1K used for pretraining
Dataset Size	NR
Dataset Balance / Split	Uses standard benchmark training/evaluation setting; exact class balance and split details NR
Hybrid Topology	Spiking Mask2Former-style architecture with Spike FPN backbone, Spike-Driven Deformable Transformer Encoder, Spike-Driven Transformer Decoder, and Spike-Driven Mask Embedding
SNN Component	Spiking neurons, Spike FPN, Spike-Driven Mask Embedding, Spike-Driven Deformable Attention, Spike-Driven Cross-Attention, Spike-Driven Self-Attention, NI-LIF neuron
Neuron Model	NI-LIF; also discusses I-LIF and generic spiking neuron layers
Transformer Component	Mask2Former-style pixel decoder, deformable Transformer encoder, Transformer decoder, cross-attention, self-attention, learnable object queries
Attention Type	Spike-driven deformable attention; spike-driven cross-attention; spike-driven self-attention
SNN–Transformer Interface Mechanism	Transformer queries, attention weights, mask embeddings, and multi-scale features are adapted into spike-driven form; the paper avoids directly spiking sparse query features in deformable attention and instead spikes attention weights to preserve query information
Model Depth / Scale	Uses Meta-SpikeFormer backbone with 15M parameters; Spike2Former total parameters reported as 34M; uses T × D settings such as 1 × 2, 2 × 2, 1 × 4, and 4 × 4 depending on experiment
Parameter Count	34M for Spike2Former; Meta-SpikeFormer backbone 15M
Design / Contribution Type	New spiking segmentation architecture; Spike-Driven Deformable Transformer Encoder; Spike-Driven Mask Embedding; Membrane Embedding Shortcut; NI-LIF neuron for stable integer training and spike-driven inference
Input Encoding Method	Direct spike-driven processing with spiking neurons; NI-LIF uses normalized integer activations during training and converts them into binary spike sequences during inference
Encoding Parameters	Virtual timestep / quantization step D; reported T × D values include 1 × 2, 2 × 2, 1 × 4, and 4 × 4; NI-LIF clips rounded membrane value to [0, D] and normalizes by D
Encoding Learned or Fixed	Partially learned/adaptive: model weights and shortcut weight are learned; NI-LIF conversion rule and virtual timestep expansion are fixed mechanisms
Event / Spike Representation	Binary spike sequences during inference; normalized integer spike activations during training; spike attention weights; spike mask embeddings
Temporal Resolution	Uses virtual timestep setting T × D; main ADE20K result uses 1 × 4; Pascal VOC2012 experiments report 1 × 2, 2 × 2, 1 × 4, and 4 × 4
Output Decoding Method	Per-mask classification and binary mask prediction; mask embedding is multiplied with per-pixel embedding to generate segmentation masks
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike degradation, information deficiency in query features, firing-rate reduction in mask embedding, NI-LIF quantization error, and the performance/energy trade-off of timestep and quantization step
Training Paradigm	Supervised direct training with pretrained spiking backbone
Training Method	Direct training using AdamW; Meta-SpikeFormer backbone pretrained on ImageNet-1K for 200 epochs; segmentation model trained with dataset-specific training steps
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	ADE20K: input size 512 × 512, learning rate 2e-4, 160k training steps; CityScapes: input size 512 × 1024, learning rate 2e-3, 90k training steps; Pascal VOC2012: input size 512 × 512, learning rate 2e-3, 80k training steps
Primary Performance Metric	Mean Intersection over Union, mIoU
Primary Metric Value	ADE20K: 46.3% mIoU; CityScapes: 75.2% mIoU; Pascal VOC2012: best 75.4% mIoU
Secondary Metrics	Parameters, power/energy estimate in mJ, ablation mIoU, spike firing rate
Baseline Models Compared	FCN, DeepLabv3, DeepLabv3+, MaskFormer, Mask2Former, Spike Calibration, SpikeFCN, SpikeDeepLab, SpikeFPN, SCGNet-L
Performance Gain Over Baseline	Compared with prior SNN methods, reports +12.7% mIoU and 5.0× efficiency on ADE20K, +14.3% mIoU and 5.2× efficiency on Pascal VOC2012, and +9.1% mIoU and 6.6× efficiency on CityScapes
Statistical Validation	Benchmark comparison and ablation study; no standard deviation, confidence interval, repeated-run analysis, or significance test reported
Ablation Study	Full. Includes direct spiking Mask2Former conversion, SDME, ME-Shortcut, SDTE, ESC, DWConv, spiking query vs spiking attention weight, multi-scale query, vanilla Transformer encoder, I-LIF, and NI-LIF normalization variants
Training Framework / Code Availability	Framework NR; code availability NR
Target Hardware	Neuromorphic hardware motivation; energy estimated under spike-driven sparse addition assumption
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Estimated power/energy consumption in mJ based on prior SNN energy formulation; no measured hardware energy
Latency / Memory / Complexity Evidence	Reports parameter count and estimated power/energy; detailed latency, memory footprint, and real hardware complexity NR
Comparison with ANN Baseline	Yes. Compares with ANN segmentation baselines including FCN, DeepLabv3, DeepLabv3+, MaskFormer, and Mask2Former
Main Contribution / Key Limitation	Contribution: extends Spiking Transformers to high-performance semantic segmentation by redesigning Mask2Former with SDTE, SDME, ME-Shortcut, and NI-LIF to reduce spike degradation and information loss. Limitation: no real hardware validation, no detailed latency/memory measurement, and limited explicit discussion of failure cases or generalization beyond segmentation benchmarks
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P053

Spikeformer: Training high-performance spiking neural network with transformer.

Full Citation	Li, Y., Lei, Y., & Yang, X. (2024). Spikeformer: Training high-performance spiking neural network with transformer. Neurocomputing, 574, Article 127279. https://doi.org/10.1016/j.neucom.2024.127279
University	Peking University
Country	China
Publication Year	2024
Publication Venue	Neurocomputing
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neucom.2024.127279
Publication Type	Journal article
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Static image classification and event-based image/gesture classification
Data Modality	Static RGB images and neuromorphic event-camera data
Signal Dimensionality	2D image frames and 2D spatiotemporal event-frame sequences
Signal Characteristics	DVS-Gesture and DVS-CIFAR10 use 128 × 128 event-frame inputs; ImageNet images are randomly cropped to 224 × 224; DVS data are processed as frame sequences across simulation timesteps
Dataset Name	DVS-Gesture, DVS-CIFAR10, ImageNet
Dataset Type	Public benchmark datasets; mixed static image and neuromorphic/event-based datasets
Dataset Size	DVS-Gesture: 1342 training records from 23 subjects and 288 test examples from 6 subjects; DVS-CIFAR10: 10,000 images, with 9000 training and 1000 testing images; ImageNet: 1.28M training images and 50,000 validation images
Dataset Balance / Split	DVS-CIFAR10: 1000 images per class across 10 classes; DVS-Gesture: 11 classes with train/test subject split; ImageNet uses standard train/validation split
Hybrid Topology	Transformer-based directly trained SNN; convolutional tokenizer produces spike-token embeddings, followed by Transformer blocks with divided spatio-temporal attention, sequence pooling, and classification head
SNN Component	Spiking neurons in convolutional tokenizer, IF/LIF/PLIF spike neurons, spike trains, spiking residual-style CT module, spiking MLP component
Neuron Model	IF, LIF, PLIF; LIAF and PLIAF are also evaluated/discussed
Transformer Component	Vision Transformer-style encoder blocks, temporal multi-head self-attention, spatial multi-head self-attention, sequence pooling, Transformer classifier
Attention Type	Divided Spatio-Temporal Attention; temporal multi-head self-attention followed by spatial multi-head self-attention
SNN–Transformer Interface Mechanism	Convolutional Tokenizer converts input frames into spike-token embeddings; tokens are given spatial and temporal positional embeddings, then processed by Transformer blocks with spatio-temporal attention
Model Depth / Scale	Spikeformer variants include Spikeformer-2, -4, -7, -10, -7L, and -14; reported settings include 2–14 Transformer blocks, 2–8 attention heads, MLP ratio 1–3, and embedding dimensions 128, 256, 384, or 512
Parameter Count	DVS-CIFAR10 Spikeformer-7: 9.28M; ImageNet Spikeformer-7L: 38.75M; DVS-Gesture parameter count NR
Design / Contribution Type	New directly trained Transformer-based SNN; Convolutional Tokenizer for stable training; divided Spatio-Temporal Attention for spatial-temporal modeling; sequence pooling for final classification
Input Encoding Method	AER event preprocessing for DVS-Gesture and DVS-CIFAR10; static ImageNet images are processed as frame inputs through spiking neurons and CT module; exact static image spike encoding rule NR
Encoding Parameters	Surrogate-gradient slope $\alpha = 2$; reset potential Vreset = 0; threshold Vth = 1; DVS-Gesture uses 16 timesteps for best result; DVS-CIFAR10 uses 4 timesteps and also reports 1 timestep; ImageNet uses 4 timesteps
Encoding Learned or Fixed	Partially learned: CT module and model weights are learned; spike neuron threshold/reset settings are fixed in experiments, while PLIF/PLIAF include learnable membrane time constant
Event / Spike Representation	Spike trains, binary spike outputs S[t], membrane potentials, event-frame sequences from DVS data
Temporal Resolution	Main reported timesteps: 16 for DVS-Gesture, 4 for DVS-CIFAR10 and ImageNet; DVS-CIFAR10 also evaluated at 1, 2, 3, and 4 timesteps
Output Decoding Method	Sequence pooling produces weighted token aggregation; classifier output is averaged across timesteps before softmax classification
Encoding / Temporal Information Analysis	Yes. The paper analyzes time-step sensitivity, spatial vs temporal attention, spatio-temporal attention, CT token number, model depth, and energy-efficiency trade-off
Training Paradigm	Supervised direct training
Training Method	Surrogate-gradient training with BPTT
Surrogate Gradient Function	Arctangent surrogate function with $\alpha = 2$
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam for DVS-Gesture and DVS-CIFAR10; SGD for ImageNet
Loss Function	Cross-entropy loss
Training Hyperparameters	DVS-Gesture: Adam, batch size 16, weight decay 0.00015, 150 epochs, 20 warm-up epochs, learning rate 0.001. DVS-CIFAR10: Adam, batch size 32, weight decay 0.0001, 600 epochs, 25 warm-up epochs, initial learning rate 0.001, decay every 192 epochs, label smoothing 0.14. ImageNet: SGD momentum 0.9, batch size 160 on 8 GPUs, weight decay 0.0001, 130 epochs, 5 warm-up epochs, initial learning rate 0.01, cosine LR schedule, label smoothing 0.1, DropPath 0.1
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	DVS-Gesture: 98.96% with 16 timesteps; DVS-CIFAR10: 81.4% with 4 timesteps and 80.3% with 1 timestep; ImageNet: 75.89% with 4 timesteps, and 78.31% with PLIAF setting
Secondary Metrics	Parameter count, time steps, energy efficiency, additions, multiplications, memory accesses, ablation accuracy
Baseline Models Compared	ANN counterpart, ResNet-17/18/19/34/50/101/152, VGG-11/16, SEW-ResNet, tdBN, TA-SNN, PLIF/LIAF baselines, ANN-to-SNN conversion methods

Performance Gain Over Baseline	Spikeformer improves DVS-Gesture accuracy by more than 16% over the vanilla Transformer design without CT; achieves 1.7× energy efficiency over ANN counterpart at 4 timesteps and 6.4× at 1 timestep on DVS-CIFAR10; outperforms ANN counterpart by 3.13% on DVS-Gesture and 0.12% on ImageNet
Statistical Validation	Benchmark evaluation and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes CT module, spiking neuron, sequence pooling, CT token number, CT architecture, model depth, time steps, spatial attention, temporal attention, and spatio-temporal attention
Training Framework / Code Availability	PyTorch and SpikingJelly; experiments use NVIDIA RTX 3090 GPUs; code stated to be publicly available
Target Hardware	Neuromorphic hardware motivation; energy analysis based on 45 nm technology assumptions
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Estimated energy using 32-bit floating-point 45 nm assumptions: MAC = 4.6 pJ, AC = 0.9 pJ, and SRAM memory-access energy interpolation
Latency / Memory / Complexity Evidence	Reports timestep latency, parameter count, additions, multiplications, memory accesses, and energy efficiency; no measured wall-clock latency or memory footprint reported
Comparison with ANN Baseline	Yes. Uses ANN counterpart with same architecture except ReLU activations and ANN-style early convolution; compares accuracy and energy efficiency
Main Contribution / Key Limitation	Contribution: proposes Spikeformer, a directly trained Transformer-based SNN using CT and spatio-temporal attention to improve low-latency SNN classification on static and event-based datasets. Limitation: self-attention still contains dense multiplications, so it is not fully spike-driven; no real neuromorphic hardware testing; large-scale evaluation beyond ImageNet remains limited
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P054

Spiking Tucker Fusion Transformer for Audio-Visual Zero-Shot Learning

Full Citation	Li, W., Wang, P., Xiong, R., & Fan, X. (2024). Spiking Tucker Fusion Transformer for Audio-Visual Zero-Shot Learning. IEEE Transactions on Image Processing, 33, 4840–4852. https://doi.org/10.1109/TIP.2024.3430080
University	Harbin Institute of Technology
Country	China
Publication Year	2024
Publication Venue	IEEE Transactions on Image Processing
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TIP.2024.3430080
Publication Type	Journal article
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Speech, Audio, and Auditory Processing
Specific Task	Audio-visual zero-shot learning and generalized zero-shot learning for video/action classification
Data Modality	Audio features, visual/video features, and textual semantic embeddings
Signal Dimensionality	Multimodal temporal sequence features; audio-visual feature vectors; text embedding space
Signal Characteristics	Audio and visual features are extracted using pretrained SeLaVi; additional experiments use C3D visual features and VGGish audio features; visual feature vector is 4096-dimensional and audio feature vector is 128-dimensional in the alternative feature setting
Dataset Name	VGGSound-GZSL, UCF-GZSL/UCF101, ActivityNet-GZSL; also VGGSound-GZSLcls, UCF-GZSLcls, ActivityNet-GZSLcls for modified split experiments
Dataset Type	Public audio-visual video benchmark datasets for zero-shot and generalized zero-shot learning
Dataset Size	VGGSound-GZSL: 271 classes, 3200 unseen test videos; UCF-GZSL: 48 classes, 845 unseen test videos; ActivityNet-GZSL: 198 classes, 4052 unseen test videos. General dataset descriptions: ActivityNet has more than 20,000 videos; UCF101 has more than 13,000 videos
Dataset Balance / Split	VGGSound-GZSL: 138 train classes, 69 unseen validation classes, 64 unseen test classes; UCF-GZSL: 30 train, 12 unseen validation, 6 unseen test; ActivityNet-GZSL: 99 train, 51 unseen validation, 48 unseen test
Hybrid Topology	Parallel multimodal hybrid topology; spatial-temporal SNN extracts temporal audio/visual representations, Transformer-based latent semantic reasoning extracts semantic information, Tucker fusion integrates SNN and Transformer outputs, and cross-modal Transformer performs joint reasoning
SNN Component	Spatial-temporal SNN, three convolutional SNN blocks, LIF-based layers, Time-Step Factor, Global-Local Pooling, dynamic threshold adjustment
Neuron Model	LIF
Transformer Component	Latent semantic reasoning module with self-attention; cross-modal Transformer with multi-head cross-attention; Transformer-based joint reasoning module
Attention Type	Self-attention in latent semantic reasoning; multi-head cross-attention in joint reasoning; Transformer attention combined with SNN temporal representations
SNN–Transformer Interface Mechanism	SNN outputs temporal binary spike representations, while Transformer modules output floating-point semantic features; temporal-semantic Tucker fusion factorizes and fuses these heterogeneous SNN and Transformer outputs while preserving second-order interactions
Model Depth / Scale	Spatial-temporal SNN has three convolutional SNN blocks; cross-modal Transformer uses 8 heads with 64 dimensions per head; hidden/embedding dimensions include ain = 512, hemb = 512, hhid = 512, hout = 300, hproj = 64
Parameter Count	STFT: 4.16M parameters on VGGSound comparison; GFLOPs 4.27
Design / Contribution Type	New Spiking Tucker Fusion Transformer for audio-visual ZSL; Time-Step Factor; Global-Local Pooling; dynamic threshold adjustment; temporal-semantic Tucker fusion for SNN–Transformer output heterogeneity
Input Encoding Method	Pretrained audio-visual features are processed by spatial-temporal SNN blocks; temporal information is encoded using LIF spike dynamics; exact raw audio/video-to-spike encoding is NR

Encoding Parameters	Time steps analyzed from 2 to 16; best/stable performance around 8 time steps; Tucker rank constraints analyzed with ranks 20, 40, 60, and 80; dynamic thresholds adjusted using SNN output entropy and pooling matrix
Encoding Learned or Fixed	Partially learned/adaptive: TSF, GLP parameter β, dynamic threshold behavior, latent knowledge combiner, and Tucker fusion parameters are learned/adaptive; pretrained feature extraction is fixed
Event / Spike Representation	Binary spike sequences from SNN; membrane potentials; SNN temporal outputs fused with floating-point Transformer semantic features
Temporal Resolution	Time-step-based SNN processing; ablation evaluates 2, 4, 8, and 16 time steps
Output Decoding Method	Audio-visual joint features are projected into textual embedding space; classification is performed by matching projected audio-visual embeddings to class-level text embeddings in ZSL/GZSL setting
Encoding / Temporal Information Analysis	Yes. The paper analyzes time-step effects, spike redundancy, dynamic threshold sensitivity, TSF stability, rank constraint effects, and output heterogeneity between SNN binary spikes and Transformer floating-point features
Training Paradigm	Supervised zero-shot / generalized zero-shot learning using seen-class training and textual semantic embeddings for unseen-class transfer
Training Method	End-to-end training of STFT with pretrained audio-visual embeddings; combines triplet loss, projection loss, and reconstruction loss
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Total loss: Lall = 0.5Lt + 0.5(Lp + Lr), combining triplet loss, projection loss, and reconstruction loss
Training Hyperparameters	Trained for 60 epochs on NVIDIA V100S GPU; learning rate 0.0001; Adam optimizer; ain = 512, hemb = 512, hhid = 512, hout = 300, hproj = 64; cross-modal Transformer uses 8 heads and 64 dimensions per head; dropout rates vary by dataset
Primary Performance Metric	Harmonic Mean for GZSL and ZSL accuracy
Primary Metric Value	VGGSound-GZSL: S = 19.22, U = 6.81, HM = 10.06, ZSL = 8.24; UCF-GZSL: S = 56.47, U = 22.89, HM = 32.58, ZSL = 29.72; ActivityNet-GZSL: S = 22.34, U = 11.73, HM = 15.38, ZSL = 12.91
Secondary Metrics	Seen-class accuracy, unseen-class accuracy, parameter count, GFLOPs, ablation HM, ablation ZSL, t-SNE visualization, qualitative retrieval comparison
Baseline Models Compared	SJE, DEVISE, APN, VAEGAN, CJME, AVGZSLNet, AVCA, TCaF, AVMST, Hyperbolic ZSL variants, MDFT, Transformer+MLP, Spikformer+SNN, only Transformer, only SNN
Performance Gain Over Baseline	Reports HM improvements of approximately 15.4% on VGGSound, 3.9% on UCF101, and 14.9% on ActivityNet; compared with MDFT on VGGSound, STFT improves HM from 8.72 to 10.06 and ZSL from 7.13 to 8.24; on UCF, STFT improves HM from 31.36 to 32.58
Statistical Validation	Benchmark evaluation and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or significance test reported
Ablation Study	Full. Includes ablation on loss items, GLP, TSF, dynamic threshold adjustment, latent knowledge combiner, time steps, Tucker rank constraint, fixed thresholds, loss weights, SNN-only vs Transformer-only, Spikformer+SNN, Transformer+MLP, and latent knowledge slots
Training Framework / Code Availability	Training uses NVIDIA V100S GPU; framework NR; code availability NR
Target Hardware	Neuromorphic hardware motivation only; no specific neuromorphic chip or accelerator targeted
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No direct energy measurement; efficiency discussed through reduced parameters/GFLOPs and SNN spike-based temporal encoding, but hardware energy is NR
Latency / Memory / Complexity Evidence	Parameter comparison on VGGSound: STFT has 4.16M parameters and 4.27 GFLOPs; MDFT has 5.51M parameters and 5.62 GFLOPs, so STFT reduces parameters by about 32% and reduces GFLOPs
Comparison with ANN Baseline	Partial. Compares with Transformer+MLP full-float model, only-Transformer model, and multiple non-spiking audio-visual ZSL baselines
Main Contribution / Key Limitation	Contribution: proposes STFT to couple SNN temporal representations with Transformer semantic reasoning for audio-visual ZSL using TSF, GLP, dynamic thresholds, and Tucker fusion. Limitation: ZSL performance slightly decreases on UCF101, likely due to fixed Tucker rank constraint; no measured neuromorphic hardware energy or deployment
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P055

Spike-SLR: An Energy-efficient Parallel Spiking Transformer for Event-based Sign Language Recognition

Full Citation	Lin, X., Liu, M., Liu, K., & Chen, H. (2024). Spike-SLR: An Energy-efficient Parallel Spiking Transformer for Event-based Sign Language Recognition. Venue NR in uploaded paper. Code: https://github.com/Arktis2022/Spike-SLR
University	Tsinghua University
Country	China
Publication Year	2024
Publication Venue	British Machine Vision Conference
Publisher	British Machine Vision Association
DOI / URL	https://www.scopus.com/pages/publications/105029630129?origin=resultslist
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Event-based isolated sign language recognition
Data Modality	DVS/event-camera data converted into event-frame sequences
Signal Dimensionality	2D spatiotemporal event-frame sequence with polarity channels
Signal Characteristics	Input DVS sequence represented as (I_0 \in R^{T_0 \times 2 \times H_0 \times W_0}); preprocessed into (I \in R^{T \times 2 \times H \times W}); experiments use spatial size 96 × 96, sample length 500 ms, and 10 time steps
Dataset Name	SL-Animals-DVS-4sets, SL-Animals-DVS-3sets, N-LSA64-Both, N-LSA64-Right

Dataset Type	Public event-based sign language datasets; N-LSA64 is converted from LSA64 using v2e
Dataset Size	SL-Animals-DVS: 59 individuals, 19 signs, 4 recording sessions/environments; N-LSA64: 3200 DVS videos, 10 subjects, 5 repetitions, 64 sign-language classes
Dataset Balance / Split	Data split into train/validation/test at 6:2:2; exact class balance NR
Hybrid Topology	SNN-based patch embedding block followed by parallel spike-driven Transformer blocks, soft-attention block, and SNN-based classification head
SNN Component	LIF spike neuron layers, SNN-based patch embedding, spike tokens, spike soft-attention block, CB-S3A block, spiking MLP, SNN classification head
Neuron Model	LIF
Transformer Component	Parallel spike-driven Transformer block; Conv-Based Simplified Spiking Self-Attention; MLP sub-block
Attention Type	Spike soft-attention mask; Conv-Based Simplified Spiking Self-Attention; simplified spike-driven self-attention without softmax
SNN–Transformer Interface Mechanism	Patch embedding converts DVS frames into spatio-temporal spike tokens; CB-S3A generates spike-form Q and K through convolution and uses V without convolution; spike Q/K/V are masked through column summation and Hadamard products; soft-attention spike fire map masks MLP features
Model Depth / Scale	Number of parallel spike-driven Transformer blocks (L = 2); input time step (T = 10); sample length 500 ms; ablation studies test 1–5 blocks and 1, 5, 10, 15, 20 time steps
Parameter Count	0.70M parameters
Design / Contribution Type	New Spike-SLR model; first introduction of spiking Transformer into event-based SLR; parallel spike-driven Transformer block; spike soft-attention block; CB-S3A simplified attention
Input Encoding Method	Event streams are converted into DVS event frames; continuous event frames are randomly selected and spatially cropped; soft-attention branch uses sum-average-repeat operation
Encoding Parameters	Sample length 500 ms; time step 10; input spatial size 96 × 96; 2 polarity/event channels; learning rate 4e–3; batch size 32
Encoding Learned or Fixed	Fixed event-frame preprocessing and cropping; learned patch embedding, attention, MLP, and classifier parameters
Event / Spike Representation	DVS event-frame sequence, binary spike tokens, spike fire-rate map, spike-form Q/K/V tensors, membrane potentials
Temporal Resolution	10 time steps over 500 ms in the main setting; ablation includes 1, 5, 10, 15, and 20 time steps
Output Decoding Method	Final membrane potential output passes through a spike neuron layer and then an SNN-based classification head to produce the sign class
Encoding / Temporal Information Analysis	Yes. The paper analyzes time-step effects, spike firing rates across blocks, soft-attention effects, parallel vs serial Transformer design, and energy consumption
Training Paradigm	Supervised direct SNN training
Training Method	Direct training using SpikingJelly SNN framework; AdamW optimizer; 1cycle learning-rate policy; spatial and temporal random cropping; repeated augmentation within each training batch
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	Single NVIDIA RTX 3090; batch size 32; 240 epochs; learning rate 4e–3; 1cycle learning-rate policy; sample length 500 ms; time step 10; (L = 2) Transformer blocks; train/validation/test split 6:2:2
Primary Performance Metric	Classification accuracy and energy consumption
Primary Metric Value	SL-Animals-DVS-4sets: 89.47%; SL-Animals-DVS-3sets: 90.06%; N-LSA64-Both: 84.69%; N-LSA64-Right: 86.90%; energy: 0.03 mJ per event-frame sequence
Secondary Metrics	Parameter count, FLOPs/SOPs, power in mJ, spike firing rate, time-step sensitivity
Baseline Models Compared	TORE + GoogLeNet, TORE + ResNet18, VoxelGrid + ResNet18, SITS + ResNet18, VK-SITS + ResNet18, EVT, SLAYER, STBP, DECOLLE, SEW ResNet18, Spike-driven EVT
Performance Gain Over Baseline	Improves over EVT by 1.35% on SL-Animals-DVS-4sets and 2.61% on SL-Animals-DVS-3sets; improves over SEW ResNet18 by 4.05% and 0.97%; reduces energy by 99.27% compared with EVT, from 4.13 mJ to 0.03 mJ
Statistical Validation	Benchmark comparison and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes time-step ablation, number of Transformer blocks, parallel vs serial Transformer design, with/without soft attention, and spike firing-rate analysis
Training Framework / Code Availability	SpikingJelly; code available at GitHub: Arktis2022/Spike-SLR
Target Hardware	Neuromorphic/edge SNN deployment motivation; energy comparison references ROLLS for SNNs and Intel Stratix 10 TX for ANN implementations
Hardware Deployment Status	No real hardware deployment reported; energy is estimated analytically
Energy Evidence	Estimated using 12.5 pJ per FLOP and 77 fJ per SOP; reports 0.03 mJ for Spike-SLR versus 4.13 mJ for EVT
Latency / Memory / Complexity Evidence	Reports 0.70M parameters, 0.44G SOPs, 0.03 mJ energy, 500 ms sample length, and 10 time steps; measured wall-clock latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares against ANN event-based SLR baselines including EVT, TORE + ResNet18, TORE + GoogLeNet, VoxelGrid + ResNet18, SITS + ResNet18, and VK-SITS + ResNet18
Main Contribution / Key Limitation	Contribution: proposes Spike-SLR, an energy-efficient parallel Spiking Transformer for event-based sign language recognition using CB-S3A and spike soft-attention, achieving strong accuracy with very low estimated energy. Limitation: no real neuromorphic hardware measurement, venue/DOI not reported in the uploaded paper, and evaluation is limited to isolated event-based SLR datasets
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P056

LDD: High-Precision Training of Deep Spiking Neural Network Transformers Guided by an Artificial Neural Network

Full Citation	Liu, Y., Zhao, C., Jiang, Y., Fang, Y., & Chen, F. (2024). LDD: High-Precision Training of Deep Spiking Neural Network Transformers Guided by an Artificial Neural Network. Biomimetics, 9(7), 413. https://doi.org/10.3390/biomimetics9070413
University	Tsinghua University
Country	China

Publication Year	2024
Publication Venue	Biomimetics
Publisher	MDPI
DOI / URL	https://doi.org/10.3390/biomimetics9070413
Publication Type	Journal article
Database Source	Scopus
Paper Category	Training-efficiency focused
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	High-precision training of deep Spiking Transformer models for image classification
Data Modality	Static RGB image data
Signal Dimensionality	2D images converted into patch/token sequences and multi-time-step spike representations
Signal Characteristics	CIFAR images are 32 × 32 RGB; ImageNet images are 224 × 224 RGB; SNN input is represented as (I \in R^{T \times C \times H \times W}); spike-based Transformer uses sparse binary spike trains across timesteps
Dataset Name	CIFAR10, CIFAR100, ImageNet
Dataset Type	Public image classification benchmark datasets
Dataset Size	CIFAR10: 60,000 images across 10 classes; CIFAR100: 60,000 images across 100 classes; ImageNet: about 1.2M training images, 50,000 validation images, and 100,000 test images
Dataset Balance / Split	CIFAR100: 500 training and 100 test images per class; CIFAR10 and ImageNet use standard benchmark splits; exact CIFAR10 split details NR
Hybrid Topology	ANN-guided SNN Transformer training topology; pretrained ANN Transformer teacher guides SNN Transformer student through layer-by-layer Dual Distillation and Dual SNN temporal projection
SNN Component	Spike-based Transformer student, spiking neurons, Spiking Patch Splitting, Spike-Driven Self-Attention, spike trains, Dual SNN, temporal layers, spike distillation token
Neuron Model	IF/LIF discussed generally; exact neuron model used in experiments NR
Transformer Component	Spikformer, Spike-driven Transformer, Meta-SpikeFormer, ViT-L/32 teacher ANN, Transformer blocks, self-attention, MLP, linear classification head
Attention Type	Spike-Driven Self-Attention in SNN Transformer; self-attention in Dual SNN temporal layers; standard ANN Transformer self-attention in teacher model
SNN–Transformer Interface Mechanism	ANN teacher features are aligned with SNN features layer by layer; Dual SNN projects the SNN's multi-time-step distillation token into the ANN single-time feature space using temporal self-attention and a linear layer
Model Depth / Scale	CIFAR models: Spikformer-4-384, Spiking Transformer-4-384, Meta-Spikformer-4-384; ImageNet models: Spikformer-8-512, Spiking Transformer-8-512, Meta-Spikformer-8-512; deep experiments include Spiking Transformer-10-512, 12-512, and 24-512
Parameter Count	NR
Design / Contribution Type	New training method: layer-by-layer Dual Distillation; Dual SNN temporal projection; layer-wise alignment loss; approximation-error analysis for deep SNN Transformer training
Input Encoding Method	Poisson encoder mentioned in training algorithm; Spiking Patch Splitting uses convolutional patch-splitting and relative position embedding to convert images into spike-token sequences
Encoding Parameters	CIFAR10 uses 2 timesteps; CIFAR100 uses 4 timesteps; ImageNet uses 4 timesteps; CIFAR uses 64 patches of size 4 × 4; ImageNet SPS divides images into 14 × 14 patches
Encoding Learned or Fixed	Partially learned: patch-splitting, relative position embedding, Transformer weights, Dual SNN temporal layers, and alignment mappings are learned; Poisson/input encoding is fixed
Event / Spike Representation	Sparse binary spike trains, multi-time-step spike features, spike-form Q/K/V, SNN distillation token, Dual SNN token
Temporal Resolution	2 timesteps for CIFAR10; 4 timesteps for CIFAR100 and ImageNet in the main experiments
Output Decoding Method	Linear classification head; Dual SNN temporal projection is used for distillation alignment; exact final spike temporal readout rule NR
Encoding / Temporal Information Analysis	Yes. The paper analyzes temporal feature alignment, Dual SNN projection, average/max/temporal projection ablation, layer-wise feature matching, and loss landscape smoothing
Training Paradigm	Supervised direct training with ANN-to-SNN knowledge distillation
Training Method	Layer-by-layer Dual Distillation from pretrained ANN teacher to SNN Transformer student; SNN student is trained from scratch with layer-wise alignment losses
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A; this is not ANN-to-SNN conversion, but direct SNN Transformer training guided by ANN knowledge distillation
Optimizer	Adam / AdamW depending on experiment
Loss Function	(L_{total} = \lambda L_{model} + (1-\lambda)\sum L_{Align}^{(k)}); alignment loss is L2 distance between teacher ANN and Dual SNN intermediate features
Training Hyperparameters	Teacher ANN: ViT-L/32; student SNN trained from scratch. General setting: batch size 256, 200 epochs, learning rate 0.05, momentum 0.9, weight decay 0.0001, attention-loss decay term $\alpha = 0.9$. ImageNet: batch size 128, AdamW, 350 epochs. CIFAR100: batch size 256, AdamW, 300 epochs, scaling factor 0.2
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	Best reported results: CIFAR10 96.1%, CIFAR100 82.3%, ImageNet 80.9% using Meta-SpikeFormer + LDD
Secondary Metrics	Accuracy gain over baseline, model depth, temporal-layer ablation accuracy, loss landscape smoothness
Baseline Models Compared	Spikformer, Spike-driven Transformer, Meta-SpikeFormer, direct training without LDD, temporal projection variants using average/max/temporal layer, deep Spiking Transformer variants
Performance Gain Over Baseline	CIFAR10: Meta-SpikeFormer improves from 95.4% to 96.1%; CIFAR100: Meta-SpikeFormer improves from 79.3% to 82.3%; ImageNet: Spikformer improves from 73.7% to 78.8%, Spike-driven Transformer from 74.6% to 79.4%, and Meta-SpikeFormer from 77.2% to 80.9%
Statistical Validation	Benchmark comparison, ablation experiments, deep-network study, and loss-landscape visualization; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes temporal-layer ablation, average/max/temporal projection comparison, CIFAR10/CIFAR100 ablation, deep SNN Transformer depth study, and loss-surface visualization with/without LDD
Training Framework / Code Availability	Framework NR; code availability NR
Target Hardware	Neuromorphic hardware motivation only; no specific hardware target

Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No direct energy measurement; energy efficiency is discussed conceptually based on SNN sparse spike computation
Latency / Memory / Complexity Evidence	Reports timestep count and ability to train deeper SNN Transformers up to 24 blocks; no measured latency, memory footprint, or computational complexity table reported
Comparison with ANN Baseline	Partial. Uses pretrained ViT-L/32 ANN as teacher and discusses ANN/SNN feature alignment, but main quantitative comparisons are against SNN Transformer baselines rather than matched ANN accuracy baselines
Main Contribution / Key Limitation	Contribution: proposes LDD, a layer-wise ANN-guided distillation method that improves deep SNN Transformer training by aligning ANN and SNN features across temporal and spatial dimensions. Limitation: no real hardware or energy validation, no code/framework details, and evaluation is limited to image classification benchmarks
RQ Mapping	Primary: RQ4; Secondary: RQ2, RQ3, RQ5, RQ1

P057

HydroSNN: Event-Driven Computer Vision with Spiking Transformers for Energy-Efficient Edge Perception in Sustainable Water Conservancy and Urban Water Utilities

Full Citation	Liu, J., Liu, H., & Li, Y. (2026). HydroSNN: Event-Driven Computer Vision with Spiking Transformers for Energy-Efficient Edge Perception in Sustainable Water Conservancy and Urban Water Utilities. Sustainability, 18(3), 1562. https://doi.org/10.3390/su18031562
University	Shanghai Dahua Surveying & Mapping Technology Co., Ltd
Country	China
Publication Year	2026
Publication Venue	Sustainability
Publisher	MDPI
DOI / URL	https://doi.org/10.3390/su18031562
Publication Type	Journal article
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Industrial Monitoring, Fault Diagnosis, and Infrastructure Inspection
Specific Task	Water-surface segmentation, maritime obstacle/object detection, debris/anomaly detection, valve/gate/pump state recognition, event-driven edge perception
Data Modality	RGB frames, asynchronous event-camera streams, optional acoustic/vibration traces, optional SCADA-related signals for weak consistency/diagnostics
Signal Dimensionality	2D RGB images, 2D spatiotemporal event streams, event-token sequences, scalar auxiliary time-series traces
Signal Characteristics	Event stream represented as (x, y, t, p) with polarity; RGB frame $F_t \in R^{H\times W\times 3}$; event packets are divided into micro-steps; default window is 200 ms with 64 micro-steps; token budget is 640 tokens per window; patch size is 16 × 16
Dataset Name	MaSTr1325, SeaDronesSee, DSEC, MVSEC, DAVIS/ECD, DDD20
Dataset Type	Public water-surface/maritime vision datasets and public event-camera benchmark datasets
Dataset Size	MaSTr1325: 1325 labeled images; DDD20: 51 h long-duration data; SeaDronesSee, DSEC, MVSEC, and DAVIS exact dataset sizes NR in the paper
Dataset Balance / Split	MaSTr1325 uses training/validation split; SeaDronesSee uses Object Detection v2 training/test split and MODS tracking split; exact class balance NR
Hybrid Topology	Event-driven spiking Transformer framework with Spiking Temporal Tokenization, Physics-Guided Spiking Attention, Cross-Modal Self-Supervision, and lightweight task decoders
SNN Component	LIF neurons, spike tokens, spiking temporal tokenization, spiking encoder, sparse spike-budget controller, spike-rate regularization, event-driven token emission
Neuron Model	LIF
Transformer Component	12-layer spiking Transformer backbone; physics-guided spiking attention; multi-head attention; task-specific segmentation/detection/state decoders
Attention Type	Physics-Guided Spiking Attention; sparse neighborhood attention over active tokens; continuity-regularized attention
SNN–Transformer Interface Mechanism	STT converts RGB/event/auxiliary inputs into sparse latency-aware spike tokens; PGSA processes active spike tokens using sparse attention with physics-guided continuity bias; task heads decode token outputs into masks, boxes, or states
Model Depth / Scale	12 spiking Transformer layers; hidden dimension 384; 6 attention heads; patch size 16 × 16; token budget 640 per 200 ms window; PGSA neighborhood size K = 12
Parameter Count	NR
Design / Contribution Type	New HydroSNN framework; Spiking Temporal Tokenization; adaptive spike-budget controller; Physics-Guided Spiking Attention; Cross-Modal Self-Supervision; operation-based energy and latency evaluation for water infrastructure monitoring
Input Encoding Method	Event streams are binned into adaptive micro-steps and patch-level currents; RGB frame differences can be thresholded or learned-gated into pseudo-events; patch currents drive LIF-style spiking tokenization
Encoding Parameters	Window Δ = 200 ms; N = 64 micro-steps; δ_{min} = 0.5 ms; δ_{max} = 8 ms; controller gain κ = 0.12; dead-zone d = 0.05; token budget C = 640; target firing rate ρ^* = 0.12; surrogate width γ = 1.0
Encoding Learned or Fixed	Partially learned: event/RGB weighting scalars, token MLP, pseudo-event gating, PGSA probes, and Transformer weights are learned; controller bounds and token-budget mechanism are fixed design choices
Event / Spike Representation	Binary address-event maps, sparse spike tokens, LIF spike emissions, active-token neighborhoods, token-level membrane/spike activity
Temporal Resolution	200 ms input window divided into 64 micro-steps; adaptive bin size constrained between 0.5 ms and 8 ms
Output Decoding Method	Segmentation decoder produces masks; detection head produces bounding boxes; state head uses pooled tokens followed by softmax classification
Encoding / Temporal Information Analysis	Yes. The paper analyzes token budget, spike-rate target, latency jitter, long-run token stability, environmental robustness, event-driven sparsity, and attention–physics agreement
Training Paradigm	Supervised fine-tuning with cross-modal self-supervised pretraining
Training Method	CM-SSL pretraining for 200 epochs, followed by supervised fine-tuning for 60 epochs using AdamW and cosine learning-rate decay

Surrogate Gradient Function	Triangular surrogate derivative with width $\gamma = 1.0$
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Combined supervised loss for segmentation/detection/classification plus masked predictive coding loss, InfoNCE contrastive loss, physics regularization, spike-rate loss, and token-budget latency loss
Training Hyperparameters	Pretraining: 200 epochs. Fine-tuning: 60 epochs, AdamW, learning rate $1 \times 10^{■3}$, weight decay 0.05, cosine decay, batch size 32, $\gamma = 1.0$, target firing rate $\rho^* = 0.12$, K = 12
Primary Performance Metric	mIoU for segmentation, F1/mAP for detection, latency, and normalized operation-based energy proxy
Primary Metric Value	MaSTr1325 mIoU: 88.9%; SeaDronesSee F1@0.5: 83.8%; SeaDronesSee mAP@0.5: 50.9%; DSEC latency/energy: 22.4 ms / 0.17; MVSEC: 23.0 ms / 0.18; DAVIS: 20.8 ms / 0.16; DDD20: 24.6 ms / 0.18
Secondary Metrics	Valve accuracy, false alarms per hour, latency median/L95/jitter, debris detection F1, attention-boundary IoU, continuity residual distribution, token throughput, environmental robustness
Baseline Models Compared	ResNet50-DeepLabV3+, ViT-B/16 with UPer-style decoder, EV-Voxel-CNN, Spikformer-S
Performance Gain Over Baseline	Improves MaSTr1325 mIoU over ResNet50-DeepLabV3+ by +5.7 and over Spikformer-S by +3.4; improves SeaDronesSee F1 over Spikformer-S by +4.5 and mAP by +5.3; reduces normalized energy versus ResNet50 from 1.00 to 0.17 on DSEC
Statistical Validation	Benchmark evaluation, ablation/sensitivity analysis, robustness testing, and qualitative diagnostics; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Partial/full sensitivity analysis. Includes token budget C, PGSA neighborhood size K, target spike rate ρ^*, pseudo-event extraction strategy, environmental regime robustness, latency jitter, false-positive causes, attention–physics agreement, and continuity residuals
Training Framework / Code Availability	Framework NR; code availability NR
Target Hardware	Edge AI / embedded deployment; latency measured on NVIDIA Jetson Orin NX; future neuromorphic hardware deployment discussed
Hardware Deployment Status	Evaluated by replay-based benchmark on embedded Jetson Orin NX; no neuromorphic chip deployment or field utility deployment reported
Energy Evidence	Operation-based energy proxy using SOP/MAC counts with synaptic-to-MAC energy ratio 1:6; no direct device-level power measurement or rail-level monitoring
Latency / Memory / Complexity Evidence	Reports wall-clock latency covering tokenization, encoding, inference, and I/O overhead; DSEC median latency 21.9 ms and L95 28.9 ms; MVSEC median 22.4 ms and L95 29.8 ms; memory footprint NR
Comparison with ANN Baseline	Yes. Compares with RGB ANN baselines including ResNet50-DeepLabV3+ and ViT-B/16, plus EV-Voxel-CNN and Spikformer-S
Main Contribution / Key Limitation	Contribution: proposes HydroSNN for event-driven water-infrastructure perception using STT, PGSA, and CM-SSL, improving accuracy, latency, and estimated energy. Limitation: evaluation mainly uses public benchmarks rather than diverse real utility-site datasets; energy is estimated by operation proxy rather than measured hardware power
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P058

Masked Spikformer: Gaussian based and Random Spike Masking for Energy-Efficient Spiking Transformers

Full Citation	Marsi, O., Ambellouis, S., Mennesson, J., Meurie, C., Fleury, A., & Tatkeu, C. (2025). Masked Spikformer: Gaussian based and Random Spike Masking for Energy-Efficient Spiking Transformers. In 2025 International Conference on Content-Based Multimedia Indexing (CBMI). IEEE. https://doi.org/10.1109/CBMI66578.2025.11339336
University	Univ. Gustave Eiffel
Country	France
Publication Year	2025
Publication Venue	International Conference on Content-Based Multimedia Indexing, CBMI 2025
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/CBMI66578.2025.11339336
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Event-based object classification and gesture recognition using masked Spikformer variants
Data Modality	Neuromorphic event-camera data / DVS event streams
Signal Dimensionality	2D spatiotemporal event streams represented across discrete timesteps
Signal Characteristics	Event-based sparse temporal data; model operates with T = 16 timesteps; spike masking regulates temporal spike activity in SSA and MLP modules
Dataset Name	CIFAR10-DVS; DVS128 Gesture
Dataset Type	Public neuromorphic/event-based benchmark datasets
Dataset Size	CIFAR10-DVS: 9000 training samples and 1000 test samples; DVS128 Gesture: 11 gesture classes from 29 subjects under three lighting conditions
Dataset Balance / Split	CIFAR10-DVS train/test split reported; exact class balance and DVS128 Gesture split NR
Hybrid Topology	Fully spiking Transformer based on Spikformer; includes Spiking Patch Splitting, masked Spiking Self-Attention, masked MLP block, and final linear classification head
SNN Component	LIF neurons, spike activations, spike trains, random spike masking, Gaussian spike masking, Gaussian spike weighting, masked SSA, masked MLP
Neuron Model	LIF
Transformer Component	Spikformer backbone; Spiking Patch Splitting; Spiking Self-Attention; MLP block; linear classifier
Attention Type	Spiking Self-Attention with Randomly Masked SSA and Gaussian-Based Masked SSA variants
SNN–Transformer Interface Mechanism	Spike-based Q and K matrices are masked before attention; attention scores are also masked; spike masking/weighting is applied inside SSA and MLP modules to reduce redundant spike transmission

Model Depth / Scale	Architecture derived from Spikformer with 2 encoder blocks and 16 timesteps
Parameter Count	NR
Design / Contribution Type	Proposes Masked Spikformer with three temporal spike-regulation strategies: Random Spike Masking, Gaussian-Based Spike Masking, and Gaussian-Based Spike Weighting
Input Encoding Method	Uses event-based DVS datasets processed through Spiking Patch Splitting; exact event preprocessing/binning details NR
Encoding Parameters	T = 16 timesteps; RSM masking ratios 20%, 50%, and 70%; GSM uses fixed Gaussian with $\mu = T/2$ and $\sigma = T/4$; GSM variants are centered and inverted; GSW learns Gaussian parameters μ and σ
Encoding Learned or Fixed	RSM is fixed random masking; GSM is fixed deterministic Gaussian-based binary masking; GSW is learned continuous temporal weighting
Event / Spike Representation	Binary spike activations; masked spike tensors; spike-based Q/K/V; temporally weighted spike activations for GSW
Temporal Resolution	16 discrete timesteps
Output Decoding Method	Final linear classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes temporal masking, timestep-dependent spike preservation, energy–accuracy trade-off, and temporal spike redundancy
Training Paradigm	Supervised direct training with surrogate gradients
Training Method	Direct training of Masked Spikformer; masking is applied during both training and inference
Surrogate Gradient Function	Surrogate-gradient training is used, but exact surrogate function NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	192 epochs; AdamW optimizer; T = 16 timesteps; 2 encoder blocks; GSM uses $\mu = T/2$ and $\sigma = T/4$; results averaged over 8 seeds for proposed variants
Primary Performance Metric	Classification accuracy and inference energy consumption
Primary Metric Value	CIFAR10-DVS best proposed result: GSW(i) 83.61 ± 1.26%; DVS128 Gesture best proposed result: RSM 70% 98.52 ± 0.34%; lowest energy with RSM 70%: 2.24 mJ on CIFAR10-DVS and 3.09 mJ on DVS128 Gesture
Secondary Metrics	Energy reduction percentage, timestep count, mean ± standard deviation accuracy, masking ratio, energy in mJ
Baseline Models Compared	MST, TA-SNN, SEW-ResNet, Spikformer
Performance Gain Over Baseline	On CIFAR10-DVS, GSW(i) improves over Spikformer from 80.9% to 83.61%; on DVS128 Gesture, RSM 70% improves over Spikformer from 98.30% to 98.52%; RSM 70% reduces energy from 8.20 mJ to 2.24 mJ on CIFAR10-DVS and from 9.92 mJ to 3.09 mJ on DVS128 Gesture
Statistical Validation	Mean ± standard deviation reported over 8 seeds for proposed variants; no formal significance test or confidence interval reported
Ablation Study	Full method ablation. Compares RSM at 20%, 50%, and 70%; GSM centered/inverted; GSW centered/inverted; reports both accuracy and energy
Training Framework / Code Availability	PyTorch and SpikingJelly; code availability NR
Target Hardware	Energy model based on 45 nm CMOS platform
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Estimated inference energy using MAC and AC operation counts; EMAC = 4.6 pJ and EAC = 0.9 pJ on 45 nm CMOS
Latency / Memory / Complexity Evidence	Reports inference energy and operation-based SOP/MAC formulation; latency and memory footprint NR
Comparison with ANN Baseline	Partial. Includes MST, an ANN-to-SNN converted model, but main comparisons are against SNN/Spiking Transformer baselines
Main Contribution / Key Limitation	Contribution: introduces Masked Spikformer and evaluates random and Gaussian temporal spike-regulation strategies that reduce energy while preserving or improving accuracy. Limitation: evaluated only on two neuromorphic classification datasets; energy is estimated rather than measured on hardware; GSW improves accuracy but provides limited energy reduction due to soft real-valued weighting
RQ Mapping	Primary: RQ3; Secondary: RQ2, RQ4, RQ5, RQ1

P059

HSTNet: Violent Action Detection

Full Citation	Meng, F., Zou, L., Lin, J., & Liu, Z. (2026). HSTNet: Violent Action Detection. Applied Sciences, 16(4), 1825. https://doi.org/10.3390/app16041825
University	Wuhan University
Country	China
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Publication Type	Journal article
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Computer Vision and Image Understanding
Specific Task	Violent action detection and general video action recognition
Data Modality	RGB video frames and residual/difference frames derived from RGB video
Signal Dimensionality	2D RGB video frame sequences; temporal clip input of 16 frames; residual frame sequence for SNN branch
Signal Characteristics	Input video is sampled into contiguous 16-frame clips with stride 16; frames are resized to 224 × 224 and normalized to [0, 1]; RGB frames are fed to the Swin Transformer branch, while residual frames are fed to the SNN branch
Dataset Name	UCF101, HMDB51, Hockey Fight, Movies Fight
Dataset Type	Public action-recognition and violence-detection video datasets

Dataset Size	UCF101: 13,320 video clips across 101 classes; HMDB51: 6766 video clips across 51 classes; Hockey Fight: 1000 clips, including 500 fight and 500 non-fight clips; Movies Fight size NR
Dataset Balance / Split	Hockey Fight is balanced with 500 violent and 500 non-violent clips; UCF101/HMDB51 use public dataset settings; Movies Fight split/balance NR
Hybrid Topology	Dual-branch hybrid architecture: SNN branch based on MS-ResNet for temporal dynamics and inter-frame motion cues; ANN/Transformer branch based on Swin Transformer for spatial structural representation; feature interaction modules enable cross-branch fusion
SNN Component	MS-ResNet branch, LIF neurons, residual-frame temporal modeling, synaptic operations, event-driven sparse computation
Neuron Model	LIF
Transformer Component	Swin Transformer branch, window multi-head self-attention, shifted-window multi-head self-attention, patch partitioning, patch merging, Transformer-based within-scale and cross-scale enhancement modules
Attention Type	Window Multi-Head Self-Attention, Shifted Window Multi-Head Self-Attention, cross-branch feature-interaction attention, cross-scale multi-head attention
SNN–Transformer Interface Mechanism	SNN temporal features and Swin Transformer spatial features interact at corresponding stages through bidirectional feature interaction modules; SNN features attend to Transformer features and Transformer features attend to SNN features, followed by concatenation and 1 × 1 convolution
Model Depth / Scale	SNN branch has 4 hierarchical stages: Conv1 64 channels, Stage1 128, Stage2 256, Stage3 512, Stage4 1024. Transformer branch uses four Swin Transformer stages. Input clip length is 16 frames
Parameter Count	HSTNet: 151.04M parameters on HMDB51 comparison
Design / Contribution Type	New hybrid SNN–Transformer architecture for video action recognition; feature interaction module; within-scale enhancement; cross-scale enhancement; FPN-based multi-scale integration; adaptive classification head
Input Encoding Method	RGB video frames are directly processed by Swin Transformer; residual/difference frames are computed from RGB input and processed by the SNN branch as motion-oriented temporal input
Encoding Parameters	Clip length 16 frames; stride 16; input resolution 224 × 224; pixel normalization to [0, 1]; batch size 8; initial learning rate 3 × 10■■
Encoding Learned or Fixed	Residual-frame generation is fixed preprocessing; SNN, Transformer, feature-interaction, enhancement, and classification modules are learned
Event / Spike Representation	Residual-frame-driven spiking representations in MS-ResNet branch; spike-based temporal activations; exact spike tensor format NR
Temporal Resolution	16-frame video clip input; exact SNN internal timestep setting NR
Output Decoding Method	Classification head fuses SNN branch features, ANN branch features, and FPN features using adaptive weighting, then combines fused prediction with ANN branch prediction to produce final action label
Encoding / Temporal Information Analysis	Partial. The paper analyzes residual-frame temporal modeling, class-wise HMDB51 recognition behavior, violent-action confusion matrices, and module ablations, but does not deeply analyze spike timing or temporal spike information
Training Paradigm	Supervised learning with pretrained backbones for UCF101 and HMDB51
Training Method	Fine-tuning from pretrained backbones; supervised training for action classification using sampled video clips
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	PyTorch 1.8.0, Python 3.9, Ubuntu 20.04, NVIDIA A100-SXM4 GPU, CUDA 11.1, AMD EPYC 7543 CPU; 16-frame clips; stride 16; 224 × 224 input; batch size 8; initial learning rate 3 × 10■■; cosine annealing schedule; training for 5 epochs
Primary Performance Metric	Classification accuracy
Primary Metric Value	HMDB51: 63.40%; UCF101: 90.93%; Hockey Fight: 96.50% accuracy and 0.97 recall; Movies Fight: 98.00% accuracy and 1.00 recall
Secondary Metrics	Recall, FLOPs, SYOps, parameter count, class-wise accuracy, confusion matrix
Baseline Models Compared	ResNet18, ResNet50, C3D, Res3D, ViT-B, ViT-L, Swin-B, MS-ResNet, ReSpike-ResNet18, ReSpike-ResNet50, ViF, ViF+OViF
Performance Gain Over Baseline	On HMDB51, HSTNet improves over ResNet18 by 12.70% and ResNet50 by 9.12%; on UCF101, it improves over ResNet18 by 11.78% and ResNet50 by 8.73%; on Hockey Fight, it improves over ResNet50 from 95.50% to 96.50%; on Movies Fight, it improves over ResNet50 from 96.00% to 98.00%
Statistical Validation	Benchmark comparison, ablation studies, class-wise accuracy plots, and confusion matrices; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes dynamic feature interaction, within-scale enhancement, cross-scale enhancement, classification head, and fusion strategy comparison using Add/Concat at three fusion positions
Training Framework / Code Availability	PyTorch; code availability NR
Target Hardware	Resource-constrained/edge surveillance deployment is discussed conceptually; no specific neuromorphic hardware target
Hardware Deployment Status	No hardware deployment reported; training/evaluation performed on GPU
Energy Evidence	No direct energy measurement; computational efficiency is discussed using FLOPs, SYOps, and parameter count
Latency / Memory / Complexity Evidence	HSTNet reports 377.56G FLOPs, 9.32G SYOps, and 151.04M parameters on HMDB51 comparison; measured latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares against 2D ANN, 3D ANN, and Transformer baselines including ResNet18, ResNet50, C3D, Res3D, ViT-B, ViT-L, and Swin-B
Main Contribution / Key Limitation	Contribution: proposes HSTNet, a dual-branch MS-ResNet/Swin Transformer hybrid that fuses temporal spiking features and spatial Transformer features for action and violence detection. Limitation: uses RGB/residual frames only rather than true event-camera input; no measured energy or hardware deployment; training details such as optimizer, loss, and surrogate function are incomplete
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P060

SpikingBrain: Spiking Brain-inspired Large Models

Full Citation	Pan, Y., Feng, Y., Zhuang, J., Ding, S., Xu, H., Liu, Z., Sun, B., Chou, Y., Qiu, X., Deng, A., Hu, A., Wang, S., Zhou, P., Yao, M., Wu, J., Yang, J., Sun, G., Xu, B., & Li, G. (2025). SpikingBrain: Spiking Brain-inspired Large Models. arXiv:2509.05276v3.

University	Chinese Academy of Sciences
Country	China
Publication Year	2025
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Publication Type	Journal article
Database Source	Scopus
Paper Category	Training-efficiency focused
E&QA; Score	4
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Efficient LLM continual pretraining, supervised fine-tuning, long-context inference, instruction following, commonsense reasoning, general benchmark evaluation, and CPU/GPU deployment
Data Modality	Text/token sequences
Signal Dimensionality	1D token sequences; training context up to 128k tokens; inference evaluation up to 4M-token sequences for SpikingBrain-7B
Signal Characteristics	Long-context language-token streams; activations in linear projection layers are converted into integer spike counts and expandable sparse spike trains; supported spike coding includes binary, ternary, and bitwise formats
Dataset Name	Matrix, Infinity Instruct foundational dataset, Infinity Instruct chat dataset, DeepSeek-R1 distilled reasoning data, MMLU, CMMLU, C-Eval, ARC-C, HellaSwag, PIQA, Winogrande, LongBench, NQ, TriviaQA, QuALITY, IFEval
Dataset Type	Public/prepared LLM pretraining, SFT, reasoning, general knowledge, commonsense reasoning, and long-context benchmark datasets
Dataset Size	Continual pretraining uses about 150B tokens for SpikingBrain-7B and about 160B tokens for SpikingBrain-76B; SFT uses 500k foundational samples, 500k chat samples, and 150k reasoning samples; calibration uses 128 text samples
Dataset Balance / Split	Reasoning SFT data uses 1:1 Chinese–English ratio; other dataset balance/split details NR
Hybrid Topology	Brain-inspired LLM architecture combining hybrid efficient attention, adaptive-threshold spiking activation coding, and MoE/FFN modules; SpikingBrain-7B uses inter-layer hybrid linear attention and sliding-window attention; SpikingBrain-76B uses intra-layer hybrid linear, sliding-window, and full attention with sparse MoE
SNN Component	Adaptive-threshold spiking neurons, integer spike counts, sparse spike trains, binary/ternary/bitwise spike coding, event-driven spike-based computation for linear projection activations
Neuron Model	Adaptive-threshold simplified IF-style spiking neuron with soft reset; LIF is discussed as background but the proposed neuron removes decay for efficient large-model use
Transformer Component	Qwen2.5-derived LLM architecture; linear attention, sliding-window attention, full softmax attention, FFN/SwiGLU, sparse MoE, sink tokens, router, experts, embedding and language-model head
Attention Type	Hybrid efficient attention: Gated Linear Attention, Sliding Window Attention, and Full Softmax Attention; SpikingBrain-7B uses LA + SWA inter-layer hybridization; SpikingBrain-76B uses LA + SWA/FA intra-layer hybridization
SNN–Transformer Interface Mechanism	Continuous activations in Transformer linear projection layers are converted into integer spike counts using adaptive thresholds, then can be expanded into sparse spike trains for event-driven hardware; GPU execution uses compact integer activation format
Model Depth / Scale	SpikingBrain-7B: 7B parameters, 14 linear-attention and 14 SWA layers implied by figure. SpikingBrain-76B-A12B: 76B total parameters, 12B activated parameters, 28 layers, 16 routed experts, 1 shared expert, top-1 routing, 128 sink tokens, 7 dense FFN layers
Parameter Count	SpikingBrain-7B: 7B; SpikingBrain-76B-A12B: 76B total / 12B activated; compressed CPU-side model: 1B
Design / Contribution Type	New family of brain-inspired large models; hybrid-linear LLM architecture; conversion-based training pipeline; adaptive-threshold spike coding; MoE upcycling; MetaX non-NVIDIA training and inference adaptation; CPU-side deployment
Input Encoding Method	Text tokenization follows LLM pipeline; internal activation encoding uses adaptive-threshold spiking to map continuous activations into integer spike counts and sparse spike trains
Encoding Parameters	Adaptive threshold (V_{th}(x)=\frac{1}{k}mean(abs(x))); spike coding formats include binary {0,1}, ternary {-1,0,1}, and bitwise coding; weights and KV cache quantized to INT8; 128 calibration samples; average spike count 1.13 for SpikingBrain-7B; 69.15% sparsity reported
Encoding Learned or Fixed	Partially fixed/adaptive: threshold is dynamically computed from activation statistics; spike coding rules are fixed; quantization parameters are calibrated; model weights are trained/converted
Event / Spike Representation	Integer spike counts, sparse spike trains, binary spikes, ternary excitatory/inhibitory spikes, bitwise spike codes, inactive channels, spike-count distribution
Temporal Resolution	Virtual spike timesteps used during spike-coding expansion; bitwise ternary coding uses a 3-step window for sparsity analysis; long-context token sequence lengths include 8k, 32k, 128k, 256k, 512k, 1M, 2M, and 4M
Output Decoding Method	Standard autoregressive LLM token decoding through language-model head; CPU-side deployment generates output tokens using quantized inference
Encoding / Temporal Information Analysis	Yes. The paper analyzes adaptive threshold behavior, spike-count distribution, binary/ternary/bitwise coding, firing rate, inactive channels, sparsity, TTFT scaling, CPU decoding speed, and long-context memory behavior
Training Paradigm	Conversion-based continual pretraining and supervised fine-tuning from an open-source Transformer checkpoint
Training Method	Multi-stage conversion pipeline: attention conversion, MoE upcycling, spike-coding calibration, continual pretraining with long-context extension, and three-stage supervised fine-tuning
Surrogate Gradient Function	N/A; spike coding uses straight-through estimation / skipped gradient through spiking approximation rather than conventional surrogate-gradient SNN training
ANN-to-SNN Conversion Method	Partial / hybrid conversion: converts Transformer attention and FFN modules into hybrid-linear/MoE form and converts activations into spike-compatible integer/spike representations; not a conventional full ANN-to-SNN conversion
Optimizer	NR
Loss Function	Standard LLM pretraining/SFT losses implied; auxiliary MoE load-balancing loss mentioned; exact loss formulas NR
Training Hyperparameters	CPT: 100B tokens at 8k sequence length, then 20B–30B tokens each at 32k and 128k; total about 150B tokens. SFT: 500k foundational samples, 500k chat samples, 150k reasoning samples, each at 8k sequence length. SpikingBrain-7B 128k training uses 32-way data parallelism and 8-way sequence parallelism
Primary Performance Metric	Downstream benchmark performance, long-context TTFT, inference speed, spike sparsity, and energy-efficiency estimate
Primary Metric Value	SpikingBrain-7B pretraining: MMLU 65.84, CMMLU 71.58, C-Eval 69.80. SpikingBrain-76B pretraining: MMLU 73.58, CMMLU 78.83, C-Eval 78.89. SpikingBrain-7B reaches over 100× estimated TTFT speedup at 4M tokens; spike sparsity about 69.15%
Secondary Metrics	ARC-C, HellaSwag, PIQA, Winogrande, LongBench tasks, NQ, TriviaQA, QuALITY, IFEval, MFU, TGS, decoding speed, average spike count, inactive-channel ratio, energy per operation

Baseline Models Compared	Qwen2.5-7B, Falcon-Mamba-7B, Mistral-7B, Zamba-v1-7B, Llama3.1-8B, Jamba-52B-A12B, Mixtral-8×7B, Llama2-70B, Gemma2-27B, Llama3.2-1B
Performance Gain Over Baseline	SpikingBrain-7B achieves 26.5× TTFT speedup over Qwen2.5 at 1M tokens and over 100× estimated speedup at 4M tokens; SpikingBrain-1B achieves 4.04×, 7.52×, and 15.39× CPU decoding speedup at 64k, 128k, and 256k output lengths; event-driven spiking estimate gives 43.48× energy gain over FP16 MAC and 6.76× over INT8 MAC
Statistical Validation	Benchmark comparisons and efficiency measurements are reported; TTFT measured over 10 runs for sequence-parallel inference; no formal statistical significance test or confidence interval reported
Ablation Study	Partial. Includes before/after spiking scheme comparison, spike-count distribution analysis, coding-scheme visualization, long-context speed comparison, CPU deployment comparison, and architecture-level comparison against hybrid/Transformer/MoE baselines
Training Framework / Code Availability	Colossal-AI for SpikingBrain-7B; Megatron for SpikingBrain-76B; HuggingFace and vLLM inference support; llama.cpp CPU deployment for 1B model; MetaX C550 GPUs; code publicly available for SpikingBrain-7B
Target Hardware	MetaX C550 non-NVIDIA GPU cluster; CPU mobile inference framework; future neuromorphic/asynchronous hardware motivated by sparse spike trains
Hardware Deployment Status	Trained and deployed on MetaX GPU cluster; CPU-side inference demonstrated for 1B model; no actual neuromorphic chip deployment reported
Energy Evidence	Estimated energy using published 45 nm operation costs: FP16 MAC 1.5 pJ, INT8 MAC 0.23 pJ, INT8 addition about 0.03 pJ; proposed event-driven spiking + INT8 scheme estimated at about 0.034 pJ per effective MAC
Latency / Memory / Complexity Evidence	Reports linear or near-linear complexity; SpikingBrain-7B constant inference memory with sequence length; TTFT around 1015–1073 ms from 256k to 4M tokens; 1B CPU model shows constant decoding throughput; MetaX training reports TGS 1558 and MFU 23.4%
Comparison with ANN Baseline	Yes. Compares against Transformer, hybrid, MoE, and LLM baselines including Qwen2.5, Llama, Mistral, Mixtral, Jamba, Gemma, and Llama3.2
Main Contribution / Key Limitation	Contribution: proposes SpikingBrain, a brain-inspired LLM family combining hybrid-linear attention, MoE, adaptive-threshold spike coding, and non-NVIDIA large-scale training. Limitation: spiking energy benefits are estimated rather than measured on neuromorphic hardware; primary model remains GPU/CPU-executed using integer activation formats; preprint status and no formal significance testing
RQ Mapping	Primary: RQ4; Secondary: RQ2, RQ3, RQ5, RQ1

P061

A hybrid Spiking Neural Network–Transformer architecture for motor imagery and sleep apnea detection

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University	University of West Bohemia in Pilsen
Country	Czechia
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Publication Type	Journal article
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	Sleep apnea detection from single-lead ECG and binary motor imagery classification from EEG
Data Modality	ECG time-series signals and EEG signals
Signal Dimensionality	1D physiological time-series; single-lead ECG segments; multi-channel EEG recordings
Signal Characteristics	ECG: RR intervals and R-peak amplitudes extracted from single-lead ECG; PhysioNet uses 5-min segments with 900 RRI and 900 R-amplitude points. EEG: BCI-IV-2a uses 22 EEG channels and 3 EOG channels sampled at 250 Hz; mu and beta bands are extracted
Dataset Name	PhysioNet Apnea-ECG, UCDDB, BCI Competition IV 2a
Dataset Type	Public biomedical signal datasets; ECG sleep-apnea datasets and EEG motor-imagery BCI dataset
Dataset Size	PhysioNet Apnea-ECG: 70 recordings, released set 17,125 segments and withheld set 17,303 segments before faulty-segment removal; after preprocessing 33,654 segments remained. UCDDB: 25 subjects, with 4 subjects excluded due to no apnea events. BCI-IV-2a: 9 subjects, 2 sessions per subject, 288 trials per session
Dataset Balance / Split	PhysioNet: released set for training/tuning and withheld set for testing; after preprocessing released set has 16,709 segments and withheld set has 16,945 segments. UCDDB: 8:1:1 train/validation/test split with oversampling for apnea class. BCI-IV-2a: evaluation session re-split into 80% training and 20% testing
Hybrid Topology	CNN front-end followed by Spiking-Transformer encoder/decoder using Spiking Multi-Head Attention; spiking neurons are inserted into attention and feed-forward components
SNN Component	Spiking neuron layer, threshold-based spike activation, surrogate-gradient spike approximation, spiking feed-forward cells, spiking attention output
Neuron Model	Threshold-based spiking neuron with sigmoid surrogate gradient; exact IF/LIF neuron type NR
Transformer Component	Transformer encoder, Transformer decoder, multi-head self-attention, positional encoding, layer normalization, feed-forward network
Attention Type	Spiking Multi-Head Attention; scaled dot-product self-attention followed by spiking neuron activation
SNN–Transformer Interface Mechanism	Attention output from standard scaled dot-product attention is passed through a spiking neuron layer, producing discrete/event-driven attention activations; feed-forward blocks also include spiking neuron cells
Model Depth / Scale	CNN front-end has 3 convolutional layers with 32, 64, and 64 filters, kernel size 17, max pooling, and batch normalization; exact number of Transformer encoder/decoder layers NR
Parameter Count	Sleep apnea SpiTranNet model: 189K parameters on PhysioNet comparison; MI model reported around 2M parameters
Design / Contribution Type	New SpiTranNet architecture integrating SNN and Transformer through SMHA for physiological signal classification across ECG and EEG tasks

Input Encoding Method	ECG: R-peak detection, RRI and R-amplitude extraction, normalization, FIR bandpass filtering, median filtering, cubic interpolation, and segmented physiological input. EEG: common average referencing and bandpass filtering of mu/beta bands
Encoding Parameters	PhysioNet: 5-min segments, 900 RRI and 900 R-amplitude points. UCDDB: 11-second windows with 10-second overlap. EEG: CAR spatial filtering, mu band 8–14 Hz, beta band 15–30 Hz, sampling rate 250 Hz
Encoding Learned or Fixed	Fixed preprocessing/encoding pipeline; CNN, SMHA, Transformer, and classifier parameters are learned
Event / Spike Representation	Threshold-generated spike outputs; smoothed spike probability during training; sparse spiking activations inside attention and feed-forward blocks
Temporal Resolution	PhysioNet ECG sampled at 100 Hz; UCDDB ECG sampled at 128 Hz; BCI-IV-2a EEG sampled at 250 Hz; UCDDB window length 11 seconds; exact SNN timestep count NR
Output Decoding Method	Softmax classification output for SA/normal and MI class prediction
Encoding / Temporal Information Analysis	Partial. The paper analyzes ECG segment/window construction, EEG frequency bands, training curves, confusion matrices, and ablation between SNN, Transformer, and SpiTranNet, but does not deeply analyze spike-timing information
Training Paradigm	Supervised learning
Training Method	Direct training of hybrid CNN–Spiking Transformer model using Adam and surrogate-gradient spike approximation
Surrogate Gradient Function	Sigmoid surrogate gradient with temperature parameter
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Binary cross-entropy for sleep apnea classification; categorical cross-entropy for motor imagery classification
Training Hyperparameters	Initial learning rate 0.001; ReduceLROnPlateau learning-rate scheduler; early stopping after 30 epochs without validation-loss improvement; batch size 16; 100 epochs; five-fold cross-validation for PhysioNet; hold-out 8:1:1 split for UCDDB
Primary Performance Metric	Classification accuracy
Primary Metric Value	PhysioNet Apnea-ECG: 95.0% per-segment accuracy and 100% per-recording accuracy. UCDDB: 99.4% accuracy. BCI-IV-2a: 88.4% average binary MI accuracy
Secondary Metrics	Sensitivity, specificity, precision, recall, F1-score, AUC, Cohen's kappa, Pearson correlation coefficient, parameters, GPU memory, training time
Baseline Models Compared	SNN, Transformer, LS-SVM, DNN-HMM, LeNet-5 CNN, CNN-LSTM, CNN-BiGRU, ResNet18, RAFNet, FT-EDBN, AlexNet+LSTM, DM-IACNN, MPCNN, CNN-Transformer, CSAC-Net, SmartMatch, SE-MSResNet, TP-CL, ATCNet, SCNet, CTNet, CLTNet, MSCFormer, CIACNet
Performance Gain Over Baseline	On PhysioNet per-segment, SpiTranNet improves over standalone Transformer from 85.7% to 95.0% and over SNN from 78.2% to 95.0%. On UCDDB, it improves over CNN-LSTM from 93.7% to 99.4%. On BCI-IV-2a, it slightly exceeds SCNet, 88.4% vs 88.2%
Statistical Validation	Uses five-fold cross-validation on PhysioNet and benchmark comparisons; no confidence interval, significance test, or repeated-seed statistical testing reported
Ablation Study	Partial. Compares standalone SNN, standalone Transformer, and SpiTranNet on PhysioNet Apnea-ECG; includes parameter, memory, training-time, per-segment, and per-recording results
Training Framework / Code Availability	Framework NR; code availability NR
Target Hardware	Real-time and resource-constrained biomedical/BCI applications are discussed; no specific neuromorphic hardware targeted
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	No measured energy result; efficiency is discussed through spiking sparsity concept, parameter count, GPU memory usage, and training time
Latency / Memory / Complexity Evidence	PhysioNet comparison reports SpiTranNet with 189K parameters, 24% GPU memory, and 5 seconds/epoch; standalone Transformer uses 3.8M parameters, 32% memory, and 23 seconds/epoch; SNN uses 231K parameters, 20% memory, and 1 second/epoch
Comparison with ANN Baseline	Yes. Compares against Transformer, CNN, CNN-RNN, CNN-attention, and classical ML baselines
Main Contribution / Key Limitation	Contribution: proposes SpiTranNet, a compact SNN–Transformer architecture using SMHA for both ECG-based sleep apnea detection and EEG-based motor imagery classification. Limitation: no real hardware or measured energy evaluation; MI task is binary rather than full four-class BCI-IV-2a classification; exact Transformer depth and code availability are NR
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P062

Quantized Spike-Driven Transformer

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Paper Category	Core architecture
E&QA; Score	5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Image classification, object detection, semantic segmentation, and transfer learning
Data Modality	Static image data and event-camera/DVS data
Signal Dimensionality	2D image/token representations; event-stream data converted into frame-like temporal slices for CIFAR10-DVS
Signal Characteristics	Uses binary spike activations and low-bit quantized weights; QSD-Transformer uses 4-bit, 3-bit, or 2-bit weights with 1-bit spike activities during inference; CIFAR10-DVS event streams are divided into temporal slices and compressed into three-channel frames representing positive, negative, and all events
Dataset Name	ImageNet-1K, COCO 2017, ADE20K, CIFAR-10, CIFAR-100, CIFAR10-DVS

Dataset Type	Public image classification, object detection, semantic segmentation, transfer-learning, and neuromorphic/event-based benchmark datasets
Dataset Size	ImageNet-1K: about 1.3M training images and 50,000 validation images; COCO 2017: 118K training images and 5K validation images; ADE20K: over 20K training images and 2K validation images; CIFAR-10/100: 50,000 training and 10,000 test images; CIFAR10-DVS: 10K event streams
Dataset Balance / Split	Uses standard benchmark train/validation/test splits; exact class-balance details NR
Hybrid Topology	Quantized Spike-Driven Transformer based on SD-Transformer v2; combines Conv-based SNN blocks, Transformer-based SNN blocks, Quantized Spike-Driven Self-Attention, Information-Enhanced LIF neurons, and fine-grained ANN-to-SNN distillation
SNN Component	LIF neurons, Information-Enhanced LIF, binary spike activities, multi-bit spike training, spike-driven inference, Conv-based SNN blocks, Transformer-based SNN blocks, spike-driven self-attention
Neuron Model	LIF and Information-Enhanced LIF
Transformer Component	Spike-driven Transformer v2 backbone; Q-SDSA module; Transformer-based SNN block; Q/K/V projections; Channel FC; classification head; Mask R-CNN detector and Semantic FPN used for downstream tasks
Attention Type	Quantized Spike-Driven Self-Attention; spike-driven linear attention; softmax-free SDSA
SNN–Transformer Interface Mechanism	Q, K, and V are generated through quantized convolution layers and converted into spike representations through IE-LIF; Q-SDSA computes spike-driven attention using binary spike activities and low-bit weights; FGD aligns Q/K/V similarity distributions with an ANN Transformer teacher
Model Depth / Scale	Three model scales: 1.8M, 3.8M, and 6.8M parameters for ImageNet classification. Architecture includes four stages with Conv-based SNN blocks in early stages and Transformer-based SNN blocks in later stages; Transformer-based stages use 6 blocks in stage 3 and 2 blocks in stage 4
Parameter Count	ImageNet QSD-Transformer: 1.8M, 3.9M, and 6.8M; COCO detection: 16.9M and 34.9M; ADE20K segmentation: 3.3M and 9.6M
Design / Contribution Type	New quantized spike-driven Transformer baseline; bi-level optimization for spike information distortion; Information-Enhanced LIF neuron; membrane-potential rectify function; fine-grained distillation between Q-SDSA and ANN attention; low-bit weight quantization for Spiking Transformers
Input Encoding Method	Static images are processed through downsampling convolution and Conv-based SNN blocks; CIFAR10-DVS event streams are split into T slices and compressed into three-channel frames for positive, negative, and all events
Encoding Parameters	Training/inference timestep = 1/4 for ImageNet and CIFAR; inference spike activity is 1-bit; weights use 4-bit, 3-bit, or 2-bit quantization; IE-LIF uses multi-bit values during training and binary spikes during inference; maximum integer value b = 4
Encoding Learned or Fixed	Partially learned: quantized weights, scaling factors, MPRF parameters, IE-LIF-related parameters, and network weights are learned; spike-driven inference rule and quantization bit-width are fixed design choices
Event / Spike Representation	Binary spike activities during inference; multi-bit spike values during training; quantized spike tensors for Q/K/V; sparse spike-driven representations; CIFAR10-DVS frame representation from event slices
Temporal Resolution	Main inference uses 4 timesteps; CIFAR10-DVS transfer learning uses 10 timesteps; IE-LIF training uses 1 timestep with multi-bit activity, then expands to 4 virtual binary spike timesteps during inference
Output Decoding Method	Classification head for ImageNet/CIFAR/CIFAR10-DVS; Mask R-CNN detection head for COCO; Semantic FPN segmentation head for ADE20K
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike information distortion, Q-SDSA bimodal distribution, IE-LIF entropy improvement, activity-bit versus timestep sensitivity, attention distribution alignment, and energy–accuracy trade-off
Training Paradigm	Supervised direct training with quantization-aware training and fine-grained knowledge distillation
Training Method	Direct SNN training using surrogate gradients and STBP; LSQ-style weight quantization; straight-through estimator for quantization and IE-LIF; bi-level optimization using IE-LIF and FGD from ANN teacher
Surrogate Gradient Function	Rectangle surrogate function for spiking neuron backpropagation; STE used for non-differentiable quantization and IE-LIF operations
ANN-to-SNN Conversion Method	N/A; this is direct training / quantization-aware training of a spike-driven Transformer, not conventional ANN-to-SNN conversion
Optimizer	LAMB for ImageNet and CIFAR classification; AdamW for COCO detection, ADE20K segmentation, and transfer learning
Loss Function	Cross-entropy loss plus fine-grained distillation loss: (L = L_{CE} + \lambda L_{FGD}), with (\lambda = 2); downstream detection/segmentation losses follow Mask R-CNN and Semantic FPN frameworks but exact formulas NR
Training Hyperparameters	ImageNet: 300 epochs, batch size 1568, LAMB, base LR 6e–4, cosine decay, 10 warmup epochs, weight decay 0.05, resolution 224×224, 8 × A100 GPUs. CIFAR: 100 epochs, batch size 256, LAMB, LR 6e–4, resolution 128×128, mixup 0.8, cutmix 1.0. COCO: 24 epochs, batch size 12, AdamW, LR 1e–4, 4 × A100 GPUs. ADE20K: 160K iterations, batch size 20, AdamW, LR 1e–4, 1500 warmup iterations. Transfer learning: 100 epochs, batch size 128, AdamW, LR 1e–4
Primary Performance Metric	Top-1 accuracy for classification; mAP@0.5 for detection; mIoU for segmentation; power/energy estimate
Primary Metric Value	ImageNet: 80.3% top-1 accuracy with 6.8M parameters and 8.7 mJ power. COCO: 57.0% mAP@0.5 with 34.9M parameters and 117.2 mJ. ADE20K: 40.5% mIoU with 9.6M parameters and 37.9 mJ. Transfer learning: CIFAR-10 98.4%, CIFAR-100 87.6%, CIFAR10-DVS 89.8%
Secondary Metrics	Parameter count, model size, weight bit-width, activity bit-width, timestep, power in mJ, accuracy loss/gain, mAP, mIoU, training time
Baseline Models Compared	CAFormer, QCFS, MST, SEW-ResNet, MS-ResNet, Spikformer, SD-Transformer, SpikingResformer, SD-Transformer v2, DETR, Spiking-Yolo, Spike Calibration, EMS-SNN, SegFormer, DeepLab-V3
Performance Gain Over Baseline	Compared with SD-Transformer v2-L on ImageNet, QSD-Transformer improves accuracy from 79.7% to 80.3% while reducing parameters from 55.4M to 6.8M and power from 52.4 mJ to 8.7 mJ. On COCO, it improves mAP@0.5 over SD-Transformer v2-M from 51.2% to 57.0%. On ADE20K, it improves mIoU over SD-Transformer v2-M from 35.3% to 40.5%
Statistical Validation	Transfer-learning experiments report mean ± standard deviation over five random seeds; main ImageNet/COCO/ADE20K results do not report formal significance testing or confidence intervals
Ablation Study	Full. Includes quantizing different modules, IE-LIF alone, IE-LIF + FGD, 4/3/2-bit weight quantization, Spikformer scalability test, activity-bit versus timestep ablation, and attention-distribution visualization before/after optimization
Training Framework / Code Availability	MMDetection and MMSegmentation converted into spiking versions for detection/segmentation; experiments use A100 and V100 GPUs; code stated to be available / released on GitHub after review
Target Hardware	Resource-constrained and neuromorphic hardware motivation; future FPGA deployment discussed
Hardware Deployment Status	No real hardware deployment reported; energy is estimated theoretically
Energy Evidence	Estimated energy using 45 nm operation costs: MAC = 4.6 pJ and AC = 0.9 pJ; memory access energy is not included
Latency / Memory / Complexity Evidence	Reports model size, parameter count, power estimate, SOP/FLOP-based energy, training time, and computational complexity. Q-SDSA complexity is (O(ND^2)), compared with MHSA complexity (O(N^2D)); no measured hardware latency or memory-access energy
Comparison with ANN Baseline	Yes. Compares against ANN Transformer/CAFormer, DETR, SegFormer, DeepLab-V3, and ANN teacher distributions; also compares against multiple SNN Transformer baselines

Main Contribution / Key Limitation	Contribution: proposes QSD-Transformer, a low-bit spike-driven Transformer with IE-LIF and FGD to reduce spike information distortion and achieve high accuracy with much lower power/model size. Limitation: largest Spikformer-v2 quantization not evaluated due to unavailable code/weights; no real hardware deployment; memory-access energy is excluded from theoretical energy calculation
RQ Mapping	Primary: RQ4; Secondary: RQ2, RQ3, RQ5, RQ1

P063

TIM: An Efficient Temporal Interaction Module for Spiking Transformer

Full Citation	Shen, S., Zhao, D., Shen, G., & Zeng, Y. (2024). TIM: An Efficient Temporal Interaction Module for Spiking Transformer. arXiv:2401.11687v3.
University	Chinese Academy of Sciences
Country	China
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Publisher	ACM
DOI / URL	https://doi.org/10.24963/ijcai.2024/347
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4
Application Domain	Speech, Audio, and Auditory Processing
Specific Task	Event-based image classification, event-based action recognition, event-based car/non-car classification, and spiking speech-digit classification
Data Modality	Event-camera data and spike-based audio data
Signal Dimensionality	2D spatiotemporal event streams; temporal spike sequences for SHD audio
Signal Characteristics	Event streams contain strong temporal dynamics; TIM is designed to integrate current query information with historical query information across timesteps using 1D convolution
Dataset Name	CIFAR10-DVS, N-CALTECH101, NCARS, UCF101-DVS, HMDB51-DVS, SHD
Dataset Type	Public neuromorphic/event-based vision datasets and spiking audio classification dataset
Dataset Size	CIFAR10-DVS: 10,000 event-stream samples. N-CALTECH101: 101 categories. NCARS: binary car/non-car event-stream dataset. UCF101-DVS: 101 action categories. HMDB51-DVS: 51 action categories. SHD: 1,000 spoken digits across 20 classes, as reported in the paper
Dataset Balance / Split	NR
Hybrid Topology	Spikformer backbone enhanced with plug-and-play Temporal Interaction Module inside Spiking Self-Attention; event stream passes through Spiking Patch Splitting, position embedding, Transformer blocks with TIM, MLP, and classification head
SNN Component	LIF neurons, spike neurons, Spiking Patch Splitting, spike-based Q/K/V, Spiking Self-Attention, Conv-BN-LIF MLP, temporal spike dynamics
Neuron Model	LIF
Transformer Component	Spikformer, Spiking Self-Attention, Transformer block, MLP, classification head
Attention Type	Spiking Self-Attention enhanced by Temporal Interaction Module; TIM-enhanced SSA
SNN–Transformer Interface Mechanism	TIM is inserted into the query branch of SSA. It combines historical query information and current query information before attention computation, allowing SSA to use temporal information from previous timesteps instead of relying only on the current timestep
Model Depth / Scale	Uses Spikformer backbone with Transformer blocks containing SSA + TIM and Conv-BN-LIF MLP; exact number of Transformer blocks NR
Parameter Count	TIM model reported as 2.59M parameters on CIFAR10-DVS comparison; parameter count for other datasets NR
Design / Contribution Type	New lightweight plug-and-play Temporal Interaction Module for Spiking Transformers; improves temporal information interaction with minimal parameter increase
Input Encoding Method	Event streams are transformed into required dimensions through Spiking Patch Splitting using convolutional processing; SHD uses spike-based audio input representation
Encoding Parameters	Simulation step length = 10; LIF time constant τ = 2; firing threshold = 1; surrogate-gradient hyperparameter a = 4; default TIM α = 0.5
Encoding Learned or Fixed	Partially learned/adaptive: TIM uses convolutional temporal extraction and α-controlled temporal fusion; network weights are learned; event-stream formatting and main LIF settings are fixed
Event / Spike Representation	Event streams, spike tensors, LIF spike outputs, spike-based Q/K/V, TIM-enhanced temporal query representation
Temporal Resolution	Main experiments use 10 simulation steps; comparisons also mention 16 steps for some Spikformer baselines; TIM can match baseline performance with 6 steps in efficiency validation
Output Decoding Method	Linear classification head
Encoding / Temporal Information Analysis	Yes. The paper directly analyzes the limitation of current SSA in using only current timestep information, validates temporal enhancement through local TIM comparison, α sensitivity, and reduced-step efficiency experiments
Training Paradigm	Supervised direct SNN training
Training Method	Direct training on BrainCog platform using surrogate-gradient backpropagation
Surrogate Gradient Function	Piecewise surrogate-gradient function with hyperparameter a = 4
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	Batch size 16; 500 epochs; AdamW optimizer; initial learning rate 0.005; cosine decay; LIF τ = 2; firing threshold = 1; simulation steps = 10; default α = 0.5
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	CIFAR10-DVS: 81.6%; N-CALTECH101: 79.0%; NCARS: 96.5%; UCF101-DVS: 63.8%; HMDB51-DVS: 58.6%; SHD: 86.3%
Secondary Metrics	Parameter count, timestep/step efficiency, α sensitivity, baseline improvement, temporal-enhancement validation

Baseline Models Compared	SALT, Rollout, SEW-ResNet, ResNet-18, EventMix, DT-SNN, NDA, Event Transformer, RM SNN, VMV-GCN, Spikformer, DISTA, Spikingformer, Spike-driven Transformer, 2D-WT-Haar, Spikeformer, Res-SNN18 + RM, Spikepoint
Performance Gain Over Baseline	On CIFAR10-DVS, TIM improves baseline from 78.5% to 81.6%. On N-CALTECH101, it improves baseline from 77.7% to 79.0%. On SHD, it improves Spikformer from 85.1% to 86.3%. On SDSA generalization test, SDSA + TIM improves CIFAR10-DVS accuracy from 77.0% to 78.5%
Statistical Validation	Benchmark comparison and ablation study; no standard deviation, confidence interval, repeated-run analysis, or formal significance test reported
Ablation Study	Full/clear. Includes baseline vs TIM, local TIM vs full TIM, α sensitivity, reduced-step efficiency validation, and generalization to Spike-driven Transformer/SDSA
Training Framework / Code Availability	BrainCog platform; code available at BrainCog-X/Brain-Cog examples/TIM
Target Hardware	Neuromorphic/event-driven hardware motivation; no specific hardware platform targeted
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	No direct energy measurement; efficiency discussed through reduced simulation steps and minimal parameter increase
Latency / Memory / Complexity Evidence	TIM can reach baseline performance with 6 steps instead of 10, suggesting approximately 40% timestep/time reduction; detailed measured latency and memory footprint NR
Comparison with ANN Baseline	Partial. Compares with ANN/event-based baselines such as Event Transformer, C3D, P3D-63, and Event Frames + I3D/3D-ResNet, but the main comparison is against SNN and Spiking Transformer methods
Main Contribution / Key Limitation	Contribution: proposes TIM, a lightweight temporal interaction plug-in that enhances Spiking Transformer attention by incorporating historical query information, improving event-based recognition and SHD speech classification. Limitation: preprint status, no measured hardware energy/latency, no full statistical validation, and exact model depth/dataset split details are NR
RQ Mapping	Primary: RQ3; Secondary: RQ2, RQ1, RQ4, RQ5

P064

Stochastic Spiking Attention: Accelerating Attention with Stochastic Computing in Spiking Networks

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University	King’s College London
Country	United Kingdom
Publication Year	2024
Publication Venue	IEEE 6th International Conference on AI Circuits and Systems, AICAS 2024
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/AICAS59952.2024.10595893
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Hardware-focused
E&QA; Score	4
Application Domain	Communication, Cybersecurity, and Network Intelligence
Specific Task	Accelerating dot-product attention in SNN-based Transformers; image classification on MNIST and CIFAR-10 using ViT-Small
Data Modality	Static image data converted into token embeddings and spike streams
Signal Dimensionality	2D image data represented as token sequence (X \in R^{N \times D}), then encoded into temporal binary spike matrices over (T) timesteps
Signal Characteristics	Input embeddings are Bernoulli-coded into binary spike streams; Q, K, and V are generated as binary spike matrices using LIF neurons; stochastic computing replaces multiplication with logical AND operations
Dataset Name	MNIST, CIFAR-10
Dataset Type	Public image classification benchmark datasets
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	ViT-Small-style SNN Transformer attention block using Bernoulli input coding, LIF-based Q/K/V spike generation, stochastic spiking attention units, binary adders, and Bernoulli encoders
SNN Component	LIF neurons, binary spikes, Bernoulli spike coding, stochastic attention units, spike-based Q/K/V, binary adders, logical AND-based spike multiplication
Neuron Model	LIF
Transformer Component	ViT-Small Transformer encoder; self-attention block; query, key, and value projections; attention-value product
Attention Type	Stochastic Spiking Attention; spike-based dot-product attention; linear attention without softmax
SNN–Transformer Interface Mechanism	Input matrix is converted into stochastic binary spike streams; LIF neurons generate spike-encoded Q, K, and V; attention is computed through stochastic computing using AND gates and Bernoulli encoders instead of real-valued MAC operations
Model Depth / Scale	ViT-Small model with 6 encoder layers and 8 attention heads per layer; evaluated with 4, 8, and 10 timesteps
Parameter Count	NR
Design / Contribution Type	New Stochastic Spiking Attention framework; custom stochastic attention unit; FPGA-validated attention-block accelerator; dataflow design eliminating repeated intermediate memory read/write
Input Encoding Method	Bernoulli coding / probabilistic rate coding converts real-valued input embeddings into temporal binary spike streams
Encoding Parameters	Timesteps (T = 4, 8, 10); Q/K/V activations are binary; model parameters are INT8-quantized; key dimension (D_K) supports up to 256 using UINT8 counters; design assumes token count (N) in the range 16–128 for edge Transformer settings
Encoding Learned or Fixed	Fixed Bernoulli stochastic encoding and hardware dataflow; model weights are trained/quantized separately
Event / Spike Representation	Bernoulli binary spike streams; stochastic bit streams; binary Q/K/V spike matrices; binary attention-score matrix sequence (S^t); binary attention output sequence
Temporal Resolution	Discrete spike timesteps; evaluated at 4, 8, and 10 timesteps
Output Decoding Method	Classification output from ViT-Small architecture; exact spike readout/classifier decoding NR

Encoding / Temporal Information Analysis	Partial. The paper analyzes stochastic spike coding, timestep-dependent accuracy, stochastic attention dataflow, and hardware latency/energy, but does not deeply analyze temporal feature semantics
Training Paradigm	Supervised end-to-end SNN Transformer training
Training Method	End-to-end training using standard surrogate-gradient methods for SNNs; stochastic spiking attention incorporated into Transformer architecture
Surrogate Gradient Function	Surrogate-gradient methods are used, but the exact surrogate function is NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	INT8-quantized parameters; SNN activations evaluated at 4, 8, and 10 timesteps; other training hyperparameters NR
Primary Performance Metric	Classification accuracy, attention-block energy consumption, latency, and power
Primary Metric Value	SSA ViT-Small achieves 98.31% accuracy on MNIST and 83.53% on CIFAR-10 at 10 timesteps; FPGA SSA block latency is (3.3 \times 10^{-3}) ms with 1.47 W power
Secondary Metrics	Processing energy, memory-access energy, total energy, timestep sensitivity, FPGA/CPU/GPU latency and power
Baseline Models Compared	ANN ViT-Small attention, Spikformer attention, CPU implementation, GPU implementation
Performance Gain Over Baseline	Compared with ANN attention, SSA reduces processing energy by 6.3× and memory-access energy by 1.7×; compared with GPU SSA implementation, FPGA SSA achieves 48× lower latency and 15× lower power; compared with ANN GPU attention, FPGA SSA has 18× lower latency and 17× lower power
Statistical Validation	Benchmark evaluation and hardware comparison; no standard deviation, confidence interval, repeated-run analysis, or significance test reported
Ablation Study	Partial. Compares ANN, Spikformer, and SSA across 4, 8, and 10 timesteps; compares processing energy, memory energy, latency, and power across architectures/devices
Training Framework / Code Availability	FPGA implementation on Xilinx Zynq-7000 SoC; CPU tested on Intel i7-12850HX; GPU tested on Nvidia RTX A2000; code availability NR
Target Hardware	Digital CMOS-based ASIC design estimate and FPGA implementation; Xilinx Zynq-7000 SoC for experimental validation
Hardware Deployment Status	FPGA implementation and measurement reported; ASIC-level energy estimated but no fabricated ASIC reported
Energy Evidence	Estimated attention-block energy using 45 nm CMOS operation and memory-access metrics; FPGA power measured experimentally
Latency / Memory / Complexity Evidence	Single attention block energy: SSA 1.23 µJ processing, 52.80 µJ memory access, 54.03 µJ total; ANN attention 97.73 µJ total; Spikformer attention 109.05 µJ total. FPGA latency (3.3 \times 10^{-3}) ms and power 1.47 W
Comparison with ANN Baseline	Yes. Compares classification accuracy, energy, latency, and power with ANN ViT-Small attention implementation
Main Contribution / Key Limitation	Contribution: proposes Stochastic Spiking Attention, replacing multiplier-heavy attention operations with Bernoulli-coded stochastic computing and AND-gate operations, with FPGA validation showing large latency and power gains. Limitation: evaluation is limited to MNIST/CIFAR-10 and a single attention block; full end-to-end hardware deployment for complete Spiking Transformer is not reported
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4, RQ1

P065

Xpikeformer: Hybrid Analog-Digital Hardware Acceleration for Spiking Transformers

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Country	United Kingdom
Publication Year	2025
Publication Venue	IEEE Transactions on Very Large Scale Integration (VLSI) Systems
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TVLSI.2025.3552534
Publication Type	Journal article
Database Source	Scopus
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Communication, Cybersecurity, and Network Intelligence
Specific Task	Accelerating SNN-based Transformer inference for image classification and in-context-learning-based wireless symbol detection
Data Modality	Static image data and wireless communication symbol sequences
Signal Dimensionality	2D image token sequences for ViT models; 1D sequential query–answer token sequences for GPT-style wireless symbol detection; temporal binary spike trains across timesteps
Signal Characteristics	Activations are encoded as binary temporal spike trains; feedforward and fully connected layers use LIF-based spike processing in AIMC; attention uses Bernoulli-neuron-based stochastic spiking attention with binary Q/K/V streams
Dataset Name	CIFAR-10, ImageNet-1K, simulated ICL wireless symbol detection task with 2 × 2 and 4 × 4 antenna settings
Dataset Type	Public image classification datasets and simulated wireless communication benchmark task
Dataset Size	NR
Dataset Balance / Split	NR
Hybrid Topology	Hybrid analog–digital accelerator with AIMC engine for embedding, feedforward, and fully connected layers, and digital stochastic spiking attention engine for MHSA; data exchange uses on-chip SRAM and residual units
SNN Component	LIF neurons, Bernoulli neuron layer, spike trains, spiking neuron tiles, spike-encoded Q/K/V, stochastic attention cells, binary spike operations
Neuron Model	LIF; Bernoulli neuron layer for stochastic spiking attention
Transformer Component	ViT encoder-only Transformer for image classification; GPT-like decoder-only Transformer for wireless symbol detection; MHSA, embedding layers, feedforward layers, fully connected output layers
Attention Type	Stochastic Spiking Attention; spike-based MHSA using Bernoulli neurons and logical AND operations

SNN–Transformer Interface Mechanism	Transformer feedforward/FC operations are mapped to PCM-based AIMC spiking neuron tiles; Q/K/V spike matrices are sent to the SSA engine, where Bernoulli-neuron-based stochastic computing replaces attention multiplications with logical AND operations and counters
Model Depth / Scale	Image classification: Xpikeformer-ViT model sizes 4–384, 6–512, and 8–768. Wireless ICL: Xpikeformer-GPT model sizes 4–256 and 8–512. SSA tile supports attention-head reuse and layer-wise reuse
Parameter Count	NR
Design / Contribution Type	First dedicated hardware accelerator design for Spiking Transformers; hybrid PCM-AIMC + stochastic spiking attention architecture; row-block-wise AIMC mapping; SSA tile and engine; hardware-aware training; global drift compensation
Input Encoding Method	Bernoulli rate coding converts real-valued activations into temporal binary spike trains; input tokens are processed through spike encoding and LIF-based spiking neuron layers
Encoding Parameters	Spike encoding length varies by task/model. Xpikeformer-ViT: CIFAR-10 uses T = 11 for 4–384 and T = 10 for 6–512; ImageNet uses T = 8 for 6–512 and T = 7 for 8–768. Xpikeformer-GPT: 2 × 2 antenna task uses T = 6 for 4–256 and T = 4 for 8–512; 4 × 4 antenna task uses T = 11 for 4–256 and T = 5 for 8–512
Encoding Learned or Fixed	Partially fixed/adaptive: Bernoulli spike encoding and SSA hardware dataflow are fixed; model weights are trained; hardware-aware training and drift compensation adapt the model to PCM-AIMC nonidealities
Event / Spike Representation	Binary temporal spike trains, Bernoulli spike streams, spike-encoded Q/K/V matrices, stochastic attention scores, binary attention outputs
Temporal Resolution	Multi-step spike encoding with task-dependent T values from 4 to 11 timesteps in the reported experiments
Output Decoding Method	ViT classification head for CIFAR-10/ImageNet; GPT-like decoder output for wireless symbol detection; training loss is computed from outputs averaged over timesteps
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike encoding length, Bernoulli stochasticity, hardware noise, long-term PCM conductance drift, global drift compensation, energy breakdown, latency breakdown, and model-size effects on robustness
Training Paradigm	Supervised training from scratch with hardware-aware training for spiking Transformer deployment
Training Method	Two-stage training: conventional full-precision training followed by hardware-aware fine-tuning with PCM hardware noise injected during forward propagation; backward propagation remains ideal
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Similar to ANN Transformer training; exact loss formula NR
Training Hyperparameters	Two-stage CT + HWAT training; outputs averaged over timesteps for loss computation; AdamW optimizer; AIHWKit used to simulate PCM nonidealities; exact learning rate, batch size, and epoch count NR
Primary Performance Metric	Classification accuracy, BER for wireless detection, energy consumption, inference latency, and chip area
Primary Metric Value	Xpikeformer-ViT: 84.74% on CIFAR-10 with 6–512 model and T = 10; 70.50% on ImageNet-1K with 8–768 model and T = 7. Xpikeformer-GPT: BER 0.067 / 0.063 on 2 × 2 antennas for 4–256 / 8–512; BER 0.205 / 0.091 on 4 × 4 antennas for 4–256 / 8–512
Secondary Metrics	Spike encoding length, long-term accuracy after one year, energy per inference, latency per inference, chip area, computation-energy breakdown, runtime memory-access energy, speedup, conductance-drift robustness
Baseline Models Compared	ANN-ViT GPU, SNN-ViT GPU, ANN-GPT GPU, SNN-GPT GPU, ANN-Quant, ANN-Quant + AIMC, SNN-Digi-Opt, SwiftTron, X-Former
Performance Gain Over Baseline	On ImageNet-1K, Xpikeformer gives 9.6×–13× energy reduction over ANN-Quant, 5.4×–5.9× over ANN-Quant + AIMC, and 1.8×–1.9× over SNN-Digi-Opt. For 4 × 4 wireless ICL, it gives 7×–14.5×, 2.6×–4.0×, and 1.6×–1.8× energy reduction over the same baselines. It also reports 2.18× speedup over ANN Transformer GPU and 6.85× speedup over SNN Transformer GPU
Statistical Validation	Hardware simulation, benchmark comparison, and long-term drift analysis; no standard deviation, confidence interval, repeated-run statistical testing, or significance test reported
Ablation Study	Partial/full hardware ablation. Includes CT vs HWAT, with/without global drift compensation, one-year conductance-drift accuracy, model-size comparison, energy breakdown, latency breakdown, and comparison with digital/AIMC baselines
Training Framework / Code Availability	SpikingJelly for spiking components; IBM AIHWKit for AIMC simulation; DNN+NeuroSim V1.4 and Cadence 45 nm PDK for hardware evaluation; code availability NR
Target Hardware	Hybrid analog–digital simulated ASIC using PCM-based AIMC engine and digital SSA engine; 45 nm technology evaluation
Hardware Deployment Status	Simulated ASIC-level hardware evaluation; no fabricated chip or physical FPGA/ASIC deployment reported
Energy Evidence	Energy estimated using DNN+NeuroSim V1.4 for AIMC crossbars and Cadence 45 nm synthesis for SSA tiles; runtime compute and SRAM memory-access energy are analyzed
Latency / Memory / Complexity Evidence	Table VI reports Xpikeformer at 45 nm, 200 MHz, 784 mm² area, 0.30 mJ energy/inference, and 2.18 ms latency/inference. AIMC engine consumes 78.4% of computational energy; SSA engine consumes 18.9%; peripheral operations account for over 92% of latency
Comparison with ANN Baseline	Yes. Compares against ANN-ViT, ANN-GPT, ANN-Quant, SwiftTron, and X-Former ANN Transformer accelerator baselines
Main Contribution / Key Limitation	Contribution: proposes Xpikeformer, a hybrid PCM-AIMC and stochastic-spiking-attention accelerator for Spiking Transformers, reducing energy and improving speed compared with ANN and SNN baselines. Limitation: results are based on simulation and synthesis rather than fabricated hardware; AIMC area is large; practical NVM crossbar implementation and real-world hardware testing are left for future work
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4, RQ1

P066

PSSD-Transformer: Powerful Sparse Spike-Driven Transformer for Image Semantic Segmentation

Full Citation	Wang, H., Liang, X., Zhang, T., Gu, Y., & Geng, W. (2024). PSSD-Transformer: Powerful Sparse Spike-Driven Transformer for Image Semantic Segmentation. In Proceedings of the 32nd ACM International Conference on Multimedia, MM ’24, Melbourne, Australia, pp. 758–767. ACM. https://doi.org/10.1145/3664647.3680870
University	Zhejiang University
Country	China
Publication Year	2024
Publication Venue	ACM International Conference on Multimedia
Publisher	ACM
DOI / URL	https://doi.org/10.1145/3664647.3680870
Publication Type	Conference paper

Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image semantic segmentation on ADE20K and image classification on ImageNet/CIFAR-10
Data Modality	Static RGB image data
Signal Dimensionality	2D image data converted into patch/token feature maps and multi-timestep spike representations; dense pixel-level segmentation output
Signal Characteristics	Images are processed through spike-driven Transformer blocks; semantic segmentation uses feature maps, spike masks, category embeddings, and upsampled segmentation masks; ADE20K experiments use 512 × 512 resolution
Dataset Name	ImageNet, CIFAR-10, ADE20K
Dataset Type	Public image classification and semantic segmentation benchmark datasets
Dataset Size	NR in main paper
Dataset Balance / Split	Uses standard dataset settings; exact class balance and split details NR
Hybrid Topology	PSSD-Transformer backbone with Spiking Patch Splitting and Embedding, PSSA-based spiking Transformer encoder blocks, Group-LIF MLP, DSMS residual/shortcut structure, and Spike Mask Transformer segmentation decoder
SNN Component	LIF neurons, Group-LIF neurons, G-HLIF, G-VLIF, spike neurons for Q/K/V, PSSA, DSMS, SPG-down, spike masks, sparse spike-driven computation
Neuron Model	LIF; Group-LIF; Group Horizontal LIF; Group Vertical LIF
Transformer Component	PSSD-Transformer backbone, Spiking Transformer encoder block, Spike Mask Transformer decoder, PSSA attention layer, MLP decoder modules
Attention Type	Pure Sparse Self-Attention; spike-driven sparse attention; softmax-free attention using mask/logical operations
SNN–Transformer Interface Mechanism	Input features are transformed into spike-form Q, K, and V; PSSA replaces dense matrix multiplication with Hadamard/logical masking and row-summation operations; Spike Mask Transformer combines decoded segmentation features and category embeddings through inner product to generate semantic masks
Model Depth / Scale	Uses PSSD-Transformer backbone with repeated spiking Transformer-based encoder blocks and repeated decoder blocks; exact number of blocks/layers NR in main paper
Parameter Count	NR
Design / Contribution Type	New sparse spike-driven Transformer for semantic segmentation; proposes PSSA, DSMS, SPG-down, and Group-LIF neurons; claims first SNN/Spiking Transformer application to large-scale semantic segmentation dataset
Input Encoding Method	Spiking Patch Splitting and Embedding using Conv2D, BN, MaxPooling, patch splitting, and relative position embedding; semantic segmentation head uses spike mask features and class/category embeddings
Encoding Parameters	LIF decay factor, firing threshold, and reset potential are preset; timestep T is used in spike computation; one ablation uses time step = 4; exact full encoding parameters NR
Encoding Learned or Fixed	Partially learned: convolutional embeddings, PSSD backbone, decoder, category embeddings, and Group-LIF-related modules are learned; LIF preset values and spike computation rules are fixed
Event / Spike Representation	Binary spike tensors, spike-form Q/K/V, spike masks, sparse spike-driven feature maps, membrane potentials, category spike embeddings
Temporal Resolution	Multi-timestep spike simulation; time step 4 reported in SPG-down/G-LIF ablation; other timestep settings NR
Output Decoding Method	Spike Mask Transformer produces decoded segmentation feature matrix and category embedding; inner product generates K mask sequences, followed by bilinear interpolation to match original image size
Encoding / Temporal Information Analysis	Partial. The paper analyzes sparse spike computation, PSSA energy reduction, DSMS information preservation, SPG-down gradient propagation, and Group-LIF local spatial representation, but does not deeply analyze temporal spike dynamics
Training Paradigm	Supervised direct SNN/Spiking Transformer training
Training Method	Direct training using surrogate-gradient-style SNN learning; SPG-down is designed to improve gradient transmission through downsampling
Surrogate Gradient Function	Uses surrogate-gradient approach; exact function partly described through amplification factor α and Heaviside approximation, but full training surrogate setting NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	ADE20K resolution 512 × 512; time step 4 in one ablation; other batch size, learning rate, optimizer, epochs, and augmentation details are NR in main paper
Primary Performance Metric	mIoU for semantic segmentation; top-1 accuracy for image classification; estimated power/energy in mJ
Primary Metric Value	ADE20K best reported mIoU: 29.1% with PSSD + DSMS + MLP decoder; ImageNet sparse-spiking ablation best accuracy: 66.8% with PSSA + DSMS + MS and 2.21 mJ power
Secondary Metrics	Power in mJ, accuracy gain/loss, Grad-CAM visualization, qualitative segmentation results
Baseline Models Compared	Spikformer, SSA, PSSA, SEW residual, MS residual, DSMS, Vanilla Downsample, SPG-Down, G-HLIF, G-MLIF, SMT decoder
Performance Gain Over Baseline	On ADE20K, PSSD + DSMS + MLP improves mIoU from Spikformer baseline 26.7% to 29.1% while reducing power from 19.12 mJ to 13.50 mJ. On ImageNet ablation, PSSA + DSMS + MS reduces power from 4.96 mJ to 2.21 mJ while improving accuracy from 65.1% to 66.8%
Statistical Validation	Benchmark comparison, ablation studies, and visualization; no standard deviation, confidence interval, repeated-run analysis, or significance test reported
Ablation Study	Full. Includes PSSA vs SSA, SEW vs MS/DSMS residuals, SPG-down vs vanilla downsampling, G-HLIF/G-MLIF neuron variants, MLP vs SMT decoder, power/accuracy trade-off, and qualitative segmentation visualization
Training Framework / Code Availability	Code stated to be made publicly available after acceptance; exact framework NR
Target Hardware	Neuromorphic-chip motivation; sparse spike-driven design intended for neuromorphic deployment
Hardware Deployment Status	No real neuromorphic hardware deployment reported; experiments conducted on GPUs
Energy Evidence	Estimated energy/power using FLOPs, SOPs, spike firing rate, timestep count, MAC energy 4.6 pJ, and AC energy 0.9 pJ; no measured hardware energy
Latency / Memory / Complexity Evidence	Reports estimated power in mJ for ImageNet and ADE20K ablations; detailed latency, memory footprint, and real hardware complexity NR

Comparison with ANN Baseline	Partial. Discusses semantic segmentation ANN methods such as FCN, SegNet, DeepLab, SETR, Segmenter, and SegFormer in related work, but main experiments compare against Spikformer-style SNN/Spiking Transformer variants
Main Contribution / Key Limitation	Contribution: proposes PSSD-Transformer, integrating PSSA, DSMS, SPG-down, and Group-LIF to enable sparse spike-driven semantic segmentation with lower estimated energy and improved ADE20K mIoU. Limitation: no neuromorphic hardware deployment, energy is estimated only, several training details are in supplementary/NR, and evaluation is mainly limited to ADE20K for segmentation
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P067

Temporal-coded Spiking Transformer

Full Citation	Sun, Q., Lu, C., Chen, W., Wei, W., Wang, J., Zhang, J., Liu, X., Ye, Y., Yang, Y., & Zhang, M. (2025). Temporal-coded Spiking Transformer. In Proceedings of the 33rd ACM International Conference on Multimedia, MM '25, Dublin, Ireland. ACM. https://doi.org/10.1145/3746027.3754545
University	University of Electronic Science and Technology of China
Country	China
Publication Year	2025
Publication Venue	ACM International Conference on Multimedia
Publisher	ACM
DOI / URL	https://doi.org/10.1145/3746027.3754545
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Static image classification and neuromorphic/event-based image classification
Data Modality	Static RGB images and DVS/event-camera data
Signal Dimensionality	2D image data converted into patch/token sequences; DVS-CIFAR10 represented as 2D spatiotemporal event samples
Signal Characteristics	Uses Time-To-First-Spike coding, where each neuron fires at most one spike and earlier firing time represents more important information; DVS-CIFAR10 samples are 128 × 128; CIFAR images are 32 × 32; ImageNet images are described as 256 × 256
Dataset Name	CIFAR-10, CIFAR-100, ImageNet, DVS-CIFAR10
Dataset Type	Public static image classification and neuromorphic/event-based classification benchmark datasets
Dataset Size	CIFAR-10/100: 60,000 images each; ImageNet: 1.3M training images and 50,000 validation images across 1,000 classes; DVS-CIFAR10: 9,000 training and 1,000 test samples
Dataset Balance / Split	Uses standard benchmark splits; DVS-CIFAR10 has 9,000 training and 1,000 test samples; exact class-balance details NR
Hybrid Topology	Temporal-coded Spiking Transformer with patch embedding, repeated encoder blocks, Temporal-coded Spiking Attention, MLP, temporal-coded batch normalization, temporal-coded residual connections using CSS module, global average pooling, and classification head
SNN Component	TTFS coding, DFT-based temporal-coded spiking neuron, tSN layer, Spike-tSN layer, tBN, tRC, CSS module, spike firing time features
Neuron Model	DFT-based temporal-coded neuron with Rectified Linear postsynaptic potential kernel and dynamic firing threshold
Transformer Component	Patch embedding, Transformer encoder blocks, Temporal-coded Spiking Attention, MLP block, classification head
Attention Type	Temporal-coded Spiking Attention; QK-inspired channel attention using standard-firing query mask, row summation, and Hadamard product with key features
SNN–Transformer Interface Mechanism	Patch embeddings are converted into spike-time features; Q and K are produced through linear layers, tBN, and temporal-coded spike-neuron layers; TSA uses the standard-firing pattern of Q to form attention scores, then applies Hadamard product with K and post-linear projection
Model Depth / Scale	CIFAR models: T-SpikeFormer-4-256, 2-384, 4-384. ImageNet models: T-SpikeFormer-8-384, 8-512, 8-768. DVS-CIFAR10 model: T-SpikeFormer-2-256
Parameter Count	CIFAR: 3.10M, 5.45M, and 6.93M; ImageNet: 17.84M, 31.65M, and 69.71M; DVS-CIFAR10: 2.43M
Design / Contribution Type	First directly trained TTFS-based Spiking Transformer; proposes temporal-coded batch normalization, temporal-coded residual connection, CSS module, Temporal-coded Spiking Attention, and T-SpikeFormer
Input Encoding Method	Time-To-First-Spike temporal coding; patch embedding converts image input into flattened spike-time-form patches
Encoding Parameters	TTFS uses permissible spike time window (T_\omega); energy table reports TTFS time step 160; patch embedding channels increase progressively as D/8, D/4, D/2, D; split ratio best at 0.5 and shuffle group best at 4
Encoding Learned or Fixed	Partially learned: patch embedding, TSA, MLP, tBN parameters, and model weights are learned; TTFS coding rule, DFT threshold mechanism, and CSS structure are fixed design mechanisms
Event / Spike Representation	Spike firing time, first-spike temporal code, spike-time feature, direct firing, standard firing, no firing, sparse TTFS spikes
Temporal Resolution	TTFS spike time window; energy comparison uses 160 timesteps for T-SpikeFormer; exact internal (T_\omega) value for all experiments NR
Output Decoding Method	Global average pooling followed by linear classification head
Encoding / Temporal Information Analysis	Yes. The paper directly analyzes TTFS coding, spike firing time preservation, tBN effects on membrane potentials and spike firing-time distributions, tRC split ratio/group number, sparsity, and normalized energy consumption
Training Paradigm	Supervised direct training of temporal-coded SNN Transformer
Training Method	Direct training from scratch using PyTorch and AdamW; temporal-coded neuron training with DFT-based event-driven backpropagation mechanism
Surrogate Gradient Function	Not conventional surrogate gradient; uses DFT-based temporal-coded neuron and event-driven temporal learning formulation; exact surrogate function NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR

Training Hyperparameters	CIFAR/ImageNet: 310 epochs, AdamW, initial learning rate 0.0005, cosine decay, scaling factor 1/16; batch size 32/64 for CIFAR and 128/256 for ImageNet. DVS-CIFAR10: 106 epochs, batch size 16, initial learning rate 0.1, cosine decay, scaling factor 1/16. Uses random augmentation, mixup, and cutmix
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	Best reported results: CIFAR-10 95.78%, CIFAR-100 77.19%, ImageNet 74.43%, DVS-CIFAR10 78.20%
Secondary Metrics	Parameter count, normalized energy consumption, spike count, sparsity, split-ratio ablation accuracy, shuffle-group ablation accuracy
Baseline Models Compared	Spikformer, Spike-driven Transformer / S-Transformer, QKFormer, SNN-WS, Park et al. temporal-coded VGG models, DTA-TTFS, STD-ED, Ana et al. 2024, Spiking-GoogleNet
Performance Gain Over Baseline	Compared with temporal-coded SNNs, T-SpikeFormer achieves the best reported CIFAR-10, CIFAR-100, and ImageNet accuracy. Compared with rate-coded Spikformer/S-Transformer under similar model settings, it achieves comparable accuracy with lower normalized energy T-SpikeFormer-2-384 reaches 95.59% CIFAR-10 with 0.0517× normalized energy versus Spikformer-2-384 95.25% with 0.1300× and S-Transformer-2-384 95.60% with 0.1217×
Statistical Validation	Benchmark comparison and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or formal significance test reported
Ablation Study	Full. Includes tBN versus without tBN, membrane-potential and firing-time distribution analysis, split-ratio ablation, shuffle-group ablation, sparsity analysis, and normalized energy comparison
Training Framework / Code Availability	PyTorch; trained on multiple NVIDIA RTX 4090 GPUs; code availability NR
Target Hardware	Neuromorphic hardware motivation; energy analysis based on TrueNorth neuromorphic architecture
Hardware Deployment Status	No real neuromorphic hardware deployment reported; experiments run on GPUs with theoretical energy estimation
Energy Evidence	Estimated normalized energy using static latency, dynamic spike count, and TrueNorth energy coefficients (E_{static}=0.6), (E_{dynamic}=0.4); no measured hardware energy
Latency / Memory / Complexity Evidence	Reports normalized energy, timestep count, spike count, sparsity, and parameter count; measured latency, memory footprint, and real hardware complexity NR
Comparison with ANN Baseline	Partial. Main comparisons are against rate-coded and temporal-coded SNNs; no matched ANN Transformer baseline is directly evaluated in the main results table
Main Contribution / Key Limitation	Contribution: proposes the first TTFS-based temporal-coded Spiking Transformer by solving BN and residual-connection incompatibility through tBN and tRC/CSS, achieving strong accuracy with lower estimated energy. Limitation: no real hardware deployment or measured energy; ImageNet accuracy remains below the strongest rate-coded S-Transformer/QKFormer baselines; code availability is NR
RQ Mapping	Primary: RQ3; Secondary: RQ2, RQ4, RQ5, RQ1

P068

Spatial-Temporal Self-Attention for Asynchronous Spiking Neural Networks

Full Citation	Wang, Y., Shi, K., Lu, C., Liu, Y., Zhang, M., & Qu, H. (2023). Spatial-Temporal Self-Attention for Asynchronous Spiking Neural Networks. In Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23, pp. 3085–3093.
University	University of Electronic Science and Technology of China
Country	China
Publication Year	2023
Publication Venue	International Joint Conference on Artificial Intelligence, IJCAI-23
Publisher	IJCAI
DOI / URL	https://www.ijcai.org/proceedings/2023/0344.pdf
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Speech, Audio, and Auditory Processing
Specific Task	Event-based gesture recognition, event-based object classification, and speech command recognition
Data Modality	DVS/event-camera streams and audio speech signals converted to MFCC frame features
Signal Dimensionality	2D spatiotemporal event frames represented as (T \times 2 \times w \times h); token sequence represented as (T \times n_w \times n_h \times d); speech MFCC resized to 128 × 128 and repeated over timesteps
Signal Characteristics	Event stream consists of ((x, y, t, p)) events with polarity (p \in {+1,-1}); event streams are divided into (T) segments and aggregated into two-channel event frames; STSA preserves asynchronous inference by allowing output at any timestep
Dataset Name	DVS128 Gesture, CIFAR10-DVS, Google Speech Commands V1, Google Speech Commands V2
Dataset Type	Public neuromorphic/event-based vision datasets and public speech command recognition datasets
Dataset Size	DVS128 Gesture: 1,464 samples, with 1,176 training and 288 test samples. CIFAR10-DVS: 10,000 event streams, 10 classes, 1,000 samples per class. GSC V1: 30 categories from 1,881 speakers. GSC V2: 35 categories from 2,618 speakers
Dataset Balance / Split	DVS128 Gesture uses official train/test split. CIFAR10-DVS uses 9:1 split with first 900 samples per class for training and remaining 100 for testing. GSC uses 12-class setting with 1,500 samples randomly selected per command and split 8:2 train/test
Hybrid Topology	Spatial-Temporal Spiking Transformer framework with event-frame tokenization, repeated STS-Transformer encoder blocks, SNN-based Spatial-Temporal Self-Attention, MLP block, STRPB, and final MLP classification head
SNN Component	LIF neurons, binary spikes, spike-form Q/K/V, asynchronous event-driven spike processing, synaptic delay-inspired STSA, triangular surrogate gradient
Neuron Model	LIF
Transformer Component	Transformer encoder block, linear Q/K/V projections, spatial-temporal self-attention, MLP block, MLP classification head
Attention Type	Spatial-Temporal Self-Attention with masked temporal attention and Spatial-Temporal Relative Position Bias
SNN–Transformer Interface Mechanism	Event frames are tokenized into spatial-temporal tokens; Q, K, and V are generated through Linear + BN + LIF layers; STSA computes pairwise spatial-temporal dependencies across current and previous timesteps while masking future information to preserve asynchronous operation
Model Depth / Scale	Reported models include STS-Transformer-1-256 and STS-Transformer-2-256; uses 1 or 2 encoder blocks with token length 256; DVS experiments use 10 or 16 timesteps; speech experiments use one block and T = 4
Parameter Count	NR

Design / Contribution Type	New SNN-based Spatial-Temporal Self-Attention mechanism; Spatial-Temporal Relative Position Bias; asynchronous STS-Transformer framework; parallel masked-matrix implementation for efficient training
Input Encoding Method	Event streams are converted to frame sequences by aggregating events into (T) temporal segments; each event frame is divided into patches and mapped to tokens using a convolutional stem; speech is preprocessed using MFCC and resized to 128 × 128
Encoding Parameters	DVS input frames are 128 × 128; four 3 × 3 convolutional layers are used in tokenization; each convolutional layer is followed by max-pooling with step size 2 to divide images into 16 × 16 patches; LIF threshold = 1; decay coefficient (\tau = 0.5); T = 10 or 16 for DVS experiments and T = 4 for speech
Encoding Learned or Fixed	Fixed event-frame aggregation and MFCC preprocessing; convolutional tokenization, Q/K/V projections, STSA, STRPB, MLP, and classifier parameters are learned
Event / Spike Representation	Event tuples ((x, y, t, p)), two-channel event frames, binary spike trains, spike-form Q/K/V tensors, spatial-temporal token embeddings
Temporal Resolution	DVS experiments use 10 or 16 timesteps; speech experiments use 4 repeated timesteps; event-frame sequence length (T) is set before event aggregation
Output Decoding Method	MLP head/classification head outputs final class prediction; asynchronous STSA design allows prediction at each timestep rather than waiting for all (T) timesteps
Encoding / Temporal Information Analysis	Yes. The paper explicitly analyzes asynchronous temporal attention, spatial-temporal token interaction, causal masking of future timesteps, synaptic-delay formulation, STRPB, and STSA versus spatial-only SSA
Training Paradigm	Supervised direct SNN training
Training Method	Surrogate-gradient training with AdamW optimizer; uses Temporal Efficient Training loss and cosine learning-rate decay
Surrogate Gradient Function	Triangular surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Temporal Efficient Training loss with L2 penalty (1 \times 10^{-4})
Training Hyperparameters	Kaiming normal initialization; AdamW optimizer; initial learning rate 0.01; cosine learning-rate decay; L2 penalty (1e^{-4}); batch size 32 for STSA/STRPB ablation; 1000 epochs in ablation; LIF threshold 1 and (\tau = 0.5)
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	DVS128 Gesture: best 98.72 ± 0.160% with STS-Transformer-2-256 and T = 16. CIFAR10-DVS: best 79.93 ± 0.132% with STS-Transformer-2-256 and T = 16. GSC V1: 96.74%. GSC V2: 95.18%
Secondary Metrics	Mean accuracy over three runs for DVS experiments, timestep count, asynchronous capability, ablation convergence curves
Baseline Models Compared	SLAYER, DECOLLE, LIAF-Net, TA-SNN, tdBN, SEW-ResNet, PLIF, Spikformer, NeuNorm, Dspike, ConvNet, Attention RNN, Residual SNN, NLIP SNN, E2E SNN
Performance Gain Over Baseline	On DVS128 Gesture, STS-Transformer reaches 98.72%, outperforming Spikformer 96.9% and TA-SNN 98.61%. On CIFAR10-DVS, it reaches 79.93%, outperforming Spikformer 78.9% and SNN baselines. On GSC V1, it reaches 96.74%, outperforming E2E SNN 92.2% and Attention RNN 95.6%
Statistical Validation	DVS experiments report average accuracy across three runs with ± variation; no formal significance test or confidence interval reported
Ablation Study	Full/clear. Includes STSA versus spatial-only SSA and STRPB versus no STRPB using DVS128 Gesture; training/test curves show improved convergence and accuracy
Training Framework / Code Availability	Code reported available at GitHub: pppps/STSA; framework NR
Target Hardware	Neuromorphic hardware motivation including TrueNorth, Loihi, and Tianjic; no specific deployment target used
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No direct energy measurement; efficiency is argued through asynchronous event-driven computation, AC operations, sparse spikes, and avoidance of waiting for all timesteps
Latency / Memory / Complexity Evidence	Asynchronous design allows output at any timestep instead of waiting for full (T); no measured latency, memory footprint, parameter count, or hardware complexity reported
Comparison with ANN Baseline	Partial. Compares against ANN speech baselines such as ConvNet, Attention RNN, and Residual SNN, but main comparisons are SNN-based
Main Contribution / Key Limitation	Contribution: proposes STSA and STRPB to enable Spiking Transformers to capture both spatial and temporal dependencies while preserving asynchronous SNN operation. Limitation: no measured energy or hardware deployment, limited architecture-scale reporting, and event data are still processed using frame-based aggregation
RQ Mapping	Primary: RQ3; Secondary: RQ2, RQ1, RQ4, RQ5

P069

Fourier or Wavelet bases as counterpart self-attention in spikformer for efficient visual classification

Full Citation	Wang, Q., Zhang, D., Cai, X., Zhang, T., & Xu, B. (2025). Fourier or Wavelet bases as counterpart self-attention in spikformer for efficient visual classification. Frontiers in Neuroscience, 18, Article 1516868. https://doi.org/10.3389/fnins.2024.1516868
University	Chinese Academy of Sciences
Country	China
Publication Year	2025
Publication Venue	Frontiers in Neuroscience
Publisher	Frontiers
DOI / URL	https://doi.org/10.3389/fnins.2024.1516868
Publication Type	Journal article
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Static image classification and event-based video/object/gesture classification
Data Modality	Static image data and event-based DVS video data
Signal Dimensionality	Static images represented as repeated spike-form image sequences (T \times C \times H \times W); event-based data represented directly as (T \times C \times H \times W); patch-token sequence represented as (T \times N \times D)

Signal Characteristics	Uses sparse binary spike-form features; replaces Spikformer's dynamic Query-Key SSA bases with fixed Fourier/Wavelet bases; DVS datasets use 128 × 128 inputs; CIFAR static images use 32 × 32 inputs
Dataset Name	CIFAR10-DVS, DvsGesture / DVS128 Gesture, CIFAR-10, CIFAR-100
Dataset Type	Public neuromorphic/event-based video datasets and static image classification benchmark datasets
Dataset Size	NR in main experimental setting; CIFAR10-DVS and DvsGesture download sources are provided in data availability statement
Dataset Balance / Split	Uses standard dataset settings; exact class-balance and split details NR
Hybrid Topology	FWformer architecture: Spiking Patch Splitting module, Conditional Position Embedding, FWformer encoder layers, FW head replacing SSA, MLP sub-layer, batch normalization, spiking neuron layer, residual/membrane shortcut, global average pooling, and spiking fully connected classification layer
SNN Component	LIF neurons, spiking neuron layers, sparse binary spike sequences, Spiking Patch Splitting, spike-form features, membrane shortcut
Neuron Model	LIF
Transformer Component	Spikformer-inspired encoder; FWformer encoder; Fourier/Wavelet head replacing Spiking Self-Attention; MLP sub-layer; classification layer
Attention Type	Fourier/Wavelet transform head as counterpart/replacement for Spiking Self-Attention; fixed-basis sequence mixing instead of Query-Key dynamic attention
SNN–Transformer Interface Mechanism	Spike-form patch features are passed to fixed Fourier or Wavelet transform heads; FW head transforms spike-form features using fixed or combined basis functions, followed by BN and spiking neuron layers, replacing SSA's Q/K/V attention calculation
Model Depth / Scale	Event-based datasets: FWformer-2-256 with 2 encoder layers, feature dimension 256, timestep 16. Static CIFAR datasets: FWformer-4-384 with 4 encoder layers, feature dimension 384, timestep 4
Parameter Count	Event-based FWformer: 2.06M parameters; static-image FWformer: 6.96M parameters
Design / Contribution Type	Replaces SSA with spike-form Fourier transform, Wavelet transform, and learnable combinations of Wavelet bases; proposes FWformer as a lower-complexity Spikformer variant
Input Encoding Method	Static images are repeated across timesteps to form image sequences; event-based DVS inputs are processed as temporal image sequences; Spiking Patch Splitting converts inputs into spike-form patch tokens
Encoding Parameters	DVS datasets: patch size 16 × 16, sequence length (N=64), feature dimension (D=256), timestep 16. CIFAR10/100: patch size 4 × 4, sequence length (N=64), feature dimension (D=384), timestep 4
Encoding Learned or Fixed	Partially fixed: Fourier/Wavelet bases are fixed for 1D-FFT, 2D-FFT, and 2D-WT; DWT-C uses learnable coefficients for combining Bior1.1, Haar, and Db1 bases; network weights are learned
Event / Spike Representation	Binary spike sequences, spike-form patch tokens, spike-form sparse features, LIF spike outputs
Temporal Resolution	16 timesteps for CIFAR10-DVS and DvsGesture; 4 timesteps for CIFAR-10 and CIFAR-100
Output Decoding Method	Global average pooling over final spike features followed by spiking fully connected classification layer
Encoding / Temporal Information Analysis	Yes. The paper analyzes fixed versus dynamic basis transformations, SSA orthogonality during training, Fourier/Wavelet basis behavior, timestep settings, energy, memory, training speed, and inference speed
Training Paradigm	Supervised direct training from scratch
Training Method	Direct SNN training using PyTorch and SpikingJelly; FWformer trained under the same settings as Spikformer/SSA baselines
Surrogate Gradient Function	Surrogate-gradient training is implied for LIF-based SNN training, but exact surrogate function NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	Event-based datasets: 2 FWformer encoder layers, timestep 16, batch size 16, AdamW, initial learning rate 0.1, cosine decay, 106 epochs for CIFAR10-DVS and 200 epochs for DvsGesture. Static CIFAR: 4 encoder layers, timestep 4, batch size 128, AdamW, initial learning rate 0.0005, cosine decay, 400 epochs, random augmentation, mixup, and cutmix
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	CIFAR10-DVS best: 81.3% with DWT-C and 81.2% with DWT-C* / 2D-WT-Haar*. DvsGesture best: 99.8% with DWT-C*. CIFAR-10 best: 95.6% with 2D-WT-Haar*. CIFAR-100 best: 78.3% with 2D-FFT*
Secondary Metrics	Parameter count, GPU memory usage, training speed, inference speed, theoretical energy consumption, OPs/SOPs, accuracy by layer number
Baseline Models Compared	SSA/Spikformer, SDSA, STBP-tdBN, TET, LIAF, TA-SNN, Rollout, DECOLLE, PLIF, D-ResNet, Dspike, SALT, DSR, Hybrid training, Diet-SNN, ANN ResNet-19, ANN Transformer-4-384
Performance Gain Over Baseline	Compared with SSA/Spikformer, FWformer improves event-based accuracy by about 0.3% on CIFAR10-DVS and up to 1.5% on DvsGesture; reduces parameters by about 20% on event datasets and 25% on static datasets; reduces theoretical energy by about 20%–25%; improves training speed by about 9%–51% and inference speed by about 19%–70%
Statistical Validation	Benchmark comparison and ablation studies; no standard deviation, confidence interval, repeated-run statistical testing, or significance test reported
Ablation Study	Full. Includes 1D-FFT, 2D-FFT, 2D-WT-Haar, membrane shortcut variants, DWT-C/DWT-C-F, different Wavelet bases, layer-number ablation, SSA orthogonality analysis, memory/speed/energy comparison
Training Framework / Code Availability	PyTorch and SpikingJelly; experiments conducted on NVIDIA A100 GPU; dataset availability links provided; code availability NR
Target Hardware	Neuromorphic hardware motivation; energy estimated using 45 nm operation energy assumptions
Hardware Deployment Status	No real neuromorphic hardware deployment reported; experiments run on GPU with theoretical energy estimation
Energy Evidence	Estimated energy using MAC = 4.6 pJ and AC = 0.9 pJ on 45 nm hardware; reports OPs/Power for DVS10, DVS128, CIFAR10, and CIFAR100
Latency / Memory / Complexity Evidence	Reports GPU memory usage, training speed, inference speed, OPs, and theoretical power. Complexity of SSA is (O(N^2d)) or (O(Nd^2)), while 1D-FFT is (O(N \log N)), and 2D-FFT/2D-WT are (O(D \log D) + O(N \log N))
Comparison with ANN Baseline	Partial. Includes ANN ResNet-19 and ANN Transformer-4-384 in static CIFAR comparison, but main comparisons are against SNN/Spiking Transformer baselines
Main Contribution / Key Limitation	Contribution: proposes FWformer, showing that fixed Fourier/Wavelet bases can replace SSA in Spikformer to reduce complexity, energy, memory, and runtime while maintaining or improving accuracy. Limitation: evaluated only on visual classification datasets; no real hardware deployment; energy is theoretical; broader tasks such as detection, segmentation, and NLP are not evaluated
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ5, RQ1

P070 SpikCommander: A High-performance Spiking Transformer with Multi-view Learning for Efficient Speech Command Recognition	
Full Citation	Wang, J., Yu, L., Shen, X., Guo, S., Zhou, C., Zhao, L., Zhong, Y., Zhang, Z., & Ma, Z. (2026). SpikCommander: A High-performance Spiking Transformer with Multi-view Learning for Efficient Speech Command Recognition. In Proceedings of the Fortieth AAAI Conference on Artificial Intelligence, AAAI-26, pp. 2119–2127.
University	Harbin Institute of Technology
Country	China
Publication Year	2026
Publication Venue	AAAI Conference on Artificial Intelligence
Publisher	Association for the Advancement of Artificial Intelligence
DOI / URL	10.1609/aaai.v40i3.37194
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Speech, Audio, and Auditory Processing
Specific Task	Speech command recognition on spiking and non-spiking speech datasets
Data Modality	Spiking auditory event data and audio speech signals converted into Mel spectrograms
Signal Dimensionality	1D temporal speech sequences; SHD/SSC spike trains with 140 binned input neurons; GSC Mel spectrograms with 140 frequency bins and temporal sequence length of 100 time steps in the main setting
Signal Characteristics	SHD and SSC are spiking speech-command datasets; original 700 input neurons are reduced to 140 by spatio-temporal binning; samples are zero-padded to fixed time steps; GSC audio is downsampled from 16 kHz to 8 kHz and converted to Mel spectrograms
Dataset Name	Spiking Heidelberg Digits, SHD; Spiking Speech Commands, SSC; Google Speech Commands V2, GSC
Dataset Type	Public spiking speech datasets and public non-spiking speech command recognition dataset
Dataset Size	SHD: 10k recordings across 20 classes. SSC: 100k recordings across 35 speech-command classes. GSC: 100k recordings across 35 speech-command classes
Dataset Balance / Split	Uses preprocessing and splits following prior SCR work; exact train/validation/test split and class balance NR
Hybrid Topology	Fully spike-driven Transformer architecture with Spiking Embedding Extractor, repeated SpikCommander blocks, Multi-view Spiking Temporal-Aware Self-Attention, Spiking Contextual Refinement MLP, and classification head
SNN Component	LIF neurons, spike-based query/key/value projections, spiking temporal-aware attention, spiking contextual refinement MLP, spike-aware channel mixing, spike trains, temporal masks
Neuron Model	LIF
Transformer Component	Spiking Transformer block, MSTASA attention module, local sliding-window STASA, long-range STASA, V-branch convolutional pathway, SCR-MLP, classification head
Attention Type	Multi-view Spiking Temporal-Aware Self-Attention; includes sliding-window local STASA, long-range global STASA, and convolutional V-branch
SNN–Transformer Interface Mechanism	Speech inputs are converted into spiking embeddings through SEE; Q/K/V are generated using pointwise convolution, BN, and spiking neuron blocks; STASA aggregates masked spiking Q and K over time to form temporal-aware attention, then gates the spiking V stream using Hadamard product
Model Depth / Scale	Reported models include SpikCommander-1L-8-128 for SHD and SpikCommander-1L/2L-16-256 for SSC and GSC; main experiments use 100 time steps; sliding-window radius (w = 20)
Parameter Count	SHD: 0.19M for 1L-8-128. SSC/GSC: 1.12M for 1L-16-256 and 2.13M for 2L-16-256
Design / Contribution Type	New fully spike-driven SCR Transformer; proposes MSTASA, dual local/global STASA, V-branch, Spiking Embedding Extractor, and Spiking Contextual Refinement MLP
Input Encoding Method	SHD/SSC: spatio-temporal binning from 700 to 140 input neurons and zero-padding to fixed time steps. GSC: waveform downsampling to 8 kHz, Mel spectrogram extraction with 140 frequency bins, then spiking embedding extraction
Encoding Parameters	Main setting uses 100 time steps; total duration 1000 ms; temporal resolutions (\Delta t \in {1,4,10}) correspond to (T \in {1000,250,100}); GSC uses window size 256 and hop lengths 8, 32, or 80; sliding-window radius (w = 20)
Encoding Learned or Fixed	Partially learned: SEE, MSTASA, SCR-MLP, Q/K/V projections, and classifier weights are learned; spatio-temporal binning, padding, Mel spectrogram preprocessing, and temporal mask design are fixed
Event / Spike Representation	Binary spike trains, spiking speech inputs, spiking embeddings, spike-based Q/K/V tensors, masked spike sequences, spiking attention maps
Temporal Resolution	Main experiments use 100 time steps; long-term learning analysis evaluates 10 to 250 time steps; SHD/SSC can be discretized into 1000, 250, or 100 time steps depending on bin size
Output Decoding Method	Per-time-step softmax outputs are summed across time steps; accumulated class scores are used for final prediction
Encoding / Temporal Information Analysis	Yes. The paper analyzes local versus long-range temporal modeling, sliding-window radius, temporal mask, long-term time-step scaling, spike-train augmentation, and energy/SOP efficiency
Training Paradigm	Supervised direct SNN training
Training Method	End-to-end training from scratch using BPTT with surrogate gradients
Surrogate Gradient Function	Surrogate gradients are used; exact surrogate function NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	Cross-entropy loss over accumulated time-step scores
Training Hyperparameters	Main evaluation uses 100 time steps and (w = 20); GSC uses 8 kHz audio and 140 Mel bins; data augmentation includes SpecAugment for Mel spectrograms and drop-by-time/drop-by-neuron for spike trains; detailed optimizer and learning-rate settings are stated to be in appendix but are NR in the uploaded main text
Primary Performance Metric	Classification accuracy
Primary Metric Value	SHD: 96.41% with 0.19M parameters. SSC: 83.49% with 2.13M parameters. GSC: 96.92% with 2.13M parameters at 100 time steps; long-term GSC result reaches 97.08% at 200 time steps
Secondary Metrics	Parameter count, FLOPs, SOPs, theoretical energy consumption, time-step scaling accuracy, model size, ablation accuracy
Baseline Models Compared	SDT, Spikformer, DH-SNN, d-cAdLIF, DCLS-Delays, SpikeSCR, SE-adLIF, Event-SSMA, Pfa-SNN, Spiking LMUFormer, LMUFormer, KWT-3, AST, KW-MLP

Performance Gain Over Baseline	On SSC, SpikCommander-2L improves over SpikeSCR-2L from 82.79% to 83.49% while reducing parameters from 3.30M to 2.13M. On GSC, it improves over SpikeSCR-2L from 95.60% to 96.92% with the same parameter reduction. On SHD, it improves over Pfa-SNN from 96.26% to 96.41%
Statistical Validation	Benchmark comparison and ablation studies; five-seed analysis is mentioned in appendix, but exact mean/std values are NR in the uploaded main text
Ablation Study	Full. Includes removing data augmentation, removing V-branch, removing SWA-STASA, replacing SCR-MLP with standard MLP, replacing SEE with Conv1D projection, temporal-mask analysis, sliding-window-radius analysis, and replacing STASA with SSA/SDSA
Training Framework / Code Availability	Code available at GitHub: JackieWang9811/SCommander; exact framework NR
Target Hardware	Neuromorphic hardware motivation including Tianjic, Loihi, and TrueNorth; no specific hardware platform targeted for deployment
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation reported. On GSC, SpikCommander-2L consumes 0.042 mJ; on SSC, 0.0180 mJ; on SHD, 0.0039 mJ. No measured hardware power/energy
Latency / Memory / Complexity Evidence	Reports SOPs, FLOPs, energy, parameter count, and time-step scaling. STASA reduces attention complexity from SSA's (O(N^2D)) to (O(ND)); no measured wall-clock latency or memory footprint reported
Comparison with ANN Baseline	Yes. Compares with ANN speech-command models such as KWT-3, AST, KW-MLP, and LMUFormerA on GSC
Main Contribution / Key Limitation	Contribution: proposes SpikCommander, a fully spike-driven Transformer for SCR using MSTASA and SCR-MLP to improve temporal-context modeling with lower parameters and energy. Limitation: no real neuromorphic hardware validation; optimizer/training details are mostly in appendix; evaluated only on speech command recognition tasks
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P071
Training-Free ANN-to-SNN Conversion for High-Performance Spiking Transformers

Full Citation	Wang, J., Deng, X., Wei, W., Zhang, D., Wang, S., Sun, Q., Zhang, J., Liu, H., Xie, N., & Zhang, M. (2026). Training-Free ANN-to-SNN Conversion for High-Performance Spiking Transformers. In The Fortieth AAAI Conference on Artificial Intelligence (AAAI-26), pp. 2128–2136. Association for the Advancement of Artificial Intelligence.
University	University of Electronic Science and Technology of China
Country	China
Publication Year	2026
Publication Venue	The Fortieth AAAI Conference on Artificial Intelligence, AAAI-26
Publisher	Association for the Advancement of Artificial Intelligence
DOI / URL	10.1609/aaai.v40i3.37195
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core conversion method
E&QA; Score	4.5
Application Domain	Natural Language Processing and Language Models
Specific Task	Training-free conversion of pretrained ANN Transformers into SNN Transformers; evaluated on image classification, text classification, subjectivity classification, and language modeling
Data Modality	Static image data and text/token sequence data
Signal Dimensionality	2D image patches for ViT/CNN models; 1D token sequences for RoBERTa and GPT-2
Signal Characteristics	Pretrained ANN activations and nonlinear Transformer operations are approximated using spike-based MBE neuron representations; main converted models use short timesteps, especially T = 16 for ViT, RoBERTa, and GPT-2
Dataset Name	ImageNet, SST-2, SST-5, MR, Subj, WikiText-2, WikiText-103
Dataset Type	Public benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Uses standard benchmark settings; exact train/test/validation split details NR
Hybrid Topology	ANN-to-SNN conversion-based Spiking Transformer; pretrained ANN Transformer components are retained while nonlinear operations, FP multiplication, Softmax, GELU, Tanh, and LayerNorm are approximated through MBE neuron-based spiking representations
SNN Component	Multi-Basis Exponential Decay neuron, spike trains, spike intensity, threshold/reset mechanism, spike-driven FP multiplication, spike-based Softmax approximation, spike-based LayerNorm approximation
Neuron Model	Multi-Basis Exponential Decay neuron; Few-Spikes neuron used as comparison/analysis baseline
Transformer Component	ViT-B/16, ViT-M/16, RoBERTa-Base, RoBERTa-Large, GPT-2; Transformer nonlinear activation, attention, Softmax, LayerNorm, and MLP components
Attention Type	Transformer self-attention with spike-based approximation of floating-point multiplication and Softmax; not a conventional binary-only spiking self-attention block
SNN–Transformer Interface Mechanism	Non-spike-friendly Transformer operations are decomposed into basic functions and approximated using MBE neurons; pretrained ANN weights are not modified; activation values are represented through spike trains and accumulated basis outputs
Model Depth / Scale	ViT-B/16: 86M parameters; ViT-M/16: 64M; RoBERTa-Base: 125M; RoBERTa-Large: 355M; GPT-2: 346M; CNN comparisons include VGG16: 138M and ResNet-34: 22M
Parameter Count	VGG16: 138M; ResNet-34: 22M; ViT-B/16: 86M; ViT-M/16: 64M; RoBERTa-Base: 125M; RoBERTa-Large: 355M; GPT-2: 346M
Design / Contribution Type	New training-free ANN-to-SNN conversion framework for Transformers; proposes MBE neuron; analyzes FS neuron limitations; supports spiking approximation of GELU, FP multiplication, Softmax, and LayerNorm
Input Encoding Method	Activation-to-spike conversion using MBE neuron with exponential decay and multi-basis encoding; raw input encoding details for image/text inputs are NR because the focus is operator-level ANN-to-SNN conversion
Encoding Parameters	MBE uses basis number N, timesteps T, spike intensity, reset value, threshold, exponential decay rates, and basis weights; N = 4 is used as the main efficient setting; T = 16 for main ViT/RoBERTa/GPT-2 conversion; GELU approximation samples M = 10,000 points over the constrained interval –120 to 10
Encoding Learned or Fixed	Partially learned/fitted for operator approximation; source ANN weights remain fixed and no additional training is applied to the pretrained ANN

Event / Spike Representation	Binary spike train s[t]; accumulated basis output o(n)(T); weighted multi-basis spike representation; spike matrix for approximating FP multiplication
Temporal Resolution	Main experiments use T = 16 for ViT, RoBERTa, and GPT-2; CNN conversion uses T = 10; ablation evaluates T = 8, 10, 12, and 16
Output Decoding Method	Weighted accumulation of MBE basis outputs approximates ANN activation/function output; final task output follows the original model head, such as classification logits or language-model perplexity
Encoding / Temporal Information Analysis	Yes. The paper gives theoretical and empirical analysis of FS neuron approximation error, excessive dependence on initialization, global suboptimality, basis-count effects, decay effects, timestep sensitivity, and energy–accuracy trade-off
Training Paradigm	Training-free ANN-to-SNN conversion; pretrained supervised ANN models are converted without retraining the source ANN
Training Method	ANN-to-SNN conversion using MBE neuron-based approximation of nonlinear Transformer operators; no surrogate-gradient direct SNN training of the converted model
Surrogate Gradient Function	N/A
ANN-to-SNN Conversion Method	MBE neuron-based training-free conversion; decomposes FP multiplication, GELU/Tanh, Softmax, and LayerNorm into spike-approximable operations while preserving pretrained ANN parameters
Optimizer	NR
Loss Function	MSE is used for function-approximation analysis/fitting; task-level training loss is N/A because the method is training-free for the source ANN
Training Hyperparameters	RTX 4090 GPU; PyTorch framework; N = 4 basis components selected as main trade-off point; T = 16 for main Transformer experiments; M = 10,000 sampled points for GELU approximation; other optimizer and learning-rate details NR
Primary Performance Metric	Conversion accuracy for CV/NLU tasks; perplexity for NLG; conversion loss relative to ANN; timestep latency
Primary Metric Value	ImageNet: ViT-B/16 achieves 83.00% vs 83.44% ANN; ViT-M/16 achieves 85.31% vs 85.95% ANN. NLU: RoBERTa-Large achieves 95.98% on SST-2, 58.31% on SST-5, 90.96% on MR, and 97.45% on Subj at T = 16. NLG: GPT-2 achieves 22.69 perplexity on WikiText-2 and 23.41 on WikiText-103 at T = 16
Secondary Metrics	Timesteps, conversion loss, perplexity, firing rate, energy ratio, MSE approximation loss, basis-count effect, decay effect
Baseline Models Compared	DSR, TET, AT-SNN, Spikingformer, SDTv1, Spikeformer, QKFormer, SDTv3, QCFS, QFFS, SNM, ECL, AdaFire, ECMT, STA, SpikeZIP-TF, SpikedAttention, Spikeformer, SpikeBERT, SpikeGPT, original ANN ViT/RoBERTa/GPT-2
Performance Gain Over Baseline	On SST-2, RoBERTa-Large conversion achieves 95.98% at T = 16, outperforming SpikeZIP-TF by 2.19% while reducing timesteps by 87.5%. On MR with RoBERTa-Base, it achieves 89.00%, exceeding SpikeZIP-TF by 2.87% with 75% fewer timesteps. On WikiText-103, it reaches 23.41 perplexity, improving over SpikeGPT's 39.75 while using T = 16 instead of T = 1024
Statistical Validation	Benchmark comparison and ablation experiments; no standard deviation, confidence interval, repeated-run analysis, or statistical significance testing reported
Ablation Study	Full. Includes MBE vs FS neuron comparison, basis-count analysis, exponential decay effect, timestep sensitivity across ImageNet/MR/WikiText-103, and energy–accuracy trade-off
Training Framework / Code Availability	PyTorch; RTX 4090 used for experiments; code availability NR
Target Hardware	No specific hardware platform; neuromorphic/energy-efficient deployment is the motivation
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	Estimated energy only. Uses energy ratio ESNN/EANN = SOPs × EAC / FLOPs × EMAC, with EMAC = 4.6 pJ and EAC = 0.9 pJ. For ViT-M/16 GELU with N = 4 and T = 16, reported firing rate is 38.22%, giving estimated energy consumption of 13.7% relative to ANN GELU computation
Latency / Memory / Complexity Evidence	Latency is mainly represented by timestep count. The method reaches near-lossless conversion at T = 16 and near-optimal performance at T = 12 for several models; no measured wall-clock latency or memory footprint reported
Comparison with ANN Baseline	Yes. Directly compares converted SNN performance with original ANN models including ViT-B/16, ViT-M/16, RoBERTa, and GPT-2
Main Contribution / Key Limitation	Contribution: introduces a training-free MBE-based ANN-to-SNN conversion framework for high-performance Spiking Transformers, supporting difficult nonlinear operations such as FP multiplication, Softmax, GELU, and LayerNorm. Limitation: no real neuromorphic hardware deployment or measured energy/latency; code availability NR; detailed limitation discussion is limited
RQ Mapping	Primary: RQ4; Secondary: RQ2, RQ3, RQ5, RQ1

P072

Spiking Point Transformer for Point Cloud Classification

Full Citation	Wu, P., Chai, B., Li, H., Zheng, M., Peng, Y., Wang, Z., Nie, X., Zhang, Y., & Sun, X. (2025). Spiking Point Transformer for Point Cloud Classification. In Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI-25), pp. 21563–21571. AAAI Press.
University	University of Science and Technology of China
Country	China
Publication Year	2025
Publication Venue	AAAI Conference on Artificial Intelligence
Publisher	Association for the Advancement of Artificial Intelligence
DOI / URL	10.1609/aaai.v39i20.35459
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	Point cloud classification
Data Modality	3D point cloud data
Signal Dimensionality	3D unordered point sets with spatial coordinates and point features
Signal Characteristics	Input point cloud contains N = 1024 points; Q-SDE samples point subsets across timesteps to reduce redundancy while retaining key support points; point cloud data are sparse and high-dimensional
Dataset Name	ModelNet10, ModelNet40, ScanObjectNN

Dataset Type	Public 3D point cloud classification benchmark datasets; ModelNet10/40 are synthetic CAD datasets; ScanObjectNN is a real-world scanned object dataset
Dataset Size	ModelNet40: 12,311 CAD models, 9,843 training and 2,468 testing instances; ModelNet10: 3,991 training and 908 testing instances; ScanObjectNN: 11,416 training and 2,882 testing samples
Dataset Balance / Split	Uses standard dataset train/test splits; exact class-balance information NR
Hybrid Topology	Spiking Point Transformer framework; point cloud is encoded using Q-SDE, then processed by MLP module, Spiking Point Transformer Block, Spiking Transition Down Block, Spiking Point Encoder Modules, and classification head
SNN Component	Queue-Driven Sampling Direct Encoding, spiking neurons, LIF neurons, HD-IF neurons, Spiking Point Transformer Block, Spiking Transition Down Block, spike-form Q/K/V
Neuron Model	LIF and Hybrid Dynamics Integrate-and-Fire neuron; HD-IF combines IF, LIF, EIF, and PLIF dynamics
Transformer Component	Point Transformer-style local self-attention; Spiking Point Transformer Block; Spiking Point Encoder Module
Attention Type	Spiking local self-attention using KNN neighborhood; spike-form Query, Key, and Value attention
SNN–Transformer Interface Mechanism	Point features are encoded into spike/membrane representations; KNN neighborhood features are used to generate spike Q, K, and V; relative position encoding is added before local attention aggregation
Model Depth / Scale	Architecture includes Q-SDE, MLP module, one initial SPTB, L Spiking Point Encoder Modules with STDB and SPTB, and final classification head; exact L value NR
Parameter Count	NR
Design / Contribution Type	First Transformer-based SNN framework for 3D point cloud classification; proposes Q-SDE and HD-IF neuron for efficient point-cloud spiking representation and robust neuron dynamics
Input Encoding Method	Queue-Driven Sampling Direct Encoding; improved direct encoding that samples point subsets across timesteps using FIFO queue-driven sampling and FPS
Encoding Parameters	N = 1024 input points; timestep T up to 4; sampled point counts tested include Q-SDE256, Q-SDE512, Q-SDE768, and Q-SDE1024; firing threshold Vth = 0.5
Encoding Learned or Fixed	Q-SDE sampling procedure is fixed; HD-IF gating and model weights are learned during training
Event / Spike Representation	Binary spike signals; membrane potential features; spike Q/K/V; sampled point-cloud spike representations over multiple timesteps
Temporal Resolution	Main experiments use 4 timesteps; ablation evaluates 1, 2, 3, and 4 timesteps
Output Decoding Method	Membrane potential is passed to the classification head to produce final class prediction
Encoding / Temporal Information Analysis	Yes. The paper analyzes Q-SDE versus direct encoding and Random-SDE, timestep effects, runtime/memory efficiency, support-point preservation, HD-IF activation behavior, and theoretical energy consumption
Training Paradigm	Supervised direct SNN training
Training Method	Surrogate-gradient-based direct training using SpikingJelly/PyTorch; point cloud classification trained end-to-end for 200 epochs
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	PyTorch 1.13; SpikingJelly; 2 × RTX 3090Ti GPUs; AdamW optimizer; momentum 0.9; weight decay 0.0001; initial learning rate 0.001; learning rate decays by factor 0.3 every 50 epochs; 200 epochs; N = 1024 input points; Vth = 0.5
Primary Performance Metric	Overall Accuracy and mean Class Accuracy
Primary Metric Value	ModelNet10: 94.76% OA and 93.69% mAcc; ModelNet40: 91.43% OA and 89.39% mAcc; ScanObjectNN: 78.03% OA without voting and 82.23% OA with voting
Secondary Metrics	Runtime, memory consumption, AC operations, MAC operations, theoretical power/energy consumption, timestep ablation accuracy
Baseline Models Compared	PointNet, PointNet++, Point Transformer, PointMLP, KPConv-SNN, Spiking PointNet, P2SResLNet-B
Performance Gain Over Baseline	On ModelNet40, SPT reaches 91.43% OA, improving over P2SResLNet-B by 0.83% OA. On ModelNet10, SPT reaches 94.76% OA, improving over Spiking PointNet by 1.45% OA. On ScanObjectNN, SPT reaches 78.03% OA without voting, improving over P2SResLNet-B by 3.57% OA
Statistical Validation	Benchmark evaluation and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes timestep ablation, encoding-method ablation, Q-SDE runtime/memory analysis, HD-IF neuron ablation, and energy-efficiency analysis
Training Framework / Code Availability	PyTorch 1.13; SpikingJelly; code available at https://github.com/PeppaWu/SPT
Target Hardware	Neuromorphic hardware motivation only; no specific hardware deployment platform
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using 32-bit MAC = 4.6 pJ and AC = 0.9 pJ; SPT consumes 3.0 mJ at 1 timestep and 13.3 mJ at 4 timesteps on ModelNet40, compared with 84.7 mJ for ANN Point Transformer
Latency / Memory / Complexity Evidence	Q-SDE512 at T = 4 reduces inference runtime to 191 ms and memory to 5.2 GB, compared with direct encoding at 234 ms and 9.3 GB; training runtime/memory also reduced from 478 ms/15.3 GB to 326 ms/9.7 GB
Comparison with ANN Baseline	Yes. Compares against ANN baselines including PointNet, PointNet++, Point Transformer, and PointMLP; also compares theoretical energy against ANN Point Transformer
Main Contribution / Key Limitation	Contribution: introduces the first Spiking Point Transformer for 3D point cloud classification, with Q-SDE for efficient temporal point encoding and HD-IF for adaptive neuron dynamics. Limitation: no real neuromorphic hardware evaluation; only classification is tested, while segmentation and detection are left as future directions
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P073

Spike-Driven Transformer V2: Meta Spiking Neural Network Architecture Inspiring the Design of Next-Generation Neuromorphic Chips

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University	Chinese Academy of Sciences
Country	China

Publication Year	2024
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Publisher	International Conference on Learning Representations
DOI / URL	https://www.scopus.com/pages/publications/85195038811?origin=resultslist
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification, event-based human activity recognition, object detection, and semantic segmentation
Data Modality	Static RGB images, event-camera/DVS data, object-detection images, semantic-segmentation images
Signal Dimensionality	2D image data; 2D event-frame representation; dense prediction maps for detection and segmentation
Signal Characteristics	Static images are repeated across timesteps; ImageNet input crop is 224 × 224; HAR-DVS uses DAVIS346 event-camera data with 346 × 260 spatial resolution; spike tensors are binary 0/1 representations
Dataset Name	ImageNet-1K, HAR-DVS, COCO val2017, ADE20K, VOC2012
Dataset Type	Public benchmark datasets for image classification, event-based human activity recognition, object detection, and semantic segmentation
Dataset Size	ImageNet-1K: about 1.3M training and 50K validation images; HAR-DVS: 107,646 samples, 300 classes, raw data over 4TB; COCO: 118K train2017 and 5K val2017 images; ADE20K: 20K training and 2K validation images, 150 categories; VOC2012: 1,460 training and 1,456 validation images, 21 categories
Dataset Balance / Split	Uses standard benchmark splits; exact class-balance information NR
Hybrid Topology	Meta-SpikeFormer uses a hybrid Conv-based SNN + Transformer-based SNN architecture; early stages use Conv-based SNN blocks and later stages use Transformer-based SNN blocks with spike-driven self-attention
SNN Component	LIF spiking neuron layers, Conv-based SNN blocks, Transformer-based SNN blocks, spike-form Q/K/V, spike-driven self-attention, membrane shortcut
Neuron Model	LIF
Transformer Component	Meta Transformer-style block, Transformer-based SNN block, Channel MLP, Spike-Driven Self-Attention, pyramid Transformer-like stages
Attention Type	Spike-Driven Self-Attention; SDSA-1, SDSA-2, SDSA-3, and SDSA-4 are analyzed; SDSA-3 is used by default
SNN–Transformer Interface Mechanism	Membrane potentials are converted into spike tensors by LIF neuron layers; spike-form Query, Key, and Value are generated using re-parameterization convolutions; SDSA performs attention using sparse addition without softmax or scale
Model Depth / Scale	Main architecture has four stages: Stage 1 with 2 Conv-based SNN blocks, Stage 2 with 2 Conv-based SNN blocks, Stage 3 with 6 Transformer-based SNN blocks, and Stage 4 with 2 Transformer-based SNN blocks; model scales include 15.1M, 31.3M, and 55.4M parameters
Parameter Count	15.1M, 31.3M, and 55.4M for ImageNet models; detection/segmentation variants include 16.5M, 16.8M, 18.3M, 34.9M, 58.9M, and 75.0M depending on task/model
Design / Contribution Type	Meta Transformer-based SNN architecture; spike-driven vision backbone; SDSA operator redesign; Conv+ViT-style SNN macro design; shortcut analysis; neuromorphic-chip-oriented design guidance
Input Encoding Method	Direct encoding for static images by repeating images across timesteps; event-frame input for HAR-DVS; spike neuron layers convert membrane potentials into binary spike tensors
Encoding Parameters	ImageNet training starts with T = 1 for 200 epochs and is fine-tuned to T = 4 for 20 epochs; HAR-DVS uses T = 8; detection and segmentation report T = 1 and T = 4 settings; LIF threshold, reset, and decay are formulated but exact numerical values NR
Encoding Learned or Fixed	Partially learned: spike neuron dynamics are fixed, while convolution, RepConv, MLP, SDSA-related weights, and task heads are learned
Event / Spike Representation	Binary spike tensors; spike-form Query, Key, and Value; membrane potential tensors; spike-driven feature maps; event-frame representation for HAR-DVS
Temporal Resolution	T = 1 and T = 4 for ImageNet/detection/segmentation experiments; T = 8 for HAR-DVS
Output Decoding Method	Classification head for ImageNet and HAR-DVS; Mask R-CNN or YOLO detection head for COCO; Semantic FPN segmentation head for ADE20K and VOC2012
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike-driven computation, SDSA complexity, shortcut effects, firing rates, theoretical energy consumption, timestep effects, and power–accuracy trade-offs
Training Paradigm	Supervised direct SNN training; transfer learning/fine-tuning for detection and segmentation
Training Method	Direct training with surrogate-gradient-based SNN learning implied; ImageNet pretraining followed by fine-tuning for downstream dense prediction tasks
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	LAMB for ImageNet classification; AdamW for COCO and ADE20K/VOC segmentation/detection fine-tuning
Loss Function	Task-specific losses are used through classification, Mask R-CNN/YOLO detection, and Semantic FPN segmentation frameworks; exact loss formulas NR
Training Hyperparameters	ImageNet: 224 × 224 resolution, 200 epochs at T = 1, then 20 epochs fine-tuning at T = 4; batch size 1568 for direct training and 336 for fine-tuning; LAMB optimizer; base learning rate 6e-4 then 2e-5; cosine decay; warmup 10 and 2 epochs; weight decay 0.05; RandAugment 9/0.5; label smoothing 0.1. COCO: 24 epochs fine-tuning, batch size 12, AdamW, learning rate 1e-4, polynomial decay. ADE20K: 160K iterations, batch size 20, AdamW, learning rate 1e-3 then T = 4 fine-tuning at 1e-4
Primary Performance Metric	Top-1 accuracy for classification/HAR; mAP@0.5 for detection; mIoU for segmentation
Primary Metric Value	ImageNet-1K: 80.0% top-1 accuracy with 55.4M parameters at T = 4 using distillation; HAR-DVS: 47.5% accuracy; COCO: 51.2% mAP@0.5 with Mask R-CNN setting and 50.3% with Meta-SpikeFormer + YOLO at T = 4; ADE20K: 35.3% mIoU; VOC2012: 61.1% mIoU
Secondary Metrics	Parameter count, theoretical power/energy in mJ, timestep count, firing rate, ablation accuracy, model scale
Baseline Models Compared	ResNet-34 ANN-to-SNN, VGG-16 ANN-to-SNN, SEW-Res-SNN, MS-Res-SNN, Att-MS-Res-SNN, RSB-CNN-152, ViT, Spikformer, Spike-driven Transformer, SlowFast, ACTION-Net, TimeSformer, Res-SNN-34, ResNet-18, PVT-Tiny, PVT-Small, Spiking-Yolo, Spike Calibration, Spiking Retina, EMS-Res-SNN, DeepLab-V3, FCN-R50, Spiking FCN, Spiking DeepLab
Performance Gain Over Baseline	On ImageNet-1K, Meta-SpikeFormer reaches 80.0%, improving over prior SOTA Spike-driven Transformer/MS-Res-SNN by about 3.7% while using fewer parameters than the 66M baseline. On HAR-DVS, it improves over Res-SNN-34 and is close to ANN video baselines. On COCO and VOC2012/ADE20K, it establishes SOTA or first strong direct-training SNN dense-prediction results

Statistical Validation	Benchmark comparison and ablation experiments; no standard deviation, confidence interval, repeated-run analysis, or statistical significance testing reported
Ablation Study	Full. Includes Conv-based SNN block ablation, Channel Conv vs Channel MLP, RepConv vs Linear Q/K/V generation, SDSA-1/2/3/4 comparison, shortcut comparison, pyramid removal, fully CNN-based SNN, fully Transformer-based SNN, and firing-rate analysis
Training Framework / Code Availability	Code and models stated as available on GitHub: BICLab/Spike-Driven-Transformer-V2; detection uses spiking version of MMDetection; segmentation uses spiking version of MMSegmentation
Target Hardware	Neuromorphic chip design is a major motivation; paper discusses inspiration for next-generation Transformer-based neuromorphic chips, but does not target one specific chip
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation only. Uses EMAC = 4.6 pJ and EAC = 0.9 pJ; reports power/energy values such as 52.4 mJ for 55.4M Meta-SpikeFormer at T = 4 on ImageNet and 13.0 mJ at T = 1
Latency / Memory / Complexity Evidence	Reports computational complexity of SDSA operators: SDSA-1/2 are O(ND), SDSA-3/4 are $O(ND^2)$ under the linear attention formulation; provides model parameters, timestep counts, power estimates, and layer-wise firing rates; measured wall-clock latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares with ANN models such as RSB-CNN-152, ViT, SlowFast, ACTION-Net, TimeSformer, ResNet-18, PVT-Tiny, PVT-Small, DeepLab-V3, and FCN-R50
Main Contribution / Key Limitation	Contribution: proposes Meta-SpikeFormer, a general spike-driven Transformer-based SNN backbone that supports classification, event-based recognition, detection, and segmentation using sparse-addition SDSA and Conv+Transformer SNN macro design. Limitation: energy is theoretical rather than measured on neuromorphic hardware; larger models, more vision/language tasks, training cost from multiple timesteps, and further accuracy–power–parameter optimization remain open
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P074

SVFormer: A Direct Training Spiking Transformer for Efficient Video Action Recognition

Full Citation	Yu, L., Huang, L., Zhou, C., Zhang, H., Ma, Z., Zhou, H., & Tian, Y. (2024). SVFormer: A Direct Training Spiking Transformer for Efficient Video Action Recognition. arXiv:2406.15034v1.
University	University of Electronic Science and Technology of China
Country	China
Publication Year	2024
Publication Venue	Communications in Computer and Information Science
Publisher	Springer Nature
DOI / URL	https://link.springer.com/chapter/10.1007/978-981-96-4001-0_11
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	RGB video action recognition and event-based gesture/action recognition
Data Modality	RGB video frames and neuromorphic DVS event data
Signal Dimensionality	2D video frames processed as temporal frame sequences; DVS event stream integrated into event frames
Signal Characteristics	RGB video inputs are resized to 224 × 224; DVS128-Gesture uses 128 × 128 event frames; SVFormer processes one video frame per SNN timestep in a frame-by-frame manner
Dataset Name	UCF101, NTU-RGBD60, DVS128-Gesture
Dataset Type	Public video action-recognition and neuromorphic gesture-recognition benchmark datasets
Dataset Size	UCF101: 101 classes, 13,320 videos, 25 fps, 320 × 240 resolution, average length 7.21 s; NTU-RGBD60: 60 action classes, 56,880 samples; DVS128-Gesture: 11 gesture classes from 29 individuals under 3 illumination conditions
Dataset Balance / Split	UCF101 uses the first official train-test split; NTU-RGBD60 uses cross-subject and cross-view splits; DVS128-Gesture uses standard benchmark setting; exact class balance NR
Hybrid Topology	Hierarchical Spiking Video Transformer with four stages and one local pathway; early stages use patch embedding and local feature extractors, later stages use global self-attention modules
SNN Component	PLIF neurons, LIF neuron formulation, spiking neuron layers, binary spikes, local feature extractor, global self-attention module, local pathway, time-dependent batch normalization
Neuron Model	Parametric LIF as the main neuron model; LIF also discussed and used in ablation
Transformer Component	Spiking Video Transformer backbone, global self-attention module, spiking self-attention, MLP module, classification head
Attention Type	Spiking self-attention / global self-attention using spike-form Q, K, and V
SNN–Transformer Interface Mechanism	Video frames are processed across SNN timesteps; spiking neuron layers convert feature maps into spikes; Linear-BN-SN motifs generate spike-form Q, K, and V for self-attention
Model Depth / Scale	Default SVFormer-base uses S = [1, 1, 3, 1] blocks and C = [128, 256, 384, 512] channels; variants include SVFormer-ss, SVFormer-st, SVFormer-dp, SVFormer-wd, and SVFormer-3stg
Parameter Count	SVFormer-base: 13.80M on UCF101 and 13.76M on NTU-RGBD60; SVFormer-3stg: 1.88M on DVS128-Gesture; variants range from 8.77M to 24.07M
Design / Contribution Type	Directly trained Spiking Transformer for video action recognition; combines local feature extraction, global self-attention, PLIF neurons, local-global fusion, time-dependent batch normalization, and efficient frame-by-frame temporal processing
Input Encoding Method	Direct frame-by-frame encoding; one video frame is fed into SVFormer at each timestep; DVS events are integrated into event-frame sequences
Encoding Parameters	RGB video input size 224 × 224; DVS input size 128 × 128; UCF101/NTU main setting uses T = 16 timesteps; DVS128-Gesture also uses 16 timesteps; ablation tests T = 8 and T = 24
Encoding Learned or Fixed	Partially learned: frame sampling/preparation is fixed, while PLIF membrane time constants and network parameters are learned
Event / Spike Representation	Binary spike tensors; PLIF-generated spike activations; event frames for DVS128-Gesture; spike-form Q/K/V tensors
Temporal Resolution	Main setting uses 16 frames / 16 SNN timesteps; ablation also evaluates 8 and 24 timesteps; clip-by-clip 3D input setting uses 4 simulation steps

Output Decoding Method	Classification head applies SN, depthwise 3D convolution, BN, squeezing, and a final linear layer to produce action-category logits
Encoding / Temporal Information Analysis	Yes. The paper analyzes frame-by-frame vs clip-by-clip processing, timestep length, average firing rates, learned PLIF membrane time constants, noise robustness, and energy consumption
Training Paradigm	Supervised direct SNN training
Training Method	End-to-end direct training using surrogate-gradient backpropagation through time
Surrogate Gradient Function	Sigmoid surrogate gradient is illustrated; exact implementation slope/settings NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Classification loss / cross-entropy implied for action recognition; exact formula NR
Training Hyperparameters	RGB datasets: input size 224 × 224, batch size 64 on 4 NVIDIA V100 GPUs, 600 epochs, warmup plus cosine learning-rate decay, base learning rate 0.006, AdamW optimizer. DVS128-Gesture: input size 128 × 128, batch size 16, 600 epochs, learning rate 0.005 with cosine decay
Primary Performance Metric	Top-1 accuracy
Primary Metric Value	UCF101: 84.03% for SVFormer-base; NTU-RGBD60: 88.12% cross-subject and 94.68% cross-view; DVS128-Gesture: 97.92%
Secondary Metrics	Parameter count, timestep count, theoretical energy consumption, firing rate, noise robustness accuracy, ablation accuracy
Baseline Models Compared	RSNN-reservoir-DA, RSNN-HeNHeS-STDP, RSNN-HeNB-BP, RSNN2s-tandem-cvt, SlowFast-SDM-cvt, SGLFormer, Meta-SpikeFormer, ConvNet-PLIF, Spikformer, Spikingformer, SD-Transformer, DVANet, π-ViT
Performance Gain Over Baseline	On UCF101, SVFormer-base achieves 84.03%, outperforming Meta-SpikeFormer implementation at 83.66% and SGLFormer at 74.70% among directly trained deep SNNs; its theoretical energy is 21.904 mJ versus 1054.148 mJ for the ANN counterpart
Statistical Validation	Benchmark evaluation, robustness testing, and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes local pathway removal, PLIF-to-LIF replacement, time-dependent BN effect, timestep length T = 8/16/24, frame-by-frame versus clip-by-clip input, noise robustness, firing-rate analysis, and membrane-time-constant analysis
Training Framework / Code Availability	SpikingJelly; implementation based on SlowFast and UniFormer repositories; code availability NR
Target Hardware	Neuromorphic hardware motivation only; theoretical 45 nm energy model used
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using MAC = 4.6 pJ and AC = 0.9 pJ; SVFormer-base consumes 21.904 mJ per UCF101 video clip, compared with 1054.148 mJ for ANN counterpart
Latency / Memory / Complexity Evidence	Reports FLOPs/SOP-based energy complexity and timestep count; SVFormer-base has 0.700G remaining FLOPs and 20.760G SOPs; measured wall-clock latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares with ANN models on NTU-RGBD60 and compares energy against an ANN counterpart of SVFormer-base
Main Contribution / Key Limitation	Contribution: proposes SVFormer, a directly trained Spiking Transformer for efficient video action recognition that processes video frame-by-frame using PLIF neurons, local-global spatiotemporal modeling, and spiking self-attention. Limitation: accuracy still lags behind strong ANN VAR models; clip length is predefined, so the model must process all frames before decision-making; no real hardware energy measurement
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P075

Spiking Transformers for Event-based Single Object Tracking

Full Citation	Zhang, J., Dong, B., Zhang, H., Ding, J., Heide, F., Yin, B., & Yang, X. (2022). Spiking Transformers for Event-based Single Object Tracking. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2022), pp. 8791–8800. IEEE. https://doi.org/10.1109/CVPR52688.2022.00860
University	Dalian University of Technology
Country	China
Publication Year	2022
Publication Venue	IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/CVPR52688.2022.00860; Code: https://github.com/Jee-King/CVPR2022_STNet
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Event-based single object tracking
Data Modality	Event-camera data represented as event frames
Signal Dimensionality	2D spatiotemporal event stream; asynchronous event data converted into binned positive/negative event-frame sequences
Signal Characteristics	Each event is represented as (x, y, t, p), where (x, y) is pixel location, t is timestamp, and p ∈ {-1, 1} is polarity; event cameras provide microsecond-level temporal resolution and high dynamic range
Dataset Name	FE240hz, EED, VisEvent
Dataset Type	Public event-based object-tracking benchmark datasets
Dataset Size	FE240hz: more than 1,132K annotations on more than 143K images and corresponding events; EED: 199 bounding boxes; VisEvent after filtering: 377 training sequences and 172 testing sequences, including 63 non-rigid-object testing sequences
Dataset Balance / Split	FE240hz used for training/testing under degraded conditions; EED used for validation with STNet trained on FE240hz; VisEvent uses 377 training and 172 testing sequences after removing invalid/misaligned sequences; exact class/object balance NR
Hybrid Topology	Siamese tracking architecture based on SiamFC++; template and search branches share STNet modules; each branch contains SNNformer Feature Extractor and Temporal-Spatial Feature Fusion module
SNN Component	LIF-based SNN branch, 3-layer conv-SNN blocks, membrane potential temporal memory, dynamic spiking threshold block
Neuron Model	LIF

Transformer Component	Reduced Swin Transformer branch for global spatial feature extraction; transformer-based spatial feature branch within SNNformer Feature Extractor
Attention Type	Swin Transformer spatial attention; temporal attention; spatial attention; cross-domain attention fusion
SNN–Transformer Interface Mechanism	Transformer branch extracts global spatial features; spatial statistical cues dynamically adjust the LIF spiking threshold of the SNN branch; temporal SNN features and spatial Transformer features are fused through Temporal-Spatial Feature Fusion
Model Depth / Scale	SNN branch has three conv-SNN blocks; Transformer branch uses the first two blocks of Swin Transformer; full tracker uses template/search Siamese branches with classifier and regressor heads
Parameter Count	NR
Design / Contribution Type	New spiking Transformer tracking architecture; dynamic LIF threshold based on spatial statistics; SNNformer Feature Extractor; Temporal-Spatial Feature Fusion with temporal attention, cross-domain integrator, and spatial attention
Input Encoding Method	Event-to-frame / voxel-grid encoding; events in a time window are split into n temporal bins; each bin generates positive and negative event frames
Encoding Parameters	Number of event bins $n = 5$; decay factor $\alpha = 0.2$; base spiking threshold $V_{th} = 0.3$; dynamic threshold computed as V_{th}■ = $(K + E)V_{th}$ using representative value K and entropy E from spatial features
Encoding Learned or Fixed	Event-frame binning is fixed; dynamic spiking threshold is adaptive based on spatial features; network weights are learned
Event / Spike Representation	Positive and negative event frames; event-frame sequence; spike trains; LIF membrane potentials; temporal feature maps from SNN branch
Temporal Resolution	Event cameras provide microsecond-level event timestamps; model discretizes each time window into 5 bins
Output Decoding Method	SiamFC++-style classifier and regressor heads produce target classification score, center/quality score, and bounding-box regression output
Encoding / Temporal Information Analysis	Yes. The paper analyzes temporal and spatial cue extraction, dynamic threshold behavior, degraded-condition robustness, sparse-event failure cases, and ablation of temporal/spatial branches
Training Paradigm	Supervised learning for single object tracking
Training Method	End-to-end training in PyTorch using SGD; SNN temporal branch trained jointly with Transformer spatial branch and SiamFC++ tracking heads
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Stochastic Gradient Descent
Loss Function	SiamFC++ tracking loss combining classification loss, cross-entropy center/quality loss, and regression loss: $L = \lambda_1 L_{cls} + \lambda_2 L_{ctr} + \lambda_3 L_{reg}$
Training Hyperparameters	20 epochs; batch size 38; SGD momentum 0.9; weight decay 5e-5; learning rate increased from 1e-7 to 2e-3, linearly for first 5 epochs and cosine annealed for remaining 15 epochs; $\lambda_1 = 1$, $\lambda_2 = 1$, $\lambda_3 = 3$; $\alpha = 0.2$; $V_{th} = 0.3$; $n = 5$
Primary Performance Metric	Representative Success Rate, Representative Precision Rate, and Overlap Precision
Primary Metric Value	FE240hz: RSR 58.5, OP0.50 74.6, OP0.75 28.1, RPR 89.6. EED: RSR 40.1, OP0.50 48.5, OP0.75 11.7, RPR 70.3. VisEvent: RSR 35.5, OP0.50 39.7, OP0.75 20.4, RPR 49.2
Secondary Metrics	Tracking speed/FPS, degraded-condition performance, precision plots, success plots, ablation metrics
Baseline Models Compared	SiamRPN, ATOM, DiMP, SiamFC++, OCEAN, KYS, PrDiMP, STARK-S, TransT, EFE
Performance Gain Over Baseline	On FE240hz, STNet improves over runner-up TransT by 1.8% RSR, 3.9% OP0.50, 2.8% OP0.75, and 0.6% RPR. On EED, STNet improves over TransT by 4.0% RSR, 15.8% OP0.50, and 1.8% OP0.75, although TransT has higher RPR. On VisEvent, STNet improves over the runner-up by 2.6% RSR, 0.2% OP0.50, and 2.1% RPR
Statistical Validation	Benchmark evaluation and ablation study; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes SNN branch only, Transformer branch only, dynamic threshold removal, removal of representative value K, removal of entropy E, replacing SNN with CNN/AlexNet/LSTM, removing TSFF, and removing TA/CI/SA components
Training Framework / Code Availability	PyTorch; trained on 20-core Intel i9-10900K CPU, 64 GB RAM, and NVIDIA RTX 3090 GPU; code and pretrained models available at GitHub: Jee-King/CVPR2022_STNet
Target Hardware	GPU implementation; neuromorphic hardware motivation through SNN/event-based efficiency, but no specific neuromorphic chip target
Hardware Deployment Status	No neuromorphic hardware deployment reported
Energy Evidence	No explicit energy measurement or theoretical energy table reported; efficiency is mainly argued through event-based sensing, SNN temporal processing, and tracking speed
Latency / Memory / Complexity Evidence	Reports tracking speed comparison and training batch time; each training batch requires 26.9 s on RTX 3090 setup; detailed latency, memory footprint, and complexity values NR
Comparison with ANN Baseline	Yes. Compares with conventional deep trackers and Transformer-based trackers such as TransT and STARK-S
Main Contribution / Key Limitation	Contribution: proposes STNet, a spiking Transformer tracker that dynamically combines SNN-based temporal cues and Transformer-based spatial cues for event-based single object tracking. Limitation: performance decreases when events are unavailable or too sparse, especially around the tracked object; no real neuromorphic hardware or direct energy evaluation is provided
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P076

SGLFormer: Spiking Global-Local-Fusion Transformer with high performance

Full Citation	Zhang, H., Zhou, C., Yu, L., Huang, L., Ma, Z., Fan, X., Zhou, H., & Tian, Y. (2024). SGLFormer: Spiking Global-Local-Fusion Transformer with high performance. Frontiers in Neuroscience, 18, Article 1371290. https://doi.org/10.3389/fnins.2024.1371290
University	University of Electronic Science and Technology of China
Country	China
Publication Year	2024
Publication Venue	Frontiers in Neuroscience
Publisher	Frontiers
DOI / URL	https://doi.org/10.3389/fnins.2024.1371290; Code: https://github.com/ZhangHanN1/SGLFormer
Publication Type	Journal article
Database Source	Scopus

Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification on static datasets and DVS/event-based classification datasets
Data Modality	Static RGB images and neuromorphic DVS event streams converted into frame sequences
Signal Dimensionality	2D image sequences represented as T × B × 3 × H × W; DVS event sequences represented as T × B × 2 × H × W
Signal Characteristics	Static images are repeated across timesteps; DVS event streams are integrated into frames; binary spike activations are processed through LIF neurons, local attention, and global attention
Dataset Name	CIFAR10, CIFAR100, ImageNet, CIFAR10-DVS, DVS128-Gesture
Dataset Type	Public static image classification and neuromorphic/event-based vision benchmark datasets
Dataset Size	CIFAR10/CIFAR100: 50,000 training and 10,000 testing images; ImageNet: more than 1.2M training images and 50,000 validation images; CIFAR10-DVS: 9,000 training and 1,000 testing samples; DVS128-Gesture: 11 gesture classes from 29 individuals under three illumination conditions
Dataset Balance / Split	Uses standard benchmark train/test or train/validation splits; exact class-balance information NR
Hybrid Topology	SGLFormer contains a Tokenizer module, Global-Local-Fusion Stage, and classification head; it combines local convolutional SNN blocks with local and global Spiking Transformer attention blocks
SNN Component	LIF neurons, ConvBN-LIF blocks, SNN-optimized Maxpooling CML, local feature extraction block, spiking self-attention, spatio-temporal block classification head
Neuron Model	LIF
Transformer Component	Local Transformer Block, Global Transformer Block, Local Spiking Self-Attention, Global Spiking Self-Attention, feedforward network
Attention Type	Local Spiking Self-Attention and Global Spiking Self-Attention
SNN–Transformer Interface Mechanism	Input features are converted into spike-form representations by LIF layers; Q, K, and V are generated using ConvBN-LIF operations; LSSA partitions feature maps into local regions before spiking self-attention, while GSSA applies global spiking self-attention
Model Depth / Scale	Main reported models include SGLFormer-4-384, SGLFormer-3-256, SGLFormer-8-384, SGLFormer-8-512, SGLFormer-8-512*, and SGLFormer-8-768*; “8-512” means 8 total Global-Local-Fusion blocks with embedding dimension 512
Parameter Count	ImageNet: 16.25M for SGLFormer-8-384, 28.67M for SGLFormer-8-512, 64.02M for SGLFormer-8-768*; CIFAR10: 8.85M; CIFAR100: 8.88M; CIFAR10-DVS: 2.48M/2.58M; DVS128-Gesture: 2.08M/2.17M
Design / Contribution Type	New Spiking Global-Local-Fusion Transformer architecture; integrates local convolution and global Transformer structures; proposes CML SNN-optimized Maxpooling; introduces STB classification head for spatial-temporal feature aggregation
Input Encoding Method	Static images are repeated across timesteps; DVS event streams are integrated into frames; Tokenizer converts inputs into spike-form feature maps
Encoding Parameters	ImageNet input size 224 × 224; CIFAR input size 32 × 32; CIFAR10-DVS and DVS128-Gesture input resolution 128 × 128; timestep 4 for static datasets; timestep 10 or 16 for DVS datasets
Encoding Learned or Fixed	Partially learned: image repetition/event-frame integration is fixed, while Tokenizer, LIF-based feature extraction, attention layers, and classification head are learned
Event / Spike Representation	Binary spike tensors; spike-form Q/K/V; DVS event frames; LIF-generated spike feature maps
Temporal Resolution	Static datasets mainly use 4 timesteps; CIFAR10-DVS and DVS128-Gesture use 10 or 16 timesteps
Output Decoding Method	Classification head uses STB to aggregate spatial and temporal spike features, followed by BN and linear classifier; ImageNet does not use STB
Encoding / Temporal Information Analysis	Yes. The paper analyzes information loss from global average pooling, gradient issues caused by Maxpooling after LIF, DVS frame integration, spike-based energy consumption, and local/global feature aggregation
Training Paradigm	Supervised direct SNN training from scratch
Training Method	Direct training using surrogate-gradient learning; most hyperparameters follow Spikformer
Surrogate Gradient Function	Surrogate-gradient method used; exact surrogate function NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW for CIFAR10/CIFAR100 and DVS experiments; ImageNet optimizer NR
Loss Function	Classification loss / cross-entropy implied; exact loss formula NR
Training Hyperparameters	ImageNet: 200 epochs, input 224 × 224, batch size 512 across 8 NVIDIA V100 GPUs, cosine-decay learning rate, initial LR 0.0012, no Mixup, no STB. CIFAR10/CIFAR100: 410 epochs, batch size 64, LR 0.001, AdamW, cosine decay, Mixup, random erase, horizontal flip. CIFAR10-DVS: 106 epochs, batch size 16, LR 0.005, cosine decay. DVS128-Gesture: 202 epochs, same main setting as CIFAR10-DVS
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	ImageNet: 83.73% with SGLFormer-8-768*; CIFAR10: 96.76%; CIFAR100: 82.26%; CIFAR10-DVS: 82.9% at 10 timesteps and 82.6% at 16 timesteps; DVS128-Gesture: 97.2% at 10 timesteps and 98.6% at 16 timesteps
Secondary Metrics	Parameter count, timestep count, theoretical energy consumption, ablation accuracy
Baseline Models Compared	Hybrid training, STBP, STBP-tdBN, TET, SEW-ResNet, Spiking ResNet, MS-ResNet, Att MS-ResNet, ANN Transformer, Spikformer, Spikingformer, S-Transformer, STSFormer
Performance Gain Over Baseline	On ImageNet, SGLFormer-8-768* achieves 83.73%, improving over S-Transformer-8-768 by 6.66%, Spikingformer-8-768 by 7.88%, and Spikformer-8-768 by 8.92%. On CIFAR100, SGLFormer-4-384 improves over Spikformer-4-384 by 4.05% and Spikingformer-4-384 by 3.05%
Statistical Validation	Benchmark evaluation and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes Loc removal/replacement, LSSA replaced by GSSA, STB replaced by GAP, CML replaced by CLM, number and position of Loc/LTB/GTB blocks, and energy-consumption analysis
Training Framework / Code Availability	PyTorch, Timm, SpikingJelly; code available at GitHub: ZhangHanN1/SGLFormer
Target Hardware	No real hardware target; theoretical 45 nm MAC/AC energy model used
Hardware Deployment Status	No neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using MAC = 4.6 pJ and AC = 0.9 pJ; ImageNet energy examples: SGLFormer-8-384 = 13.04 mJ, SGLFormer-8-512 = 20.95 mJ, SGLFormer-8-512* = 10.63 mJ, SGLFormer-8-768* = 19.93 mJ
Latency / Memory / Complexity Evidence	Reports parameter counts and theoretical energy; no measured latency, memory footprint, or wall-clock inference speed reported

Comparison with ANN Baseline	Yes. Compares with ANN Transformer-8-512 and reports that SGLFormer-8-512/8-512* have lower energy consumption while achieving higher accuracy
Main Contribution / Key Limitation	Contribution: proposes SGLFormer, a high-performance directly trained Spiking Transformer that fuses convolutional local feature extraction with Transformer-based global attention, while improving SNN maxpooling and classification-head design. Limitation: no real neuromorphic hardware testing; limitation discussion is mostly future-oriented, focusing on more vision tasks, pretraining/fine-tuning, and reducing parameters for edge deployment
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P077

GT-SNT: A Linear-Time Transformer for Large-Scale Graphs via Spiking Node Tokenization

Full Citation	Zhang, H., Li, J., Zhu, Y., Zhong, H., & Chen, L. (2026). GT-SNT: A Linear-Time Transformer for Large-Scale Graphs via Spiking Node Tokenization. In Proceedings of the Fortieth AAAI Conference on Artificial Intelligence (AAAI-26), pp. 16298–16306. AAAI Press.
University	Sun Yat-sen University
Country	China
Publication Year	2026
Publication Venue	The Fortieth AAAI Conference on Artificial Intelligence, AAAI-26
Publisher	Association for the Advancement of Artificial Intelligence
DOI / URL	https://ojs.aaai.org/index.php/AAAI/article/download/38667/42629
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Communication, Cybersecurity, and Network Intelligence
Specific Task	Node classification on small, medium, and large-scale graph datasets
Data Modality	Graph-structured data with node features and edge topology
Signal Dimensionality	Graph data represented by node feature matrix $X \in R^{(N\times d)}$ and adjacency matrix $A \in R^{(N\times N)}$; spiking node tokens represented as spike count vectors
Signal Characteristics	Sparse graph topology; node embeddings are generated across propagation steps and converted into binary spike outputs and integer spike-count embeddings
Dataset Name	Cora, CiteSeer, PubMed, Co-CS, Co-Physics, Actor, Deezer, ogbn-arXiv, ogbn-Products
Dataset Type	Public node-classification graph benchmarks, including citation networks, co-authorship/co-purchase-style graphs, heterophilic graphs, and OGB large-scale graphs
Dataset Size	Cora: 2,708 nodes / 10,556 edges; CiteSeer: 3,327 / 9,104; PubMed: 19,717 / 88,648; Co-CS: 18,333 / 163,788; Co-Physics: 34,493 / 495,924; Actor: 7,600 / 30,019; Deezer: 28,281 / 185,504; arXiv: 169,343 / 1,166,243; Products: 2,449,029 / 61,859,140
Dataset Balance / Split	Uses standard benchmark settings; exact train/validation/test split and class balance NR
Hybrid Topology	GT-SNT combines Spiking Node Tokenization, auxiliary message-passing neural network encoder, Codebook Guided Self-Attention, residual connection, and output classification head
SNN Component	Spiking node tokenizer, spiking neurons, binary spike outputs, spike count embeddings, spike-count codebook reconstruction
Neuron Model	PLIF used as the main flexible neuron setting; IF and LIF are also evaluated in ablation
Transformer Component	Graph Transformer with Codebook Guided Self-Attention and node-to-token attention
Attention Type	Codebook Guided Self-Attention; linear-time node-to-token attention guided by spike-count codewords
SNN–Transformer Interface Mechanism	SNT converts propagated node embeddings into spike-count node tokens; reconstructed codebook C and one-hot assignment matrix U guide the Transformer attention by forming codebook-based keys for CGSA
Model Depth / Scale	Uses L GT-SNT layers; each layer contains SNT, a single-layer GCN-style auxiliary MPNN, CGSA, and residual linear update. Exact final depth per dataset NR
Parameter Count	NR
Design / Contribution Type	New spiking node tokenization framework for Graph Transformers; proposes SNT and CGSA to reduce Graph Transformer attention from quadratic to linear complexity
Input Encoding Method	Random feature propagation followed by spiking node tokenization; node embeddings over T propagation steps are treated as sequential inputs for spiking neurons
Encoding Parameters	Learnable random feature dimension D; propagation steps T; spike count latent space size $(T + 1)^D$; reconstructed codebook size B; truncation by Bmax; exact selected hyperparameters per dataset NR
Encoding Learned or Fixed	Partially learned: random feature matrix and spiking neuron parameters are learned; graph propagation operator and spike-count decomposition/codebook reconstruction are algorithmic mechanisms
Event / Spike Representation	Binary spike output $S^t \in \{0,1\}^{(N\times D)}$; accumulated spike count embedding ■ $\in Z^{(N\times D)}$; reconstructed codebook $C \in Z^{(B\times D)}$; one-hot assignment matrix $U \in \{0,1\}^{(N\times B)}$
Temporal Resolution	Uses T graph propagation steps as pseudo-temporal sequence length for SNN tokenization; tested T values include several small settings, but exact final timestep per dataset NR
Output Decoding Method	Final node embeddings from the last GT-SNT layer are passed to a fully connected classification head for node-label prediction
Encoding / Temporal Information Analysis	Yes. The paper analyzes latent space size, propagation step T, random feature dimension D, codebook usage, codebook collapse, spike-count tokenization, and inference/energy efficiency
Training Paradigm	Supervised node classification
Training Method	End-to-end training with spiking node tokenization, surrogate-gradient learning for spiking neurons, auxiliary MPNN encoding, and CGSA-based global aggregation
Surrogate Gradient Function	Surrogate gradient is used; general form $\Theta'(x) = \theta'(\alpha x)$ is given, but the exact surrogate function is NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	Classification loss / cross-entropy implied for node classification; exact formula NR
Training Hyperparameters	Hyperparameter search conducted for GT-SNT and baselines; each model is run 5 times with different random seeds; experiments conducted on a single NVIDIA RTX 4090 GPU; detailed learning rate, batch size, dropout, and optimizer settings NR

Primary Performance Metric	Node classification accuracy and inference latency
Primary Metric Value	Cora: 84.7 ± 0.8; CiteSeer: 74.0 ± 0.5; PubMed: 80.6 ± 0.4; Co-CS: 93.7 ± 0.4; Co-Physics: 96.2 ± 0.0; Actor: 39.1 ± 0.2; Deezer: 65.7 ± 0.1; arXiv: 72.4 ± 0.3; Products: 74.8 ± 0.4
Secondary Metrics	Peak training memory, theoretical energy consumption, inference latency, codebook usage, used codewords, latent space size, mean ± standard deviation
Baseline Models Compared	GCN, GAT, SGC, VQGraph, SpikingGCN, SpikeNet, SpikeGCL, SpikeGT, NAGphormer, GOAT, NodeFormer, SGFormer
Performance Gain Over Baseline	GT-SNT improves over spike-based baselines by an average of about 3.9%; achieves top or near-top accuracy on most datasets; reaches up to 130× lower inference latency compared with average Graph Transformer baselines
Statistical Validation	Yes. Reports mean ± standard deviation over 5 random seeds; no formal significance test reported
Ablation Study	Full. Includes Performer/Linformer attention replacement, IF/LIF neuron replacement, LayerNorm/STFNorm comparison, VQ-VAE/FSQ tokenization replacement, latent-space size analysis, random feature dimension and propagation-step analysis, codebook usage comparison, and energy/latency analysis
Training Framework / Code Availability	Code available at GitHub: Zhhuizhe/GT-SNT; implementation framework NR
Target Hardware	No specific neuromorphic hardware; theoretical energy model and GPU evaluation used
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	Theoretical energy estimation only. Reported GT-SNT energy: CS 0.16 J, Physics 0.37 J, arXiv 0.18 J, Products 3.69 J
Latency / Memory / Complexity Evidence	Reports linear complexity O(NBdv) with B << N; inference latency/memory examples: CS 0.01 s / 1,638 MB; Physics 0.02 s / 3,036 MB; arXiv 0.08 s / 7,132 MB; Products 20.83 s / 13,494 MB
Comparison with ANN Baseline	Yes, indirectly. Compares against non-spiking GNNs and Graph Transformers, including GCN, GAT, NodeFormer, SGFormer, NAGphormer, GOAT, and VQGraph; no matched ANN-to-SNN counterpart is used
Main Contribution / Key Limitation	Contribution: introduces GT-SNT, a linear-time Graph Transformer that uses spiking neurons as node tokenizers rather than only low-power computation units, producing spike-count codebooks for efficient node-to-token attention. Limitation: energy is theoretical, not hardware-measured; no neuromorphic deployment; evaluation is limited to node classification, and Products accuracy is not the best among all baselines
RQ Mapping	Primary: RQ3; Secondary: RQ2, RQ1, RQ4, RQ5

P078

TTFSFormer: A TTFS-based Lossless Conversion of Spiking Transformer

Full Citation	Zhao, L., Huang, Z., Ding, J., & Yu, Z. (2025). TTFSFormer: A TTFS-based Lossless Conversion of Spiking Transformer. In Proceedings of the 42nd International Conference on Machine Learning, PMLR 267, Vancouver, Canada.
University	Peking University
Country	China
Publication Year	2025
Publication Venue	International Conference on Machine Learning, ICML 2025
Publisher	PMLR
DOI / URL	https://www.scopus.com/pages/publications/105023583655?origin=resultslist
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core conversion method
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	Lossless or near-lossless ANN-to-SNN conversion of pretrained Vision Transformer and EVA models for ImageNet-1K image classification
Data Modality	Static RGB image data
Signal Dimensionality	2D image data processed as Transformer patch/token sequences; TTFS spike timing used to represent layer activations
Signal Characteristics	Input activations are encoded using time-to-first-spike coding, where each neuron emits at most one spike; spike time represents activation value, including extended positive/negative value ranges through zero reference time
Dataset Name	ImageNet-1K
Dataset Type	Public large-scale image classification benchmark dataset
Dataset Size	NR
Dataset Balance / Split	Uses ImageNet-1K evaluation setting; exact train/validation split details NR
Hybrid Topology	ANN-to-SNN converted Spiking Transformer; converts pretrained ViT and EVA Transformer architectures into TTFS-coded SNN equivalents, including nonlinear activations, Softmax, LayerNorm, and multiplication operators
SNN Component	TTFS-based spiking neurons, generalized nonlinear neurons, input transform kernel, output transform kernel, zero reference time mechanism, receiving and emitting stages
Neuron Model	TTFS-based generalized spiking neuron; previous TTFS neurons and current-based TTFS neurons are discussed as background
Transformer Component	Vision Transformer and EVA Transformer models; Softmax attention, LayerNorm, GELU, SiLU, matrix multiplication, and Transformer activation layers are converted
Attention Type	Standard Transformer self-attention converted into TTFS-based spiking operations; Softmax is implemented through log-sum-exp and exponential spiking-neuron constructions
SNN–Transformer Interface Mechanism	ANN Transformer activations are represented by TTFS spike times; each Transformer layer is replaced with a TTFS spiking counterpart using designed spiking implementations of activations, Softmax, LayerNorm, and multiplication
Model Depth / Scale	Evaluated models include ViT-S/16, ViT-B/16, ViT-L/16, EVA-G, EVA02-S, and EVA02-L
Parameter Count	ViT-S/16: 22M; ViT-B/16: 86M; ViT-L/16: 307M; EVA-G: 1074M; EVA02-S: 22.1M; EVA02-L: 305M
Design / Contribution Type	First TTFS-based Transformer-to-SNN conversion framework; proposes generalized TTFS neurons with flexible input/output transform kernels and zero-reference-time mechanism; provides spiking constructions for Transformer nonlinear operators
Input Encoding Method	Time-to-first-spike encoding; pixel/input values are converted to spike times before entering the first layer
Encoding Parameters	TTFS spike time range uses receiving and emitting stages; time step δ selected according to hardware; layer-wise range [a, b] monitored; time constant $\tau = \delta/(b - a)$; zero reference time $T_{ref} = T_{emit} + b\tau$; robustness tested with time precision values including precise, 4096, 1024, 512, and 384

Encoding Learned or Fixed	Fixed conversion-based encoding; pretrained ANN weights are reused; no retraining is required
Event / Spike Representation	TTFS spike timing; each neuron emits at most one spike per time window; activation values are represented by spike latency rather than firing rate
Temporal Resolution	Layer-wise TTFS time windows; robustness analysis shows comparable performance at time precision 1024 or higher; exact timestep δ depends on hardware setting
Output Decoding Method	For image classification, prediction is given by the class index corresponding to the earliest output spike; no need to convert final spikes back to real values
Encoding / Temporal Information Analysis	Yes. The paper analyzes TTFS representational range, zero reference time, nonlinear transform kernels, receiving/emitting stages, time precision robustness, and energy–accuracy trade-off
Training Paradigm	Training-free ANN-to-SNN conversion using pretrained ANN Transformer models
Training Method	Layer-wise TTFS-based conversion; no SNN retraining or surrogate-gradient direct training is used
Surrogate Gradient Function	N/A
ANN-to-SNN Conversion Method	TTFSFormer: generalized TTFS-based lossless Transformer-to-SNN conversion with spiking implementations of GELU/SiLU, Softmax, LayerNorm, and multiplication
Optimizer	N/A
Loss Function	N/A for conversion; no model training loss is used
Training Hyperparameters	N/A for SNN training; conversion monitors layer output ranges and sets layer-wise τ and Tref; pretrained weights are sourced from public Hugging Face repositories
Primary Performance Metric	Top-1 ImageNet-1K classification accuracy and ANN-to-SNN accuracy loss
Primary Metric Value	ViT-S/16: 81.40% SNN vs 81.38% ANN; ViT-B/16: 85.07% vs 85.10%; ViT-L/16: 85.78% vs 85.83%; EVA-G: 88.90% vs 88.88%; EVA02-S: 85.68% vs 85.72%; EVA02-L: 90.03% vs 90.05%
Secondary Metrics	Estimated energy consumption, energy ratio, operation count, TTFS time precision robustness, accuracy under reduced time precision
Baseline Models Compared	Spikingformer, Spike-Driven Transformer, E-Spikeformer, SRP, Lossless TTFS CNN-to-SNN, MST, STA, ECMT, SpikeZIP-TF, AdaFire
Performance Gain Over Baseline	Achieves below 0.1% accuracy loss relative to ANN models across tested ViT/EVA architectures; ViT-B/16 reaches 85.07%, exceeding STA ViT-B/32 at 82.79% and SpikeZIP-TF ViT-B/16 at 82.71%; EVA02-L reaches 90.03%, higher than ECMT EVA-G at 88.60%
Statistical Validation	Benchmark comparison and robustness analysis; no repeated-run standard deviation, confidence interval, or significance test reported
Ablation Study	Partial. Includes robustness study under different TTFS time precision values and energy estimation; no broad architectural ablation table reported
Training Framework / Code Availability	Code available at GitHub: ForestOnTheLand/TTFSFormer; pretrained ANN weights from Hugging Face; implementation framework NR
Target Hardware	Neuromorphic hardware motivation; energy estimated using 45 nm MAC/AC operation energy model
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation only. ViT-S/16: 4.9 mJ, 22.3% of ANN energy; ViT-B/16: 17 mJ, 21.0%; ViT-L/16: 59 mJ, 20.7%
Latency / Memory / Complexity Evidence	Reports operation counts for SNN layers and discusses time precision/latency trade-off; ViT-S/16 OPSNN = 5.42 × 10■, ViT-B/16 = 19.2 × 10■, ViT-L/16 = 65.7 × 10■; measured wall-clock latency and memory footprint NR
Comparison with ANN Baseline	Yes. Directly compares converted SNN accuracy with original ANN accuracy for ViT-S/16, ViT-B/16, ViT-L/16, EVA-G, EVA02-S, and EVA02-L
Main Contribution / Key Limitation	Contribution: proposes the first TTFS-based Spiking Transformer conversion framework, enabling near-lossless conversion of pretrained Transformer models while using only one spike per neuron and substantially reducing estimated energy. Limitation: evaluation is limited to ImageNet-1K classification; energy is theoretical rather than hardware-measured; lower TTFS time precision causes accuracy degradation; future work is needed for direct TTFS-based SNN training
RQ Mapping	Primary: RQ4; Secondary: RQ3, RQ2, RQ5, RQ1

P079

Energy-based jamming pattern open set recognition via spiking wavelet transformer

Full Citation	Zhong, X., Cui, J., & Ji, Z. (2025). Energy-based jamming pattern open set recognition via spiking wavelet transformer. PLOS ONE, 20(6), e0325381. https://doi.org/10.1371/journal.pone.0325381
University	Changzhi University
Country	China
Publication Year	2025
Publication Venue	PLOS ONE
Publisher	Public Library of Science
DOI / URL	https://doi.org/10.1371/journal.pone.0325381; Dataset: Kaggle RF Jamming Dataset and Zenodo synthetic dataset DOI: 10.5281/zenodo.15347076
Publication Type	Journal article
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Communication, Cybersecurity, and Network Intelligence
Specific Task	Jamming pattern open-set recognition: classify known jamming patterns and detect unknown jamming types
Data Modality	RF signal waveforms transformed into STFT-based time-frequency images / pseudo-color heatmaps
Signal Dimensionality	1D sampled time-domain RF waveforms converted into 2D time-frequency matrices/images
Signal Characteristics	Jamming signals cover 10–20 MHz frequency range; JSR ranges from –4 dB to 10 dB in 2 dB steps; signals include non-stationary, overlapping, pulsed, hopping, tone, noise, and tracking jamming patterns
Dataset Name	Synthetic jamming dataset; Kaggle Radio Frequency Jamming Dataset
Dataset Type	Synthetic RF jamming dataset and public real-world RF jamming dataset
Dataset Size	Test setup uses 500 test samples per jamming type in Combination 1; full dataset size NR

Dataset Balance / Split	Known/unknown split varies by combination. Combination 1: known CW, MTJ, LFM, PBNJ, PPNJ, FHJ; unknown PCFJ, TJ. Combination 2: unknown NLFM, TJ. Combination 3: unknown NLFM, PCFJ. Combination 4 includes real-world known CJ/DJ and unknown RJ. Exact train/validation split NR
Hybrid Topology	JP-OSR framework combining STFT preprocessing, Spiking Wavelet Transformer feature extractor, Maximum Class Separation classifier, and adaptive energy-based unknown detection
SNN Component	Spiking Patch Splitting, spiking layers, Spiking Encoder Blocks, trinary spike encoding, membrane-potential residual connection
Neuron Model	Spiking neuron layers are used, but exact neuron model NR
Transformer Component	Spiking Wavelet Transformer with Spiking Encoder Blocks, Frequency-Aware Token Mixer, MLP block, wavelet-based frequency learner, convolutional spatial learner, and channel mixer
Attention Type	Attention-free architecture; uses Frequency-Aware Token Mixing instead of conventional self-attention
SNN–Transformer Interface Mechanism	STFT time-frequency images are processed by SWT; spiking patch splitting converts image-like inputs into spike-driven representations; spiking encoder blocks use FATM and MLP operations for feature extraction before MCS and energy scoring
Model Depth / Scale	SWT hyperparameter Blocks set to 4; MCS hypersphere radius $\rho = 1.0$; detailed layer depth and embedding dimensions NR
Parameter Count	NR; model size reported as 17.5 MB
Design / Contribution Type	New JP-OSR framework integrating SWT feature extraction, hyperspherical MCS classification, and adaptive energy-based open-set detection
Input Encoding Method	STFT-based time-frequency image encoding; raw time-domain signal is segmented with a Hanning window, transformed into STFT matrix, normalized, and mapped to pseudo-color heatmap
Encoding Parameters	STFT with Hanning window; JSR range –4 to 10 dB; SWT Blocks = 4; MCS $\rho = 1.0$; energy temperature T = 1.0; threshold scaling $\lambda = 2.0$; EMA coefficient $\alpha = 0.9$; confidence cutoff pi > 0.9
Encoding Learned or Fixed	STFT preprocessing is fixed; SWT feature extractor and classifier are learned; adaptive energy threshold updates during testing
Event / Spike Representation	STFT image features are transformed into spike-driven representations inside SWT; trinary spike encoding is mentioned; exact spike representation details NR
Temporal Resolution	Original RF signals retain temporal structure through STFT windows; exact sampling rate and SNN timestep NR
Output Decoding Method	MCS logits are used for known-class classification; energy score threshold determines whether a sample is known or unknown
Encoding / Temporal Information Analysis	Yes. The paper analyzes STFT time-frequency representation, JSR-dependent recognition accuracy, adaptive threshold sensitivity, SWT/MCS ablation, and known/unknown energy-based separation
Training Paradigm	Supervised training with known jamming classes plus auxiliary OOD data for energy-based open-set detection
Training Method	Joint optimization using cross-entropy loss for known-class classification and energy-based loss for ID/OOD separation
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	Total loss: L = LCE + λLenergy; LCE for known-class classification; energy loss penalizes high energy for ID samples and low energy for auxiliary OOD samples
Training Hyperparameters	Energy temperature T = 1.0; $\lambda = 2.0$; EMA $\alpha = 0.9$; confidence cutoff pi > 0.9; SWT Blocks = 4; MCS $\rho = 1.0$; grid search over $\lambda \in \{1.0, 1.5, 2.0, 2.5\}$ and $\alpha \in \{0.5, 0.7, 0.9, 0.99\}$; other training details NR
Primary Performance Metric	TNR, TPR, Precision, F1-score, known/unknown recognition accuracy
Primary Metric Value	Proposed method: TNR 95.36 ± 0.15%, TPR 99.48 ± 0.25%, Precision 99.23 ± 0.19%, F1-score 99.35 ± 0.16%; average known JP accuracy 99.5% when JSR > –2 dB; unknown JP accuracy average 96.4%
Secondary Metrics	Confusion matrix accuracy, latency, FPS, model size, threshold sensitivity, F1-score standard deviation across JSR values
Baseline Models Compared	SVDD, OCSVM, ViT-ARPL, VAEs, Mahalanobis, OpenMax, MSP+ODIN, G-OpenMax, ZSL-JPR, JCGAN
Performance Gain Over Baseline	Proposed method achieves the best F1-score of 99.35%, outperforming JCGAN at 98.43%, ViT-ARPL at 98.41%, VAEs at 98.31%, and MSP+ODIN at 97.65%; ablation shows SWT+MCS average F1 = 99.30% versus 95.47% without SWT and 96.82% without MCS
Statistical Validation	Yes. Experiments repeated five times; results reported with mean ± standard deviation; paired t-tests on F1-scores show $p < 0.01$ against baselines
Ablation Study	Full. Includes w/o SWT, w/o MCS, SWT+MCS comparison; threshold sensitivity over λ and α; JSR robustness; computational efficiency comparison
Training Framework / Code Availability	Dataset links provided; code availability NR; inference evaluation conducted on NVIDIA RTX 3090 GPU
Target Hardware	NVIDIA RTX 3090 GPU for inference evaluation; neuromorphic/low-power deployment is discussed as motivation
Hardware Deployment Status	No neuromorphic hardware deployment reported
Energy Evidence	No direct energy measurement reported; energy efficiency is argued conceptually through SWT's spiking/event-driven design and reduced model size
Latency / Memory / Complexity Evidence	Inference-only, batch size 1: latency 6.3 ms, FPS 159, model size 17.5 MB. Compared with ARPL ViT-Base/16: 9.0 ms, 111 FPS, 344 MB; OpenMax ResNet-50: 7.2 ms, 139 FPS, 102 MB
Comparison with ANN Baseline	Yes. Compares with ANN/deep-learning OSR baselines including ViT-ARPL, ResNet-50 OpenMax, JCGAN, VAEs, SVDD, and OCSVM
Main Contribution / Key Limitation	Contribution: proposes an energy-based JP-OSR framework using SWT for time-frequency feature extraction, MCS for hyperspherical class separation, and adaptive energy thresholding for unknown jamming detection. Limitation: performance may degrade under extremely low JSR below –4 dB; no real neuromorphic hardware testing; further compression and robustness against stealthy low-energy jammers are left for future work
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P080

Spikformer: When Spiking Neural Network Meets Transformer

Full Citation	Zhou, Z., Zhu, Y., He, C., Wang, Y., Yan, S., Tian, Y., & Yuan, L. (2023). Spikformer: When Spiking Neural Network Meets Transformer. Published as a conference paper at International Conference on Learning Representations (ICLR 2023).
University	Peking University
Country	China
Publication Year	2023
Publication Venue	International Conference on Learning Representations

Publisher	International Conference on Learning Representations
DOI / URL	https://www.scopus.com/pages/publications/85199857620?origin=resultslist
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Static image classification and neuromorphic/event-based image or gesture classification
Data Modality	Static RGB images and event-camera / DVS data
Signal Dimensionality	2D image data represented as spike-form patch sequences; neuromorphic data represented as spatiotemporal event-frame sequences T × C × H × W
Signal Characteristics	Static images are repeated across timesteps to form spike sequences; CIFAR images are 32 × 32; ImageNet images are 224 × 224; CIFAR10-DVS and DVS128 Gesture use 128 × 128 event-frame inputs
Dataset Name	CIFAR10, CIFAR100, ImageNet, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public static image classification and neuromorphic/event-based vision benchmark datasets
Dataset Size	CIFAR10/CIFAR100: 50,000 training and 10,000 test images; ImageNet: around 1.3M training images and 50,000 validation images; CIFAR10-DVS: 9,000 training and 1,000 test samples; DVS128 Gesture: 11 gesture classes from 29 individuals under 3 illumination conditions
Dataset Balance / Split	Uses standard benchmark train/test or train/validation splits; exact class-balance details NR
Hybrid Topology	Spikformer consists of Spiking Patch Splitting, spike-form relative position embedding, L-block Spikformer encoder, Spiking Self-Attention, MLP block, global average pooling, and linear classification head
SNN Component	LIF spike neurons, spike neuron layers, spike sequences, spike-form patch embeddings, spike-form Q/K/V, spike-form relative position embedding
Neuron Model	LIF
Transformer Component	Spikformer encoder block, Spiking Self-Attention, MLP block, multi-head SSA, patch-splitting module, position embedding, classification head
Attention Type	Spiking Self-Attention; softmax-free spike-form self-attention using binary Query, Key, and Value
SNN–Transformer Interface Mechanism	Spiking Patch Splitting converts image/event inputs into spike-form patches; Linear-BN-spike-neuron layers generate spike-form Q, K, and V; SSA computes attention through spike-friendly dot products/logical AND and addition without softmax
Model Depth / Scale	ImageNet models include Spikformer-8-384, 6-512, 8-512, 10-512, and 8-768; CIFAR models include Spikformer-4-256, 2-384, and 4-384; neuromorphic models use shallow 2-block Spikformer with 256 embedding dimension
Parameter Count	ImageNet: 16.81M, 23.37M, 29.68M, 36.01M, and 66.34M depending on model scale; CIFAR: 4.15M, 5.76M, and 9.32M; neuromorphic model: 2.59M
Design / Contribution Type	First directly trained Spiking Transformer architecture with a dedicated Spiking Self-Attention mechanism; introduces softmax-free spike-form attention to avoid multiplication-heavy vanilla self-attention
Input Encoding Method	Static images are repeated over T timesteps; neuromorphic event data are directly represented as spatiotemporal input sequences; SPS converts input into spike-form patch embeddings
Encoding Parameters	ImageNet uses 224 × 224 input, 196 patches, 4 timesteps; CIFAR uses 64 patches from 32 × 32 inputs, 4 timesteps; neuromorphic datasets use 10 or 16 timesteps; SSA scaling factor is 0.125 for ImageNet/CIFAR and learnable for neuromorphic datasets
Encoding Learned or Fixed	Partially learned: static image repetition is fixed, while SPS, spike-form relative position embedding, SSA, MLP, and classification weights are learned
Event / Spike Representation	Binary spike sequences; spike-form Q, K, V; spike-form image patches; sparse spike activations; spike-form relative position embeddings
Temporal Resolution	Main ImageNet/CIFAR setting uses 4 timesteps; CIFAR10-DVS and DVS128 Gesture use 10 or 16 timesteps; time-step ablation also evaluates 1, 2, 4, and 6 timesteps
Output Decoding Method	Global average pooling over final Spikformer encoder features followed by a fully connected classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes SSA operation count, theoretical energy, sparse Q/K/V firing rates, timestep effects, attention-map behavior, and convergence behavior of attention variants
Training Paradigm	Supervised direct SNN training from scratch
Training Method	Backpropagation through time using surrogate-gradient learning; directly trained Spiking Transformer
Surrogate Gradient Function	Sigmoid surrogate function with $\alpha = 4$
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Classification loss / cross-entropy implied; exact loss formula NR
Training Hyperparameters	ImageNet: AdamW, batch size 128 or 256, 310 epochs, initial learning rate 0.0005 with cosine decay, input 224 × 224, standard augmentation including random augmentation, Mixup, and CutMix. CIFAR: batch size 128, 4 timesteps, 400-epoch variant reported. DVS128 Gesture: 200 epochs; CIFAR10-DVS: 106 epochs; batch size 16; learning rate 0.1 with cosine decay
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	ImageNet: best 74.81% with Spikformer-8-768 at 4 timesteps. CIFAR10: 95.51% with Spikformer-4-384 trained 400 epochs. CIFAR100: 78.21% with Spikformer-4-384 trained 400 epochs. CIFAR10-DVS: 80.9% at 16 timesteps. DVS128 Gesture: 98.3% at 16 timesteps
Secondary Metrics	Parameter count, SOPs, theoretical energy/power, timestep count, firing rate, operation count, attention variant accuracy
Baseline Models Compared	Hybrid training, Diet-SNN, STBP, STBP-NeuNorm, TSSL-BP, STBP-tdBN, TET, Spiking ResNet, SEW-ResNet, ANN Transformer, LIAF-Net, TA-SNN, Rollout, DECOLLE, PLIF, Dspike, SALT, DSR
Performance Gain Over Baseline	On ImageNet, Spikformer-8-768 reaches 74.81%, improving over SEW-ResNet-152 by 5.55%. Spikformer-8-384 achieves 70.24% with 16.81M parameters, outperforming SEW-ResNet-152 with 60.19M parameters. On CIFAR100, Spikformer-4-384 achieves 77.86%, improving over ANN ResNet-19 by 2.51%
Statistical Validation	Benchmark comparison and ablation studies; no repeated-run standard deviation, confidence interval, or formal significance test reported
Ablation Study	Full. Includes SSA versus VSA, VSA with floating-point value, identity/ReLU/LeakyReLU attention variants, timestep ablation, IF neuron comparison, transfer-learning experiments, and convergence analysis of attention variants
Training Framework / Code Availability	PyTorch, SpikingJelly, and Timm; code stated as available through Spikformer/supplementary material, exact repository URL NR in uploaded paper
Target Hardware	Neuromorphic hardware motivation; theoretical 45 nm MAC/AC energy model used

Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation only. ImageNet examples: Spikformer-8-384 = 7.734 mJ, Spikformer-8-512 = 11.577 mJ, Spikformer-8-768 = 21.477 mJ. SSA uses substantially lower theoretical energy than vanilla self-attention
Latency / Memory / Complexity Evidence	Reports SOPs/FLOPs and theoretical energy; ImageNet Spikformer-8-512 uses 11.09G SOPs and 11.577 mJ. SSA complexity can be reduced to $O(Nd^2)$ when sequence length is larger than head dimension; measured wall-clock latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares against ANN Transformer and ANN ResNet-19; ANN Transformer has higher accuracy but much higher theoretical energy than Spikformer
Main Contribution / Key Limitation	Contribution: proposes Spikformer, the first directly trained SNN Transformer with softmax-free Spiking Self-Attention using spike-form Q/K/V, achieving strong results on static and neuromorphic vision datasets. Limitation: accuracy remains below ANN Transformer on ImageNet; energy evidence is theoretical rather than hardware-measured; no real neuromorphic deployment is reported
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P081

Spikingformer: A Key Foundation Model for Spiking Neural Networks

Full Citation	Zhou, C., Yu, L., Zhou, Z., Zhang, H., Wang, J., Zhou, H., Ma, Z., & Tian, Y. (2026). Spikingformer: A Key Foundation Model for Spiking Neural Networks. In Proceedings of the Fortieth AAAI Conference on Artificial Intelligence (AAAI-26), pp. 2236–2244. AAAI Press.
University	Peking University
Country	China
Publication Year	2026
Publication Venue	The Fortieth AAAI Conference on Artificial Intelligence, AAAI-26
Publisher	Association for the Advancement of Artificial Intelligence
DOI / URL	https://ojs.aaai.org/index.php/AAAI/article/download/37207/41169
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Natural Language Processing and Language Models
Specific Task	Static image classification, neuromorphic/event-based classification, and natural language understanding on GLUE tasks
Data Modality	Static RGB images, neuromorphic/event data, and text/token sequences
Signal Dimensionality	2D image sequences represented as T × C × H × W; neuromorphic data with C = 2; text/token sequences for GLUE/NLU tasks
Signal Characteristics	Static images use RGB channels and are encoded through the first Spiking Tokenizer layer; neuromorphic inputs use event-frame/spike-form representations; language tasks use tokenized text after pretraining/fine-tuning
Dataset Name	ImageNet-1K, CIFAR10, CIFAR100, DVS128 Gesture, CIFAR10-DVS, GLUE datasets including MNLI, QQP, QNLI, SST-2, CoLA, STS-B, MRPC, and RTE
Dataset Type	Public static image classification, neuromorphic/event-based vision, and natural language understanding benchmark datasets
Dataset Size	ImageNet-1K: NR in this paper; CIFAR/DVS dataset sizes NR in this paper; GLUE subset sizes NR
Dataset Balance / Split	Uses standard benchmark settings; exact class balance and train/test/validation split details NR
Hybrid Topology	Spikingformer contains a Spiking Tokenizer, multiple Spiking Transformer Blocks, Pre-activation Spiking Self-Attention, Spiking MLP blocks, MS residual connections, global average pooling, and classification head
SNN Component	LIF neurons, multistep LIF layers, spike-driven MS residual connection, spiking tokenizer, spike-form Q/K/V, spike-form feature maps, spiking MLP
Neuron Model	LIF / multistep LIF
Transformer Component	Spiking Transformer Block, Pre-activation Spiking Self-Attention, Spiking MLP, Transformer-style global modeling backbone
Attention Type	Pre-activation Spiking Self-Attention; global spike-driven self-attention using spike-form Query, Key, and Value
SNN–Transformer Interface Mechanism	Spiking Tokenizer projects input into spike-form patches; pre-activation LIF/SN layers generate spike-form inputs before ConvBN and Q/K/V projections; PSSA computes spike-based attention while MS residual connections preserve spike-driven computation
Model Depth / Scale	Main ImageNet models: Spikingformer-8-384, Spikingformer-8-512, Spikingformer-8-768; variants include Spikingformer†-8-384, †-8-512, and †-8-768; CIFAR/neuro models include 5.76M, 9.32M, and 2.57M parameter variants
Parameter Count	ImageNet: 16.81M, 29.68M, and 66.34M; CIFAR: 5.76M and 9.32M; neuromorphic models: 2.57M
Design / Contribution Type	New spike-driven Spiking Transformer foundation backbone; integrates MS residual connections with self-attention to avoid non-spike integer–float multiplications while retaining global attention capability
Input Encoding Method	Static images are encoded into spike-form through the first layer of the Spiking Tokenizer; neuromorphic inputs are processed as spike/event-frame sequences; text is pretrained/fine-tuned through Spikingformer for NLU
Encoding Parameters	ImageNet input/test resolution 224 × 224; ImageNet models use 4 timesteps; CIFAR models use 4 timesteps; DVS/CIFAR10-DVS use 10 or 16 timesteps; GLUE Spikingformer uses 4 timesteps
Encoding Learned or Fixed	Partially learned: spike generation follows fixed LIF dynamics, while tokenizer, ConvBN layers, PSSA, MLP, and task heads are learned
Event / Spike Representation	Binary spike data {0,1} before ConvBN layers; spike-form patches; spike-form Q/K/V; spike-driven feature maps; avoids non-spike residual values such as {0,1,2,...} found in Spikformer
Temporal Resolution	4 timesteps for ImageNet/CIFAR/GLUE; 10 or 16 timesteps for neuromorphic datasets
Output Decoding Method	Image/neuromorphic tasks use global average pooling plus fully connected classification head; GLUE tasks use task-specific NLU prediction heads after pretraining and fine-tuning
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike-driven residual behavior, SEW vs MS residual connections, firing rate, spike/non-spike activation distributions, theoretical energy, and performance across timesteps/tasks
Training Paradigm	Supervised direct SNN training for classification; masked language pretraining followed by supervised fine-tuning for GLUE/NLU
Training Method	Direct training with surrogate-gradient-based SNN learning; BPTT implied for SNN direct training; Wikipedia-English masked language pretraining for NLU followed by GLUE fine-tuning
Surrogate Gradient Function	Surrogate-gradient training is used, but exact surrogate function NR
ANN-to-SNN Conversion Method	N/A

Optimizer	NR for ImageNet/CIFAR; NLU optimizer NR
Loss Function	Classification loss implied for image/neuromorphic tasks; masked language modeling loss implied for pretraining; task-specific GLUE losses implied; exact formulas NR
Training Hyperparameters	ImageNet: 224 × 224 test resolution, 4 timesteps. CIFAR/DVS: CIFAR10-DVS training uses 106 epochs, learning rate 0.1 with cosine schedule, 10 or 16 timesteps; other detailed batch size/optimizer settings NR
Primary Performance Metric	Top-1 accuracy for classification; average accuracy for GLUE/NLU; theoretical energy consumption
Primary Metric Value	ImageNet-1K: Spikingformer-8-768 = 75.85%, Spikingformer†-8-768 = 77.64%. CIFAR10: Spikingformer† = 95.95%. CIFAR100: Spikingformer† = 80.37%. DVS128: Spikingformer = 98.3%, Spikingformer† = 98.6%. CIFAR10-DVS: Spikingformer† = 81.4%. GLUE average accuracy: 66.8%
Secondary Metrics	Parameter count, timestep count, theoretical energy, firing rate, spike-driven/non-spike behavior, model size, GLUE task-level scores
Baseline Models Compared	SEW-ResNet, MS-ResNet, Spikformer, SD-Transformer, ANN Transformer, QKFormer, BERTbase, Q2BERT, ELMo, SpikeBERT
Performance Gain Over Baseline	On ImageNet, Spikingformer-8-512 improves over Spikformer-8-512 by 1.41% and reduces theoretical energy by 60.36%; Spikingformer-8-768 improves over Spikformer-8-768 by 1.04% and reduces theoretical energy by 57.34%; Spikingformer†-8-768 reaches 77.64%, improving over Spikformer-8-768 by 2.83%; on GLUE, Spikingformer improves over SpikeBERT by 7.1% average accuracy
Statistical Validation	Benchmark comparison across 13 datasets; no repeated-run standard deviation, confidence interval, or formal statistical significance test reported
Ablation Study	Partial/full architectural analysis. Includes SEW vs MS residual comparison, spike-behavior visualization, energy/performance comparison, classification-head discussion, and Spikingformer vs Spikingformer† comparison
Training Framework / Code Availability	Code available at GitHub: TheBrainLab/Spikingformer; implementation framework NR
Target Hardware	Neuromorphic hardware motivation; theoretical 45 nm MAC/AC energy model used
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation only using MAC = 4.6 pJ and AC = 0.9 pJ. ImageNet examples: Spikingformer-8-512 = 7.46 mJ, Spikingformer-8-768 = 13.68 mJ, Spikingformer†-8-768 = 16.30 mJ; GLUE Spikingformer = 6.76 mJ
Latency / Memory / Complexity Evidence	Reports theoretical energy, timestep count, parameters, and spike-driven computation behavior; measured wall-clock latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares with ANN Transformer on ImageNet/CIFAR and BERTbase/ELMo/Q2BERT on GLUE; ANN Transformer still has higher accuracy on ImageNet/CIFAR but higher theoretical energy
Main Contribution / Key Limitation	Contribution: proposes Spikingformer, a spike-driven Transformer SNN backbone that combines MS residual connections with global self-attention to eliminate non-spike residual computation while improving accuracy and energy efficiency across vision, neuromorphic, and language tasks. Limitation: energy is theoretical rather than hardware-measured; no neuromorphic deployment; some training details and statistical validation are not reported
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P082
Finding Visual Saliency in Continuous Spike Stream

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University	Beijing Institute of Technology
Country	China
Publication Year	2024
Publication Venue	AAAI Conference on Artificial Intelligence
Publisher	Association for the Advancement of Artificial Intelligence
DOI / URL	https://ojs.aaai.org/index.php/AAAI/article/download/28610/29185
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Visual saliency detection / salient-object detection from continuous spike streams
Data Modality	Continuous binary spike stream captured by spike camera
Signal Dimensionality	3D spatiotemporal spike array; spike stream with spatial resolution 250 × 400 and temporal resolution 20,000 Hz
Signal Characteristics	Binary spike stream with values 0/1; visual saliency is encoded in spatiotemporal spike patterns; spike camera uses fovea-like sampling and emits spikes according to luminance accumulation
Dataset Name	SVS: Spike-based Visual Saliency dataset
Dataset Type	Newly constructed real-world spike-camera visual saliency dataset with object masks
Dataset Size	130 spike sequences; each sequence divided into 200 subsequences; total 25B spikes; 100 training sequences with 20,000 annotated frames; 30 validation sequences with 6,000 annotated frames
Dataset Balance / Split	Training: 24 high-light and 76 low-light sequences; validation: 8 high-light and 22 low-light sequences; annotated frames have 20 ms interval, corresponding to 400 spike frames
Hybrid Topology	Full spiking Recurrent Spiking Transformer framework with spike-based spatiotemporal feature extraction, recurrent feature aggregation, multi-scale refinement, and multi-step loss
SNN Component	LIF neurons, CBS blocks, spike neuron layers, binary spike communication, recurrent spike aggregation, element-wise OR spike residual operation
Neuron Model	LIF
Transformer Component	Recurrent Spiking Transformer, Recurrent Feature Aggregation unit, spiking self-attention, MLP block, multi-head attention between adjacent spike steps
Attention Type	Spiking self-attention / recurrent spiking attention using current-step Query and adjacent-step Key/Value

SNN–Transformer Interface Mechanism	Temporal spike representation is processed by CBS modules into spike features; RFA uses current step feature as Query and next-step feature as Key/Value; attention output is projected and fused through element-wise OR to preserve binary spike communication
Model Depth / Scale	Four-block spike-based spatiotemporal feature extractor; RFA modules tested in different counts; final model uses 6 RFA modules; timestep T = 5 in multi-step experiments
Parameter Count	NR
Design / Contribution Type	First investigation of visual saliency in continuous spike streams; proposes full SNN-based Recurrent Spiking Transformer; introduces SVS dataset, recurrent feature aggregation, multi-scale refinement, and multi-step loss
Input Encoding Method	Inter-spike interval representation: $S = C / \Delta txy$; spike data interval is 0.02 s, corresponding to 400 spike frames per input
Encoding Parameters	Spike camera resolution 250 × 400; temporal resolution 20,000 Hz; input size 256 × 256 for training; time interval 0.02 s; multi-step mode uses T = 5; maximum grayscale C used in inter-spike interval encoding
Encoding Learned or Fixed	Inter-spike interval representation and spike-camera sampling are fixed; network feature extraction, RFA, refinement, and prediction layers are learned
Event / Spike Representation	Binary spike stream; inter-spike interval representation; spike plane; spike-based feature maps; binary spike communication through OR operations
Temporal Resolution	Original spike-camera temporal resolution is 20,000 Hz; annotated interval is 20 ms / 400 spike frames; multi-step SNN training uses 5 steps
Output Decoding Method	Multi-scale refinement output is passed through 1 × 1 convolution and sigmoid function to generate saliency map
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike-camera sampling, light-intensity scale, inter-spike interval representation, recurrent temporal aggregation, single-step vs multi-step prediction, and continuous 20,000 Hz spike-stream inference
Training Paradigm	Supervised learning for spike-based saliency segmentation
Training Method	Direct SNN training using AdamW; full spiking architecture trained on SVS saliency masks
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Multi-step loss combining binary cross-entropy loss, IoU loss, and SSIM loss: $L = \Sigma \alpha i(Lbce + Liou + Lssim)$
Training Hyperparameters	20 epochs; initial learning rate 2 × 10■■, linearly decayed to 2 × 10■■; input size 256 × 256; spike interval 0.02 s / 400 spike frames; multi-step mode uses T = 5; same settings used for comparison methods
Primary Performance Metric	MAE, maximum F-measure, mean F-measure, Structure-measure
Primary Metric Value	Single-step RST: MAE 0.0784, Fmaxβ 0.6313, mFβ 0.6171, Sm 0.6970. Multi-step RST: MAE 0.0554, Fmaxβ 0.6981, mFβ 0.6882, Sm 0.7591
Secondary Metrics	Energy consumption, continuous detection performance, recurrent-mode ablation, RFA/refine ablation, multi-step loss ablation, element-wise operation ablation
Baseline Models Compared	Spiking Deeplab, Spiking FCN, EVSNN, Spikformer-ADD, Spikformer-OR
Performance Gain Over Baseline	In multi-step mode, RST achieves best MAE 0.0554 and Sm 0.7591, outperforming Spikformer-ADD MAE 0.0717 and Sm 0.7563, Spiking Deeplab MAE 0.0726 and Sm 0.7125, and EVSNN MAE 0.0945 and Sm 0.7023
Statistical Validation	Benchmark comparison and ablation studies; no repeated-run standard deviation, confidence interval, or formal significance test reported
Ablation Study	Full. Includes recurrent mode comparison, RFA/refine module analysis, multi-step loss, element-wise operation comparison, single-step versus multi-step evaluation, and qualitative comparison under different light intensity scales
Training Framework / Code Availability	Code and dataset available at GitHub: BIT-Vision/SVS; implementation framework NR
Target Hardware	Neuromorphic / low-power spike-camera processing motivation; no specific hardware accelerator used
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Estimated/comparative energy reported: RST consumes 5.8 mJ per inference compared with 167 mJ for ANN counterpart, giving 28.7× lower energy; no measured chip-level energy reported
Latency / Memory / Complexity Evidence	Demonstrates continuous single-step inference on 20,000 Hz spike input; measured wall-clock latency, parameter count, and memory footprint NR
Comparison with ANN Baseline	Partial. Main comparisons are with SNN-based methods, but the paper reports energy comparison against an ANN-based counterpart
Main Contribution / Key Limitation	Contribution: proposes the first full SNN-based Recurrent Spiking Transformer for visual saliency detection from continuous spike-camera streams and introduces the SVS dataset. Limitation: dataset is relatively task-specific and limited to saliency detection; no real neuromorphic hardware deployment; parameter count and measured latency are NR
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P083

SpikeGPT: Generative Pre-trained Language Model with Spiking Neural Networks

Full Citation	R. J. Zhu, Q. Zhao, G. Li and J. K. Eshraghian. (2024). SpikeGPT: Generative Pre-trained Language Model with Spiking Neural Networks. Under review as a conference paper at ICLR 2024.
University	University of California
Country	United States
Publication Year	2024
Publication Venue	International Conference on Learning Representations
Publisher	International Conference on Learning Representations
DOI / URL	https://www.scopus.com/pages/publications/85210043033?origin=resultslist
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Core architecture
E&QA; Score	4
Application Domain	Natural Language Processing and Language Models
Specific Task	Autoregressive language generation, language modeling, and text classification

Data Modality	Text/token sequences represented through binary spiking activations
Signal Dimensionality	1D sequential language tokens; sequence dimension is aligned with SNN temporal processing instead of adding an extra SNN timestep dimension
Signal Characteristics	Token sequences are streamed sequentially; binary event-driven spike activations are used between layers; model reduces quadratic self-attention complexity to linear sequence processing
Dataset Name	Enwik8, WikiText-2, WikiText-103, OpenWebText2, MR, SST-5, SST-2, Subj
Dataset Type	Public natural language generation and natural language understanding benchmark datasets
Dataset Size	Enwik8: first 100 million bytes of English Wikipedia dump; WikiText-2: approximately 2 million tokens; WikiText-103: more than 100 million tokens; SST-5: 11,855 sentences; other dataset sizes NR
Dataset Balance / Split	Enwik8 split: 90% training, 5% validation, 5% testing; for NLU datasets without standard split, 10% randomly selected as test set following prior work; exact class balance NR
Hybrid Topology	Decoder-only SpikeGPT architecture with Binary Embedding, stacked SpikeGPT blocks, Spiking RWKV token mixer, Spiking Receptance Feed-Forward Network channel mixer, residual connections, and generation/classification head
SNN Component	LIF neurons, binary embedding, binary spike activations, spiking RWKV, spiking RFFN, spike-based residual processing, sparse event-driven activations
Neuron Model	LIF
Transformer Component	GPT-style autoregressive language model; RWKV-style recurrent token mixer replacing self-attention; feed-forward/channel-mixing block; generation head and classification head
Attention Type	Self-attention is replaced by Spiking RWKV / recurrent linear attention-like token mixing; no conventional multi-head softmax self-attention
SNN–Transformer Interface Mechanism	Continuous token embeddings are converted into binary spikes using Binary Embedding or first-layer spiking neurons; Spiking RWKV processes tokens recurrently using spike inputs and hidden states; SRFFN preserves block-level binary spiking characteristics
Model Depth / Scale	45M model: 12 layers, feature dimension 512, sequence length 1024 or 3072; 216M model trained with pretraining and fine-tuning; exact depth/dimension of 216M model NR
Parameter Count	SpikeGPT 45M: 45.1M parameters; SpikeGPT 216M: 216M parameters
Design / Contribution Type	First generative pre-trained SNN language model; replaces Transformer self-attention with Spiking RWKV to enable recurrent spike-based language modeling with linear complexity
Input Encoding Method	Binary Embedding converts continuous token embeddings into binary spikes; 45M model uses character-level tokenizer; 216M model uses BPE tokenizer and first-layer neurons for encoding
Encoding Parameters	LIF threshold = 1; reset = 0; decay $\beta = 0.5$; sequence lengths $T = 1024$ and 3072 for 45M Enwik8 experiments; firing rate used in energy estimate = 0.15
Encoding Learned or Fixed	Partially learned: token embeddings and model parameters are learned; binarization/spiking conversion uses fixed Heaviside/LIF dynamics with surrogate-gradient training
Event / Spike Representation	Binary spike activations; sparse event-driven spike tensors; LIF membrane potentials; spike outputs from SRWKV and SRFFN layers
Temporal Resolution	Uses language sequence position as SNN temporal dimension; 45M experiments use sequence lengths 1024 and 3072; no additional SNN temporal dimension is introduced
Output Decoding Method	NLG: generation head predicts next-token distribution using softmax. NLU: average pooling over final token embeddings followed by classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes binary embedding, sequence-as-time processing, spike and membrane-potential visualization, firing-rate behavior, and operation/energy reduction from sparse spiking activations
Training Paradigm	Direct SNN training; autoregressive pretraining; downstream fine-tuning for NLG and NLU
Training Method	Backpropagation with surrogate-gradient learning; decoder-only autoregressive training; fine-tuning for generation and classification tasks
Surrogate Gradient Function	Arctangent surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	NLG: autoregressive language modeling objective / next-token likelihood. NLU: classification cross-entropy-style objective using pooled sequence embedding
Training Hyperparameters	45M model: learning rate 6 × 10■■, dropout 0.03, trained for 12 hours, sequence length 1024 or 3072. 216M model: learning rate 4 × 10■■, warmup 500 steps, trained for 48 hours, no dropout, BPE tokenizer. Experiments used four NVIDIA V100 GPUs
Primary Performance Metric	BPC for Enwik8; perplexity for WikiText-2/WikiText-103; classification accuracy for NLU tasks; theoretical energy/operation reduction
Primary Metric Value	Enwik8: SpikeGPT 45M with $T = 3072$ achieves test BPC 1.262. WikiText-103: SpikeGPT 216M with pretraining achieves test PPL 39.75. WikiText-2: test PPL 18.01. NLU with 216M pretraining: SST-2 88.76%, SST-5 51.27%, MR 85.63%, Subj 95.30%
Secondary Metrics	Train BPC, validation perplexity, complexity per layer, parameter count, energy ratio, firing rate, spike/membrane-potential visualization
Baseline Models Compared	Transformer, Reformer, Synthesizer, Linear Transformer, Performer, Stacked LSTM, SHA-LSTM, GPT-2 Small, GPT-2 Medium, TextCNN, TextSCNN, LSTM, BERT
Performance Gain Over Baseline	SpikeGPT 45M outperforms Stacked LSTM and SHA-LSTM on Enwik8 BPC and is competitive with efficient Transformer variants. SpikeGPT 216M outperforms GPT-2 Small and GPT-2 Medium on WikiText-2 test perplexity but underperforms GPT-2 on WikiText-103. Energy analysis reports 32.2× lower overall operation/energy cost than Vanilla GPT-style computation under the assumed neuromorphic model
Statistical Validation	Benchmark comparison and ablation study; no repeated-run standard deviation, confidence interval, or formal significance test reported
Ablation Study	Full/clear ablation. Compares Vanilla RWKV, Heaviside RWKV, SpikeGPT-S, and SpikeGPT-R; also visualizes spike and membrane-potential patterns in SRWKV and SRFFN layers
Training Framework / Code Availability	PyTorch and SpikingJelly; open-source repository and fine-tuned models stated to be released after anonymous review / included in supplementary material
Target Hardware	Neuromorphic/asynchronous hardware motivation; theoretical analysis assumes sparse spike execution on neuromorphic hardware
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy analysis only. Assumes MAC = 4.5 pJ and AC = 0.9 pJ; reports Vanilla GPT overall energy 3.29 × 10¹■ pJ versus SpikeGPT 1.02 × 10■ pJ, giving 32.2× reduction

Latency / Memory / Complexity Evidence	Reduces self-attention-style quadratic complexity $O(T^2 \cdot d)$ to recurrent linear complexity $O(T \cdot d)$; no measured wall-clock latency, memory footprint, or hardware runtime reported
Comparison with ANN Baseline	Yes. Compares with ANN Transformer, GPT-2, BERT, LSTM, TextCNN, and efficient Transformer variants
Main Contribution / Key Limitation	Contribution: proposes SpikeGPT, the first large-scale directly trained generative SNN language model, using Spiking RWKV and SRFFN to support autoregressive NLP with sparse binary activations and linear sequence complexity. Limitation: still behind vanilla Transformer/GPT-2 on some large language-modeling settings, especially WikiText-103; energy evidence is theoretical; no real neuromorphic hardware deployment; paper is under review with anonymous authors
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P084

Towards Efficient Spiking Transformer: A Token Sparsification Framework for Training and Inference Acceleration

Full Citation	Zhuge, Z., Wang, P., Yao, X., & Cheng, J. (2024). Towards Efficient Spiking Transformer: A Token Sparsification Framework for Training and Inference Acceleration. In Proceedings of the 41st International Conference on Machine Learning, PMLR 235, Vienna, Austria.
University	Chinese Academy of Sciences
Country	China
Publication Year	2024
Publication Venue	International Conference on Machine Learning, ICML 2024
Publisher	PMLR
DOI / URL	https://www.scopus.com/pages/publications/85203836031?origin=resultslist
Publication Type	Conference paper
Database Source	Scopus
Paper Category	Efficiency-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Token sparsification for accelerating Spiking Transformer training and reducing inference energy on image and event-based classification tasks
Data Modality	Static image data and neuromorphic/event-based vision data
Signal Dimensionality	2D images represented as spiking token sequences across timesteps; static images use 32 × 32 or 224 × 224 input; neuromorphic datasets use 128 × 128 input
Signal Characteristics	Images are split into tokens over multiple timesteps; CIFAR uses 64 patches; ImageNet uses 196 patches; DVS datasets use 16 × 16 patch size and 16 timesteps
Dataset Name	CIFAR-10, CIFAR-100, ImageNet, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public static image classification and neuromorphic/event-based vision benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Uses standard benchmark settings; exact train/test/validation split and class balance NR
Hybrid Topology	STATA is a plug-and-play token sparsification framework inserted into Spiking Transformer architectures; it uses Timestep-wise Anchor Token, token pruning, Spiking Self-Attention, FFN layers, and dual attention-map alignments
SNN Component	LIF neurons, binary spikes, spiking neurons for Q/K/V, Spiking Self-Attention, spike-based token representations, timestep-wise spiking tokens
Neuron Model	LIF
Transformer Component	Spikformer, Spike-driven Transformer, Spikingformer, Spiking Self-Attention, FFN, token sequence processing, anchor-token attention
Attention Type	Spiking Self-Attention; anchor-token-guided attention for token importance ranking; attention-map alignment across timestep, head, and layer dimensions
SNN–Transformer Interface Mechanism	Image tokens are mapped into spike-form Q, K, and V through BN, linear weights, and spiking neurons; a timestep-wise Anchor Token computes attention scores against image tokens at each timestep, then informative tokens are retained while non-informative tokens are pruned
Model Depth / Scale	Evaluated on Spikformer-8-768 for ImageNet; Spikingformer ST-4-384 and ST-4-256 for CIFAR100 analysis; also tested on Spikformer, Spike-driven Transformer, and Spikingformer variants across CIFAR and DVS datasets
Parameter Count	Spikformer-8-768 parameter count NR in this paper; ST-4-384 and ST-4-256 parameter counts NR
Design / Contribution Type	New token sparsification framework for Spiking Transformers; proposes Timestep-wise Anchor Token, Intra-Alignment, Inter-Alignment, and training/inference acceleration strategy
Input Encoding Method	Standard Spiking Transformer input processing: image input is passed through Conv2d, BN, spiking neuron, and max-pooling, then split into spiking token sequences; STATA adds Anchor Token for token scoring
Encoding Parameters	ImageNet input 224 × 224 with 196 patches; CIFAR input 32 × 32 with 64 patches; CIFAR/DVS timesteps: $T = 4$ for static datasets and $T = 16$ for neuromorphic datasets; $\alpha = \beta = 1e{-}2$ for alignment losses; sparsity factor γ controls retained token ratio
Encoding Learned or Fixed	Partially learned: Anchor Token and network weights are learned; token pruning rule and alignment losses are algorithmic training mechanisms
Event / Spike Representation	Binary spike tokens; spike-form Q/K/V; timestep-wise token sequences; informative and non-informative token sets selected by Anchor Token attention
Temporal Resolution	$T = 4$ for CIFAR/ImageNet static datasets; $T = 16$ for CIFAR10-DVS and DVS128 Gesture; token importance is computed separately at each timestep
Output Decoding Method	Global average pooling followed by classification head, inherited from the underlying Spiking Transformer
Encoding / Temporal Information Analysis	Yes. The paper analyzes token importance across timesteps, attention-map quality, timestep-wise variation, head-wise variation, layer-wise alignment, sparsity factor γ, training speed, memory, OPs, and theoretical energy
Training Paradigm	Supervised direct SNN training with token sparsification
Training Method	Direct training of Spiking Transformers with STATA; cross-entropy loss plus Intra-Alignment and Inter-Alignment attention losses
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW

Loss Function	Total loss: L = LCE + αLintra + βLinter, where LCE is cross-entropy, Lintra aligns timestep/head attention maps, and Linter aligns shallow and deep layer attention maps
Training Hyperparameters	ImageNet: 224 × 224 input, 196 patches, AdamW, 8 GPUs, batch size 128, 310 epochs, random augmentation and Mixup. CIFAR10/100: 32 × 32 input, 64 patches, single GPU, batch size 192, 310 epochs, T = 4. DVS datasets: 128 × 128 input, patch size 16 × 16, T = 16. Token sparsification inserted every 3 blocks from block L/3; Pa and Pr = {0, L/3 + 1, L – 1}; α = β = 1e–2
Primary Performance Metric	Top-1 accuracy, training speedup, memory reduction, theoretical energy reduction
Primary Metric Value	ImageNet STATA with Spikformer-8-768: 74.03% top-1 accuracy, 10.70G OPs, 11.16 mJ energy. CIFAR100 STATA: 79.86% accuracy, 12.16 GB memory, 1.48× training speedup, 0.71 mJ energy
Secondary Metrics	Top-5 accuracy, OPs ratio, power ratio, pruning ratio γ, training memory, theoretical energy, timestep count
Baseline Models Compared	Original Spiking Transformer, Random Token pruning, l1-norm pruning, SSA-based pruning, Hybrid training, TET, Spiking ResNet, STBP-tdBN, SEW ResNet, Spikformer, Spike-driven Transformer, Spikingformer, Dspike, PLIF, TA-SNN, DSR, TCJA
Performance Gain Over Baseline	Compared with original Spiking Transformer, STATA achieves up to about 1.53× training speedup and about 48% energy reduction with comparable performance. On CIFAR100, STATA achieves 79.86% versus 78.47% for SSA-based pruning at similar 1.48× speedup. On ImageNet, STATA reaches 74.03%, much higher than Random Token 70.13% and SSA-based 70.45% under similar acceleration
Statistical Validation	Benchmark comparison and ablation studies; no repeated-run standard deviation, confidence interval, or formal significance test reported
Ablation Study	Full. Includes different token pruning metrics, sparsity factor γ, Timestep-wise Anchor Token, Anchor Token without timestep-wise design, Intra-Alignment, Inter-Alignment, OPs/power ratio under different γ, and training speed/memory analysis
Training Framework / Code Availability	Experiments conducted on RTX 3090 GPUs; software framework and code availability NR
Target Hardware	GPU training acceleration and neuromorphic-style inference motivation; theoretical 45 nm MAC/AC energy model used
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation only. On ImageNet, STATA reduces Spikformer-8-768 energy from 21.48 mJ to 11.16 mJ. On CIFAR100, STATA reduces energy from 1.29 mJ to 0.71 mJ
Latency / Memory / Complexity Evidence	Reports training speedup and memory reduction: CIFAR100 STATA uses 12.16 GB versus 18.74 GB for original and reaches 1.48× speedup. ImageNet STATA achieves around 1.53×–1.6× training speedup depending on γ. OPs ratio can be reduced to 54.87% at γ = 0.2
Comparison with ANN Baseline	Partial. The paper motivates Spiking Transformer efficiency relative to ANN Transformers and reports SNN/ANN operation-energy assumptions, but experiments mainly compare SNN and Spiking Transformer baselines
Main Contribution / Key Limitation	Contribution: proposes STATA, a token sparsification framework that accelerates Spiking Transformer training and inference using timestep-wise anchor-token scoring and dual attention alignment while preserving accuracy. Limitation: energy is theoretical rather than hardware-measured; no neuromorphic deployment; method is evaluated mainly on classification datasets
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4, RQ1

P085

Spiking Transformer with Experts Mixture

Full Citation	Zhou, Z., Lu, Y., Jia, Y., Che, K., Niu, J., Huang, L., Shi, X., Zhu, Y., Li, G., Yu, Z., & Yuan, L. (2024). Spiking Transformer with Experts Mixture. 38th Conference on Neural Information Processing Systems, NeurIPS 2024.
University	Peking University
Country	China
Publication Year	2024
Publication Venue	Conference on Neural Information Processing Systems
Publisher	NeurIPS
DOI / URL	https://www.webofscience.com/wos/woscc/full-record/WOS:001633245800088
Publication Type	Conference paper
Database Source	WoS
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	Image classification on static image datasets and DVS/event-based datasets
Data Modality	Static image data and neuromorphic/event-camera data
Signal Dimensionality	2D image sequence represented as (I \in R^{T \times C \times H \times W}), then converted into flattened spike patches with token and channel dimensions
Signal Characteristics	Static images and neuromorphic event data are processed as spike sequences over timesteps; patch splitting produces (N) flattened spike patches with (D)-dimensional channels
Dataset Name	ImageNet-1K, CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Standard benchmark splits used; exact balance/split details NR
Hybrid Topology	Spiking Transformer with MoE-style spike-driven expert routing; SEMM is embedded into Spiking Transformer backbones by replacing/augmenting SSA and MLP with EMSA and EMSP
SNN Component	LIF neurons, spiking router, spike-form experts, spike sequences, spiking self-attention, spiking MLP/perceptron
Neuron Model	LIF
Transformer Component	Spiking Transformer block with Patch Splitting, Relative Position Embedding, Spiking Self-Attention, MLP, Global Average Pooling, and Classification Head
Attention Type	Spiking self-attention; Spike-driven self-attention; Experts Mixture Spiking Attention
SNN–Transformer Interface Mechanism	Experts and routers both output spike sequences; SEMM combines router spikes and expert spike outputs through element-wise masking/Hadamard product and addition, avoiding softmax and Top-K routing
Model Depth / Scale	ImageNet models include 8-block, 384-channel and 8-block, 512-channel variants; CIFAR uses Spiking Transformer-4-384; CIFAR10-DVS and DVS128 Gesture use Spiking Transformer-2-256; final EMSA expert number is 4
Parameter Count	ImageNet SEMM models: 16.05M parameters for 8-384 and 28.22M for 8-512; small-dataset parameter counts NR

Design / Contribution Type	Proposes Spiking Experts Mixture Mechanism, Experts Mixture Spiking Attention, and Experts Mixture Spiking Perceptron for SNN-compatible MoE-style dynamic sparse conditional computation
Input Encoding Method	Patch Splitting converts image/event inputs into spike patch sequences; Relative Position Embedding is generated by Conv-BN-LIF block
Encoding Parameters	ImageNet and CIFAR use T = 4 timesteps; CIFAR10-DVS and DVS128 Gesture use T = 10 and T = 16 settings; LIF threshold and reset dynamics defined, but exact threshold values NR
Encoding Learned or Fixed	Partially learned: patch/linear/router/expert weights are learned; LIF spike-generation dynamics are fixed
Event / Spike Representation	Binary spike sequences; spike-form Q/K/V; spiking router (R \in {0,1}^{T \times N \times m}); spike-form expert outputs
Temporal Resolution	4 timesteps for ImageNet/CIFAR; 10 and 16 timesteps for neuromorphic datasets
Output Decoding Method	Global Average Pooling over final spiking Transformer output followed by linear classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes router average spike rate, dynamic sparse conditional computation, spatial-temporal router masks, expert routing behavior, and spike sparsity across different images
Training Paradigm	Supervised direct training
Training Method	Surrogate-gradient-based direct training; SEMM inserted into Spikformer, Spike-driven Transformer, and Spikingformer baselines
Surrogate Gradient Function	Sigmoid surrogate function with $\alpha = 4$
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW for ImageNet-1K; optimizer for smaller datasets follows baseline settings and is not fully reported
Loss Function	Cross-entropy classification loss implied; exact loss formulation NR
Training Hyperparameters	ImageNet: T = 4, AdamW, batch size 128 or 256, 310 epochs, initial learning rate 0.0004 with cosine decay, random augmentation, mixup, cutmix. CIFAR/neuromorphic settings vary by backbone: CIFAR epochs 210–410, DVS128 Gesture 200 epochs, CIFAR10-DVS 106 epochs, batch size 16–128, cosine learning-rate decay
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	Best reported values: ImageNet-1K 76.62% with SD-Transformer-8-512 + SEMM; CIFAR-10 96.16%; CIFAR-100 80.26%; CIFAR10-DVS 82.90%; DVS128 Gesture 99.30%
Secondary Metrics	Parameter count, timestep count, router average spike rate, theoretical synaptic operations, ablation accuracy
Baseline Models Compared	Spikformer, Spike-driven Transformer, Spikingformer, SEW-ResNet, MS-ResNet, tdBN, PLIF, DIET-SNN, Dspike, DSR
Performance Gain Over Baseline	On ImageNet, SD-Transformer-8-512 + SEMM improves from 74.57% to 76.62%; Spikformer-8-512 + SEMM improves from 73.38% to 75.93%; Spikingformer-8-512 + SEMM improves from 74.79% to 76.03%
Statistical Validation	Benchmark comparison and ablation experiments; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes EMSA-only, EMSP-only, EMSA+EMSP, router removal, expert-number sensitivity, parameter–accuracy comparison with ANN-MoE variants, and router spike-rate analysis
Training Framework / Code Availability	Uses public baseline code and follows baseline credentials; 8 NVIDIA RTX 4090 GPUs for ImageNet and 1 NVIDIA RTX 4090 GPU for other datasets; proposed SEMM code availability NR
Target Hardware	Neuromorphic hardware motivation discussed; no specific hardware platform used for deployment
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	No measured energy result; efficiency is argued through spike-driven computation, avoidance of softmax/Top-K, sparse router activation, and theoretical synaptic-operation analysis
Latency / Memory / Complexity Evidence	Provides theoretical parameter and synaptic-operation comparison for SSA, EMSA, MLP, and EMSP; reports SEMM achieves approximate computational overhead relative to baselines, but measured latency and memory are NR
Comparison with ANN Baseline	Partial. Discusses ANN-MoE and compares against ANN-style heavy MoE routing conceptually, but main experiments compare with SNN and Spiking Transformer baselines
Main Contribution / Key Limitation	Contribution: introduces SEMM, an SNN-compatible MoE paradigm using spike-driven router–expert interactions, plus EMSA and EMSP for Spiking Transformers. Limitation: no real neuromorphic hardware deployment, no measured energy/latency, and limited discussion of failure cases or statistical significance
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ5, RQ1

P086

QKFormer: Hierarchical Spiking Transformer using Q-K Attention

Full Citation	Zhou, C., Zhang, H., Zhou, Z., Yu, L., Huang, L., Fan, X., Yuan, L., Ma, Z., Zhou, H., & Tian, Y. (2024). QKFormer: Hierarchical Spiking Transformer using Q-K Attention. arXiv preprint arXiv:2403.16552v2.
University	Peking University
Country	China
Publication Year	2024
Publication Venue	ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEM
Publisher	NeurIPS
DOI / URL	https://dl.acm.org/doi/10.5555/3737916.3738332
Publication Type	Conference paper
Database Source	WoS
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification on ImageNet-1K, CIFAR-10, CIFAR-100, CIFAR10-DVS, and DVS128 Gesture
Data Modality	Static RGB images and event-based neuromorphic vision data
Signal Dimensionality	2D image sequences represented as spike tensors; input form (T \times H \times W \times n), then converted into hierarchical spike-token representations
Signal Characteristics	Static RGB datasets use (T_0 = 1), (n = 3); neuromorphic datasets use (T_0 = T), (n = 2). CIFAR images are 32 × 32; ImageNet uses 224 × 224, 288 × 288, and 384 × 384 inference settings; DVS data are represented as temporal event inputs
Dataset Name	ImageNet-1K, CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public static image and neuromorphic/event-based benchmark datasets

Dataset Size	ImageNet-1K: 1.28M training images and 50k validation images, 1000 classes; CIFAR-10/100: 50,000 training and 10,000 test images; CIFAR10-DVS: 9,000 training and 1,000 test samples; DVS128 Gesture: 1,342 samples, 11 gesture classes, 29 individuals
Dataset Balance / Split	Standard benchmark splits used; exact class-balance details NR
Hybrid Topology	Hierarchical Spiking Transformer using Q-K Attention; QKFormer uses SPEDS modules for hierarchical patch embedding, Q-K attention blocks in early stages, SSA in the final stage, SMLP blocks, and classifier head
SNN Component	LIF neurons, spike-form Q and K, spiking patch embedding, spike-form attention vectors, spike masks, SMLP, binary spike activations
Neuron Model	LIF; ablation also tests IF and PLIF
Transformer Component	Hierarchical Spiking Transformer; QKFormer blocks; Spikformer blocks in final stage; Q-K Token Attention; Q-K Channel Attention; Spiking Self-Attention; SMLP
Attention Type	Q-K Attention; Q-K Token Attention; Q-K Channel Attention; mixed spiking attention; Spiking Self-Attention in later stage
SNN–Transformer Interface Mechanism	Input images/events are converted into spike patches through SPEDS; Q and K are produced by learnable linear layers followed by BN and spiking neurons; Q-K attention generates spike-form token or channel attention vectors through row/column summation and LIF activation, then applies mask operations to K
Model Depth / Scale	ImageNet: HST-10-384, HST-10-512, HST-10-768; stage blocks set as {1, 2, 7}. CIFAR: HST-4-384 with blocks {1, 1, 2}. DVS: HST-2-256 with blocks {0, 1, 1}. Timesteps: mostly 4 for static datasets, 10/16 for neuromorphic datasets
Parameter Count	ImageNet: 16.47M, 29.08M, 64.96M depending on scale; CIFAR: 6.74M; DVS/CIFAR10-DVS: 1.50M
Design / Contribution Type	New Q-K attention mechanism with linear complexity; hierarchical Spiking Transformer architecture; Spiking Patch Embedding with Deformed Shortcut; mixed spiking attention integration
Input Encoding Method	Static images are encoded into spike-form patches through SPEDS-1 with an additional Conv-BN-LIF spike encoder; DVS/event inputs are processed as temporal spike/event representations; patch size 4 × 4 at the first stage
Encoding Parameters	Main static experiments use T = 4; neuromorphic experiments use T = 10 or 16; LIF dynamics use membrane time constant τ, threshold (V_{th}), and reset potential (V_{reset}); surrogate sigmoid uses $\alpha = 4$
Encoding Learned or Fixed	Partially learned: patch embedding, Q/K projections, SPEDS, and Transformer weights are learned; LIF spike-generation rule is fixed
Event / Spike Representation	Binary spike tensors; spike-form Q and K; token attention vector ($A_t \in \{0,1\}^{N \times 1}$); channel attention vector ($A_c \in \{0,1\}^{1 \times D}$); spike-form hierarchical feature maps
Temporal Resolution	Static datasets: 4 timesteps in main experiments; ImageNet also reports 1-step and 4-step models; CIFAR100 ablation tests T = 1, 2, 4, and 6; neuromorphic datasets use 10 and 16 timesteps
Output Decoding Method	Final fully connected classifier after the last QKFormer/Spikformer block
Encoding / Temporal Information Analysis	Yes. The paper analyzes Q-K attention memory consumption, spike firing rates, variance/expectation of QKTA versus SSA, timestep effects, spike sparsity visualization, and theoretical energy consumption
Training Paradigm	Supervised direct training
Training Method	Direct SNN training using surrogate gradients; hierarchical spiking Transformer training with AdamW on ImageNet and baseline-style training on CIFAR/DVS datasets
Surrogate Gradient Function	Sigmoid surrogate function with $\alpha = 4$
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW for ImageNet; optimizer for CIFAR/DVS NR
Loss Function	Cross-entropy classification loss implied; exact loss formula NR
Training Hyperparameters	ImageNet: AdamW, base learning rate (6×10^{-4}), batch size 512, 200 epochs, 8 NVIDIA V100 GPUs, RandAugment, random erasing, stochastic depth. CIFAR: 400 epochs, batch size 64. DVS128 Gesture: 200 epochs. CIFAR10-DVS: 106 epochs
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	ImageNet-1K best: 85.65% top-1 and 97.74% top-5 with HST-10-768 at 384 × 384 input; CIFAR-10: 96.18%; CIFAR-100: 81.15%; DVS128 Gesture: 98.6%; CIFAR10-DVS: 84.0%
Secondary Metrics	Top-5 accuracy, parameter count, theoretical power/energy in mJ, timestep count, memory consumption, firing rate, training/inference time
Baseline Models Compared	RMP, QCFS, MST, SEW-ResNet, Spikformer, Spikingformer, Spike-driven Transformer/S-Transformer, ViT, DeiT, Swin Transformer, STBP-tdBN, TET, CML, STSA
Performance Gain Over Baseline	On ImageNet, QKFormer HST-10-768 reaches 85.65%, outperforming Spikformer-8-768 by 10.84%. On CIFAR100, QKFormer reaches 81.15%, improving over Spikformer by 2.94%. On CIFAR10-DVS, QKFormer reaches 84.0%, improving over Spikformer by 3.1% while using fewer parameters
Statistical Validation	Benchmark comparison, ablation, and training/testing curves; no repeated-run standard deviation, confidence interval, or statistical significance test reported
Ablation Study	Full. Includes SPEDS ablation, Q-K attention variants, QKTA/QKCA/SSA combinations, residual-connection type, neuron model type, timestep sensitivity, memory comparison, firing-rate analysis, and training/inference-time comparison
Training Framework / Code Availability	PyTorch, SpikingJelly, Timm; code and models stated as available at GitHub: zhouchenlin2096/QKFormer
Target Hardware	Neuromorphic hardware motivation and theoretical 45 nm energy model; no specific hardware deployment platform used
Hardware Deployment Status	No hardware deployment reported
Energy Evidence	Theoretical energy estimation using SOPs, firing rates, ($E_{MAC}=4.6$) pJ and ($E_{AC}=0.9$) pJ under a 45 nm hardware assumption; no measured hardware energy
Latency / Memory / Complexity Evidence	Reports theoretical complexity: QKTA ($O(D)$), QKCA ($O(N)$), compared with SSA ($O(N^2D)$). Memory comparison shows SSA can consume about 10× GPU memory than QKTA when ($\sqrt{N}=50$). Training/inference time comparison with Spikformer is reported
Comparison with ANN Baseline	Yes. Compares with ViT, DeiT, and Swin Transformer on ImageNet in terms of accuracy, parameters, and theoretical energy/power
Main Contribution / Key Limitation	Contribution: proposes QKFormer, a hierarchical directly trained Spiking Transformer using linear-complexity Q-K attention and SPEDS, achieving very high ImageNet accuracy for directly trained SNNs. Limitation: limited to image/DVS classification tasks; no real neuromorphic hardware deployment; measured energy/latency on hardware not reported
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ5, RQ1

P087

Combining aggregated attention and transformer architecture for accurate and efficient performance of Spiking Neural Networks

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University	Tianjin University
Country	China
Publication Year	2025
Publication Venue	Neural Networks
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.neunet.2025.107789
Publication Type	Journal article
Database Source	WoS
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification on static and neuromorphic datasets
Data Modality	Static RGB images and event-based neuromorphic vision data
Signal Dimensionality	2D image sequences represented as (I \in R^{T \times C \times H \times W}), then converted into spike patch features (U_0 \in R^{T \times N \times D})
Signal Characteristics	Static image datasets use repeated image sequences across timesteps; neuromorphic datasets use event-based temporal inputs. SAFormer processes binary sparse spike matrices through SPS, SASA, MLP, and classification head
Dataset Name	CIFAR-10, CIFAR-100, Tiny-ImageNet, CIFAR10-DVS, DVS128-Gesture
Dataset Type	Public static image and neuromorphic/event-based benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Standard benchmark protocols followed; exact train/test split and class balance NR
Hybrid Topology	Spiking Transformer architecture named SAFormer; SPS module followed by stacked SAFormer encoder blocks combining Spike Aggregated Self-Attention and MLP, then classification head
SNN Component	LIF neurons, spike neural layers, binary spike matrices, spike patch splitting, spike-form Q and K, spike-based SASA, spiking MLP
Neuron Model	LIF
Transformer Component	Transformer-style encoder blocks; self-attention-like SASA module; MLP block; classification head
Attention Type	Spike Aggregated Self-Attention; linear-complexity spike attention using Q and K only; no value matrix; no softmax
SNN–Transformer Interface Mechanism	Input images are transformed into spike patch features through SPS; Q and K are generated from learnable linear transformations, aggregated by adaptive average pooling, converted to spikes by BN and spike neuron layers, then used in SASA through Hadamard product, summation, broadcast, DWC, and linear projection
Model Depth / Scale	Static datasets: SAFormer-4-384 with 4 encoder blocks and 384 embedding dimensions. Neuromorphic datasets: SAFormer-2-256 with 2 encoder blocks and 256 embedding dimensions. Ablation also uses SAFormer-2-512
Parameter Count	NR
Design / Contribution Type	Proposes Spike Aggregated Self-Attention and SAFormer; removes the value matrix from attention; uses aggregation function to reduce sparsity and sequence length; adds depthwise convolution module to improve feature diversity
Input Encoding Method	Spike Patch Splitting with five convolutional layers; first layer encodes static RGB image into spike form; adaptive average pooling is used inside SASA to aggregate Q and K
Encoding Parameters	Static datasets mainly use T = 4; neuromorphic datasets use T = 16; LIF uses membrane time constant (\tau), firing threshold (V_{th}), and reset potential (V_{reset}=0); aggregation reduced sequence length (n) is treated as a hyperparameter
Encoding Learned or Fixed	Partially learned: convolutional SPS, Q/K projections, DWC, MLP, and classifier are learned; LIF firing rule and adaptive average pooling aggregation are fixed operations
Event / Spike Representation	Binary spike signals; spike patch feature maps; spike-form Q and K matrices; aggregated spike matrices; sparse attention map; DWC-enhanced spike features
Temporal Resolution	T = 4 for CIFAR/Tiny-ImageNet experiments; T = 16 for CIFAR10-DVS and DVS128-Gesture; timestep sensitivity is analyzed for T = 1, 2, 4, and 6 on CIFAR
Output Decoding Method	Global Average Pooling after the final encoder block, followed by a fully connected classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes attention sparsity, attention-map visualization, impact of timestep, aggregation sequence length (n), energy consumption, GPU memory, GPU execution time, and computational complexity
Training Paradigm	Supervised direct training
Training Method	Direct SNN training using spiking neurons and backpropagation; follows Spikformer and S-Transformer experimental protocols
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Cross-entropy classification loss implied; exact loss formula NR
Training Hyperparameters	Static datasets: 400 epochs, batch size 64, AdamW, SAFormer-4-384. Neuromorphic datasets: SAFormer-2-256, T = 16, 300 epochs for CIFAR10-DVS, 400 epochs for DVS128-Gesture
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	CIFAR-10: 95.80%; CIFAR-100: 79.07%; Tiny-ImageNet: 67.28%; CIFAR10-DVS: 81.3%; DVS128-Gesture: 98.3%
Secondary Metrics	Energy consumption, error rate, relative difference, GPU memory consumption, GPU execution time, timestep sensitivity, ablation accuracy
Baseline Models Compared	Hybrid training, SALT, PLIF, STBP, STBP NeuNorm, TSSL-BP, Dspike, tdBN, TET, DIET-SNN, MS-ResNet, SEW-ResNet, DSR, Spikformer, S-Transformer, ANN ResNet-19, DCT, Online-LTL, Offline-LTL, ASGL, LM-H, LM-HT, LIAF-Net, TA-SNN, Rollout, DECOLLE
Performance Gain Over Baseline	On CIFAR-10, SAFormer improves over Spikformer by 0.30% and over S-Transformer by 0.20%. On CIFAR-100, it improves over Spikformer by 0.87% and over S-Transformer by 0.67%. On Tiny-ImageNet, it improves over Spikformer by 1.25% and S-Transformer by 1.57%. On CIFAR10-DVS, it improves over Spikformer by 0.4% and S-Transformer by 1.3%
Statistical Validation	Reports averages and standard deviations over three independent runs for comparative plots; no formal significance test reported
Ablation Study	Full. Includes DWC module ablation, aggregation-function ablation, core SASA-only variant, SASA-Mul vs SASA-Add vs final SASA, timestep sensitivity, aggregation sequence length (n), GPU memory, and GPU execution time

Training Framework / Code Availability	Code available at GitHub: 98ming/SAFormer; specific software framework NR
Target Hardware	Neuromorphic and edge hardware motivation; theoretical 45 nm energy model; no specific hardware deployment platform
Hardware Deployment Status	No real hardware deployment reported
Energy Evidence	Theoretical energy estimation using SOPs, firing rate, timestep, (E_{MAC}=4.6) pJ, and (E_{AC}=0.9) pJ under a 45 nm hardware assumption. Reported energy: CIFAR-10 0.49 mJ, CIFAR-100 0.58 mJ, Tiny-ImageNet 2.23 mJ, CIFAR10-DVS 1.67 mJ, DVS128-Gesture 1.23 mJ
Latency / Memory / Complexity Evidence	SASA has (O(nD)) time and space complexity compared with (O(N^2D)) for VSA/SSA and (O(ND)) for SDSA. The paper also evaluates GPU memory and execution time as aggregation length (n) changes
Comparison with ANN Baseline	Partial. Includes ANN ResNet-19 baseline on CIFAR-10/CIFAR-100, but main comparisons are against SNN and Spiking Transformer models
Main Contribution / Key Limitation	Contribution: proposes SAFormer and SASA, reducing attention complexity by using aggregated Q and K without a value matrix, while DWC improves feature diversity and energy efficiency. Limitation: no real neuromorphic hardware benchmark; mainly evaluated on classification tasks; object detection and broader applications are left for future work
RQ Mapping	Primary: RQ2; Secondary: RQ3, RQ4, RQ5, RQ1

P088

An Energy-Efficient Neuromorphic Accelerator Based on Deformable Spiking Transformer for Dynamic Vision Sensor

Full Citation	Zhang, H., Zhang, S., Mao, W., & Wang, Z. (2025). An Energy-Efficient Neuromorphic Accelerator Based on Deformable Spiking Transformer for Dynamic Vision Sensor. IEEE Transactions on Circuits and Systems I: Regular Papers, 72(12), 7860–7873. https://doi.org/10.1109/TCSI.2025.3573092
University	Nanjing University
Country	China
Publication Year	2025
Publication Venue	IEEE Transactions on Circuits and Systems I: Regular Papers
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TCSI.2025.3573092
Publication Type	Journal article
Database Source	WoS
Paper Category	Hardware-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	DVS event-stream classification / gesture and object classification on DVS128 Gesture and CIFAR10-DVS
Data Modality	Dynamic Vision Sensor event data streams converted into integrated event frames
Signal Dimensionality	2D spatiotemporal event-frame sequence represented as (I \in R^{T \times C \times H \times W})
Signal Characteristics	DVS events contain pixel position, timestamp, and polarity. Integrated input uses two polarity channels, so (C = 2). DVS128 Gesture input size is 128 × 128; CIFAR10-DVS is downscaled to 64 × 64
Dataset Name	DVS128 Gesture; CIFAR10-DVS
Dataset Type	Public neuromorphic/event-based vision benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Standard dataset protocols used; exact split and class balance NR
Hybrid Topology	Deformable Spiking Transformer with encoder block, three hierarchical stages, patch embedding, DST blocks, Spike-Driven Deformable Attention, Spike-Driven MLP, membrane-potential residual connections, and classification head
SNN Component	LIF neurons, spike-driven convolution, spike-driven deformable attention, spike-driven MLP, binary spike activations, sparse spiking computation flow
Neuron Model	LIF
Transformer Component	Deformable Spiking Transformer; DST block; Spike-Driven Deformable Attention; SDMLP; hierarchical Transformer-style stages
Attention Type	Spike-Driven Deformable Attention; deformable attention with sampled important tokens; spike-driven attention without multipliers
SNN–Transformer Interface Mechanism	DVS event frames are encoded into spikes by encoder and patch-embedding blocks; SDDA generates query, sampled key/value features, offsets, shifted points, and spike-form attention outputs; SAC hardware performs token-dimension summation, LIF processing, and channel masking
Model Depth / Scale	Three-stage DST; DST block numbers are {0, 1, 1}; maximum channel number is 256; two DST blocks total; offset groups (G = 16); timesteps T = 10 and T = 16
Parameter Count	DST: 1.66M parameters
Design / Contribution Type	Algorithm–hardware co-design. Proposes Deformable Spiking Transformer, Spike-Driven Deformable Attention, Spiking Convolution Core, Spiking Attention Core, Auxiliary Computing Core, and sparse spiking computing flow
Input Encoding Method	DVS events are integrated into frame-based representation; encoder block converts integrated frames into spikes using Conv-BN-MaxPooling and LIF neuron processing
Encoding Parameters	Two event-polarity channels; T = 10 or 16; DVS128 input 128 × 128; CIFAR10-DVS input 64 × 64; weight quantization 8-bit; some membrane-shortcut activations quantized to 8-bit; offset groups = 16
Encoding Learned or Fixed	Partially learned: convolution, projection, offset-generation, SDDA, and SDMLP parameters are learned; DVS event integration, reference-point generation, and post-training quantization are fixed procedures
Event / Spike Representation	Integrated DVS event frames; binary spike tensors; spike-form query/key/value-like features; sampled spiking feature map; membrane-potential residual features
Temporal Resolution	T = 10 and T = 16 timesteps in algorithm experiments; DVS events originally have high temporal resolution, but exact event-window duration NR
Output Decoding Method	Fully connected classification head after the final DST stage
Encoding / Temporal Information Analysis	Yes. The paper analyzes DVS sparsity, event-frame integration, SDDA sampling, offset groups, max-pooling/downsampling choices, kernel size of offset-generation convolution, reference-point downsampling, sparse convolution speed-up, and SAC power breakdown
Training Paradigm	Supervised direct training for DVS classification; hardware inference acceleration

Training Method	Direct SNN training using surrogate-gradient-based LIF models implied; cosine schedule and AdamW optimizer; post-training quantization for deployment
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Cross-entropy classification loss implied; exact loss formula NR
Training Hyperparameters	240 training epochs with 10 warmup epochs on NVIDIA GeForce RTX 3090; random flip, random cutout, and TrivialAugment. DVS128 Gesture: batch size 16, learning rate 0.0024 for T=10 and 0.002 for T=16. CIFAR10-DVS: batch size 32, learning rate 0.015 for T=10 and 0.014 for T=16
Primary Performance Metric	Classification accuracy and hardware energy efficiency
Primary Metric Value	Algorithm accuracy: DVS128 Gesture 98.3% at T=10 and 99.0% at T=16; CIFAR10-DVS 84.3% at T=10 and 85.5% at T=16. Hardware energy efficiency: up to 2.15 TOPS/W when compared with Transformer accelerators
Secondary Metrics	Parameter count, inference latency, peak throughput, normalized energy efficiency, power, on-chip memory, speed-up from sparse convolution, SAC power breakdown
Baseline Models Compared	Spikformer, Spikingformer, Spike-driven Transformer, QKFormer, SpikingResformer; SNN hardware designs; Transformer accelerators; DVS processing hardware baselines
Performance Gain Over Baseline	On CIFAR10-DVS, DST reaches 85.5%, outperforming QKFormer's 84.0% and Spikingformer's 81.3%. Hardware achieves 247 GOPS peak throughput and at least 1.11× higher energy efficiency than prior Transformer accelerators
Statistical Validation	Benchmark comparisons and ablation studies; no repeated-run standard deviation, confidence interval, or statistical significance test reported
Ablation Study	Full. Includes offset group number, SDDA Version 1 vs Version 2, max-pooling/downsampling choices, Conv1 kernel size, reference-point downsampling factor, sparse convolution speed-up, and SAC power analysis
Training Framework / Code Availability	Training framework NR; hardware synthesized with Synopsys Design Compiler, placed/routed with Synopsys IC Compiler, simulated using Synopsys Verilog Compiled Simulator, and power estimated using PrimeTime PX; code availability NR
Target Hardware	TSMC 28 nm CMOS neuromorphic accelerator with Spiking Convolution Core, Spiking Attention Core, Auxiliary Computing Core, global controller, off-chip memory interface, and 302 KB on-chip memory
Hardware Deployment Status	ASIC implementation through synthesis, place-and-route, and power simulation under TSMC 28 nm; no fabricated silicon measurement reported
Energy Evidence	Energy/power estimated from post-synthesis/place-route simulation and PrimeTime PX waveform-based power analysis; reported power around 120 mW and energy efficiency up to 2.15 TOPS/W
Latency / Memory / Complexity Evidence	Frequency 200 MHz; supply voltage 0.81 V; 302 KB on-chip memory; peak throughput 247 GOPS; throughput 259 GOPS; power 120–120.4 mW; DVS128 Gesture latency 5.38 ms; average latency per layer 0.23 ms
Comparison with ANN Baseline	Partial. Main comparison is with SNN, Spiking Transformer, hardware SNN, hardware Transformer, and DVS accelerator baselines; no direct matched ANN model baseline is central
Main Contribution / Key Limitation	Contribution: proposes a DVS-focused Deformable Spiking Transformer and a 28 nm neuromorphic accelerator with sparse spiking convolution and attention cores, achieving high DVS accuracy and competitive energy efficiency. Limitation: hardware is not reported as fabricated/measured silicon; SRAM access dominates SAC power; future work is needed for lower-power process and customized SRAM
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4, RQ1

P089

Memory-Efficient Reversible Spiking Neural Networks

Full Citation	Zhang, H., & Zhang, Y. (2024). Memory-Efficient Reversible Spiking Neural Networks. AAAI-format paper / arXiv preprint arXiv:2312.07922v1.
University	Zhejiang University
Country	China
Publication Year	2024
Publication Venue	AAAI Conference on Artificial Intelligence
Publisher	Association for the Advancement of Artificial Intelligence
DOI / URL	https://www.webofscience.com/wos/woscc/full-record/WOS:001239314700059
Publication Type	Conference paper
Database Source	WoS
Paper Category	Training-efficiency focused
E&QA; Score	4
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification and event-based classification using reversible SNNs, including RevSFormer and RevSResNet
Data Modality	Static images and neuromorphic/event-camera data
Signal Dimensionality	2D static images and 2D spatiotemporal event-frame/spike sequences processed over simulation timesteps
Signal Characteristics	Static datasets use 32 × 32 images; neuromorphic datasets use spike trains/event streams split into 10 or 16 temporal slices; SNNs are unfolded over simulation timesteps during BPTT training
Dataset Name	CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public static image and neuromorphic/event-based benchmark datasets
Dataset Size	CIFAR-10/100: 60,000 images each, with 50,000 training and 10,000 test images. CIFAR10-DVS: 10,000 event-stream samples. DVS128 Gesture: 11 gesture classes from 29 people under three lighting conditions
Dataset Balance / Split	CIFAR-10: 10 classes with 5,000 training and 1,000 test images per class; CIFAR-100: 100 classes with 500 training and 100 test images per class; exact DVS balance NR
Hybrid Topology	Reversible SNN framework applied to both ResNet-like SNNs and Transformer-like SNNs; RevSFormer uses a spiking tokenizer, reversible spiking Transformer blocks, SSA, MLP, and classification head
SNN Component	LIF neurons, IF neurons, binary spikes, membrane potentials, reversible spiking blocks, RevSResNet, RevSFormer, BPTT training
Neuron Model	LIF and IF
Transformer Component	RevSFormer; spiking tokenizer; reversible spiking Transformer block; Spiking Self-Attention; spiking MLP; classification head

Attention Type	Spiking Self-Attention
SNN–Transformer Interface Mechanism	In RevSFormer, the reversible block splits input into two residual streams; Spiking Self-Attention is used as function F and spiking MLP as function G, allowing intermediate activations and membrane potentials to be reconstructed during reverse computation
Model Depth / Scale	RevSFormer-2-384, RevSFormer-4-384, RevSFormer-2-256, and deeper RevSFormer-16-384 memory analysis; RevSResNet21, RevSResNet37, RevSResNet24 also evaluated
Parameter Count	RevSFormer-2-384: 5.76M; RevSFormer-4-384: 9.32M; RevSFormer-2-256: NR in table; RevSResNet21: 11.05M; RevSResNet37: 23.59M; RevSResNet24: 0.26M
Design / Contribution Type	Proposes reversible spiking block, RevSResNet, and RevSFormer to reduce GPU memory during SNN training by reconstructing intermediate activations and membrane potentials instead of storing them
Input Encoding Method	Static images are processed through convolutional/spiking tokenizer or first convolutional encoding layer; CIFAR10-DVS uses AER preprocessing split into 10 or 16 slices; DVS128 Gesture uses the same main AER preprocessing
Encoding Parameters	Static datasets use T = 4; neuromorphic datasets use T = 10 and T = 16; CIFAR images are 32 × 32; DVS128 Gesture resolution is 128 × 128
Encoding Learned or Fixed	Partially learned: tokenizer/convolutional layers and Transformer parameters are learned; AER slicing and reversible reconstruction mechanism are fixed
Event / Spike Representation	Binary spike signals, spike trains, membrane potentials, intermediate activations, two-stream reversible spike features
Temporal Resolution	T = 4 for CIFAR-10/CIFAR-100; T = 10 and T = 16 for CIFAR10-DVS and DVS128 Gesture
Output Decoding Method	Average pooling and fully connected classifier for RevSResNet; spiking neuron plus fully connected classification head for RevSFormer
Encoding / Temporal Information Analysis	Yes. The paper analyzes temporal reversibility, memory growth with timesteps, membrane-potential reconstruction, GPU memory versus depth, GPU memory versus timestep, and memory versus embedding dimension
Training Paradigm	Supervised direct SNN training
Training Method	BPTT with surrogate-gradient training and reversible forward-reset-reverse-backward computation
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	SGD for ResNet models on CIFAR; AdamW for Transformer models and neuromorphic datasets
Loss Function	Cross-entropy classification loss implied; exact formula NR
Training Hyperparameters	CIFAR ResNet: IF neuron, 200 epochs, learning rate 0.1, SGD, batch size 32. CIFAR Transformer: LIF neuron, 310 epochs, learning rate 0.0005, AdamW, batch size 64. CIFAR10-DVS and DVS128 Gesture: LIF neuron, AdamW, learning rate 0.001, batch size 16, 106 epochs for CIFAR10-DVS and 202 epochs for DVS128 Gesture
Primary Performance Metric	GPU memory per image during training; Top-1 accuracy
Primary Metric Value	RevSFormer-4-384: 41.74 MB/image, 95.34% on CIFAR-10, 79.04% on CIFAR-100; RevSFormer-2-256: 82.20% on CIFAR10-DVS at T=16 and 97.57% on DVS128 Gesture at T=16
Secondary Metrics	Parameters, FLOPs, timestep, maximum batch size, maximum timestep, training time per epoch, GPU utilization, memory-saving ratio
Baseline Models Compared	MS ResNet, Spikingformer, Spikformer, Hybrid training, Diet-SNN, STBP, STBP NeuNorm, TSSL-BP, STBP-tdBN, TET, DS-ResNet, LIAF-Net, TA-SNN, Rollout, tdBN, PLIF, SEW-ResNet, Dspike, DSR
Performance Gain Over Baseline	RevSFormer-4-384 reduces memory from 125.06 MB/image to 41.74 MB/image, a 3.00× reduction, while maintaining similar accuracy. RevSResNet37 reduces memory from 89.33 MB/image to 23.58 MB/image, a 3.79× reduction. RevSFormer-16-384 saves 9.1× GPU memory per image in depth analysis
Statistical Validation	Benchmark comparison and ablation/memory analysis; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full for memory behavior. Includes memory usage versus network depth, timestep, embedding dimension, computational overhead, maximum batch size, maximum timestep, and GPU-model comparison
Training Framework / Code Availability	PyTorch and SpikingJelly; code available at GitHub: mi804/RevSNN
Target Hardware	GPU training environment; neuromorphic hardware motivation discussed but no neuromorphic deployment
Hardware Deployment Status	No hardware deployment reported; evaluated on GPUs including RTX3080, RTX3090, RTX4090, Tesla V100, and Tesla A100
Energy Evidence	No direct energy measurement. Energy efficiency is discussed conceptually through SNN sparsity and neuromorphic potential, but the paper focuses on GPU training memory
Latency / Memory / Complexity Evidence	Strong memory evidence. RevSFormer-4-384 reduces memory 3.00×; RevSResNet37 reduces memory 3.79×; reversible models require about 1.27×–1.33× longer training time but support larger batch sizes and timesteps
Comparison with ANN Baseline	Partial. The paper discusses ANNs and reversible ANN architectures conceptually, but experiments mainly compare SNN baselines
Main Contribution / Key Limitation	Contribution: introduces reversible spiking blocks and applies them to RevSResNet and RevSFormer, reducing training memory by reconstructing activations and membrane potentials. Limitation: memory reduction comes with extra training computation; no real neuromorphic hardware deployment; contribution is training-efficiency focused rather than a new attention mechanism
RQ Mapping	Primary: RQ4; Secondary: RQ2, RQ3, RQ5, RQ1

P090

ST-RTNet: An energy-efficient spike temporal residual transformer network for rock segmentation in deep space exploration

Full Citation	Zha, K., Yuan, J., Fan, L., Shen, Y., & Liu, X. (2026). ST-RTNet: An energy-efficient spike temporal residual transformer network for rock segmentation in deep space exploration. Computers in Industry, 174, 104417. https://doi.org/10.1016/j.compind.2025.104417
University	Nanjing University of Aeronautics and Astronautics
Country	China
Publication Year	2026
Publication Venue	Computers in Industry, Volume 174
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.compind.2025.104417
Publication Type	Journal article
Database Source	WoS

Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Remote Sensing, Geospatial, and Environmental Monitoring
Specific Task	Pixel-level rock segmentation for Mars, Moon, and Tianwen deep-space imagery
Data Modality	RGB planetary surface images converted into spike sequences
Signal Dimensionality	2D semantic segmentation input represented as spike tensor with temporal dimension, generally ((T, B, C, H, W))
Signal Characteristics	MoonData uses RGB lunar images of 480 × 720; MarsData images are resized to 256 × 256; TWMARS images are originally 1024 × 1024 and resized to 256 × 256; spike representation uses temporal unfolding with (T = 1) to (6), with (T = 4) selected as the best trade-off
Dataset Name	MoonData, MarsData, Tianwen Mars Rock Dataset / TWMARS
Dataset Type	Public / curated planetary rock segmentation datasets
Dataset Size	MoonData: 9,766 RGB images; MarsData: 30,912 images after augmentation/refinement; TWMARS: 336 images
Dataset Balance / Split	MoonData split 6:2:2 for train/validation/test; MarsData split 18,547 training, 6,182 validation, and 6,183 testing images; TWMARS split 268 training, 33 validation, and 34 testing images
Hybrid Topology	Encoder–decoder SNN–Transformer segmentation network; ST-RTNet consists of spike generator, encoder, decoder, and segmentation head; encoder combines spike-driven ResNet block, spike spatio-temporal attention module, and spike multilayer perceptron
SNN Component	Spike generator, LIF neurons, spike-driven ResNet block, binary spike matrices, spike-driven convolution, SMLP, spike temporal residual computation
Neuron Model	LIF
Transformer Component	Spike Spatio-Temporal Attention Module, Transformer-style encoder block, SMLP, segmentation head
Attention Type	Spike Spatio-Temporal Attention Module using (\langle Q, K, V, T \rangle) attention; temporal attention based on membrane-potential dynamics
SNN–Transformer Interface Mechanism	Input image is converted to spike sequence by the spike generator; encoder generates Q, K, V, and temporal tensor T; SSTAM combines Q–T and K–T attention maps with V, then passes outputs through LIF neurons to preserve binary spike propagation
Model Depth / Scale	Exact layer depth NR; architecture includes spike generator, SRB, SSTAM, SMLP, MSF decoder blocks, and segmentation head; main setting uses (T = 4)
Parameter Count	ST-RTNet: 12.98M parameters
Design / Contribution Type	New spike-driven segmentation architecture for deep-space rock segmentation; proposes ST-RTNet and SSTAM; combines CNN local feature extraction, Transformer-style global attention, and SNN temporal membrane-potential dynamics
Input Encoding Method	Spike generator converts floating-point image input into discrete spike sequence; LIF neurons replace ReLU activations in convolutional layers
Encoding Parameters	Time steps tested from 1 to 6; main model uses (T = 4); thresholds tested at 0.5, 1.0, and dynamic 0.5/1.0; best setting uses threshold 0.5 in encoding layer and 1.0 in subsequent layers; surrogate sigmoid parameter (\alpha = 4)
Encoding Learned or Fixed	Partially learned: convolutional, attention, SMLP, and decoder weights are learned; spike thresholding and temporal spike generation follow fixed LIF dynamics
Event / Spike Representation	Binary spike signals; spike matrices; membrane-potential tensors; temporal spike feature maps; output decoder tensor ((T, B, C, H, W))
Temporal Resolution	Main reported setting (T = 4); hyperparameter analysis evaluates (T = 1, 2, 3, 4, 5, 6)
Output Decoding Method	Decoder output is processed by segmentation head; softmax computes pixel-level class probabilities; final output is obtained by averaging across temporal dimension
Encoding / Temporal Information Analysis	Yes. The paper analyzes membrane-potential temporal dynamics, (T)-based attention, timestep effects, threshold effects, energy–accuracy trade-off, and visualization of temporal feature maps
Training Paradigm	Supervised direct SNN training for semantic segmentation
Training Method	Surrogate-gradient-based direct training with LIF neurons; gradient clipping; hybrid Dice and cross-entropy optimization
Surrogate Gradient Function	Sigmoid surrogate gradient with (\alpha = 4)
ANN-to-SNN Conversion Method	N/A
Optimizer	Stochastic Gradient Descent
Loss Function	Hybrid Dice loss and cross-entropy loss
Training Hyperparameters	Learning rate (1 \times 10^{-3}); batch size 8; 100 epochs; weight decay (1 \times 10^{-4}); momentum 0.9; thresholds 0.5 and 1.0; timesteps 1–6; two NVIDIA RTX 4090 GPUs
Primary Performance Metric	Segmentation accuracy, F1-Score, MIoU, and energy consumption
Primary Metric Value	MarsData: Acc 98.79%, F1-Score 98.81%, MIoU 97.03%. MoonData: Acc 98.10%, F1-Score 95.56%, MIoU 77.40%. TWMARS transfer learning MIoU: 89.36%
Secondary Metrics	AC operations, MAC operations, INT-8 energy, Float-32 energy, energy-accuracy coefficient, loss curves, visualization results
Baseline Models Compared	FCN, UNet, Swin-UNet, SCG-TransNet, RockSeg, Spike-driven-Rock, Spikingformer-Rock, Spikformer-Rock, SGLFormer-Rock
Performance Gain Over Baseline	Compared with RockSeg, ST-RTNet reduces energy by 10.08× on INT-8 and 6.17× on Float-32 in the reported complexity table; abstract reports up to 90.13% INT-8 and 83.76% Float-32 energy reduction compared with ANN models. It also improves MIoU over Spikformer-Rock by 2.22% on MarsData and 0.87% on MoonData
Statistical Validation	Benchmark comparison, ablation, loss curves, MIoU curves, feature-map visualization, and transfer-learning evaluation; no confidence interval, repeated-run standard deviation, or formal significance test reported
Ablation Study	Full. Includes spike-driven mechanism, SSTAM contribution, timestep (T), threshold (V_{th}), attention-output variants, upsampling methods, energy consumption, and visualization analysis
Training Framework / Code Availability	PyTorch and SpikingJelly; code availability NR; data stated as available on request
Target Hardware	Theoretical energy evaluation for INT-8 and Float-32 chips; neuromorphic chip deployment discussed as future work
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using AC/MAC counts, firing rate, timestep, and Horowitz-style operation energy. INT-8 energy for ST-RTNet is reported as 1.04 mJ; Float-32 energy is 34.00 mJ in the complexity table
Latency / Memory / Complexity Evidence	Reports 3.69G AC operations and 6.67G MAC operations for ST-RTNet; discusses GPU latency/memory limitations due to temporal unfolding, but measured inference latency and memory footprint are NR
Comparison with ANN Baseline	Yes. Compares against ANN segmentation models including FCN, UNet, Swin-UNet, SCG-TransNet, and RockSeg

Main Contribution / Key Limitation	Contribution: introduces the first directly trained spike-driven SNN–Transformer rock segmentation model for deep-space exploration, using SSTAM to capture spatio-temporal membrane-potential dynamics while reducing energy consumption. Limitation: energy evaluation is theoretical rather than measured on neuromorphic hardware; energy rises as timestep increases; fine-grained segmentation still has room for improvement
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P091

Spiking neural networks for EEG signal analysis using wavelet transform

Full Citation	Yuan, L., Wei, J., & Liu, Y. (2025). Spiking neural networks for EEG signal analysis using wavelet transform. Frontiers in Neuroscience, 19, Article 1652274. https://doi.org/10.3389/fnins.2025.1652274
University	Academy of Military Sciences
Country	China
Publication Year	2025
Publication Venue	Frontiers in Neuroscience
Publisher	Frontiers
DOI / URL	https://doi.org/10.3389/fnins.2025.1652274
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	EEG emotion recognition and auditory attention decoding
Data Modality	EEG signal
Signal Dimensionality	1D multichannel temporal EEG transformed into 2D spatial-topological spectro-spatial maps and token sequences
Signal Characteristics	DEAP uses 32-channel EEG recorded at 512 Hz; KUL uses EEG from 16 subjects in auditory attention tasks; all EEG was re-referenced, band-pass filtered from 1–32 Hz, downsampled to 128 Hz, normalized, segmented using sliding windows, and projected into 2D topographic maps
Dataset Name	DEAP; KUL
Dataset Type	Public EEG / BCI benchmark datasets
Dataset Size	DEAP: 32 participants, 40 trials per participant, each trial 1 min with 3 s baseline; KUL: 16 subjects, 8 trials per subject, each trial 6 min
Dataset Balance / Split	NR
Hybrid Topology	Spiking Wavelet Transformer architecture: EEG preprocessing → 2D topographic map projection → Spiking Patch Splitting → L spiking wavelet encoder blocks → global average pooling → MLP / classification head
SNN Component	LIF neurons, spike-based Q/K/V, spiking patch splitting, spiking wavelet self-attention, spiking encoder blocks
Neuron Model	LIF
Transformer Component	Spiking Transformer encoder; Spiking Wavelet Self-Attention; MLP block; classification head
Attention Type	Spiking self-attention integrated with Haar discrete wavelet transform; softmax-free spike-based Q/K/V attention
SNN–Transformer Interface Mechanism	EEG topographic maps are divided into patches and tokenized; Q, K, and V are generated through Linear + BN + LIF layers; Haar DWT decomposes features into wavelet sub-bands, IDWT reconstructs features, and wavelet features are concatenated with spiking attention output
Model Depth / Scale	Uses L spiking wavelet encoder blocks; exact number of layers, heads, embedding size, and parameter scale NR; SNN simulation timestep = 4
Parameter Count	NR
Design / Contribution Type	New SpikeWavformer architecture; new Spiking Wavelet Self-Attention module; integration of Haar DWT with spiking self-attention for EEG time-frequency feature extraction; theoretical convergence analysis; theoretical energy-efficiency analysis
Input Encoding Method	EEG preprocessing, α-band cortical activity projection into 2D topological maps, patch/token embedding through SPS, and spike generation through LIF neurons
Encoding Parameters	Band-pass filter 1–32 Hz; downsampled to 128 Hz; Haar DWT; wavelet sub-bands LL, LH, HL, HH; timestep = 4; average spiking rate = 12.3%; sigmoid surrogate-gradient parameter $\alpha = 4$
Encoding Learned or Fixed	Partially learned: EEG filtering, Haar DWT, and topographic projection are fixed; patch projection, Q/K/V projection, convolution, BN, and classifier parameters are learned
Event / Spike Representation	Binary spike sequences; spike-based Q, K, V tensors; wavelet coefficient representations; 2D EEG topographic token representations
Temporal Resolution	EEG downsampled to 128 Hz; KUL decision windows tested at 0.1, 0.2, 0.5, 1, 2, 5, and 10 s; SNN timestep = 4
Output Decoding Method	Global average pooling followed by MLP / fully connected classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes EEG time-frequency decomposition using DWT, convergence properties of SWSA, spiking rate, theoretical energy efficiency, and saliency-map interpretability for DEAP and KUL
Training Paradigm	Supervised learning
Training Method	End-to-end direct training with cross-entropy loss and surrogate-gradient learning
Surrogate Gradient Function	Sigmoid surrogate gradient with $\alpha = 4$
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Cross-entropy loss
Training Hyperparameters	Adam optimizer; initial learning rate 1 × 10■■; 200 epochs; two NVIDIA RTX 4090 GPUs; EEG re-referenced to average electrode response; 6th-order Chebyshev Type II band-pass filter from 1–32 Hz; downsampled to 128 Hz; normalized to zero mean and unit variance per trial; timestep = 4
Primary Performance Metric	Classification accuracy
Primary Metric Value	DEAP: Arousal accuracy 76.51% ± 5.48%, Valence accuracy 77.10% ± 5.68%; KUL: 80.5% at 0.1 s, 86.7% at 0.2 s, 94.2% at 0.5 s, 96.5% at 1 s, 97.1% at 2 s, 97.3% at 5 s, and 98.6% at 10 s decision window
Secondary Metrics	Standard deviation, decision-window accuracy, spiking rate, theoretical energy-efficiency ratio, saliency-map interpretability

Baseline Models Compared	DEAP: EEGNet, SCN, DCN, Tsception; KUL: Linear CCA, nonlinear CNN, STAnet
Performance Gain Over Baseline	On DEAP, SpikeWavformer improves over Tsception by +14.94 percentage points for arousal and +17.96 percentage points for valence; on KUL, it improves over STAnet by +6.4 percentage points at 1 s and +4.7 percentage points at 10 s; theoretical energy efficiency is over 7× compared with ANN counterpart
Statistical Validation	Reports accuracy and standard deviation on DEAP; no formal statistical significance test, confidence interval, or repeated-run significance analysis reported
Ablation Study	No formal ablation study reported; includes comparative study, theoretical convergence analysis, theoretical energy analysis, and saliency-map interpretability
Training Framework / Code Availability	Experiments conducted on two NVIDIA RTX 4090 GPUs; exact software framework and code availability NR
Target Hardware	Portable / resource-constrained BCI and neuromorphic hardware are discussed as target deployment scenarios
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimate only; uses AC/MAC ratio = 1/7, average spiking rate = 12.3%, timestep = 4, and reports over 7× energy efficiency compared with ANN counterpart
Latency / Memory / Complexity Evidence	Low computational overhead is discussed conceptually through Haar wavelet efficiency and event-driven spike computation, but measured latency, memory footprint, FLOPs, and runtime complexity are NR
Comparison with ANN Baseline	Yes. Compares with EEGNet, SCN, DCN, Tsception, CNN, CCA, STAnet, and an ANN counterpart for theoretical energy estimation
Main Contribution / Key Limitation	Contribution: proposes SpikeWavformer, an end-to-end SNN–Transformer EEG framework combining Haar DWT and spiking self-attention for automatic time-frequency EEG feature extraction, emotion recognition, and auditory attention decoding. Limitation: no real neuromorphic hardware validation; energy evidence is theoretical; parameter count, latency, memory, and code availability are NR; formal ablation study is not reported
RQ Mapping	Primary: RQ1, RQ3; Secondary: RQ2, RQ4, RQ5

P092

SpikingViT: A Multiscale Spiking Vision Transformer Model for Event-Based Object Detection

Full Citation	Yu, L., Chen, H., Wang, Z., Zhan, S., Shao, J., Liu, Q., & Xu, S. (2025). SpikingViT: A Multiscale Spiking Vision Transformer Model for Event-Based Object Detection. IEEE Transactions on Cognitive and Developmental Systems, 17(1), 130–146. https://doi.org/10.1109/TCDS.2024.3422873
University	Zhejiang University
Country	China
Publication Year	2025
Publication Venue	IEEE Transactions on Cognitive and Developmental Systems
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TCDS.2024.3422873
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Event-based object detection
Data Modality	Event-camera / DVS data
Signal Dimensionality	2D spatiotemporal event data represented as temporal tensors and multiscale feature maps
Signal Characteristics	Event camera data are represented by polarity, timestamp, and spatial coordinates. The model converts events into tensors with temporal dimension T, height H, and width W. Gen1 has 240 × 304 resolution; 1Mpx has 720 × 1280 resolution and is downsampled to 360 × 640 for memory and efficiency.
Dataset Name	Gen1; 1Mpx / Gen4; DSEC-detection
Dataset Type	Public event-based object detection benchmark datasets
Dataset Size	Gen1: 228,000 car bounding boxes and 28,000 pedestrian bounding boxes. 1Mpx / Gen4: about 25 million bounding boxes for cars, pedestrians, and two-wheelers. DSEC-detection size NR.
Dataset Balance / Split	Gen1 sampling frequency: 1, 2, and 4 Hz. 1Mpx sampling frequency: 30 and 60 Hz. For 1Mpx and DSEC-detection, only pedestrian, rider/two-wheeler, and car categories are retained. Exact train/test split and class balance NR.
Hybrid Topology	Hierarchical multiscale Spiking Vision Transformer. Pipeline: event data conversion module → temporal extension embedding → multiscale feature extraction module with four SNN-based learning stages → non-spiking YOLOX detection head
SNN Component	LIF neurons, spiking self-attention block, LIF-activated MLP block, temporal memory spiking neuron block, residual voltage memory
Neuron Model	LIF
Transformer Component	Spiking Vision Transformer; multiscale spiking Transformer stages; spiking self-attention; patch merging; YOLOX detection head
Attention Type	Spiking self-attention; softmax-free spike-based Q/K/V attention
SNN–Transformer Interface Mechanism	Event tensors are converted into temporal feature maps and passed through patch merging. Q, K, and V are generated by linear projection, batch normalization, and LIF neurons. SSA processes spike-form Q/K/V and passes multiscale features to a YOLOX detection head.
Model Depth / Scale	Four learning stages. SpikingViT-T: layer numbers 1,1,2,1 with timesteps 32,64,128,256. SpikingViT-S: layer numbers 1,2,2,1 with timesteps 48,96,192,384. SpikingViT-B: layer numbers 2,2,2,2 with timesteps 64,128,256,512. Sequence length is set to 5 in main experiments.
Parameter Count	SpikingViT-T: 9.2M; SpikingViT-S: 13.5M; SpikingViT-B: 21.5M
Design / Contribution Type	New multiscale Spiking Vision Transformer for event-based object detection; event data conversion module; temporal extension embedding; multiscale feature extraction; temporal memory spiking neuron using residual voltage memory; SNN–Transformer + YOLOX hybrid detector
Input Encoding Method	Event-to-tensor conversion with polarity merging; temporal extension embedding using convolutional kernel mapping; multiscale temporal expansion through patch merging and LIF-based spike generation

Encoding Parameters	Event tensor initially E ∈ [2, T, H, W], then polarity channels are merged to E ∈ [T, H, W]. Temporal extension maps E(T,H,W) to X(T*, H/2, W/2). Sequence length = 5. LIF reset voltage = 0, membrane time constant τ = 2.0, attention-map LIF threshold = 0.5, other LIF thresholds = 1.0.
Encoding Learned or Fixed	Partially learned. Event quantization and polarity merging are fixed preprocessing choices; temporal extension kernel, projections, SSA, LAMLP, and TMSN parameters are learned.
Event / Spike Representation	Event tensors; temporal spike feature maps; binary spike outputs from LIF neurons; residual membrane voltage memory; spike-form Q/K/V
Temporal Resolution	Variable timesteps across stages. For SpikingViT-B: 64, 128, 256, and 512 timesteps across stages. Sequence length = 5; event interval length = sequence length × 10 ms.
Output Decoding Method	Multiscale feature maps are fed into a non-spiking YOLOX detection head to output bounding boxes and object classes
Encoding / Temporal Information Analysis	Yes. The paper analyzes event polarity merging, temporal extension, variable timesteps, sequence length effects, residual voltage memory, firing rate, and energy efficiency. It also visualizes success/failure cases on Gen1, 1Mpx, and DSEC-detection.
Training Paradigm	Supervised learning
Training Method	Direct SNN training using surrogate-gradient learning through LIF neurons; mixed precision training; YOLOX-based object detection training
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	YOLOX object detection loss; exact full loss formula NR
Training Hyperparameters	Gen1: batch size 24, sequence length 5, maximum learning rate 2e-4, minimum learning rate 0, 300,000 iterations, trained 3–4 days on one NVIDIA RTX 3090. 1Mpx: batch size 10 per GPU, sequence length 5, maximum learning rate 3e-4, minimum learning rate 0, 500,000 iterations, trained about 6 days on two NVIDIA RTX 3090 GPUs. CosineAnnealingWarmRestarts schedule; random horizontal flipping, zoom-in, and zoom-out augmentation.
Primary Performance Metric	mAP and AP50
Primary Metric Value	SpikingViT-B: Gen1 mAP 0.394 and AP50 0.616; 1Mpx / Gen4 mAP 0.362 and AP50 0.622; DSEC-detection mAP 0.386 and AP50 0.669
Secondary Metrics	AP75, parameter count, runtime, firing rate, FLOPs, theoretical energy efficiency
Baseline Models Compared	Asynet, AEGNN, Inception, RRC-Events, MatrixLSTM, YOLOv3 Events, RED, ASTMNet, RVT, Spiking DenseNet, LT-SNN, KD-SNN, EMS-ResNet, TR-YOLO, EAS-SNN, ANN benchmark model
Performance Gain Over Baseline	Compared with EMS-ResNet34 on Gen1, SpikingViT-B improves mAP from 0.310 to 0.394 and AP50 from 0.590 to 0.616. Compared with ANN benchmark, it improves mAP from 0.381 to 0.394 and AP50 from 0.603 to 0.616 while achieving 2.28× theoretical energy efficiency.
Statistical Validation	Benchmark evaluation and ablation experiments; no standard deviation, confidence interval, repeated-run analysis, or formal statistical significance test reported
Ablation Study	Full. Includes data conversion module vs general implementation, sequence length variation, TMSN placement, with/without TMSN, offset learning in TMSN, DSEC-detection validation, and energy comparison with ANN benchmark
Training Framework / Code Availability	SpikingJelly with CuPy acceleration; source code provided at GitHub: Chrazqee/SpikingViT
Target Hardware	Neuromorphic hardware is discussed as a target, but the model is trained/evaluated on GPUs
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimate based on SOP, firing rate, timesteps, FLOPs, AC energy 0.9 pJ, and MAC energy 4.6 pJ. Reports 2.28× energy-efficiency improvement over ANN benchmark.
Latency / Memory / Complexity Evidence	SpikingViT-B runtime: 26.88 ms on Gen1 with sequence length 5; 65.36 ms on 1Mpx with sequence length 5. FLOPs: 5.92G. Parameters: 21.48M in energy comparison. The DC module reduces runtime from 165.84 ms to 26.88 ms compared with general implementation.
Comparison with ANN Baseline	Yes. Compares with a standard Transformer-based ANN benchmark using frame-based inputs, non-spiking attention, ReLU activations, and similar parameter scale
Main Contribution / Key Limitation	Contribution: proposes SpikingViT, a multiscale SNN–Transformer object detector for event-camera data, combining event data conversion, temporal extension, SSA, LAMLP, and residual-voltage TMSN memory. Limitation: YOLOX detection head is non-spiking; no real neuromorphic deployment; SSA requires buffering events across the full time window, creating memory and latency challenges for current neuromorphic chips.
RQ Mapping	Primary: RQ1, RQ2, RQ3; Secondary: RQ4, RQ5

P093

Spike-driven Transformer

Full Citation	Yao, M., Hu, J., Zhou, Z., Yuan, L., Tian, Y., Xu, B., & Li, G. (2023). Spike-driven Transformer. In Advances in Neural Information Processing Systems / 37th Conference on Neural Information Processing Systems, NeurIPS 2023.
University	Chinese Academy of Sciences
Country	China
Publication Year	2023
Publication Venue	NeurIPS 2023
Publisher	NeurIPS
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Publication Type	Conference paper
Database Source	WoS
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Static image classification and neuromorphic event-based classification
Data Modality	Static RGB images and event-camera / DVS data converted into frame sequences
Signal Dimensionality	2D images represented as spike patch sequences; event streams represented as temporal frame sequences

Signal Characteristics	ImageNet input size 224 × 224 in default experiments; 288 × 288 used for enlarged inference setting; images are divided into N = 196 patches by the SPS module; static images are repeated over timesteps; DVS event streams are converted into frame sequences before processing
Dataset Name	ImageNet-1K, CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public static image and neuromorphic/event-based benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Uses standard benchmark settings; exact train/test split and class balance NR
Hybrid Topology	Fully spike-driven Transformer architecture with Spiking Patch Splitting, L-block Spike-driven Transformer encoder, Spike-Driven Self-Attention, spike-driven MLP, membrane-potential shortcuts, global average pooling, and classification head
SNN Component	LIF neurons, spike encoding layer, binary spike communication, spike-driven convolution, spike-driven MLP, SDSA, membrane shortcut, spike-form Q/K/V
Neuron Model	LIF
Transformer Component	Vision Transformer-style encoder; Spiking Patch Splitting; Spike-Driven Self-Attention; MLP block; linear classification head
Attention Type	Spike-Driven Self-Attention; mask-and-add attention; linear spike-based self-attention; softmax-free and scale-free attention
SNN–Transformer Interface Mechanism	Spiking Patch Splitting converts image/event input into spike patch tokens. Linear + BN + LIF layers generate spike-form Q, K, and V. SDSA replaces matrix multiplication, softmax, and scaling with Hadamard masking, sparse addition, column summation, and a spiking neuron layer. Membrane-potential shortcuts ensure that transmitted spike tensors remain binary.
Model Depth / Scale	Reported ImageNet models include Spiking Transformer-8-384, 6-512, 8-512, 10-512, and 8-768. Encoder blocks range from 6 to 10; channel dimensions include 384, 512, and 768. Main timestep is T = 4, with T = 1 also tested.
Parameter Count	8-384: 16.81M; 6-512: 23.37M; 8-512: 29.68M; 10-512: 36.01M; 8-768: 66.34M
Design / Contribution Type	New Spike-driven Transformer architecture; new SDSA module; residual connection rearrangement using membrane shortcuts; energy-efficient spike-driven Transformer design with only sparse addition operations
Input Encoding Method	Static images are repeated across timesteps and converted into spike patch tokens through Spiking Patch Splitting; event datasets are converted into frame sequences; first SNN layer acts as the spike encoding layer
Encoding Parameters	ImageNet input size 224 × 224; inference variant 288 × 288; N = 196 patches; T = 4 for main ImageNet/CIFAR experiments; T = 16 for CIFAR10-DVS and DVS128 Gesture; SDSA spike firing rates reported for 8-512: QS 0.0091, KS 0.0090, g(QS,KS) 0.0713, VS 0.1649, SDSA output 0.0209
Encoding Learned or Fixed	Partially learned: patch splitting, convolution, linear projections, BN, and model weights are learned; LIF spike-generation dynamics are fixed
Event / Spike Representation	Binary spike tensors; spike patch sequences; spike-form Q, K, V; membrane potential tensors; binary attention vectors; spike firing rate maps
Temporal Resolution	T = 4 for main ImageNet/CIFAR static image experiments; T = 1 tested for ultra-low-latency ImageNet inference; T = 16 for CIFAR10-DVS and DVS128 Gesture
Output Decoding Method	Global average pooling over the final spike-driven encoder output followed by a fully connected classification head
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike firing rate, binary spike communication, SDSA attention maps, theoretical energy consumption, linear attention complexity, and the effect of membrane shortcuts on binary spike preservation
Training Paradigm	Supervised direct training
Training Method	Direct SNN training with surrogate-gradient learning; membrane-potential shortcut training; Spike-driven Transformer trained end-to-end
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Lamb for ImageNet experiments
Loss Function	NR
Training Hyperparameters	ImageNet: batch size 128 or 256; 310 training epochs; cosine-decay learning rate with initial value 0.0005; Lamb optimizer; input size 224 × 224; N = 196 patches; random augmentation and mixup. Smaller dataset training settings mostly follow Spikformer; detailed settings NR in main paper
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	ImageNet-1K best result: 77.07% / approximately 77.1% top-1 accuracy using Spiking Transformer-8-768 with 288 × 288 inference. CIFAR-10: 95.6%; CIFAR-100: 78.4%; CIFAR10-DVS: 80.0%; DVS128 Gesture: 99.3%
Secondary Metrics	Theoretical power / energy consumption, parameter count, timestep count, spike firing rate, attention energy ratio, ablation accuracy
Baseline Models Compared	Hybrid training ResNet-34, TET, Spiking ResNet, tdBN, SEW-ResNet, MS-ResNet, Att-MS-ResNet, Res-CNN, vanilla Transformer, Spikformer, tdBN, PLIF, Dspike, DSR, DIET-SNN, ANN ResNet19, ANN Transformer4-384
Performance Gain Over Baseline	On ImageNet, Spike-driven Transformer-8-512 reaches 74.57% compared with Spikformer-8-512 at 73.38%. Spike-driven Transformer-8-768 reaches about 77.1%, higher than Spikformer-8-768 at 74.81%. SDSA self-attention energy is reported up to 87.2× lower than vanilla self-attention; ANN Transformer-8-512 uses 41.77 mJ compared with 4.50 mJ for the spike-driven counterpart at T = 4
Statistical Validation	Benchmark comparison and ablation study; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full/clear ablation. Includes baseline Spikformer, SDSA-only, membrane shortcut-only, and full model on CIFAR-10/100. Also reports spike firing-rate attention maps and energy analysis
Training Framework / Code Availability	Source code and models available at https://github.com/BICLab/Spike-Driven-Transformer
Target Hardware	Neuromorphic chips are discussed as the target deployment direction, including TrueNorth, Loihi, and Tianjic-style spike-driven hardware
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation only. Energy is computed from MAC/AC operation assumptions, spike firing rates, timestep T, and FLOPs. Reports self-attention energy reduction up to 87.2× and strong reduction versus ANN Transformer power
Latency / Memory / Complexity Evidence	SDSA has linear complexity in both token and channel dimensions; VSA complexity $O(N^2D + N^2D)$, while SDSA is reduced to approximately $O(ND)$. Reports theoretical power: 8-512 at 4.50 mJ with T = 4 and 1.13 mJ with T = 1. No measured latency or memory footprint reported
Comparison with ANN Baseline	Yes. Compares with vanilla ANN Transformer-8-512, Res-CNN-104, ANN ResNet19, and ANN Transformer4-384
Main Contribution / Key Limitation	Contribution: proposes the first fully spike-driven Transformer design using SDSA and membrane shortcuts so that Transformer computation is reduced to sparse addition, mask operations, and binary spike communication. Limitation: energy results are theoretical rather than measured on neuromorphic hardware; no real hardware deployment; some training and loss details are not fully reported in the main paper

RQ Mapping	Primary: RQ2, RQ5; Secondary: RQ1, RQ3, RQ4

P094
A Y-shaped spiking neural network for automatic retinal segmentation

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Country	China
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Publisher	Elsevier
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Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	Automatic retinal OCT layer segmentation and pathological fluid/lesion segmentation
Data Modality	Optical Coherence Tomography images
Signal Dimensionality	2D grayscale OCT B-scan images expanded into temporal spike representation
Signal Characteristics	Retinal OCT images with layer boundaries, noise, blurriness, artifacts, and pathological regions such as DME, IRF, SRF, and PED. Input images are resized/cropped to dataset-specific training sizes.
Dataset Name	HCMS, Duke DME, AROI
Dataset Type	Public retinal OCT segmentation datasets
Dataset Size	HCMS: 35 subjects, 3234 images, 3038 training / 196 test. Duke DME: 10 subjects, 110 images, 88 training / 22 test. AROI: 24 subjects, 1136 annotated images, 938 training / 198 test.
Dataset Balance / Split	HCMS: 3038/196 train/test; Duke DME: 88/22 train/test; AROI: 938/198 train/test. Exact class balance NR.
Hybrid Topology	Y-shaped SOCT-Net with dual-path encoder and single-path decoder. One encoder path uses Spike ResBlocks for local/detail extraction, while the parallel Spike TransBlock path uses self-attention for global retinal boundary modeling.
SNN Component	LIF neurons, Spike ResBlock, spike residual learning, Spike TransBlock, spiking tensors Q/K/V, temporal and spatial-channel attention, spike-based encoder-decoder processing
Neuron Model	LIF
Transformer Component	Spike TransBlock with multi-head self-attention; self-attention branch integrated into the encoder
Attention Type	Multi-head spike self-attention; temporal attention; spatial-channel attention
SNN–Transformer Interface Mechanism	OCT image is expanded along a time dimension and passed through LIF layers. Spike TransBlock generates spike-form Q, K, and V through convolution and LIF layers, then applies multi-head self-attention to model global retinal-layer dependencies.
Model Depth / Scale	Encoder has three layers with residual block numbers [3, 3, 2]. Channels are [64, 128, 256]. Number of attention heads = 8. MLP expansion ratio = 4. Input temporal dimension T = 2.
Parameter Count	9.4650M
Design / Contribution Type	New Y-shaped SNN–Transformer segmentation network for OCT retinal layer segmentation; combines Spike ResBlocks, Spike TransBlock, cross convolution, dual-path encoder, and decoder with skip connections
Input Encoding Method	Direct image-to-spike temporal expansion; OCT image is expanded by one time dimension with T = 2, then processed through convolution, group normalization, LIF, temporal attention, and spatial-channel attention
Encoding Parameters	T = 2; initial convolution kernel 3 × 3 with stride 1; cross convolutions 1 × 9 and 9 × 1; LIF threshold and membrane parameters mostly NR
Encoding Learned or Fixed	Partially learned: temporal expansion is fixed; convolutional encoding, SRB, STB, and attention weights are learned
Event / Spike Representation	Binary spike tensors; membrane potential representation; spike-form Q, K, V; temporal spike feature maps
Temporal Resolution	2 timesteps
Output Decoding Method	Single-path decoder with upsampling and skip connections outputs pixel-wise retinal segmentation maps
Encoding / Temporal Information Analysis	Partial. The paper discusses temporal image expansion, LIF dynamics, sparse spike processing, residual spike representation, and spike-based global modeling, but does not deeply quantify temporal information loss
Training Paradigm	Supervised learning
Training Method	End-to-end training using BPTT-style surrogate-gradient learning for SNN layers
Surrogate Gradient Function	Rectangular / piecewise surrogate gradient based on the sign function around the firing threshold; exact parameter value NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Dice loss + MSE loss
Training Hyperparameters	Learning rate 1e-3; weight decay 5e-4; Cyclic Learning Rate scheduler; 100 epochs; horizontal/vertical flipping with 50% probability; random rotation between –20° and 20° with 50% probability
Primary Performance Metric	Dice score
Primary Metric Value	Duke DME Dice = 0.8252; HCMS Dice = 0.9086; AROI Dice = 0.8612
Secondary Metrics	IoU / MIoU, per-class Dice, parameter count, FLOPs, inference time
Baseline Models Compared	MGU-Net, ReLayNet, Swin-UNet, TransUNet, MT-UNet, U2Net-P, U2Net, H2Former, Y-Net
Performance Gain Over Baseline	On Duke DME, SOCT-Net achieves Dice 0.8252 and IoU 0.7326, outperforming H2Former Dice 0.8205. On HCMS, SOCT-Net achieves Dice 0.9086 and IoU 0.8364, slightly above MGU-Net Dice 0.9082. On AROI, SOCT-Net achieves Dice 0.8612 and IoU 0.8040, outperforming H2Former Dice 0.8555 and IoU 0.7993.
Statistical Validation	Benchmark comparison and ablation study; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes SOCT-Net without STB, without SRB, without data enhancement, and full SOCT-Net on AROI dataset

Training Framework / Code Availability	PyTorch; code available at GitHub: YBYopen/soctopen
Target Hardware	Neuromorphic hardware and resource-constrained deployment are discussed conceptually
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No measured energy result. The paper discusses potential energy and latency advantages of SNNs on neuromorphic hardware, but energy is not experimentally measured
Latency / Memory / Complexity Evidence	Parameters: 9.4650M; FLOPs: 110.1137G; inference time: 0.2298 s for 384 × 384 image size. The paper notes GPU FLOPs overestimate SNN cost because dense GPU computation does not reflect sparse event-driven SNN execution.
Comparison with ANN Baseline	Yes. Compares against CNN-based, Transformer-based, and CNN–Transformer medical segmentation baselines
Main Contribution / Key Limitation	Contribution: proposes SOCT-Net, a Y-shaped SNN–Transformer network for retinal OCT segmentation, combining Spike ResBlocks for local detail extraction and Spike TransBlock for global boundary modeling. Limitation: no real neuromorphic hardware validation; energy efficiency is discussed conceptually rather than measured; FLOPs and inference time are GPU-based and may not reflect true SNN efficiency
RQ Mapping	Primary: RQ1, RQ2; Secondary: RQ3, RQ4, RQ5

P095

Spike Memory Transformer: An Energy-Efficient Model in Distributed Learning Framework for Autonomous Depression Detection

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Country	China
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Publisher	IEEE
DOI / URL	https://doi.org/10.1109/JIOT.2025.3539850
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	Binary depression detection from multimodal vlog data
Data Modality	Audiovisual behavioral data: facial landmark visual features and acoustic low-level descriptors
Signal Dimensionality	Multimodal time-series data; time-aligned visual and acoustic sequences concatenated along the feature/spatial dimension
Signal Characteristics	D-Vlog samples are extracted from YouTube vlog videos. Visual modality uses 69 facial key points; acoustic modality uses 25 low-level audio descriptors extracted with OpenSmile/eGeMAPS. Frames are extracted every 0.06 s and standardized to length L = 300 using truncation or polynomial interpolation.
Dataset Name	D-Vlog
Dataset Type	Public multimodal depression detection dataset
Dataset Size	961 samples total
Dataset Balance / Split	Train: 647 samples, 375 depressed / 272 normal. Validation: 102 samples, 57 depressed / 45 normal. Test: 212 samples, 123 depressed / 89 normal.
Hybrid Topology	Spike Memory Transformer with positional encoding, stacked multiscale attention module, SNN-based memory/feedforward component, residual connection, layer normalization, and final SNN classifier
SNN Component	Delta spike encoding, first-order LIF neurons, SNN feedforward/memory module, SNN classifier, spike sequence output, membrane memory sequence
Neuron Model	First-order Leaky Integrate-and-Fire
Transformer Component	Learnable positional encoding, temporal attention, spatial attention, multihead self-attention, residual connection, layer normalization
Attention Type	Multiscale attention combining temporal attention and spatial attention; two variants: serial MSA-Model 1 and parallel MSA-Model 2
SNN–Transformer Interface Mechanism	Transformer-style multiscale attention first computes temporal and spatial attention scores. The attention output is Delta-encoded into spikes and passed through an SNN memory/feedforward module. The resulting spiking multiscale attention weights are combined with the original input using Hadamard product, followed by residual connection and layer normalization.
Model Depth / Scale	Stacked MSA blocks with depth N; exact number of stacked layers NR. Two model variants are evaluated: SMT-Model 1 using serial temporal-to-spatial attention and SMT-Model 2 using parallel temporal/spatial attention.
Parameter Count	NR
Design / Contribution Type	New SNN–Transformer depression detection model; SNN-based memory/feedforward module inside Transformer pipeline; computation-oriented hierarchical IoT framework; multiscale weighted federated learning scheme
Input Encoding Method	Early fusion of time-aligned audiovisual features; learnable temporal and spatial positional encoding; Delta modulation spike encoding before SNN feedforward/memory module
Encoding Parameters	Fixed sequence length L = 300; frame interval 0.06 s; visual feature dimension 69; audio feature dimension 25; fused feature dimension 94; Delta modulation threshold ε is defined but exact value NR
Encoding Learned or Fixed	Partially learned: positional encodings and attention projections are learned; Delta modulation rule is fixed; SNN parameters are learned
Event / Spike Representation	Binary spike sequence from Delta modulation; spike output sequence Yspk; membrane memory sequence Ymem; spike count used for class prediction
Temporal Resolution	0.06 s sampling/frame interval; standardized sequence length 300
Output Decoding Method	Final SNN classifier outputs spike and memory sequences. Class prediction is determined by spike counts across time; the class with the larger total spike count is selected.
Encoding / Temporal Information Analysis	Yes. The paper analyzes time-aligned multimodal fusion, temporal/spatial positional encoding, Delta spike encoding, LIF memory dynamics, PCA visualization of sample distribution, and power variation during SMT training.

Training Paradigm	Supervised learning; distributed/federated learning simulation
Training Method	Direct training with surrogate-gradient backpropagation through SNN components; decentralized federated learning is simulated for distributed SMT training
Surrogate Gradient Function	Arctangent surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	Cross-entropy loss computed over the SNN memory sequence
Training Hyperparameters	PyTorch 2.1.1; NVIDIA Tesla M60 8GB; batch size 28; learning rate 0.001; fixed sequence length 300; experiments repeated multiple times and averaged
Primary Performance Metric	Classification accuracy and inference energy consumption
Primary Metric Value	SMT-Model 1 accuracy: 70.73%; SMT-Model 2 accuracy: 69.81%. Inference energy: 142.47 J for SMT-Model 1 and 154.23 J for SMT-Model 2.
Secondary Metrics	Training energy, training time, training memory usage, inference time, federated-learning client energy
Baseline Models Compared	GaussianNB, Logistic Regression, AdaBoost, ResNet152, VGG19, DenseNet201, SqueezeNet1.0, MobileNet, Xception, EfficientNet, Depression Detector, STST, DepMamba
Performance Gain Over Baseline	SMT-Model 1 achieves the best accuracy among reported methods at 70.73%, slightly higher than STST at 70.70% and DepMamba at 68.87%. Compared with classical deep learning models, SMT reduces inference energy consumption by an average of 38%.
Statistical Validation	Experiments repeated multiple times and averaged; no standard deviation, confidence interval, or formal statistical significance test reported
Ablation Study	Full. Includes replacing SNN with MLP, removing spatial attention, removing temporal attention, removing both spatiotemporal attention branches, comparing SMT-Model 1 and SMT-Model 2, and federated-learning simulation
Training Framework / Code Availability	PyTorch 2.1.1; code availability NR
Target Hardware	IoT terminal devices, edge devices, intelligence agents, distributed IoT depression-detection framework
Hardware Deployment Status	Evaluated on GPU; no real IoT terminal or neuromorphic hardware deployment reported
Energy Evidence	GPU-based energy measurement during training and inference. SMT-Model 1 reports training energy 73,527 J and inference energy 142.47 J; SMT-Model 2 reports training energy 72,432 J and inference energy 154.23 J.
Latency / Memory / Complexity Evidence	SMT-Model 1: training time 965 s, training memory 5534 MB, inference time 2.36 s. SMT-Model 2: training time 889 s, training memory 5521 MB, inference time 2.72 s.
Comparison with ANN Baseline	Yes. Compares with conventional ML, CNN/deep-learning models, and D-Vlog-specific deep multimodal depression detection baselines
Main Contribution / Key Limitation	Contribution: proposes SMT, an SNN–Transformer model for low-inference-energy multimodal depression detection, and embeds it within a computation-oriented hierarchical IoT and federated-learning framework. Limitation: no neuromorphic or real IoT deployment; SNN training remains costly on von Neumann hardware; large-scale FL validation is limited by the small D-Vlog dataset.
RQ Mapping	Primary: RQ1, RQ2, RQ4, RQ5; Secondary: RQ3

P096

ReSpike: residual frames-based hybrid spiking neural networks for efficient action recognition

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Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	Frame-based video action recognition/classification
Data Modality	RGB video frames transformed into key frames and residual/event-like frames
Signal Dimensionality	2D video frame sequences with temporal dimension; key frames processed as dense RGB images and residual frames processed as sparse temporal sequences
Signal Characteristics	Video clips use 16 frames. Clips are divided into key frames carrying spatial information and residual frames carrying temporal motion information. Residual frames are computed by subtracting each frame from its segment key frame.
Dataset Name	HMDB-51, UCF-101, Kinetics-400
Dataset Type	Public video action recognition benchmark datasets
Dataset Size	HMDB-51: 6766 clips from 3312 videos, 51 action classes. UCF-101: 13,320 clips from 2500 videos, 101 action classes. Kinetics-400: over 300k trimmed clips, 400 action classes; about 240k training clips and 20k validation clips used.
Dataset Balance / Split	Kinetics-400 uses training and validation splits. UCF-101 reports 3-fold mean accuracy using all three provided splits. HMDB-51 uses the test subset. Exact class balance NR.
Hybrid Topology	ANN–SNN hybrid architecture. ANN branch processes RGB key frames for spatial features; SNN branch processes event-like residual frames for temporal dynamics; four multi-scale cross-attention modules fuse ANN and SNN features; final classification head combines both branches.
SNN Component	LIF neurons, MS-ResNet-based SNN branch, direct coding, residual-frame temporal processing, spike-based temporal feature extraction, spike-rate/SyOP energy analysis
Neuron Model	LIF

Transformer Component	Transformer-style cross-attention fusion modules; self-attention layer; cross-attention layer; MLP and normalization layers inside fusion block
Attention Type	Multi-scale cross-attention between ANN spatial features and SNN temporal features; self-attention on SNN features before cross-attention
SNN–Transformer Interface Mechanism	Intermediate SNN feature maps are tokenized and used as Query features, while ANN feature maps from key frames are tokenized and used as Key and Value features. Cross-attention injects ANN spatial context into the SNN temporal branch at multiple ResNet stages.
Model Depth / Scale	Uses four cross-attention fusion modules corresponding to four ResNet blocks. ANN backbones include ResNet-18, ResNet-50, ResNet-101, and ResNet-152. Cross-attention heads across four Transformer layers are 2, 4, 8, and 16. Clip length is 16 frames.
Parameter Count	NR
Design / Contribution Type	New ReSpike framework; key-residual video representation; hybrid ANN–SNN feature extractor; multi-scale cross-attention fusion for spatial-temporal integration; energy-efficient action recognition design
Input Encoding Method	Key-residual input representation. Each segment uses the first RGB frame as key frame; residual frames are generated by subtracting the key frame from later frames in the same segment. Direct coding is used for SNN spike generation.
Encoding Parameters	Clip length T = 16 frames; stride s controls key/residual partition. s = 2 gives 8 key + 8 residual frames; s = 4 gives 4 key + 12 residual frames. Residual frames per segment = s – 1.
Encoding Learned or Fixed	Fixed key-residual frame encoding; ANN, SNN, and cross-attention parameters are learned
Event / Spike Representation	Residual frames are treated as sparse event-like frames; SNN branch produces spike-based temporal features; spike rates are used for SyOP and energy estimation
Temporal Resolution	16-frame clips; SNN timesteps equal the number of residual frames per segment, T′ = s – 1. For s = 4, T′ = 3 residual-frame timesteps per segment.
Output Decoding Method	ANN global spatial feature and temporally averaged SNN feature are concatenated, followed by average pooling and a fully connected classification layer. Segment-level predictions are averaged for final video prediction.
Encoding / Temporal Information Analysis	Yes. The paper analyzes key-residual representation, stride effects, residual-frame sparsity, temporal dynamics, attention-map visualization, spike rate, SyOPs, and energy–accuracy tradeoff.
Training Paradigm	Supervised learning
Training Method	End-to-end joint training of ANN branch, SNN branch, and cross-attention modules using spatio-temporal backpropagation for SNN weights
Surrogate Gradient Function	Arctangent surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	Stochastic Gradient Descent
Loss Function	Cross-entropy loss
Training Hyperparameters	PyTorch implementation; SGD optimizer; momentum 0.9; weight decay 1 × 10■■; 100 epochs; clip length 16; batch size 32; trained on NVIDIA A100 GPU
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	Kinetics-400: ReSpike ResNet-50 with s = 2 achieves 70.1% Top-1 and 89.4% Top-5. UCF-101: ReSpike ResNet-50 with s = 2 achieves 85.6% Top-1 and 97.3% Top-5. HMDB-51: ReSpike ResNet-50 with s = 4 achieves 58.2% Top-1 and 83.9% Top-5.
Secondary Metrics	Top-5 accuracy, inference GFLOPs, energy consumption, SyOPs, spike rate, accuracy–energy tradeoff
Baseline Models Compared	ResNet-18, ResNet-50, C3D, I3D, Res3D, ViT-B, MViT-S, MViT-B, MS-ResNet, SNN-ResNet, TET-ResNet, AdaScan, Residual Two-stream
Performance Gain Over Baseline	Compared with prior SNN baselines, ReSpike improves by over 30% absolute accuracy on HMDB-51, UCF-101, and Kinetics-400. On Kinetics-400, ReSpike reaches 70.1% versus SNN baselines around 34.4%–37.7%. Compared with ANN baselines, ReSpike achieves comparable accuracy with up to 6.8× lower energy than 3D ANN models.
Statistical Validation	Benchmark evaluation and ablation studies; UCF-101 reports 3-fold mean accuracy; no confidence interval, standard deviation, or formal significance test reported
Ablation Study	Full. Includes key-only, residual-only, all-frame input, ANN-only, SNN-only, hybrid with/without cross-attention, stride s analysis, backbone scaling, and energy comparison
Training Framework / Code Availability	PyTorch; code shared at GitHub: Intelligent-Computing-Lab-Yale/ReSpike
Target Hardware	Energy-constrained devices and neuromorphic hardware are discussed as motivation; energy estimated using 45 nm CMOS operation costs
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimate using 45 nm CMOS assumptions: ANN FLOP = 4.6 pJ and SNN SyOP = 0.9 pJ. Reports energy for ReSpike variants, ANN baselines, and SNN baselines.
Latency / Memory / Complexity Evidence	Reports inference GFLOPs and energy. ReSpike ResNet-50 with s = 4 uses 23.57 GFLOPs and about 110.15–110.21 mJ. ReSpike ResNet-50 with s = 2 uses 37.47 GFLOPs and about 173.65–173.67 mJ. Memory and measured runtime NR.
Comparison with ANN Baseline	Yes. Compares with 2D CNNs, 3D CNNs, Transformer video models, optical-flow-based ANN methods, and ResNet ANN backbones
Main Contribution / Key Limitation	Contribution: proposes ReSpike, an ANN–SNN hybrid video action recognition framework that uses RGB key frames for ANN spatial extraction, residual frames for SNN temporal modeling, and Transformer-style cross-attention for multi-scale fusion. Limitation: not a fully spiking Transformer; ANN branch and cross-attention remain non-spiking; energy evidence is theoretical, with no real neuromorphic deployment or measured hardware energy.
RQ Mapping	Primary: RQ1, RQ2, RQ3, RQ5; Secondary: RQ4

P097

An Energy-Efficient Mechanical Fault Diagnosis Method Based on Neural-Dynamics-Inspired Metric SpikingFormer for Insufficient Samples in Industrial Internet of Things

Full Citation	Wang, C., Yang, J., Jie, H., Zhao, Z., & Wang, W. (2025). An Energy-Efficient Mechanical Fault Diagnosis Method Based on Neural-Dynamics-Inspired Metric SpikingFormer for Insufficient Samples in Industrial Internet of Things. IEEE Internet of Things Journal, 12(1), 1081–1097. https://doi.org/10.1109/JIOT.2024.3476034
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Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Industrial Monitoring, Fault Diagnosis, and Infrastructure Inspection
Specific Task	Few-sample mechanical fault classification for gear faults, aircraft engine bearing faults, and ship waterjet propulsion machinery faults
Data Modality	Vibration signals converted into 2D time-frequency images
Signal Dimensionality	1D vibration time series transformed into 2D time-frequency representations; then encoded as spiking image patch sequences
Signal Characteristics	Raw vibration signals are collected from rotating machinery sensors. FSWT converts 1D vibration samples into 2D time-frequency images. UConn uses 20 kHz sampling and 3600 points/sample; AEB uses 25 kHz sampling and 2048 points/sample; SWPMS uses 51.2 kHz sampling, 8192 raw points per acquisition, and 1024 processed points/sample.
Dataset Name	UConn gear dataset; AEB aircraft engine bearing dataset; SWPMS ship waterjet propulsion machinery system dataset; CWRU dataset for generalization verification
Dataset Type	Public benchmark dataset, laboratory-acquired dataset, and real-world engineering dataset
Dataset Size	UConn: 9 classes, 30 samples per class. AEB: 12 labels/classes, 30 samples per class. SWPMS: 12 operating/fault labels, with 300 samples per state/condition reported. CWRU: 300 samples per state in generalization experiments.
Dataset Balance / Split	Few-sample setting. UConn and AEB use 30 samples per class; generalization tasks use 60 training / 30 testing samples. Exact main train/test split for all datasets NR.
Hybrid Topology	Metric SpikingFormer architecture consisting of Spiking Patch Splitting, L stacked Spikformer Encoder Blocks, Multiscale Mask Spiking Self-Attention, MLP block, Global Average Pooling + Normalization module, and Rate Encoding Metric Classifier
SNN Component	LIF spiking neurons, spiking tokenizer, spiking patch sequence, spike-form Q/K/V, spiking residual connections, MMSSA, firing-rate encoding, spiking metric classifier
Neuron Model	LIF
Transformer Component	Spikformer Encoder Block; spiking self-attention; MLP block; relative position embedding; Transformer-style Q/K/V attention mechanism
Attention Type	Multiscale Mask Spiking Self-Attention; sparse mask-based and sparse-addition spiking attention; softmax-free attention
SNN–Transformer Interface Mechanism	FSWT time-frequency images are split into spiking patches by SPS. Linear projections generate Q, K, and V, which are converted into spike tensors by spiking neuron layers. MMSSA performs Hadamard masking between spiking Q and K, column summation, spike-neuron activation, and masking of spiking V. REMC converts spike sequences into firing-rate feature vectors for prototype-based metric classification.
Model Depth / Scale	Main MSF architecture uses SEB-4-256: 4 Spikformer Encoder Blocks with 256-dimensional feature embedding. SPS uses four convolutional layers. Time-step sensitivity tested at 1, 2, 4, and 6; final main setting uses 4 timesteps.
Parameter Count	4.21M
Design / Contribution Type	New Metric SpikingFormer for few-sample IIoT fault diagnosis; MMSSA mechanism; Rate Encoding Metric Classifier; neural-dynamics-inspired backpropagation loss; customized data acquisition validation on aircraft engine and ship propulsion platforms
Input Encoding Method	Frequency Sliced Wavelet Transform converts 1D vibration signals into 2D time-frequency images; Spiking Patch Splitting tokenizes images into spiking patches; firing-rate encoding converts final spike sequences into metric-learning features
Encoding Parameters	Time steps explored: 1, 2, 4, 6; main model uses 4 timesteps. SPS output has N spiking patches and D = 256 channels. REMC uses average firing rate over T timesteps. Neural-chaos loss uses P0 = 0.6. Exact LIF threshold and reset values NR.
Encoding Learned or Fixed	Partially learned: FSWT preprocessing and rate encoding are fixed; convolutional tokenizer, relative position embedding, MMSSA, MLP, and metric prototypes are learned
Event / Spike Representation	Binary spike tensors; spike-form Q/K/V; spiking patch tensors; firing-rate vector R; class prototypes in continuous metric space
Temporal Resolution	Model timestep T mainly set to 4 after sensitivity analysis; raw signal sampling rates are 20 kHz for UConn, 25 kHz for AEB, and 51.2 kHz for SWPMS
Output Decoding Method	Global average pooling and normalization produce D-dimensional features; REMC converts spike activity to rate-encoded vectors and classifies by Euclidean distance to class prototypes
Encoding / Temporal Information Analysis	Yes. The paper analyzes timestep influence, firing-rate-based energy, MMSSA operation, spike-to-prototype conversion, robustness under noise, and t-SNE feature separability
Training Paradigm	Supervised few-sample learning; metric learning with spiking neural dynamics
Training Method	Direct SNN training using surrogate-gradient backpropagation with a neural-dynamics-inspired loss; joint optimization of metric loss, spiking loss, cross-entropy backpropagation loss, and neural-chaos dynamics loss
Surrogate Gradient Function	Gaussian surrogate gradient approximation
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	LossMSF = Lossms + Lossbp + Lossnc, where Lossms = Lossmetric + Lossspiking; Lossmetric is prototype metric loss, Lossspiking enforces spike-output consistency, Lossbp uses cross-entropy, and Lossnc is neural-chaos dynamics loss
Training Hyperparameters	Initial learning rate 1e-5; batch size 64; 100 epochs; Python 3.7; Intel Core i7-11800H CPU; NVIDIA GeForce RTX 3050 GPU
Primary Performance Metric	Fault diagnosis accuracy and energy consumption
Primary Metric Value	UConn: 99.14 ± 0.13% accuracy, 0.29 mJ energy. AEB: 90.07 ± 0.53% accuracy, 0.32 mJ energy. SWPMS: 91.63 ± 0.47% accuracy, 0.28 mJ energy.
Secondary Metrics	Parameter count, SOPs, 95% confidence interval, robustness under SNR noise, generalization accuracy, confusion matrix, t-SNE visualization
Baseline Models Compared	ResNet, ViT, C-ECAFormer, LiConvFormer, Spiking ResNet, SEW ResNet, SpikingFormer
Performance Gain Over Baseline	On UConn, MSF achieves 99.14% versus SpikingFormer 96.28%. On AEB, MSF reaches 90.07%, improving over SpikingFormer by 4.52 percentage points. On SWPMS, MSF reaches 91.63%, improving over SpikingFormer by 6.91 percentage points. Parameter count is 7.04× lower than ResNet and 20.46× lower than ViT.
Statistical Validation	Accuracy is reported with 95% confidence intervals; robustness and generalization tests are included. No formal statistical significance test is reported.
Ablation Study	Full. Includes MMSSA versus SSA, improved spiking residual connection versus common residual connection, with/without neural-chaos loss, timestep sensitivity, robustness under noise, and generalization across mechanical systems

Training Framework / Code Availability	Python 3.7 implementation; code availability NR
Target Hardware	IIoT edge devices and resource-constrained industrial diagnostic platforms
Hardware Deployment Status	No real neuromorphic or IIoT edge-device deployment reported; data acquisition platforms are built for AEB and SWPMS experiments
Energy Evidence	Theoretical energy estimation based on SOPs; each SOP is assumed to consume 77 fJ. Reported energy: 0.29 mJ on UConn, 0.32 mJ on AEB, and 0.28 mJ on SWPMS.
Latency / Memory / Complexity Evidence	Parameter count 4.21M. SOPs: 3.72G on UConn, 4.18G on AEB, and 3.70G on SWPMS. Measured latency and memory footprint NR.
Comparison with ANN Baseline	Yes. Compares against ResNet, ViT, C-ECAFormer, and LiConvFormer, in addition to SNN-based baselines
Main Contribution / Key Limitation	Contribution: proposes MSF, an energy-efficient Metric SpikingFormer for few-sample mechanical fault diagnosis, combining MMSSA, REMC, improved spiking residual connections, and neural-dynamics-inspired training. Limitation: energy evidence is theoretical; no real neuromorphic/edge deployment; future work is needed for compression, quantization, mixed precision, distributed computing, and integration of domain knowledge.
RQ Mapping	Primary: RQ1, RQ2, RQ4, RQ5; Secondary: RQ3

P098

ASG-TDM: A Graph-Enhanced Transformer With Spiking Preprocessing and a Dendritic Head for Multi-Task Facial Analysis

Full Citation	Vasudeva, K., Chandran, S., & Chauhan, S. S. (2026). ASG-TDM: A Graph-Enhanced Transformer With Spiking Preprocessing and a Dendritic Head for Multi-Task Facial Analysis. IEEE Access, 14, 20916–20926. https://doi.org/10.1109/ACCESS.2026.3660866
University	National Institute of Technology Durgapur
Country	India
Publication Year	2026
Publication Venue	IEEE Access
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/ACCESS.2026.3660866
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Computer Vision and Image Understanding
Specific Task	Multi-task facial analysis, including facial expression recognition, facial attribute recognition, age estimation, and driver action/monitoring classification
Data Modality	Facial images and facial video frames; facial landmarks; frame-difference motion signals; patch tokens and landmark graph nodes
Signal Dimensionality	2D facial images and short temporal frame sequences; image patch tokens; facial landmark graph nodes; spike-rate features from temporal frame differences
Signal Characteristics	Input faces are resized to 224 × 224. FER2013 images are originally 48 × 48 grayscale. Video/frame inputs are converted into short-term frame differences, then into LIF-based spike-rate features at patch and landmark levels.
Dataset Name	FER2013, RAF-DB, AffectNet eight-class subset, CelebA, AgeDB, SDD driver monitoring set; CK+ is also mentioned for extended expression evaluation, but detailed CK+ results are NR
Dataset Type	Public facial expression, facial attribute, age-estimation, and driver-monitoring datasets
Dataset Size	FER2013: 35,887 images. CelebA: more than 200,000 face images. AgeDB: 12,800 age-labelled faces. AffectNet: more than 450,000 in-the-wild facial images. RAF-DB: 30,000 aligned facial images. SDD and CK+ sizes NR
Dataset Balance / Split	FER2013 uses standard train/validation/test split. CelebA uses official train/validation/test partitions. AgeDB uses common five-fold split. AffectNet and RAF-DB use official/standard splits. SDD follows the original paper split. Exact class balance NR
Hybrid Topology	Unified ASG-TDM pipeline: spiking preprocessing front-end → dynamic facial-landmark graph → graph-enhanced vision Transformer backbone with patch and graph branches → cross-attention fusion → dendritic multi-task head
SNN Component	LIF spiking front-end, binary spike generation from frame differences, patch-level spike rates, landmark-node spike rates, spike token, rate regularization, optional surrogate-gradient spike training
Neuron Model	Leaky Integrate-and-Fire
Transformer Component	Vision Transformer-style patch branch, multi-head self-attention, graph attention branch, node-to-patch cross-attention, feed-forward network, task-specific output heads
Attention Type	Multi-head self-attention, graph attention, node-to-patch cross-attention, spike-rate-biased cross-attention
SNN–Transformer Interface Mechanism	Short-term frame differences are encoded by LIF neurons into patch and landmark spike rates. These rates update the dynamic landmark graph, bias node-to-patch cross-attention, and gate the dendritic head. The Transformer itself remains mostly real-valued rather than fully spiking.
Model Depth / Scale	ASG-TDM has ViT-Base-level scale. Reported model size: 92.1M parameters, 19.1 GFLOPs, 12.5 ms latency on V100. Exact number of Transformer layers, heads, landmark nodes, and dendritic compartments NR
Parameter Count	92.1M
Design / Contribution Type	New unified spiking–graph–Transformer–dendritic architecture for multi-task facial analysis; adaptive spiking preprocessing; dynamic facial-landmark graph; graph-enhanced Transformer fusion; gated dendritic head for interpretability and robustness
Input Encoding Method	Absolute frame-difference encoding followed by exponential smoothing, LIF spike generation, patch-level and landmark-level spike-rate pooling, and compact spike-token projection
Encoding Parameters	Frame difference xt = |It – It–1|; exponential smoothing β defined but value NR; membrane time constant 20 ms; leak factor λ = 0.95; refractory constant ρ = 0.1; temporal rate window = 5 frames; target firing rate ≈ 10%; adaptive threshold formula defined but θ0 and θ1 values NR
Encoding Learned or Fixed	Mostly fixed spiking preprocessing and rate coding; Transformer, graph attention, cross-attention, dendritic gates, and task heads are learned
Event / Spike Representation	Binary pixel-level spikes; patch spike rates; landmark-node spike rates; compact spike token; spike-rate-driven graph and dendritic gates

Temporal Resolution	Short temporal window of 5 frames for spike-rate estimation; SDD sampled at 5 frames/s; membrane time constant 20 ms; exact temporal setup for static-image datasets N/A
Output Decoding Method	Dendritic head pools patch, node, and spike-rate descriptors into gated compartment outputs; soma integration produces shared representation; task-specific heads output expression, attributes, age, or driver-action predictions
Encoding / Temporal Information Analysis	Partial. The paper analyzes frame-difference spike-rate encoding, dynamic graph co-activation, spike-rate-guided cross-attention, robustness under occlusion/noise, and dendritic interpretability, but does not deeply analyze spike temporal information loss
Training Paradigm	Supervised multi-task learning with self-supervised pretraining followed by multi-task fine-tuning
Training Method	End-to-end training of graph-enhanced Transformer and dendritic head; default rate-coded spikes do not require gradients through the binary spike function; optional end-to-end spike training uses surrogate gradients
Surrogate Gradient Function	Optional sigmoid surrogate gradient with temperature annealed from 2 to 0.5; default setting uses rate coding without backpropagation through binary spike events
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Cross-entropy for expression, attribute, and driver-action tasks; MAE for age estimation; learned task-weighted multi-task loss; dendritic compartment regularizer with L1 and KL terms; firing-rate regularization
Training Hyperparameters	PyTorch; four NVIDIA V100 GPUs; AdamW; initial learning rate 3 × 10■■; weight decay 0.05; cosine decay; 5 warm-up epochs; batch size 128 for FER2013, CelebA, and AgeDB; batch size 64 for SDD; 100 epochs for FER2013, CelebA, RAF-DB, and AffectNet; 50 epochs for AgeDB and SDD; label smoothing 0.1; five random seeds
Primary Performance Metric	Top-1 accuracy for classification tasks; mean absolute error for AgeDB age estimation
Primary Metric Value	FER2013: 88.1 ± 0.2%; RAF-DB: 92.4 ± 0.3%; AffectNet-8: 65.7 ± 0.3%; CelebA mean attribute accuracy: 96.2 ± 0.1%; AgeDB MAE: 3.95 ± 0.03; SDD: 92.3 ± 0.3%
Secondary Metrics	Standard deviation, 95% confidence interval, paired t-test p-value, robustness under occlusion/noise, macro-F1, parameters, GFLOPs, latency, power × time energy proxy
Baseline Models Compared	ResNet-50, ViT-Base, GraphViT, Spikformer, G2V2Former, graph/landmark-aware models, graph-enhanced Transformer without dendritic head, and ablated ASG-TDM variants
Performance Gain Over Baseline	Compared with ViT-Base, ASG-TDM improves FER2013 from 81.0% to 88.1%, RAF-DB from 89.9% to 92.4%, AffectNet-8 from 63.4% to 65.7%, CelebA from 94.2% to 96.2%, AgeDB MAE from 4.75 to 3.95, and SDD from 86.5% to 92.3%. Improvements are reported as statistically significant in most cases.
Statistical Validation	Yes. Experiments are repeated five times with different seeds; mean ± standard deviation and 95% confidence intervals are reported; paired t-tests against the strongest baseline are reported, with $p < 0.01$ for all datasets except one
Ablation Study	Full. Includes removal of spiking front-end, landmark graph, cross-attention fusion, and dendritic head. Full model performs best: FER2013 88.1%, RAF-DB 92.4%, CelebA 96.2%, AgeDB MAE 3.95
Training Framework / Code Availability	PyTorch; code availability NR
Target Hardware	General GPU deployment; possible future edge and neuromorphic deployment for the spiking front-end
Hardware Deployment Status	Evaluated on NVIDIA V100 GPU; no direct neuromorphic hardware deployment reported
Energy Evidence	Energy proxy only, reported as power × latency on V100. ASG-TDM reports power × time = 2.20 J. No measured neuromorphic or edge-device energy result
Latency / Memory / Complexity Evidence	ASG-TDM: 92.1M parameters, 19.1 GFLOPs, 12.5 ms latency at 224 × 224 input on one NVIDIA V100. Memory footprint NR. The paper contains placeholder text for real-time FPS/latency in the neuromorphic-hardware discussion, so deployment timing claims should be treated cautiously
Comparison with ANN Baseline	Yes. Compares against CNN, ViT, graph-guided Transformer, and non-spiking facial analysis baselines
Main Contribution / Key Limitation	Contribution: proposes ASG-TDM, a unified facial-analysis pipeline combining LIF-based spiking preprocessing, landmark graph attention, Transformer patch reasoning, cross-attention fusion, and a dendritic multi-task head. Limitation: not a fully spiking Transformer; spiking is mainly used as input preprocessing and gating; no neuromorphic hardware validation; energy is a GPU proxy; some implementation details and SDD/CK+ details are NR
RQ Mapping	Primary: RQ1, RQ2, RQ3; Secondary: RQ4, RQ5

P099

SDformerFlow: Spatiotemporal Swin Spikeformer for Event-Based Optical Flow Estimation

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University	Universitat Politecnica de Catalunya
Country	Spain
Publication Year	2024
Publication Venue	PATTERN RECOGNITION
Publisher	Springer Nature
DOI / URL	https://link.springer.com/chapter/10.1007/978-3-031-78354-8_30
Publication Type	Conference paper
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Dense event-based optical flow estimation
Data Modality	Event-camera data / asynchronous event streams
Signal Dimensionality	2D spatiotemporal event streams represented as voxel/event volumes with temporal bins and polarity channels
Signal Characteristics	Event stream represented as events E = {(x■, y■, t■, p■)}. Events are discretized into voxel/event-volume representation using temporal bins. For SNN input, B = 10 and n = 2 are commonly used, producing T = B/n = 5 timesteps and C = 4 channels. DSEC resolution is 640 × 480; training uses cropped resolutions such as 288 × 384.
Dataset Name	DSEC; MVSEC; MDR synthetic/rendered dataset for MVSEC training
Dataset Type	Public event-camera optical-flow benchmark datasets; synthetic/rendered event dataset
Dataset Size	MDR: 80,000 training samples and 6,000 validation samples. DSEC and MVSEC full dataset sizes NR in the paper.

Dataset Balance / Split	DSEC training sequences are split into training and validation because the official test set lacks ground truth. MVSEC evaluation uses standard sequences including outdoor_day1 and indoor_flying1/2/3. MDR is used for training before MVSEC evaluation. Exact split counts for DSEC/MVSEC NR.
Hybrid Topology	Fully spiking encoder-decoder Spikeformer architecture for SDformerFlow; ANN counterpart STTFlowNet uses spatiotemporal Swin Transformer encoders. SDformerFlow pipeline: event input → Spiking Feature Generator → Shortcut Patch Embedding → Spatiotemporal Swin Spikeformer encoders → spike decoder blocks → optical-flow prediction.
SNN Component	LIF neurons in SDformerFlow-v1; PSN neurons in SDformerFlow-v2; spiking feature generator; spiking patch embedding; spiking MLP; spiking patch merging; spike decoder; membrane-potential shortcuts
Neuron Model	LIF for SDformerFlow-v1; Parallel Spiking Neuron for SDformerFlow-v2
Transformer Component	Spatiotemporal Swin Spikeformer encoder; 3D window-based shifted spiking self-attention; patch merging; spiking MLP; Swin-style hierarchical stages
Attention Type	Spatiotemporal Swin Spikeformer attention; 3D window spike-driven self-attention; shifted 3D window SDSA; spiking dot-product attention in v1; spiking QK linear attention in v2
SNN–Transformer Interface Mechanism	Event voxels are split into temporal spike representations and processed by spiking convolutional feature generation. Spike-form Q/K/V or Q/K tensors are generated through Linear + BN + spiking neuron layers. SDSA performs attention inside 3D spatiotemporal windows, preserving spike-driven processing.
Model Depth / Scale	STSF encoder has four Swin stages with block depths 2–2–6–2. Attention heads across stages are 3, 6, 12, and 24. Window size is 2 × 9 × 9 for cropped resolution and 2 × 15 × 15 for full-resolution fine-tuning. Main SDformerFlow-v2 uses 4 encoders, SPE, QK attention, PSN neurons, and 10 timesteps / 2 channels in best configuration.
Parameter Count	SDformerFlow-v1-en4: 56.48M; SDformerFlow-v2-en4: 54.92M; SDformerFlow-SPE-QK-s10-c2: 54.92M; STTFlowNet-en3: 20.30M; STTFlowNet-en4: 57.51M
Design / Contribution Type	First use of Spikeformer / spatiotemporal Swin Spikeformer for event-based optical flow estimation; proposes SDformerFlow and ANN counterpart STTFlowNet; introduces SNN variants using LIF and PSN neurons; uses membrane-potential shortcuts, shortcut patch embedding, and spiking QK linear attention
Input Encoding Method	Event discretized volume representation. Events are binned using bilinear sampling over x, y, and normalized timestamp. For SNN, one event voxel chunk is split into temporal blocks and polarity channels, producing T × 2n × H × W representation.
Encoding Parameters	B = 10 bins; n = 2 temporal blocks; SNN input commonly T = 5 timesteps and C = 4 channels; SDformerFlow-v2 best variant uses s10-c2; LIF settings in v1: Vth = 0.1 and τm = 2; inverse-tangent surrogate width = 2
Encoding Learned or Fixed	Event voxelization and temporal binning are fixed; spiking feature generator, patch embedding, STSF encoders, attention, and decoder weights are learned; PSN neuron parameters are learnable
Event / Spike Representation	Event voxel volume; ON/OFF polarity channels; binary spike tensors; spike-form Q/K/V; spike-form attention vectors; membrane-potential shortcut features
Temporal Resolution	DSEC optical-flow ground truth rate is 10 Hz; event chunks are divided according to optical-flow ground-truth rate; common SNN timestep setting T = 5, with 10 timesteps used in the best SDformerFlow-v2 configuration
Output Decoding Method	Spike decoder uses transposed convolutional layers and skip connections from STSF encoders. Multi-scale flow predictions are generated and upsampled to full resolution; final output is dense optical-flow map with horizontal and vertical flow components.
Encoding / Temporal Information Analysis	Yes. The paper analyzes event voxel representation, temporal partitioning, timestep/channel choices, spatiotemporal shifted-window attention, LIF versus PSN neuron dynamics, spike rates, and theoretical energy consumption.
Training Paradigm	Supervised learning
Training Method	End-to-end supervised training with backpropagation through time and surrogate gradients for SNN models
Surrogate Gradient Function	Inverse tangent surrogate gradient with width 2
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Mean absolute error between predicted optical flow and ground-truth optical flow over valid pixels
Training Hyperparameters	DSEC: trained on three NVIDIA RTX 2080 Ti GPUs; AdamW; 80 epochs; initial learning rate 0.001; weight decay 0.01; multistep scheduler halves learning rate every 10 epochs; random horizontal/vertical flips and random crops at 288 × 384; full-resolution fine-tuning for 30 epochs with batch size 1 or 2. MVSEC: trained from scratch on MDR with 256 × 256 crops for 50 epochs.
Primary Performance Metric	Average Endpoint Error
Primary Metric Value	DSEC benchmark: SDformerFlow-v2 AEE = 1.602, outlier = 10.051%, AAE = 4.871. MVSEC average: SDformerFlow-v2 AEE = 0.66, outlier = 1.57%. Best ablation variant on DSEC validation: SDformerFlow-SPE-QK-s10-c2 with PSN gives cropped EPE 0.93 and full-resolution EPE 1.61.
Secondary Metrics	Outlier percentage over 3 pixels, Average Angular Error, EPE, parameter count, FLOPs, average spiking rate, theoretical energy/power consumption
Baseline Models Compared	E-RAFT, EV-FlowNet / EVFlowNet retrained, IDNet, TMA, E-Flowformer, TamingCM, MultiCM, OF_EV_SNN, Spike-FlowNet, XLIF-EV-FlowNet, Adaptive-SpikeNet, Spatiotemporal_SNN, ADM-Flow, ET-FlowNet, STE-FlowNet, GRU-EV-FlowNet
Performance Gain Over Baseline	On DSEC, SDformerFlow-v2 improves over OF_EV_SNN among SNNs: AEE 1.602 versus 1.707 and outlier 10.051% versus 10.308%. On MVSEC average, SDformerFlow-v2 reaches AEE 0.66, outperforming several SNN baselines and approaching or exceeding the ANN counterpart in some sequences. Energy is about one-tenth of STTFlowNet ANN counterpart and about one-third of EVFlowNet baseline in theoretical estimate.
Statistical Validation	Benchmark evaluation and ablation studies; no standard deviation, confidence interval, repeated-run analysis, or formal statistical significance test reported
Ablation Study	Full. Includes event voxel vs count input, temporal block number, patch size, window size, training resolution, timestep/channel setting, shortcut type, encoder number, SPE, dot-product vs QK attention, and LIF vs PSN neuron models
Training Framework / Code Availability	Code available at GitHub: yitian97/SDformerFlow; framework details NR
Target Hardware	Neuromorphic/event-driven hardware is discussed as the intended efficiency direction; experiments run on GPUs
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using FLOPs, spike rate, timestep count, MAC energy 4.6 pJ, and AC energy 0.9 pJ under 45 nm technology assumptions. SDformerFlow-v2-en4 reports 36.83 mJ compared with STTFlowNet-en3 at 399.65 mJ and EVFlowNet retrained at 102.95 mJ.
Latency / Memory / Complexity Evidence	Reports parameter count, FLOPs, and theoretical power/energy. SDformerFlow-v2-en4: 54.92M parameters, 42.63G FLOPs, average spiking rate 0.36, 36.83 mJ. Measured latency and memory footprint NR.
Comparison with ANN Baseline	Yes. Compares against ANN counterpart STTFlowNet, EVFlowNet, E-RAFT, IDNet, TMA, E-Flowformer, ADM-Flow, and other ANN event-based optical-flow methods

Main Contribution / Key Limitation	Contribution: proposes SDformerFlow, the first fully spiking Spatiotemporal Swin Spikeformer for event-based optical-flow estimation, achieving state-of-the-art SNN performance on DSEC and MVSEC with large theoretical energy savings. Limitation: performance still trails top ANN correlation-volume methods; full event-camera asynchrony is not fully exploited because entire chunks are fed into spatiotemporal attention; transformer scalability across resolutions remains challenging; no hardware implementation is reported.
RQ Mapping	Primary: RQ1, RQ2, RQ3, RQ4, RQ5

P100

Spiking the Transformer: NeuViT with Single-Step Attention for Efficient High-Resolution UAV Vision

Full Citation	Teoh, J. S., Tan, C. K., Baskaran, V. M., & Wong, W. P. (2026). Spiking the Transformer: NeuViT with Single-Step Attention for Efficient High-Resolution UAV Vision. Results in Engineering, 29, 108955. https://doi.org/10.1016/j.rineng.2025.108955
University	Monash University Malaysia
Country	Malaysia
Publication Year	2026
Publication Venue	Results in Engineering
Publisher	Elsevier
DOI / URL	https://doi.org/10.1016/j.rineng.2025.108955
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Real-time high-resolution UAV object detection under power and latency constraints
Data Modality	High-resolution RGB aerial images converted internally into spike-token activations
Signal Dimensionality	2D high-resolution image patches/tokens; hierarchical window-based feature maps
Signal Characteristics	Input resolutions range from 960 × 1280 to 1500 × 2000 pixels. Main evaluation uses 1500 × 2000 UAV imagery with many small, low-contrast, and occluded objects.
Dataset Name	VisDrone2019 / VisDrone2019-DET-test-dev
Dataset Type	Public UAV object detection benchmark dataset
Dataset Size	VisDrone2019 contains 10,209 UAV images and approximately 2.8 million annotated bounding boxes across 10 object categories. Test-dev evaluation uses 1,610 high-resolution images.
Dataset Balance / Split	Uses VisDrone2019 standard detection benchmark setting; exact train/validation/test split counts NR in the paper.
Hybrid Topology	Neuromorphic Vision Transformer backbone integrated with Cascade R-CNN and FPN. Architecture follows a four-stage Swin-style hierarchy with Spiking Patch Embedding, NeuroSpike Blocks, Spiking W-MSA / SW-MSA, Spiking-MSSA, LayerNorm residual integration, and detection heads.
SNN Component	Single-step stateless spiking neurons, binary spike activations, spike-driven attention, NeuroSpike Window Selection, spike-rate saliency, spike-token representation, spike generation energy model
Neuron Model	Single-step stateless spiking neuron using Heaviside threshold activation; no membrane potential tracking or temporal recurrence
Transformer Component	Swin Transformer-style hierarchical backbone; shifted-window attention; multi-head self-attention reformulated as Spiking-MSSA; FPN and Cascade R-CNN detection framework
Attention Type	Spike-driven multi-head self-attention; Spiking-MSSA; binary spike-coincidence attention; shifted-window and window-based attention
SNN–Transformer Interface Mechanism	Image patches are embedded into spike tokens using single-step spiking activation. Q, K, and V are generated through linear projection, batch normalization, and Heaviside spike activation. Attention is computed through binary spike coincidence instead of dense QK dot-products.
Model Depth / Scale	Four-stage hierarchical backbone. Stage 1 and 2 use 2 blocks each, Stage 3 uses 6 blocks, and Stage 4 uses 2 blocks. NeuViT is ViT/Swin-T-scale with 27.56M parameters.
Parameter Count	27.56M parameters
Design / Contribution Type	New NeuViT architecture for UAV vision; single-step stateless spiking neuron; Spiking-MSSA; NeuroSpike Window Selection; operation-aware and memory-aware energy modeling; UAV deployment feasibility analysis
Input Encoding Method	4 × 4 patch embedding followed by single-step stateless spike activation; spike-rate saliency used for optional window selection
Encoding Parameters	Input resolution up to 1500 × 2000; patch size 4 × 4; sigmoid surrogate steepness k = 4; target average spike rate about 30–31%; attention threshold Vth = 0.4; stage/block-specific FFN thresholds manually calibrated
Encoding Learned or Fixed	Partially learned: patch embedding, Transformer weights, Q/K/V projections, detection head, and residual gains are learned; spike thresholding rule and manually calibrated threshold schedule are fixed
Event / Spike Representation	Binary spike tokens; spike-form Q/K/V tensors; spike-rate saliency maps; pre-threshold activation distributions; spike-coincidence maps
Temporal Resolution	Single-step / one forward pass per frame; no recurrent temporal timesteps; N/A for conventional SNN temporal simulation
Output Decoding Method	Cascade R-CNN detection head with FPN outputs bounding boxes and object classes
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike-rate stability, per-stage spike and coincidence rates, spike distribution, pre-threshold activation entropy, small-object failure modes, and window-pruning behavior.
Training Paradigm	Supervised object detection training
Training Method	End-to-end training using surrogate-gradient learning for stateless spike activations within the detection framework
Surrogate Gradient Function	Sigmoid surrogate gradient; k = 4 selected after comparison with ATAN and piecewise-quadratic alternatives
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Cascade R-CNN object detection losses; exact full loss formula NR
Training Hyperparameters	72 epochs; AdamW; initial learning rate 1 × 10■■; weight decay 0.05; β terms 0.9 and 0.999; cosine annealing; 1000 warmup iterations; multi-scale augmentation using 960 × 1280, 1200 × 1600, 1080 × 1920, and 1500 × 2000; horizontal flip probability 0.5; photometric distortion; validation every 5 epochs; PyTorch 2.4.1; NVIDIA A100 80GB GPUs
Primary Performance Metric	COCO-style Average Precision and deployment efficiency under UAV constraints

Primary Metric Value	At 1500 × 2000 resolution, NeuViT achieves AP 21.43, AP50 36.01, AP75 22.69, APS 13.86, latency 45.21 ms, 22.13 FPS, total energy 0.18 J/frame, and power 3.98 W.
Secondary Metrics	MACs/ACs, compute energy, memory energy, peak memory, spike rate, FPS, power, AP by scale, per-class AP, confusion matrix, spike/coincidence statistics
Baseline Models Compared	Swin-T, SparseViT, EfficientNet-B0, EfficientNet-B5, SEW-ResNet18, SEW-ResNet50
Performance Gain Over Baseline	Compared with Swin-T, NeuViT reduces total energy by 88.8%, compute energy by 98.4%, operation count by 69.3%, latency by 31.1%, and power by 83.8%, while retaining 79.4% of Swin-T detection accuracy. Compared with SparseViT, NeuViT reduces total energy by 86.4% and power by 82.9%.
Statistical Validation	Partial. Reports five random seeds for training setup and extensive scenario/failure analysis, but formal repeated-run statistical testing for the main AP comparison is NR. Spike distribution analysis reports $p < .001$ and Cohen's d for small-object detection differences.
Ablation Study	Full/strong. Includes surrogate-gradient comparison, surrogate steepness k analysis, multi-scale resolution analysis, per-stage spike/coincidence statistics, window-pruning analysis, scenario-based failure analysis, technology-node scaling, and memory-access sensitivity.
Training Framework / Code Availability	PyTorch 2.4.1; code and models publicly available at GitHub: jayshent/NeuViT
Target Hardware	UAV onboard edge AI platforms; projected deployment on embedded GPUs, ARM SoCs, FPGA accelerators, custom ASICs, and neuromorphic accelerators such as Loihi 2
Hardware Deployment Status	No real UAV, neuromorphic chip, or embedded hardware deployment reported; profiling is GPU-based with analytical deployment projections
Energy Evidence	Analytical energy modeling using operation and memory counts with 45 nm CMOS constants: MAC 4.6 pJ, AC/ADD 0.9 pJ, spike 0.4 pJ, L1 2 pJ, L2 10 pJ, DRAM 1280 pJ. Also projects energy across 45 nm, 16 nm, and 7 nm nodes.
Latency / Memory / Complexity Evidence	Measured GPU latency: 45.21 ms at 1500 × 2000; 22.13 FPS; peak memory 2.06 GB; activation memory about 6.48 GB at maximum resolution; 83.45 GACs compared with Swin-T 271.89 GMACs; average spike rate 31.23%; projected power remains below 10 W across tested nodes.
Comparison with ANN Baseline	Yes. Directly compares against dense ANN Transformer Swin-T, SparseViT, EfficientNet CNN baselines, and non-spiking deployment assumptions.
Main Contribution / Key Limitation	Contribution: proposes NeuViT, a single-step stateless spiking vision Transformer for high-resolution UAV detection that replaces dense attention with spike-coincidence computation and demonstrates strong power/latency efficiency under UAV constraints. Limitation: AP is lower than dense Transformer baselines, especially for small and low-contrast objects; NeuroSpike window pruning can create blind spots; energy is analytically modeled rather than measured on neuromorphic or UAV hardware.
RQ Mapping	Primary: RQ1, RQ2, RQ3, RQ4, RQ5

P101

Efficient Spiking Transformer Based on Temporal Multi-Scale Processing and Cross-Time-Step Distillation

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University	Zibo Power Supply Company
Country	China
Publication Year	2025
Publication Venue	Electronics
Publisher	MDPI
DOI / URL	https://doi.org/10.3390/electronics14244918
Publication Type	Journal article
Database Source	WoS
Paper Category	Efficiency-focused
E&QA; Score	4
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification on static and dynamic/event-based datasets
Data Modality	Static RGB images and DVS/event-camera data
Signal Dimensionality	2D image data and 2D spatiotemporal event-stream data
Signal Characteristics	CIFAR images are 32 × 32 color images; ImageNet-1K uses large-scale natural images; DVS128-Gesture and CIFAR10-DVS contain asynchronous event streams with high temporal resolution
Dataset Name	CIFAR-10, CIFAR-100, ImageNet-1K, DVS128-Gesture, CIFAR10-DVS
Dataset Type	Public benchmark datasets
Dataset Size	CIFAR-10: 60,000 images, 10 classes; CIFAR-100: 60,000 images, 100 classes; ImageNet-1K: approximately 1.28 million images, 1000 classes; CIFAR10-DVS: approximately 1000 event sequences per class; DVS128-Gesture size NR
Dataset Balance / Split	CIFAR-10: 50,000 train / 10,000 test, 6000 images per class; CIFAR-100: 600 images per class; ImageNet-1K standard benchmark split implied; DVS128-Gesture and CIFAR10-DVS split details NR
Hybrid Topology	Efficient Spiking Transformer optimization framework applied to existing Spiking Transformer backbones; combines multi-scale resolution processing, cross-time-step distillation, attention-based token pruning, feature reuse, and confidence-based early output
SNN Component	LIF neurons, spike feature maps, spiking self-attention, spike-form Q/K/V, multi-time-step spike features
Neuron Model	LIF
Transformer Component	Spikformer, S-D Transformer / Spike-driven Transformer, SGLFormer, S-D Transformer V2, QKFormer
Attention Type	Spiking self-attention; spike-driven self-attention; attention-based token pruning
SNN–Transformer Interface Mechanism	Input is processed through LIF neurons to produce spike feature maps; Transformer attention uses spike-form Query, Key, and Value; cross-time-step distillation transfers information from later time steps to earlier time steps; token pruning removes low-attention tokens
Model Depth / Scale	Evaluated on five Spiking Transformer backbones: Spikformer, S-D Transformer, SGLFormer, S-D Transformer V2, and QKFormer; energy comparison uses models with final dimension 384; exact layer/block depth NR
Parameter Count	NR

Design / Contribution Type	Unified efficiency framework combining cross-time-step knowledge distillation, multi-scale resolution processing, cross-scale feature reuse, confidence-based classification evaluation, and attention-guided token pruning
Input Encoding Method	Static images are processed through multi-scale resolution downsampling and LIF-based spike feature extraction; DVS event data are used for dynamic benchmarks, but exact event encoding pipeline is NR
Encoding Parameters	Multi-scale image sizes include 56 × 56, 112 × 112, and 224 × 224; token pruning keeps tokens with attention score above γ times the global maximum; $\gamma \in (0,1)$; cross-scale fusion uses learnable α and β weights; confidence threshold controls whether higher-resolution processing is triggered
Encoding Learned or Fixed	Partially learned: LIF/spike dynamics and multi-scale processing are fixed design mechanisms, while feature fusion weights and pruning-related parameters are learnable/adaptive
Event / Spike Representation	Binary spike feature maps, spike-form Q/K/V, attention-score-based token representation, DVS event streams
Temporal Resolution	Multi-time-step SNN processing is used, but exact number of simulation time steps is NR
Output Decoding Method	Average pooling and linear classification head; confidence-based evaluation decides whether to output prediction early or trigger higher-resolution processing
Encoding / Temporal Information Analysis	Partial. The paper analyzes cross-time-step consistency, temporal distillation, token redundancy, pruning efficiency, power reduction, and inference speedup, but does not deeply analyze spike information loss or event-encoding fidelity
Training Paradigm	Supervised learning
Training Method	Surrogate-gradient-based SNN training with cross-time-step knowledge distillation; no ANN-to-SNN conversion used
Surrogate Gradient Function	Sigmoid surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	Cross-entropy loss plus cross-time-step KL-divergence distillation loss
Training Hyperparameters	ImageNet-1K setting: learning rate 6 × 10■■; warm-up epochs 5; weight decay 0.05; batch size 256; epochs 200
Primary Performance Metric	Top-1 classification accuracy and inference efficiency
Primary Metric Value	Static datasets: Spikformer-Ours 96.73 / 78.62 / 75.84 on CIFAR-10 / CIFAR-100 / ImageNet-1K; S-D Transformer-Ours 96.82 / 79.12 / 78.62; SGLFormer-Ours 97.12 / 83.07 / 84.33; S-D Transformer V2-Ours 97.37 / 82.07 / 80.75; QKFormer-Ours 97.02 / 81.92 / 84.76. Dynamic datasets: Spikformer-Ours 98.74 on DVS128-Gesture and 81.56 on CIFAR10-DVS; S-D Transformer-Ours 99.42 and 80.73; SGLFormer-Ours 99.00 and 83.72
Secondary Metrics	Power consumption, training speed, inference speed, ablation accuracy, module-level efficiency
Baseline Models Compared	Spikformer, S-D Transformer / Spike-driven Transformer, SGLFormer, S-D Transformer V2, QKFormer
Performance Gain Over Baseline	Static accuracy gains reach up to +1.55% on ImageNet-1K for S-D Transformer; dynamic accuracy improves across DVS128-Gesture and CIFAR10-DVS. Efficiency: Spikformer power reduced from 7.73 mJ to 5.43 mJ with 1.42× inference speedup; S-D Transformer power reduced from 3.90 mJ to 3.37 mJ with 1.57× inference speedup; SGLFormer power reduced from 13.04 mJ to 10.47 mJ with 1.63× inference speedup
Statistical Validation	Benchmark comparison and ablation study reported; no standard deviation, confidence interval, repeated-run analysis, or significance test reported
Ablation Study	Yes. Ablates time-step distillation, attention pruning, classification evaluation, and full model using Spikformer baseline
Training Framework / Code Availability	Ubuntu 20.04; Python 3.7; PyTorch 1.13.0; CUDA 11.8; 8 × V100 GPUs; code availability NR
Target Hardware	GPU-based experimental environment; neuromorphic/edge deployment discussed conceptually
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Reports power/energy-style values in mJ and speedup comparisons; no measured hardware energy from neuromorphic chip or FPGA reported
Latency / Memory / Complexity Evidence	Reports training and inference speedups: Spikformer-Ours 1.82× training and 1.42× inference; S-D Transformer-Ours 2.01× training and 1.57× inference; SGLFormer-Ours 1.97× training and 1.63× inference
Comparison with ANN Baseline	No direct matched ANN baseline; comparison is mainly against Spiking Transformer/SNN-Transformer baselines
Main Contribution / Key Limitation	Contribution: proposes an efficiency-oriented Spiking Transformer framework combining temporal distillation, adaptive multi-scale processing, cross-scale feature reuse, and attention-based token pruning. Limitation: many implementation details are limited; exact event encoding and backbone-specific integration are not fully explained; no real hardware validation; sensitivity to pruning/confidence thresholds is not deeply analyzed
RQ Mapping	Primary: RQ4 and RQ5; Secondary: RQ1, RQ2, RQ3

P102

Spiking transformer with learnable threshold mechanism for underwater image dehazing to aid vision-based navigation

Full Citation	Sudevan, V., Kausar, R., Javed, S., Karki, H., De Masi, G., & Dias, J. (2026). Spiking transformer with learnable threshold mechanism for underwater image dehazing to aid vision-based navigation. Neuromorphic Computing and Engineering, 6, 014010. https://doi.org/10.1088/2634-4386/ae41e1
University	Khalifa University
Country	United Arab Emirates
Publication Year	2026
Publication Venue	Neuromorphic Computing and Engineering
Publisher	IOP Publishing
DOI / URL	https://doi.org/10.1088/2634-4386/ae41e1
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	5
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Underwater image dehazing / underwater image enhancement
Data Modality	RGB underwater images; RGB-LAB transformed image sequences
Signal Dimensionality	2D image data converted into time-dependent spiking image sequences
Signal Characteristics	Raw underwater images are repeatedly passed across a user-defined timestep value to form time-dependent sequences; RGB images are converted to LAB color space; all input images are resized to 512 × 512 for training and inference

Dataset Name	UIEB, EUVP, and real-world underwater mission images for qualitative testing
Dataset Type	Public underwater image enhancement/dehazing benchmark datasets; additional real-world mission images for qualitative evaluation
Dataset Size	NR
Dataset Balance / Split	Benchmark datasets used for supervised training/testing; exact train/test split NR; real-world mission data used only for qualitative evaluation
Hybrid Topology	Lightweight spiking Transformer-based dehazing network; RGB and LAB branches are spike-coded and processed through convolutional SNN layers, spiking Transformer blocks, K estimator, background light estimator, and soft image reconstruction module
SNN Component	LIF neurons, LTMPLIF neurons, spike coding module, convolutional SNN layers, spike-based Q/K/V, binary spike feature maps
Neuron Model	LIF and learnable-threshold membrane-potential LIF neuron, LTMPLIF
Transformer Component	Spiking Transformer block; adaptive spike-based self-attention; spike-based Q/K/V transformation module
Attention Type	Adaptive spike-based self-attention, ASBSA; softmax-free and scale-free spike-based self-attention
SNN–Transformer Interface Mechanism	Static RGB images are repeated across T timesteps, converted to RGB-LAB sequences, spike-coded using convolution + LTMPLIF neurons, then processed by spiking Transformer blocks where Q, K, and V are generated through spike-based transformation modules and LTMPLIF neurons
Model Depth / Scale	Shallow architecture with depth of two layers; feature channels up to 64; timestep T = 10; image size 512 × 512 in main experiments
Parameter Count	0.5670M / approximately 0.57M parameters
Design / Contribution Type	New lightweight Spiking Transformer architecture for underwater dehazing; learnable-threshold LTMPLIF neurons; RGB-LAB hybrid processing; adaptive spike-based self-attention; custom dehazing loss; energy-efficient image restoration framework
Input Encoding Method	Direct coding / repeated static-image temporal coding; RGB image is repeated over T timesteps, converted to LAB, then transformed into binary spike representations using convolutional layers followed by LIF/LTMPLIF neurons
Encoding Parameters	Timestep T = 10; image resolution 512 × 512; surrogate smoothing parameter $\lambda = 4$; layer-wise learnable threshold Vth; spike-coded RGB/LAB branch outputs use 16 channels after downsampling
Encoding Learned or Fixed	Partially learned: image repetition and RGB-to-LAB conversion are fixed; convolutional spike coding and LTMPLIF thresholds are learned
Event / Spike Representation	Binary spike feature maps; spike-coded RGB and LAB tensors; spike-based Q, K, and V; membrane potential representation used for reconstruction
Temporal Resolution	10 timesteps; temporal dynamics arise from repeated static input processed through membrane potential accumulation, leakage, and threshold-based firing
Output Decoding Method	K estimate and background light estimate B are produced from spiking modules and passed into a soft image reconstruction equation; final LIF membrane potential without reset is used for reconstruction loss/output formation
Encoding / Temporal Information Analysis	Yes. The paper analyzes learnable threshold effects, spike rate, SOPs, analytical energy, RGB-LAB processing, and resolution-dependent efficiency; it also explains that temporal dynamics are induced internally by spiking neurons rather than external temporal variation
Training Paradigm	Supervised learning
Training Method	Direct SNN training using surrogate-gradient backpropagation through time
Surrogate Gradient Function	Fast-sigmoid surrogate gradient
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Composite dehazing loss: Lnet = LMSE + α(1 – LSSIM) + βLTV, with $\alpha = 0.5$ and $\beta = 0.25$
Training Hyperparameters	Trained for 100 epochs on UIEB and 50 epochs on EUVP; validation starts at epoch 20; learning rate 0.001; timestep T = 10; image size 512 × 512; $\lambda = 4$; trained on NVIDIA A100 GPU with 40GB RAM
Primary Performance Metric	PSNR, SSIM, energy consumption, and computational efficiency
Primary Metric Value	UIEB: PSNR 21.6773 dB, SSIM 0.8795, 7.42 GSOPs, 0.0151 J. EUVP: PSNR 23.4562 dB, SSIM 0.8439, 8.43 GSOPs, 0.0159 J
Secondary Metrics	Parameter count, SOPs, energy ratio, image resolution scalability, qualitative underwater mission enhancement, color restoration, structural clarity, noise/artifact suppression
Baseline Models Compared	UIE-SNN, WaterNet, Shallow-UWNet, Ucolor, PUIE-Net, PDCFNet, UIEConv, PUGAN, LFT-DGAN, LANet, Deep-WaveNet, DICAM, U-Trans, MFMN, MMIETransformer, WPFNet, DDFormer, Ghost-UNet, Dynamic SpectraFormer, CFPNet
Performance Gain Over Baseline	Compared with UIE-SNN, snnTrans-DHZ uses 0.57M vs 31.03M parameters, reduces energy by 8.79× on UIEB and 4.60× on EUVP, and reduces SOPs from 147.49G to 7.42G on UIEB and from 84.65G to 8.43G on EUVP. Against Ghost-UNet, it uses about 3.26× less energy on UIEB
Statistical Validation	Benchmark comparison, ablation study, resolution analysis, and qualitative real-world testing are reported; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes adaptive vs fixed threshold, RGB-LAB vs RGB-only processing, and image-resolution analysis at 256 × 256, 512 × 512, and 1024 × 1024
Training Framework / Code Availability	PyTorch and SpikingJelly; source code stated as available at snnTrans-DHZ
Target Hardware	Underwater robotic platforms, edge devices, and potential Loihi neuromorphic chip deployment
Hardware Deployment Status	No real neuromorphic hardware deployment reported; edge-device and Loihi testing proposed as future work
Energy Evidence	Analytical energy estimation based on SOPs, spike rate, and 45 nm CMOS operation energy values; not measured on physical neuromorphic hardware
Latency / Memory / Complexity Evidence	0.57M parameters; 7.4156 GSOPs and 0.0150 J at 512 × 512; 1.6711 GSOPs and 0.0017 J at 256 × 256; 25.8353 GSOPs and 0.0266 J at 1024 × 1024; no direct latency measurement reported
Comparison with ANN Baseline	Yes. Compares against CNN-based, GAN-based, attention-based, and Transformer-based non-spiking underwater image enhancement methods
Main Contribution / Key Limitation	Contribution: proposes snnTrans-DHZ, a lightweight Spiking Transformer for underwater image dehazing using LTMPLIF neurons, RGB-LAB spike coding, ASBSA, and image-formation-based reconstruction. Limitation: energy values are analytical rather than measured on real hardware; absolute PSNR/SSIM is lower than some large CNN/Transformer dehazing models; real neuromorphic deployment remains future work
RQ Mapping	Primary: RQ1, RQ2, RQ3, RQ5; Secondary: RQ4

P103

SpikingResformer: Bridging ResNet and Vision Transformer in Spiking Neural Networks	
Full Citation	Shi, X., Hao, Z., & Yu, Z. (2024). SpikingResformer: Bridging ResNet and Vision Transformer in Spiking Neural Networks. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024), pp. 5610–5619. https://doi.org/10.1109/CVPR52733.2024.00536
University	Shanghai AI Laboratory
Country	China
Publication Year	2024
Publication Venue	IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/CVPR52733.2024.00536
Publication Type	Conference paper
Database Source	WoS
Paper Category	Core architecture
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification and transfer learning on static and neuromorphic vision datasets
Data Modality	Static RGB images; event-stream data converted into frame-like representations for transfer learning
Signal Dimensionality	2D image data; neuromorphic event streams stacked into 2D frame channels
Signal Characteristics	Main ImageNet experiments use 224 × 224 input resolution, with additional 288 × 288 inference setting; neuromorphic event data are stacked over time into frame representations using positive events, negative events, and summed-event channels
Dataset Name	ImageNet, ImageNet100, CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS-Gesture
Dataset Type	Public static image and neuromorphic/event-based vision benchmark datasets
Dataset Size	ImageNet100 contains 100 ImageNet categories; other exact sample counts NR in the paper
Dataset Balance / Split	Standard benchmark splits implied; exact split and class-balance details NR
Hybrid Topology	Spiking Vision Transformer architecture that combines ResNet-style multi-stage convolutional backbone with Transformer-style Dual Spike Self-Attention; each Spiking Resformer block contains MHDSSA and GWSFFN modules
SNN Component	LIF neurons, spiking neuron layers, binary spike tensors, spike-driven residual blocks, Dual Spike Transformation, Group-Wise Spiking Feed-Forward Network
Neuron Model	LIF
Transformer Component	Dual Spike Self-Attention, Multi-Head DSSA, Transformer-style attention block, feed-forward network block
Attention Type	Dual Spike Self-Attention; Multi-Head Dual Spike Self-Attention; spike-driven self-attention without softmax
SNN–Transformer Interface Mechanism	Spike features are processed through Dual Spike Transformation; DSSA generates spiking attention maps using spike inputs and convolution/BN-based transformations, then applies spike-driven attention with scaling factors based on firing rates
Model Depth / Scale	SpikingResformer has stem + 3 stages. Stage blocks: 1, 2, and 3 blocks. Main models: Ti, S, M, L. T = 4 for main ImageNet experiments. Stage dimensions: Ti 64/192/384; S 64/256/512; M 64/384/768; L 128/512/1024. Head counts increase across stages up to 16 heads in SpikingResformer-L
Parameter Count	SpikingResformer-Ti: 11.14M; S: 17.76M; M: 35.52M; L: 60.38M
Design / Contribution Type	New Spiking Vision Transformer architecture; new DSSA mechanism; new scaling strategy for DSSA; ResNet-style multi-stage Spiking Transformer design; GWSFFN for local feature extraction
Input Encoding Method	Static images are processed through stem convolution, max-pooling, and spiking neuron layers; neuromorphic event streams are stacked into frame channels for transfer learning
Encoding Parameters	Main timestep T = 4; transfer learning uses T = 4 for CIFAR-10/CIFAR-100 and T = 10 for CIFAR10-DVS/DVS-Gesture; DSSA scaling factors use firing-rate-based terms $c1 = 1/\sqrt{(fX\ d)}$ and $c2 = 1/\sqrt{(fAttn\ HW/p^2)}$; p values across stages are 4, 2, and 1
Encoding Learned or Fixed	Partially learned: convolutional/stage features and attention transformations are learned; LIF neuron dynamics and spike-generation mechanism are fixed
Event / Spike Representation	Binary spike tensors; spike attention maps; spike-form feature maps; neuromorphic event streams represented as stacked positive, negative, and summed event frames
Temporal Resolution	4 timesteps for main ImageNet/static transfer experiments; 10 timesteps for CIFAR10-DVS and DVS-Gesture transfer experiments
Output Decoding Method	Global average pooling followed by a linear classification head
Encoding / Temporal Information Analysis	Yes. The paper provides theoretical analysis of DSSA scaling, mean/variance of Dual Spike Transformation, spike-driven computation proof, and ablation showing that removing DSSA scaling prevents convergence
Training Paradigm	Supervised direct training and transfer learning
Training Method	Direct training of SpikingResformer; ImageNet pretraining followed by transfer learning/fine-tuning on CIFAR-10, CIFAR-100, CIFAR10-DVS, and DVS-Gesture
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	Main experiments use T = 4 and 224 × 224 input resolution, with 288 × 288 inference also reported; the paper states it generally follows Spike-driven Transformer training setup, but optimizer, learning rate, batch size, and epoch details are NR in the main text
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	ImageNet: SpikingResformer-Ti 74.34%, S 75.95%, M 77.24%, L 78.77% at 224 × 224; SpikingResformer-L reaches 79.40% at 288 × 288 inference
Secondary Metrics	Parameter count, SOPs, estimated energy, transfer-learning accuracy, ablation accuracy
Baseline Models Compared	Spiking ResNet, STBP-tdBN, TET, SEW ResNet, Spikformer, Spikingformer, Spike-driven Transformer, PLIF, Dspike
Performance Gain Over Baseline	SpikingResformer-Ti outperforms Spike-driven Transformer-8-384 by 2.06% while saving 5.67M parameters and 1.44 mJ energy. SpikingResformer-M outperforms Spike-driven Transformer-8-768 by 0.92% while saving 30.82M parameters. Transfer learning improves over Spikformer by 0.37% on CIFAR-10 and 2.15% on CIFAR-100
Statistical Validation	Benchmark comparison and ablation study reported; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported

Ablation Study	Full. Includes removing multi-stage architecture, removing group-wise convolution, replacing/removing DSSA, removing p × p convolution, and removing DSSA scaling
Training Framework / Code Availability	Code available at GitHub: xyshi2000/SpikingResformer
Target Hardware	Neuromorphic hardware is discussed as motivation; no specific hardware platform targeted experimentally
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Estimated energy consumption based on SOPs, following prior Spiking Transformer evaluation practice; no measured hardware energy reported
Latency / Memory / Complexity Evidence	ImageNet 224 × 224: Ti 2.73G SOPs / 2.46 mJ; S 3.74G SOPs / 3.37 mJ; M 6.07G SOPs / 5.46 mJ; L 9.74G SOPs / 8.76 mJ. Direct latency and memory footprint NR
Comparison with ANN Baseline	Partial. The paper motivates comparison against ANN ViTs/ResNets but experimental comparison is mainly against SNN and Spiking Transformer baselines
Main Contribution / Key Limitation	Contribution: proposes SpikingResformer, combining ResNet-style multi-stage local feature extraction with DSSA-based global attention, achieving strong ImageNet accuracy with lower parameters and estimated energy than prior Spiking Vision Transformers. Limitation: no real hardware validation; energy is estimated; experiments focus mainly on classification; transfer to DVS-Gesture is weaker than direct training baselines
RQ Mapping	Primary: RQ2; Secondary: RQ1, RQ3, RQ4, RQ5

P104

Advanced SpikingYOLOX: Extending Spiking Neural Network on Object Detection with Spike-based Partial Self-Attention and 2D-Spiking Transformer

Full Citation	Miao, W., Shen, J., Xu, H., Kärkkäinen, T., Xu, Q., Xu, Y., & Cong, F. (2025). Advanced SpikingYOLOX: Extending Spiking Neural Network on Object Detection with Spike-based Partial Self-Attention and 2D-Spiking Transformer. In Proceedings of the 33rd ACM International Conference on Multimedia (MM ’25), October 27–31, 2025, Dublin, Ireland. ACM, 10 pages. https://doi.org/10.1145/3746027.3754854
University	Dalian University of Technology
Country	China
Publication Year	2025
Publication Venue	Proceedings of the 33rd ACM International Conference on Multimedia, MM ’25
Publisher	ACM
DOI / URL	https://doi.org/10.1145/3746027.3754854
Publication Type	Conference paper
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	Static-image object detection using SNN-based YOLOX architecture
Data Modality	Static RGB images
Signal Dimensionality	2D image data represented as spatial feature maps; input defined as X ∈ R^(B×C×H×W)
Signal Characteristics	Static image input; single-timestep SNN inference; object detection output includes confidence, class, center coordinates, width, and height
Dataset Name	MS-COCO 2017 / COCO2017val and COCO2017test
Dataset Type	Public object detection benchmark dataset
Dataset Size	NR
Dataset Balance / Split	COCO2017 validation and test sets used; exact split size and class-balance details NR
Hybrid Topology	Advanced SpikingYOLOX extends SpikingYOLOX with PSA-SNN and 2D-Spiking Transformer; follows backbone–neck–head detector structure with CSP-FFC-SNN, SPP-SNN, PSA-SNN in deepest backbone layer, FPN neck, and decoupled detection head
SNN Component	LIF neurons, spike-driven self-attention, signed spiking neurons inherited from SpikingYOLOX, PSA-SNN, 2D-Spiking Transformer, 2D-SDSA, spike-based Value sequence
Neuron Model	LIF; signed spiking neurons also mentioned in the SpikingYOLOX backbone context
Transformer Component	PSA-SNN module; 2D-Spiking Transformer; 2D-Spike-Driven Self-Attention; 2D-MLP
Attention Type	Spike-based partial self-attention; 2D spike-driven self-attention; linear softmax-free self-attention
SNN–Transformer Interface Mechanism	LIF converts feature maps into spike form; RepConv generates spike-form K, Q, and V; an additional spiking layer is applied only to the Value sequence; 2D-SDSA computes Q(K■V) with scale factor 0.125 and 2D convolutional projection
Model Depth / Scale	Five model scales: Nano, Tiny, S, M, and L; timestep T = 1; exact layer/block depth NR
Parameter Count	NR
Design / Contribution Type	New SNN-based object detector; Advanced SpikingYOLOX; PSA-SNN; 2D-Spiking Transformer; 2D-SDSA; analysis of where to place spiking neurons in K/Q/V sequences
Input Encoding Method	Direct static-image input with single-timestep spike processing; input is processed through SNN-based YOLOX backbone and LIF-based spike layers
Encoding Parameters	Timestep T = 1; 2D-SDSA scale factor = 0.125; additional spike layer applied only to V sequence after RepConv
Encoding Learned or Fixed	Partially learned: convolutional/Transformer weights are learned; spike-generation mechanism and single-timestep setting are fixed
Event / Spike Representation	Binary spike activations; spike-form Value sequence; spike-driven K/Q/V feature representations; sparse spike-based feature maps
Temporal Resolution	Single timestep, T = 1
Output Decoding Method	YOLOX-style decoupled detection head outputs object confidence, class label, bounding-box center coordinates, width, and height
Encoding / Temporal Information Analysis	Yes, partially. The paper analyzes mutual information and perturbation effects of applying spikes to K, Q, and V sequences, and shows through ablation that applying the extra spike layer only to V gives the best performance
Training Paradigm	Supervised direct training

Training Method	End-to-end direct training using surrogate-gradient-based SNN training
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Binary Cross Entropy loss for classification/object detection branches; IoU loss for regression branch
Training Hyperparameters	300 epochs on MS-COCO 2017; initial learning rate depends on model scale; step-wise learning-rate degradation; Mosaic augmentation during first 50% of epochs; MixUP used in the last 50% of epochs with 50% probability
Primary Performance Metric	mAP / object detection accuracy on COCO2017val
Primary Metric Value	Advanced SpikingYOLOX-Nano: mAP 45.00; Tiny: 55.41; S: 59.21; M: 62.89; L: 64.93 on COCO2017val
Secondary Metrics	AP@50:95, AP@50:75, AP-small, AP-medium, AP-large, GFLOPs, firing rate, energy, latency, FPS
Baseline Models Compared	SpikingYOLOX, Spiking-YOLO, Bayesian Optimization SNN detector, Spike Calibration, EMS-YOLO, SpikeYOLO, Salience DETR, YOLOv10, YOLOv11
Performance Gain Over Baseline	Advanced SpikingYOLOX improves over SpikingYOLOX at all scales: Nano 45.00 vs 6.62 mAP; Tiny 55.41 vs 54.28; S 59.21 vs 57.20; M 62.89 vs 61.76; L 64.93 vs 62.14. It also outperforms prior directly trained SNN detectors, including SpikingYOLOX-L 59.6 and SpikeYOLO T=1,D=4 64.1
Statistical Validation	Benchmark comparison, qualitative detection examples, and ablation studies reported; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes consecutive vs non-consecutive spiking neurons, PSA-SNN vs vanilla PSA vs no PSA-SNN, and K/Q/V spike-placement settings in the 2D-Spiking Transformer
Training Framework / Code Availability	PyTorch and SpikingJelly; code availability NR
Target Hardware	Neuromorphic hardware and real-time edge deployment discussed conceptually; experiments run on GPU
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Energy estimated/reported from GFLOPs and firing rate comparison; not measured on neuromorphic hardware
Latency / Memory / Complexity Evidence	Latency and FPS tested on a single NVIDIA RTX 4070, but exact values are shown only in figure form and not fully reported in text. Energy table reports: Nano 3.27 GFLOPs, firing rate 0.3762, 1.05E–05 J; Tiny 16.53 GFLOPs, 5.97E–05 J; S 28.98 GFLOPs, 1.06E–04 J; M 78.49 GFLOPs, 2.96E–04 J; L 163.89 GFLOPs, 6.38E–04 J
Comparison with ANN Baseline	Yes. Compares against ANN object detectors including Salience DETR, YOLOv10, and YOLOv11
Main Contribution / Key Limitation	Contribution: proposes Advanced SpikingYOLOX with PSA-SNN and 2D-Spiking Transformer to improve SNN-based object detection under single-timestep direct training. Limitation: evaluated mainly on static COCO object detection; no real neuromorphic hardware deployment; energy values are not measured on neuromorphic chips; parameter counts are not reported
RQ Mapping	Primary: RQ1 and RQ2; Secondary: RQ3, RQ4, RQ5

P105

Decoding Listener's Identity: Person Identification from EEG Signals Using a Lightweight Spiking Transformer

Full Citation	Lin, Z., Cai, S., & Li, H. (2025). Decoding Listener's Identity: Person Identification from EEG Signals Using a Lightweight Spiking Transformer. In Proceedings of Interspeech 2025, Rotterdam, The Netherlands, pp. 5548–5552. https://doi.org/10.21437/Interspeech.2025-1637
University	The Chinese University of Hong Kong
Country	China
Publication Year	2025
Publication Venue	Interspeech 2025
Publisher	International Speech Communication Association
DOI / URL	https://doi.org/10.21437/Interspeech.2025-1637
Publication Type	Conference paper
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	EEG-based person identification / listener identity classification
Data Modality	EEG signals recorded during music listening
Signal Dimensionality	1D multichannel temporal signal; EEG window represented as E ∊ R^(C×T), where C is number of EEG channels and T is samples within a window
Signal Characteristics	32-channel EEG recorded with EPOC Flex saline sensors; sampling rate 128 Hz; 10-second EEG windows with 1280 time points; signals recorded during personal and non-personal music-listening trials
Dataset Name	EEG-Music Emotion Recognition Challenge dataset
Dataset Type	Public EEG benchmark dataset
Dataset Size	26 subjects; each subject participated in 16 trials of 90 seconds; training dataset contains 294 trials; test dataset contains 104 held-out trials plus 44 additional trials
Dataset Balance / Split	10% validation set extracted from training data; test set includes held-out trials from the same 26 subjects; class-count balancing used in CrossEntropyLoss
Hybrid Topology	Lightweight Spiking Transformer with two Conv-based SNN stages followed by two Transformer-based SNN stages; combines convolutional SNN blocks and spike-driven Transformer blocks
SNN Component	LIF neuron layers, Conv-based SNN blocks, separable convolution block, channel convolution block, spike-driven self-attention, spike-form Q/K/V
Neuron Model	LIF
Transformer Component	Lightweight Spiking Transformer; Spike-Driven Self-Attention block; Transformer-based SNN block; MLP block
Attention Type	Spike-Driven Self-Attention, SDSA; softmax-free spike-form attention
SNN–Transformer Interface Mechanism	EEG features are first processed by Conv-based SNN blocks; Transformer-based SNN blocks then form spike tensors Qs, Ks, and Vs using LIF neurons and RepConv; SDSA computes Qs(Ks■Vs) using spike-form additions rather than dense dot-product attention

Model Depth / Scale	Four-stage model: first two stages are Conv-based SNN blocks and last two stages are Transformer-based SNN blocks; MLP expansion ratio r = 4
Parameter Count	3.91M parameters
Design / Contribution Type	Lightweight Spiking Transformer for EEG person identification; parameter-reduced adaptation of Spike-driven Transformer V2 / Spikformer-style architecture for EEG signals
Input Encoding Method	Direct EEG-window input processed through LIF-based spiking layers; no handcrafted EEG features required
Encoding Parameters	EEG window length 10 seconds; 1280 samples per window; 32 channels; sampling rate 128 Hz; LIF leakage factor β and firing threshold Vth are defined but exact numerical values NR
Encoding Learned or Fixed	Partially learned: convolutional, RepConv, MLP, and classifier weights are learned; LIF spike-generation dynamics are fixed/defined by neuron model
Event / Spike Representation	Spike tensors; spike-form Q, K, and V; binary spike activations from LIF neurons
Temporal Resolution	EEG sampled at 128 Hz; 10-second windows; 1280 time points per window
Output Decoding Method	Refined feature map is passed to a linear classification layer for multi-class person identity prediction
Encoding / Temporal Information Analysis	Partial. The paper motivates SNNs for EEG temporal dynamics and reports low firing rate, but does not deeply analyze spike information loss or temporal encoding quality
Training Paradigm	Supervised learning
Training Method	Surrogate-gradient-based direct SNN training
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	CrossEntropyLoss with class-count balancing
Training Hyperparameters	300 epochs; Adam optimizer; ReduceLROnPlateau learning-rate scheduler; 10% validation split from training data; class-balanced loss
Primary Performance Metric	Person identification accuracy
Primary Metric Value	100% classification accuracy
Secondary Metrics	Parameter count, energy consumption, firing rate
Baseline Models Compared	Compact Convolutional Transformer, reduced-size CCT, Spike-driven Transformer V2
Performance Gain Over Baseline	Achieves the same 100% accuracy as Spike-driven Transformer V2 while reducing parameters from 30.9M to 3.91M and energy from 6065.1 µJ to 760.7 µJ; compared with reduced CCT, improves accuracy from 98.65% to 100% while using much less energy
Statistical Validation	Benchmark comparison and ablation study reported; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Partial. Includes parameter-reduction/downsampling strategy and effect of reducing Conv-based SNN blocks from two to one
Training Framework / Code Availability	Source code available at GitHub: PatrickZLin/Decode-Listener-Identity; exact framework NR
Target Hardware	Energy-constrained BCI systems, wearable EEG devices, and edge devices discussed conceptually
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Analytical energy estimate based on 45 nm CMOS process; reported inference energy is 760.7 µJ; average firing rate is 7.18%
Latency / Memory / Complexity Evidence	3.91M parameters; 760.7 µJ inference energy; average firing rate 7.18%; direct latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compared with CCT and reduced-size CCT ANN baselines
Main Contribution / Key Limitation	Contribution: proposes a lightweight Spiking Transformer for EEG-based person identification that reaches 100% accuracy with much lower energy than ANN and larger SNN baselines. Limitation: evaluated on a relatively small 26-subject dataset; generalization to larger and more diverse populations is uncertain; no real neuromorphic hardware test
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P106

Spike-HAR++: an energy-efficient and lightweight parallel spiking transformer for event-based human action recognition

Full Citation	Lin, X., Liu, M., & Chen, H. (2024). Spike-HAR++: an energy-efficient and lightweight parallel spiking transformer for event-based human action recognition. Frontiers in Computational Neuroscience, 18, 1508297. https://doi.org/10.3389/fncom.2024.1508297
University	Tsinghua University
Country	China
Publication Year	2024
Publication Venue	Frontiers in Computational Neuroscience
Publisher	Frontiers
DOI / URL	https://doi.org/10.3389/fncom.2024.1508297
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Event-based human action recognition and sign/sign-language classification from DVS event streams
Data Modality	Event-camera / DVS data
Signal Dimensionality	2D spatiotemporal event-frame sequences represented as T × 2 × H × W, where 2 denotes event polarity channels
Signal Characteristics	Sparse asynchronous event streams converted into frame-based DVS sequences; SL-Animals-DVS and DVS128 Gesture use 128 × 128 DVS data; DailyAction-DVS uses 346 × 260 DAVIS event data
Dataset Name	SL-Animals-DVS, N-LSA64, DVS128 Gesture, DailyAction-DVS
Dataset Type	Public event-based human action recognition / gesture / sign-language datasets
Dataset Size	SL-Animals-DVS: 59 individuals and 19 signs; N-LSA64: 3,200 DVS videos, 10 subjects, 64 signs, 5 repetitions; DVS128 Gesture: 1,342 recordings, 29 subjects, 11 actions; DailyAction-DVS: 1,440 recordings, 15 subjects, 12 actions

Dataset Balance / Split	N-LSA64 split into train/validation/test at 6:2:2; SL-Animals-DVS evaluated under 4-set and 3-set settings; other exact train/test split details NR
Hybrid Topology	Spike-HAR family uses SNN-based patch embedding, parallel spike-driven Transformer blocks, spike attention branch, and SNN-based classification head; Spike-HAR++ modifies the attention branch to use higher-dimensional PE features
SNN Component	LIF neurons, SNN patch embedding block, spike attention branch, spike firing-rate map, spike tokens, SNN-based classification head
Neuron Model	LIF
Transformer Component	Parallel spike-driven Transformer block; Conv-based Simplified Spiking Self-Attention, CB-S3A; MLP block
Attention Type	Conv-based simplified spiking self-attention; spike attention mask; spike firing-rate attention branch
SNN–Transformer Interface Mechanism	Event frames are converted into spike tokens by the PE block; CB-S3A extracts spike-form Q and K using Conv2D and spike neuron layers, while V is obtained directly; spike attention maps perform mask operations on MLP features to guide local feature focus
Model Depth / Scale	L = 2 parallel spike-driven Transformer blocks; PE block has 4 Conv2D layers, 3 BN layers, 3 SNN layers, and 2 max-pooling layers; only two MLP layers are used
Parameter Count	Spike-HAR: 0.70M; Spike-HAR++: 1.80M
Design / Contribution Type	New lightweight Spiking Transformer family for event-based HAR; parallel spike-driven Transformer block; CB-S3A; spike attention branch; high-dimensional attention branch in Spike-HAR++
Input Encoding Method	Frame-based event representation; continuous event frames are randomly selected, cropped, linearly sampled if longer than T, or zero-padded if shorter than T
Encoding Parameters	SL-Animals-DVS and N-LSA64: sample length 500 ms, T = 10, learning rate 1e–4; DVS128 Gesture: sample length 6000 ms, T = 20, learning rate 1e–3; DailyAction-DVS: sample length 1200 ms, T = 10, learning rate 1e–3
Encoding Learned or Fixed	Fixed event-frame preprocessing with learned PE, Transformer, attention, and classifier parameters
Event / Spike Representation	DVS event frames, binary spike tensors, spike tokens, spike firing-rate maps, attention masks
Temporal Resolution	T = 10 for SL-Animals-DVS, N-LSA64, and DailyAction-DVS; T = 20 for DVS128 Gesture
Output Decoding Method	SNN-based classification head maps final spike features to action/sign class labels
Encoding / Temporal Information Analysis	Yes. The paper analyzes event-frame timestep settings, sample length, spike firing-rate maps, attention masks, and the effect of attention branch placement in Spike-HAR vs Spike-HAR++
Training Paradigm	Supervised direct SNN training
Training Method	End-to-end training using SpikingJelly SNN modules; direct training of LIF-based Spiking Transformer
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	Batch size 32; 240 epochs; one-cycle learning-rate policy; trained on a single NVIDIA GeForce RTX 3090; spatial and temporal random crop; each training sample repeated twice with different augmentations
Primary Performance Metric	Classification accuracy
Primary Metric Value	Spike-HAR++: 91.93% on SL-Animals-DVS-4sets, 92.82% on SL-Animals-DVS-3sets, 85.78% on N-LSA64-Both, 88.33% on N-LSA64-Right, 97.92% on DVS128 Gesture, and 98.47% on DailyAction-DVS
Secondary Metrics	Model parameters, FLOPs/SOPs, energy consumption, timestep, sample length
Baseline Models Compared	EVT, Spike-driven EVT, TORE + GoogLeNet, TORE + ResNet18, VoxelGrid + ResNet18, SITS + ResNet18, VK-SITS + ResNet18, SLAYER, STBP, DECOLLE, SEW ResNet18, EventDrop + SEW ResNet18, NDA + SEW ResNet18, EventRPG + SEW ResNet18, PLIF-SNN, Res-SNN-18, ASA-SNN, Motion-SNN
Performance Gain Over Baseline	On SL-Animals-DVS, Spike-HAR++ improves over EVT by 3.81% on 4sets and 5.37% on 3sets. On N-LSA64, Spike-HAR++ improves over EVT by 1.72% on Both and over Spike-driven EVT by 5.71% on Right. Energy is reduced by 99.27% for Spike-HAR and 98.55% for Spike-HAR++ compared with EVT
Statistical Validation	Benchmark comparison, ablation study, and visualization are reported; no standard deviation, confidence interval, repeated-run analysis, or statistical significance test reported
Ablation Study	Full. Includes timestep analysis, number of MLP blocks, attention branch, serial vs parallel Transformer block, and architecture comparison between Spike-HAR and Spike-HAR++
Training Framework / Code Availability	SpikingJelly; code stated as available at Spike-HAR++
Target Hardware	Neuromorphic / edge HAR devices discussed; energy comparison references ROLLS for SNNs and Intel Stratix 10 TX for ANNs
Hardware Deployment Status	No real deployment on neuromorphic hardware reported
Energy Evidence	Analytical energy estimate using 77 fJ per SOP and 12.5 pJ per FLOP; Spike-HAR consumes 0.03 mJ and Spike-HAR++ consumes 0.06 mJ per event-frame sequence
Latency / Memory / Complexity Evidence	Spike-HAR: 0.70M parameters, 0.44G SOPs, 0.03 mJ; Spike-HAR++: 1.80M parameters, 0.74G SOPs, 0.06 mJ; direct latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compared with ANN-based HAR models such as EVT, TORE + GoogLeNet, TORE + ResNet18, VoxelGrid + ResNet18, SITS + ResNet18, and VK-SITS + ResNet18
Main Contribution / Key Limitation	Contribution: proposes Spike-HAR and Spike-HAR++, lightweight parallel Spiking Transformer models for event-based HAR with strong accuracy and very low estimated energy. Limitation: energy is analytically estimated rather than measured on real neuromorphic hardware; future work requires larger event-based HAR benchmarks and practical deployment validation
RQ Mapping	Primary: RQ1; Secondary: RQ2, RQ3, RQ4, RQ5

P107

SpikingDynamicMaskFormer: Enhancing Efficiency in Spiking Neural Networks with Dynamic Masking

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University	Shanghai University
Country	China
Publication Year	2026
Publication Venue	Electronics

Publisher	MDPI
DOI / URL	https://doi.org/10.3390/electronics15010189
Publication Type	Journal article
Database Source	WoS
Paper Category	Efficiency-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Event-based classification on neuromorphic vision datasets
Data Modality	Event-camera / neuromorphic vision data
Signal Dimensionality	2D spatiotemporal event-stream data represented as spike/event frames and patch/token features
Signal Characteristics	Sparse event-driven spike streams; binary spike representations; channel-wise spike firing rates; inactive channels after SSA; timestep-based event-frame processing
Dataset Name	DVS128 Gesture, CIFAR10-DVS, N-Caltech101
Dataset Type	Public neuromorphic/event-based vision benchmark datasets
Dataset Size	DVS128 Gesture: 1342 event streams, 11 hand gestures, 29 subjects; CIFAR10-DVS: 10 classes with 10,000 samples per class; N-Caltech101: 100 object classes plus background
Dataset Balance / Split	CIFAR10-DVS uses first 9000 samples per class for training and remaining 1000 for testing; exact splits for DVS128 Gesture and N-Caltech101 NR
Hybrid Topology	Spikformer-based architecture enhanced with Dynamic Mask Encoder Blocks and Lightweight Position Embedding; consists of SPS × 4, LPE, stacked DMEB modules, channel-wise SSA, DMConv MLP modules, and classification head
SNN Component	LIF neurons, MultiStepLIFNode, binary spike activations, dynamic mask layer, DMConv, spike firing-rate analysis
Neuron Model	LIF
Transformer Component	Spiking Transformer backbone; Spikformer-style spiking self-attention; channel-wise spiking self-attention; Dynamic Mask Encoder Block
Attention Type	Channel-wise Spiking Self-Attention; spike-form Q/K/V attention; softmax-free spiking self-attention
SNN–Transformer Interface Mechanism	Q, K, and V are generated through 1 × 1 convolution, batch normalization, learnable channel-wise mask layer, and spiking neuron activation; masked spike features are then processed by channel-wise SSA and DMConv modules
Model Depth / Scale	Evaluated mainly as SDMFormer-2-256 and SDMFormer-1-512; four SPS blocks; patch size 16; 16 detection heads by default; 16 timesteps
Parameter Count	SDMFormer-CIFAR10-DVS: 1.09M; SDMFormer-DVS128: 1.04M; SDMFormer-N-Caltech101: 0.98M
Design / Contribution Type	Dynamic masking / structured channel pruning for Spiking Transformers; Dynamic Mask Encoder Block; Lightweight Position Embedding; parameter-efficient Spikformer optimization
Input Encoding Method	Direct encoding of neuromorphic event data into spike/event-frame representation; first convolution uses floating-point pixel/event-frame input before spike-based processing
Encoding Parameters	16 timesteps; patch size 16; four SPS blocks; batch size 16; model configurations SDMFormer-2-256 and SDMFormer-1-512; 45 nm CMOS energy constants: MAC = 4.6 pJ, AC = 0.9 pJ
Encoding Learned or Fixed	Partially learned: mask parameters, convolution weights, LPE tensor, and model weights are learned; LIF dynamics and binary spike-generation process are fixed
Event / Spike Representation	Binary spike tensors; spike-form Q/K/V; channel-wise masked spike features; spike firing-rate maps; inactive-channel ratio
Temporal Resolution	16 timesteps in all main experiments
Output Decoding Method	Classification head produces class prediction from final SDMFormer features
Encoding / Temporal Information Analysis	Yes. The paper analyzes inactive-channel ratios, per-channel spike firing rates, mask ratios, channel-averaged spike firing rates, and LPE heatmaps across timesteps
Training Paradigm	Supervised direct SNN training
Training Method	End-to-end surrogate-gradient training with learnable dynamic masks and SpikingJelly LIF modules
Surrogate Gradient Function	SurrogateHeaviside piecewise-polynomial surrogate for the mask layer; exact LIF neuron surrogate gradient NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	NR
Training Hyperparameters	Batch size 16; 16 timesteps; 200 epochs; AdamW optimizer; initial learning rate 1 × 10■³; experiments repeated with three independent random seeds
Primary Performance Metric	Classification accuracy, parameter reduction, energy-consumption reduction
Primary Metric Value	CIFAR10-DVS: 81.5 ± 0.1%; DVS128: 98.61 ± 0.03%; N-Caltech101: 83.54 ± 0.46%
Secondary Metrics	Parameter count, inference throughput, training throughput, training memory usage, energy ratio, mask ratio, spike firing rate
Baseline Models Compared	Spikformer, PLIF, Dspike, CML, Spikingformer, Spike-driven Transformer, Auto-Spikformer, STSA, TCJA-SNN, QKFormer, STBP-tdBN, SEW-ResNet
Performance Gain Over Baseline	Compared with Spikformer, SDMFormer reduces parameters to 37.93–42.69% and energy consumption to 42.79–50.13% of the baseline. It achieves 196.20 img/s on CIFAR10-DVS, improving inference throughput over STSA, QKFormer, and TCJA-SNN
Statistical Validation	Three independent random seeds used for accuracy reporting; mean ± variation reported for main accuracy results; no formal statistical significance test reported
Ablation Study	Full/strong. Includes mask-rate analysis, parameter-ratio comparison, energy-ratio comparison, spike firing-rate analysis, LPE visualization, inference throughput, training throughput, and training memory comparison
Training Framework / Code Availability	PyTorch 2.2.2; SpikingJelly 0.0.0.0.14; single NVIDIA GeForce RTX 2080 Ti GPU with 22 GB VRAM; code available at GitHub: KovT000/SDMFormer
Target Hardware	Resource-constrained neuromorphic / low-power SNN deployment discussed conceptually; experiments run on GPU
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Algorithm-level energy estimation using operation counts, spike activity, and 45 nm CMOS MAC/AC energy values; no physical neuromorphic chip power measurement
Latency / Memory / Complexity Evidence	Inference throughput on CIFAR10-DVS: 196.20 img/s; training throughput: 57.37 img/s; training memory usage: 8771 MiB; parameter count reduced to around 1M depending on dataset
Comparison with ANN Baseline	No direct ANN baseline in main experiments; comparison focuses on SNN and Spiking Transformer baselines

Main Contribution / Key Limitation	Contribution: proposes SDMFormer, which reduces Spikformer redundancy through learnable channel-wise dynamic masking and lightweight position embedding while maintaining competitive neuromorphic classification accuracy. Limitation: energy results are operation-level estimates rather than hardware measurements; quantitative evaluation is limited to neuromorphic datasets; mask-learning stability in deeper layers is identified as future work
RQ Mapping	Primary: RQ5; Secondary: RQ2, RQ3, RQ4, RQ1

P108

Multi-Timescale Motion-Decoupled Spiking Transformer for Audio-Visual Zero-Shot Learning

Full Citation	Li, W., Wang, P., Wang, X., Zuo, W., Fan, X., & Tian, Y. (2025). Multi-Timescale Motion-Decoupled Spiking Transformer for Audio-Visual Zero-Shot Learning. IEEE Transactions on Circuits and Systems for Video Technology, 35(11), 10772–10786. https://doi.org/10.1109/TCSVT.2025.3574499
University	Harbin Institute of Technology
Country	China
Publication Year	2025
Publication Venue	IEEE Transactions on Circuits and Systems for Video Technology
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/TCSVT.2025.3574499
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Speech, Audio, and Auditory Processing
Specific Task	Audio-visual zero-shot learning and generalized zero-shot learning for video classification
Data Modality	RGB video frames, audio clips, synthetic event streams, and textual class-label embeddings
Signal Dimensionality	Multimodal temporal data: visual frame sequences, audio feature sequences, sparse event streams, and text embeddings
Signal Characteristics	RGB frames are converted into synthetic event streams to emphasize motion and suppress background scene bias; events are represented as (x, y, t, p) with polarity p ∈ {-1, 1}; audio and visual embeddings are projected into shared semantic spaces
Dataset Name	VGGSound-GZSL, UCF-GZSL, ActivityNet-GZSL; also VGGSound-GZSLcls, UCF-GZSLcls, ActivityNet-GZSLcls
Dataset Type	Public audio-visual video classification and zero-shot learning benchmark datasets
Dataset Size	ActivityNet: 200 categories, over 20,000 clips, more than 849 hours; UCF101: 101 action classes and 13,320 video clips; VGGSound: over 200,000 10-second clips across 309 sound-event categories
Dataset Balance / Split	Uses seen/unseen class splits for ZSL and GZSL; adjusted cls splits are used to avoid overlap with YouTube-8M classes; exact per-class split counts are shown in the paper but not fully detailed in text
Hybrid Topology	Dual-stream MDST++ architecture decoupling contextual semantic information and motion information; uses RJLU for semantic modeling, EGM + SNN/SpikeFormer for motion modeling, DAB for audio motion, dynamic threshold block, CRM fusion, and projection/reconstruction layers
SNN Component	LIF neurons, SNN motion branch, event-based motion processing, dynamic threshold adjustment, SpikeFormer, multi-stage timestep shrinkage
Neuron Model	LIF
Transformer Component	Spiking Transformer / SpikeFormer, self-attention, cross-modal transformer, Recurrent Joint Learning Unit, Cross-Modal Reasoning Module
Attention Type	Spiking self-attention, self-attention, cross-attention, multi-head cross-attention
SNN–Transformer Interface Mechanism	RGB frames are converted into sparse synthetic events by EGM; event streams are processed by SNN/SpikeFormer to extract motion features; semantic features from RJLU and motion features from SNN are fused through CRM and cross-modal Transformer layers
Model Depth / Scale	MDST uses a three-layer spiking MLP; MDST++ replaces this with SpikeFormer and divides SNN processing into m stages with decreasing timesteps T1 > T2 > ... > Tm; cross-modal Transformer uses 8 heads with head dimension 64
Parameter Count	MDST++: 6.21M parameters on UCF-GZSL; MDST: 5.51M parameters
Design / Contribution Type	New multimodal Spiking Transformer framework for audio-visual ZSL; Event Generation Model; motion-decoupled dual-stream learning; Recurrent Joint Learning Unit; Discrepancy Analysis Block; dynamic LIF threshold; multi-stage timestep shrinkage
Input Encoding Method	RGB-to-event conversion using EGM; audio and visual feature extraction using pretrained encoders; audio motion modeled through discrepancy between visual events and audio features
Encoding Parameters	Event defined as (xk, yk, tk, pk); event generated when logarithmic brightness change exceeds threshold C; model settings include Tin = 512, Thid = 512, Tproj = 64, Tfin = 300; dropout rates vary by dataset; average timestep analysis uses values from 5 to 12
Encoding Learned or Fixed	Partially learned: EGM event thresholding is rule-based, while audio/visual encoders, DAB parameter βa, SNN/SpikeFormer parameters, dynamic thresholds, and fusion layers are learned/adaptively updated
Event / Spike Representation	Synthetic event streams, sparse event tensors, LIF spikes, spike-form Q/K/V representations, timestep-dependent spike features
Temporal Resolution	Multi-stage timestep shrinkage uses decreasing timesteps across stages; exact final timestep schedule NR
Output Decoding Method	Audio-visual joint embeddings are projected into textual class-label embedding space and reconstructed; ZSL/GZSL prediction is made by matching projected audio-visual features to textual class embeddings
Encoding / Temporal Information Analysis	Yes. The paper analyzes event generation thresholds, dynamic LIF thresholds, average timestep effects, multi-stage timestep shrinkage, motion-semantic decoupling, and qualitative event-based motion visualization
Training Paradigm	Supervised training for audio-visual ZSL/GZSL using seen-class training data and textual label embeddings
Training Method	End-to-end training of MDST/MDST++ with event generation, RJLU semantic learning, SNN/SpikeFormer motion learning, cross-modal reasoning, projection, and reconstruction
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	Adam
Loss Function	Total loss Lall = Ln + Lp + Lr, combining joint triplet loss, projection loss, and reconstruction loss
Training Hyperparameters	Trained for 50 epochs; learning rate 0.0001; single NVIDIA V100S GPU; dataset-specific dropout rates: denc = 0.20/0.25/0.10, ddec = 0.25/0.20/0.15, dwproj = 0.1/0.1/0.1 for VGGSound, UCF, and ActivityNet respectively

Primary Performance Metric	GZSL harmonic mean and ZSL accuracy
Primary Metric Value	MDST++ achieves VGGSound-GZSL: HM 9.25, ZSL 8.48; UCF-GZSL: HM 33.38, ZSL 33.81; ActivityNet-GZSL: HM 14.07, ZSL 14.12
Secondary Metrics	Seen-class accuracy, unseen-class accuracy, parameter count, GFLOPs, ablation HM/ZSL, average timestep performance
Baseline Models Compared	SJE, DEVISE, APN, VAEGAN, CJME, AVGZSLNet, AVCA, TCaF, Hyper-alignment, AVMST, MDST
Performance Gain Over Baseline	On UCF-GZSL, MDST++ improves over MDST from HM 31.36 to 33.38 and ZSL 31.53 to 33.81. Compared with strong baselines, MDST++ achieves the best HM/ZSL on VGGSound-GZSL, UCF-GZSL, and ActivityNet-GZSL. The paper also states that incorporating motion and multi-timescale information improves HM and ZSL accuracy by 26.2% and 39.9%
Statistical Validation	Benchmark comparison and extensive ablation studies reported; no standard deviation, confidence interval, repeated-run analysis, or formal statistical significance test reported
Ablation Study	Full. Includes SNN vs MLP for event data, unimodal vs multimodal input, removal of RJLU/MIM/CRM, loss component analysis, dynamic threshold, EGM threshold, SpikeFormer vs SpikingMLP, timestep shrinkage, average timestep, parameter analysis, and qualitative retrieval visualization
Training Framework / Code Availability	Training performed on NVIDIA V100S GPU; exact software framework and code availability NR
Target Hardware	Neuromorphic hardware suitability discussed conceptually, including possible deployment on Intel Loihi or IBM TrueNorth
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No direct measured energy consumption reported; efficiency is argued through event-driven sparse SNN computation, timestep shrinkage, parameter count, and GFLOPs
Latency / Memory / Complexity Evidence	MDST++ has 6.21M parameters and 7.14 GFLOPs on UCF-GZSL; MDST has 5.51M parameters and 5.62 GFLOPs; direct latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compared against non-spiking audio-visual ZSL methods such as SJE, DEVISE, APN, VAEGAN, CJME, AVGZSLNet, AVCA, TCaF, Hyper-alignment, and AVMST
Main Contribution / Key Limitation	Contribution: proposes MDST++, a dual-stream audio-visual ZSL framework that decouples semantic context from motion using synthetic events, SNN/SpikeFormer motion modeling, dynamic LIF thresholds, and multi-stage timestep shrinkage. Limitation: no physical neuromorphic hardware validation; energy evidence is not directly measured; model depends on pretrained audio/visual feature extractors and synthetic event generation
RQ Mapping	Primary: RQ1 and RQ2; Secondary: RQ3, RQ4, RQ5

P109

ECA-ATCNet: Efficient EEG Decoding with Spike Integrated Transformer Conversion for BCI Applications

Full Citation	Li, X., Shen, Q., Wang, H., & Wang, Z. (2025). ECA-ATCNet: Efficient EEG Decoding with Spike Integrated Transformer Conversion for BCI Applications. Applied Sciences, 15(4), 1894. https://doi.org/10.3390/app15041894
University	Donghua University
Country	China
Publication Year	2025
Publication Venue	Applied Sciences
Publisher	MDPI
DOI / URL	https://doi.org/10.3390/app15041894
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Biomedical and Healthcare Signal Processing
Specific Task	Motor imagery EEG classification; ANN-to-SNN conversion of Transformer self-attention for low-energy inference
Data Modality	Multichannel EEG time-series; additional image datasets used for SIT-conversion generalization experiments
Signal Dimensionality	1D multichannel temporal EEG signal; 2D RGB images for auxiliary conversion validation
Signal Characteristics	BCI-2a EEG: 22 channels, 4 s trials, 250 Hz sampling rate, 0.5–100 Hz filtering, 4 MI classes. PhysioNet EEG: 64-channel EEG, 18 motor-cortex electrodes used, 4 s trials, 160 Hz, 640 time points. Image experiments use 224 × 224 inputs
Dataset Name	BCI Competition IV-2a, PhysioNet MI-EEG, CIFAR10, CIFAR100, ImageNet-1K
Dataset Type	Public EEG motor imagery benchmark datasets; public image classification datasets for conversion generalization
Dataset Size	BCI-2a: 5184 MI-EEG trials, 9 subjects, 576 trials per subject. PhysioNet MI-EEG: 109 subjects, 84 trials per subject. CIFAR10: 60,000 images. CIFAR100: 60,000 images. ImageNet-1K: 1.2M training images and 50,000 validation images
Dataset Balance / Split	BCI-2a: two sessions per subject, one session for training and one for evaluation. PhysioNet: 11-fold cross-validation; 100 subjects for training and 10 for evaluation per fold. CIFAR10/100 use standard train/test splits. ImageNet-1K uses standard train/validation split
Hybrid Topology	ECA-ATCNet combines ECA-Conv, sliding-window segmentation, multi-head attention, temporal convolution, and FC Softmax classifier; SIT-conversion converts Transformer self-attention into a Spike Integrated Transformer using spiking Q/K/V and Spiking–Softmax
SNN Component	Spike Integrated Transformer, spiking neurons for Q/K/V, SI-exp neuron, SI-norm neuron, Spiking–Softmax, spike sequences, ANN-to-SNN conversion
Neuron Model	SI-exp neuron, SI-norm neuron, spiking neurons for Q/K/V; IF/LIF-style neuron details for Q/K/V are NR
Transformer Component	Multi-head self-attention in ATCNet/ECA-ATCNet; Spike Integrated Transformer self-attention; EdgeNeXt, Next-ViT, and UniFormer used for generalization experiments
Attention Type	Efficient Channel Attention, multi-head self-attention, scaled dot-product attention, Spiking–Softmax-based self-attention
SNN–Transformer Interface Mechanism	ANN Transformer self-attention is converted by replacing Q/K/V computation with spiking neurons and replacing Softmax with Spiking–Softmax; SI-exp approximates exponential operation and SI-norm approximates normalization
Model Depth / Scale	ECA-ATCNet has ECA-Conv block, sliding-window block, multiple parallel ATC blocks, and FC classifier. SIT-conversion uses 12 time steps. Generalization models: EdgeNeXt has 3 Transformer blocks, Next-ViT has 4, and UniFormer has 10
Parameter Count	ECA-ATCNet: 0.44 Mb parameters. EdgeNeXt: 5.59M; Next-ViT: 31.76M; UniFormer: 10.21M
Design / Contribution Type	New EEG decoding architecture ECA-ATCNet; Efficient Channel Attention Convolution for spatial/spectral feature selection; Spike Integrated Transformer conversion; Spiking–Softmax using SI-exp and SI-norm neurons

Input Encoding Method	EEG input is normalized multichannel raw time series; ECA-Conv extracts temporal, spatial, and spectral features. For SIT-conversion, ANN activations/self-attention are converted into spike-based representations using spiking neurons and rate/spike-sequence style encoding
Encoding Parameters	SIT-conversion uses 12 time steps; SI-exp internal parameters trained on 10,000 values in range [−1, 1]; SI-exp RMSE for exponential approximation = 0.0015413; Spiking–Softmax MSE at K = 12 is 0.013991
Encoding Learned or Fixed	Partially learned: ECA-ATCNet weights and attention modules are trained; SI-exp internal parameters are optimized/pretrained; ANN-to-SNN conversion rules and Spiking–Softmax mechanism are fixed after parameter optimization
Event / Spike Representation	Spike sequences, spike counts, SI-exp spike outputs, SI-norm normalized spike outputs, spiking Q/K/V, spike-based Softmax approximation
Temporal Resolution	EEG trials are 4 s; BCI-2a sampled at 250 Hz; PhysioNet sampled at 160 Hz. SIT-conversion uses 12 SNN time steps
Output Decoding Method	ECA-ATCNet uses FC layer with Softmax for class probability output; converted SNN uses Spike Integrated Transformer output after Spiking–Softmax-based self-attention for classification
Encoding / Temporal Information Analysis	Yes. The paper analyzes Softmax approximation across different SNN time steps, SI-exp/SI-norm behavior, time-step–accuracy trade-off, EEG temporal features, and energy reduction from spike-based inference
Training Paradigm	Supervised learning for ECA-ATCNet; ANN-to-SNN conversion for SIT-converted models
Training Method	ECA-ATCNet is trained as an ANN, then self-attention is converted using SIT-conversion; image-domain Transformer variants are fine-tuned as ANNs and converted to SNNs
Surrogate Gradient Function	N/A for main SIT-conversion method; surrogate gradient NR for any directly trained SNN comparisons
ANN-to-SNN Conversion Method	SIT-conversion based on Spiking–Softmax; converts Transformer self-attention into SNN form using SI-exp and SI-norm neurons
Optimizer	Adam for ECA-ATCNet; AdamW for EdgeNeXt, Next-ViT, and UniFormer experiments
Loss Function	Categorical cross-entropy for ECA-ATCNet; image experiments use standard classification training loss, exact loss NR
Training Hyperparameters	ECA-ATCNet: Glorot uniform initialization, Adam optimizer, learning rate 0.0009, batch size 64, 1000 epochs, early stopping patience 300, TensorFlow, two NVIDIA RTX 4090 GPUs. Image experiments: batch size 256, AdamW, cosine learning rate schedule; EdgeNeXt fine-tuned 100 epochs; Next-ViT 20 epochs on ImageNet and 100 on CIFAR; UniFormer 25 epochs on ImageNet and 100 on CIFAR
Primary Performance Metric	EEG classification accuracy; κ-score; conversion loss; energy reduction
Primary Metric Value	ECA-ATCNet ANN: 87.89% on BCI-2a and 71.88% on PhysioNet. SIT-converted ECA-ATCNet: 87.29% on BCI-2a and 71.88% on PhysioNet
Secondary Metrics	Time steps, theoretical energy consumption, energy reduction, FLOPs, conversion loss, standard deviation, κ-score
Baseline Models Compared	ATCNet, EEG-TCNet, D-ATCNet, CSANet, ConTraNet, EEGNet, HR-SNN, LeNet, EdgeNeXt, Next-ViT, UniFormer, MST, SNM, QCFS, Spikformer, Spikingformer
Performance Gain Over Baseline	On BCI-2a, ECA-ATCNet improves over ATCNet by about 2.5% and over other EEG baselines by 0.8%–4.1%. SIT-converted ECA-ATCNet reduces energy by 53.52% on BCI-2a and 52.84% on PhysioNet with only 0.60%–0.73% accuracy loss
Statistical Validation	Reports standard deviation across BCI-2a subjects and notes statistical significance test result $p > 0.05$; no strong significant difference reported
Ablation Study	Full. Includes ECA attention ablation, efficient spatial vs spectral attention, SIT-conversion time-step analysis, Softmax approximation loss, model conversion comparison, and energy-efficiency comparison
Training Framework / Code Availability	TensorFlow for ECA-ATCNet; two NVIDIA RTX 4090 GPUs; code availability NR
Target Hardware	Low-power portable BCI devices and neuromorphic hardware are discussed conceptually
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation using FLOPs, spike activity, and 45 nm CMOS operation energy values: ANN MAC = 4.6 pJ and SNN AC = 0.9 pJ
Latency / Memory / Complexity Evidence	SIT-conversion uses only 12 time steps; ECA-ATCNet has 0.44 Mb parameters; energy reduction of 52.84%–53.52% for EEG models and 18.18%–45.13% for image Transformer models; direct measured latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares ANN ECA-ATCNet with SIT-converted SNN version and compares converted SNN Transformer variants against their ANN counterparts
Main Contribution / Key Limitation	Contribution: proposes ECA-ATCNet for accurate MI-EEG decoding and SIT-conversion with Spiking–Softmax for low-latency near-lossless Transformer ANN-to-SNN conversion. Limitation: energy results are theoretical rather than measured on neuromorphic hardware; public MI-EEG datasets are small; statistical significance is not strong; hardware deployment remains future work
RQ Mapping	Primary: RQ1 and RQ4; Secondary: RQ2, RQ3, RQ5

P110

Decision SpikeFormer: Spike-Driven Transformer for Decision Making

Full Citation	Huang, W., Gu, Q., & Ye, N. (2025). Decision SpikeFormer: Spike-Driven Transformer for Decision Making. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025), pp. 19241–19250. https://doi.org/10.1109/CVPR52734.2025.01792
University	Shanghai AI Laboratory
Country	China
Publication Year	2025
Publication Venue	IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/CVPR52734.2025.01792
Publication Type	Conference paper
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Offline RL policy learning through sequence modeling and next-action prediction
Data Modality	State, action, reward/return-to-go trajectory sequences from offline RL datasets
Signal Dimensionality	1D sequential decision trajectories with state, action, and return-to-go tokens

Signal Characteristics	Input sequence contains historical action, return-to-go, and state tuples; sequence is embedded, repeated across SNN timesteps, and processed autoregressively to predict the next action
Dataset Name	D4RL MuJoCo, D4RL Adroit, AntMaze, D4RL-Hopper sparse-reward variants
Dataset Type	Public offline reinforcement learning benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Uses standard D4RL offline datasets and evaluation protocol; mean and variance reported over five random seeds
Hybrid Topology	Decision Transformer-style spike-driven autoregressive architecture; embedding layer, M stacked decoder blocks, spike-driven self-attention, spike-driven MLP, PTBN, membrane shortcut, and prediction head
SNN Component	LIF neurons, spike neuron layers, membrane potential dynamics, spike-driven MLP, TSSA, PSSA, PTBN, membrane shortcut
Neuron Model	LIF
Transformer Component	Decision SpikeFormer / DSFormer; Decision Transformer-style decoder blocks; temporal and positional spike-driven self-attention
Attention Type	Temporal Spiking Self-Attention, Positional Spiking Self-Attention, Step-by-Step Spiking Self-Attention; causal masking
SNN–Transformer Interface Mechanism	State–action–return sequences are embedded with fully connected layers, repeated for T SNN timesteps, transformed into spike-form Q/K/V through Linear + PTBN + spike neuron layers, then processed by TSSA or PSSA
Model Depth / Scale	M stacked decoder blocks; SNN timestep T = 4 for all tasks; PSSA local window size S = 8; exact block count/channel size NR
Parameter Count	Exact count NR; paper states DSFormer has nearly the same parameter count as Decision Transformer, with TSSA about 0.3% fewer and PSSA about 0.08% more parameters at inference
Design / Contribution Type	First spike-driven Transformer for offline RL; introduces TSSA, PSSA, and Progressive Threshold-dependent Batch Normalization for sequence decision-making
Input Encoding Method	Offline RL trajectory tokens are concatenated as action, return-to-go, and state tuples; embedded into token vectors and repeated across SNN timesteps
Encoding Parameters	SNN timestep T = 4; context length N; channel dimension D; PSSA local window size S = 8; exact context length and hidden dimension NR
Encoding Learned or Fixed	Partially learned: embedding, Q/K/V projections, positional bias, decoder blocks, MLP, and prediction head are learned; spike threshold dynamics and timestep repetition are fixed design mechanisms
Event / Spike Representation	Binary spikes, spike-form Q/K/V matrices, membrane potentials, timestep-concatenated spike representations
Temporal Resolution	4 SNN timesteps; offline RL sequence lengths also evaluated from 50 to 200 in AntMaze analysis
Output Decoding Method	Final decoder output is normalized along the temporal dimension and passed to a prediction head to generate the next action
Encoding / Temporal Information Analysis	Yes. The paper analyzes temporal concatenation, entropy reduction in TSSA, local positional dependency in PSSA, long-range temporal dependency on AntMaze, and sparse-reward behavior on Hopper
Training Paradigm	Offline reinforcement learning as supervised conditional sequence modeling
Training Method	Direct training of spike-driven Transformer policy on offline datasets; no environment interaction during training
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	Action prediction / sequence-modeling loss implied by Decision Transformer framework; exact loss formula NR
Training Hyperparameters	T = 4 SNN timesteps; five random seeds; PSSA local window size S = 8; other optimizer, learning rate, batch size, context length, and training epoch details NR in main paper
Primary Performance Metric	D4RL normalized score and energy consumption
Primary Metric Value	MuJoCo mean: TSSA 75.7, PSSA 78.8. Adroit mean: TSSA 47.9, PSSA 48.6. Reported overall energy saving: 78.4%
Secondary Metrics	Mean/variance over five seeds, task-level normalized score, power/energy in µJ, attention complexity, ablation average score
Baseline Models Compared	Behavior Cloning, CQL, Decision Transformer, FCNet, SpikeGPT, SpikeBERT, SDSA-1, SDSA-3
Performance Gain Over Baseline	On MuJoCo, PSSA improves over Decision Transformer from 74.7 to 78.8 mean score while reducing energy from 410.5 µJ to 88.8 µJ. On Adroit, PSSA improves over Decision Transformer from 27.8 to 48.6 mean score
Statistical Validation	Reports mean and variance scores from five random seeds; no formal statistical significance test reported
Ablation Study	Full. Includes attention mechanism comparison, normalization comparison, SNN timestep analysis, PSSA local-window analysis, AntMaze sequence-length analysis, and sparse-reward Hopper analysis
Training Framework / Code Availability	Code and models stated to be public at project page; exact framework NR
Target Hardware	Energy-constrained embodied AI and neuromorphic hardware are discussed conceptually
Hardware Deployment Status	No real neuromorphic chip deployment reported
Energy Evidence	Energy is estimated/computed for model comparison; no physical neuromorphic hardware power measurement reported
Latency / Memory / Complexity Evidence	TSSA complexity $O(TDN^2)$; PSSA complexity $O(STDN)$, approximated as $O(TDN)$ when S is constant; PSSA self-attention energy reported as 31.0 µJ compared with 44.6 µJ for TSSA and 168.2 µJ for Decision Transformer
Comparison with ANN Baseline	Yes. Compares directly with ANN baselines including Decision Transformer, FCNet, BC, and CQL
Main Contribution / Key Limitation	Contribution: proposes DSFormer, the first spike-driven Transformer for offline RL, with TSSA, PSSA, and PTBN to model temporal/positional dependencies efficiently. Limitation: no neuromorphic chip deployment; some implementation details are in appendix rather than main text; evaluation is limited to simulated offline RL benchmarks
RQ Mapping	Primary: RQ1 and RQ2; Secondary: RQ3, RQ4, RQ5

P111

Transformer-Based Spiking Neural Networks for Multimodal Audiovisual Classification.

Full Citation	Guo, L., Gao, Z., Qu, J., Zheng, S., Jiang, R., Lu, Y., & Qiao, H. (2024). Transformer-Based Spiking Neural Networks for Multimodal Audiovisual Classification. IEEE Transactions on Cognitive and Developmental Systems, 16(3), 1077–1086. https://doi.org/10.1109/TCDS.2023.3327081
University	Chinese Academy of Sciences
Country	China
Publication Year	2024
Publication Venue	IEEE Transactions on Cognitive and Developmental Systems
Publisher	IEEE

DOI / URL	https://doi.org/10.1109/TCDS.2023.3327081
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Robotics, Autonomous Systems, and UAV Vision
Specific Task	Audiovisual classification using image/audio modalities and spiking cross-attention fusion
Data Modality	Visual images, audio signals, event-based visual data, neuromorphic audio spike streams
Signal Dimensionality	2D image/event-frame data plus 1D audio waveform converted to 2D log-mel spectrogram features
Signal Characteristics	CIFAR10-AV samples contain 32 × 32 color images and 5-second audio; UrbanSound8K-AV samples contain color images and 4-second audio; audio is converted to log-mel spectrogram using STFT, 64 mel bins, and maximum frequency 8000 Hz
Dataset Name	CIFAR10-AV, UrbanSound8K-AV, MNIST-DVS + N-TIDIGITS
Dataset Type	Self-constructed audiovisual datasets and public event-based audiovisual benchmark dataset
Dataset Size	CIFAR10-AV: 60,000 multimodal samples; UrbanSound8K-AV: 8732 multimodal samples; MNIST-DVS + N-TIDIGITS size NR
Dataset Balance / Split	70:30 train/test split; results averaged over five independent trials
Hybrid Topology	Spiking Multimodel Transformer combining visual and audio subnetworks with Spiking Cross Attention, MLP, and classification layer
SNN Component	LIF neurons, spike neuron layers, spiking Q/K/V computation, spiking cross-attention, spike-based audiovisual fusion
Neuron Model	LIF
Transformer Component	Spiking Cross Attention module; Transformer-style cross-attention between audio and visual modalities
Attention Type	Spiking cross-attention; audio-to-visual attention; visual-to-audio attention; relative-position-bias-based attention
SNN–Transformer Interface Mechanism	Visual and audio features are transformed into Q, K, and V through Linear + BatchNorm + Dropout + LIF spike neuron layers; SCA computes cross-modal attention between audio and visual spike features and concatenates the attended outputs
Model Depth / Scale	SMMT uses visual/audio convolutional feature extractors, SCA module, MLP layer, and classifier; exact layer depth NR
Parameter Count	SMMT: 9.99M parameters
Design / Contribution Type	New Spiking Multimodel Transformer; new Spiking Cross Attention module; construction of CIFAR10-AV and UrbanSound8K-AV datasets; mobile robot validation
Input Encoding Method	Image features extracted through convolutional layers; audio waveform converted into log-mel spectrogram through STFT, mel filtering, and power-to-decibel conversion; features then converted to spike representations using LIF neuron layers
Encoding Parameters	Learning rate 1e–5; membrane time constant τm = 2.0; drop rate = 0.1; initial membrane potential = 0; threshold = 1; Sigmoid surrogate parameter α = 4; audio uses 64 mel bins and 8000 Hz maximum frequency
Encoding Learned or Fixed	Partially learned: convolutional, SCA, MLP, scaling factor, and RPB parameters are learned; log-mel preprocessing and LIF spike-generation mechanism are fixed
Event / Spike Representation	Binary spike sequences, LIF spike activations, event-based MNIST-DVS visual streams, N-TIDIGITS cochlea spike streams, spike-form Q/K/V
Temporal Resolution	Main SMMT experiments use timestep = 1 and timestep = 4; MNIST-DVS and N-TIDIGITS use event/spike temporal streams, exact temporal window NR
Output Decoding Method	Classification layer outputs final audiovisual class prediction
Encoding / Temporal Information Analysis	Partial. The paper visualizes audio and image attention maps and analyzes multimodal fusion, but does not deeply quantify spike temporal information preservation
Training Paradigm	Supervised learning
Training Method	Direct SNN training using backpropagation through time
Surrogate Gradient Function	Sigmoid surrogate function with α = 4
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	Learning rate 1e–5; batch size 4; τm = 2.0; dropout rate 0.1; initial membrane potential 0; neuron threshold 1; five independent trials; NVIDIA A100 40GB GPU
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	CIFAR10-AV: 93.53% at T = 1 and 97.01% at T = 4 in Table II; UrbanSound8K-AV: 94.30% at T = 1 and 96.85% at T = 4; MNIST-DVS + N-TIDIGITS: 99.82%
Secondary Metrics	Power/energy in mJ, timestep, parameter count, modality ablation accuracy, fusion-method ablation accuracy, mobile-robot classification accuracy
Baseline Models Compared	STBP, STBP NeuNorm, Hybrid Training, Diet-SNN, TSSL-BP, STBP-tdBN, TET, Dspike, Spikformer, Transformer-4-384, DARTS, ResNet-18, PLIF, Two-Stream CNN, Liu's multimodal SNN work, concatenation fusion, additive fusion, attention without KQV
Performance Gain Over Baseline	On UrbanSound8K-AV, SMMT reaches 96.85%, outperforming Spikformer at 94.47% and Diet-SNN at 95.3%. On MNIST-DVS + N-TIDIGITS, SMMT reaches 99.82%, improving over Liu's multimodal work at 99.10%. On CIFAR10-AV, multimodal fusion improves over image-only 92.74% and audio-only 91.10%
Statistical Validation	Results averaged over five independent trials; no standard deviation, confidence interval, or formal significance test reported
Ablation Study	Full. Includes modality ablation, dropout removal, RPB removal, dropout + RPB removal, and comparison with concatenation/addition/attention fusion methods
Training Framework / Code Availability	PyTorch; NVIDIA A100 40G GPU; constructed datasets available at the GitHub link reported in the paper
Target Hardware	Mobile robot platform and energy-constrained multimodal perception applications
Hardware Deployment Status	Deployed and tested on a mobile robot platform for two UrbanSound8K classes; no neuromorphic hardware deployment reported
Energy Evidence	Energy consumption estimated using prior operation-energy methods; Table II reports CIFAR10-AV SMMT power of 1.08 mJ at T = 1 and 3.39 mJ at T = 4; Table VIII reports UrbanSound8K-AV SMMT power of 1.08 mJ at T = 1 and 3.39 mJ at T = 4
Latency / Memory / Complexity Evidence	Reports parameter count and energy/power values; SMMT has 9.99M parameters; direct latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares against ANN baselines including Transformer-4-384, DARTS, ResNet-18, DCNN, and Two-Stream CNN

Main Contribution / Key Limitation	Contribution: proposes SMMT with Spiking Cross Attention for energy-efficient multimodal audiovisual classification and introduces CIFAR10-AV and UrbanSound8K-AV datasets. Limitation: no neuromorphic hardware measurement; energy is estimated; robot deployment is limited to two classes; dataset construction relies partly on paired data from separate sources
RQ Mapping	Primary: RQ1 and RQ2; Secondary: RQ3, RQ4, RQ5

P112
ASFNOformer: A Superior Frequency Domain Token Mixer in Spiking Transformer

Full Citation	Gao, S., Hong, Z., Gu, Y., Wu, J., Yang, Y., & Huang, R. (2025). ASFNOformer: A Superior Frequency Domain Token Mixer in Spiking Transformer. Electronics, 14(24), 4860. https://doi.org/10.3390/electronics14244860
University	Shanghai University
Country	China
Publication Year	2025
Publication Venue	Electronics
Publisher	MDPI
DOI / URL	https://doi.org/10.3390/electronics14244860
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Static and event-based image classification using frequency-domain token mixing in Spiking Transformers
Data Modality	Static image data and neuromorphic/event-camera data
Signal Dimensionality	Static 2D image data; 2D spatiotemporal event data represented with timestep dimension T, channel C, height H, and width W
Signal Characteristics	Neuromorphic datasets contain binary spikes triggered by spatiotemporal pixel-intensity changes; the method emphasizes high-frequency components and temporal dynamics in binary spike streams
Dataset Name	CIFAR-10, CIFAR10-DVS, DVS128 Gesture, N-Caltech101
Dataset Type	Public static image and neuromorphic/event-based vision benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Standard benchmark evaluation implied; exact train/test split and class balance NR
Hybrid Topology	ASFNOformer replaces Spikformer/Spiking Transformer self-attention token mixer with Adaptive Spiking Fourier Neural Operator token mixer; pipeline includes Spike Patch Splitting, Relative Position Embedding, ASFNOformer encoder blocks, MLP, Global Average Pooling, and Classification Head
SNN Component	LIF neurons, LWP-LIF neurons, spike patch splitting, spiking neuron layers, binary spike signals, spike-form patch tokens
Neuron Model	LIF and LWP-LIF
Transformer Component	Spiking Transformer / Spikformer-style encoder; QKFormer baseline; ASFNO token mixer; MLP block; relative position embedding
Attention Type	Attention-free frequency-domain token mixing using ASFNO; baseline comparisons include Spike Self-Attention and QK attention
SNN–Transformer Interface Mechanism	Input spike patches are processed through SPS and RPE, then ASFNO applies FFT/DFT across spatiotemporal dimensions, uses block-diagonal MLP channel mixing with LWP-LIF, and applies inverse FFT before residual and MLP processing
Model Depth / Scale	CIFAR-10 ASFNOformer: 9.12M parameters, 4 timesteps. Dynamic dataset ASFNOformer: 2.12M parameters, 16 timesteps. Exact block depth NR
Parameter Count	CIFAR-10: 9.12M; CIFAR10-DVS/DVS128-Gesture: 2.12M
Design / Contribution Type	New Adaptive Spiking Fourier Neural Operator token mixer; frequency-domain analysis for binary spike signals; FFT over spatial and temporal dimensions; block-diagonal weight matrix; weight sharing; LWP-LIF neuron design
Input Encoding Method	Spike Patch Splitting converts input image/event data into spike-form patch tokens; neuromorphic data are processed as binary spike/event streams; static images are repeatedly presented across timesteps
Encoding Parameters	Static CIFAR-10 uses 4 timesteps; CIFAR10-DVS and DVS128 Gesture use 16 timesteps; FFT/DFT applied over spatiotemporal dimensions; block-diagonal weight matrix reduces parameter complexity from $O(d^2)$ to $O(d^2/k)$
Encoding Learned or Fixed	Partially learned: block-diagonal MLP weights, LWP-LIF leakage parameters, RPE, and classifier are learned; FFT/IFFT transform operation is fixed
Event / Spike Representation	Binary spike signals, spike-form patches, time-series spike events, frequency-domain spike representations, LWP-LIF membrane-potential dynamics
Temporal Resolution	4 timesteps for CIFAR-10; 16 timesteps for CIFAR10-DVS and DVS128 Gesture
Output Decoding Method	Global Average Pooling followed by fully connected Classification Head
Encoding / Temporal Information Analysis	Yes. The paper provides theoretical and visual analysis of binary spike signals in the Fourier domain, time-frequency comparisons, feature-map visualization, and frequency-distribution analysis showing stronger high-frequency preservation by ASFNO
Training Paradigm	Supervised direct SNN training
Training Method	Direct training with BPTT-style SNN training; ASFNO integrated into Spikformer and QKFormer baselines
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	Python 3.10; PyTorch 2.0.1+cu118; SpikingJelly 0.0.0.0.14; timm 0.9.12; 200 training epochs for all datasets; other optimizer, learning rate, batch size, and augmentation settings NR
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	CIFAR-10: 96.04%; CIFAR10-DVS: 82.80%; DVS128 Gesture: 99.10%; QKFormer-based ASFNO on CIFAR10-DVS: 85.5%
Secondary Metrics	Parameter count, timestep count, FLOPs, accuracy improvement, frequency-domain visualization, standard deviation within 0.2%
Baseline Models Compared	TET, tdBN, TEBN, Real Spike, DSR, Spikformer, Spikingformer, CML, S-Transformer, QKFormer

Performance Gain Over Baseline	On CIFAR-10, ASFNOformer improves over Spikformer from 95.51% to 96.04% while reducing parameters from 9.32M to 9.12M. On CIFAR10-DVS, it improves over Spikformer from 80.90% to 82.80% while reducing parameters from 2.57M to 2.12M. With QKFormer baseline, ASFNO improves CIFAR10-DVS accuracy from 84.0% to 85.5%
Statistical Validation	Paper states models converged across random seeds and standard deviation of error was within 0.2%; no formal statistical significance test reported
Ablation Study	Yes. Includes ASFNO vs Spikformer on CIFAR10-DVS, ASFNO integrated into QKFormer, FLOPs comparison, feature-map visualization, and frequency-domain visualization
Training Framework / Code Availability	PyTorch, SpikingJelly, timm; paper states code/data are not open-source due to privacy/review-stage restrictions, but may be provided separately
Target Hardware	Neuromorphic hardware is discussed as a future deployment target
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No measured energy result; efficiency is argued through reduced parameter count, binary sparse spike computation, FFT-based token mixing, and suitability for neuromorphic hardware
Latency / Memory / Complexity Evidence	Reports reduced parameter count of about 25% / 24%; ASFNO increases FLOPs from 2.79G to 3.19G in one ablation, about 15% overhead; block-diagonal weight matrix reduces parameter complexity; direct latency and memory footprint NR
Comparison with ANN Baseline	Partial. The paper discusses ANN AFNO/Fourier token mixers and static dataset competitiveness, but main experiments compare against SNN and Spiking Transformer baselines rather than matched ANN models
Main Contribution / Key Limitation	Contribution: proposes ASFNOformer, a frequency-domain Spiking Transformer token mixer that applies FFT across spatial and temporal dimensions, uses block-diagonal adaptive spectral mixing, and introduces LWP-LIF for improved temporal adaptability. Limitation: no hardware deployment or measured energy; several training details are NR; code/data are not openly released; static-dataset gains are modest
RQ Mapping	Primary: RQ2 and RQ3; Secondary: RQ1, RQ4, RQ5

P113

Hybrid Attention Spike Transformer

Full Citation	Fan, X., Zhang, H., & Zhang, Y. (2025). Hybrid Attention Spike Transformer. IET Cyber-Systems and Robotics, 7, e70010. https://doi.org/10.1049/csy2.70010
University	Zhejiang University
Country	China
Publication Year	2025
Publication Venue	IET Cyber-Systems and Robotics
Publisher	Wiley
DOI / URL	https://doi.org/10.1049/csy2.70010
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Image classification on static and neuromorphic datasets using a hybrid-attention Spiking Transformer
Data Modality	Static RGB images and DVS/event-camera data
Signal Dimensionality	Static 2D image data and 2D spatiotemporal DVS data represented as T × C × H × W
Signal Characteristics	Static images use C = 3; DVS data use C = 2; input image sequence is transformed into spike tokens X ∊ R^(T×N×D) through convolutional patch splitting
Dataset Name	Tiny ImageNet, CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public static image and neuromorphic/event-based vision benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Standard dataset protocols implied; exact train/test split and class balance NR
Hybrid Topology	HAST follows a ViT-like architecture with Convolutional Patch Splitting, stacked hybrid Transformer encoder blocks, Composite Tokens Self-Attention, Channel Attention Feed Forward, residual connections, and classification head
SNN Component	LIF neurons, spike tokens, convolution + LIF layers, CTSA spike attention, CAFF spike feedforward block, membrane-potential residual shortcut
Neuron Model	LIF
Transformer Component	Hybrid Attention Spike Transformer encoder; Composite Tokens Self-Attention; Channel Attention Feed Forward; ViT-style classification head
Attention Type	Hybrid attention combining composite-token self-attention and channel attention; softmax-free spike self-attention
SNN–Transformer Interface Mechanism	CPS converts analog images/event frames into spike tokens using convolution and LIF neurons; CTSA adds channel tokens to patch/class tokens, generates spike-form Q/K/V through Linear + LIF, and performs softmax-free attention; CAFF uses local convolutional features and channel attention to weight class tokens
Model Depth / Scale	HASTS, HASTM, and HASTL variants; 4 timesteps for static datasets; 10 or 16 timesteps for DVS datasets; patch size 4 for CIFAR, patch size 16 for DVS experiments
Parameter Count	Reported as model size: HASTS 6.00 MB, HASTM 10.70 MB, HASTL 23.50 MB; exact number of parameters NR
Design / Contribution Type	New hybrid-attention Spiking Transformer; Convolutional Patch Splitting; Composite Tokens Self-Attention; Channel Attention Feed Forward; convolution-enhanced local inductive bias for training spike Transformers from scratch
Input Encoding Method	CPS acts as both patch embedding and spike encoder; it uses affine transformation, convolution, and LIF neurons to convert image/event inputs into spike-form tokens
Encoding Parameters	LIF threshold Uth = 0.5; reset potential Ureset = 0; decay factor τ = 0.25; surrogate function = Sigmoid; surrogate window size a = 1; patch size 4 for CIFAR and 16 for DVS
Encoding Learned or Fixed	Partially learned: convolutional layers, affine parameters α and β, attention modules, channel attention, and classifier are learned; LIF spike-generation rule is fixed
Event / Spike Representation	Binary spike signals, spike tokens, class tokens, patch tokens, channel tokens, spike-form Q/K/V matrices
Temporal Resolution	Static datasets use 4 timesteps; CIFAR10-DVS and DVS128 Gesture use 10 and 16 timestep settings
Output Decoding Method	Final encoded tokens are passed to a fully connected classification head

Encoding / Temporal Information Analysis	Partial. The paper analyzes spike sparsity, information loss in spike Transformer attention, channel-token fusion, local information extraction, and timestep effects on DVS performance, but does not deeply analyze spike temporal information preservation
Training Paradigm	Supervised direct SNN training from scratch
Training Method	Surrogate-gradient-based direct training u
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	Python 3.10; PyTorch 2.0.1+cu118; SpikingJelly 0.0.0.0.14; timm 0.9.12; 200 training epochs for all datasets; other optimizer, learning rate, batch size, and augmentation settings NR
Primary Performance Metric	Top-1 classification accuracy
Primary Metric Value	CIFAR-10: 96.04%; CIFAR10-DVS: 82.80%; DVS128 Gesture: 99.10%; QKFormer-based ASFNO on CIFAR10-DVS: 85.5%
Secondary Metrics	Parameter count, timestep count, FLOPs, accuracy improvement, frequency-domain visualization, standard deviation within 0.2%
Baseline Models Compared	TET, tdBN, TEBN, Real Spike, DSR, Spikformer, Spikingformer, CML, S-Transformer, QKFormer
Performance Gain Over Baseline	On CIFAR-10, ASFNOformer improves over Spikformer from 95.51% to 96.04% while reducing parameters from 9.32M to 9.12M. On CIFAR10-DVS, it improves over Spikformer from 80.90% to 82.80% while reducing parameters from 2.57M to 2.12M. With QKFormer baseline, ASFNO improves CIFAR10-DVS accuracy from 84.0% to 85.5%
Statistical Validation	Paper states models converged across random seeds and standard deviation of error was within 0.2%; no formal statistical significance test reported
Ablation Study	Yes. Includes ASFNO vs Spikformer on CIFAR10-DVS, ASFNO integrated into QKFormer, FLOPs comparison, feature-map visualization, and frequency-domain visualization
Training Framework / Code Availability	PyTorch, SpikingJelly, timm; paper states code/data are not open-source due to privacy/review-stage restrictions, but may be provided separately
Target Hardware	Neuromorphic hardware is discussed as a future deployment target
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No measured energy result; efficiency is argued through reduced parameter count, binary sparse spike computation, FFT-based token mixing, and suitability for neuromorphic hardware
Latency / Memory / Complexity Evidence	Reports reduced parameter count of about 25% / 24%; ASFNO increases FLOPs from 2.79G to 3.19G in one ablation, about 15% overhead; block-diagonal weight matrix reduces parameter complexity; direct latency and memory footprint NR
Comparison with ANN Baseline	Partial. The paper discusses ANN AFNO/Fourier token mixers and static dataset competitiveness, but main experiments compare against SNN and Spiking Transformer baselines rather than matched ANN models
Main Contribution / Key Limitation	Contribution: proposes ASFNOformer, a frequency-domain Spiking Transformer token mixer that applies FFT across spatial and temporal dimensions, uses block-diagonal adaptive spectral mixing, and introduces LWP-LIF for improved temporal adaptability. Limitation: no hardware deployment or measured energy; several training details are NR; code/data are not openly released; static-dataset gains are modest
RQ Mapping	Primary: RQ2 and RQ3; Secondary: RQ1, RQ4, RQ5

P114
Dynamic SpikFormer: Low-Latency & Energy-Efficient Spiking Neural Networks with Dynamic Time Steps for Vision Transformers

Full Citation	Datta, G., Liu, Z., Li, A., & Beerel, P. A. (2025). Dynamic SpikFormer: Low-Latency & Energy-Efficient Spiking Neural Networks with Dynamic Time Steps for Vision Transformers. In Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2025). IEEE. https://doi.org/10.1109/ICASSP49660.2025.10888589
University	Case Western Reserve University
Country	United States
Publication Year	2025
Publication Venue	IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2025
Publisher	IEEE
DOI / URL	https://doi.org/10.1109/ICASSP49660.2025.10888589
Publication Type	Conference paper
Database Source	WoS
Paper Category	Efficiency-focused
E&QA; Score	4
Application Domain	Computer Vision and Image Understanding
Specific Task	Image classification using Spikformer-based spiking Vision Transformers with dynamic time-step allocation
Data Modality	Static image data
Signal Dimensionality	2D image data processed as patch/token sequences in Vision Transformer architecture
Signal Characteristics	Static images are processed by compact ViT/Spikformer architecture; spike activity is dynamically masked across timesteps to reduce redundant computation
Dataset Name	CIFAR-10, CIFAR-100, ImageNet
Dataset Type	Public image classification benchmark datasets
Dataset Size	NR
Dataset Balance / Split	Standard benchmark protocol implied; exact train/test split and class balance NR
Hybrid Topology	Spikformer-based spiking Vision Transformer with Spiking Patch Splitting, spiking multi-head self-attention, spiking MLP, LIF layers, and Dynamic Time Step Spiking mask layers
SNN Component	LIF neurons, spiking blocks, spike outputs, binary timestep masks, surrogate-gradient training, dynamic timestep gating
Neuron Model	LIF
Transformer Component	Spikformer / compact ViT architecture; multi-head self-attention block; MLP block; patch splitting module
Attention Type	Spiking multi-head self-attention without softmax; dot-product attention with scaling

SNN–Transformer Interface Mechanism	Linear/convolutional layers generate spike activations through LIF layers; the DTSS layer applies trainable binary timestep masks to filter LIF spike outputs, allowing each ViT module to use a different effective number of timesteps
Model Depth / Scale	Spikformer-4-384 for CIFAR-10/100; Spikformer-8-512 for ImageNet; maximum timestep setting up to 4 or 6 depending on experiment; average timestep dynamically reduced
Parameter Count	NR
Design / Contribution Type	Dynamic Time Step Spiking training framework; trainable timestep score and binary mask; layer/module sensitivity analysis; mask loss for latency–accuracy trade-off
Input Encoding Method	Direct image input through spiking patch-splitting/embedding layer; subsequent LIF layers generate spikes
Encoding Parameters	Maximum timestep Tmax; trainable timestep parameter tensor TP; score tensor TS; binary timestep mask TM; mask loss weight λm; initial timestep settings from 1 to 4 and maximum timestep settings up to 6
Encoding Learned or Fixed	Partially learned: timestep masks/scores and network weights are learned; LIF spike-generation rule and architectural modifications are fixed
Event / Spike Representation	Binary spikes, timestep masks, masked spike outputs, LIF membrane potentials, sparse spike activations
Temporal Resolution	Dynamic average timesteps: CIFAR-10 up to 4.79; CIFAR-100 up to 3.81; ImageNet 1.30 in reported comparison
Output Decoding Method	Classification head outputs image class prediction
Encoding / Temporal Information Analysis	Yes. The paper analyzes activation-map distributions and membrane-potential sensitivity across ViT modules, grouping layers into SPS, QKV, attention, and MLP parts to decide where timestep masking should be applied
Training Paradigm	Supervised direct SNN training
Training Method	Surrogate-gradient direct training with Dynamic Time Step Spiking and combined cross-entropy plus mask loss
Surrogate Gradient Function	Surrogate-gradient function for timestep mask and spiking neurons; exact spiking-neuron surrogate function NR
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Total loss = cross-entropy loss + timestep mask loss
Training Hyperparameters	AdamW optimizer; initial learning rate 0.004; batch size 256; mask loss weight examples include 1e–4 and 1e–6; initial timestep values tested from 1 to 4; maximum timestep values tested from 4 to 6
Primary Performance Metric	Classification accuracy and average number of timesteps
Primary Metric Value	CIFAR-10: 95.97% with 4.79 average timesteps; CIFAR-100: 78.90% with 3.81 average timesteps; ImageNet: 68.04% with 1.30 average timesteps
Secondary Metrics	Spiking activity, SOPs, normalized compute energy, normalized memory energy, normalized total energy, latency reduction
Baseline Models Compared	Spikformer, Compact Convolutional Transformer / CCT, Diet-SNN, STBP-tdBN, Hoyer-regularized SNN, DSR, TET, non-spiking ViT
Performance Gain Over Baseline	On CIFAR-100, dynamic Spikformer achieves 78.90% with 3.81 timesteps versus Spikformer 78.85% with 6 timesteps. On CIFAR-10, it achieves 95.97% with 4.79 timesteps versus Spikformer 95.94% with 6 timesteps. On ImageNet, it improves over replicated 1-step Spikformer from 63.12% to 68.04%
Statistical Validation	Benchmark comparison and ablation study reported; no standard deviation, confidence interval, repeated-run analysis, or formal significance test reported
Ablation Study	Full. Includes mask loss combinations over QKV, attention, MLP, and SPS modules; mask-loss weight analysis; initial/max timestep analysis; energy comparison against Spikformer and non-spiking ViT
Training Framework / Code Availability	Framework/code availability NR
Target Hardware	Edge AI and neuromorphic-style hardware with spike routing / timestep skipping support; Loihi-style zero-gating support discussed conceptually
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Analytical/simulation-based energy comparison using normalized compute, memory, and total energy; energy models use FPGA simulation and prior operation-energy estimates
Latency / Memory / Complexity Evidence	Reports dynamic timestep reduction, spiking activity, SOPs, and normalized energy. CIFAR-100 ablation reports 78.90% accuracy, 3.81 timesteps, 6.54% spiking activity, and 319.51M SOPs. Compared with Spikformer, dynamic timestep SNN reduces total energy to 1× baseline while Spikformer uses 2.98× on CIFAR-10 and 1.89× on CIFAR-100
Comparison with ANN Baseline	Yes. Compares normalized energy against non-spiking ViT and discusses accuracy/efficiency trade-off relative to standard ViT-style models
Main Contribution / Key Limitation	Contribution: proposes Dynamic SpikFormer, a dynamic timestep training framework that assigns different effective timesteps to ViT modules based on sensitivity, reducing latency and energy while preserving accuracy. Limitation: evaluated only on classification benchmarks; no real neuromorphic chip deployment; energy values are normalized estimates rather than measured hardware power
RQ Mapping	Primary: RQ4 and RQ5; Secondary: RQ2, RQ3, RQ1

P115

Event-Driven Spatiotemporal Computing for Robust Flight Arrival Time Prediction: A Probabilistic Spiking Transformer Approach

Full Citation	Chen, Q., & Le, M. (2026). Event-Driven Spatiotemporal Computing for Robust Flight Arrival Time Prediction: A Probabilistic Spiking Transformer Approach. Aerospace, 13(2), 203. https://doi.org/10.3390/aerospace13020203
University	Nanjing University of Aeronautics and Astronautics
Country	China
Publication Year	2026
Publication Venue	Aerospace
Publisher	MDPI
DOI / URL	https://doi.org/10.3390/aerospace13020203
Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Communication, Cybersecurity, and Network Intelligence
Specific Task	Estimated Time of Arrival prediction for aircraft in terminal airspace, with probabilistic uncertainty quantification

Data Modality	ADS-B trajectory data, aircraft kinematic time-series, neighboring-aircraft states, and meteorological features
Signal Dimensionality	Multivariate 1D temporal trajectory sequence with multi-agent spatial context
Signal Characteristics	Each aircraft state includes longitude, latitude, barometric altitude, ground speed, heading, and vertical rate; sequence window length is 60 s; data are resampled to 1 Hz; neighbor aircraft are selected within 50 km; meteorological features include wind speed, wind direction, and temperature
Dataset Name	Real-world ADS-B arrival trajectory dataset from a major Terminal Control Area in Eastern China
Dataset Type	Real-world operational aviation trajectory dataset
Dataset Size	12,450 arrival trajectories sampled from 36 days across January–December 2018
Dataset Balance / Split	Chronological split: 80% training, 10% validation, 10% testing; three days randomly sampled per month across one year
Hybrid Topology	Probabilistic Spiking Transformer with Adaptive Spiking Temporal Encoding, Spiking Encoder with Event-Driven Self-Attention, Multi-Aircraft Cross-Attention, and Gaussian probabilistic decoder
SNN Component	LIF neurons, adaptive spike generation, binary spike masks, sparse spike trains, event-driven self-attention gating, surrogate-gradient learning
Neuron Model	LIF
Transformer Component	Spiking Transformer encoder, Event-Driven Self-Attention, Feed-Forward Network, Multi-Aircraft Cross-Attention, Gaussian output head
Attention Type	Event-Driven Self-Attention; Distance-Biased Multi-Aircraft Cross-Attention; logits-level temporal gating
SNN–Transformer Interface Mechanism	Continuous ADS-B features are projected into latent currents using learnable linear projection, converted into spike trains through LIF neurons, then used as spike masks for event-driven attention and trajectory feature extraction
Model Depth / Scale	Sliding window L = 60; model dimension D = 128; Spiking Encoder has 2 layers; K = 10 nearest neighboring aircraft; LIF parameters τm = 2.0 and Vth = 0.5
Parameter Count	NR
Design / Contribution Type	New Probabilistic Spiking Transformer for ETA regression; adaptive spiking temporal encoding; event-driven self-attention; distance-biased multi-aircraft cross-attention; Gaussian uncertainty head
Input Encoding Method	Adaptive Spiking Temporal Encoding using learnable linear projection from continuous trajectory features to latent currents, followed by LIF spike generation
Encoding Parameters	Window length 60 s; 1 Hz sampling; model dimension 128; LIF τm = 2.0; threshold Vth = 0.5; surrogate-gradient steepness α = 4.0; K = 10 neighbors within 50 km
Encoding Learned or Fixed	Partially learned: input projection weights, Transformer parameters, MACA parameters, and Gaussian head are learned; LIF firing rule and spike threshold mechanism are fixed
Event / Spike Representation	Sparse binary spike trains, LIF membrane potentials, spike activity masks, event-driven temporal activity masks, sparse spatiotemporal trajectory embeddings
Temporal Resolution	ADS-B trajectories resampled at 1 Hz; 60 s input window
Output Decoding Method	Gaussian decoder head outputs predicted ETA mean μ and uncertainty σ; final ETA prediction is derived from the probabilistic distribution
Encoding / Temporal Information Analysis	Yes. The paper analyzes input starvation, dead-neuron risk, firing rate, spike sparsity, temporal trajectory dynamics, prediction-error evolution over remaining time, and uncertainty calibration
Training Paradigm	Supervised probabilistic regression
Training Method	Direct SNN training with surrogate gradients and end-to-end optimization of adaptive spiking projection, Spiking Encoder, MACA, and Gaussian decoder
Surrogate Gradient Function	Sigmoid-style surrogate gradient: $\partial S/\partial V \approx 1 / (1 + \alpha\|V - Vth\|)^2$, with $\alpha = 4.0$
ANN-to-SNN Conversion Method	N/A
Optimizer	AdamW
Loss Function	Hybrid probabilistic loss: negative log-likelihood plus MAE; total training objective includes Gaussian likelihood-based ETA uncertainty modeling
Training Hyperparameters	PyTorch 2.1; NVIDIA RTX 4090 GPU; AdamW optimizer; weight decay 1 × 10■■; initial learning rate 1 × 10■³; cosine annealing scheduler; 100 epochs; D = 128; 2 encoder layers
Primary Performance Metric	MAE, RMSE, CRPS, and PICP
Primary Metric Value	PST achieves MAE 49.27 s, RMSE 71.15 s, CRPS 33.85, and PICP 97.12% on the held-out test set; ablation section reports full-model MAE 48.85 s and RMSE 70.42 s
Secondary Metrics	Prediction interval coverage, PIT calibration, firing rate, inference latency, theoretical energy, robustness under wind speed, conflict-condition error, traffic-density error
Baseline Models Compared	LSTM, GRU, TCN, vanilla Transformer, GAT-LSTM, ANN-Transformer / non-spiking Transformer
Performance Gain Over Baseline	PST reduces MAE from 128.45 s for LSTM to 49.27 s, approximately 61.6% improvement; improves over vanilla Transformer from 68.25 s to 49.27 s; improves over GAT-LSTM from 85.6 s to 49.27 s
Statistical Validation	Chronological train/validation/test split, ablation studies, robustness analysis, and probabilistic calibration reported; no standard deviation, confidence interval across repeated training runs, or formal statistical significance test reported
Ablation Study	Full. Includes w/o Spiking Dynamics, w/o MACA, w/o Distance Bias, w/o Adaptive Encoding, and ANN-Transformer variants; reports MAE, RMSE, CRPS, and firing-rate effects
Training Framework / Code Availability	PyTorch 2.1; NVIDIA RTX 4090; data availability on request; code availability NR
Target Hardware	Neuromorphic and low-power edge aviation/ATC systems discussed conceptually, including Intel Loihi and IBM TrueNorth
Hardware Deployment Status	No real neuromorphic hardware deployment reported; evaluated on GPU with theoretical neuromorphic energy analysis
Energy Evidence	Theoretical energy estimation based on sparse AC operations versus dense MAC operations; PST average firing rate is 13.8%; reported theoretical energy is 2.7% of dense Transformer energy, equivalent to about 37× energy-efficiency improvement
Latency / Memory / Complexity Evidence	GPU inference latency: 1.35 ms/sample on RTX 4090; theoretical complexity $Osnn \approx O(\rho^2 L^2 D + \rho L D^2)$; average firing rate 0.138; input projection accounts for less than 2.1% of total operations
Comparison with ANN Baseline	Yes. Compares with vanilla Transformer and ANN-Transformer/non-spiking Transformer variants, as well as LSTM, TCN, and GAT-LSTM baselines
Main Contribution / Key Limitation	Contribution: proposes PST for probabilistic ETA prediction, resolving SNN regression input starvation through adaptive spiking encoding and improving multi-aircraft interaction modeling through distance-biased MACA. Limitation: dataset is from one regional TMA; energy results are theoretical rather than measured on neuromorphic hardware; current model predicts ETA only, not full 4D trajectory waypoints
RQ Mapping	Primary: RQ1 and RQ2; Secondary: RQ3, RQ4, RQ5

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Enhancing Robustness Against Adversarial Attacks in Multimodal Emotion Recognition With Spiking Transformers

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Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Computer Vision and Image Understanding
Specific Task	Robust multimodal emotion classification under no attack and adversarial attacks including FGSM, BIM, and PGD
Data Modality	Facial image/video, speech/audio, text, and EEG signals represented as heatmap features
Signal Dimensionality	Multimodal 2D heatmap representations derived from image/video attention maps, speech features, text features, and EEG time-frequency maps
Signal Characteristics	Video/image/text/audio/EEG features are converted into heatmaps; EEG uses STFT to transform 1D time-series into 2D representations; wavelet thresholding aligns multimodal heatmaps and suppresses high-frequency noise; adversarial perturbations are evaluated on the input features
Dataset Name	MELD; MER Tri-modal dataset; MER Quadri-modal dataset
Dataset Type	Public / constructed multimodal emotion recognition datasets; code and data stated as available through Zenodo
Dataset Size	MELD: 1,433 dialogue segments, approximately 13,000 utterances/dialogues; class counts include anger 1,607, disgust 351, fear 358, happiness 2,308, neutrality 6,436, sadness 1,002, surprise 1,636. MER sizes are reported inconsistently in the paper: Tri-modal and Quadri-modal settings are described with 2,160/194,400 and 2,781/166,860 original/fused samples, but the text appears to swap these values in different paragraphs
Dataset Balance / Split	MELD has imbalanced emotion categories; MER experiments use 80% training and 20% validation; exact MELD train/validation/test split NR
Hybrid Topology	SPSNCVT pipeline: multimodal heatmap feature extraction, wavelet threshold alignment, cross-modal semantic fusion, Spiking Convolutional Patch Embedding, Spiking Convolutional Condition Position Encoding, Spiking Self-Attention, convolutional MLP, and head classifier
SNN Component	Sliding Parallel Spiking Neuron, LIF-based spike mechanism, spiking patch embedding, spiking convolutional position encoding, spiking self-attention, spiking MLP, spike raster/excitation dynamics
Neuron Model	Sliding Parallel Spiking Neuron based on LIF; comparison also includes CLIF, Izhikevich, KLIF, MultiStep LIF, SimpleLIF, and GLIF
Transformer Component	Spiking Convolutional Vision Transformer; Vision Transformer feature extraction; Spiking Self-Attention; convolutional Transformer fusion block; head classification module
Attention Type	Weighted cross-modal self-attention; Spiking Self-Attention; spiking convolutional attention; multimodal adaptive attention weighting
SNN–Transformer Interface Mechanism	Multimodal heatmaps are fused and copied across T timesteps, then passed through SPSN-based spiking patch embedding; Q, K, and V are generated using SPSN plus convolution/batch-normalization layers, and attention is computed as spike-enhanced QK■V with convolutional normalization
Model Depth / Scale	SPSNCVT uses repeated Spiking Transformer blocks denoted by L ×; input is copied across T timesteps; exact number of layers, heads, embedding dimension, and timesteps NR
Parameter Count	NR
Design / Contribution Type	New SPSNCVT framework for adversarially robust multimodal emotion recognition; heatmap-based modality unification; multiscale wavelet alignment; sliding parallel spiking neuron classifier; convolutional Spiking Transformer fusion
Input Encoding Method	Each modality is converted into heatmap features using pretrained feature extractors: VideoMAE for video, Wav2Vec2 for speech, CLIP for text/image pairs, ViT for images, and STFT + ViT-style extraction for EEG; fused heatmaps are then spike-encoded by SPSN modules
Encoding Parameters	ViT heatmap output: [1, 12, 197, 197]; Wav2Vec2 output: [1, 12, 152, 152]; CLIP image output: [1, 16, 257, 257]; CLIP text heatmap dimension depends on text length; six wavelet threshold functions are evaluated; threshold parameter λ; timestep T and sliding-window size k defined but exact numerical values NR
Encoding Learned or Fixed	Partially learned: pretrained feature extractors and SPSNCVT classifier weights are learned/used; wavelet thresholding and heatmap construction are preprocessing/alignment mechanisms; SPSN weights within sliding window are learned/shared
Event / Spike Representation	Spike outputs, membrane potentials, spike raster maps, SPSN spike sequences, heatmap-to-spike representations, spike-form Q/K/V
Temporal Resolution	SPSN uses a sliding temporal window of size k; multimodal classifier copies input across T timesteps; exact timestep value NR
Output Decoding Method	Global Average Pooling followed by fully connected layer, Mish activation, and final fully connected classification layer
Encoding / Temporal Information Analysis	Yes. The paper analyzes spike raster changes under original and adversarial inputs, excitation-rate dynamics across emotion classes, sliding-window temporal dependency capture, and the effect of adversarial attacks on neuron firing patterns
Training Paradigm	Supervised learning for multimodal emotion classification and adversarial robustness evaluation
Training Method	End-to-end training/evaluation of SPSNCVT classifier with multimodal heatmap fusion; adversarial robustness tested under FGSM, BIM, and PGD attacks
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	Label Smoothing Cross-Entropy for training; standard Cross-Entropy Loss for validation

Training Hyperparameters	Label smoothing parameter λ is defined but numerical value NR; MER split 80% training / 20% validation; other optimizer, learning rate, batch size, epoch count, and hardware details NR
Primary Performance Metric	Accuracy and F1 score under clean and adversarial attack settings
Primary Metric Value	MELD: F1 = 81.56% for 3-class and 70.95% for 7-class emotion recognition. SPSNCVT clean accuracy = 71.98%; under attacks: PGD 69.79%, FGSM 70.21%, BIM 70.85%. MER Tri-modal spiking-attention model: 80.99% clean accuracy. MER Quadri-modal spiking-attention model: 86.34% clean accuracy and 86.95% under FGSM
Secondary Metrics	MAE, correlation, inference time, loss, spike firing/excitation patterns, robustness drop under attacks
Baseline Models Compared	CNN, KETransformer, ConGCN, DialogueRNN, Transformer-based models, attention-based models, RoBERTa, RoBERTa DialogueRNN, COSMIC, Mamba, Spiking Transformer, conventional SNN, and multiple LIF neuron variants
Performance Gain Over Baseline	Abstract reports SPSNCVT gains of 3.60%–4.01% and 7.03%–13.73% over baseline models under adversarial attacks. In clean MELD 7-class setting, SPSNCVT reaches 70.95% F1, higher than COSMIC at 63.21 and RoBERTa DialogueRNN at 63.61. Under PGD, SPSNCVT keeps 69.79% accuracy, while CNN drops to 53.37%
Statistical Validation	Benchmark comparison and ablation analysis reported; no standard deviation, confidence interval, repeated-run statistics, or formal significance test reported
Ablation Study	Full. Ablates wavelet threshold module, spiking convolutional condition position encoding, and SPSN. Full model reports Acc-7 71.98%, F1 70.95%, MAE 0.506, Corr 0.631, and inference time 1.3 ms
Training Framework / Code Availability	Code and data are stated as available at Zenodo; exact framework NR
Target Hardware	Resource-constrained real-time emotion recognition systems; neuromorphic chips discussed as suitable future deployment hardware
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Qualitative/indirect. The paper claims improved energy efficiency due to sparse spiking computation and SPSN efficiency, but no measured energy value or neuromorphic hardware power result is reported
Latency / Memory / Complexity Evidence	Inference time from MELD ablation: full model 1.3 ms; without wavelet threshold 1.1 ms; without position encoding 1.2 ms; without SPSN 2.5 ms. Parameter count and memory footprint NR
Comparison with ANN Baseline	Yes. Compares against CNN, Transformer, Mamba, RoBERTa-based models, COSMIC, and other non-spiking multimodal emotion recognition baselines
Main Contribution / Key Limitation	Contribution: proposes SPSNCVT, combining multimodal heatmap feature extraction, wavelet alignment, SPSN spiking computation, and convolutional Spiking Transformer fusion to improve robustness against adversarial attacks in multimodal emotion recognition. Limitation: performance may decline under high-intensity attacks; no measured neuromorphic hardware deployment; several training hyperparameters and parameter counts are NR
RQ Mapping	Primary: RQ1 and RQ2; Secondary: RQ3, RQ4, RQ5

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Auto-Spikformer: Spikformer architecture search

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Publication Type	Journal article
Database Source	WoS
Paper Category	Application-focused
E&QA; Score	4.5
Application Domain	Event-Based / Neuromorphic Vision
Specific Task	Architecture search for efficient Spikformer models on static and neuromorphic image classification datasets
Data Modality	Static RGB images and neuromorphic/event-camera data
Signal Dimensionality	2D image data and 2D spatiotemporal event-stream data represented through spike/token sequences
Signal Characteristics	CIFAR images are 32 × 32; CIFAR10-DVS and DVS128 Gesture use 128 × 128 event-based data; SNN models use sparse spike activations, firing rates, timesteps, and spike-based accumulate operations
Dataset Name	CIFAR-10, CIFAR-100, CIFAR10-DVS, DVS128 Gesture
Dataset Type	Public static image and neuromorphic/event-based vision benchmark datasets
Dataset Size	CIFAR-10/100: 50,000 training and 10,000 test images; CIFAR10-DVS: 9,000 training and 1,000 test images; DVS128 Gesture: 11 hand-gesture categories
Dataset Balance / Split	CIFAR and CIFAR10-DVS standard train/test splits reported; DVS128 Gesture split details NR
Hybrid Topology	Auto-Spikformer uses Spikformer as the base architecture and searches both Transformer architecture parameters and SNN inner parameters through a one-shot supernet and evolutionary search
SNN Component	LIF neurons, Spiking Patch Splitting, Spiking Self-Attention, spike sequences, threshold search, decay search, timestep search, firing-rate optimization
Neuron Model	LIF
Transformer Component	Spikformer / Spiking Vision Transformer; Transformer encoder blocks; SSA block; SPS embedding module; MLP block
Attention Type	Spiking Self-Attention
SNN–Transformer Interface Mechanism	Static images are encoded into spike form through the first SNN convolution/SPS module; Spikformer blocks process spike tokens through SSA; Auto-Spikformer jointly searches Transformer dimensions and SNN neuron parameters such as threshold, decay, and timestep
Model Depth / Scale	Search space includes embedding dimension, MLP ratio, head number, depth, threshold, decay, and timestep. STs: embed 336–384, depth 2–4. STl: embed 336–480, depth 2–6. SNN search: threshold 0.6–2.0, decay 1.25–10, timestep 2–4
Parameter Count	CIFAR searched models: Auto-Spikformer STs1 4.69M, STs2 4.64M, STl1 7.09M, STl2 9.20M, STl3 8.46M. Neuromorphic optimal model: 2.48M

Design / Contribution Type	One-shot Spiking Transformer Architecture Search; Discrete Spiking Parameters Search; weight-entanglement supernet training; alternate-choice SNN parameter training; energy–accuracy balanced fitness function
Input Encoding Method	Direct image/event input through Spiking Patch Splitting and SNN first-layer spike encoding; SNN inner parameters are searched to control spike generation and firing rate
Encoding Parameters	LIF baseline uses $\tau = 2$ and threshold uth = 0.5; SNN search range: threshold 0.6–2.0 with step 0.2, decay τ 1.25–10 with step 0.25, timestep 2–4 with step 1. CIFAR experiments use timestep 4; neuromorphic experiments use timestep 16
Encoding Learned or Fixed	Partially learned/searched: network weights are learned through supernet training, while SNN inner parameters are selected through discrete evolutionary search; LIF spike rule is fixed
Event / Spike Representation	Binary spike sequences, spike feature maps, firing-rate-based representations, spike-form inputs to SSA, DVS event streams
Temporal Resolution	CIFAR searched models use 4 timesteps; CIFAR10-DVS and DVS128 Gesture use 16 timesteps
Output Decoding Method	Classification head outputs image/event class prediction
Encoding / Temporal Information Analysis	Yes. The paper analyzes channel redundancy using SSIM, firing rate, threshold/decay effects, timestep choice, Pareto energy–accuracy trade-offs, and spike-operation energy
Training Paradigm	Supervised direct SNN training combined with one-shot neural architecture search
Training Method	Supernet training based on Spikformer training strategy; weight entanglement for Transformer search space; alternate-choice method for SNN search space; evolutionary search for final subnet selection
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	NR
Training Hyperparameters	Supernet trained for 1000 epochs. Evolutionary search starts with 50 random SNN parameter sets, selects top 20 parents, uses mutation probability 0.4 and crossover probability 0.3, and runs for 20 generations. FAEB uses $\alpha = 0.5$
Primary Performance Metric	Top-1 classification accuracy and theoretical energy consumption
Primary Metric Value	Best CIFAR results: Auto-Spikformer STl3 achieves 96.39% on CIFAR-10 and 78.22% on CIFAR-100 with 8.46M parameters and 0.925 µJ energy. Neuromorphic results: 81.2% on CIFAR10-DVS and 98.6% on DVS128 Gesture
Secondary Metrics	Parameter count, timestep, firing rate, energy in µJ, Pareto frontier, SOPs, CIFAR and DVS accuracy
Baseline Models Compared	Hybrid Training, Diet-SNN, STBP, STBP NeuNorm, TSSL-BP, STBP-tdBN, TET, AutoSNN, SNASNet, SpikeDHSD, ANN ResNet-19, Transformer-4-384, Spikformer, AutoST, LIAF-Net, TA-SNN, Rollout, DECOLLE, tdBN, PLIF, SEW-ResNet, Dspike, SALT, DSR
Performance Gain Over Baseline	Auto-Spikformer STs2 reaches 77.91% on CIFAR-100 with 4.64M parameters and 0.509 µJ, compared with Spikformer-4-384 at 77.86%, 9.32M parameters, and 0.952 µJ. Auto-Spikformer STl3 improves CIFAR-10/CIFAR-100 accuracy to 96.39%/78.22%. On neuromorphic datasets, Auto-Spikformer improves CIFAR10-DVS from Spikformer's 80.9% to 81.2% and DVS128 Gesture from 98.3% to 98.6%
Statistical Validation	Search analysis includes Pareto-front evaluation and Kendall's tau rank correlations for energy–accuracy behavior; no standard deviation, confidence interval, repeated-run statistics, or formal significance test reported
Ablation Study	Strong search/ablation analysis. Includes DSPS effectiveness, FAEB vs random sampling, SNN-parameter-only search, Transformer + SNN mixed search, Pareto-front comparison, and CIFAR/neuromorphic benchmark comparisons
Training Framework / Code Availability	Computing support from Pengcheng Cloudbrain; software framework NR; code availability NR
Target Hardware	Energy-efficient SNN / neuromorphic hardware discussed conceptually; theoretical 45 nm CMOS energy model used
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	Theoretical energy estimation based on SOPs and 45 nm CMOS operation costs: MAC = 4.6 pJ and AC = 0.9 pJ; no physical hardware energy measurement
Latency / Memory / Complexity Evidence	Reports parameters, timestep, firing rate, SOPs, and theoretical energy. Example: STs2 uses 4.64M parameters and 0.509 µJ compared with Spikformer-4-384 at 9.32M and 0.952 µJ. Direct latency and memory footprint NR
Comparison with ANN Baseline	Yes. Compares against ANN ResNet-19 and non-spiking Transformer-4-384
Main Contribution / Key Limitation	Contribution: proposes Auto-Spikformer, the first one-shot NAS framework for Spiking Vision Transformers, jointly optimizing Transformer architecture and SNN inner neuron parameters using DSPS and FAEB. Limitation: energy is theoretically estimated rather than measured on hardware; experiments focus mainly on image/event classification; larger-scale benchmark extension is left for future work
RQ Mapping	Primary: RQ2 and RQ5; Secondary: RQ3, RQ4, RQ1

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Binary Event-Driven Spiking Transformer

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Country	China
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Publisher	IJCAI
DOI / URL	https://www.ijcai.org/proceedings/2025/0458.pdf
Publication Type	Conference paper
Database Source	WoS
Paper Category	Efficiency-focused
E&QA; Score	4.5
Application Domain	Neuromorphic Hardware and Edge AI Acceleration
Specific Task	Static image classification and neuromorphic event-based image classification using a binary Spiking Transformer
Data Modality	Static RGB images and neuromorphic/event-camera data
Signal Dimensionality	2D image data and 2D spatiotemporal event-stream data represented as spike/token sequences

Signal Characteristics	Static images and event streams are processed using binary spike activities; weights and attention maps are binarized to 1-bit; CIFAR10-DVS uses event-stream data; ImageNet uses large-scale static image data
Dataset Name	CIFAR-10, CIFAR-100, ImageNet-1K, CIFAR10-DVS
Dataset Type	Public static image classification and neuromorphic/event-based vision benchmark datasets
Dataset Size	CIFAR-10 and CIFAR-100 standard datasets used; ImageNet-1K used; CIFAR10-DVS used. Exact sample counts NR in the paper
Dataset Balance / Split	Standard benchmark splits implied; exact train/test split and class-balance details NR
Hybrid Topology	BESTformer combines binary Spiking Patch Splitting, Binary Spiking Self-Attention, Binary Spiking Transformer Encoder Blocks, reversible encoder connections, classification head, and distillation head
SNN Component	LIF neurons, hard-reset LIF, soft-reset LIF, binary spike activities, binary weights, binary attention maps, Binary Spiking Patch Splitting, Binary Spiking Self-Attention
Neuron Model	LIF; HR-LIF for standard spiking dynamics; SR-LIF for attention-map binarization
Transformer Component	BESTformer; Spikformer-style Transformer-based SNN; Binary Spiking Self-Attention; Binary Spiking Encoder Blocks; reversible Transformer encoder
Attention Type	Binary Spiking Self-Attention; event-driven self-attention; 1-bit binarized attention map
SNN–Transformer Interface Mechanism	Binary spike activities are processed through binary convolution and batch normalization; Query and Key generate the attention map Attn = QK■; the attention map is binarized using SR-LIF and then used in Binary Spiking Self-Attention
Model Depth / Scale	BESTformer-2-384, BESTformer-4-384, BESTformer-6-384, BESTformer-8-512, and BESTformer-2-256 are evaluated; ImageNet uses BESTformer-8-512; CIFAR10-DVS uses BESTformer-2-256
Parameter Count	Parameter count NR; model size reported: BESTformer-2-384 0.73–0.86 MB; BESTformer-4-384 1.18–1.31 MB; BESTformer-8-512 5.57 MB; BESTformer-2-256 0.34 MB
Design / Contribution Type	Binary Event-Driven Spiking Transformer; 1-bit weight binarization; 1-bit attention-map binarization; Coupled Information Enhancement; reversible framework; information-enhancement distillation
Input Encoding Method	Static images/event inputs are processed through Binary Spiking Patch Splitting and spike-based Transformer blocks; 1-bit spike activities serve as inputs to binary convolution and B-SSA operations
Encoding Parameters	Weights are binarized to {-1, +1}; spike activities are 1-bit; attention maps are binarized by SR-LIF; Table 1 example uses N = 1, T = 4, threshold Vth = 1, and time constant $\tau = 0.5$. Main experiments use time steps 1, 2, 4, 10, or 16 depending on dataset/model
Encoding Learned or Fixed	Partially learned: model weights, layer-wise attention binarization factors λ, classifier, and distillation heads are learned; binary spike rules and binarization functions are fixed mechanisms
Event / Spike Representation	1-bit binary spike activities, 1-bit binary weights, binarized attention maps, membrane potentials, spike-form Q/K/V features
Temporal Resolution	CIFAR-10/100: mainly 2 or 4 time steps; ImageNet: 1 or 4 time steps; CIFAR10-DVS: 10 or 16 time steps
Output Decoding Method	Dual-head output: classification head predicts ground-truth class, while distillation head learns from the full-precision teacher model's hard decision
Encoding / Temporal Information Analysis	Yes. The paper analyzes limited representation capability caused by binarization, value-set size of feature maps, entropy preservation through reversible connections, and attention-map binarization using SR-LIF
Training Paradigm	Supervised direct SNN training with knowledge distillation from a full-precision teacher
Training Method	Direct training of binary Spiking Transformer with Coupled Information Enhancement, reversible encoder framework, and information-enhancement distillation
Surrogate Gradient Function	NR
ANN-to-SNN Conversion Method	N/A
Optimizer	NR
Loss Function	Global loss: average of cross-entropy loss for classification head and cross-entropy loss for distillation head using the teacher model's hard decision
Training Hyperparameters	Detailed implementation settings are stated to be in supplementary materials; optimizer, learning rate, batch size, and epoch count NR in main paper
Primary Performance Metric	Classification accuracy, model size, SOPs, and NS-ACE
Primary Metric Value	CIFAR-10: BESTformer-4-384 achieves 95.73% with 1-bit weights, 4 timesteps, and 1.18 MB. CIFAR-100: BESTformer-4-384 achieves 79.80% with 1-bit weights, 4 timesteps, and 1.31 MB. ImageNet: BESTformer-8-512 achieves 63.46% with 1-bit weights and 4 timesteps; 62.39% with 1 timestep. CIFAR10-DVS: BESTformer-2-256 achieves 80.80% with 16 timesteps
Secondary Metrics	Weight bits, time steps, model size, NS-ACE, SOPs, information representation capability, ablation accuracy gain
Baseline Models Compared	CBP-QSNN / Yoo and Jeong, BitSNNs / Hu et al., Quantized Spikformer / Shen et al., SEW-ResNet18, SEW-ResNet34, Spikformer-4-384, Spikformer-8-512, Wide-7B-Net, full-precision Spikingformer
Performance Gain Over Baseline	On ImageNet with 1 timestep, BESTformer-8-512 reaches 62.39%, improving by 7.85% over 1-bit Quantized Spikformer-8-512 at 54.54%. On CIFAR-100, BESTformer-4-384 reaches 79.80%, improving over 2-bit Quantized Spikformer-4-384 at 75.91%. On CIFAR10-DVS, BESTformer-2-256 improves over 1-bit Spikformer-2-256 from 79.80% to 80.80%
Statistical Validation	Benchmark comparison and ablation study reported; no standard deviation, confidence interval, repeated-run statistics, or formal statistical significance test reported
Ablation Study	Full. Includes reversible framework, information-enhancement distillation, dual-head configuration, CIE effects across BESTformer-2-384/4-384/6-384, and information-representation analysis across encoder depth
Training Framework / Code Availability	Code available at GitHub: CaoHLin/BESTFormer; exact training framework NR
Target Hardware	Resource-constrained edge devices and neuromorphic computing environments; neuromorphic hardware motivation includes TrueNorth, Loihi, and Tianjic
Hardware Deployment Status	No real neuromorphic hardware deployment reported
Energy Evidence	No measured chip/FPGA energy; efficiency is evaluated using model size, SOPs, and NS-ACE. ImageNet BESTformer-8-512 reports 5.57 MB model size, 2.13G SOPs, and 2.13G NS-ACE
Latency / Memory / Complexity Evidence	BESTformer-8-512 on ImageNet reduces model size from full-precision 59.36 MB to 5.57 MB and NS-ACE from 104.32G to 2.13G; direct measured latency and memory footprint beyond model size NR
Comparison with ANN Baseline	Partial. Compares against full-precision Transformer-based SNN counterpart and quantized SNN baselines; no direct matched ANN Transformer baseline in the main tables
Main Contribution / Key Limitation	Contribution: proposes BESTformer, a 1-bit binary event-driven Spiking Transformer that binarizes both weights and attention maps, and introduces CIE to recover information lost from binarization through reversible modeling and distillation. Limitation: energy/efficiency evidence is metric-based rather than measured on physical hardware; training details are mostly in supplementary materials; experiments focus on classification tasks
RQ Mapping	Primary: RQ2 and RQ5; Secondary: RQ3, RQ4, RQ1